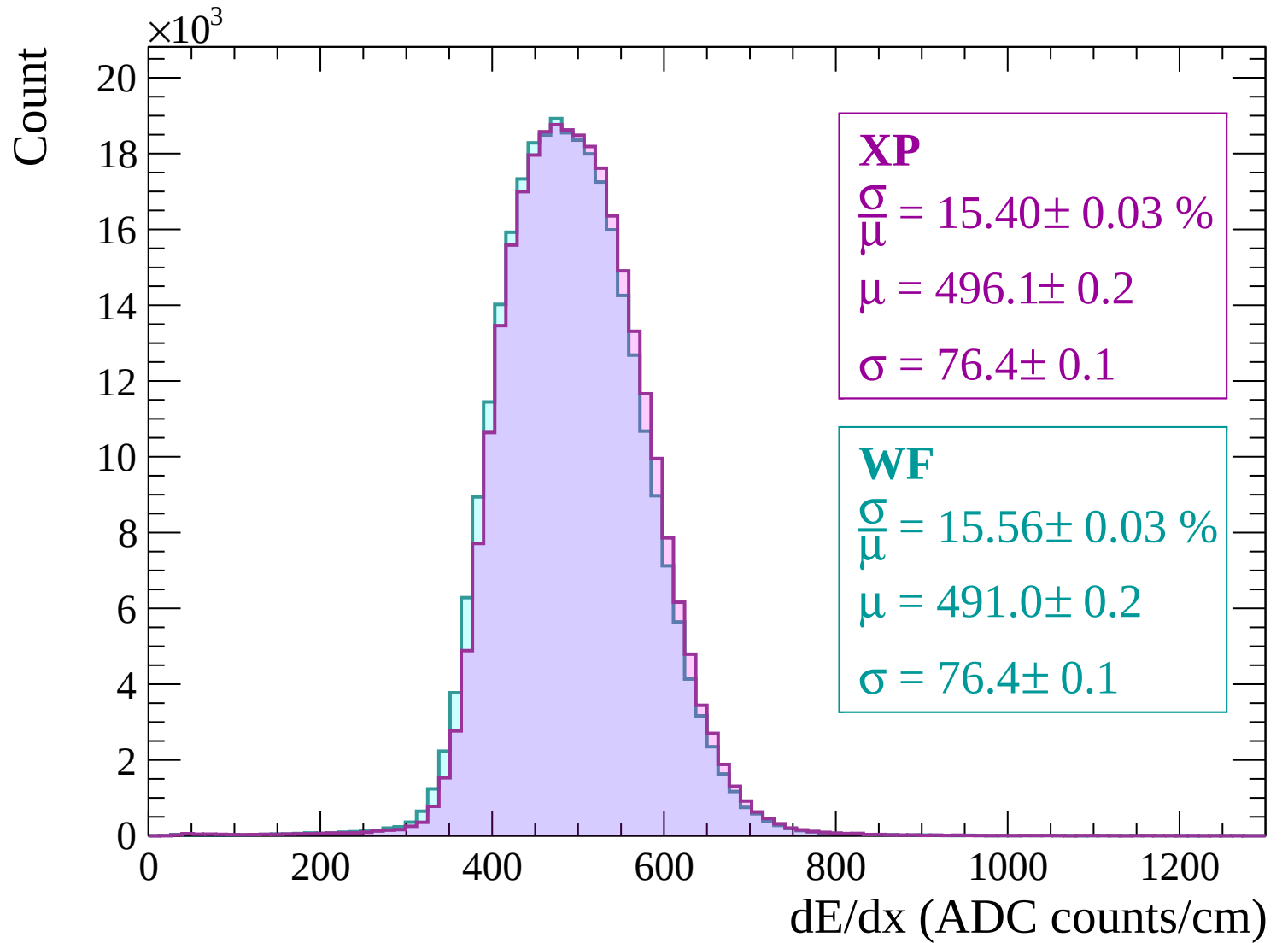
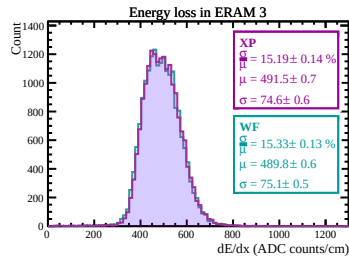
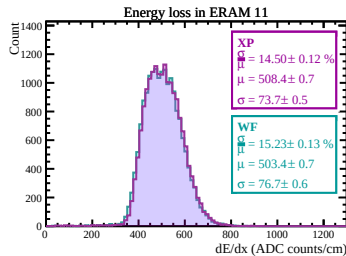
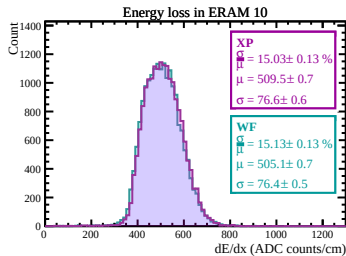
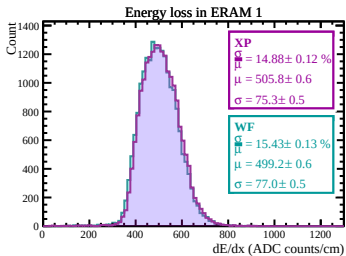
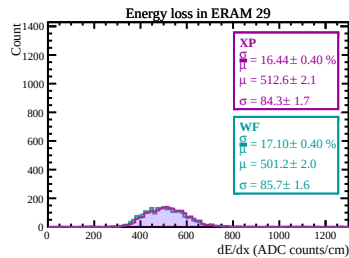
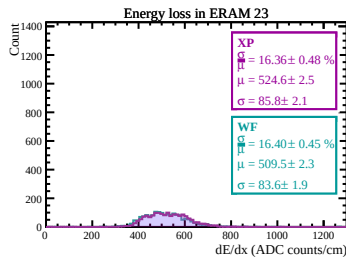
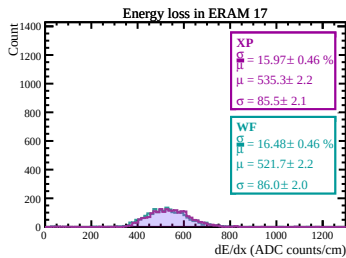
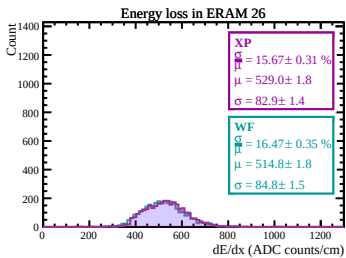
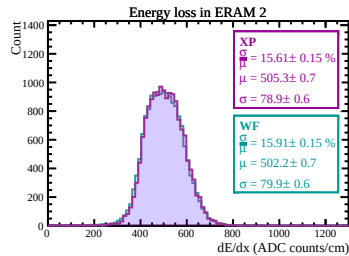
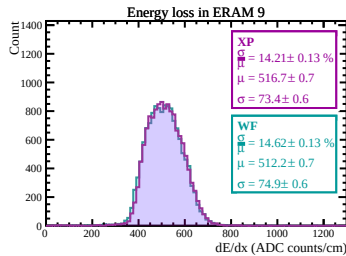
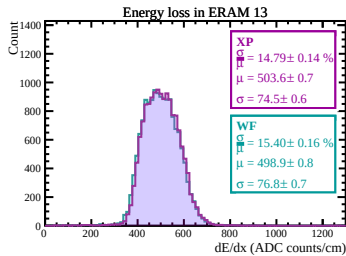
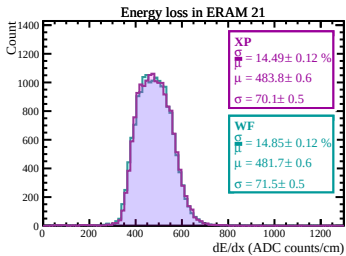
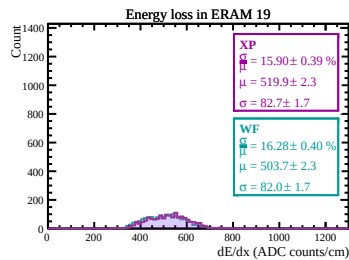
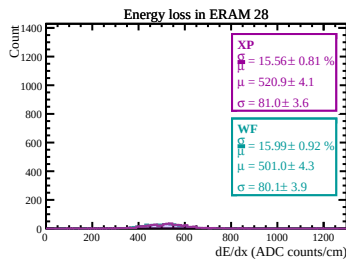
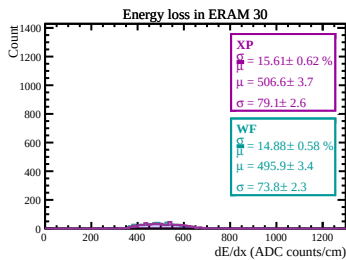
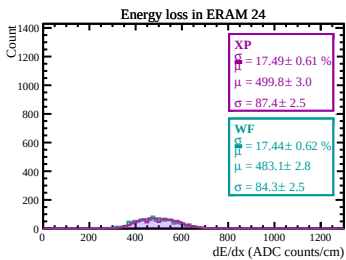
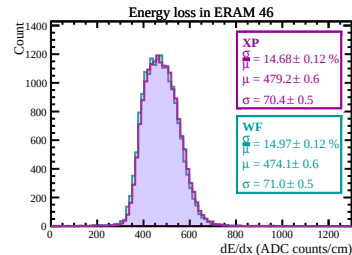
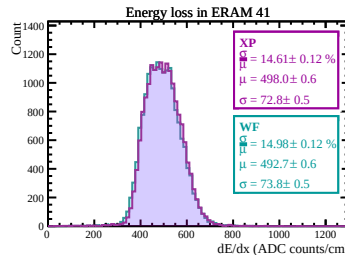
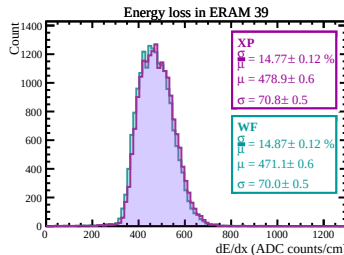
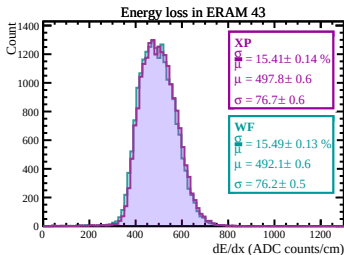
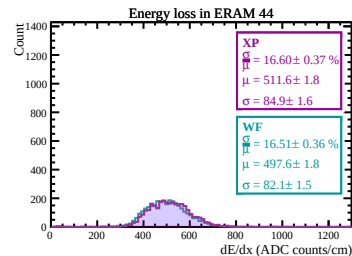
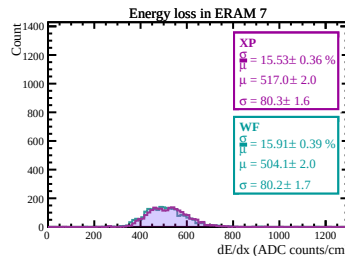
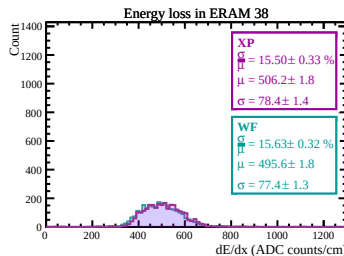
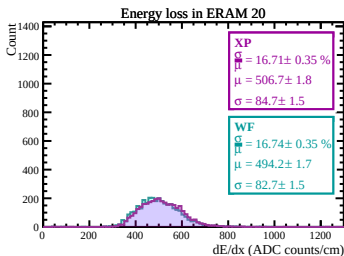
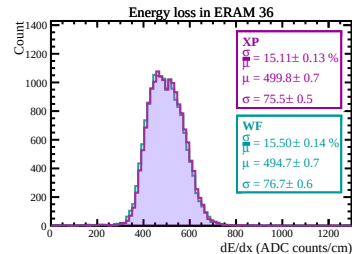
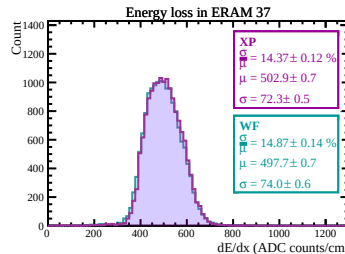
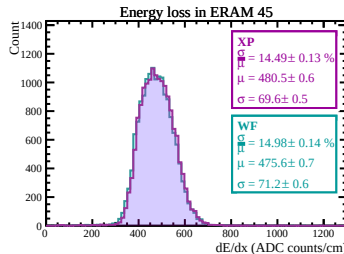
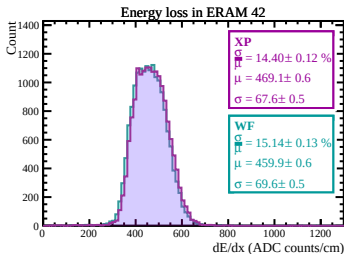
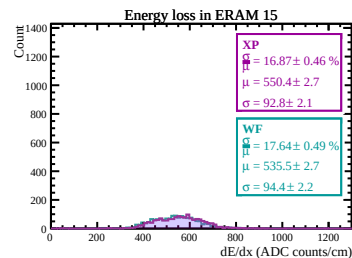
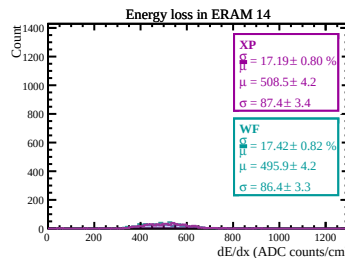
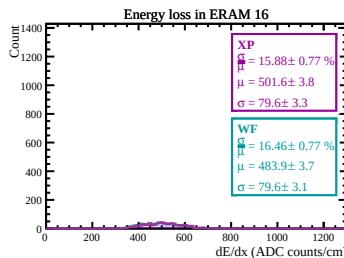
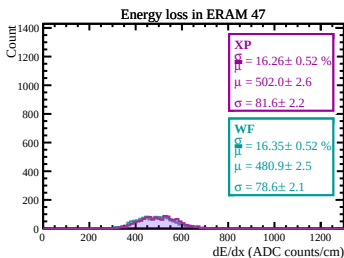
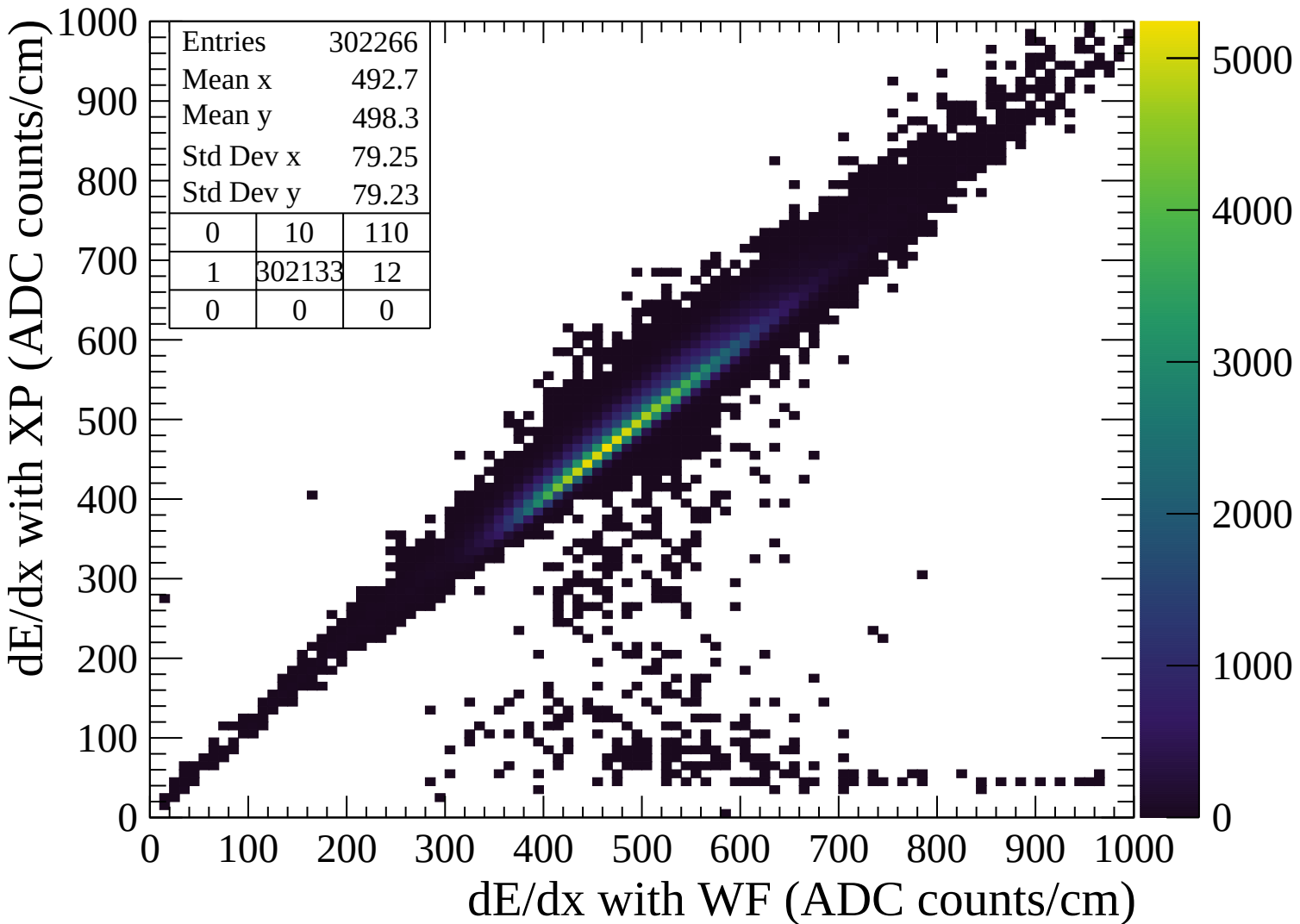
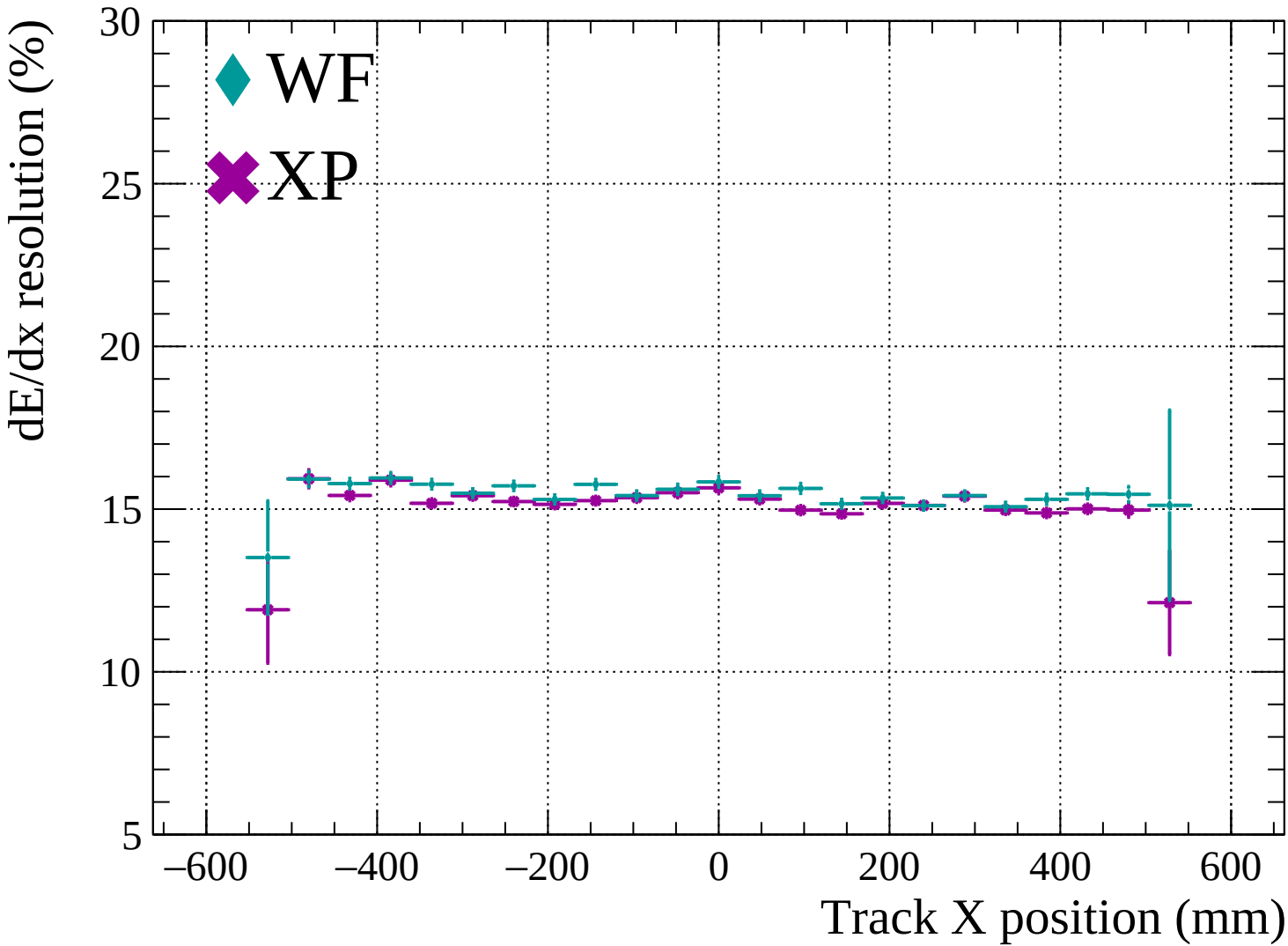


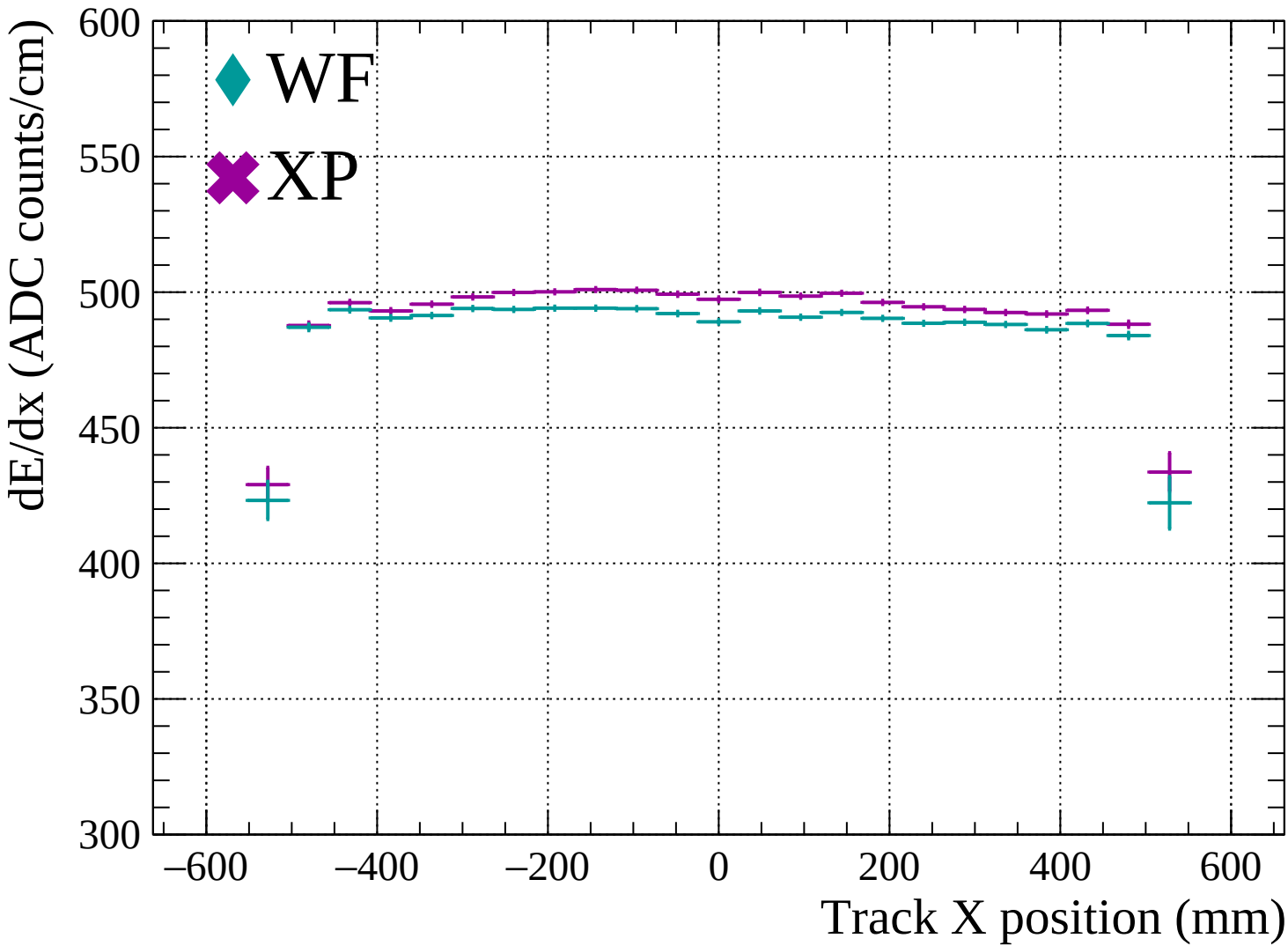


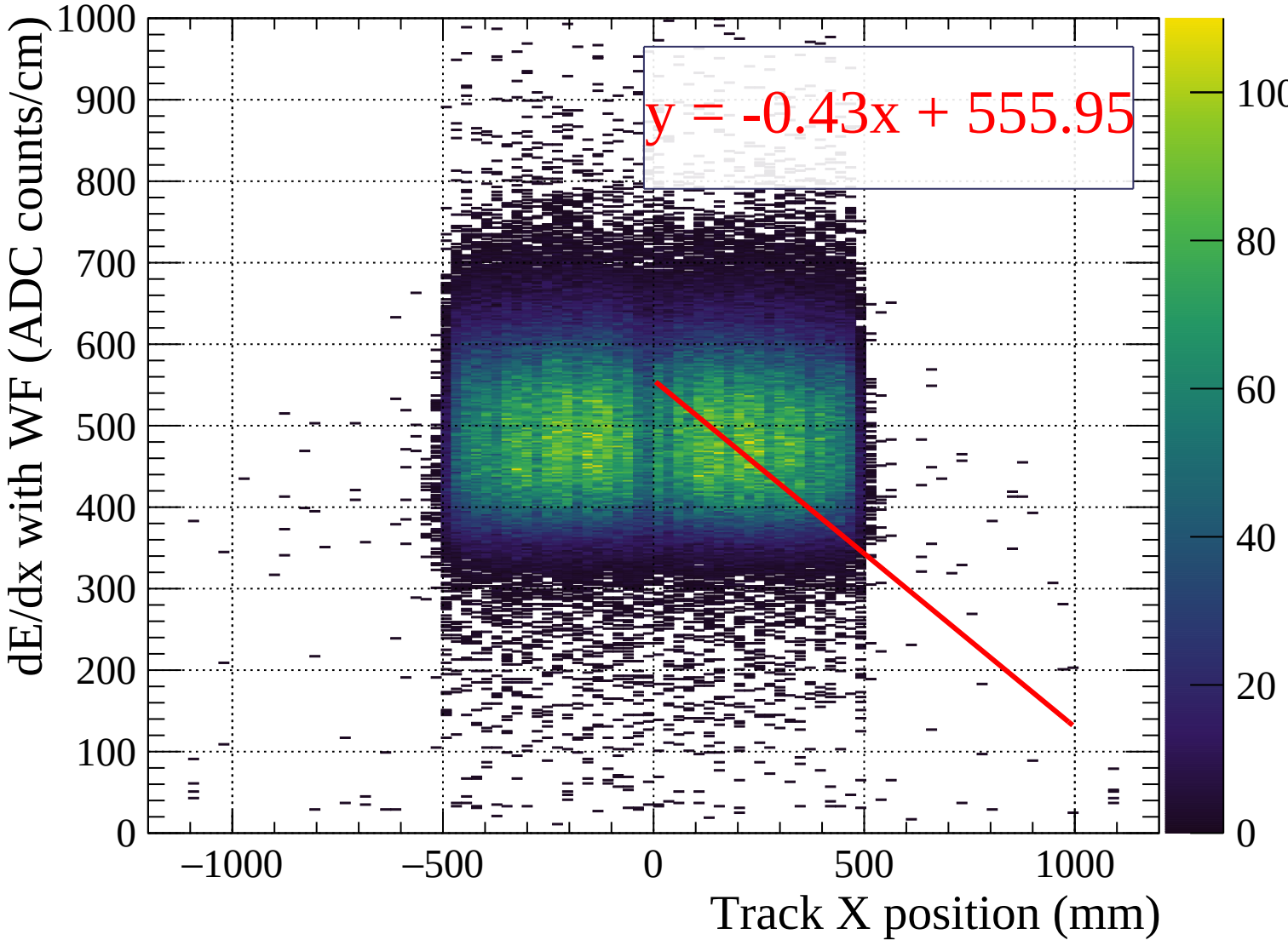


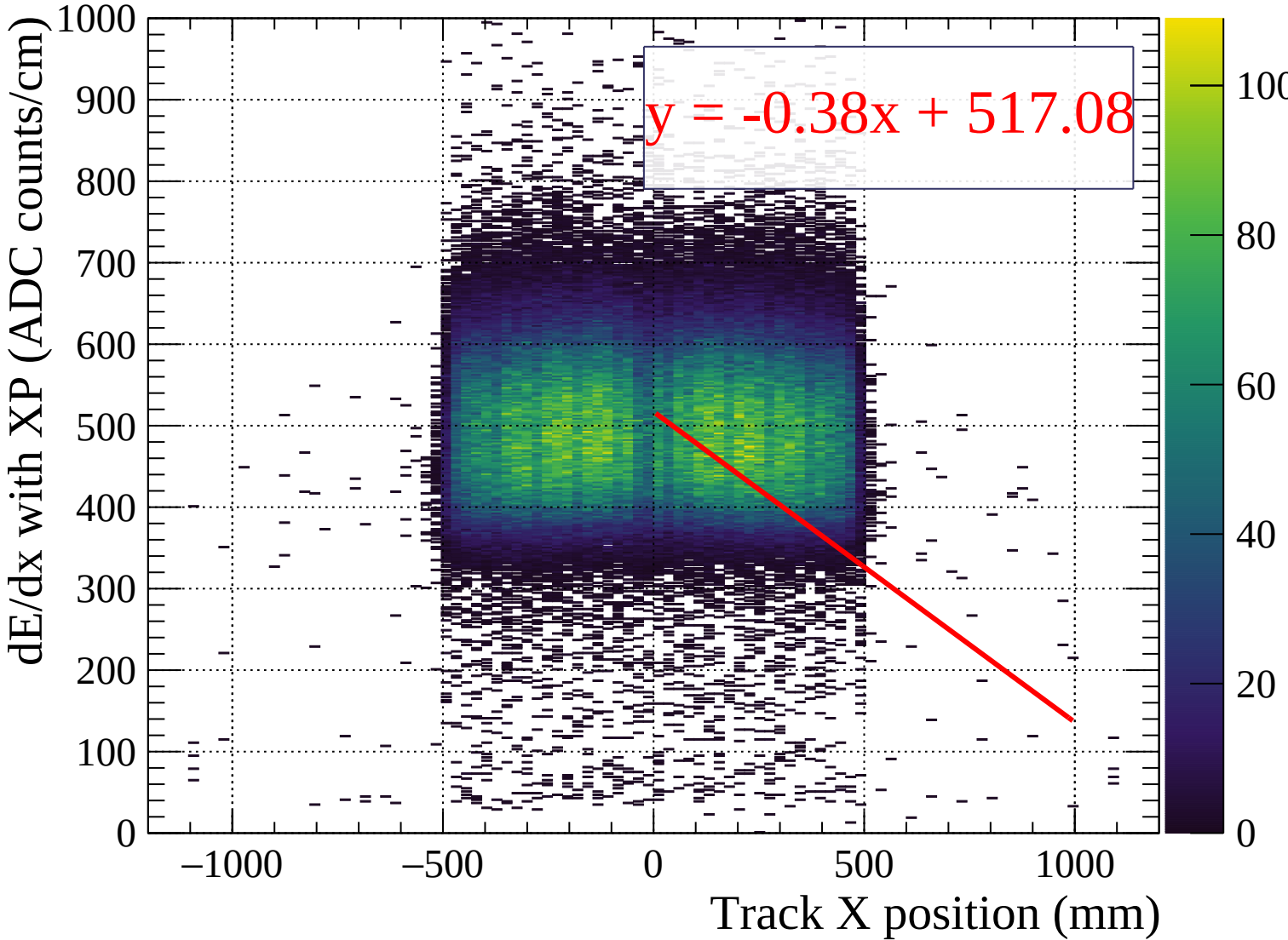


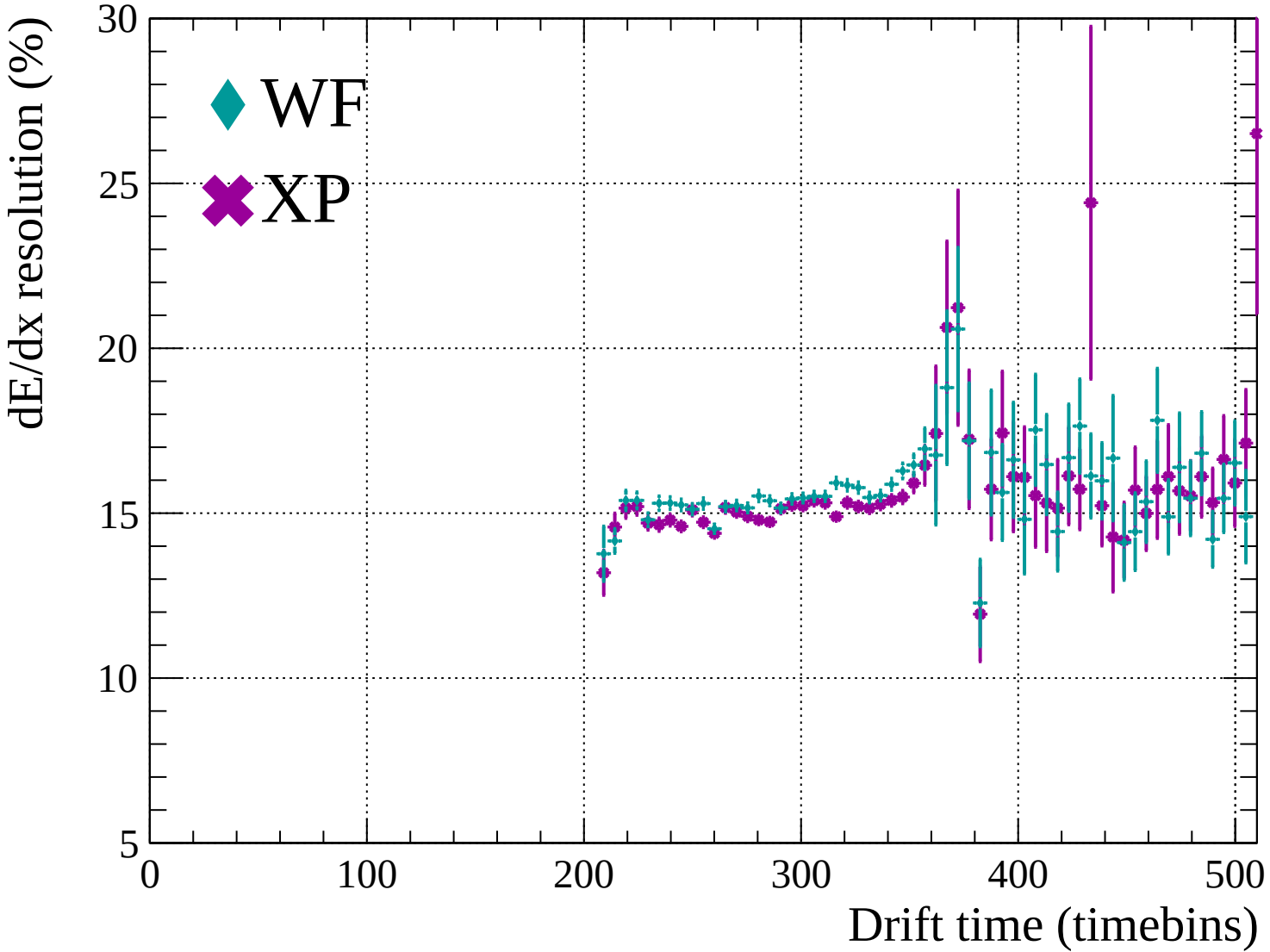


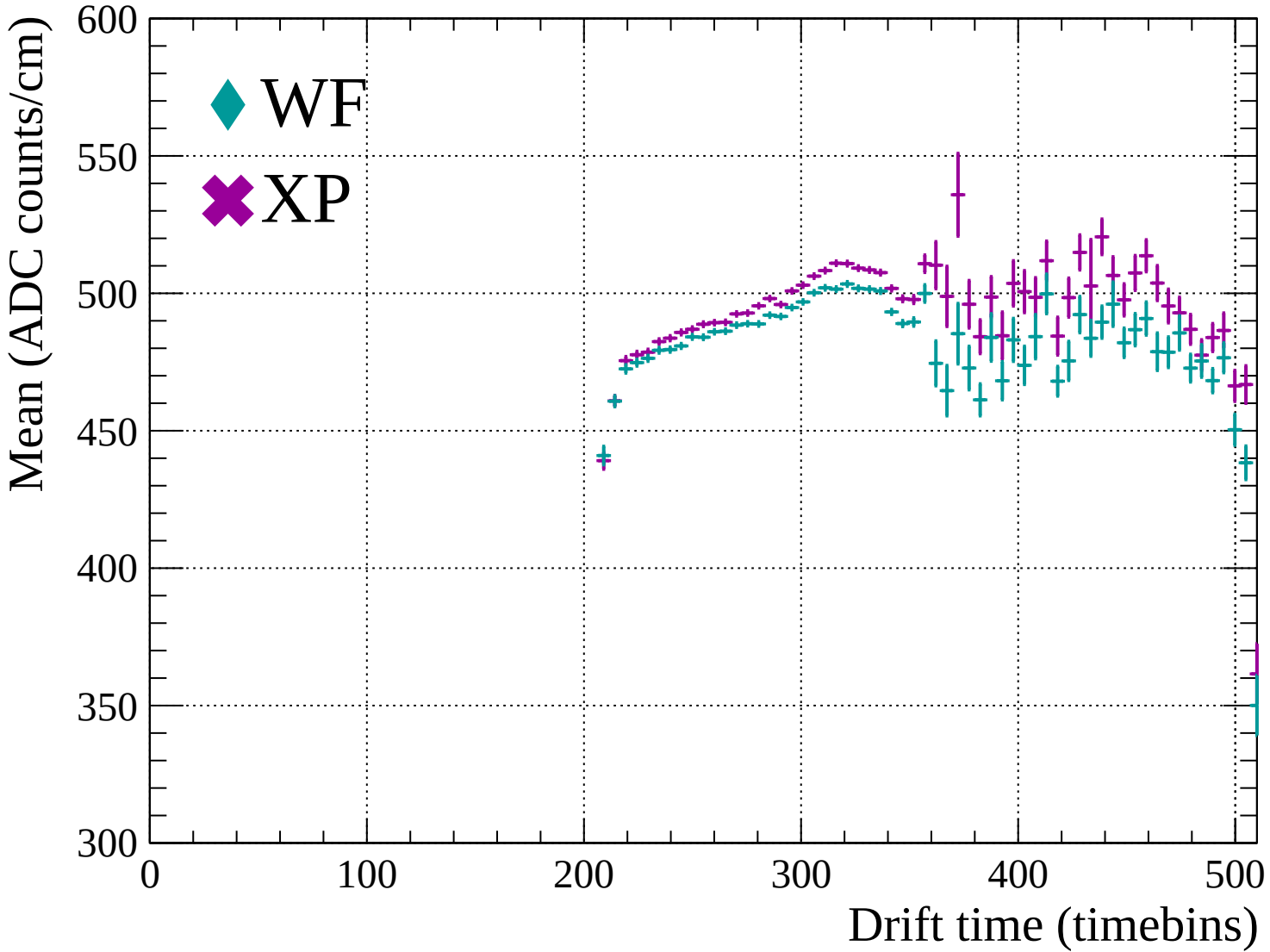


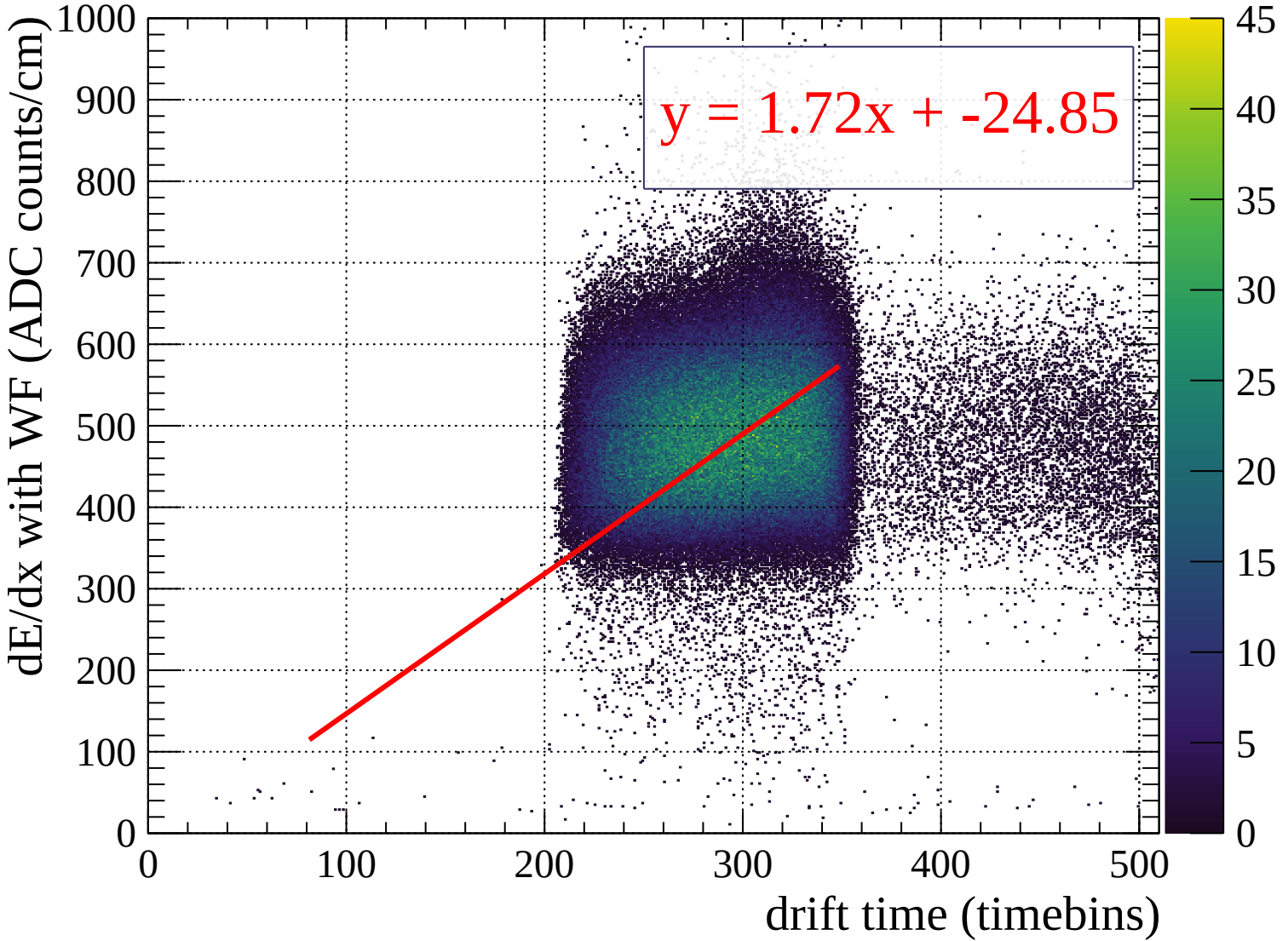


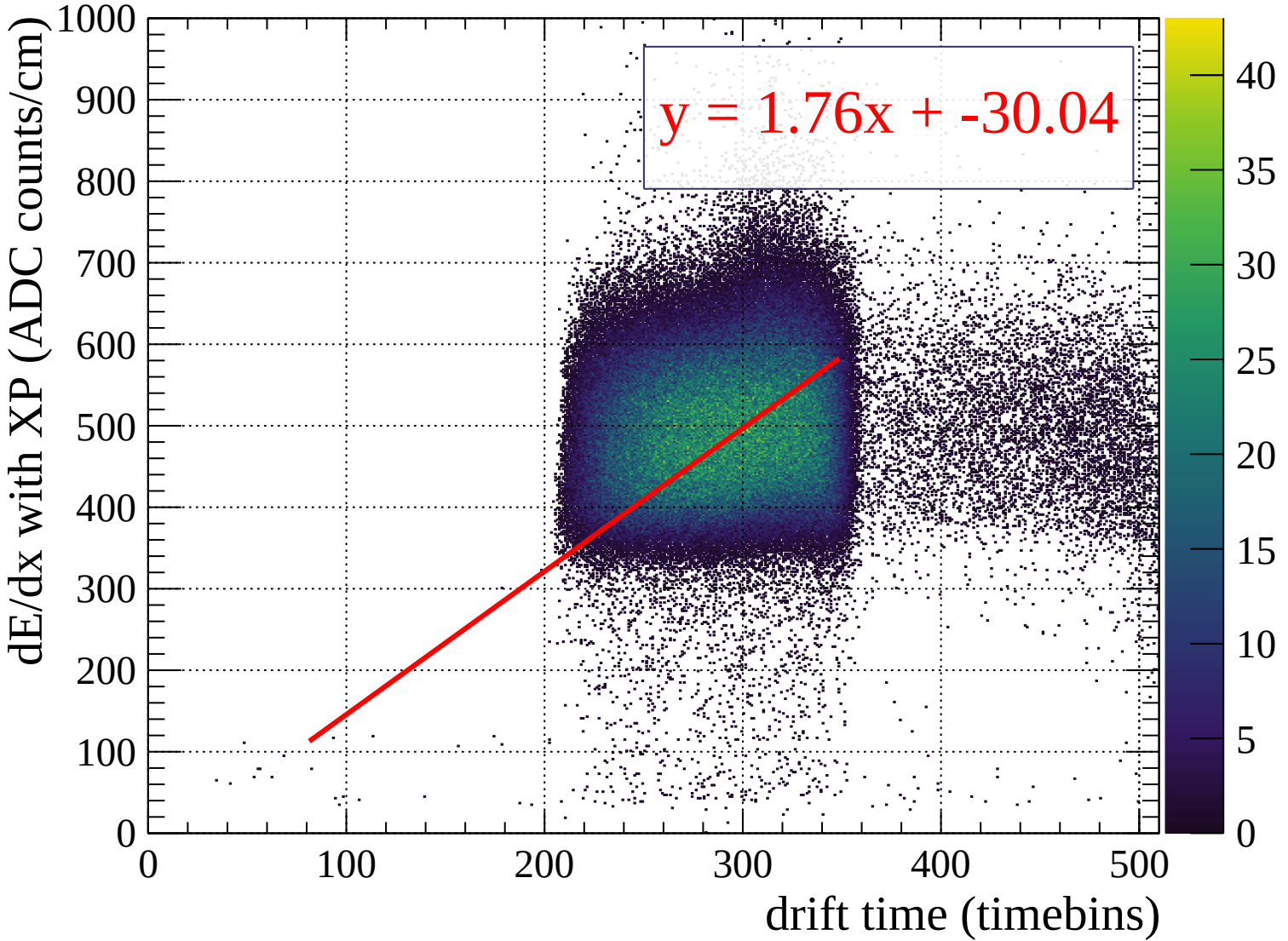


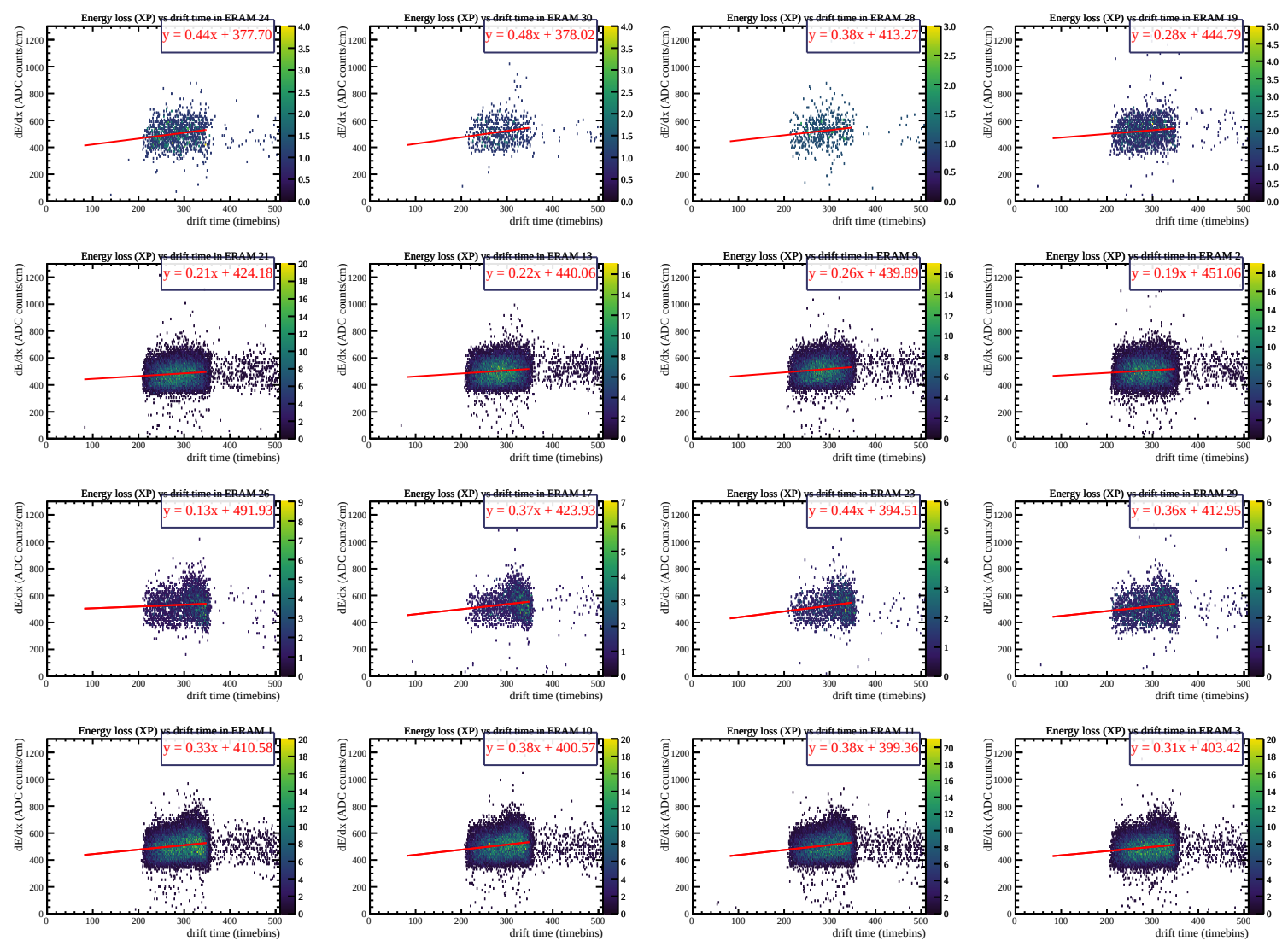


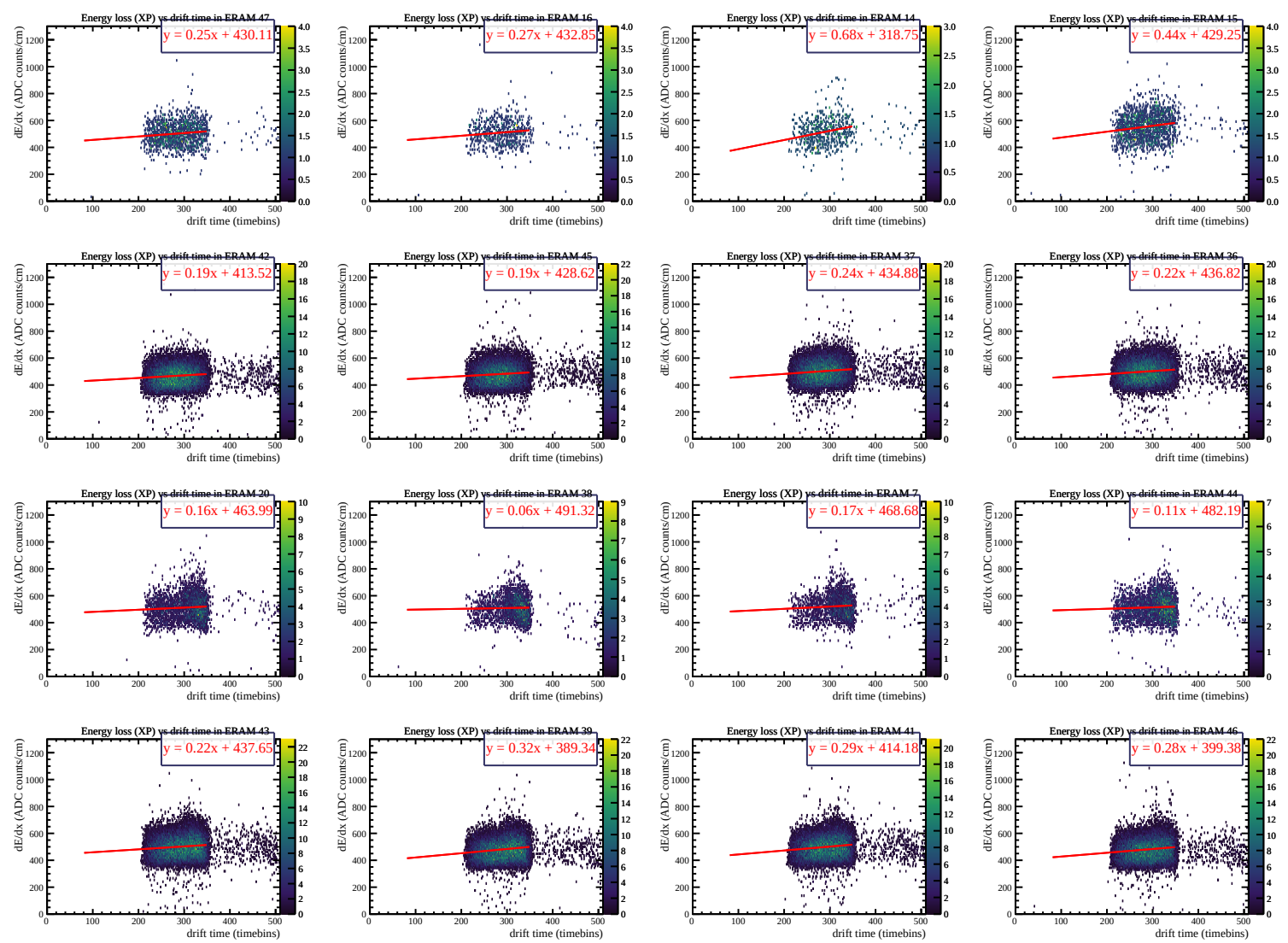


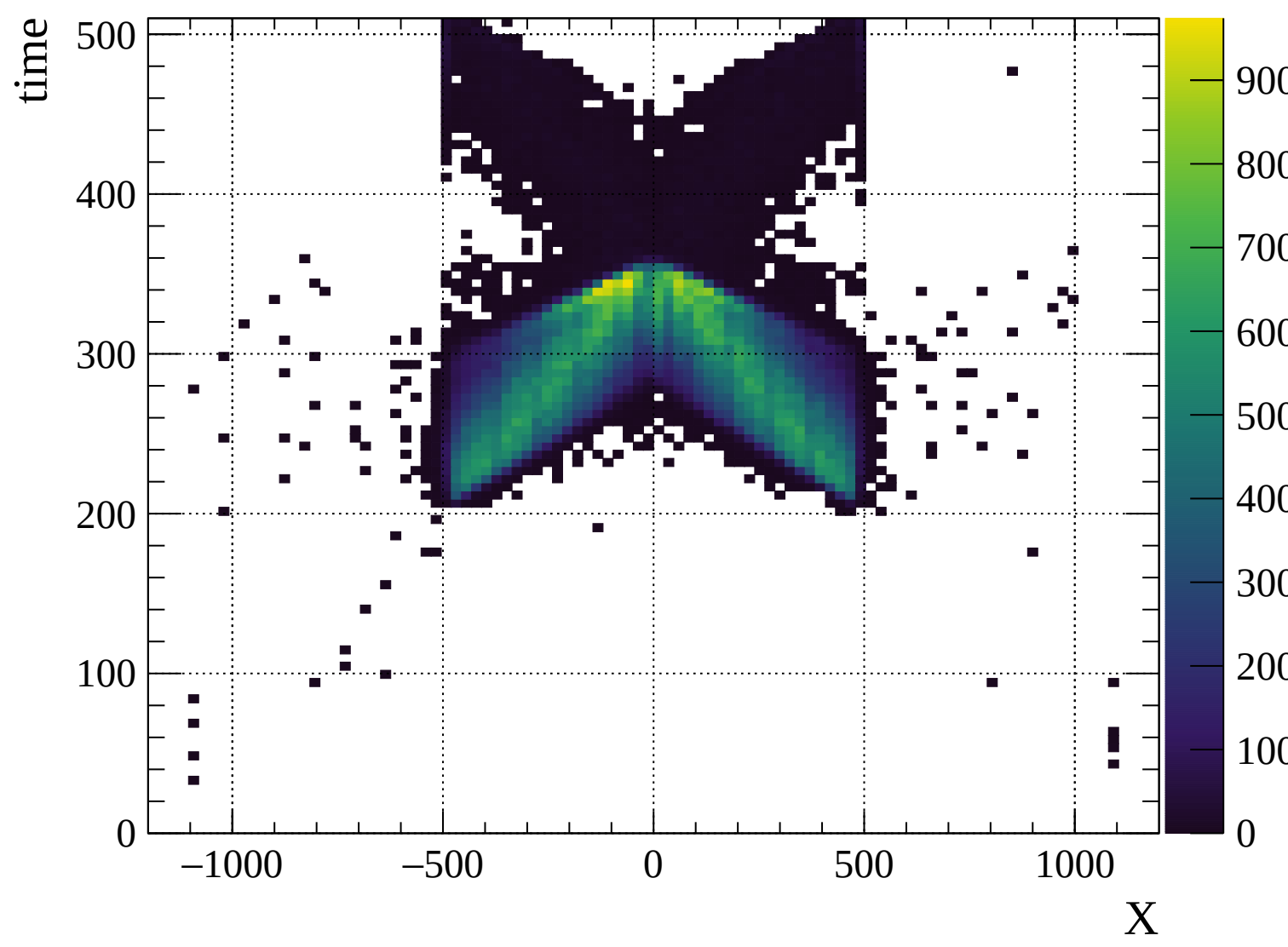


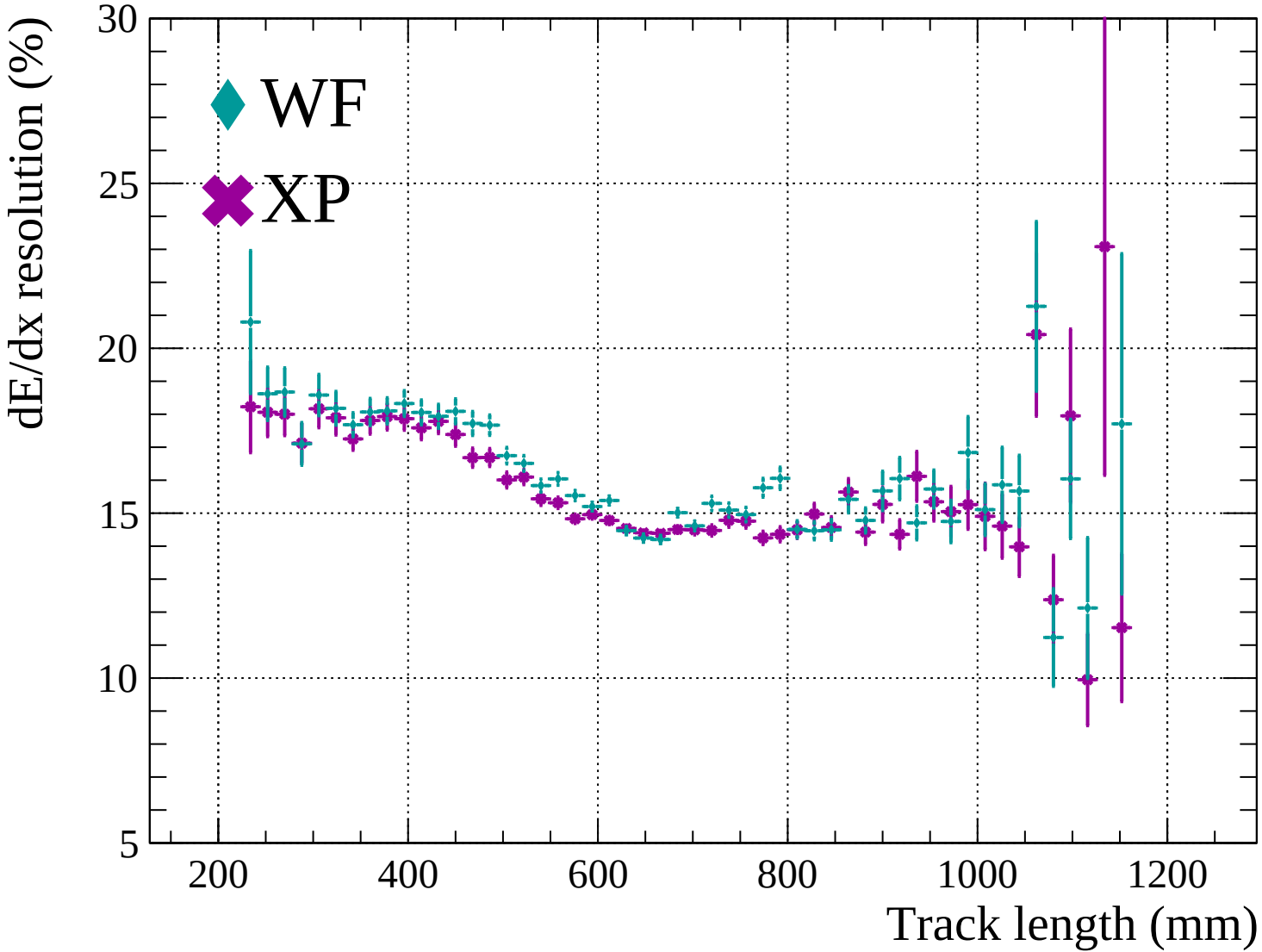


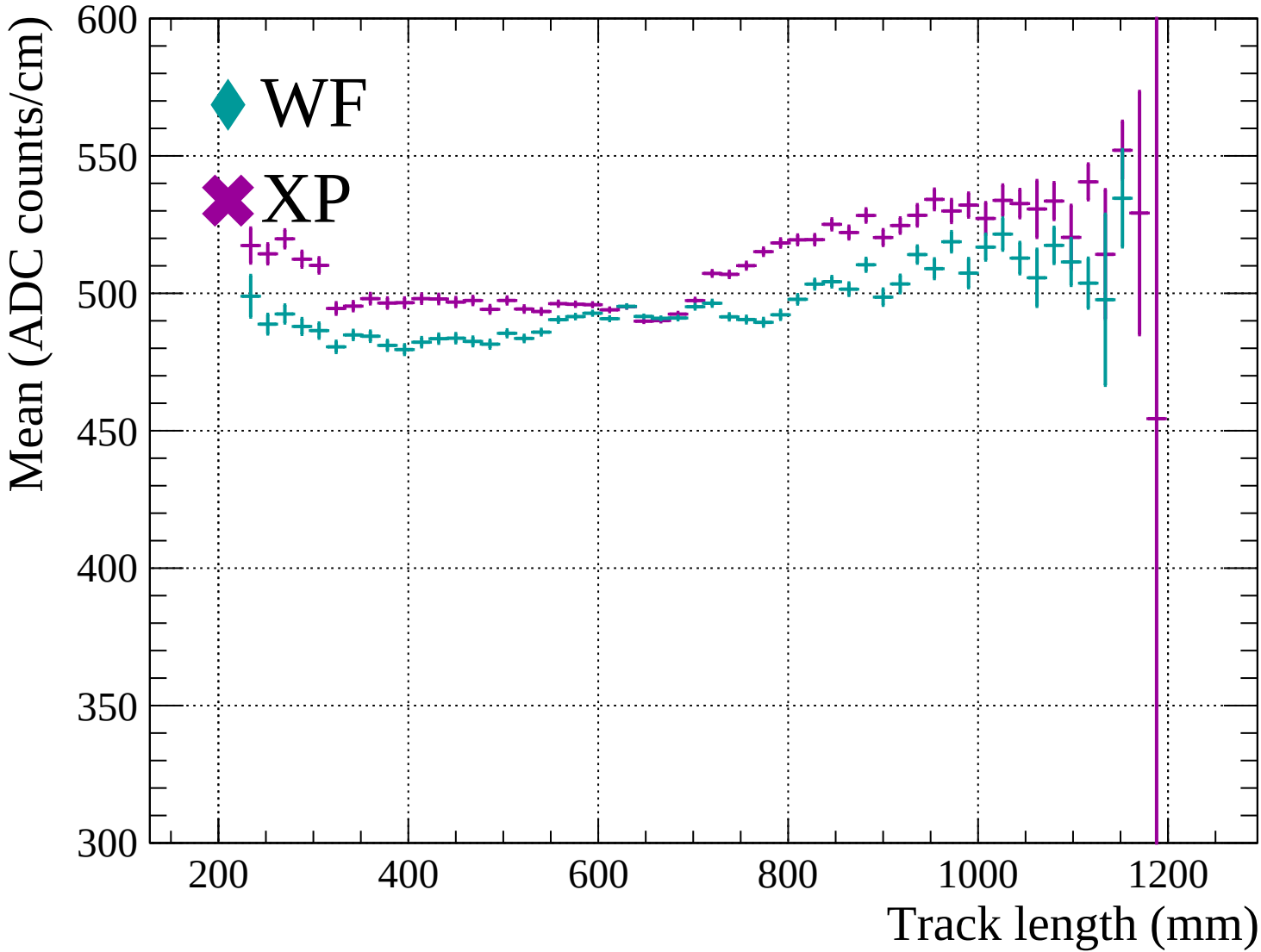


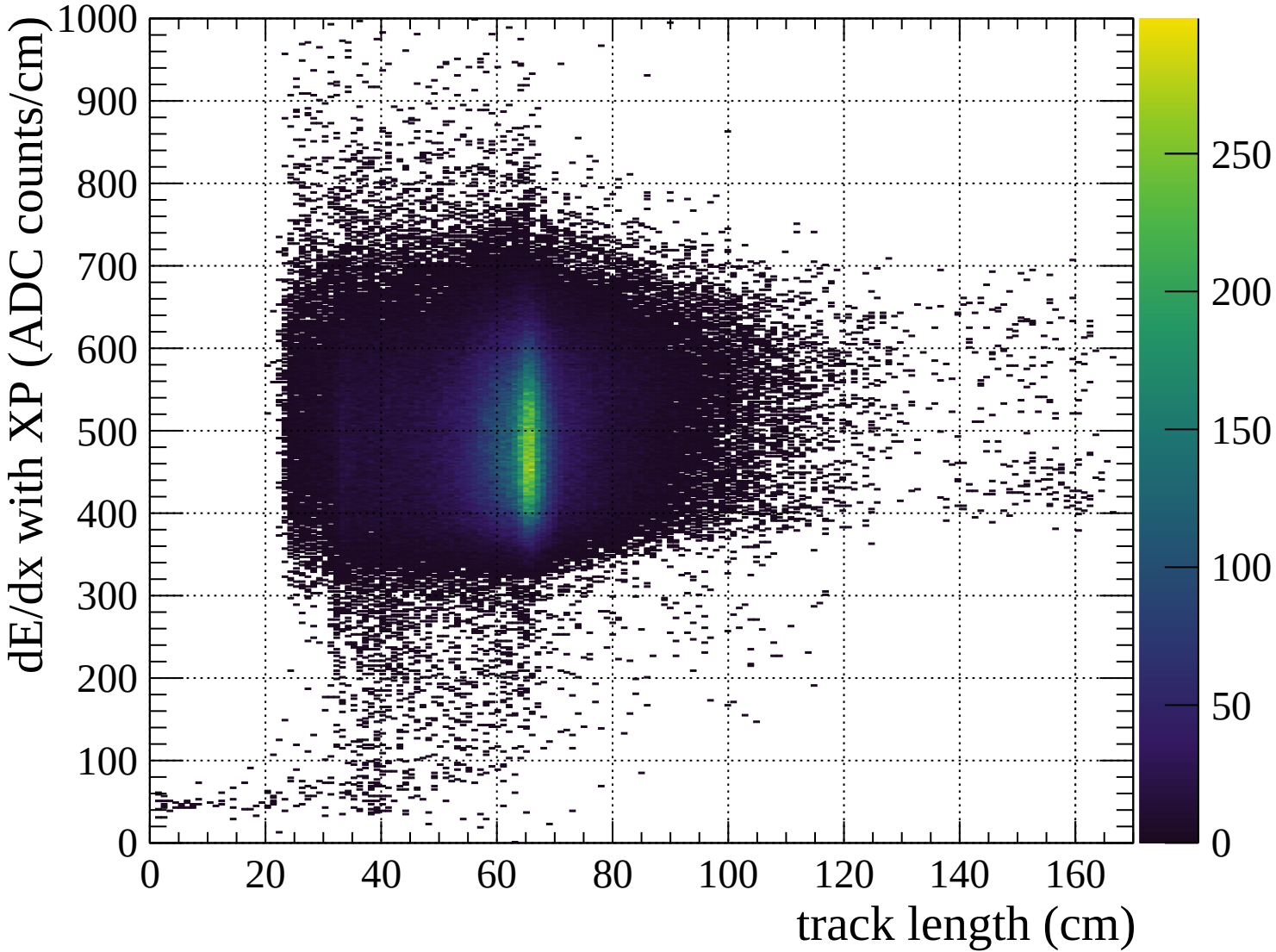


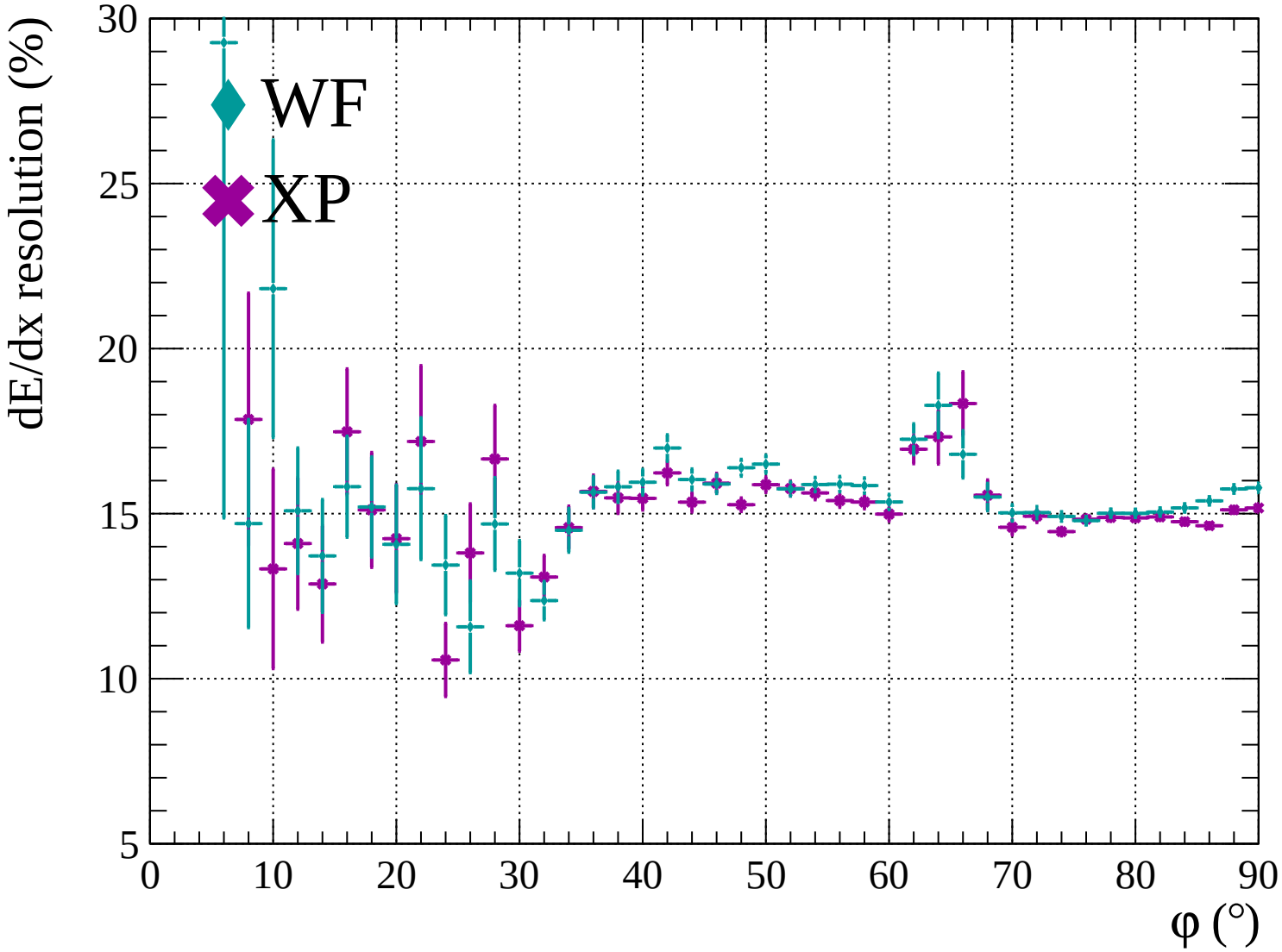


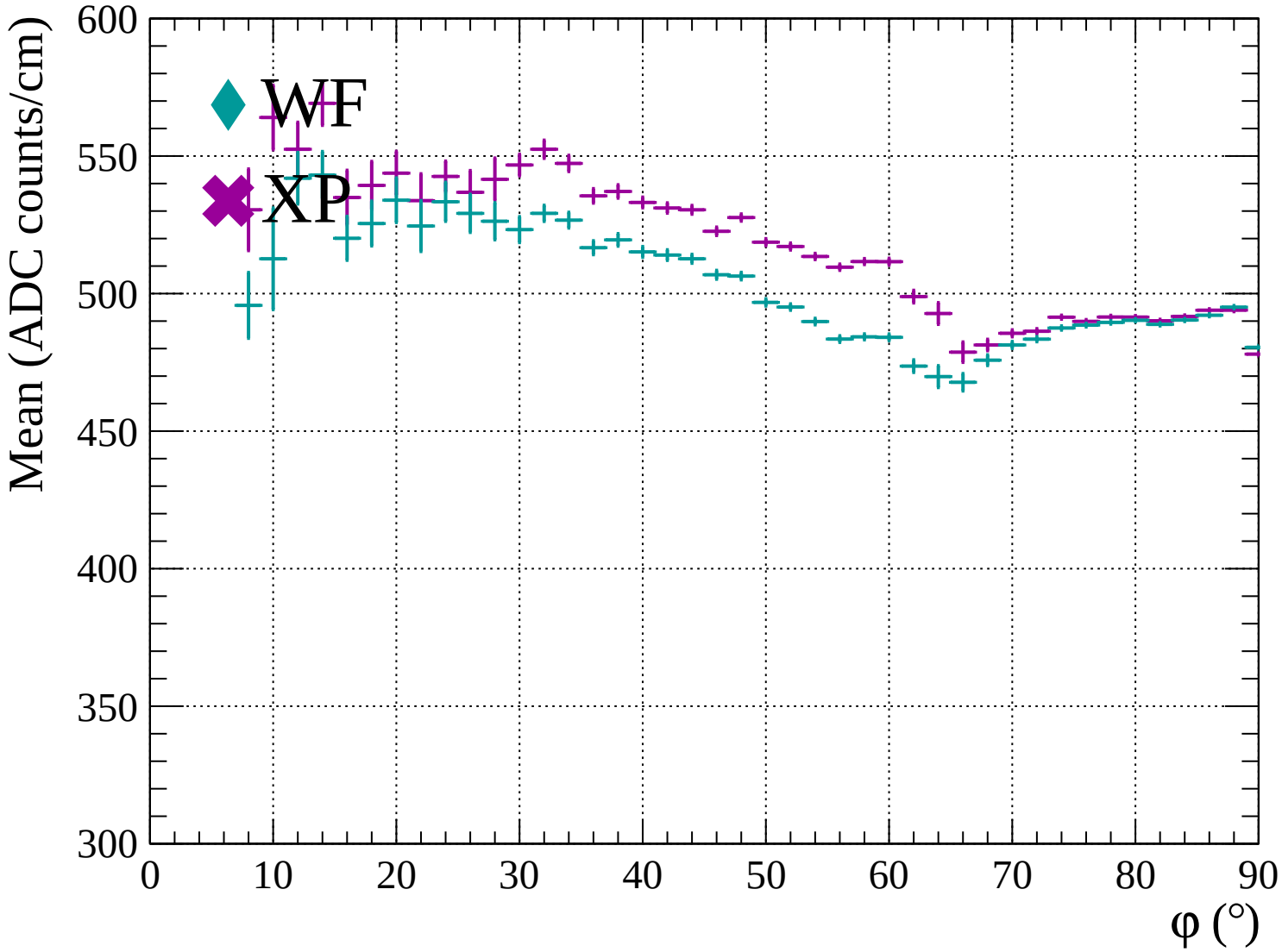


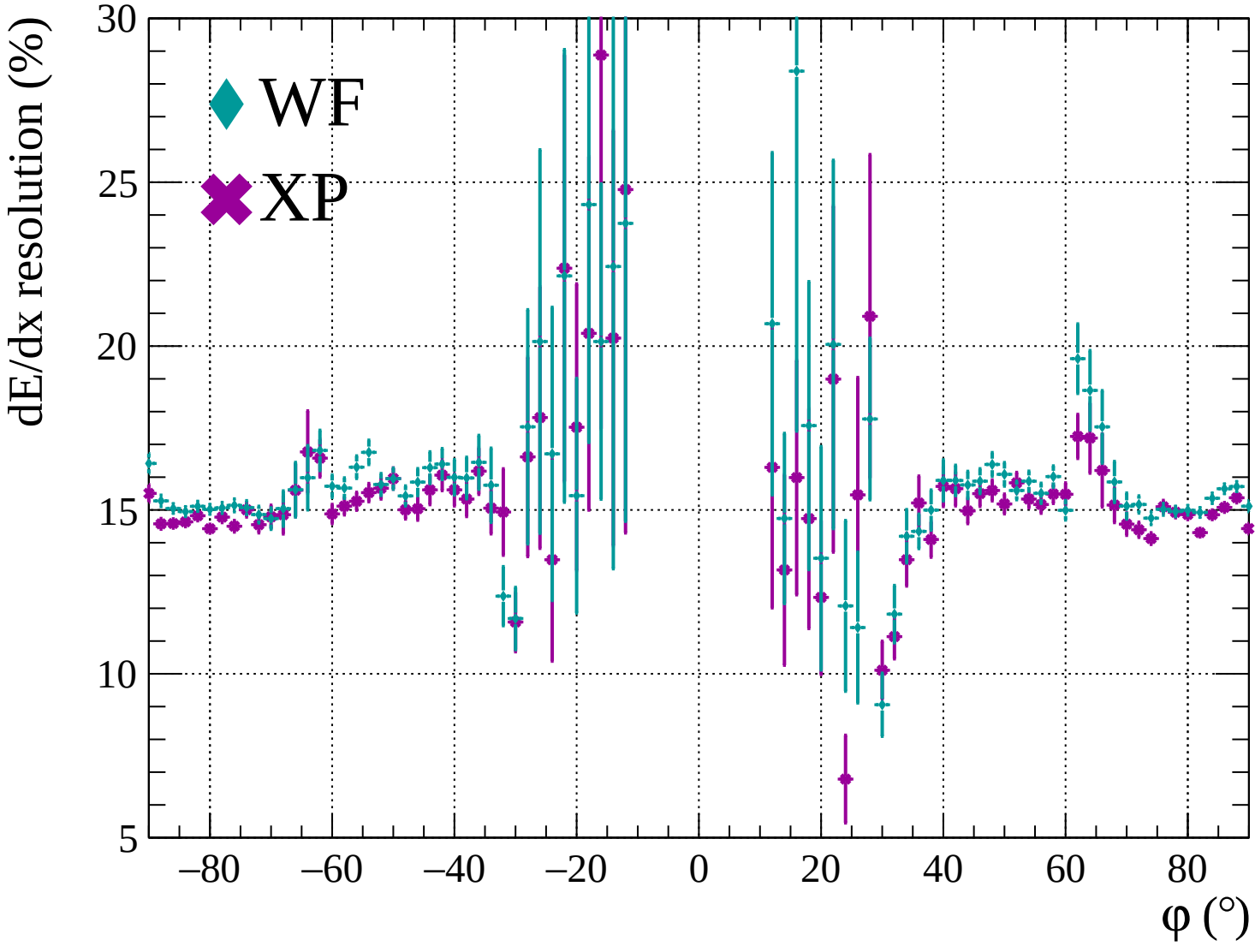


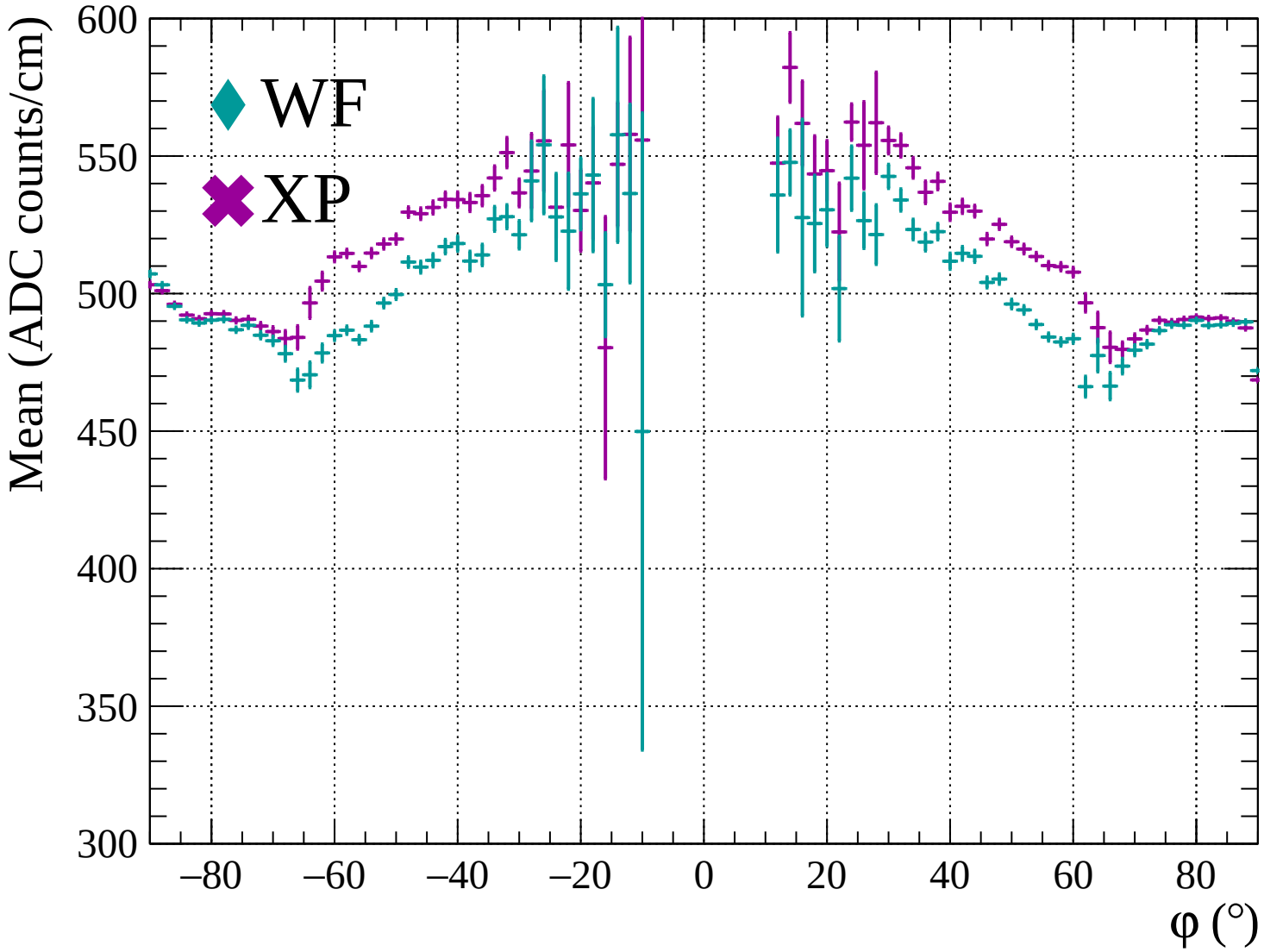


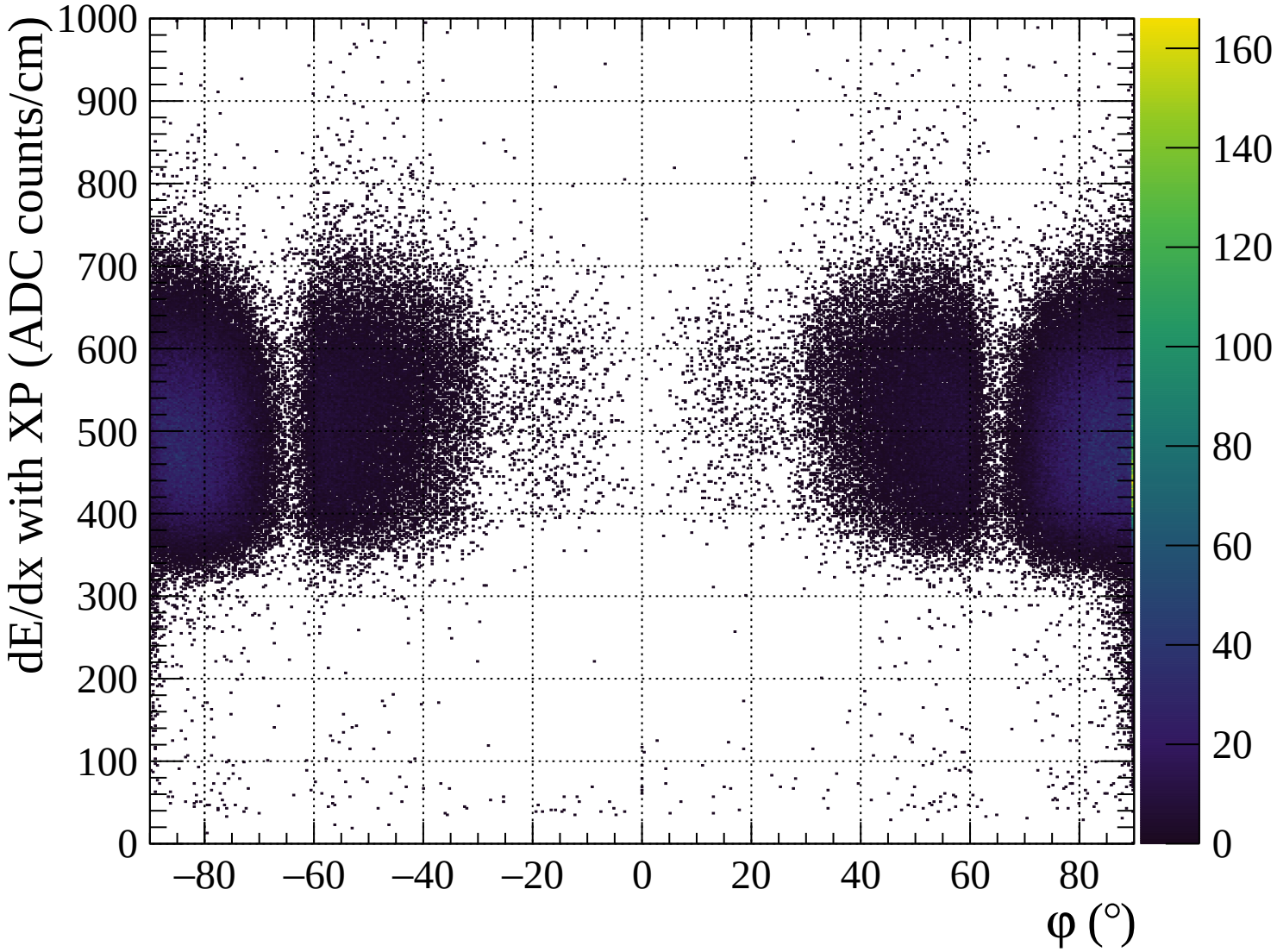


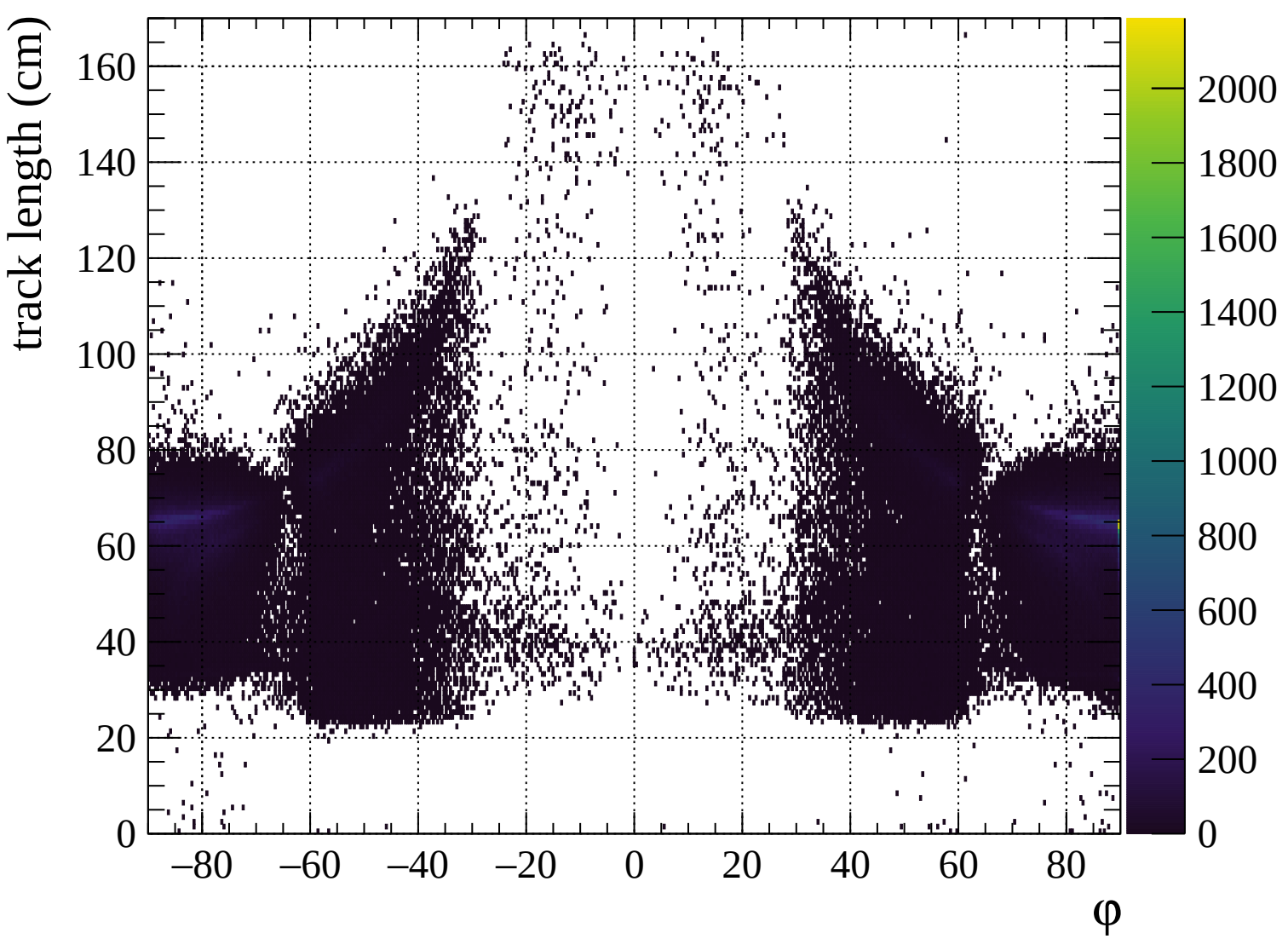


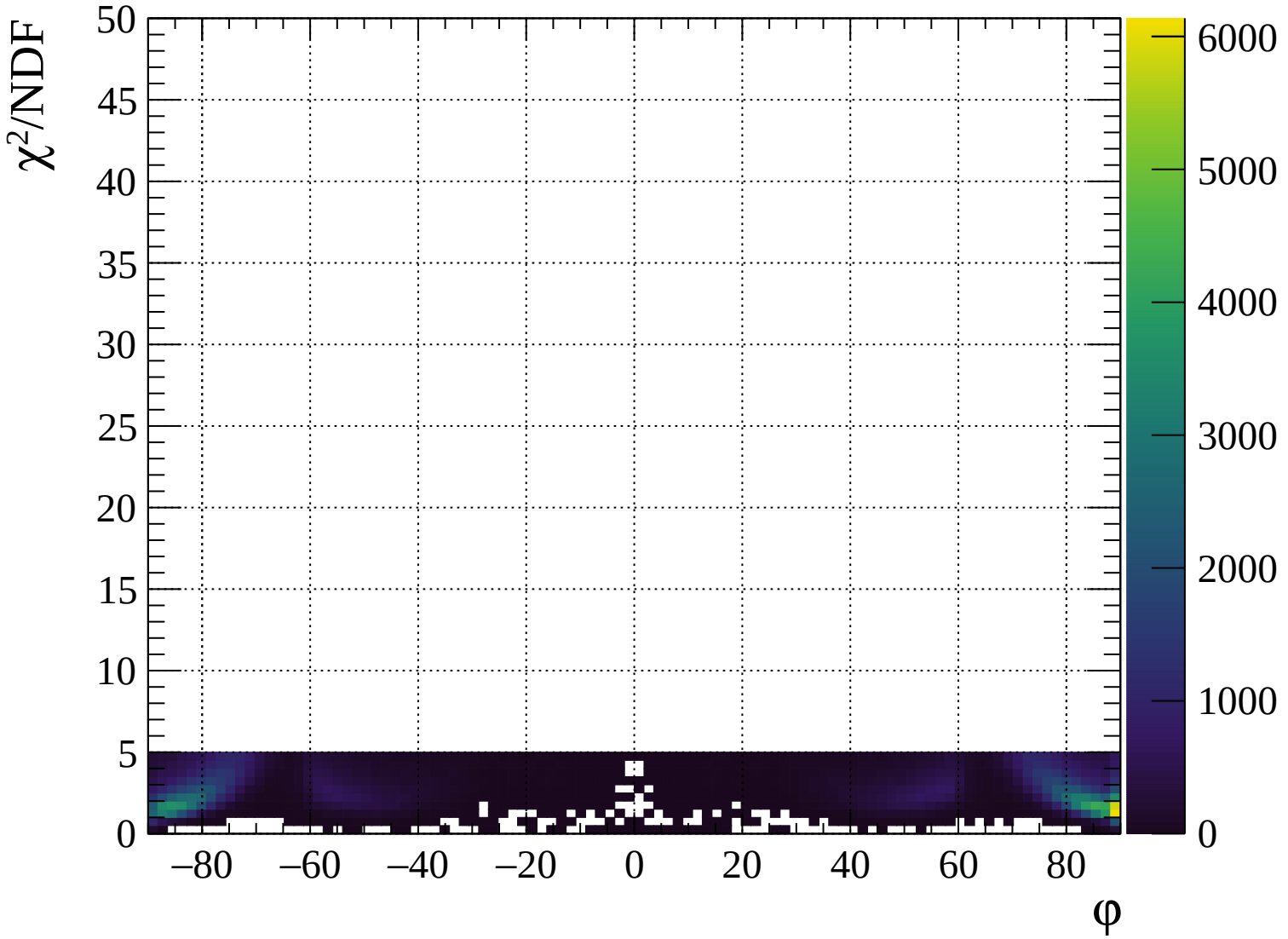


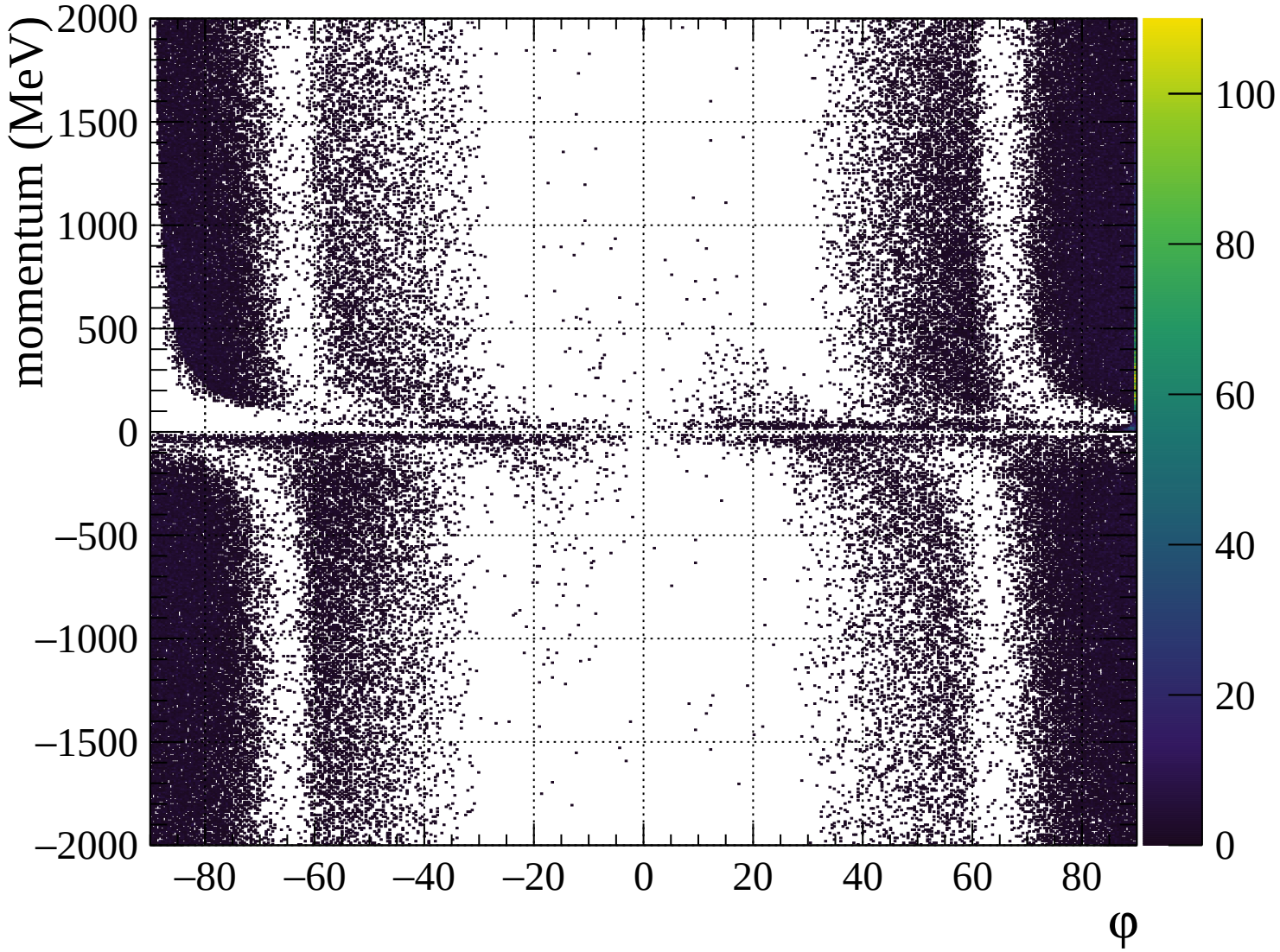


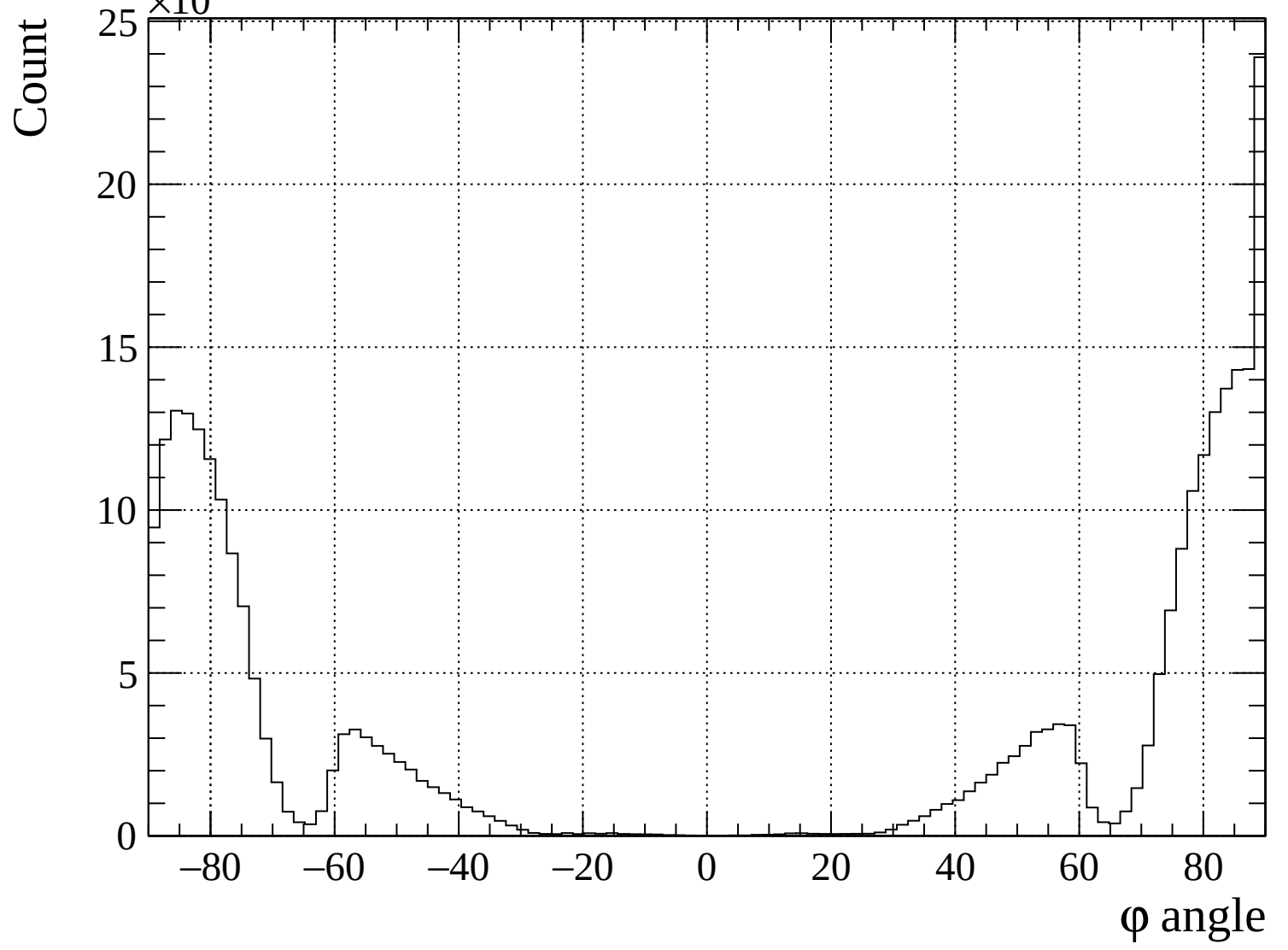


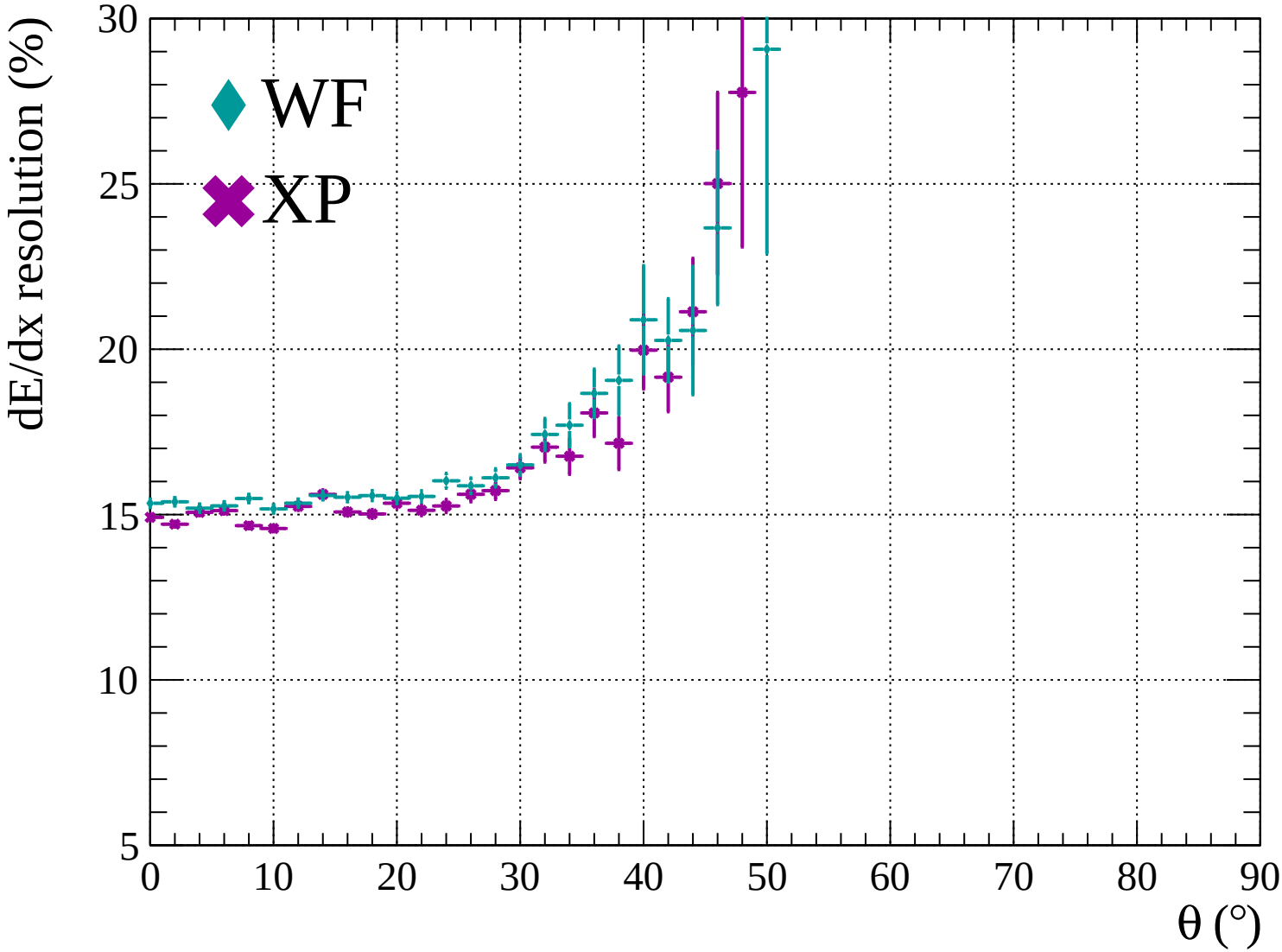


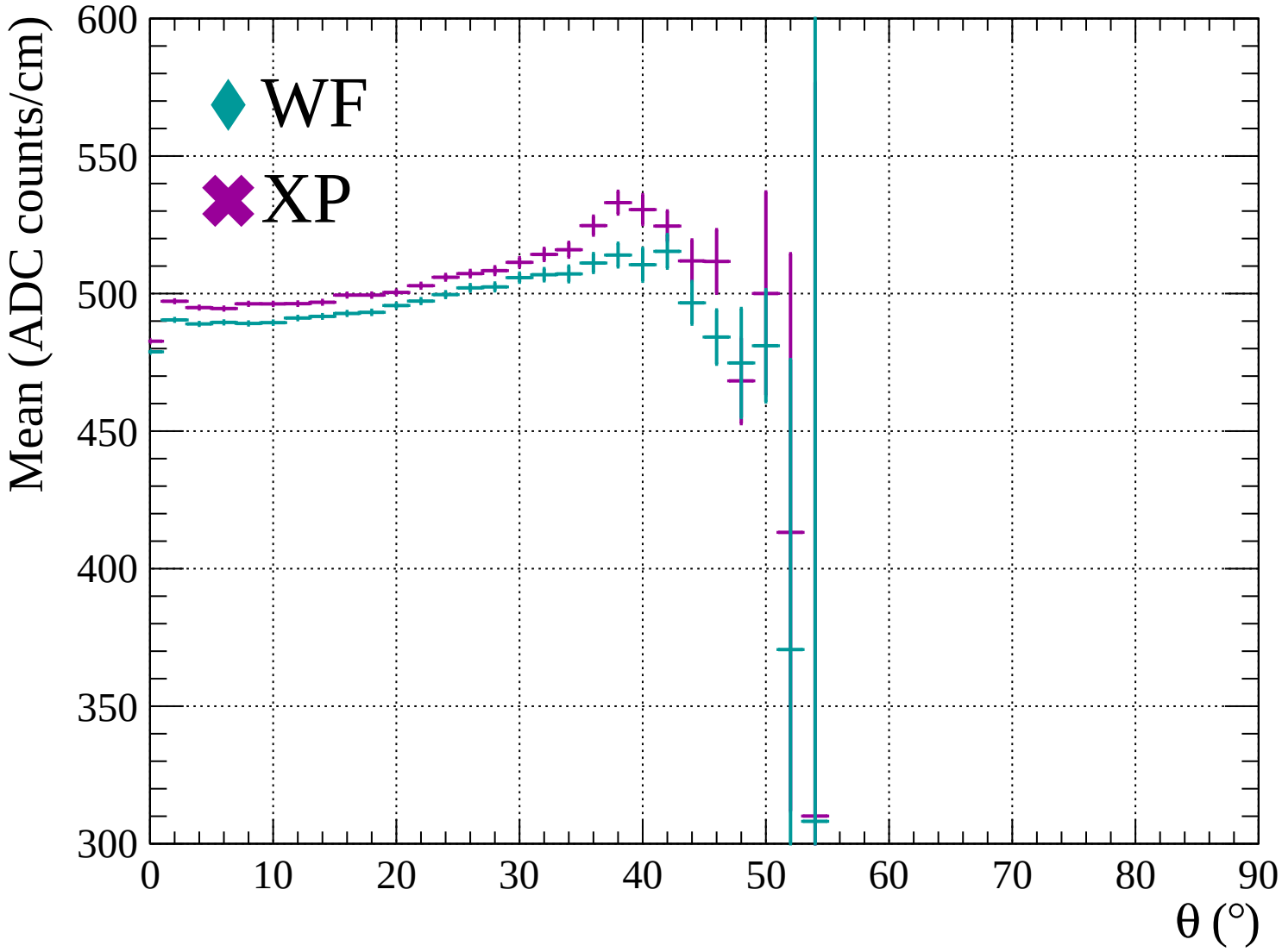


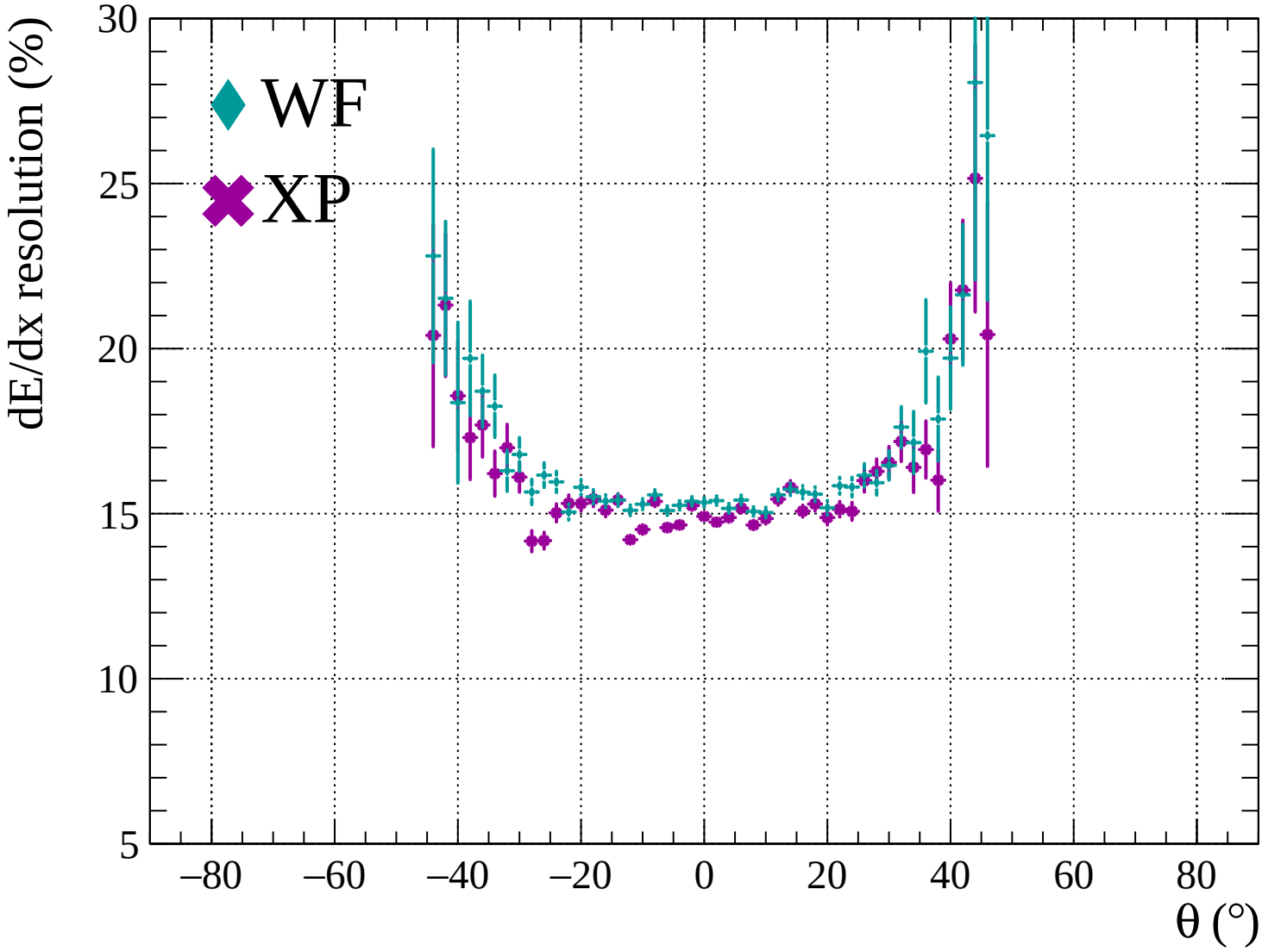


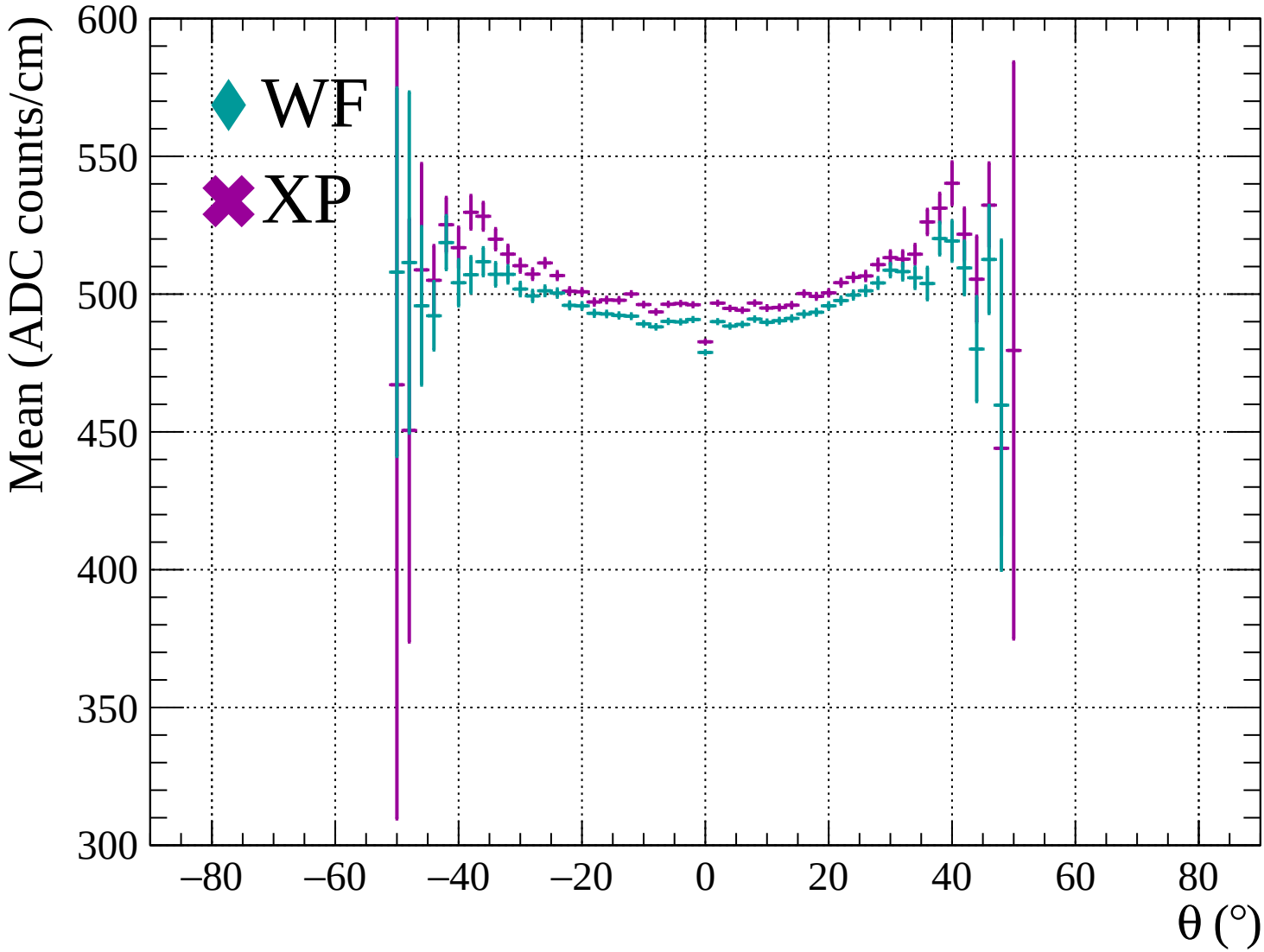


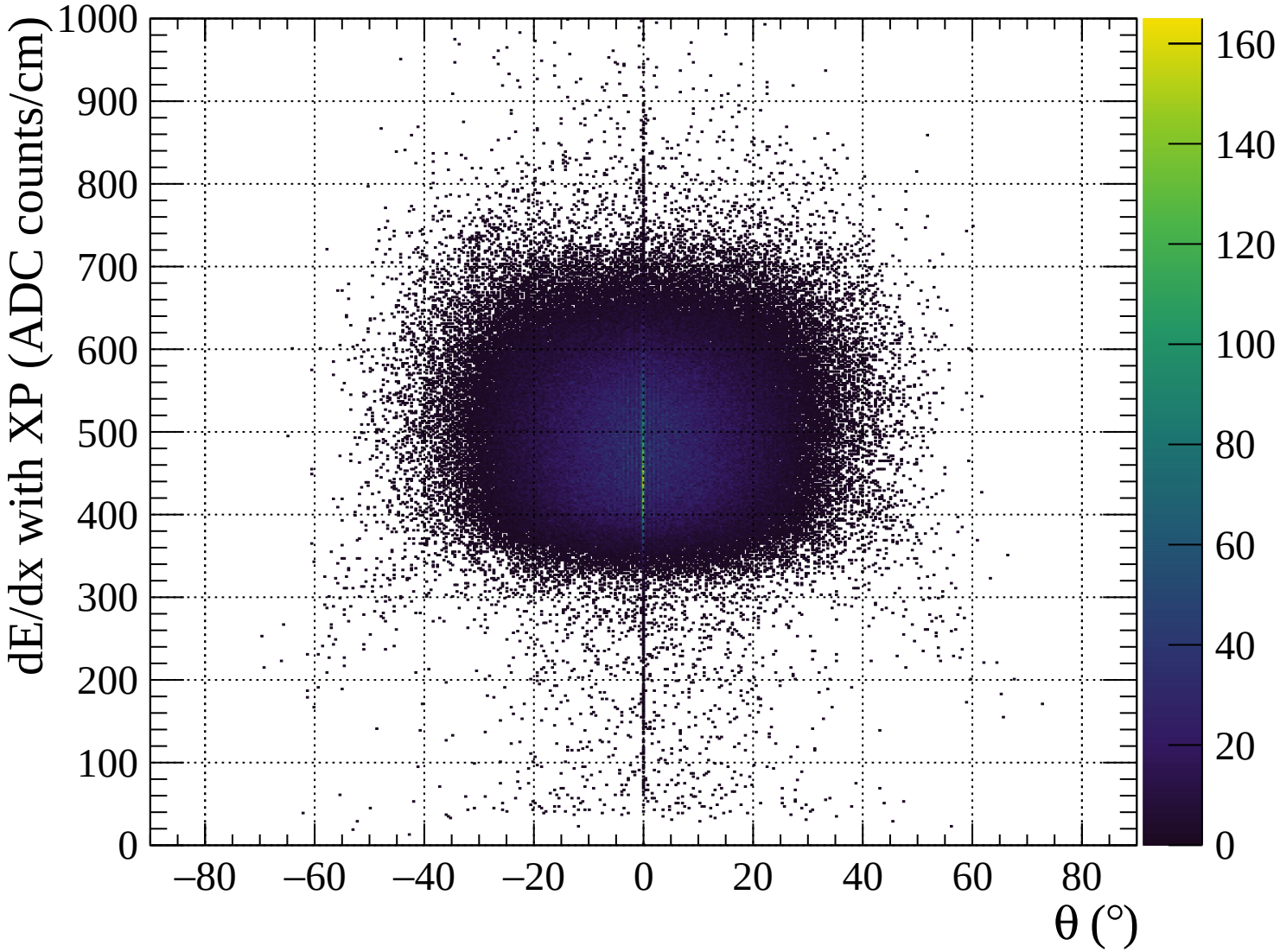


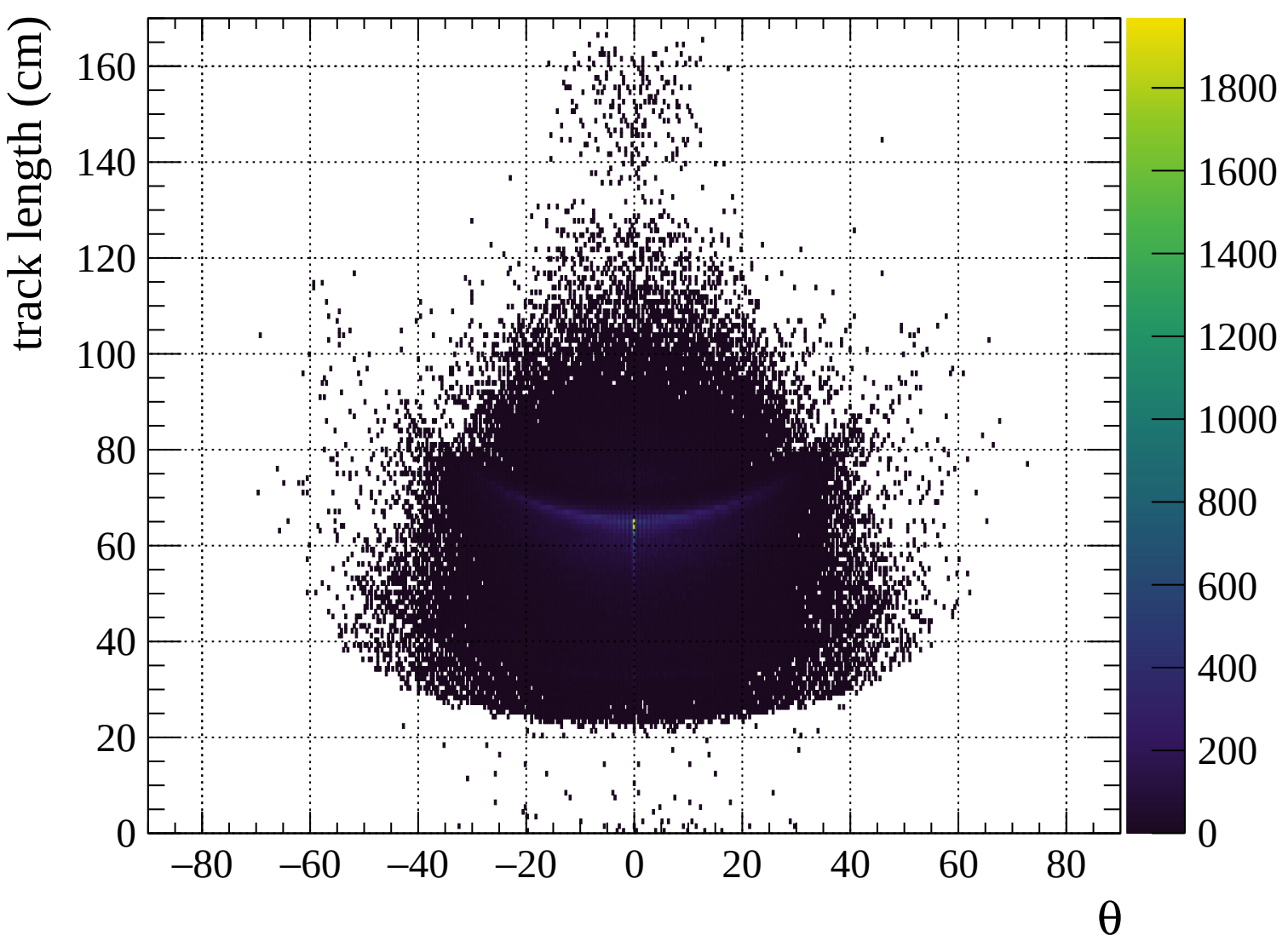


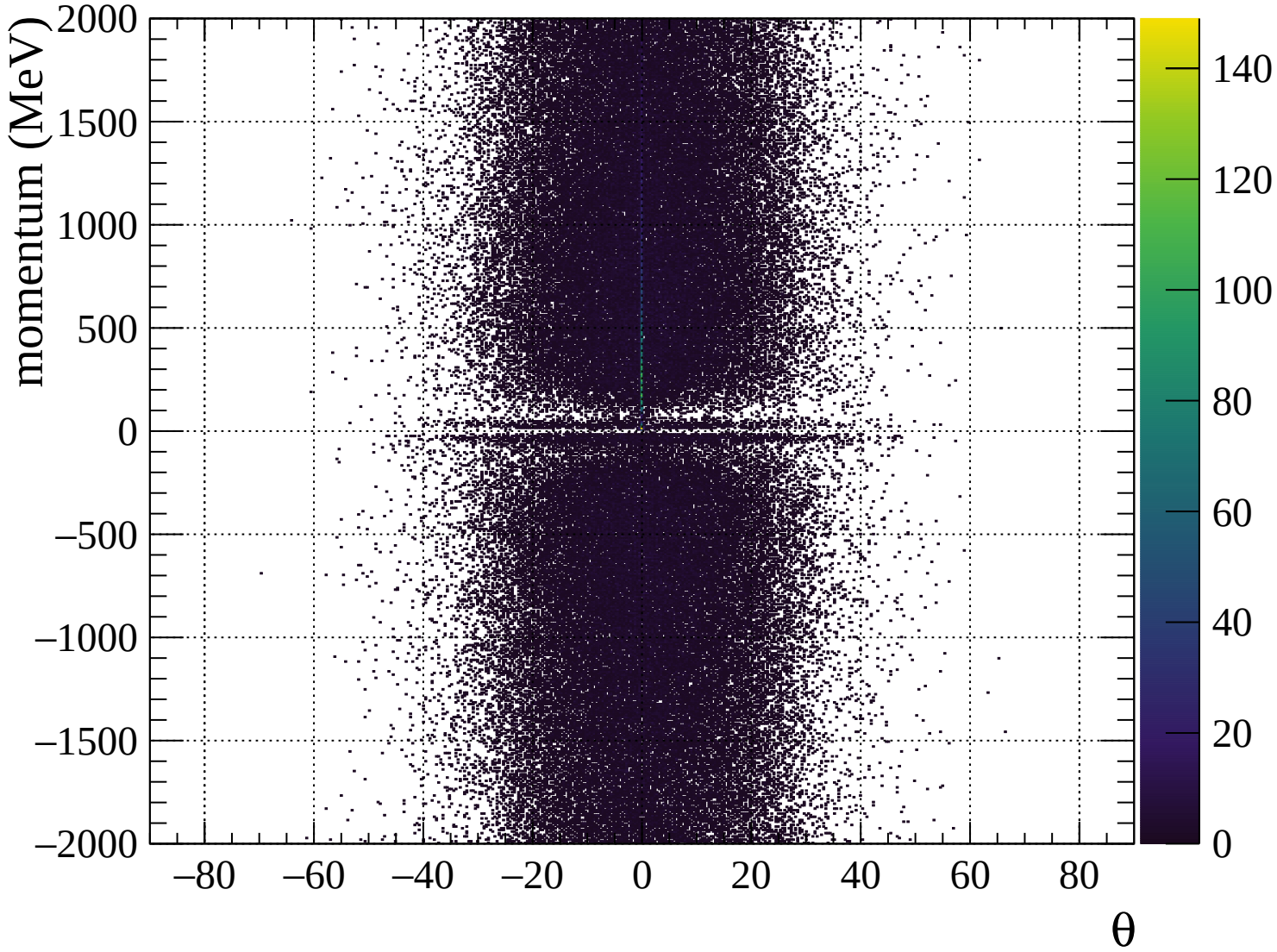


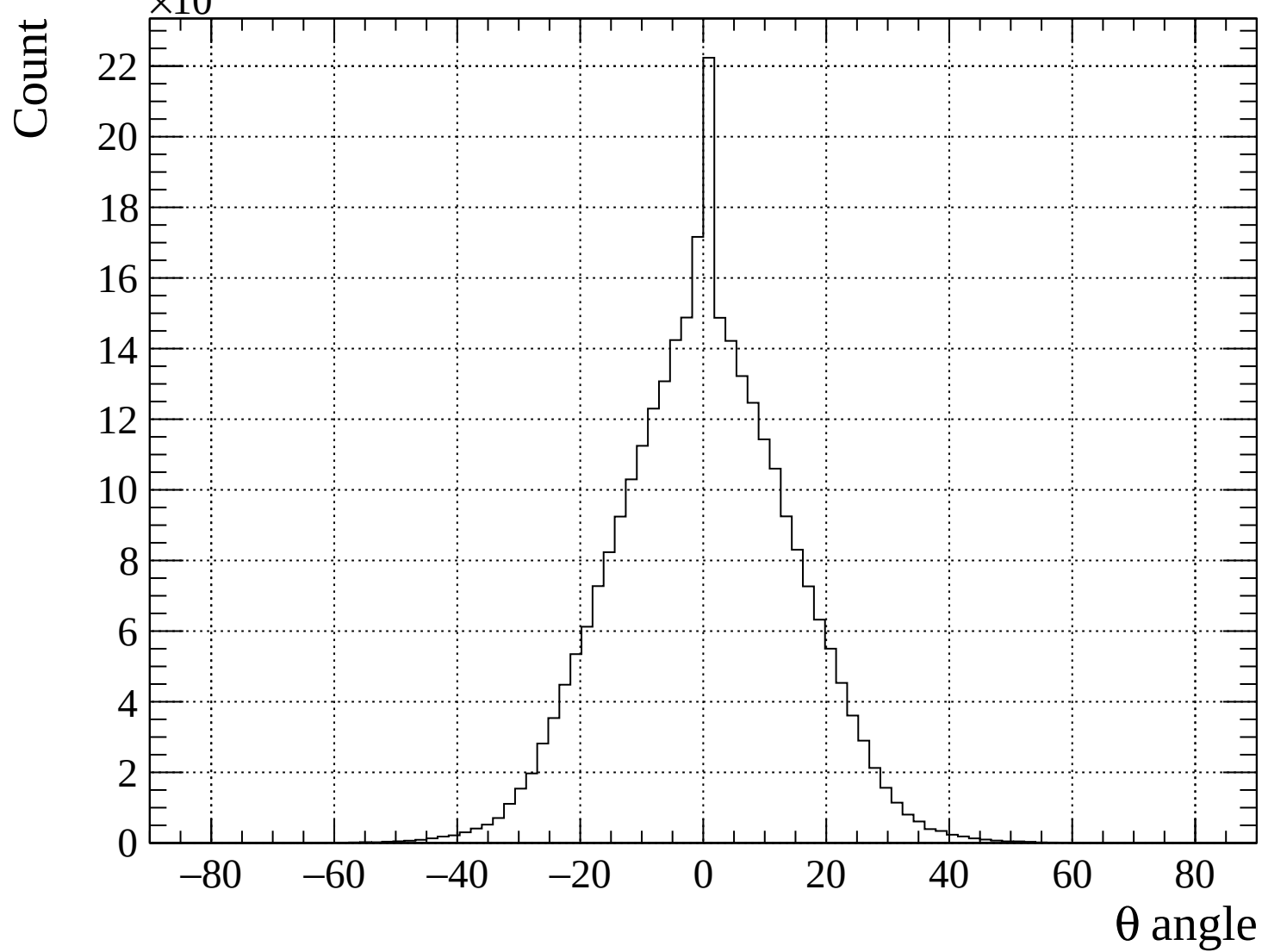


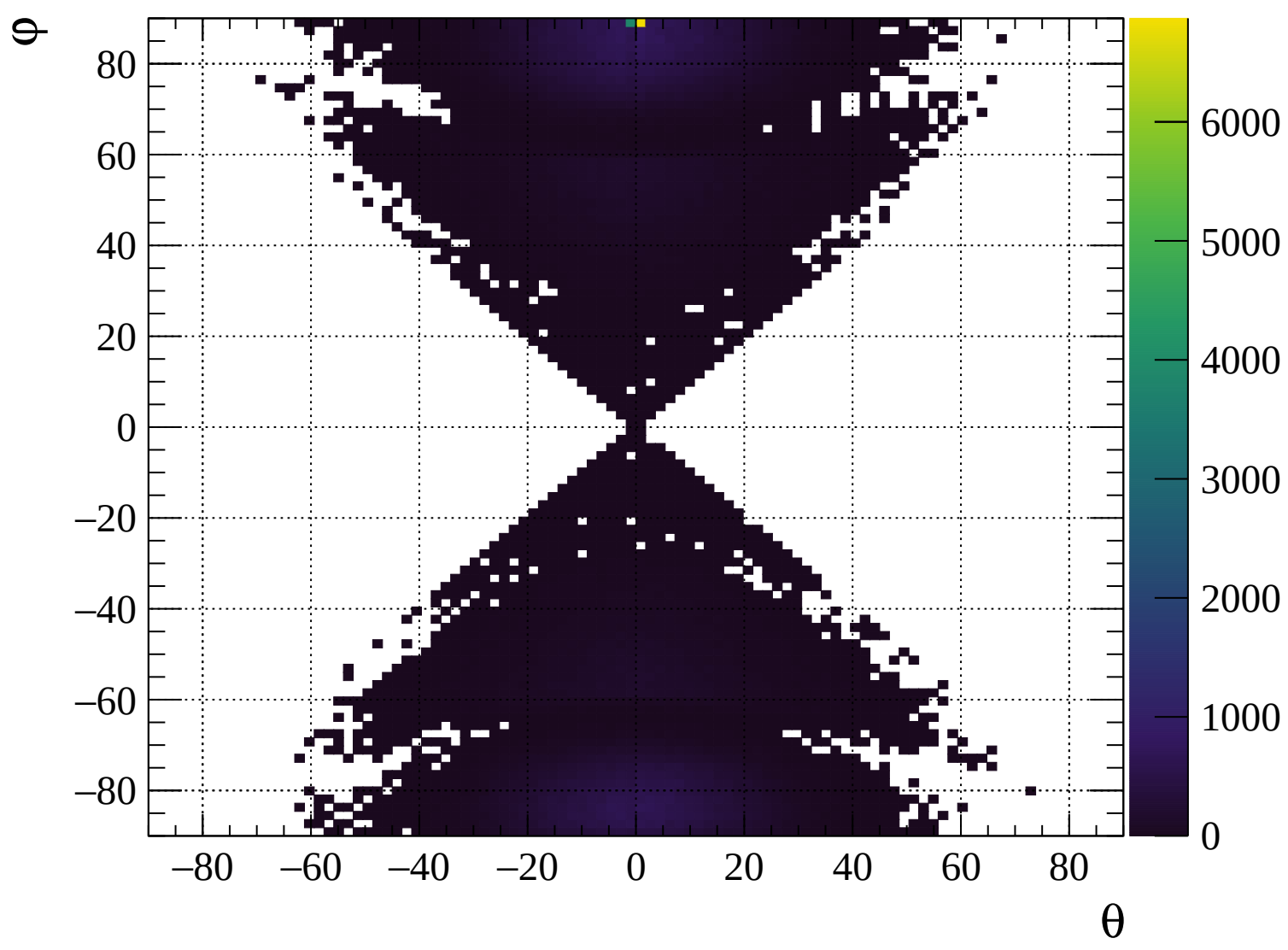


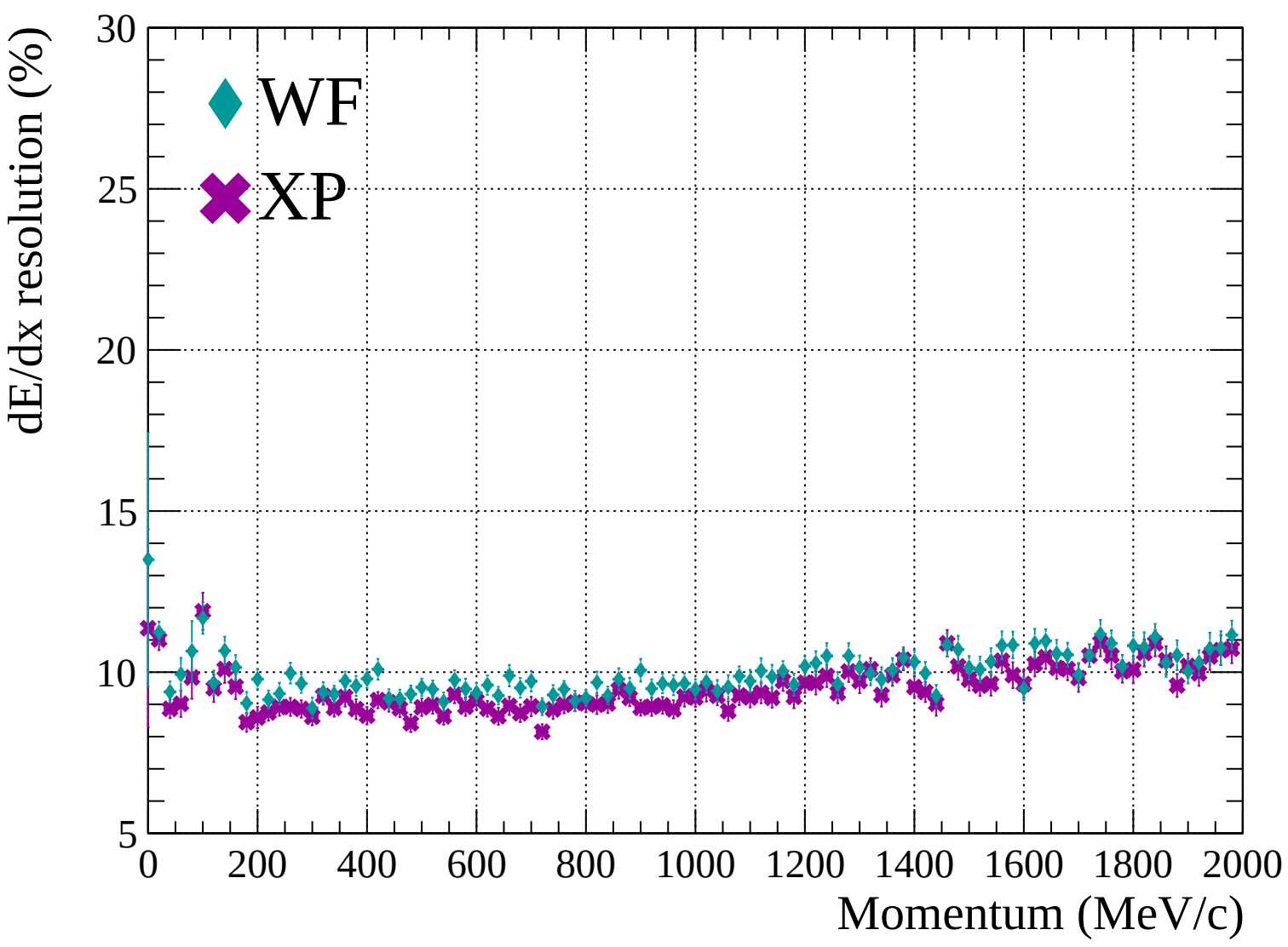


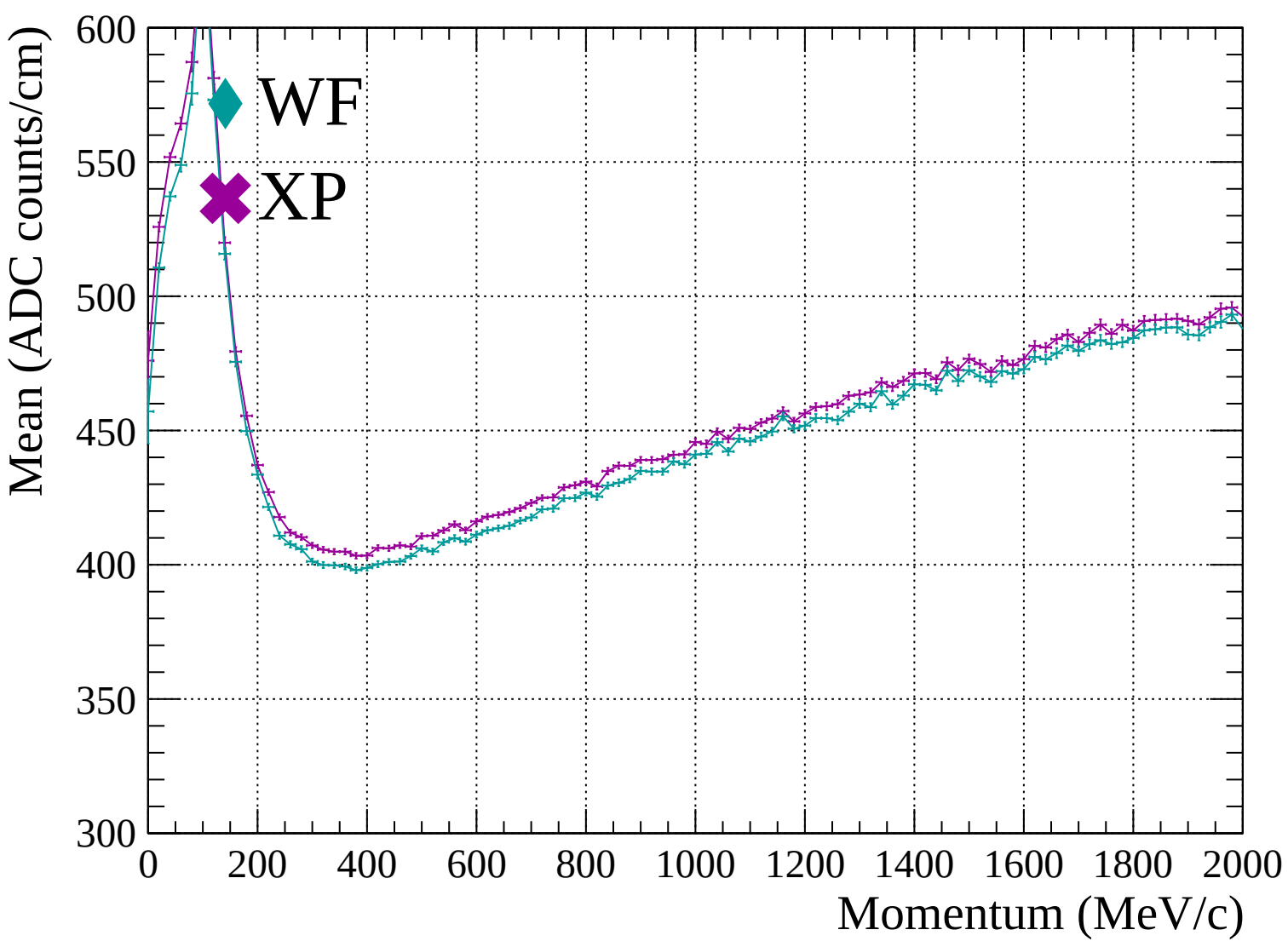


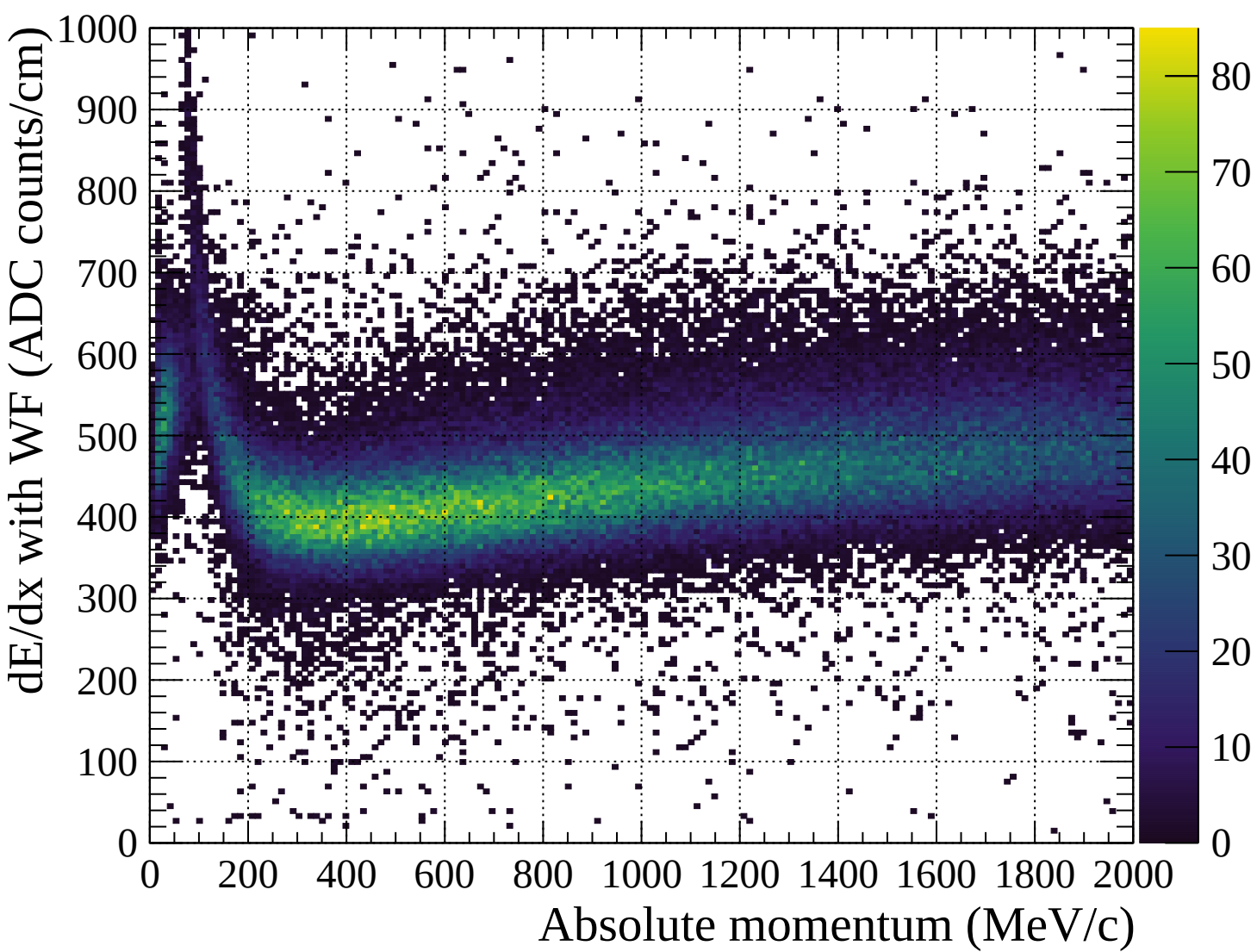


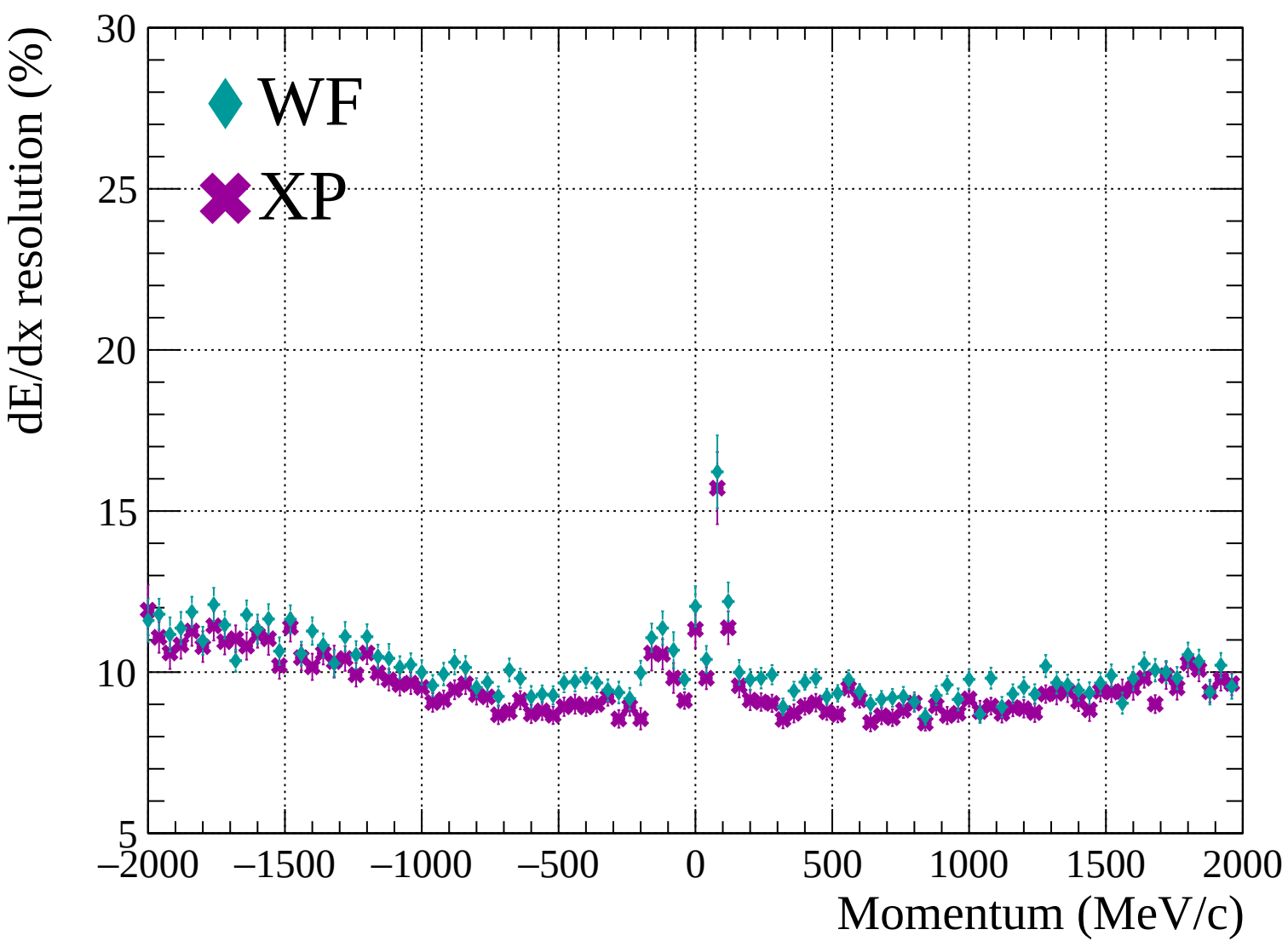


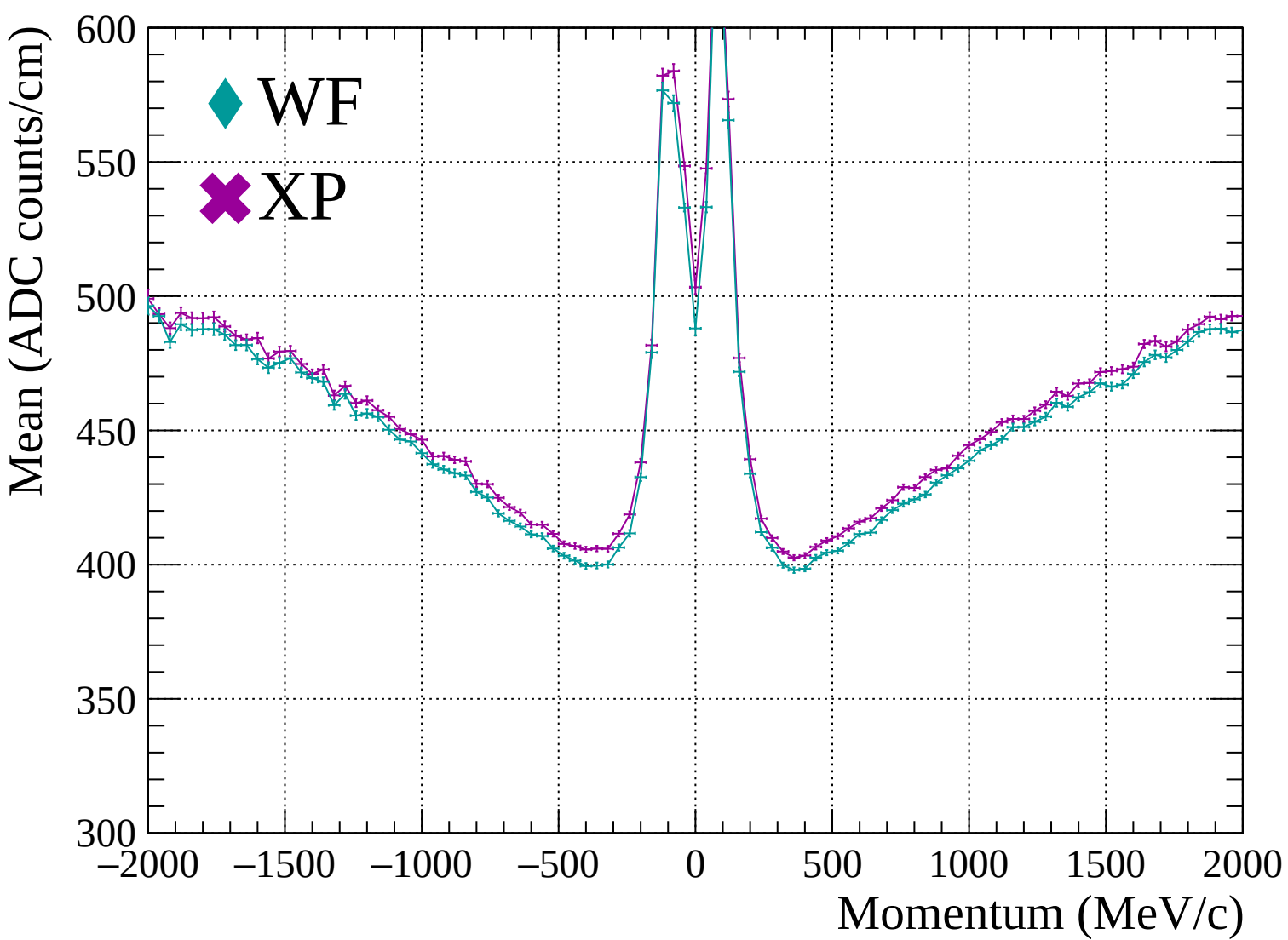


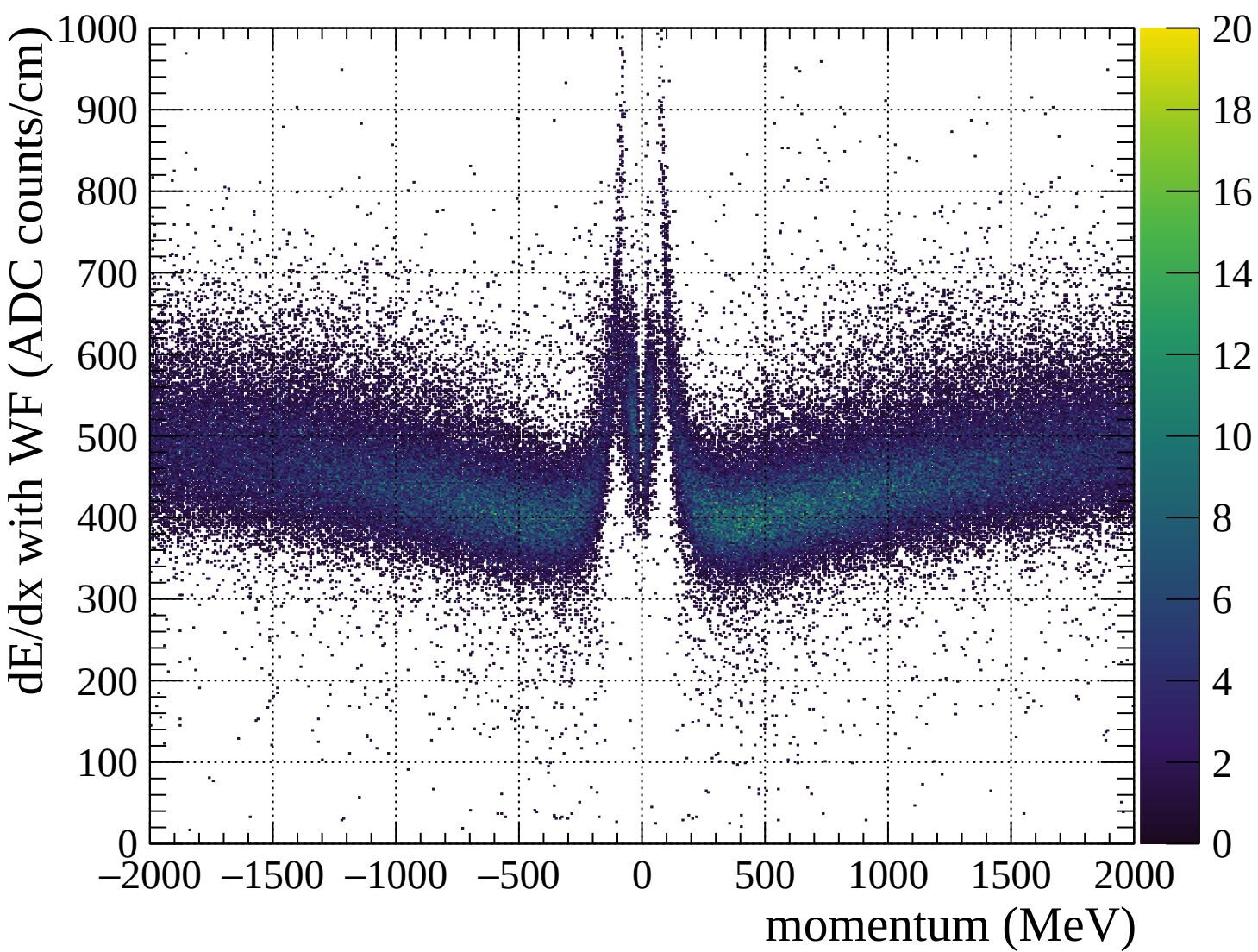


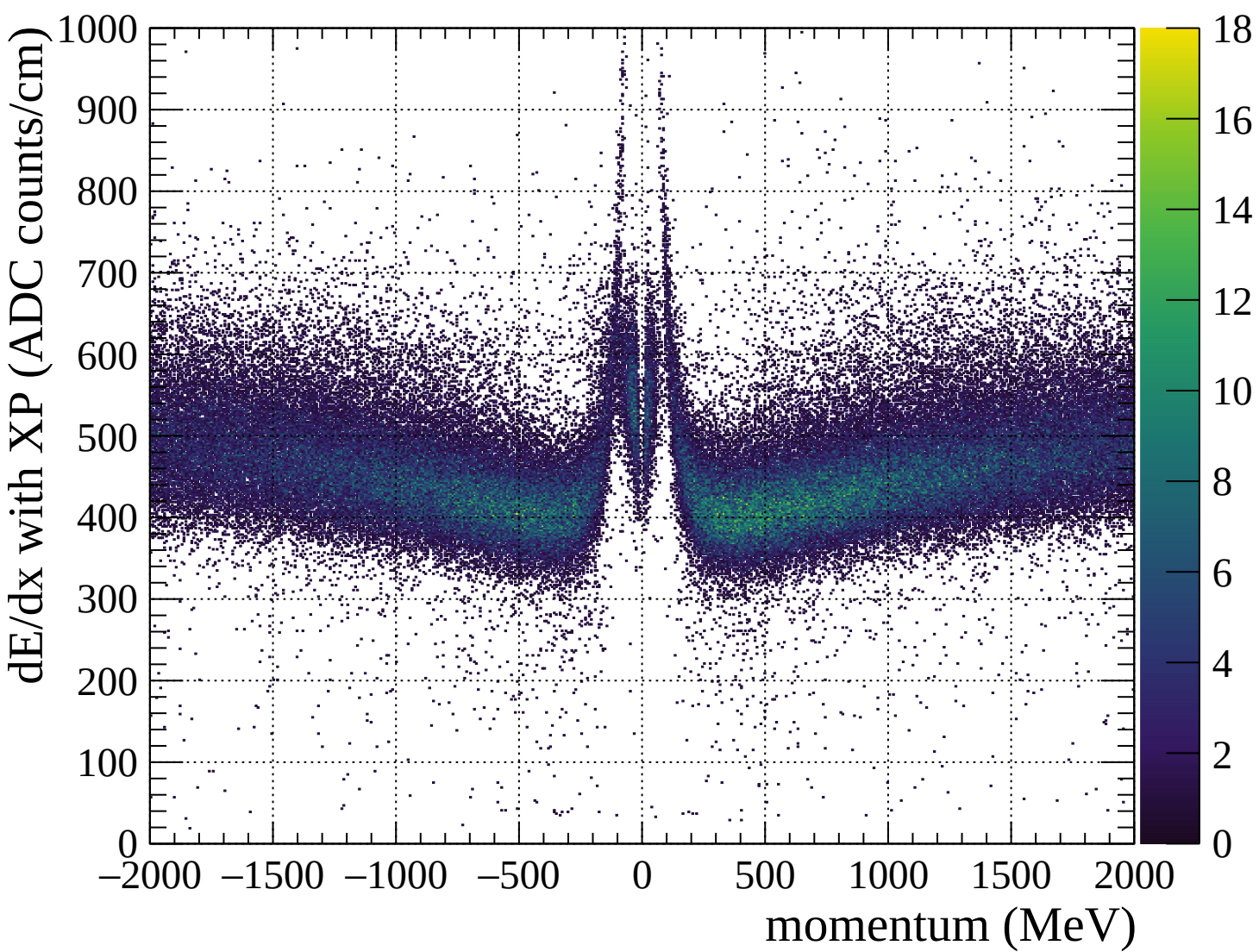




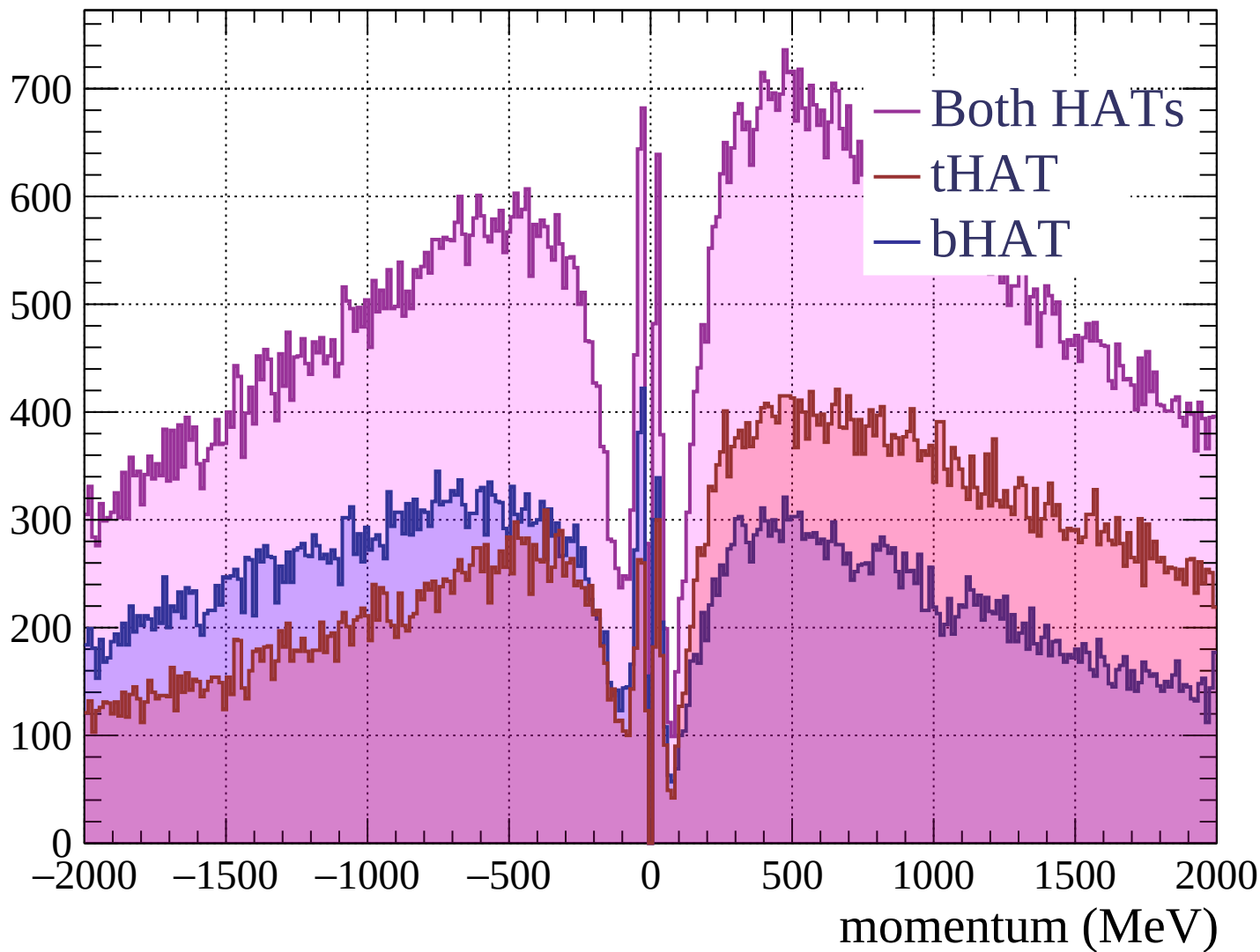


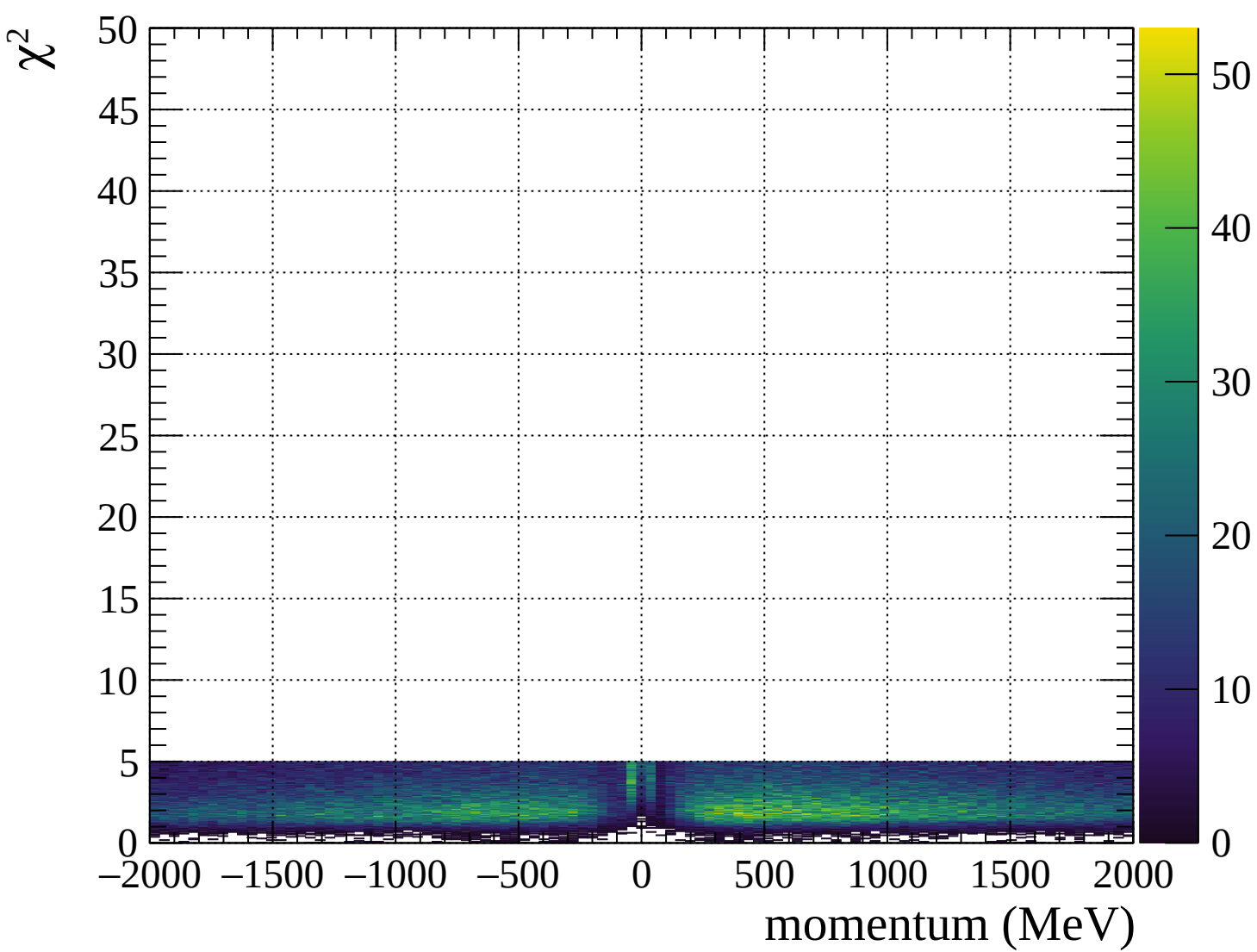




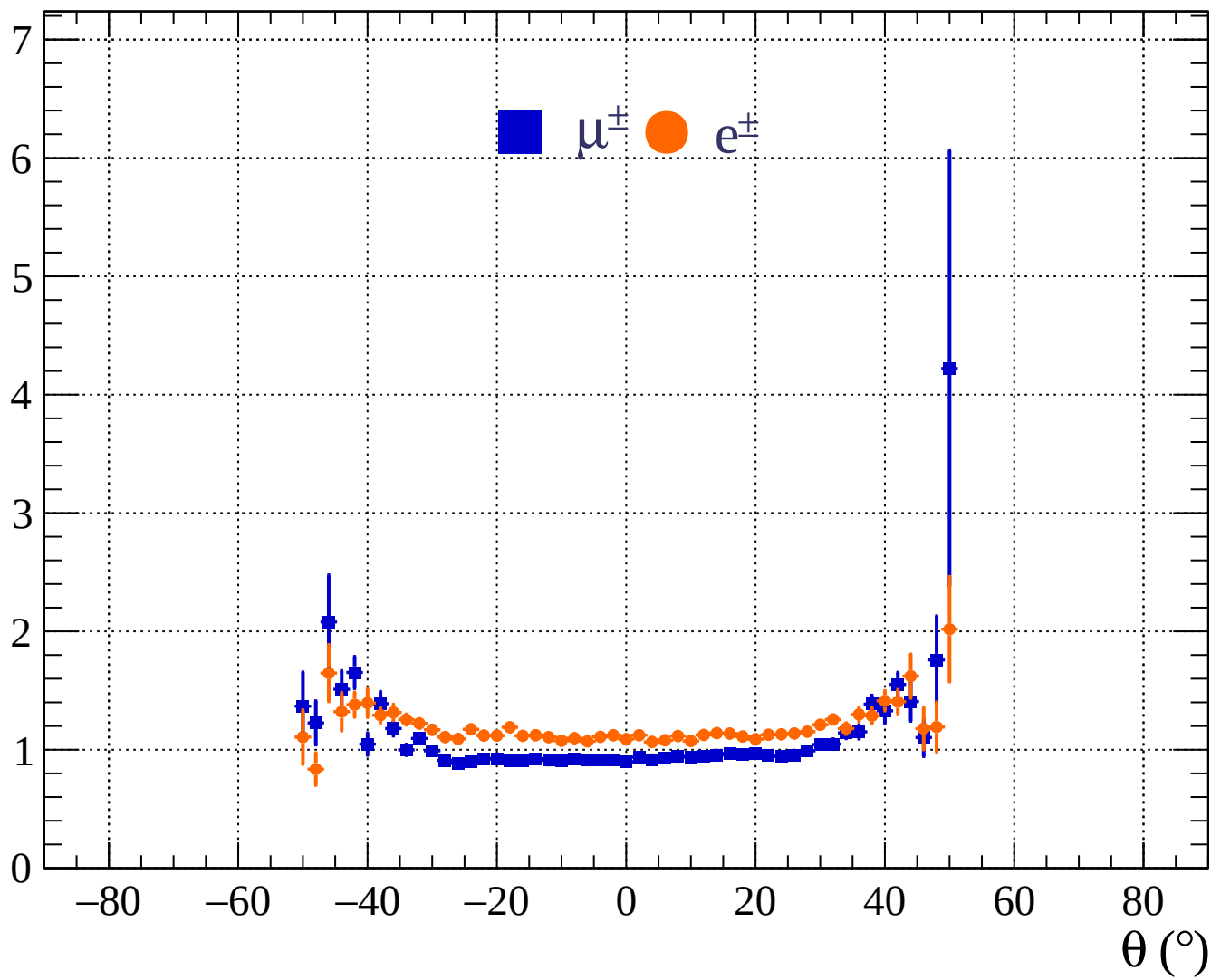


Count

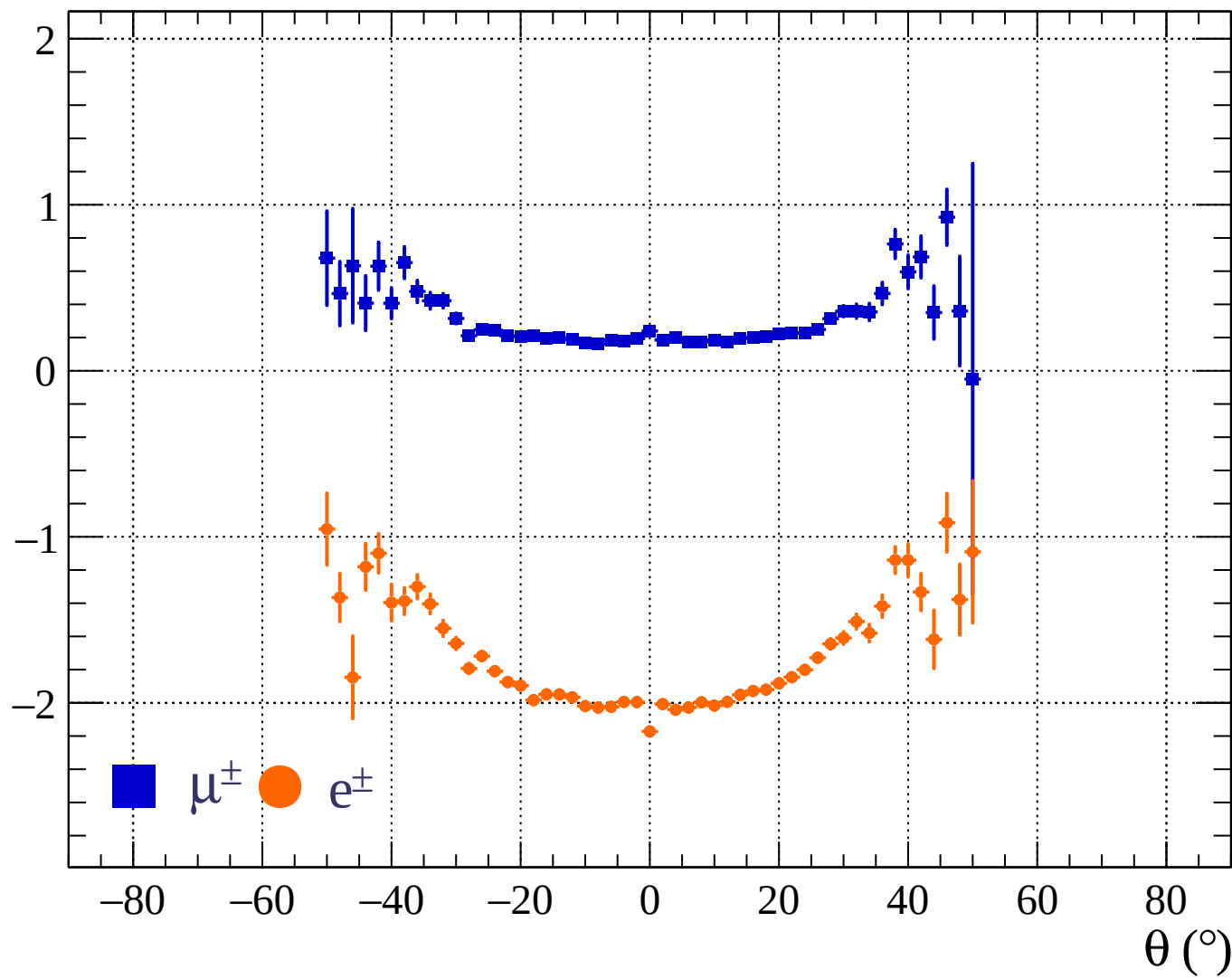




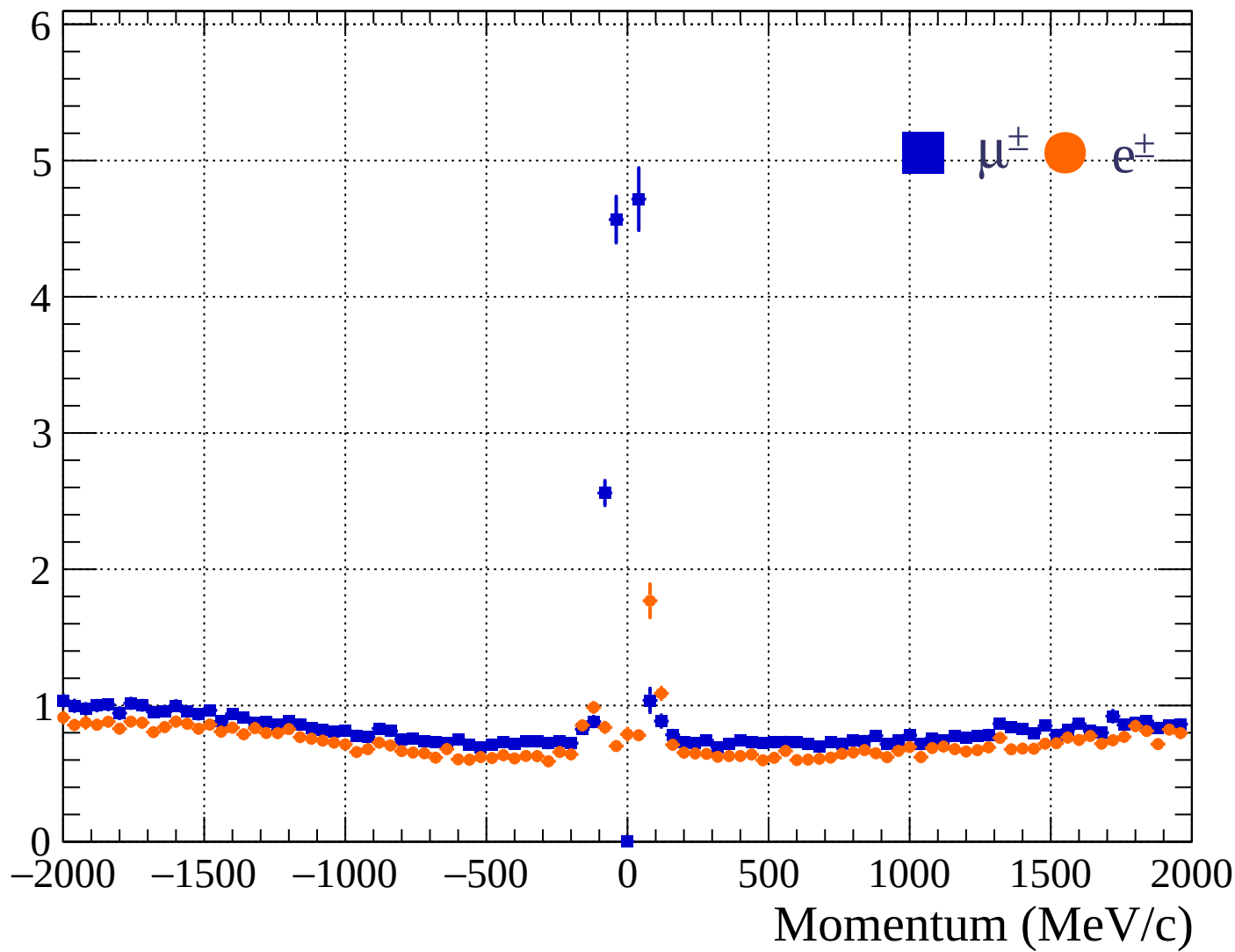
Pulls standard deviation



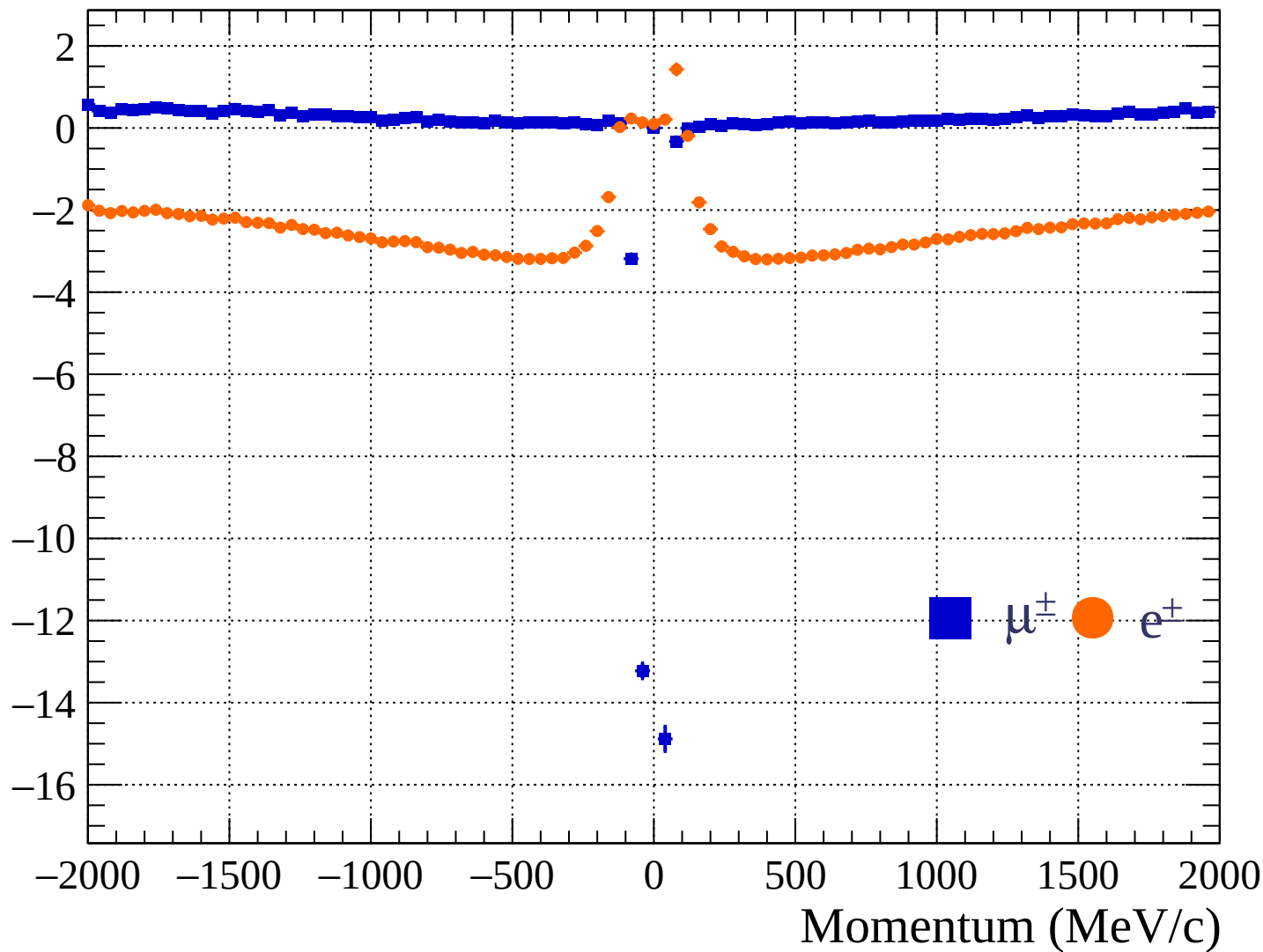
Pulls mean

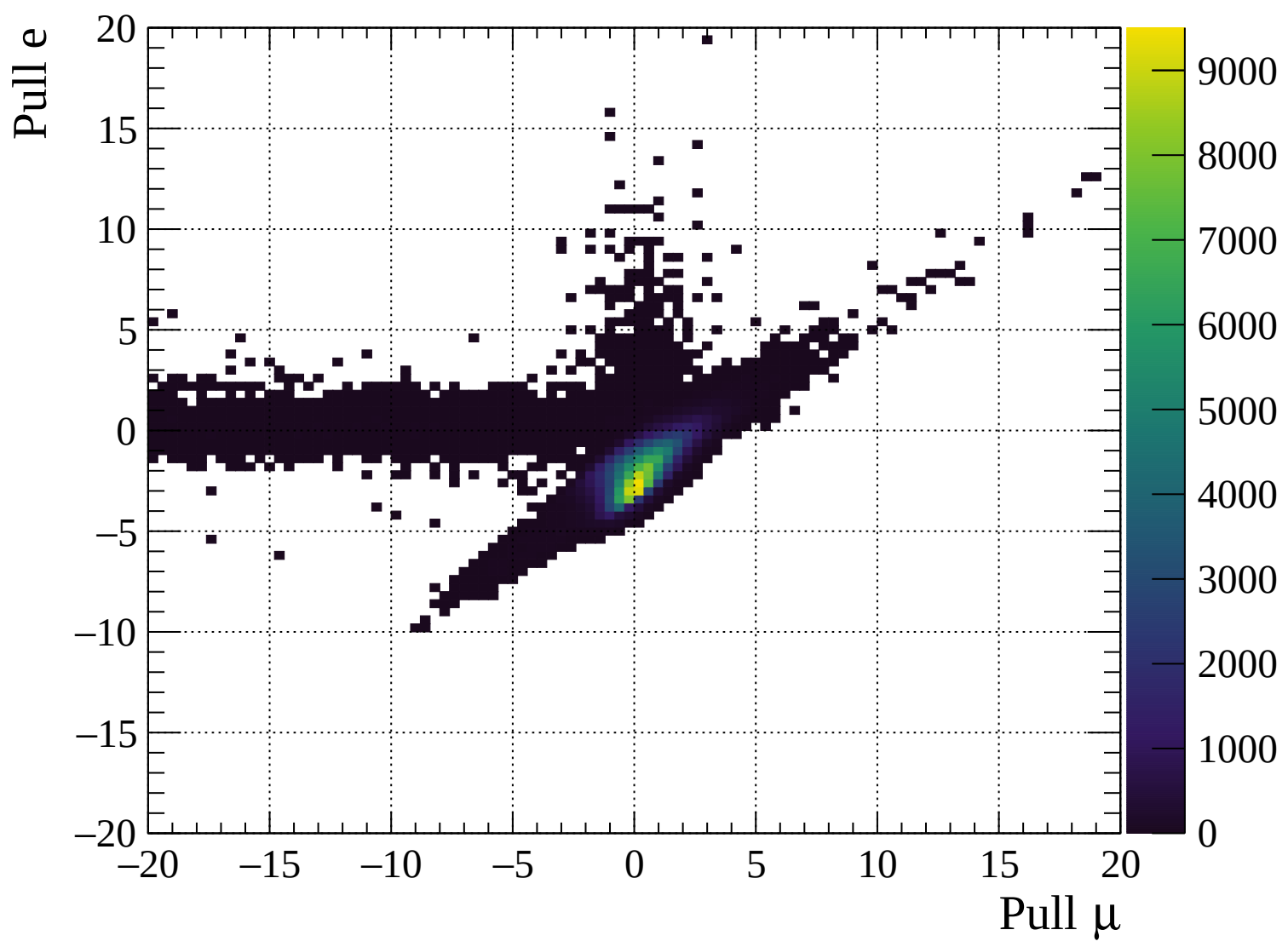


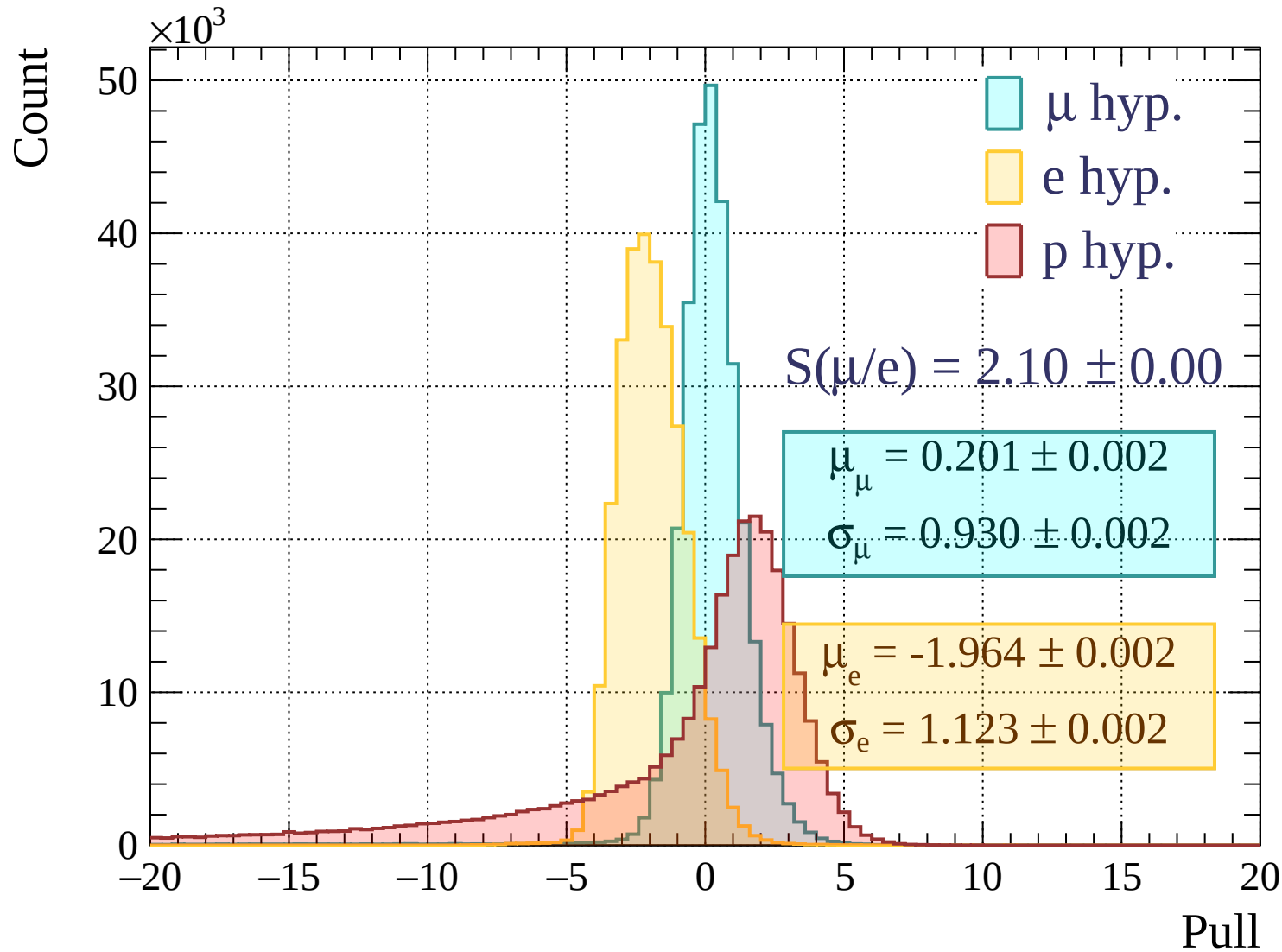
Pulls standard deviation

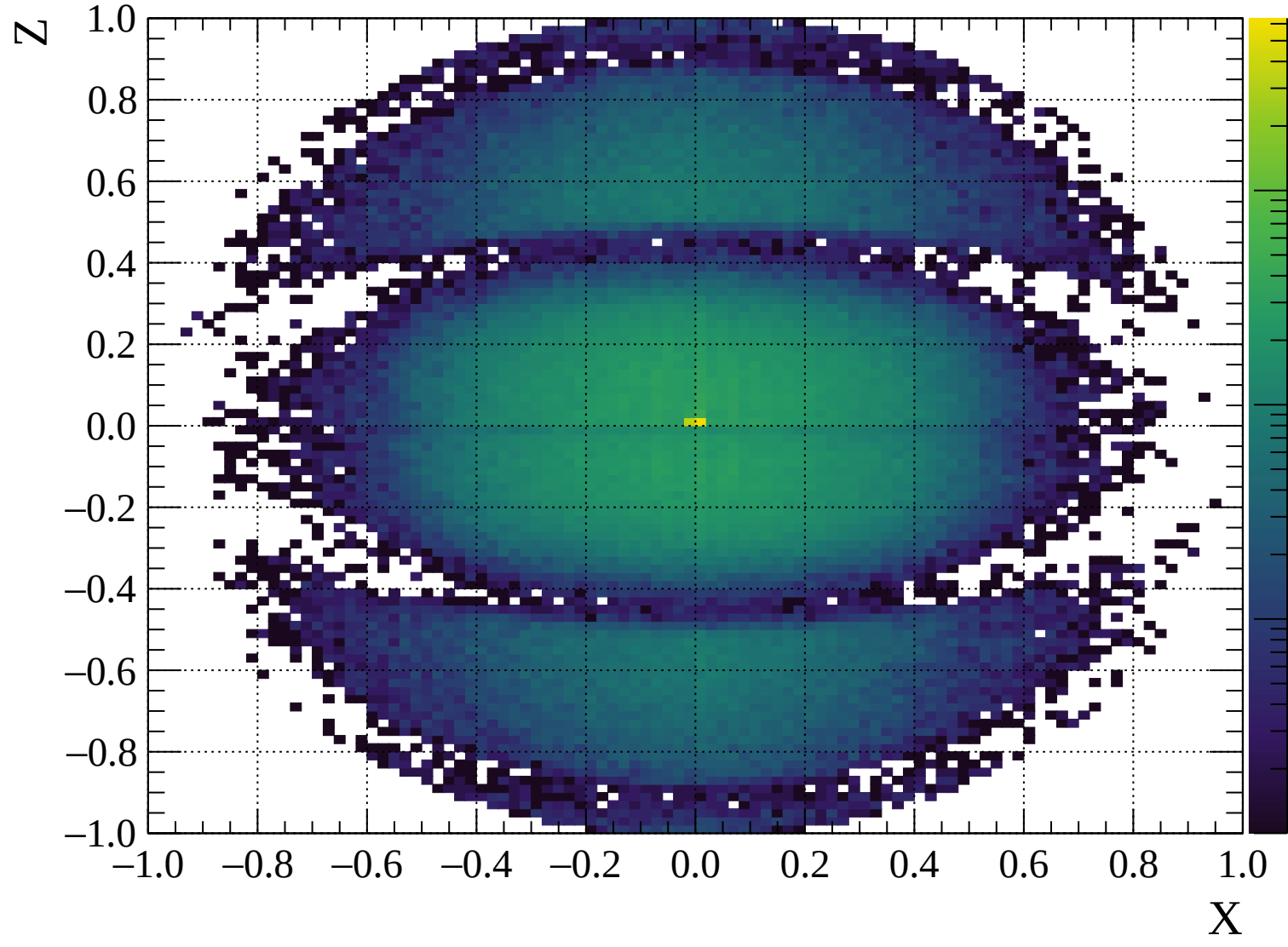


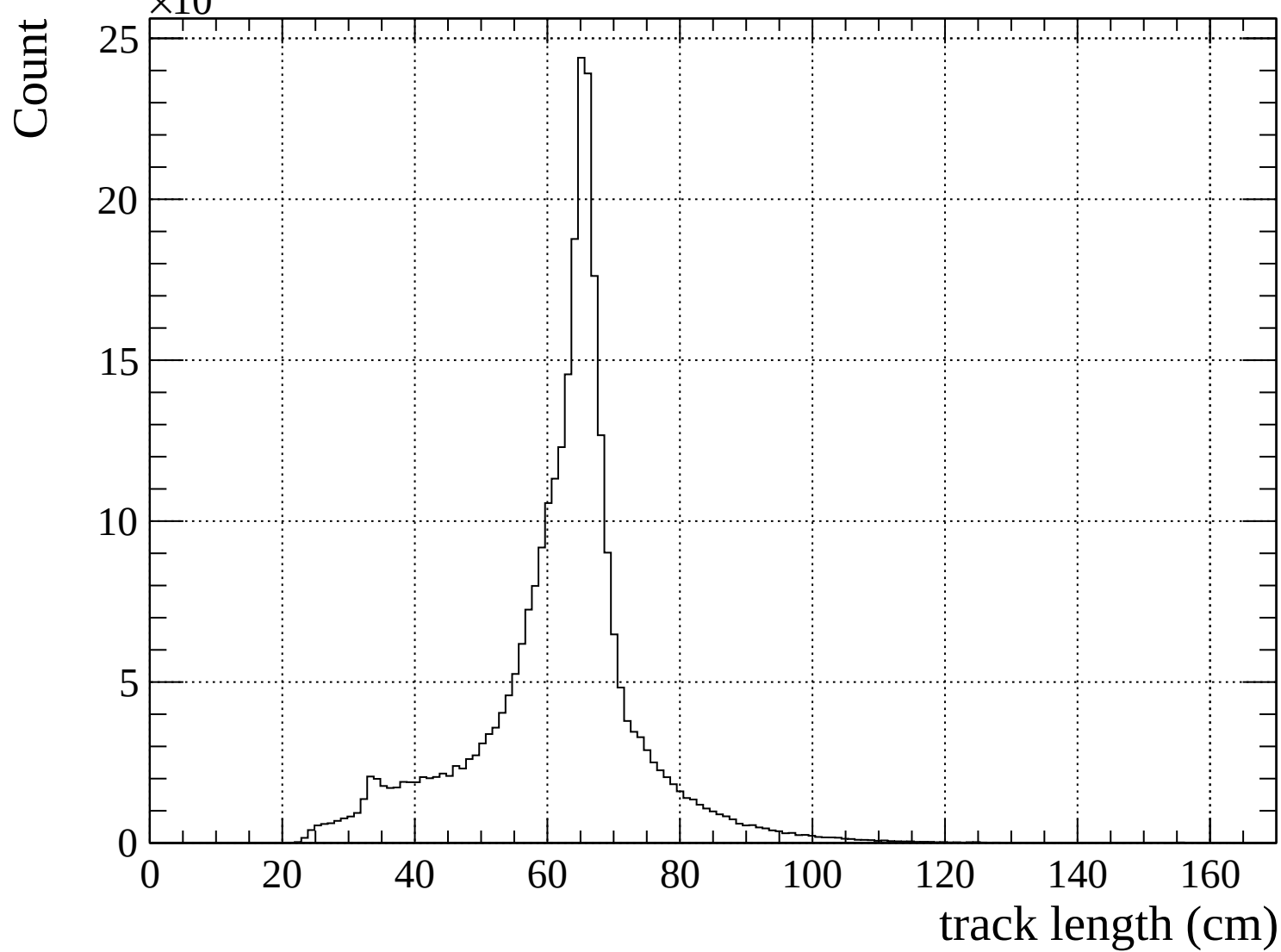
Pulls mean

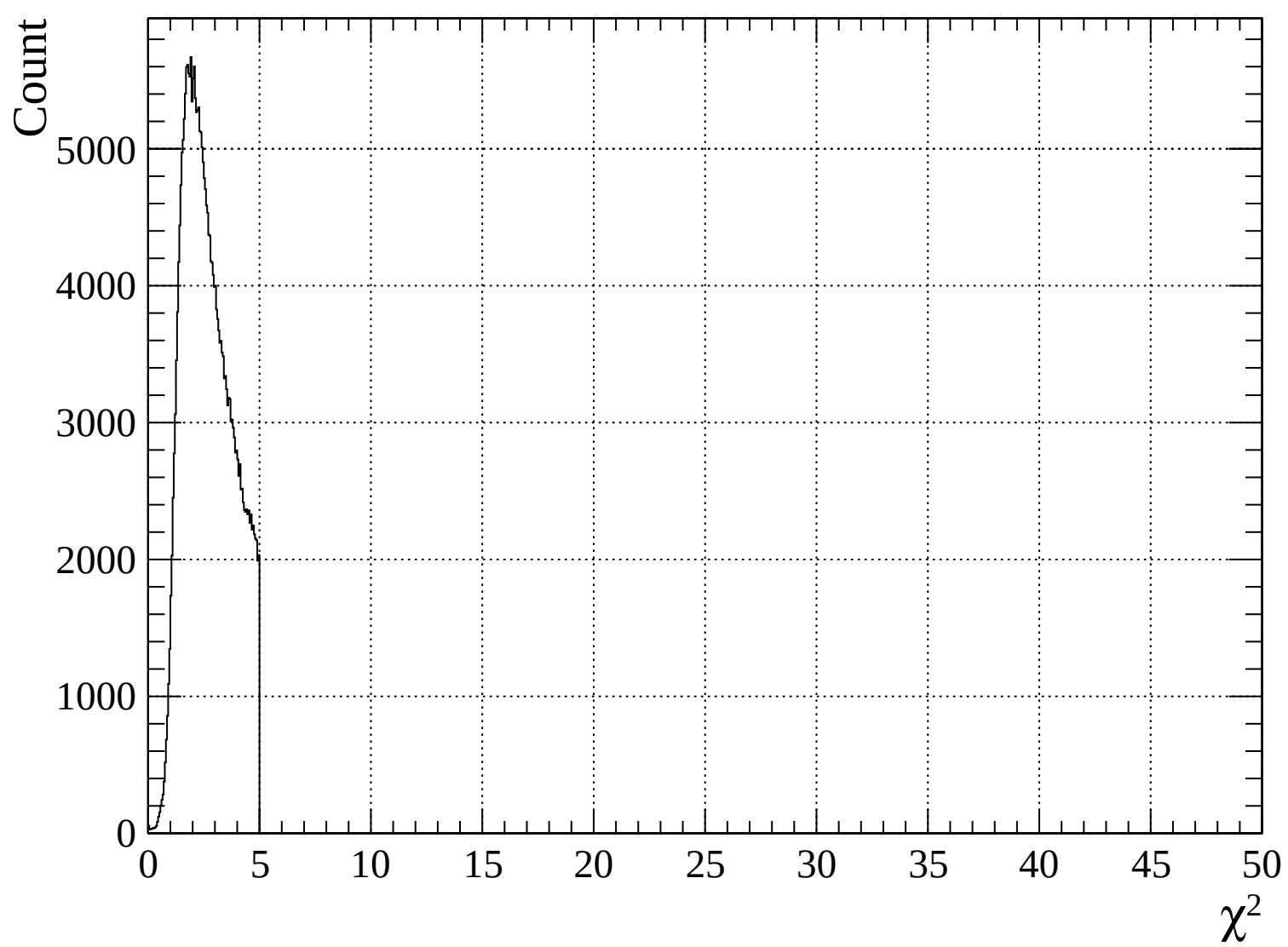


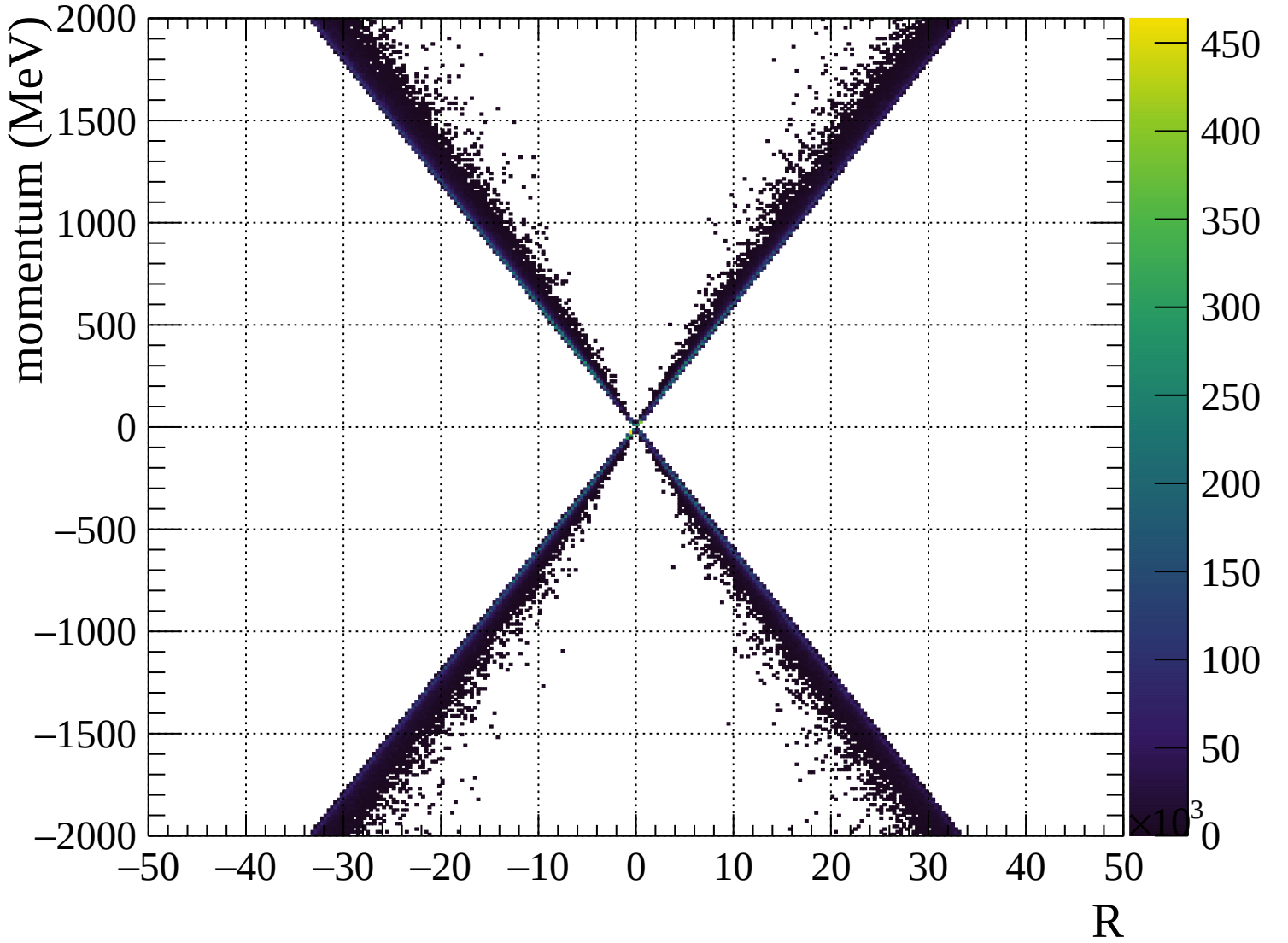


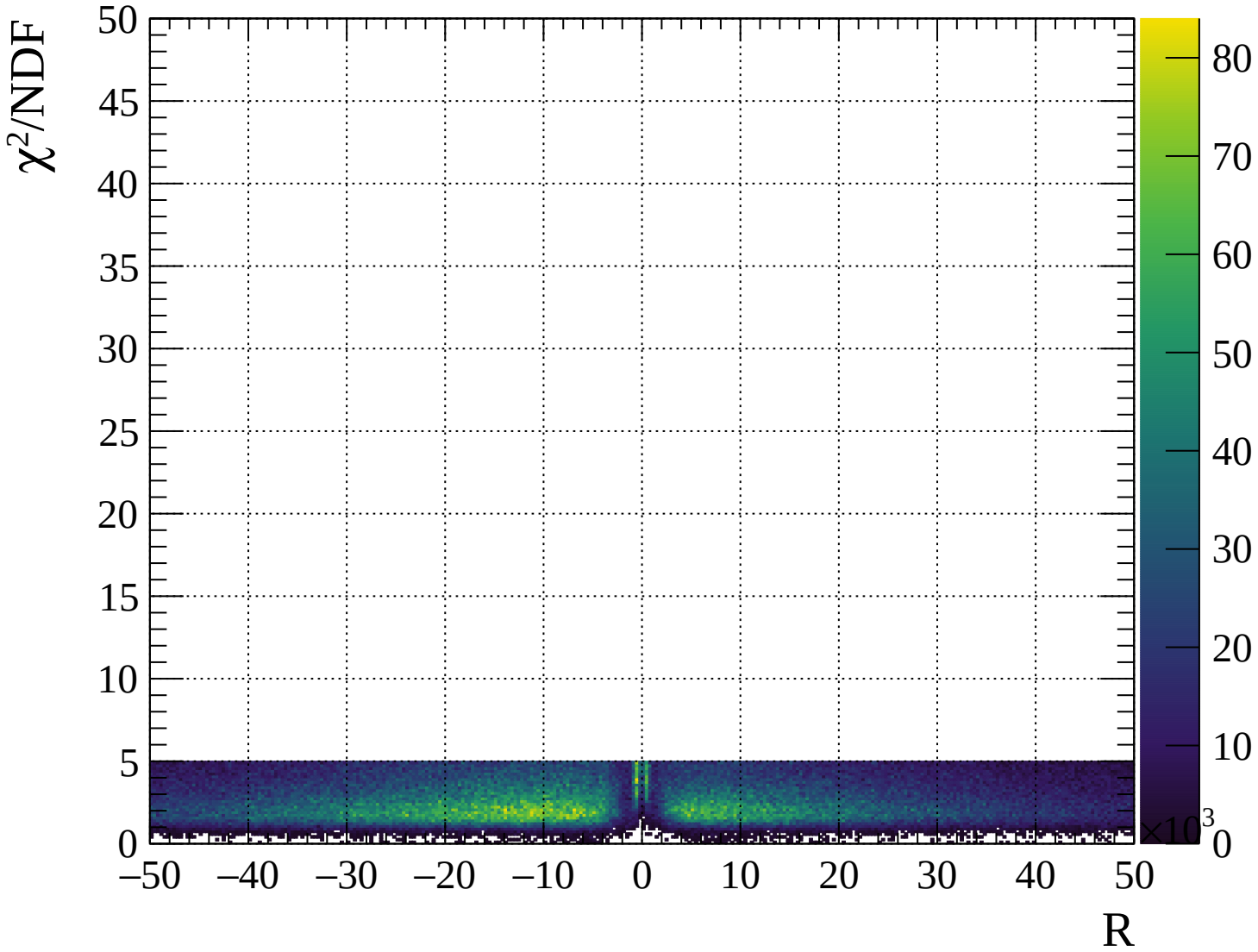






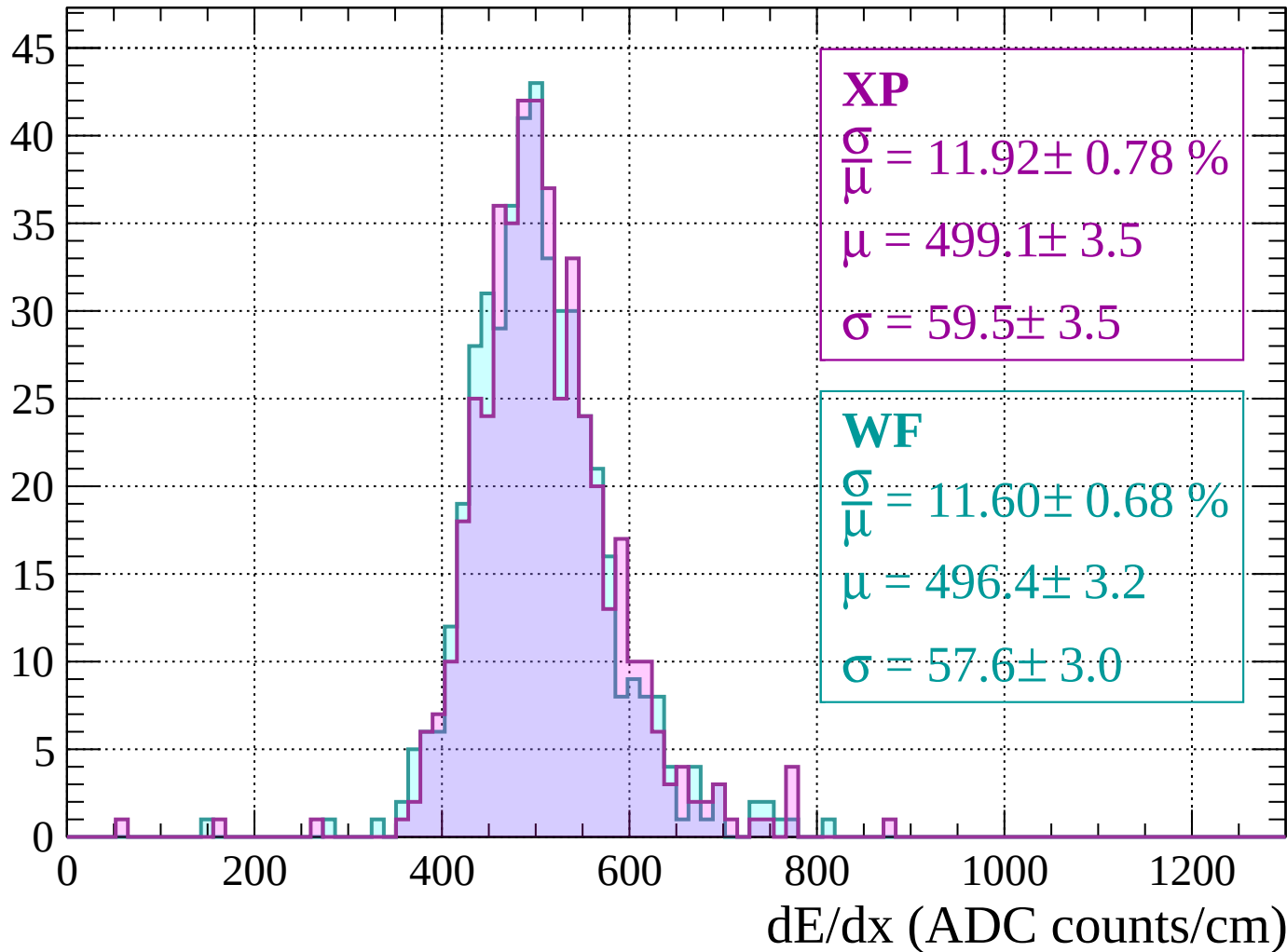






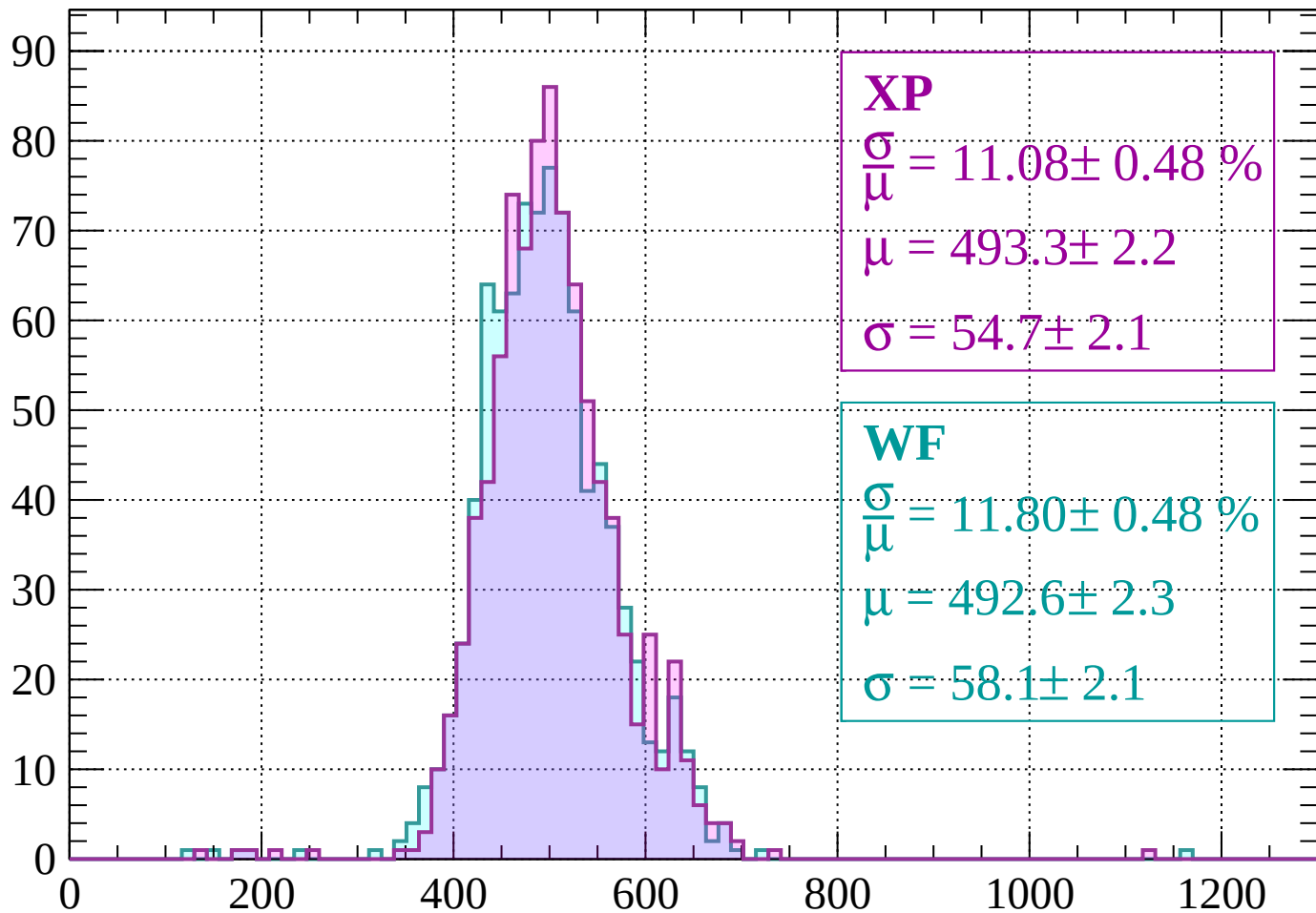
Energy loss | -2000 < p < -1960

Count



Energy loss | -1960 < p < -1920

Count



XP

$$\frac{\sigma}{\mu} = 11.08 \pm 0.48 \%$$

$$\mu = 493.3 \pm 2.2$$

$$\sigma = 54.7 \pm 2.1$$

WF

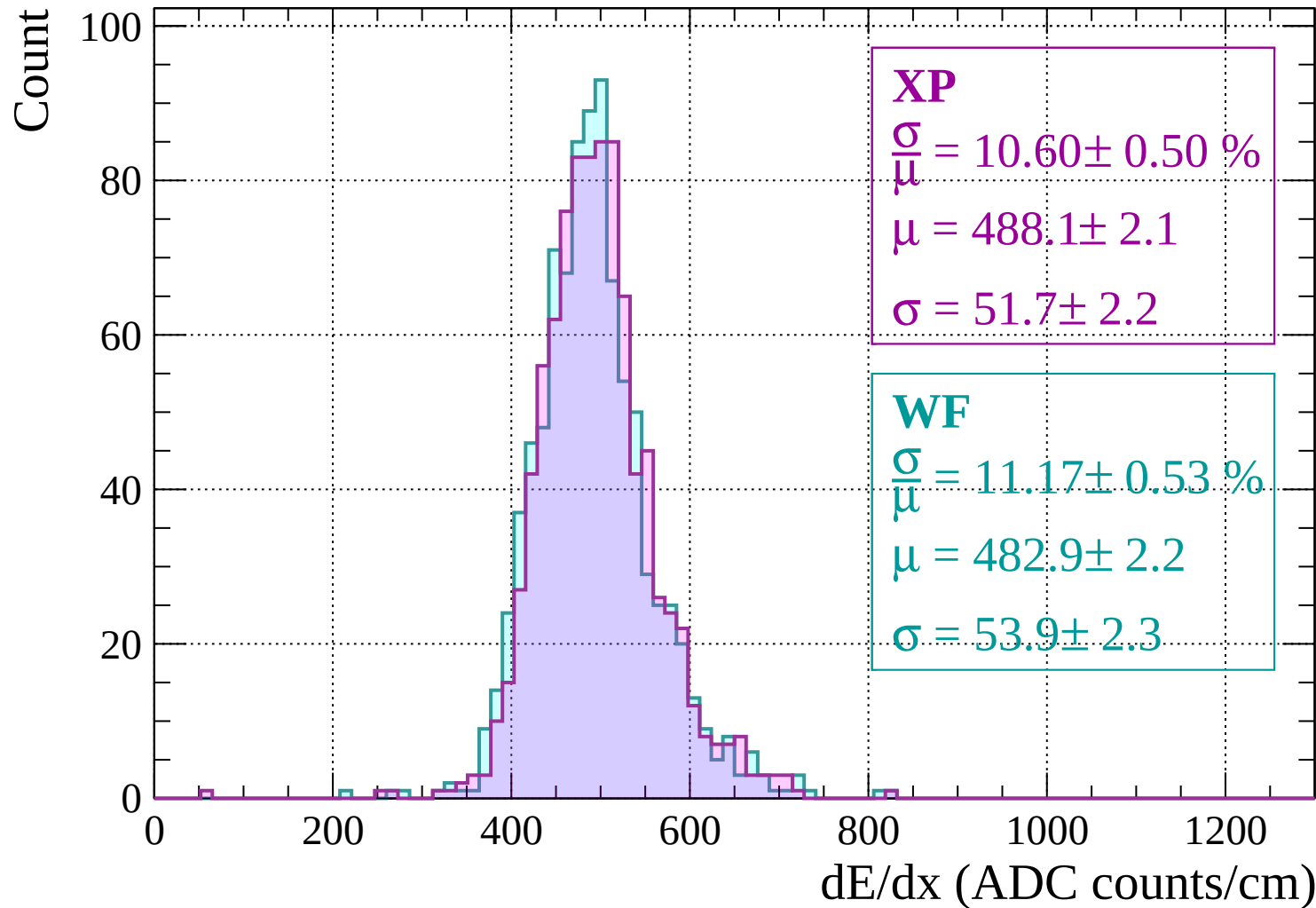
$$\frac{\sigma}{\mu} = 11.80 \pm 0.48 \%$$

$$\mu = 492.6 \pm 2.3$$

$$\sigma = 58.1 \pm 2.1$$

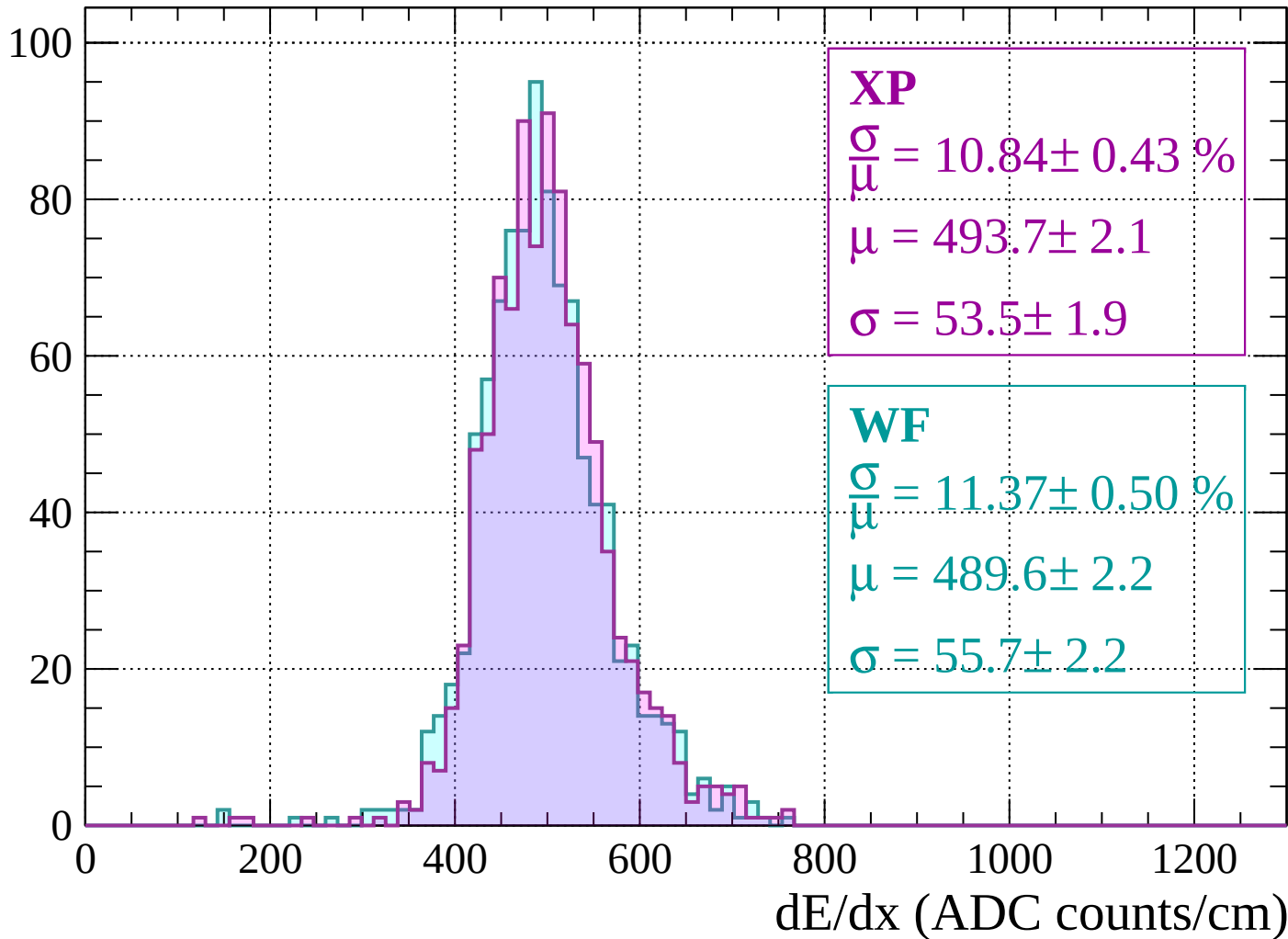
dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880



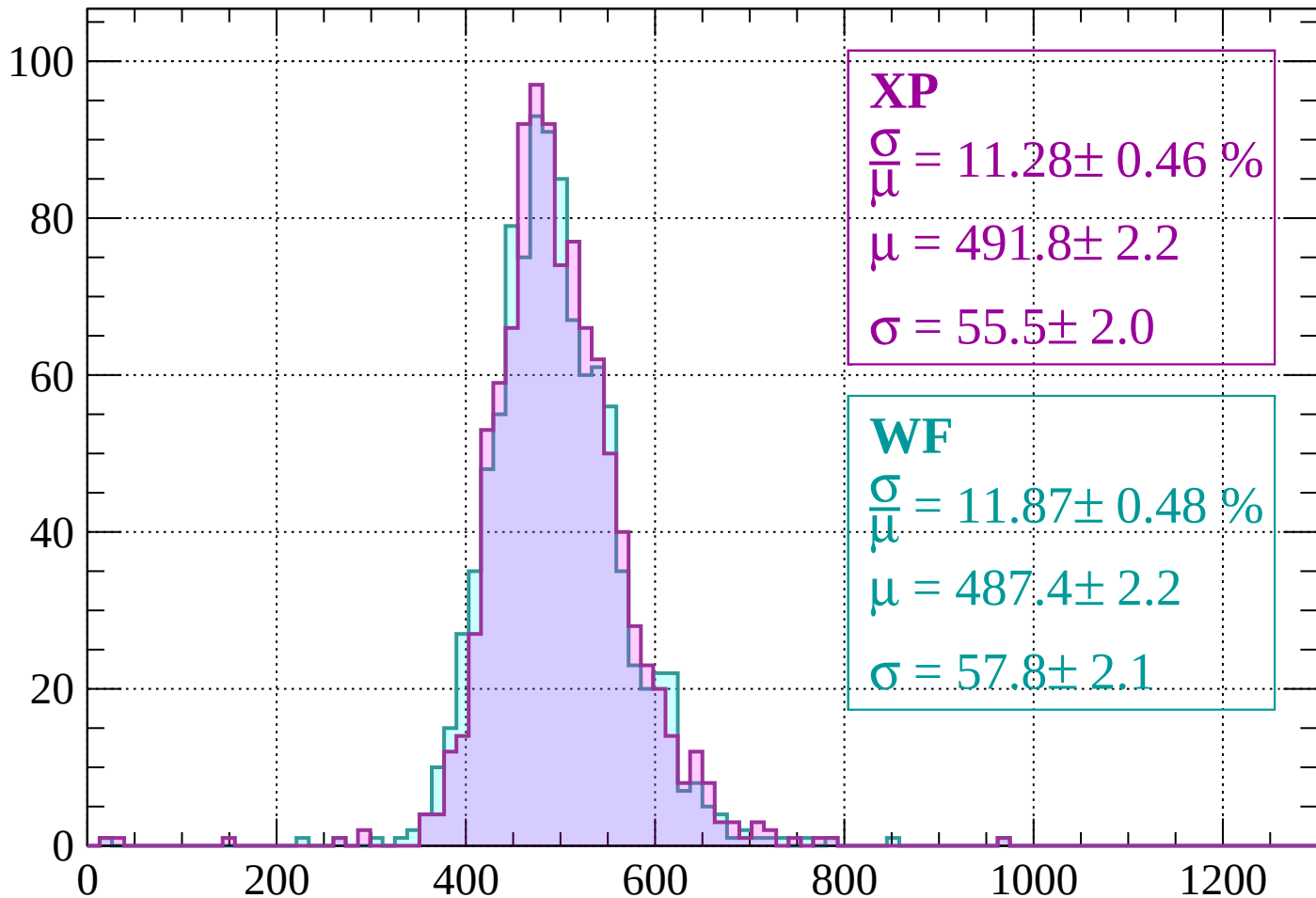
Energy loss | $-1880 < p < -1840$

Count



Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 11.28 \pm 0.46 \%$$

$$\mu = 491.8 \pm 2.2$$

$$\sigma = 55.5 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 11.87 \pm 0.48 \%$$

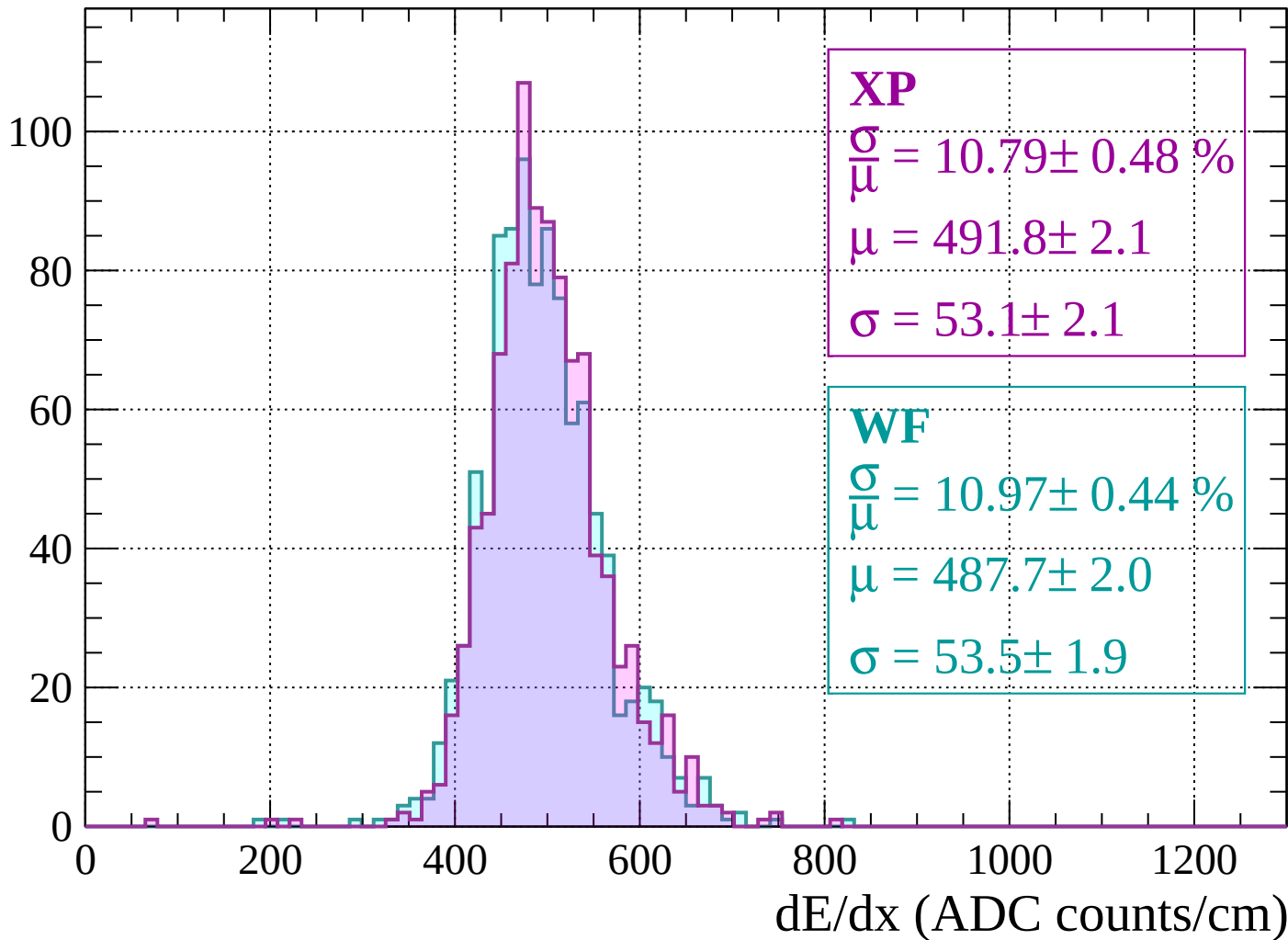
$$\mu = 487.4 \pm 2.2$$

$$\sigma = 57.8 \pm 2.1$$

dE/dx (ADC counts/cm)

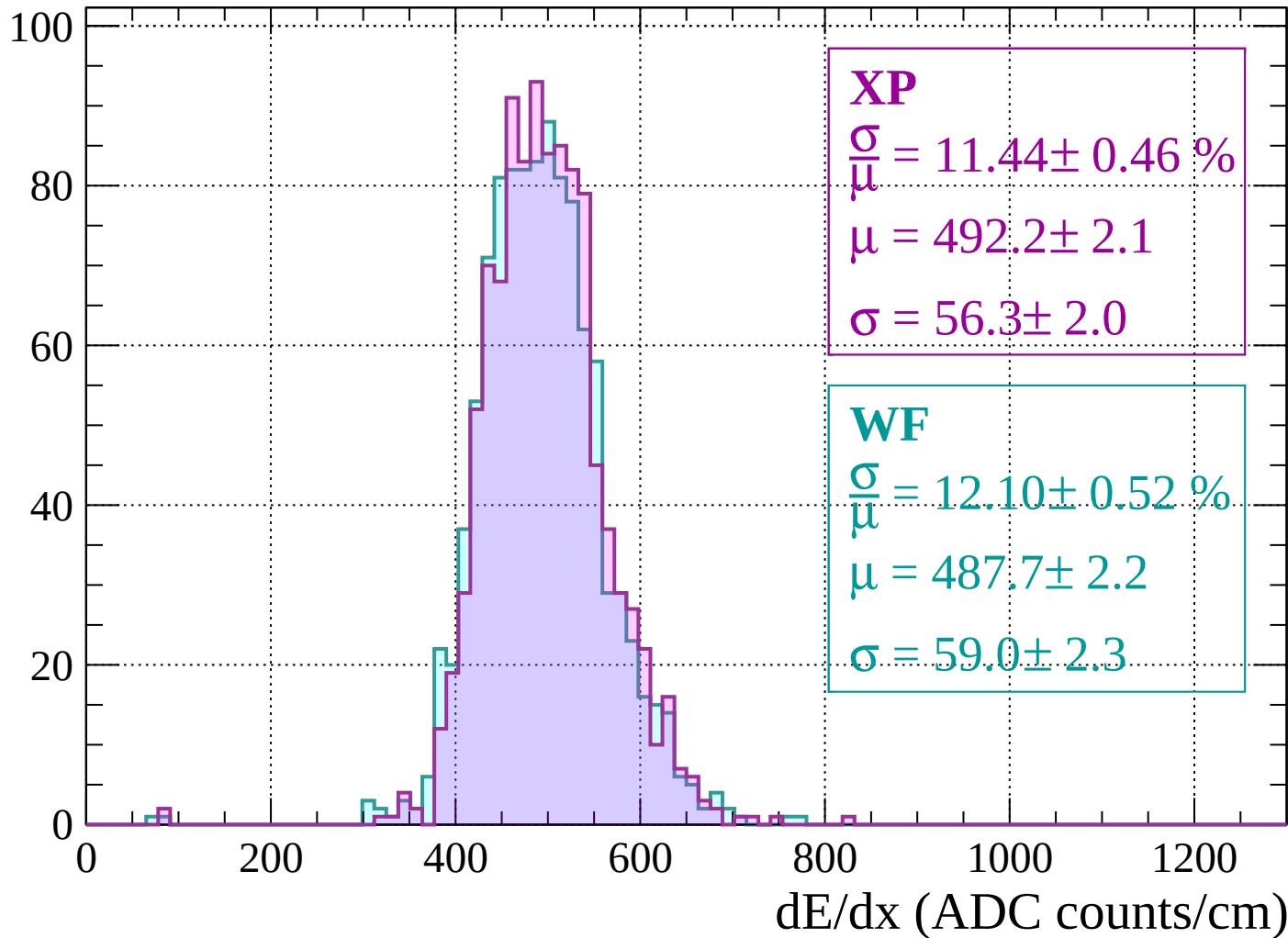
Energy loss | $-1800 < p < -1760$

Count



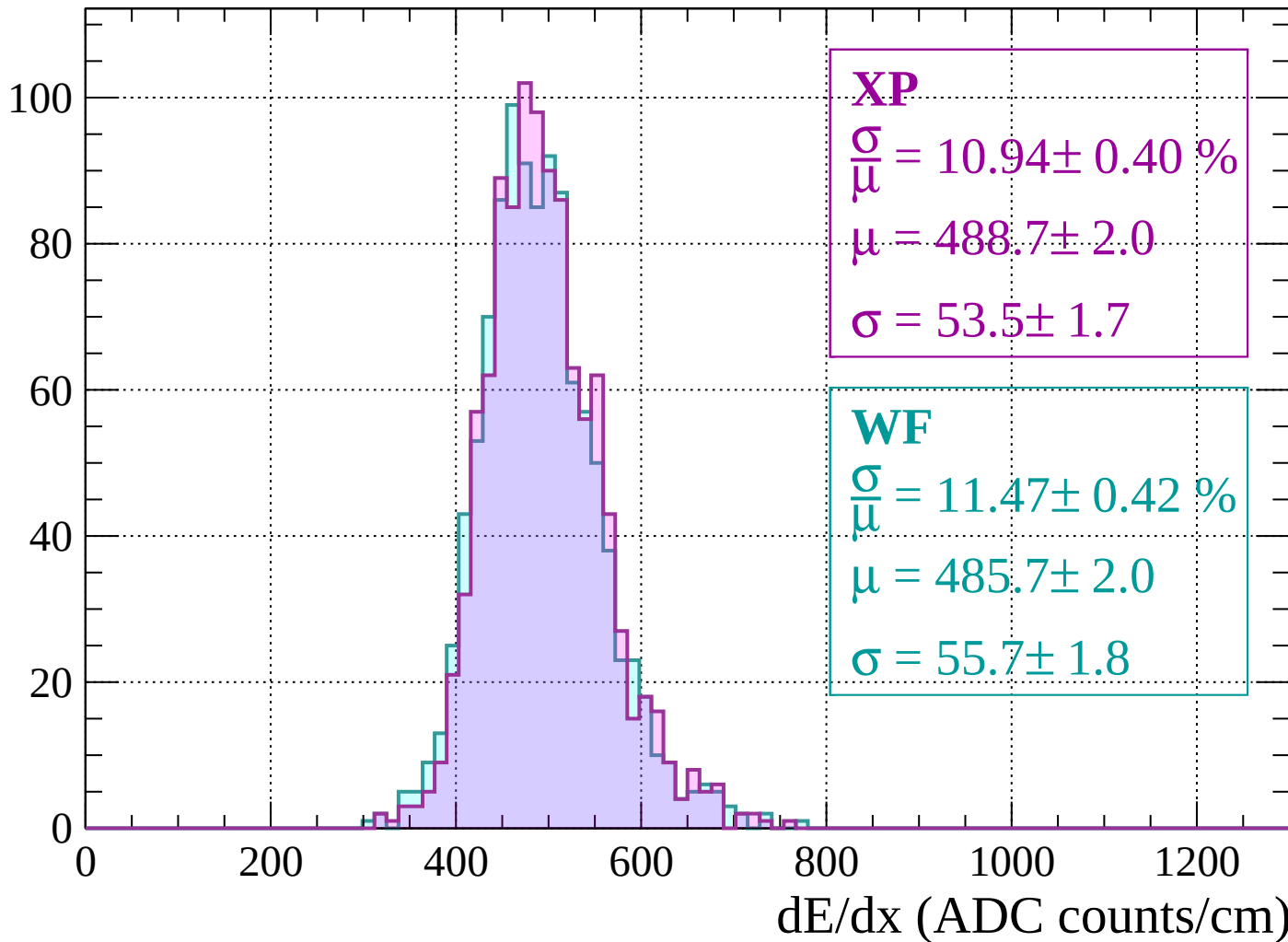
Energy loss | $-1760 < p < -1720$

Count



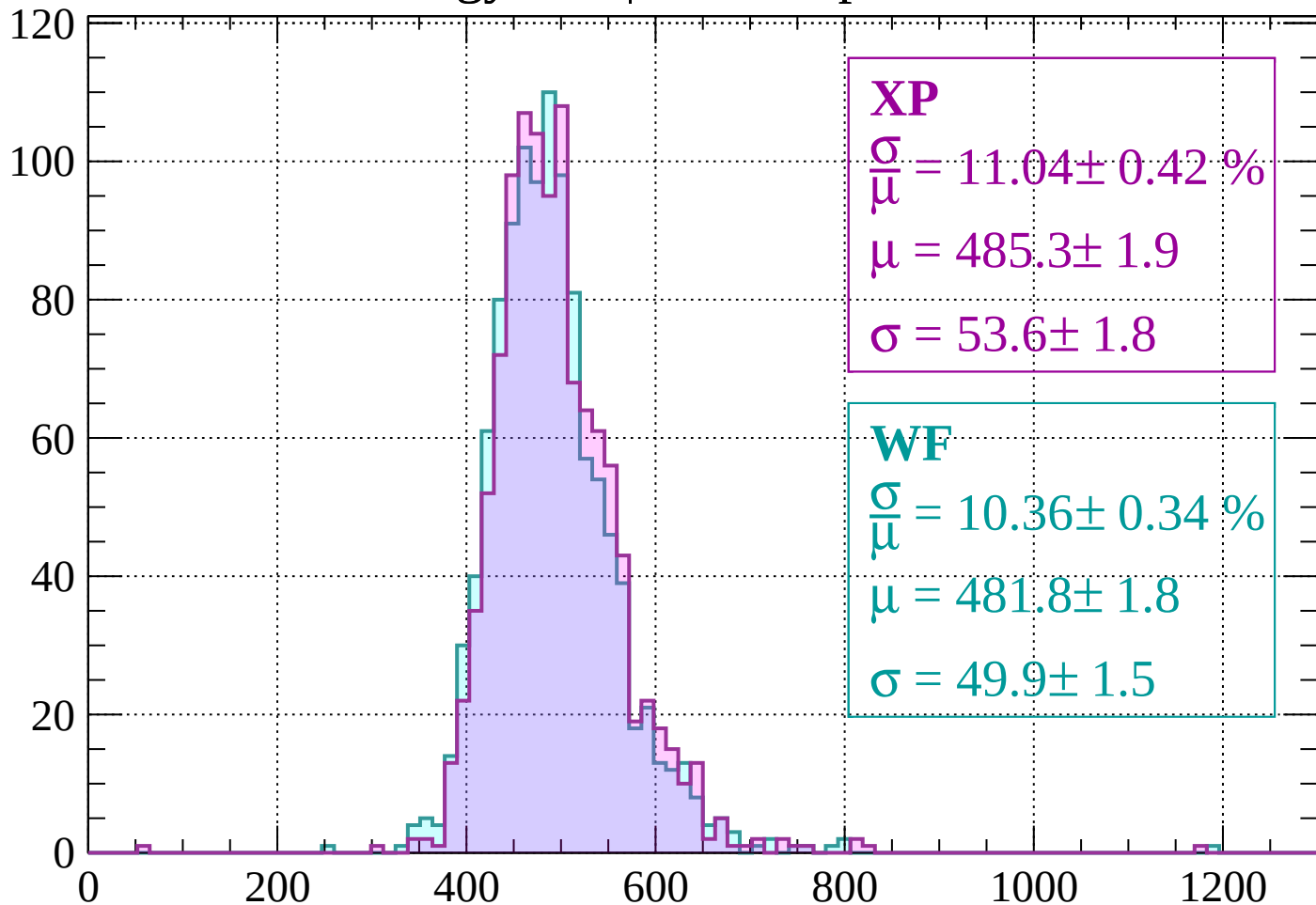
Energy loss | $-1720 < p < -1680$

Count



Energy loss | -1680 < p < -1640

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 11.04 \pm 0.42 \%$$

$$\mu = 485.3 \pm 1.9$$

$$\sigma = 53.6 \pm 1.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 10.36 \pm 0.34 \%$$

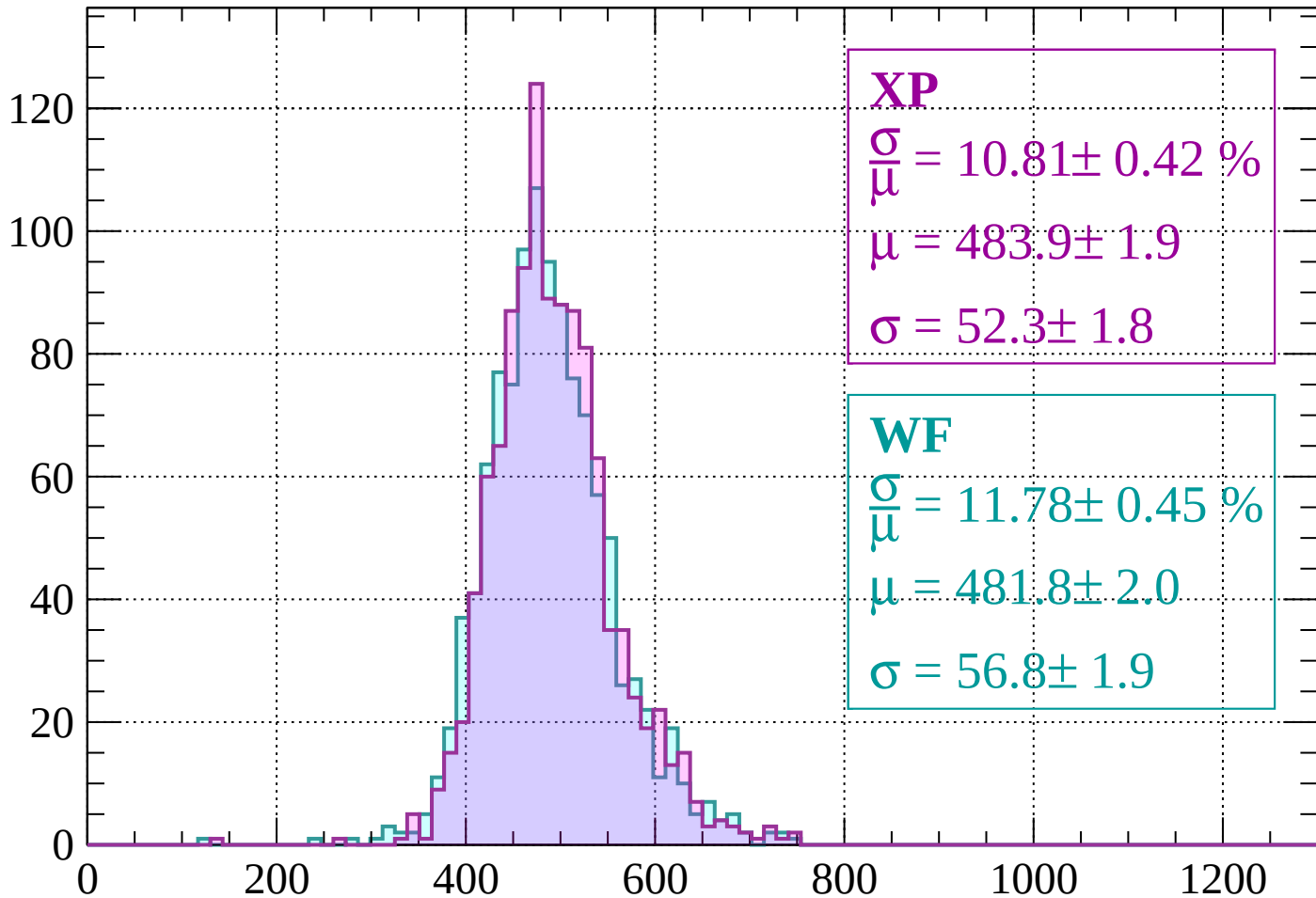
$$\mu = 481.8 \pm 1.8$$

$$\sigma = 49.9 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 10.81 \pm 0.42 \%$$

$$\mu = 483.9 \pm 1.9$$

$$\sigma = 52.3 \pm 1.8$$

WF

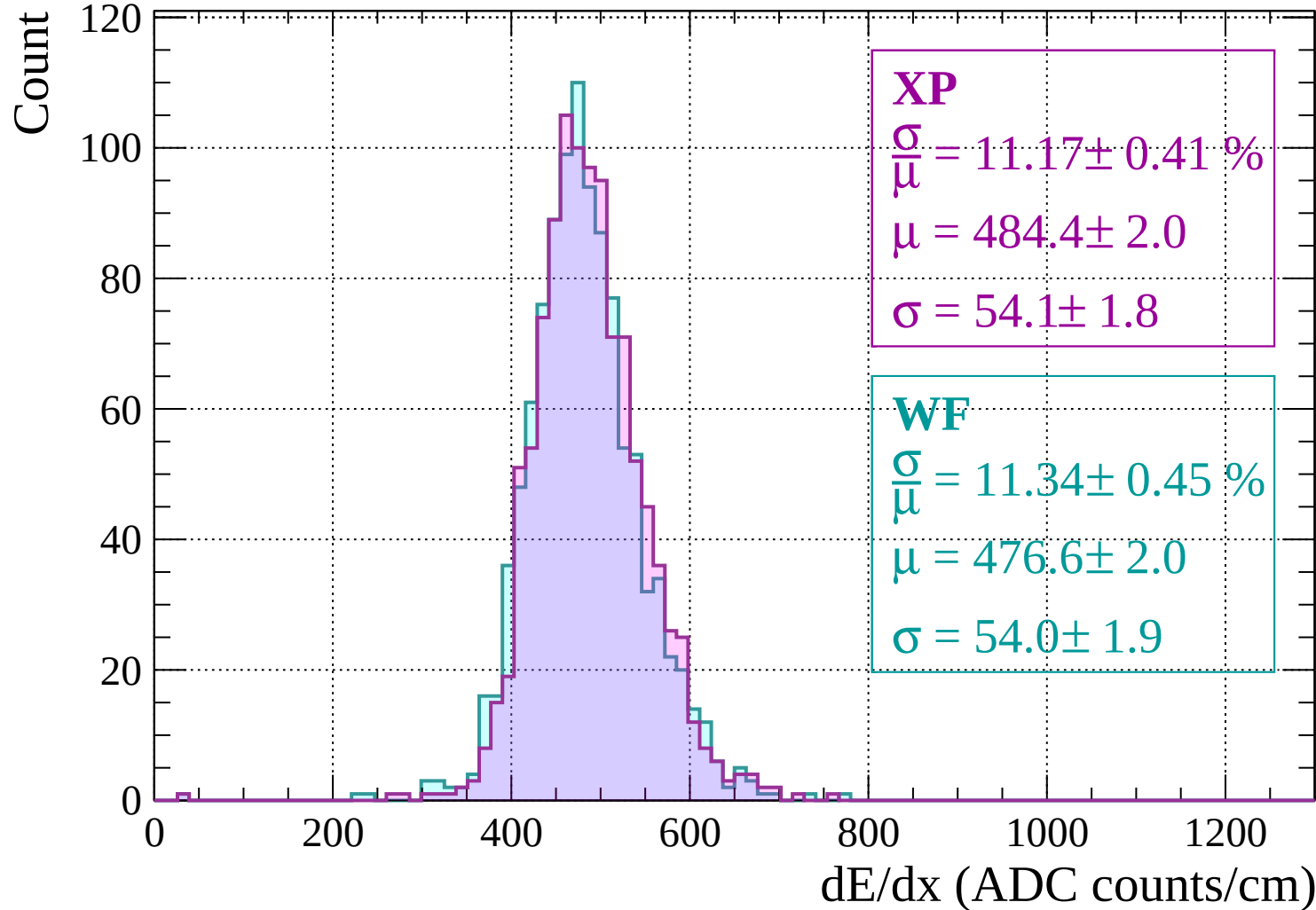
$$\frac{\sigma}{\mu} = 11.78 \pm 0.45 \%$$

$$\mu = 481.8 \pm 2.0$$

$$\sigma = 56.8 \pm 1.9$$

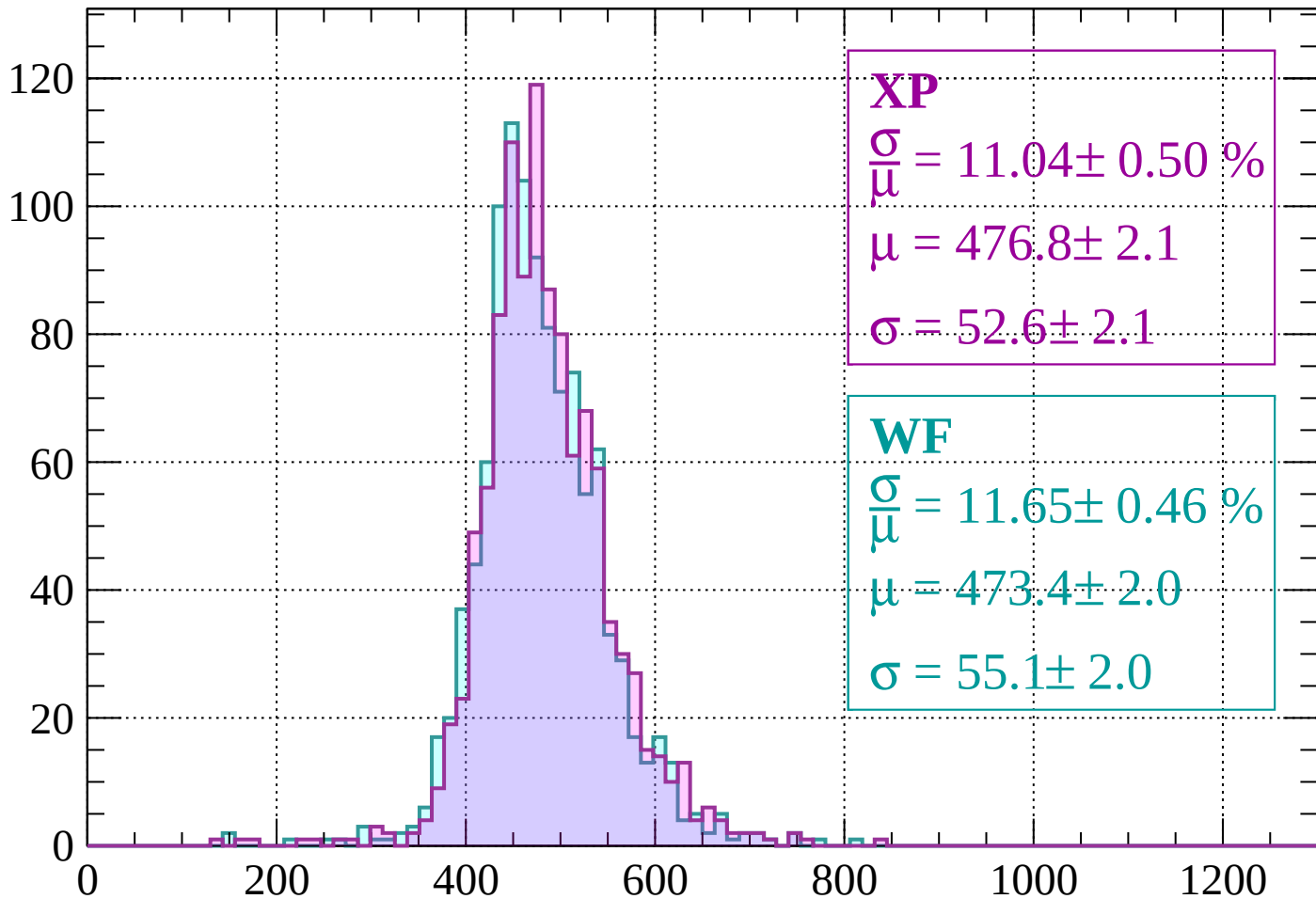
dE/dx (ADC counts/cm)

Energy loss | -1600 < p < -1560



Energy loss | $-1560 < p < -1520$

Count



XP

$$\frac{\sigma}{\mu} = 11.04 \pm 0.50 \%$$

$$\mu = 476.8 \pm 2.1$$

$$\sigma = 52.6 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 11.65 \pm 0.46 \%$$

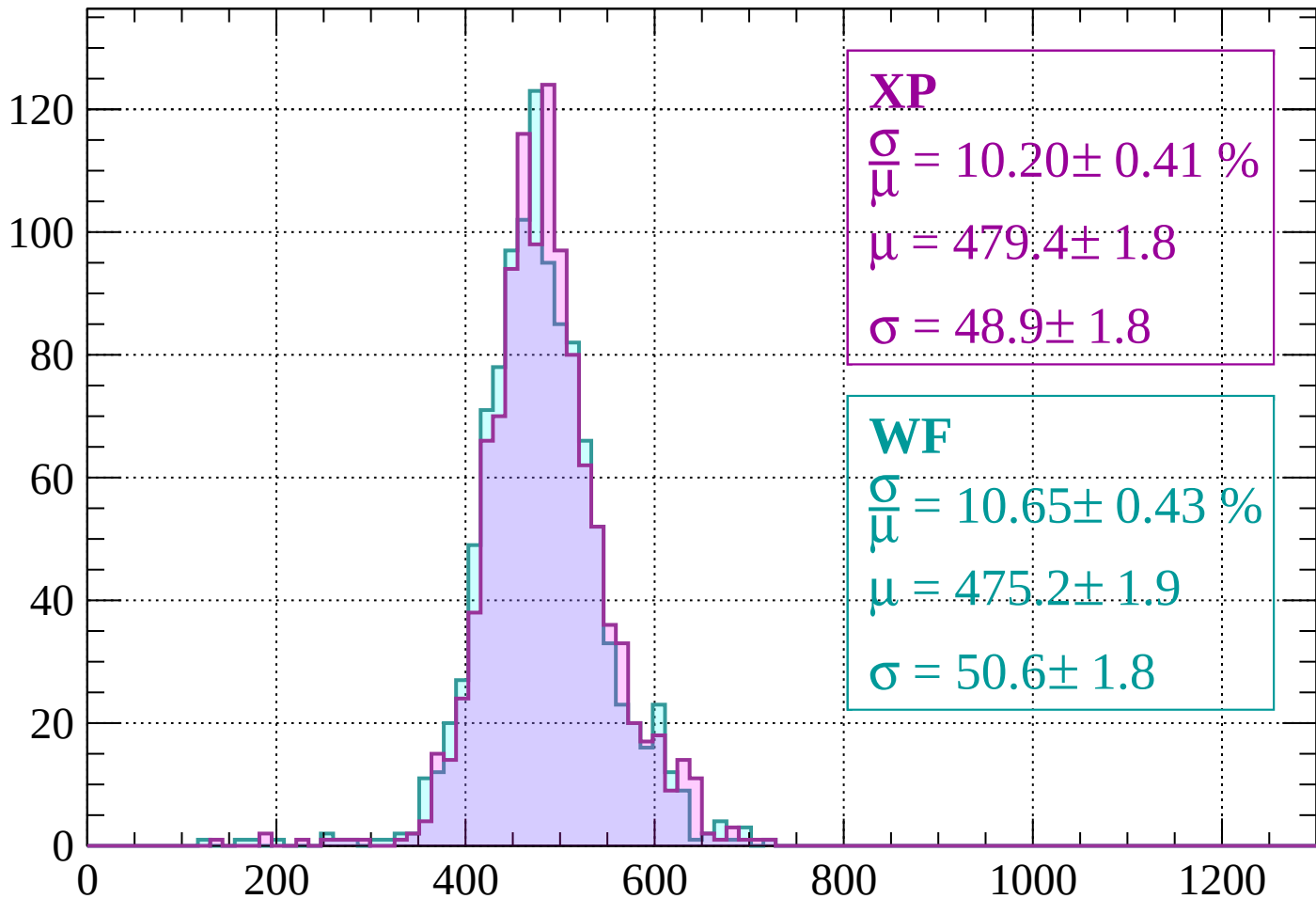
$$\mu = 473.4 \pm 2.0$$

$$\sigma = 55.1 \pm 2.0$$

dE/dx (ADC counts/cm)

Energy loss | -1520 < p < -1480

Count



XP

$$\frac{\sigma}{\mu} = 10.20 \pm 0.41 \%$$

$$\mu = 479.4 \pm 1.8$$

$$\sigma = 48.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 10.65 \pm 0.43 \%$$

$$\mu = 475.2 \pm 1.9$$

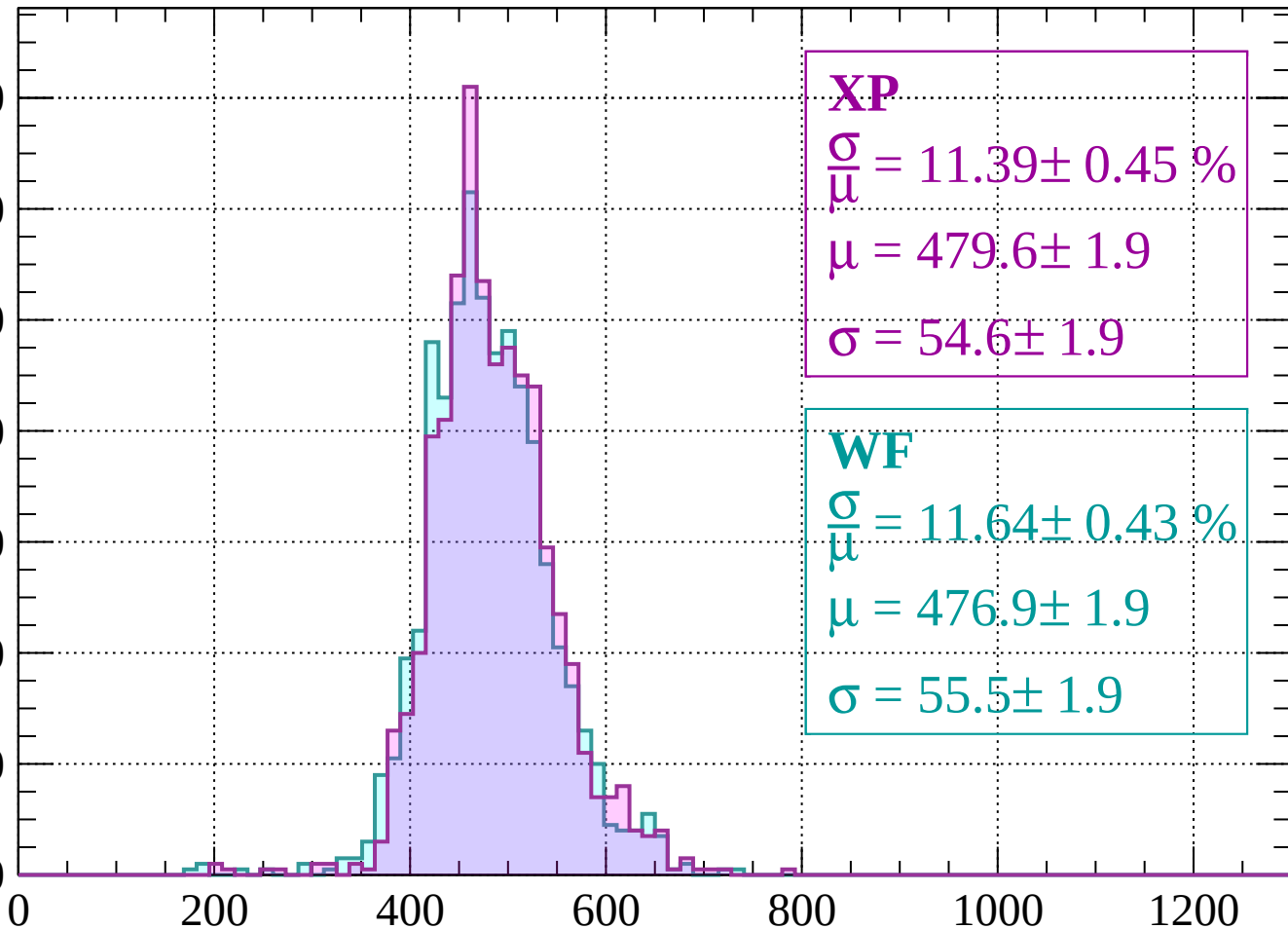
$$\sigma = 50.6 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 11.39 \pm 0.45 \%$$

$$\mu = 479.6 \pm 1.9$$

$$\sigma = 54.6 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 11.64 \pm 0.43 \%$$

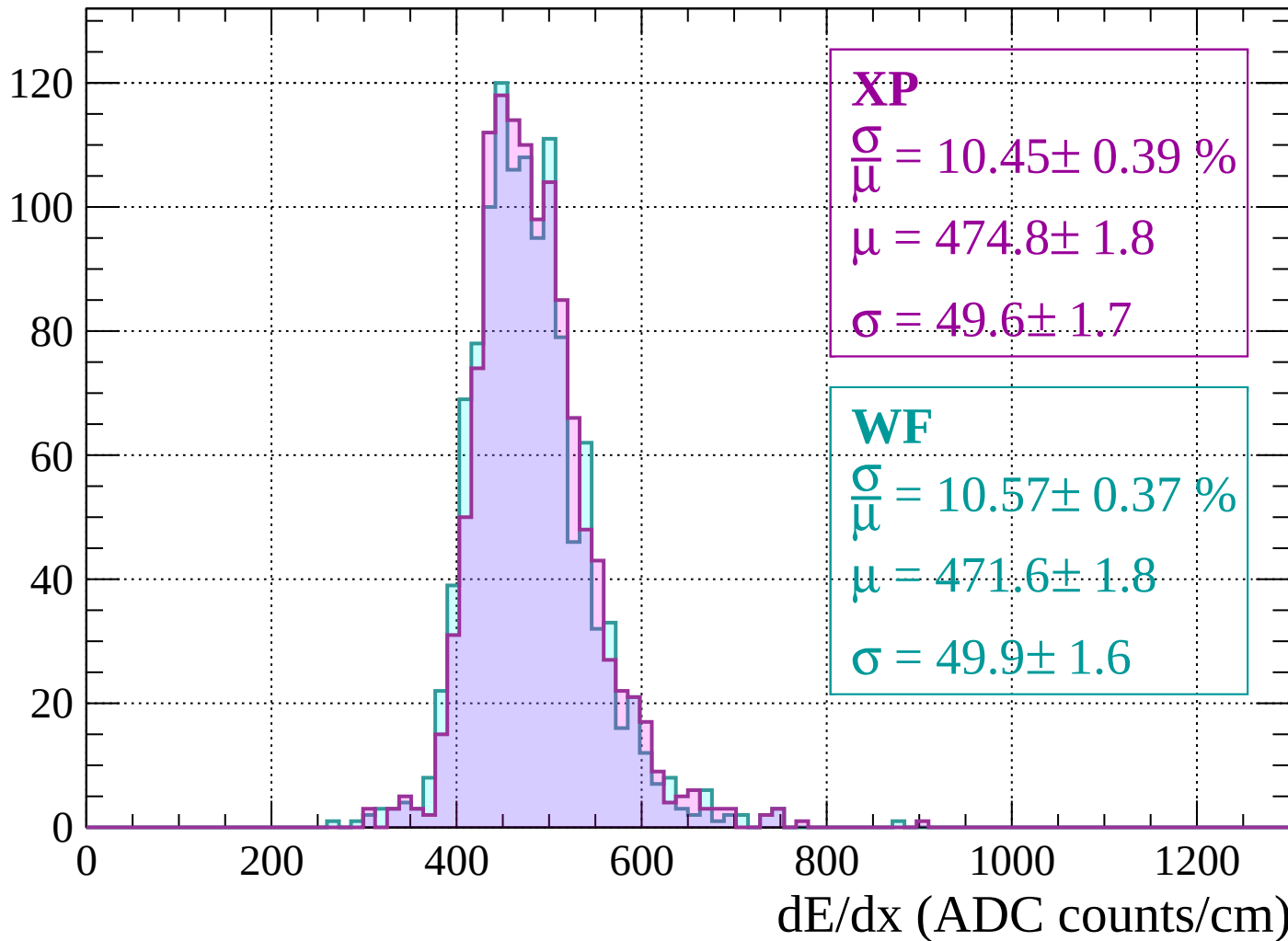
$$\mu = 476.9 \pm 1.9$$

$$\sigma = 55.5 \pm 1.9$$

dE/dx (ADC counts/cm)

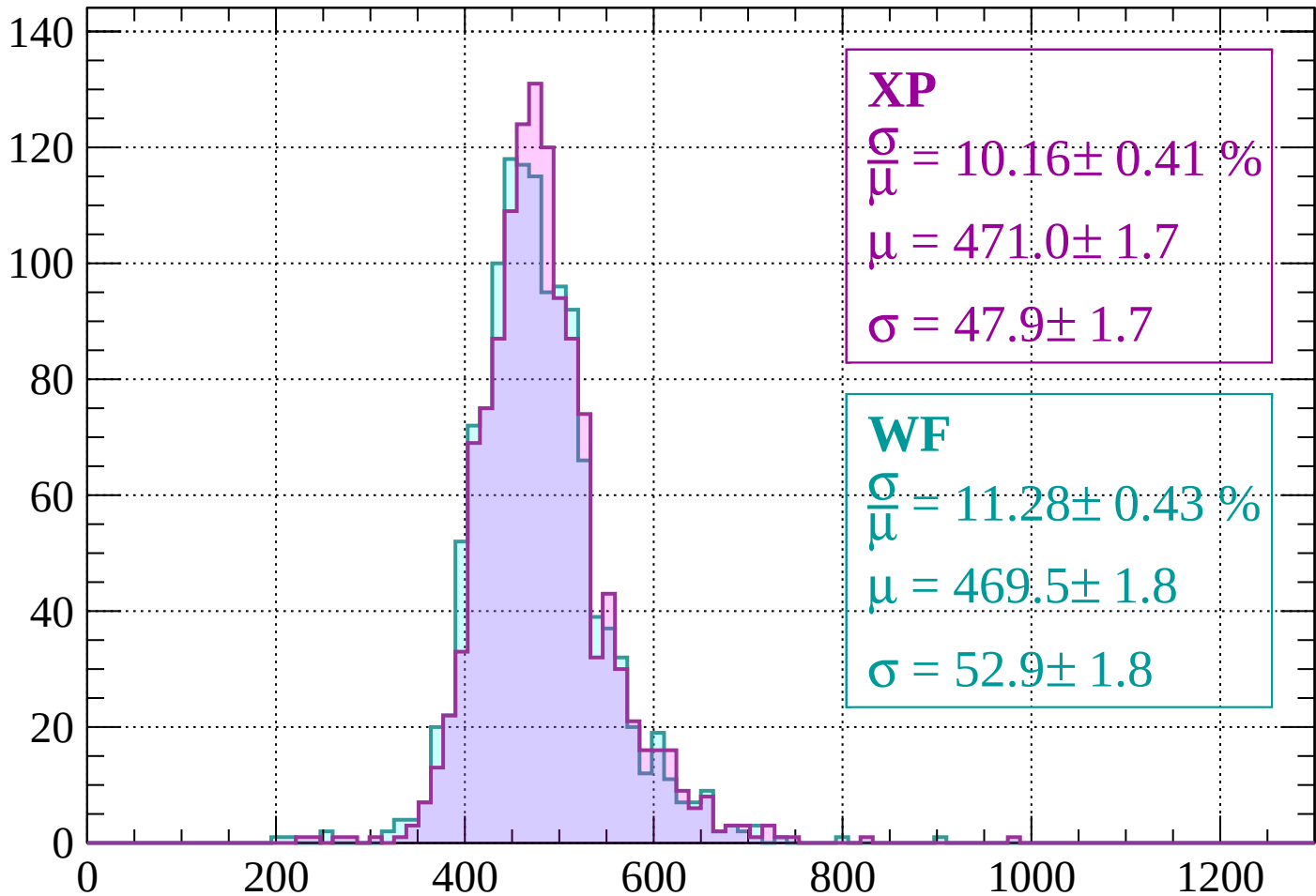
Energy loss | $-1440 < p < -1400$

Count



Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 10.16 \pm 0.41 \%$$

$$\mu = 471.0 \pm 1.7$$

$$\sigma = 47.9 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 11.28 \pm 0.43 \%$$

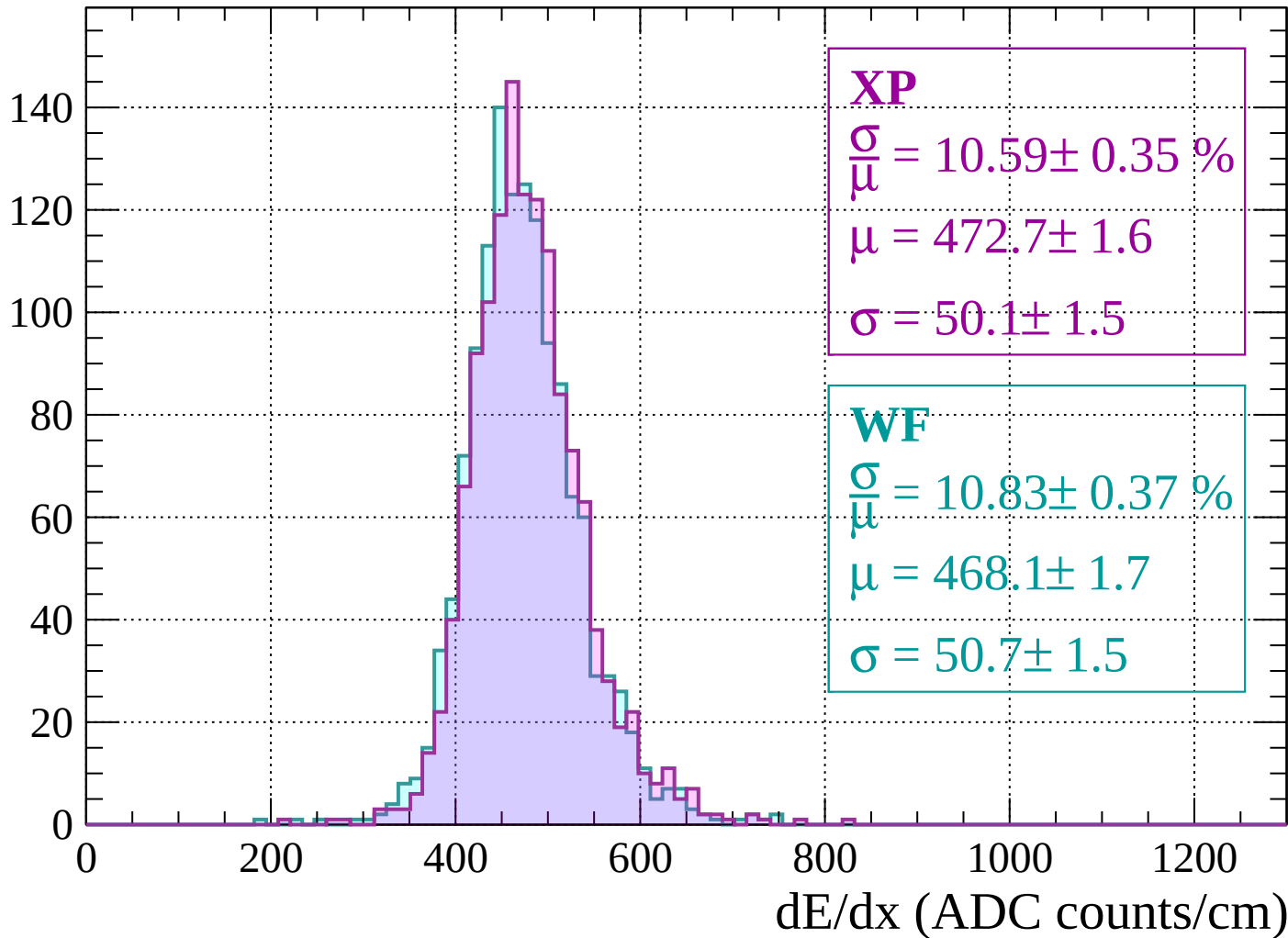
$$\mu = 469.5 \pm 1.8$$

$$\sigma = 52.9 \pm 1.8$$

dE/dx (ADC counts/cm)

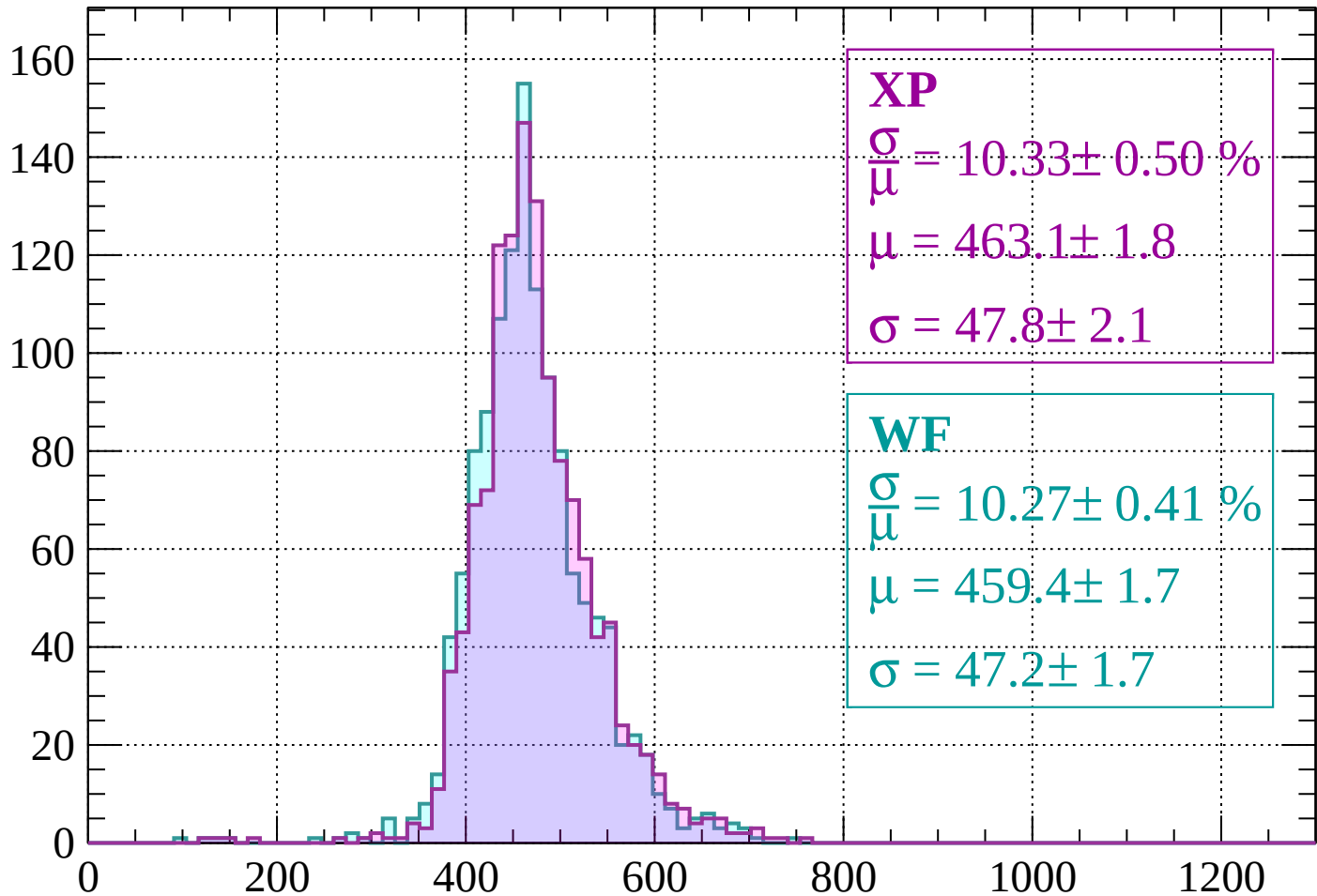
Energy loss | -1360 < p < -1320

Count



Energy loss | -1320 < p < -1280

Count



XP

$$\frac{\sigma}{\mu} = 10.33 \pm 0.50 \%$$

$$\mu = 463.1 \pm 1.8$$

$$\sigma = 47.8 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 10.27 \pm 0.41 \%$$

$$\mu = 459.4 \pm 1.7$$

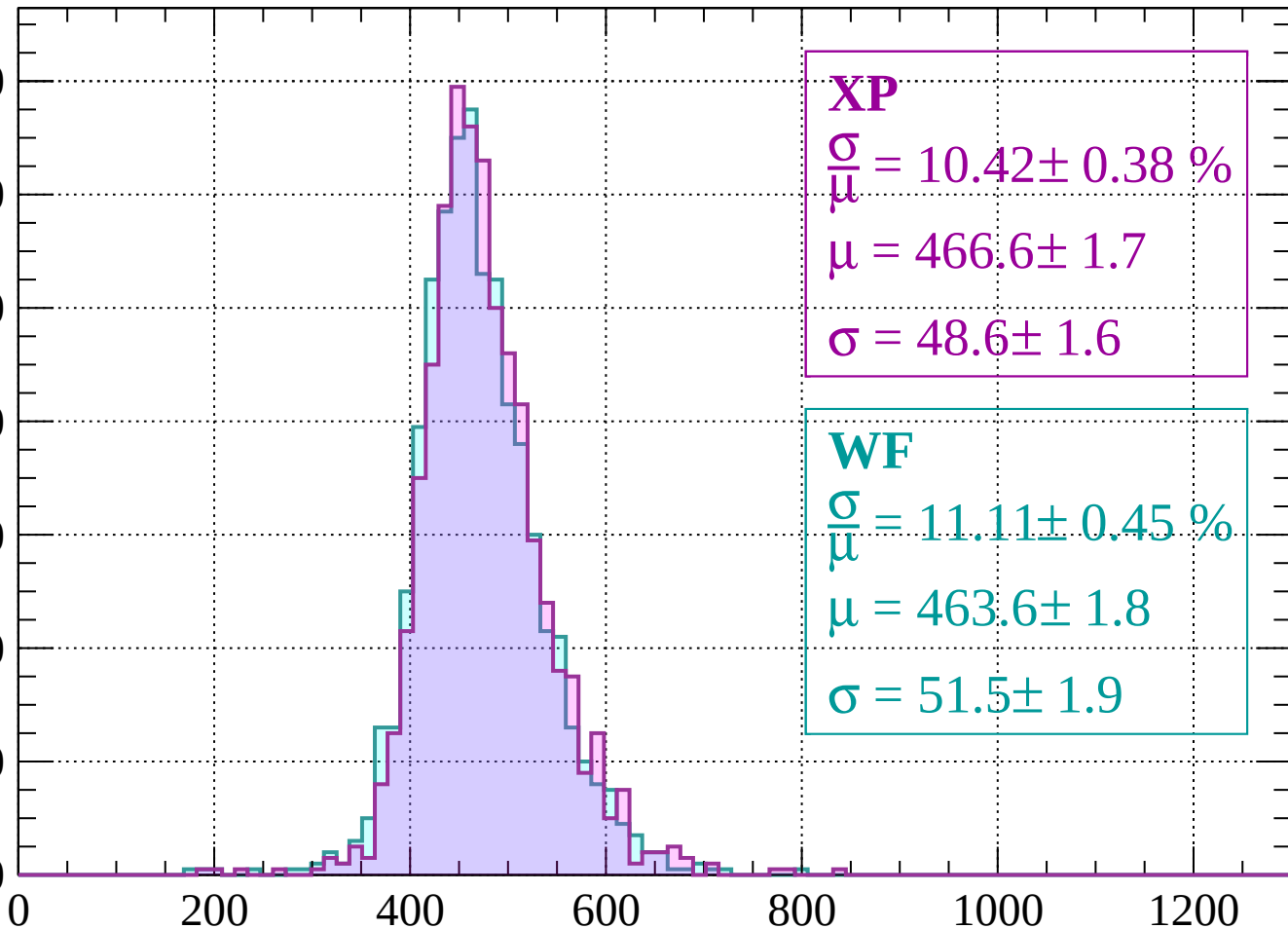
$$\sigma = 47.2 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | -1280 < p < -1240

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 10.42 \pm 0.38 \%$$

$$\mu = 466.6 \pm 1.7$$

$$\sigma = 48.6 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 11.11 \pm 0.45 \%$$

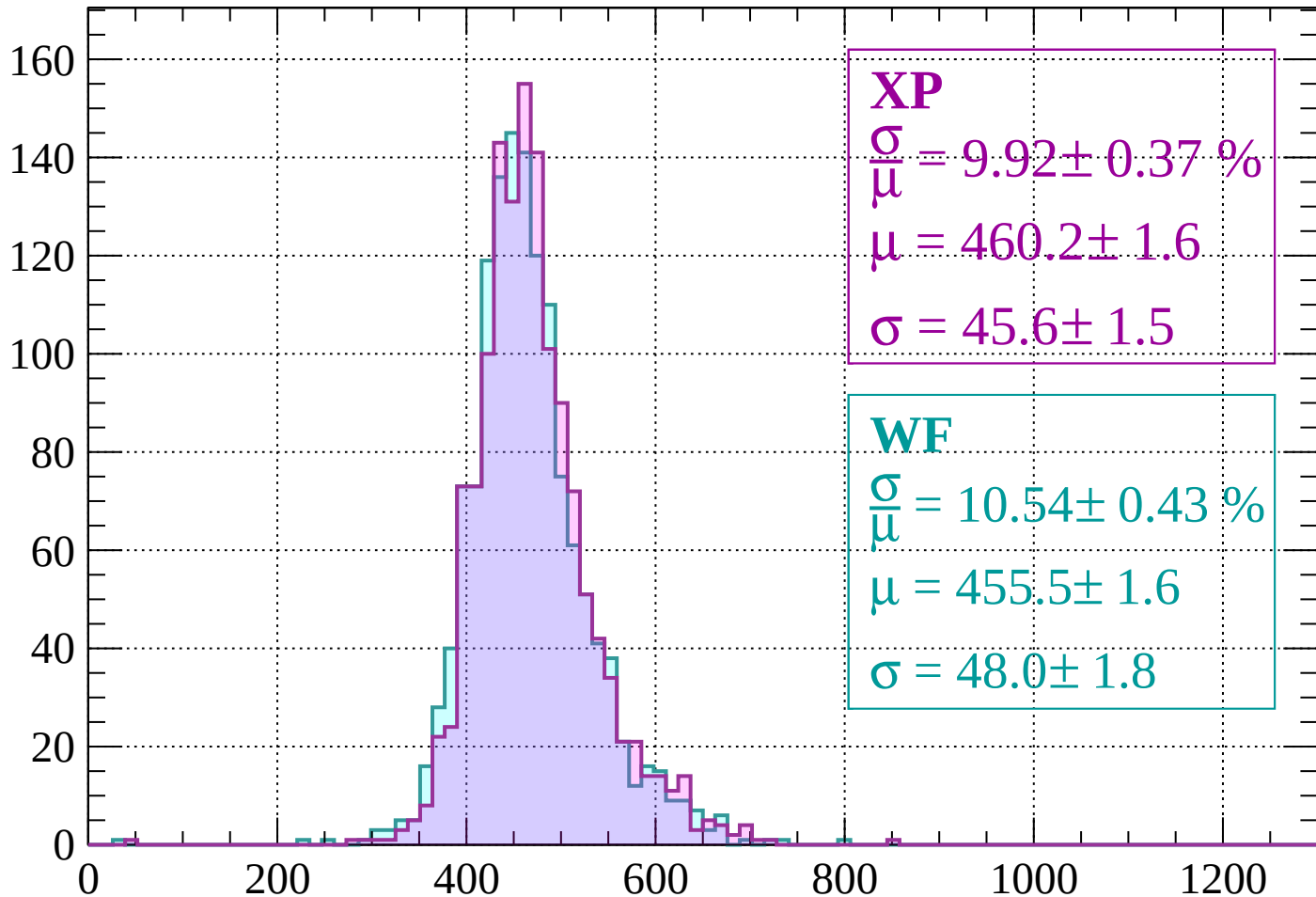
$$\mu = 463.6 \pm 1.8$$

$$\sigma = 51.5 \pm 1.9$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.92 \pm 0.37 \%$$

$$\mu = 460.2 \pm 1.6$$

$$\sigma = 45.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 10.54 \pm 0.43 \%$$

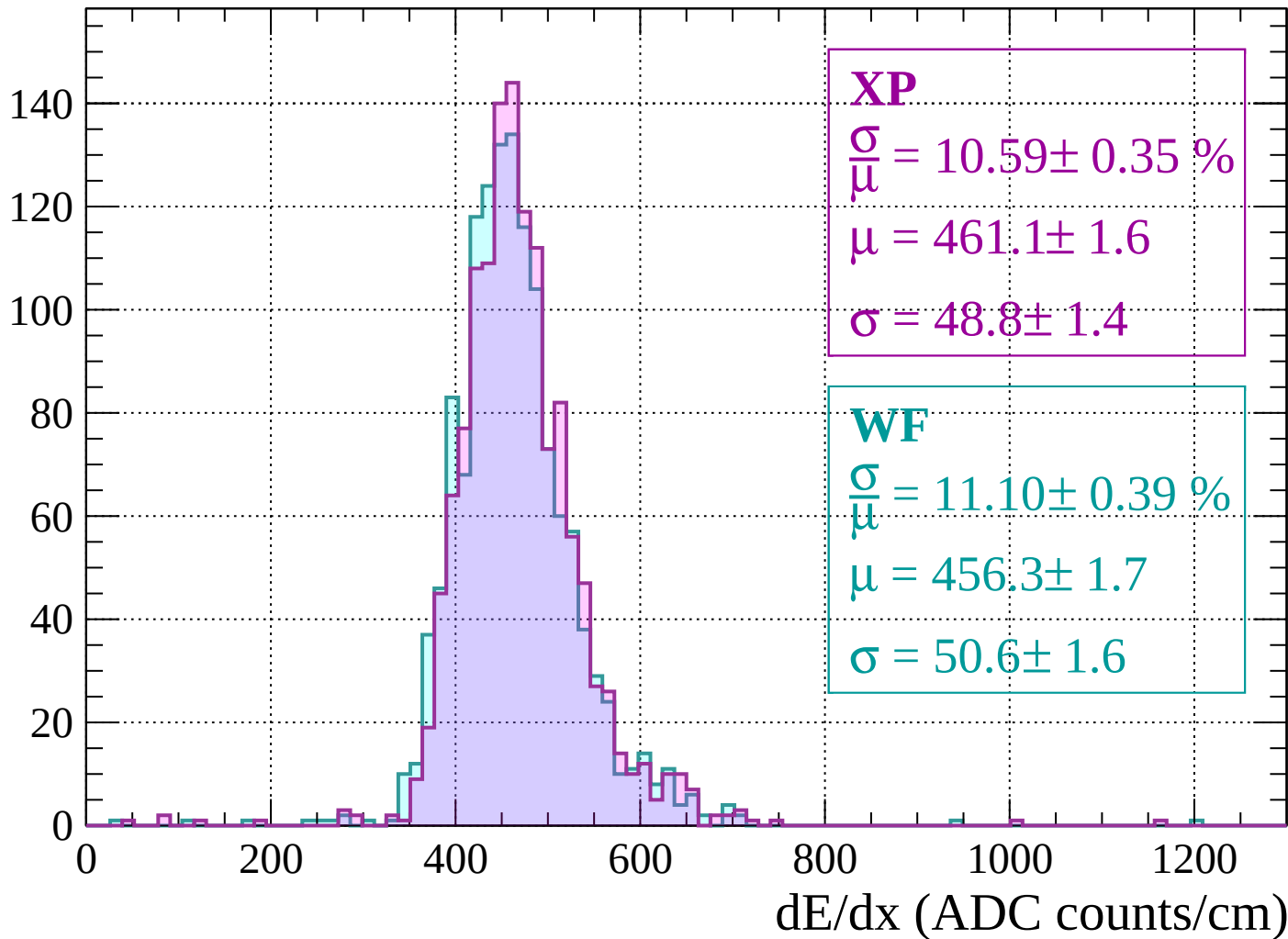
$$\mu = 455.5 \pm 1.6$$

$$\sigma = 48.0 \pm 1.8$$

dE/dx (ADC counts/cm)

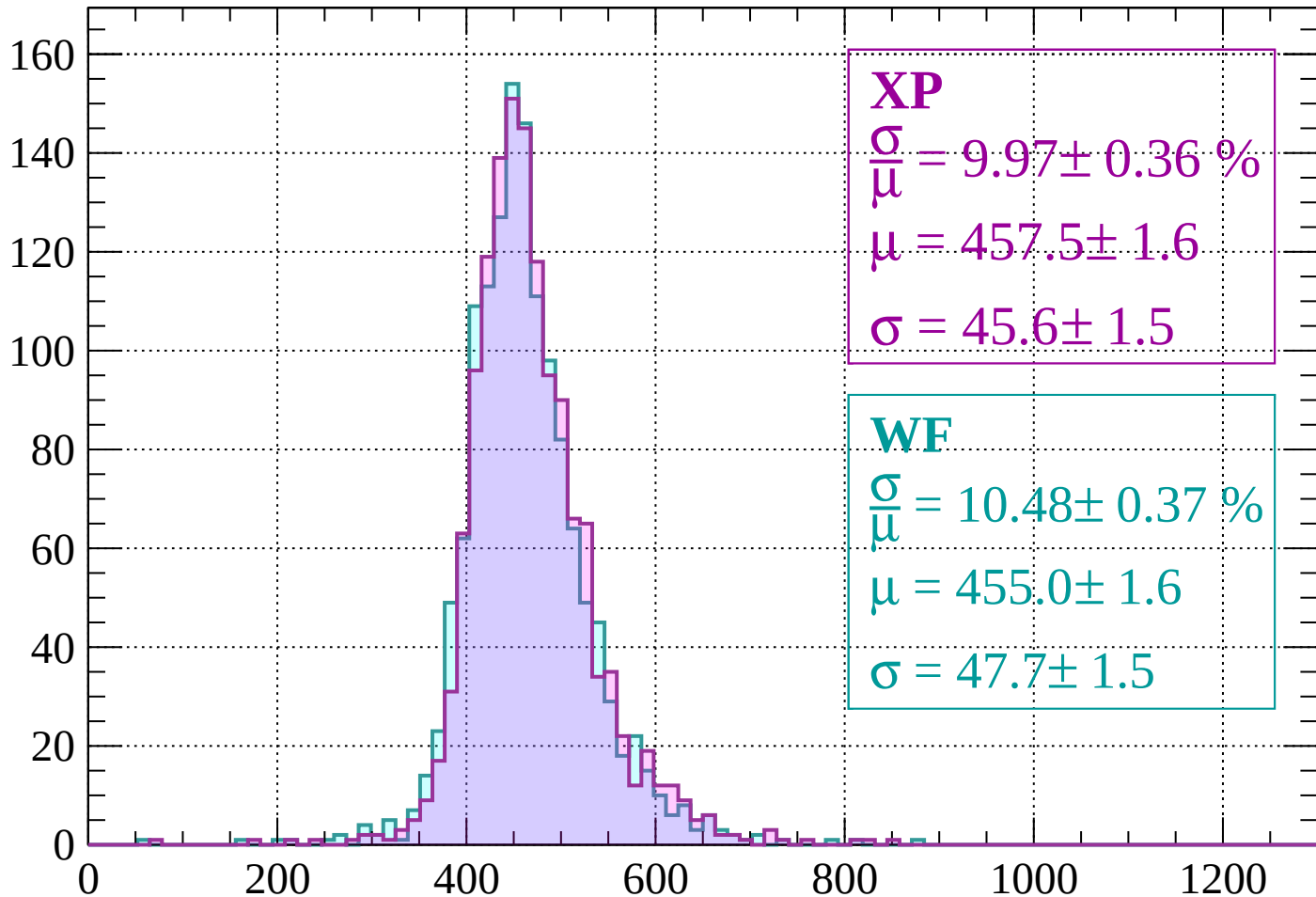
Energy loss | -1200 < p < -1160

Count



Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\mu} = 9.97 \pm 0.36 \%$$

$$\mu = 457.5 \pm 1.6$$

$$\sigma = 45.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 10.48 \pm 0.37 \%$$

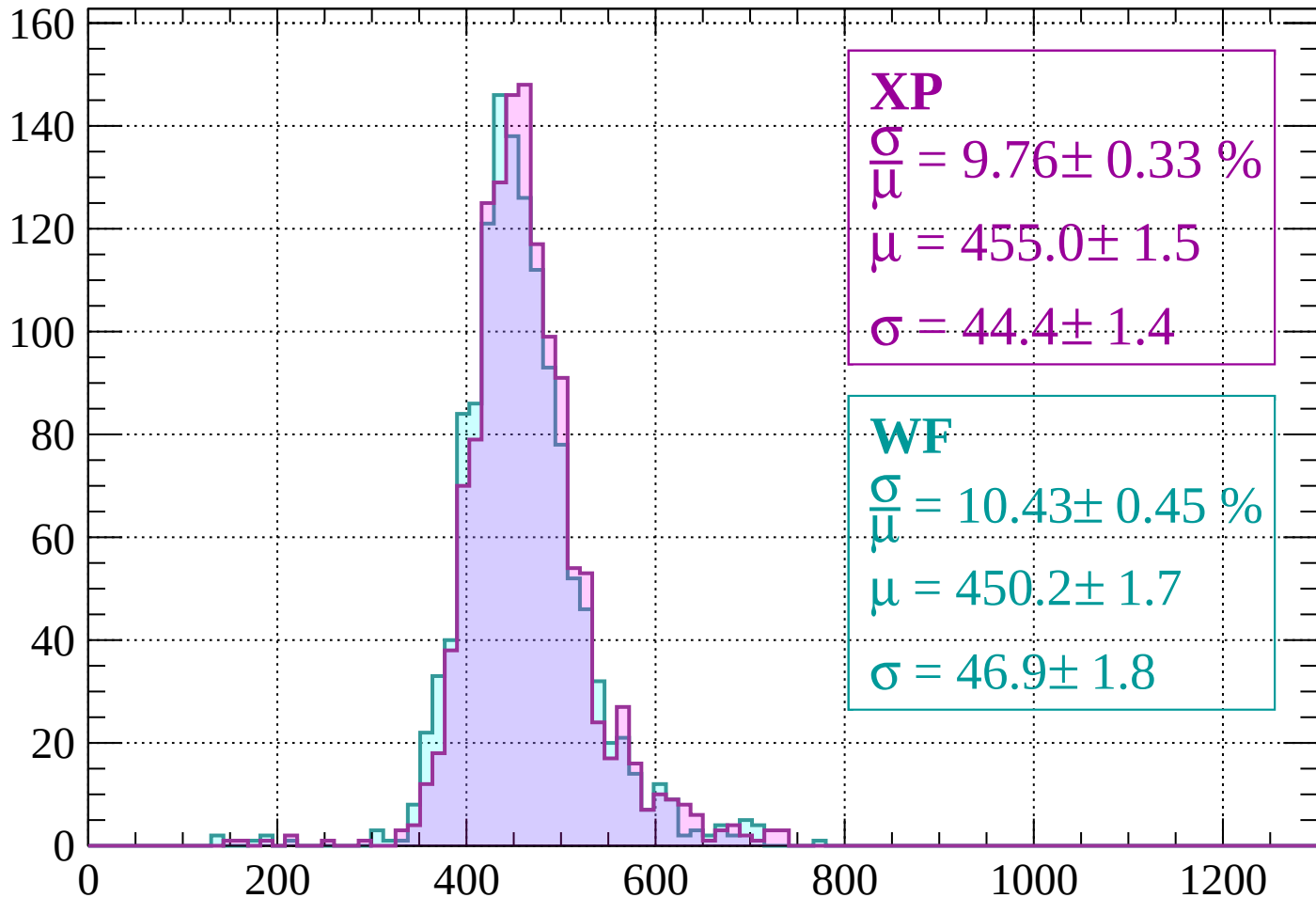
$$\mu = 455.0 \pm 1.6$$

$$\sigma = 47.7 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -1120 < p < -1080

Count



XP

$$\frac{\sigma}{\mu} = 9.76 \pm 0.33 \%$$

$$\mu = 455.0 \pm 1.5$$

$$\sigma = 44.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.43 \pm 0.45 \%$$

$$\mu = 450.2 \pm 1.7$$

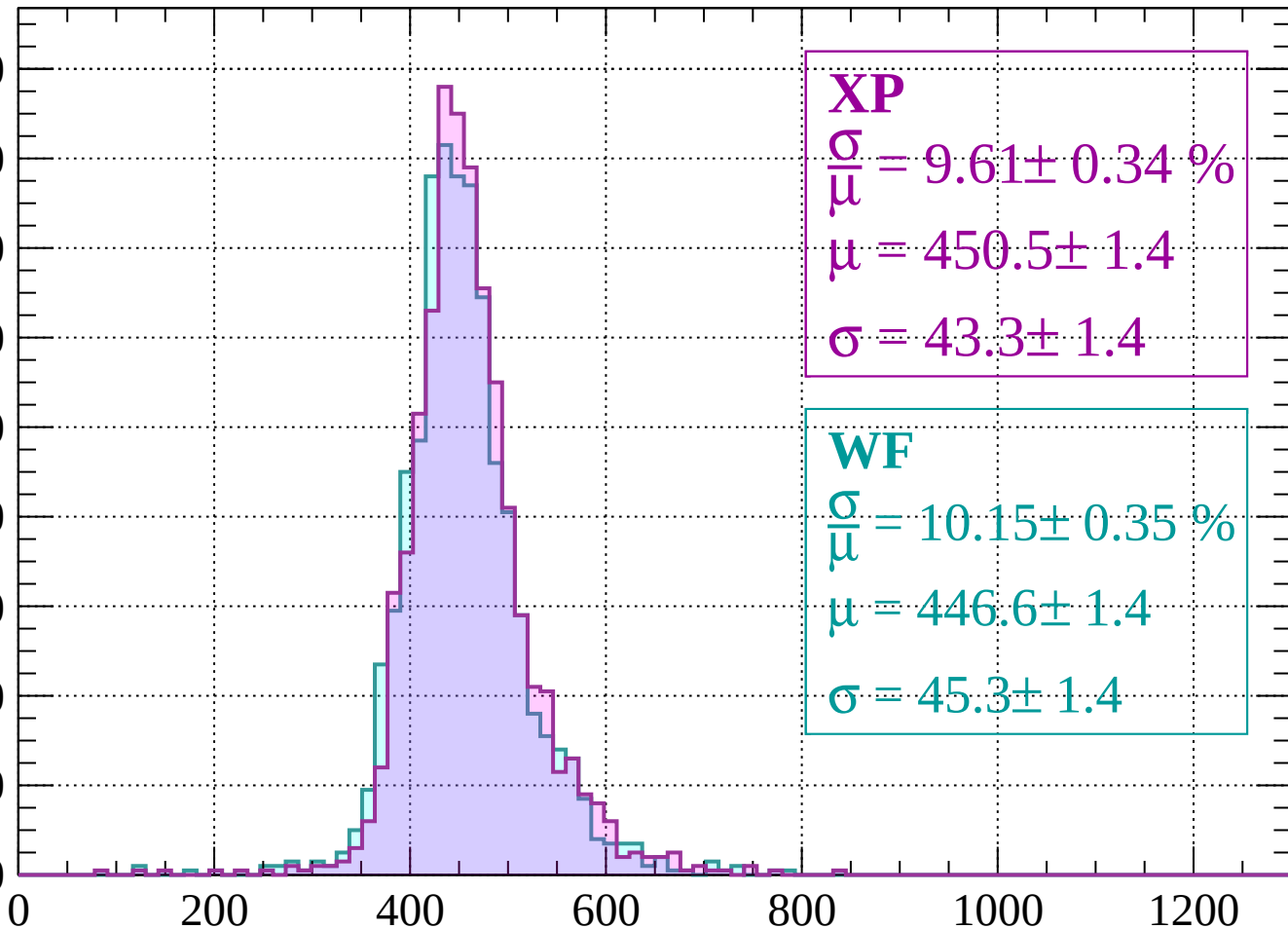
$$\sigma = 46.9 \pm 1.8$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.61 \pm 0.34 \%$$

$$\mu = 450.5 \pm 1.4$$

$$\sigma = 43.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 10.15 \pm 0.35 \%$$

$$\mu = 446.6 \pm 1.4$$

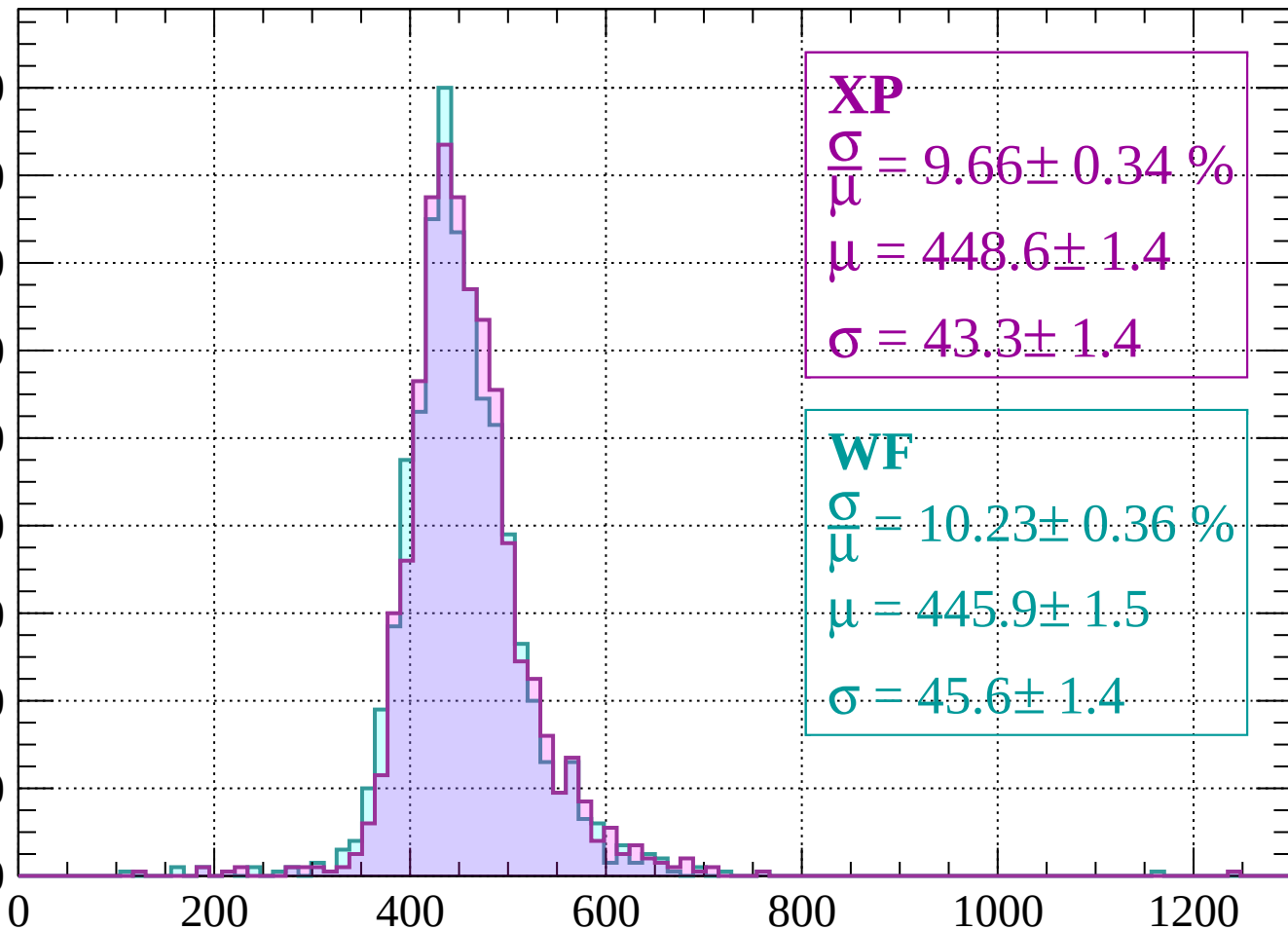
$$\sigma = 45.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | -1040 < p < -1000

Count

180
160
140
120
100
80
60
40
20
0

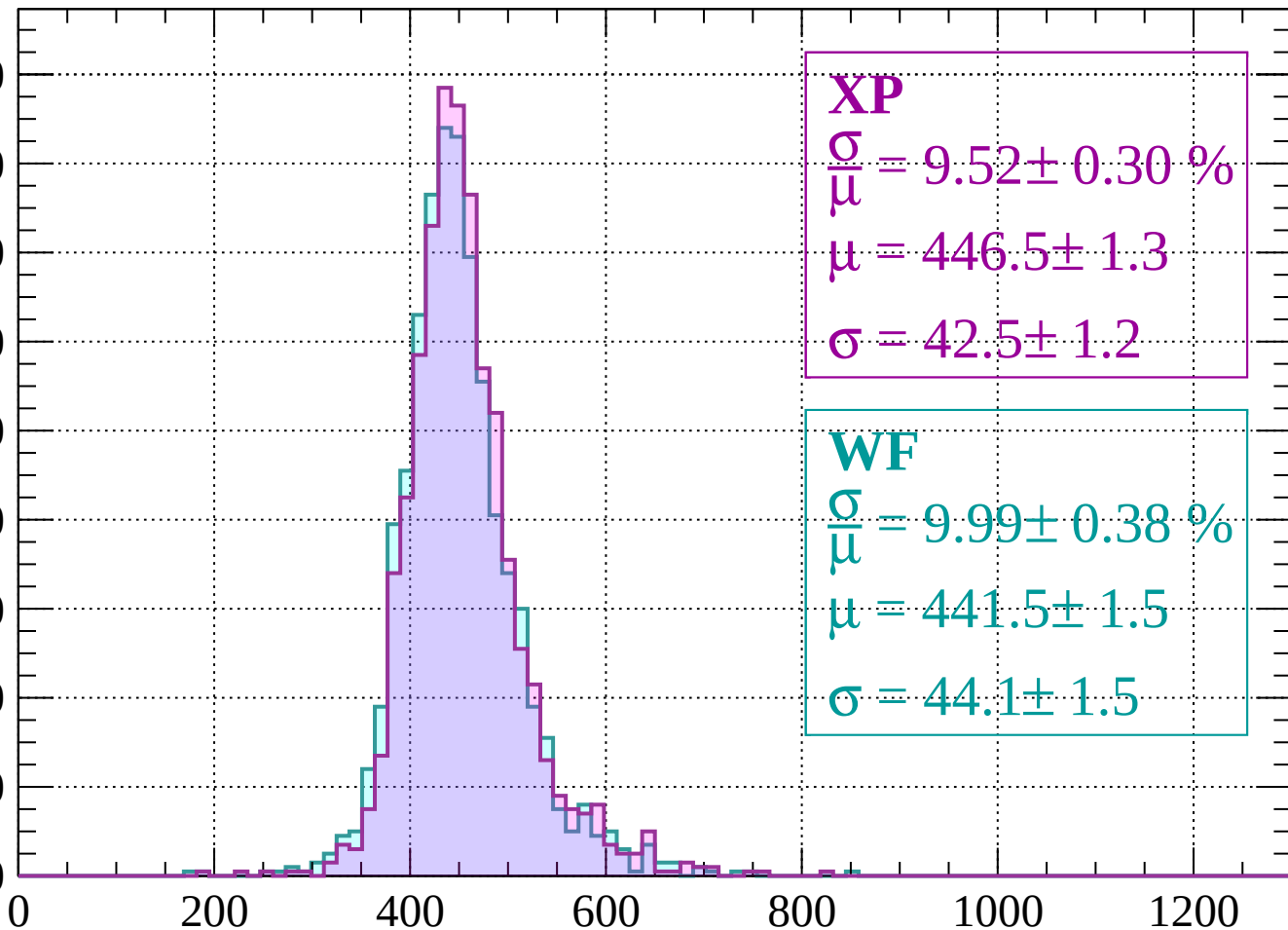


dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.52 \pm 0.30 \%$$

$$\mu = 446.5 \pm 1.3$$

$$\sigma = 42.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.99 \pm 0.38 \%$$

$$\mu = 441.5 \pm 1.5$$

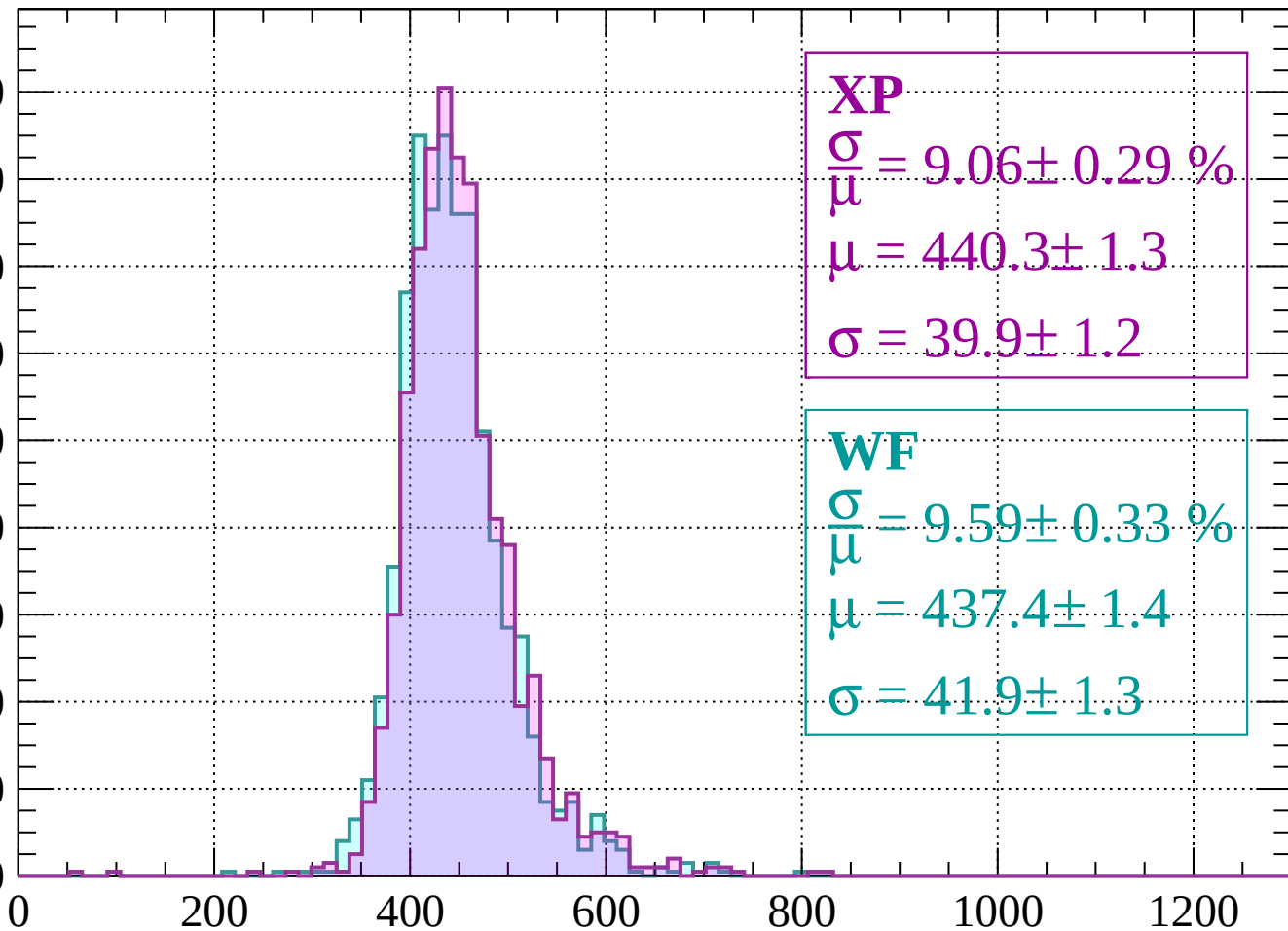
$$\sigma = 44.1 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.29 \%$$

$$\mu = 440.3 \pm 1.3$$

$$\sigma = 39.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.33 \%$$

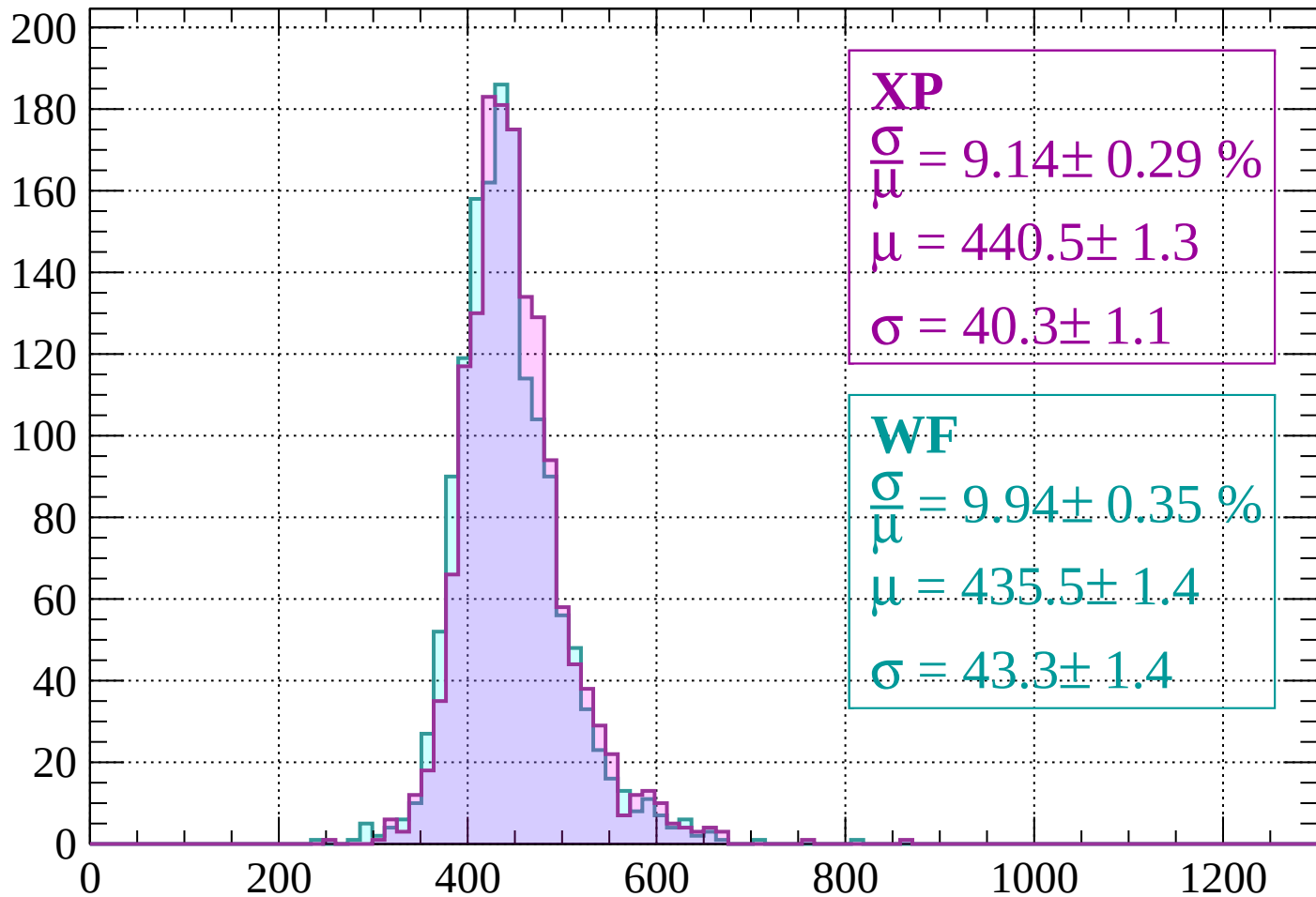
$$\mu = 437.4 \pm 1.4$$

$$\sigma = 41.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 9.14 \pm 0.29 \%$$

$$\mu = 440.5 \pm 1.3$$

$$\sigma = 40.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.94 \pm 0.35 \%$$

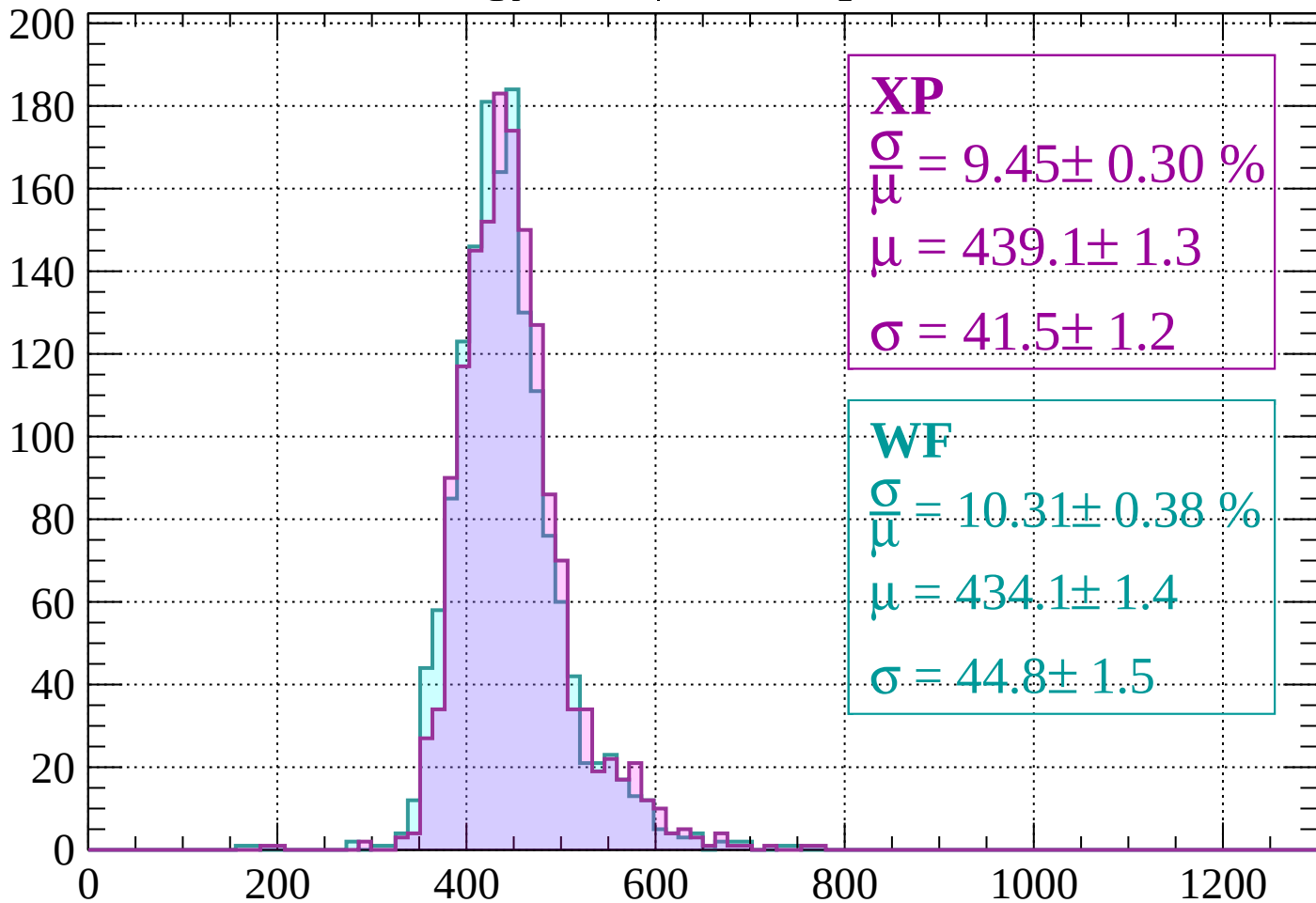
$$\mu = 435.5 \pm 1.4$$

$$\sigma = 43.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 9.45 \pm 0.30 \%$$

$$\mu = 439.1 \pm 1.3$$

$$\sigma = 41.5 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.31 \pm 0.38 \%$$

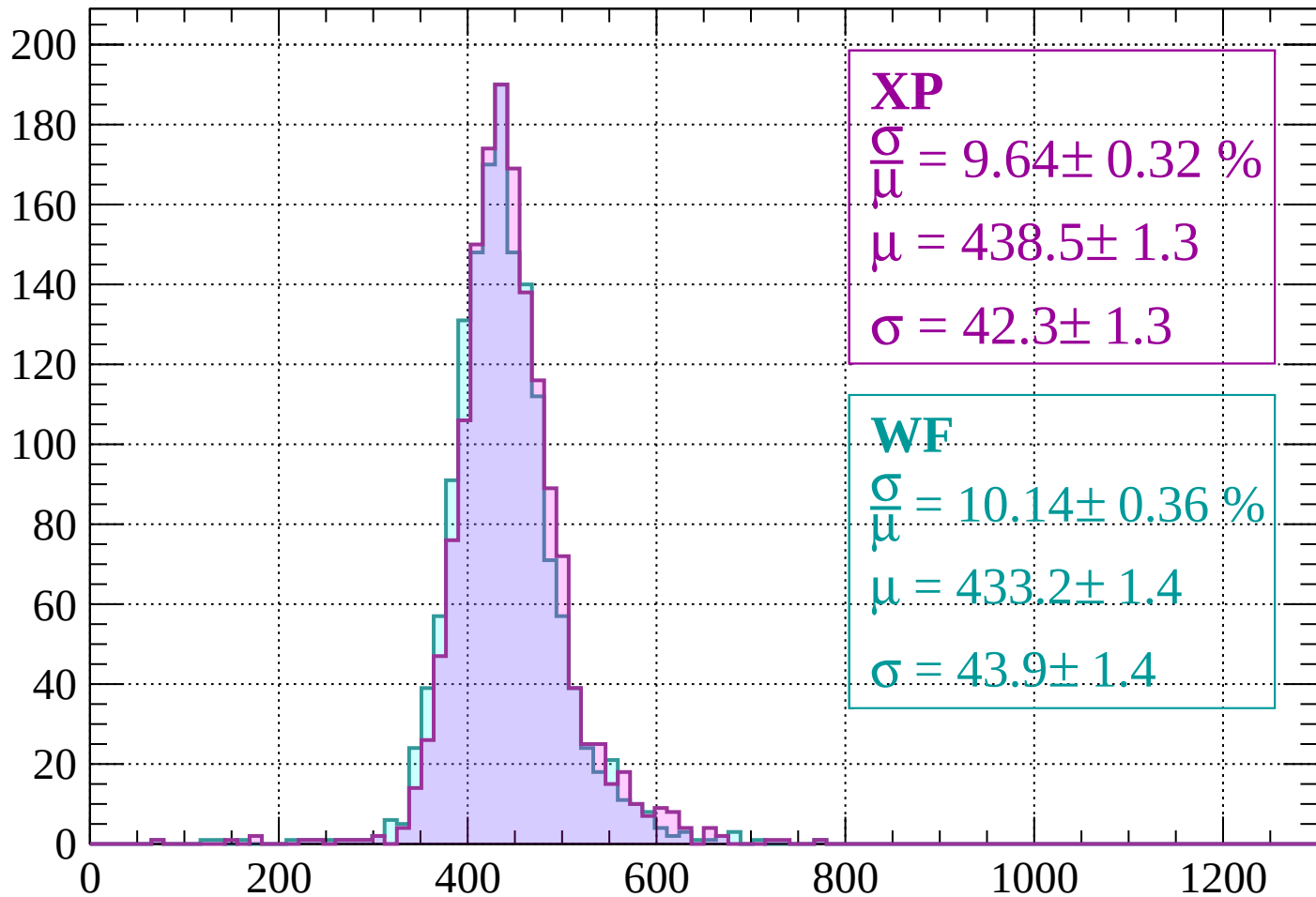
$$\mu = 434.1 \pm 1.4$$

$$\sigma = 44.8 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 9.64 \pm 0.32 \%$$

$$\mu = 438.5 \pm 1.3$$

$$\sigma = 42.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.14 \pm 0.36 \%$$

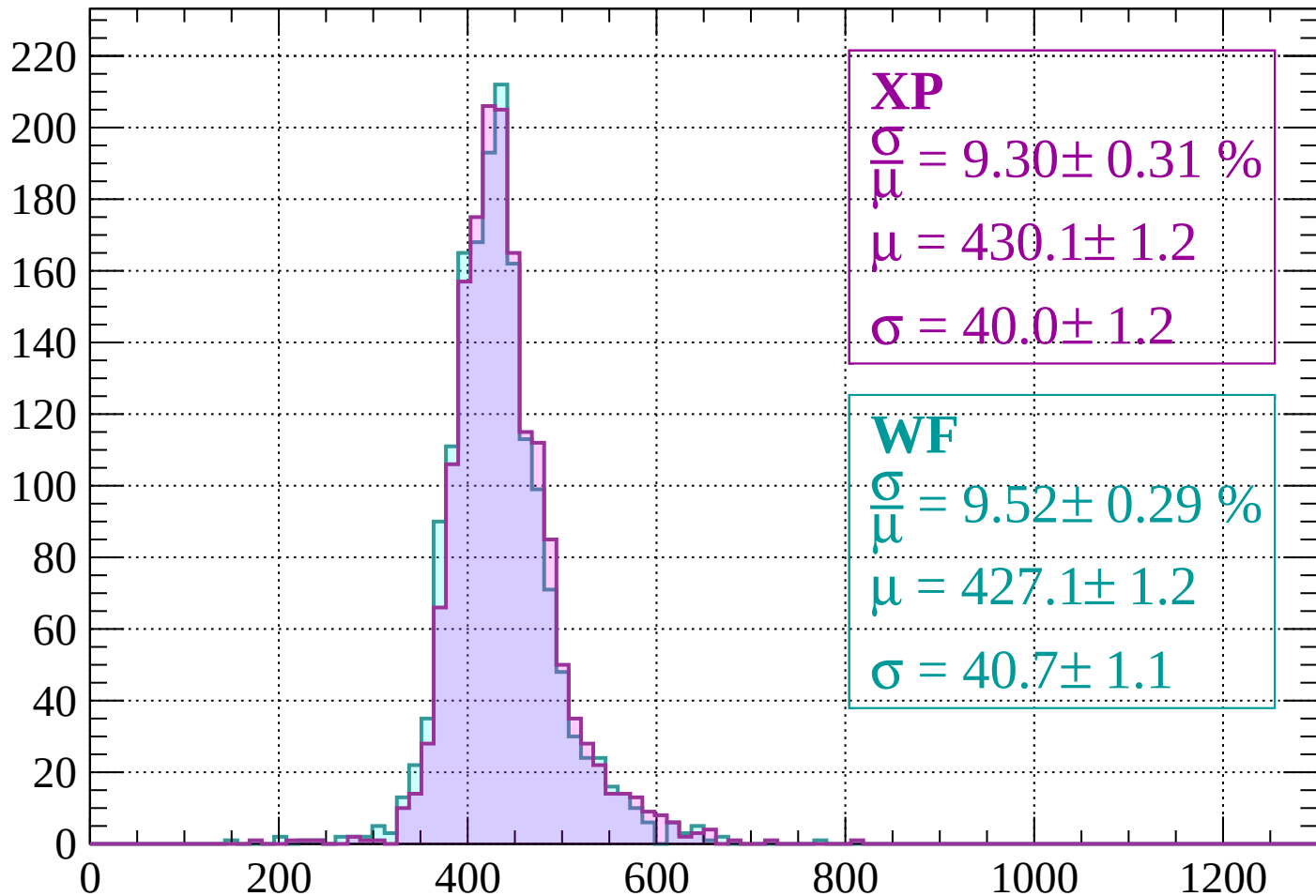
$$\mu = 433.2 \pm 1.4$$

$$\sigma = 43.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.31 \%$$

$$\mu = 430.1 \pm 1.2$$

$$\sigma = 40.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.52 \pm 0.29 \%$$

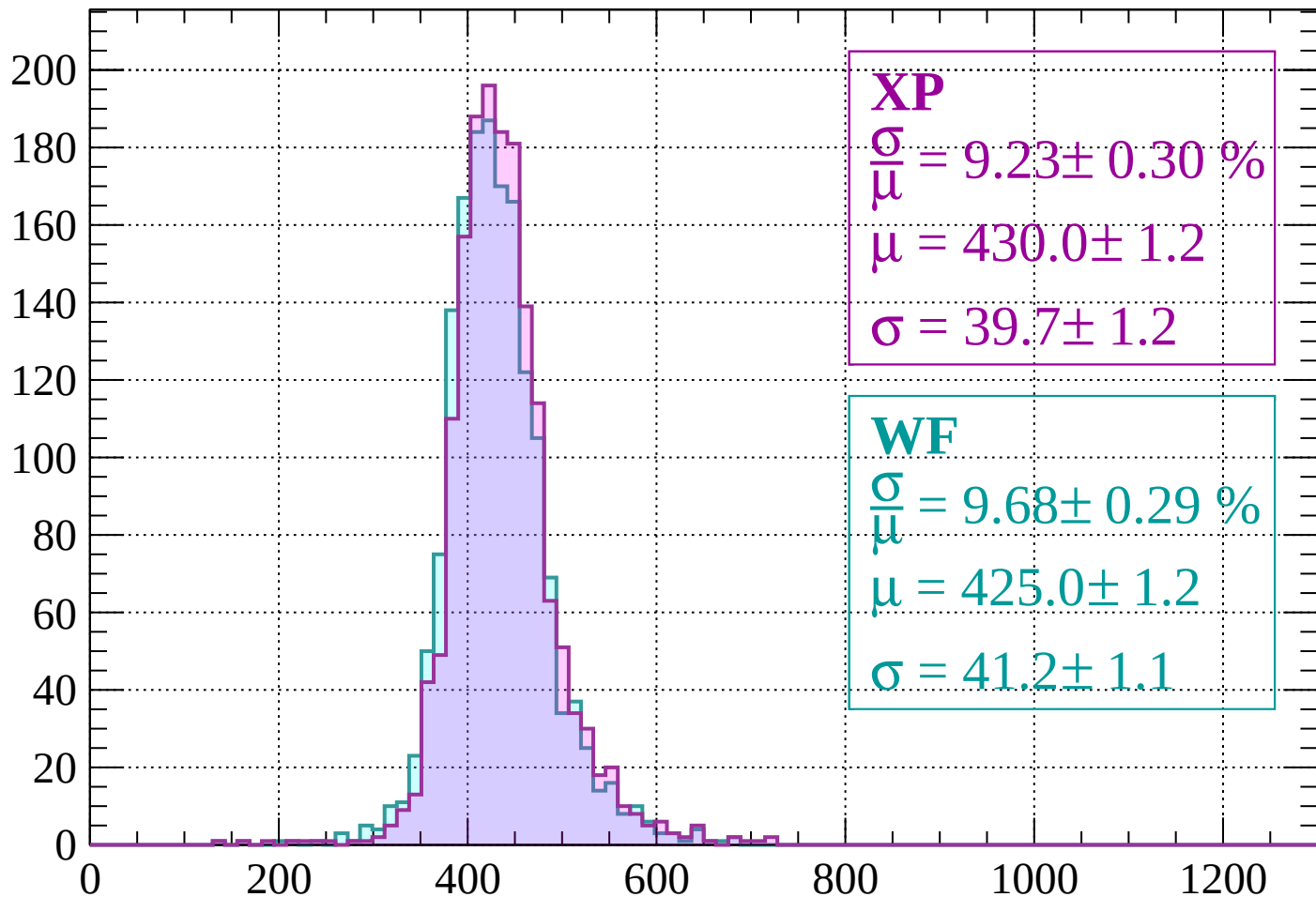
$$\mu = 427.1 \pm 1.2$$

$$\sigma = 40.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.23 \pm 0.30 \%$$

$$\mu = 430.0 \pm 1.2$$

$$\sigma = 39.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.68 \pm 0.29 \%$$

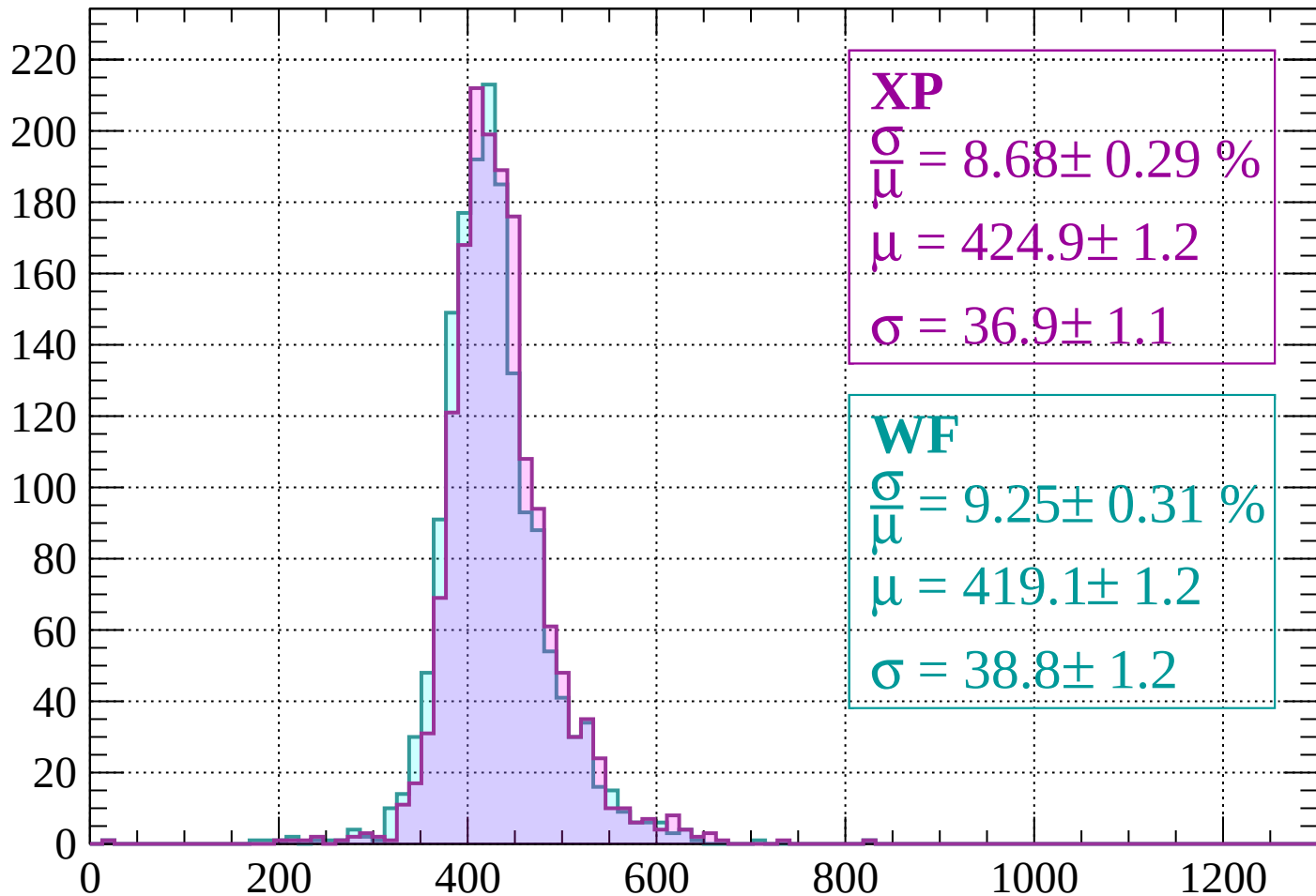
$$\mu = 425.0 \pm 1.2$$

$$\sigma = 41.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.68 \pm 0.29 \%$$

$$\mu = 424.9 \pm 1.2$$

$$\sigma = 36.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.25 \pm 0.31 \%$$

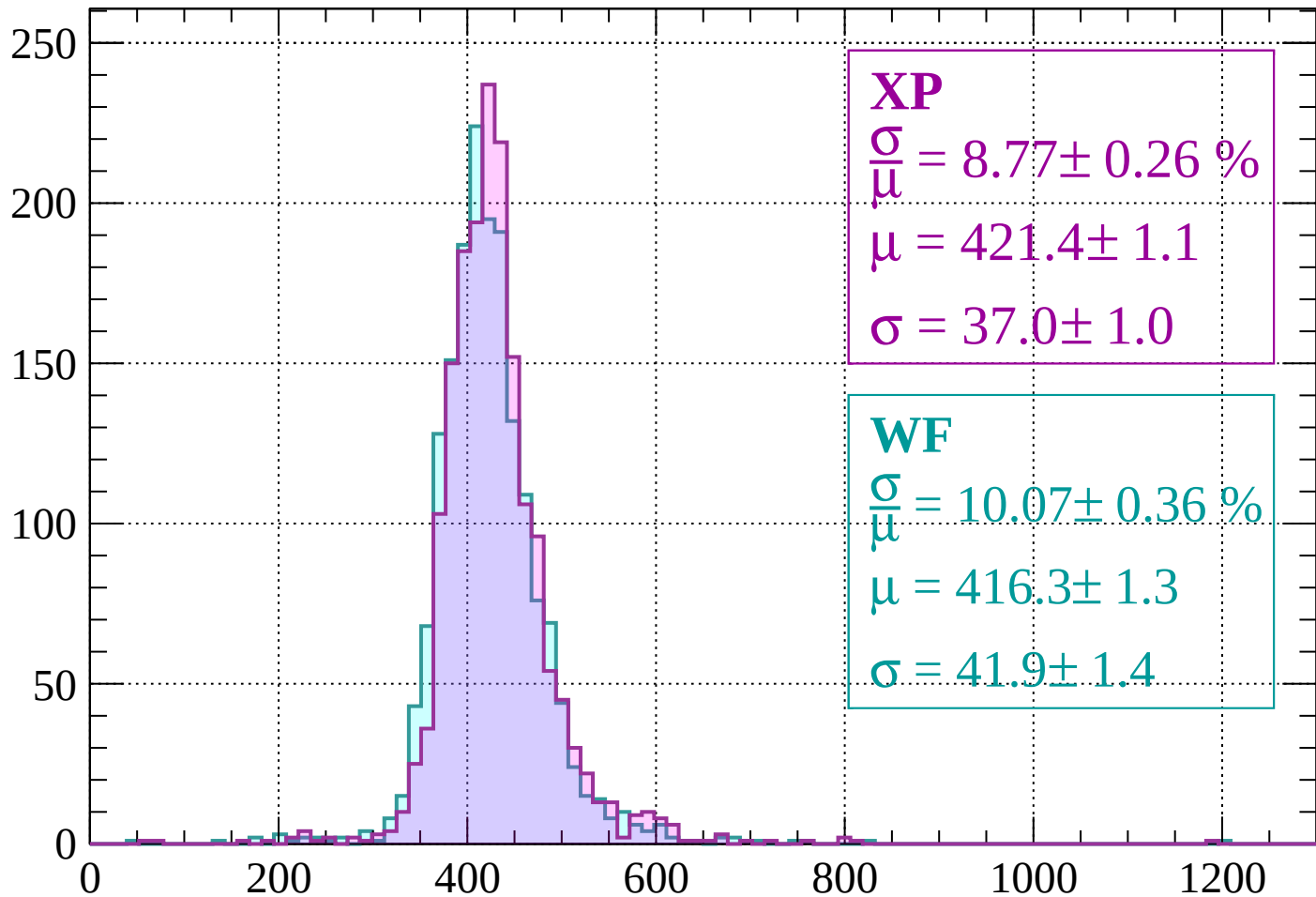
$$\mu = 419.1 \pm 1.2$$

$$\sigma = 38.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 8.77 \pm 0.26 \%$$

$$\mu = 421.4 \pm 1.1$$

$$\sigma = 37.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 10.07 \pm 0.36 \%$$

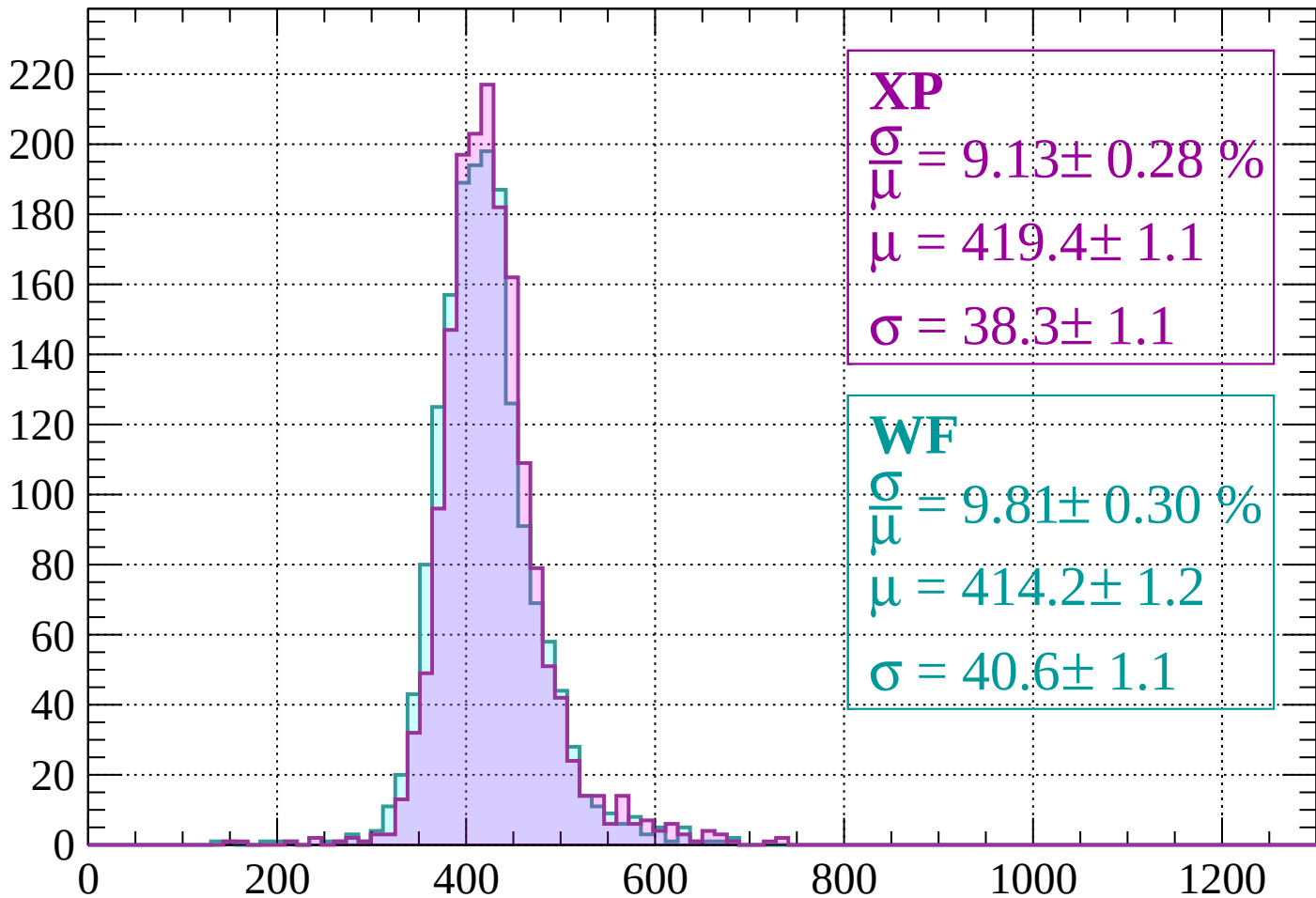
$$\mu = 416.3 \pm 1.3$$

$$\sigma = 41.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.28 \%$$

$$\mu = 419.4 \pm 1.1$$

$$\sigma = 38.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.81 \pm 0.30 \%$$

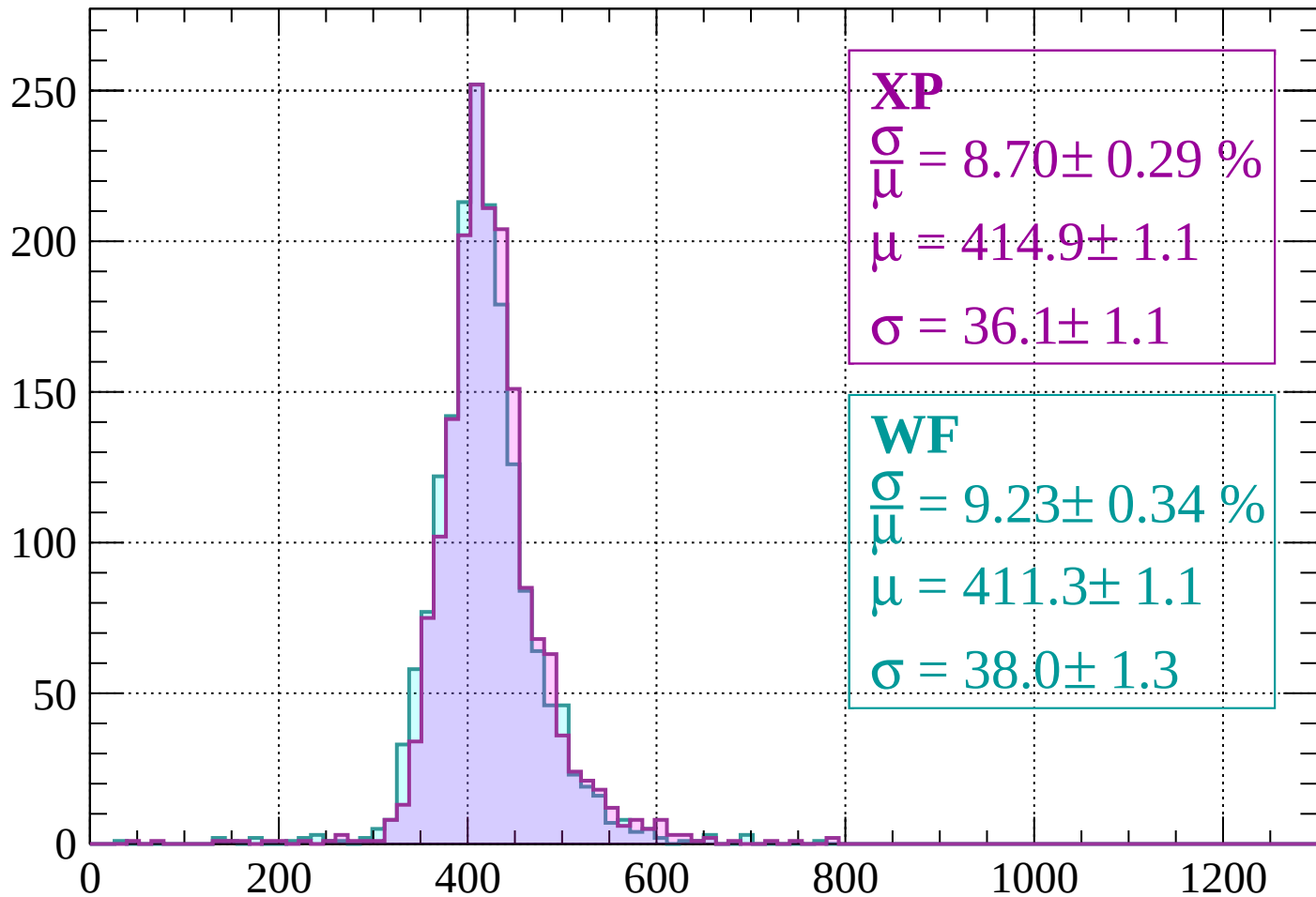
$$\mu = 414.2 \pm 1.2$$

$$\sigma = 40.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.70 \pm 0.29 \%$$

$$\mu = 414.9 \pm 1.1$$

$$\sigma = 36.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.23 \pm 0.34 \%$$

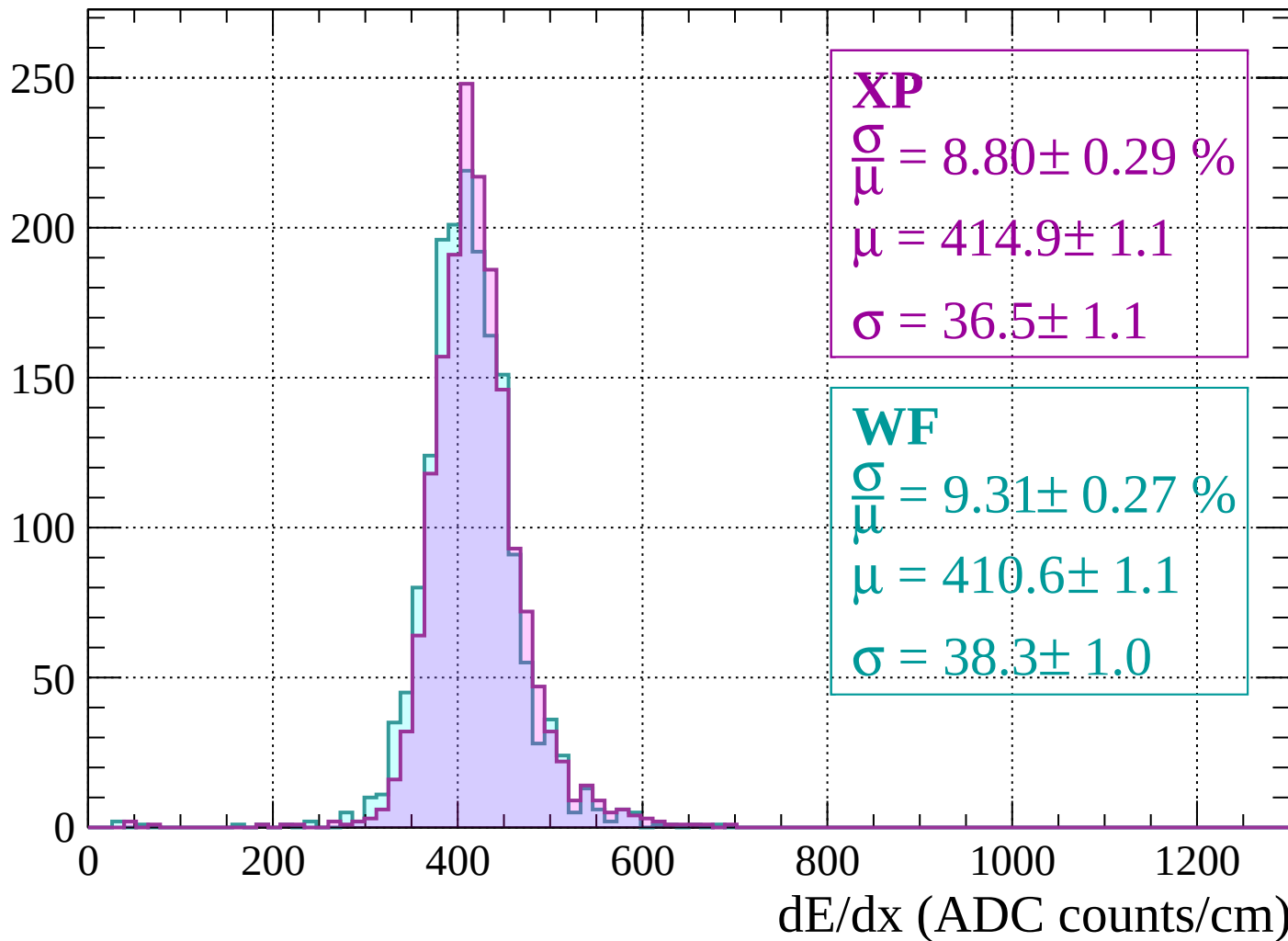
$$\mu = 411.3 \pm 1.1$$

$$\sigma = 38.0 \pm 1.3$$

dE/dx (ADC counts/cm)

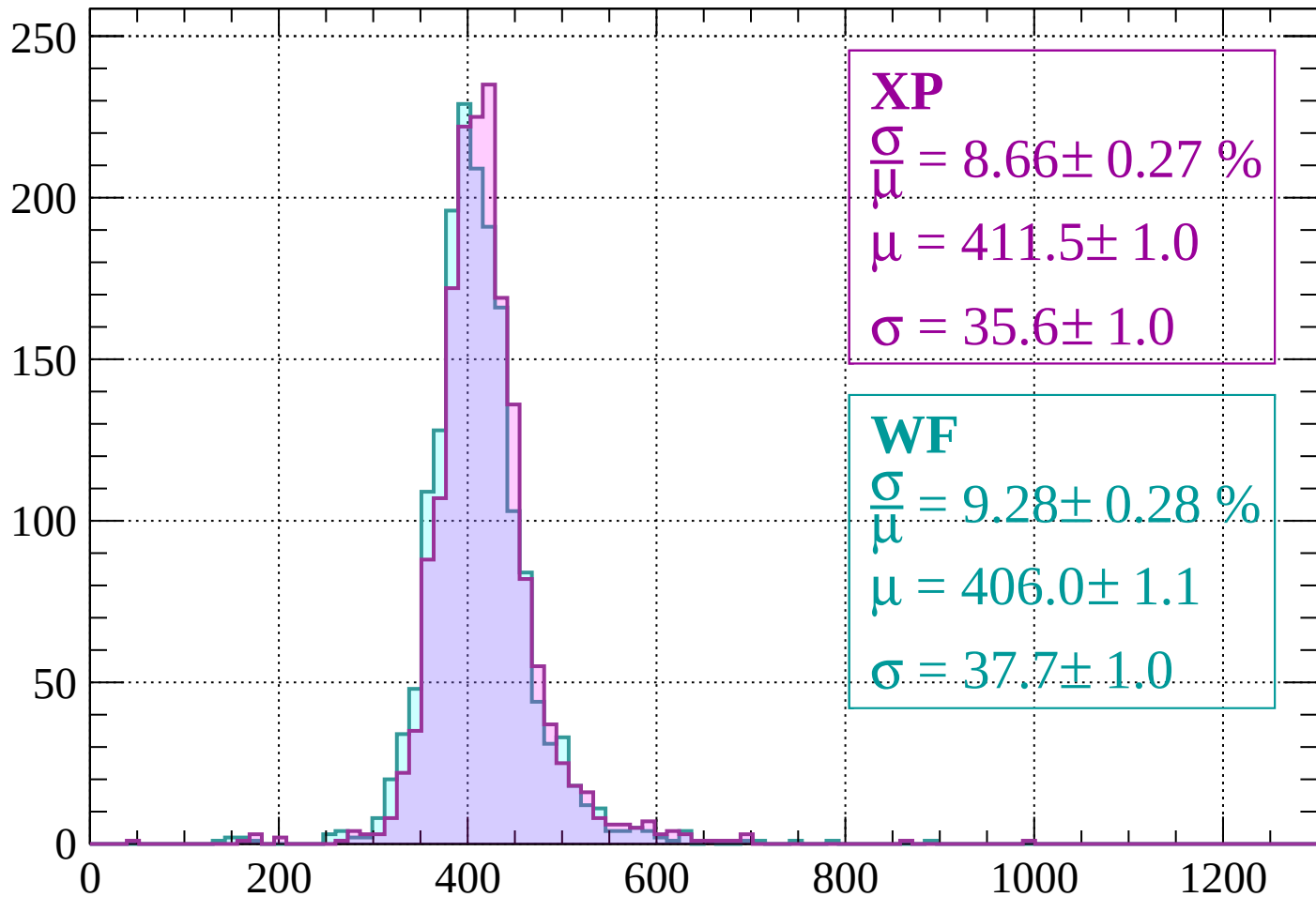
Energy loss | $-560 < p < -520$

Count



Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.27 \%$$

$$\mu = 411.5 \pm 1.0$$

$$\sigma = 35.6 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.28 \pm 0.28 \%$$

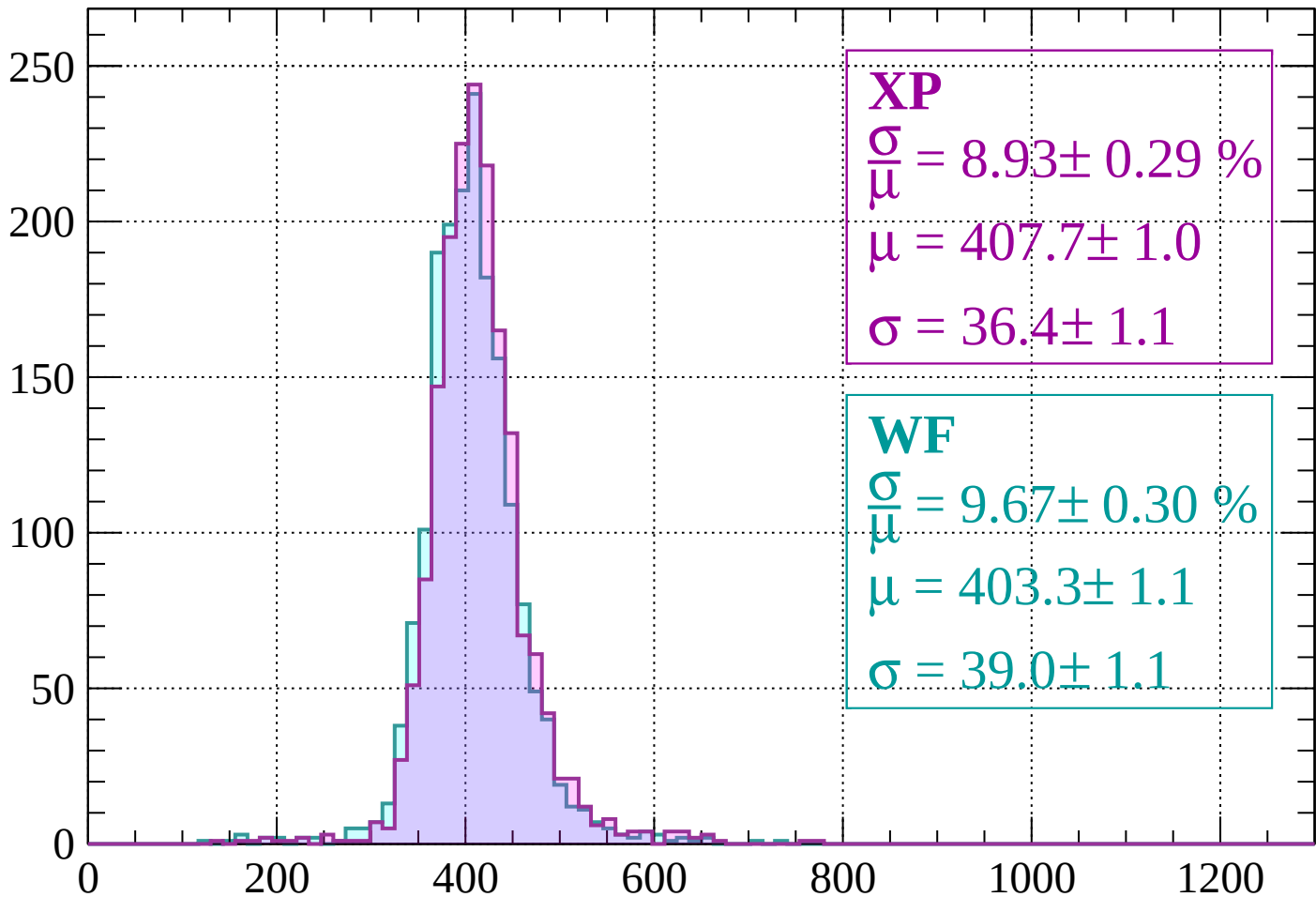
$$\mu = 406.0 \pm 1.1$$

$$\sigma = 37.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 8.93 \pm 0.29 \%$$

$$\mu = 407.7 \pm 1.0$$

$$\sigma = 36.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.67 \pm 0.30 \%$$

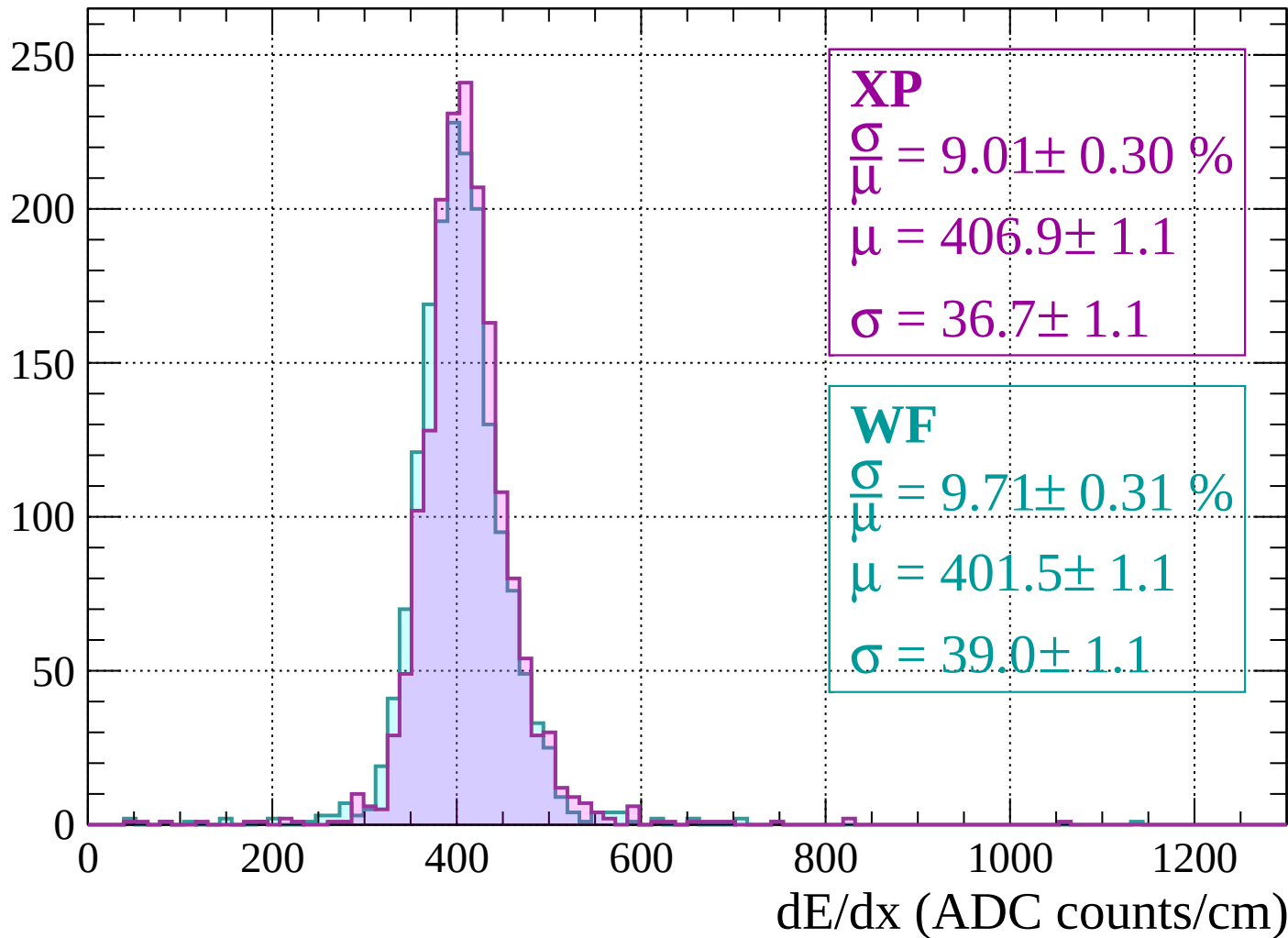
$$\mu = 403.3 \pm 1.1$$

$$\sigma = 39.0 \pm 1.1$$

dE/dx (ADC counts/cm)

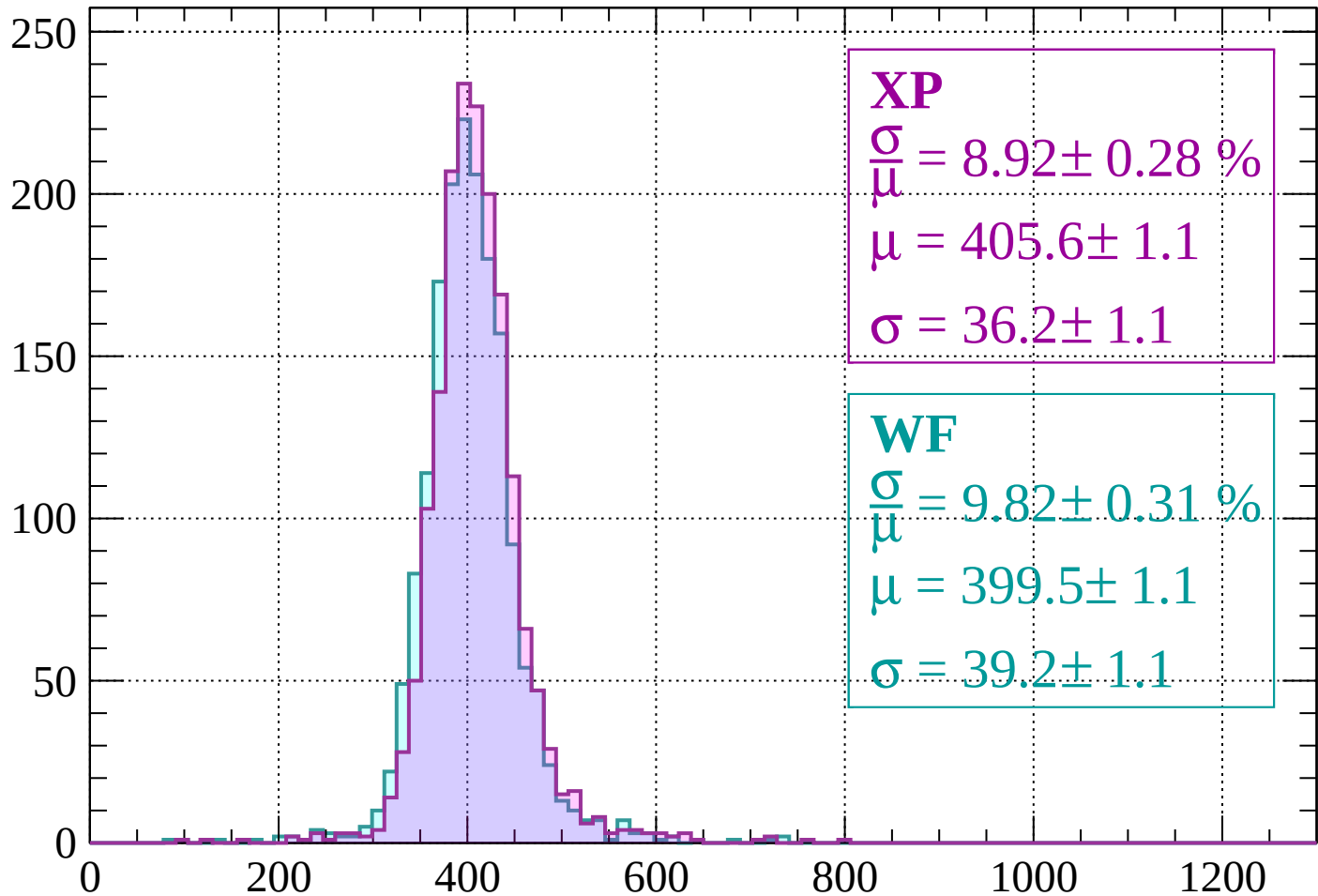
Energy loss | $-440 < p < -400$

Count



Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.28 \%$$

$$\mu = 405.6 \pm 1.1$$

$$\sigma = 36.2 \pm 1.1$$

WF

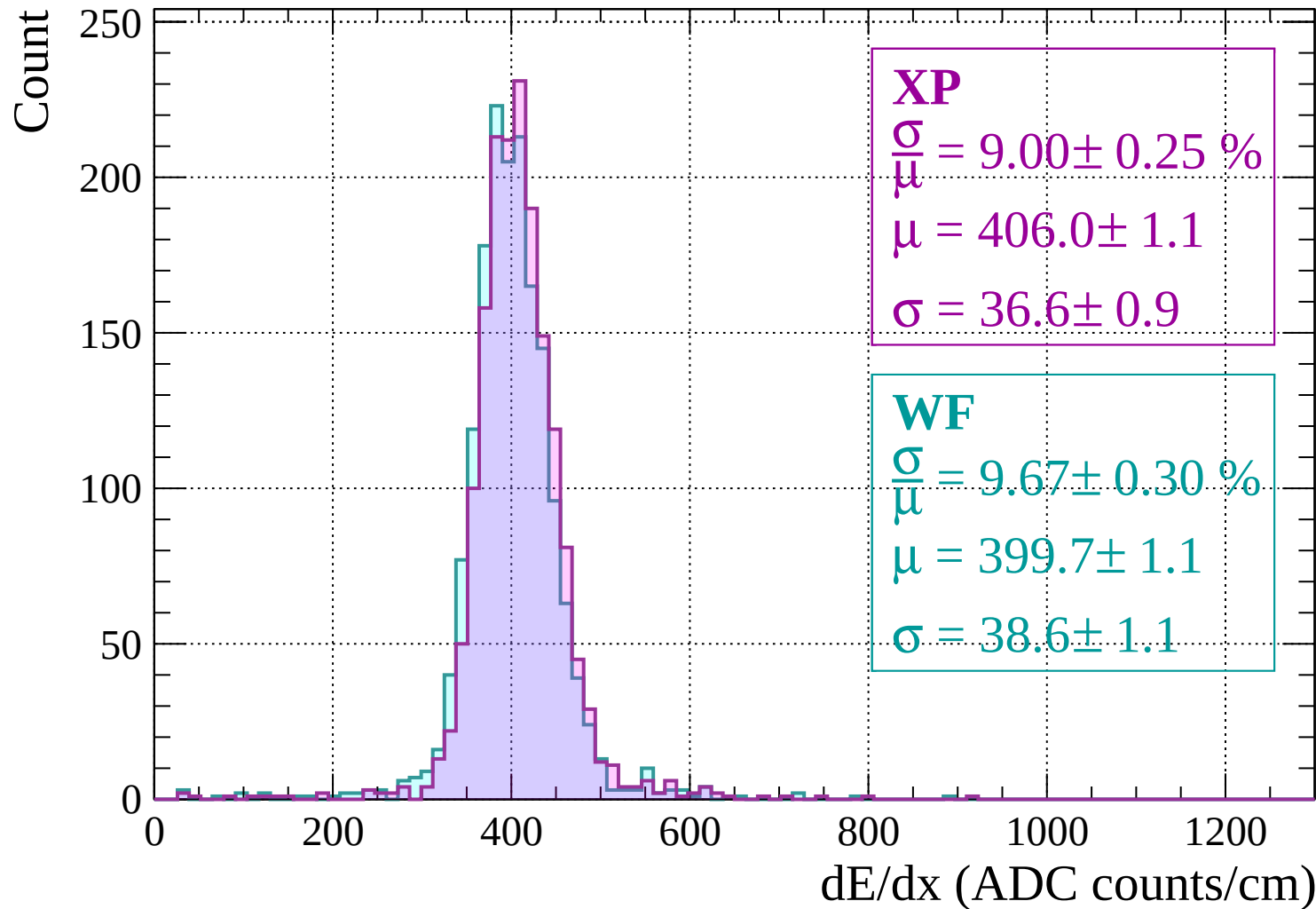
$$\frac{\sigma}{\mu} = 9.82 \pm 0.31 \%$$

$$\mu = 399.5 \pm 1.1$$

$$\sigma = 39.2 \pm 1.1$$

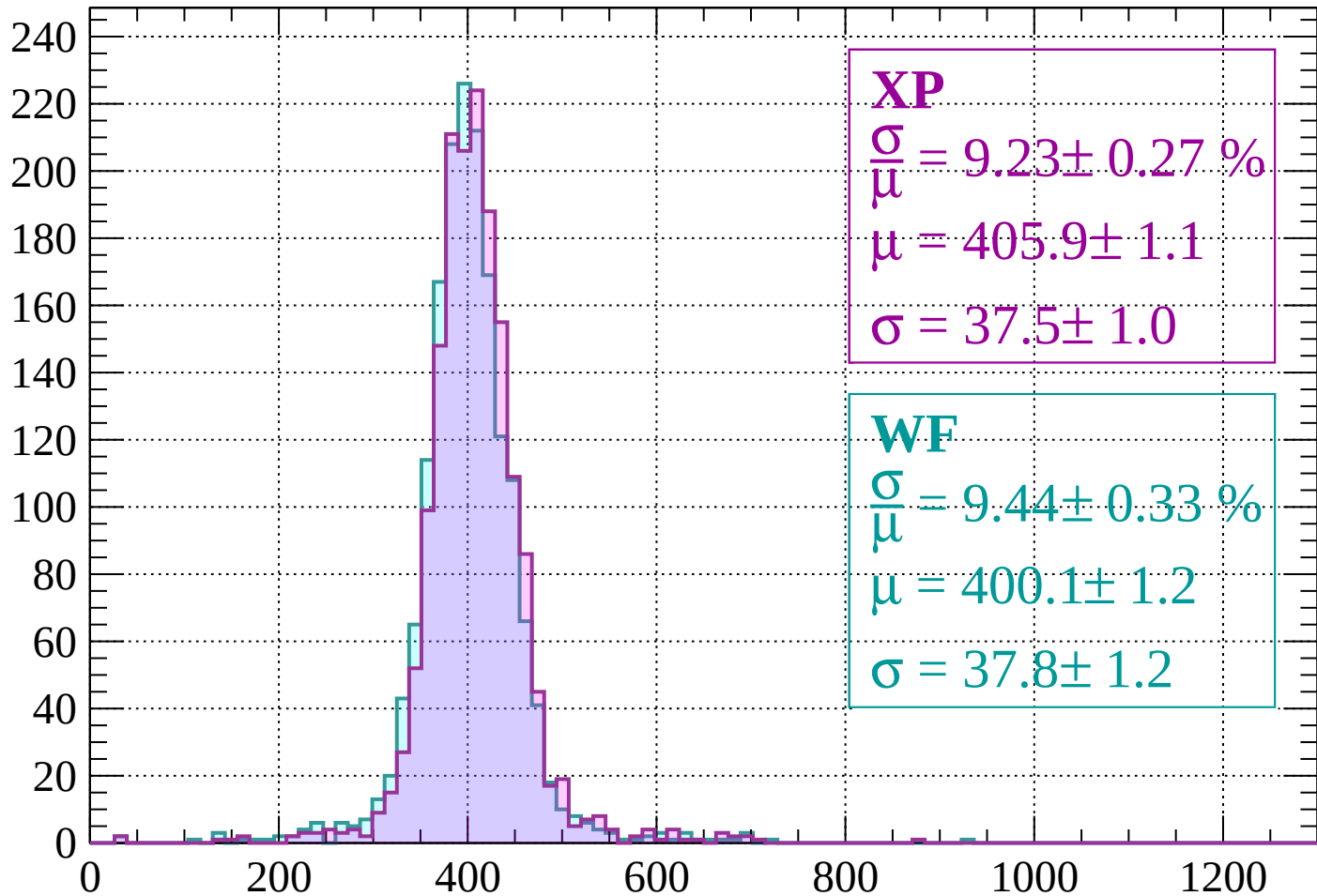
dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$



Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 9.23 \pm 0.27 \%$$

$$\mu = 405.9 \pm 1.1$$

$$\sigma = 37.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.44 \pm 0.33 \%$$

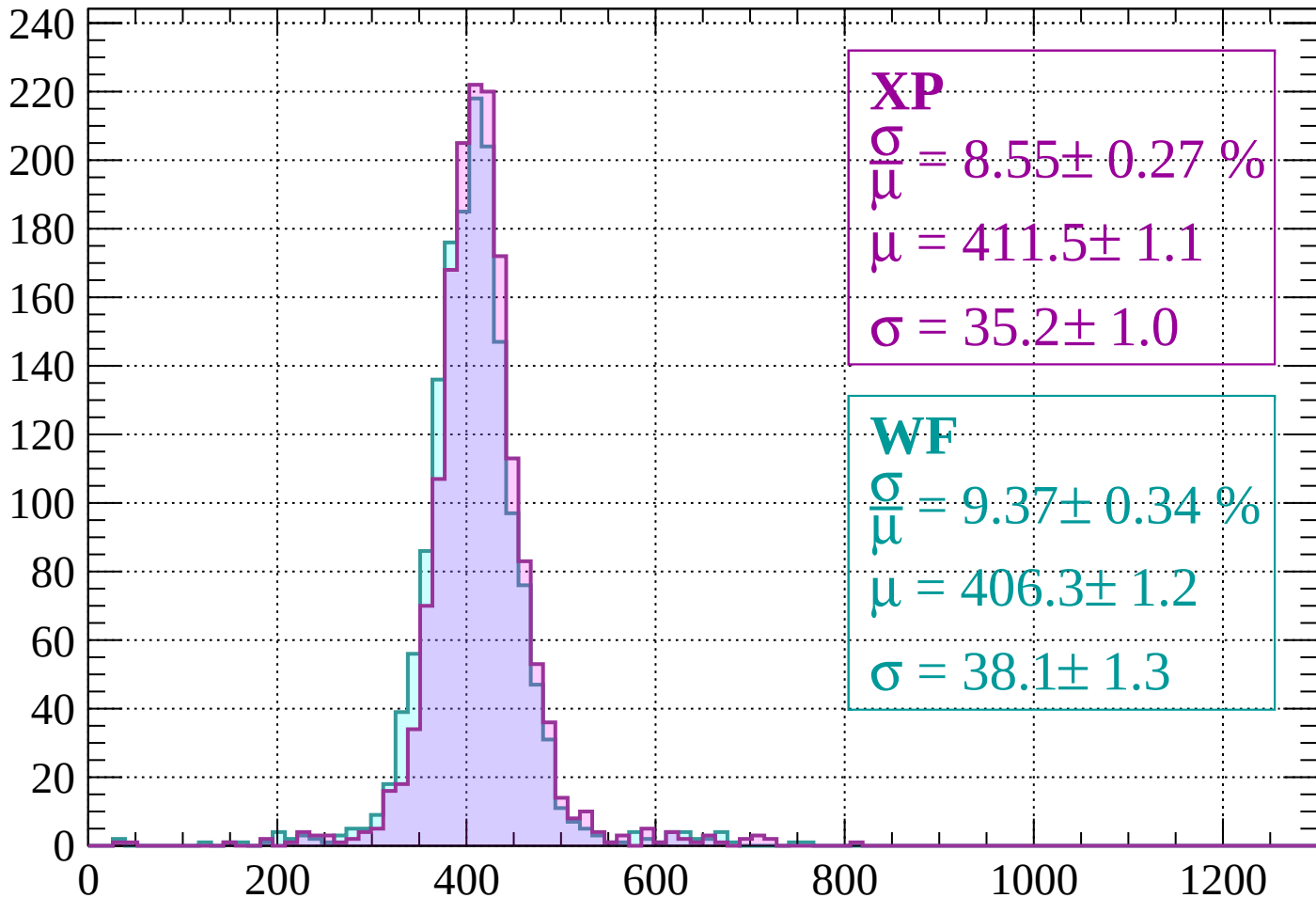
$$\mu = 400.1 \pm 1.2$$

$$\sigma = 37.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.27 \%$$

$$\mu = 411.5 \pm 1.1$$

$$\sigma = 35.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.34 \%$$

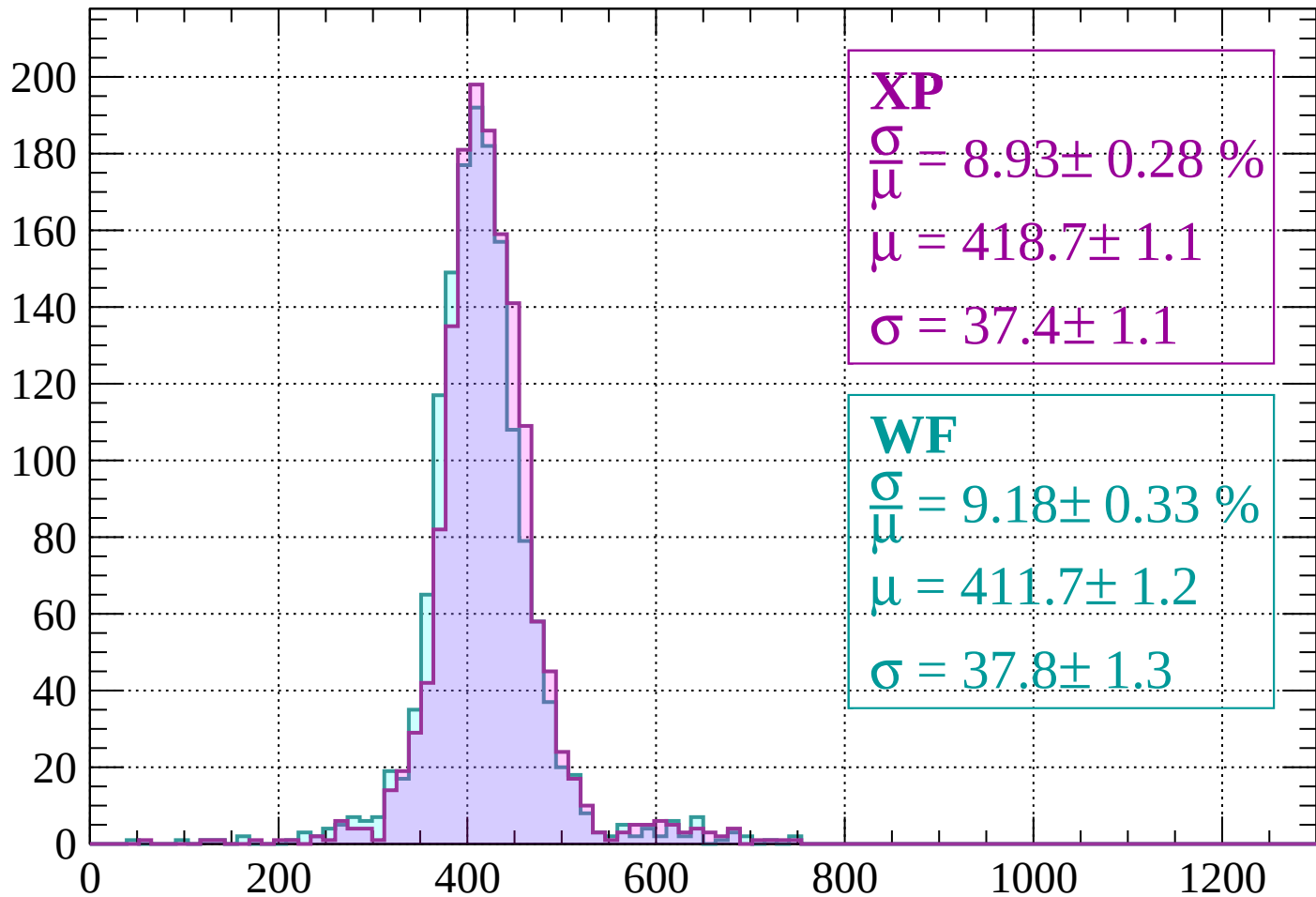
$$\mu = 406.3 \pm 1.2$$

$$\sigma = 38.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.93 \pm 0.28 \%$$

$$\mu = 418.7 \pm 1.1$$

$$\sigma = 37.4 \pm 1.1$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.18 \pm 0.33 \%$$

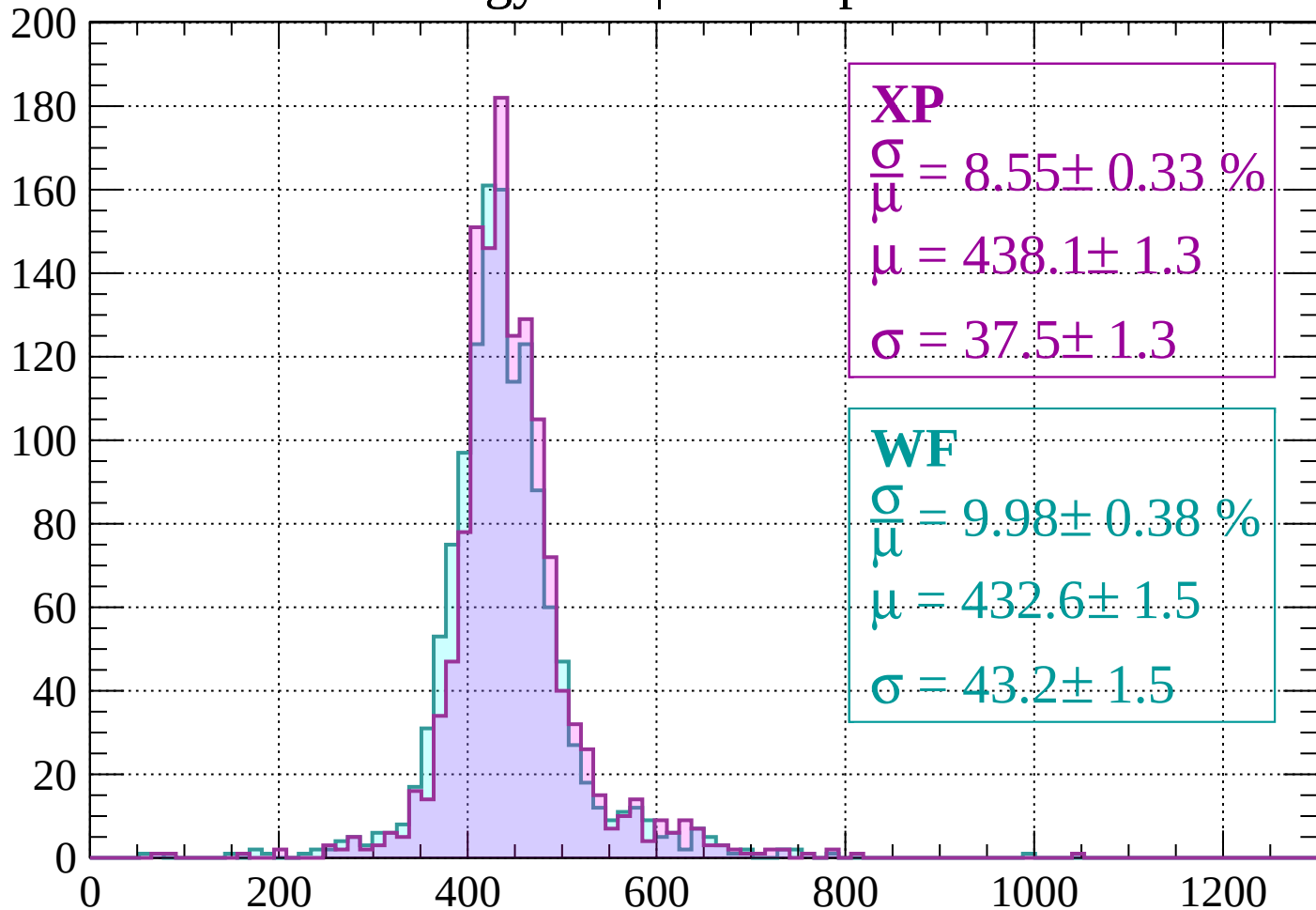
$$\mu = 411.7 \pm 1.2$$

$$\sigma = 37.8 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.33 \%$$

$$\mu = 438.1 \pm 1.3$$

$$\sigma = 37.5 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.98 \pm 0.38 \%$$

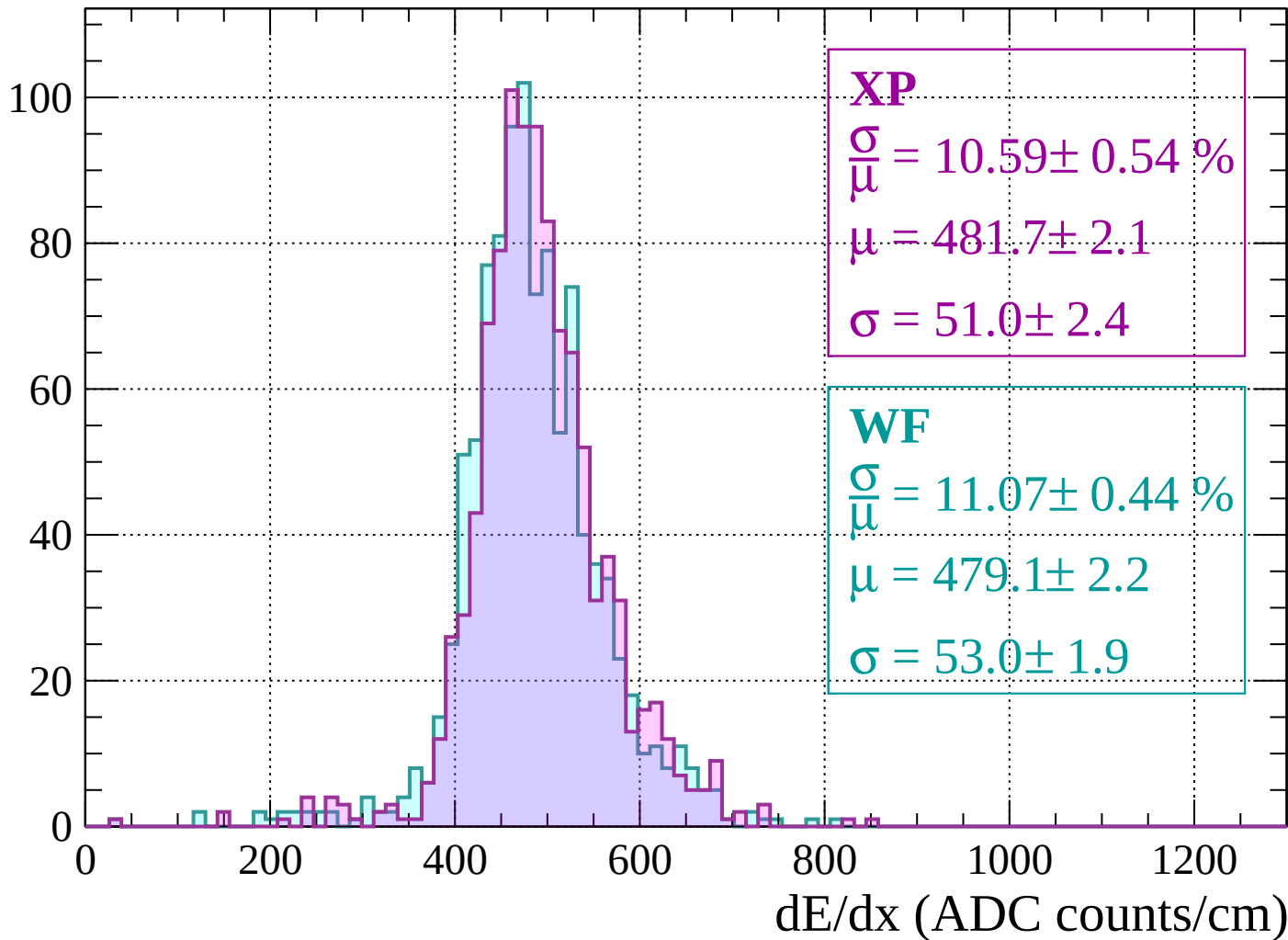
$$\mu = 432.6 \pm 1.5$$

$$\sigma = 43.2 \pm 1.5$$

dE/dx (ADC counts/cm)

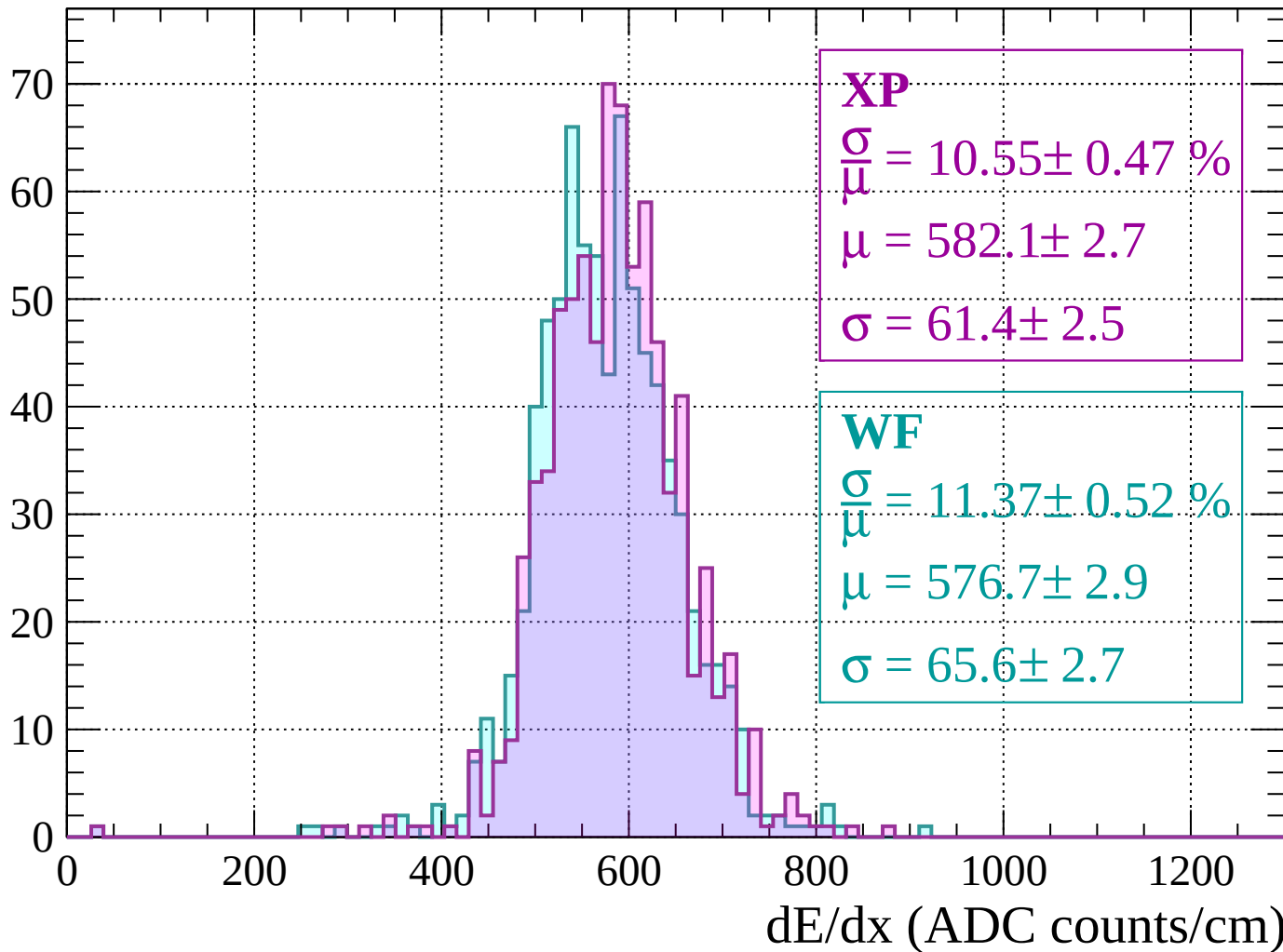
Energy loss | $-160 < p < -120$

Count



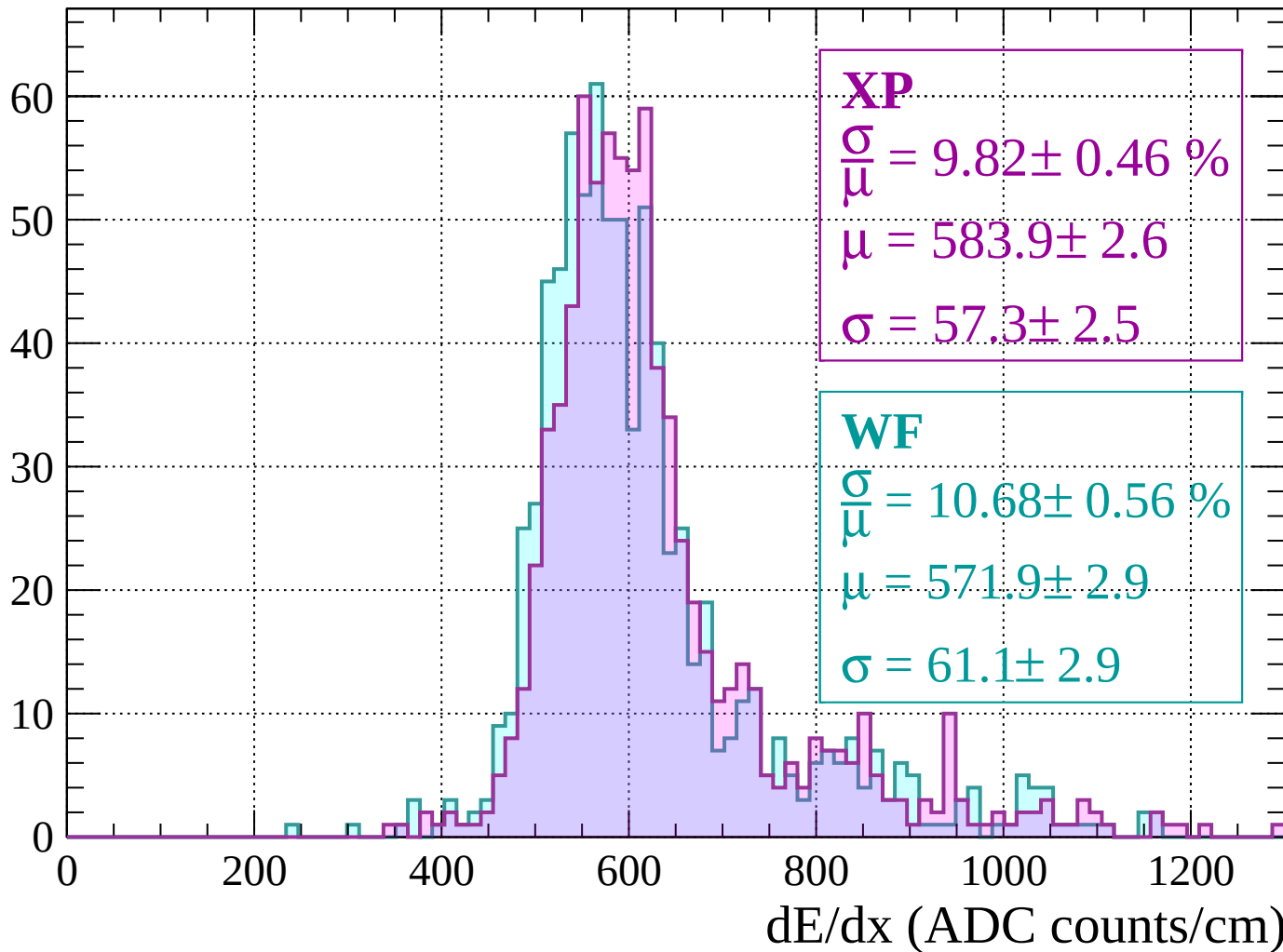
Energy loss | $-120 < p < -80$

Count



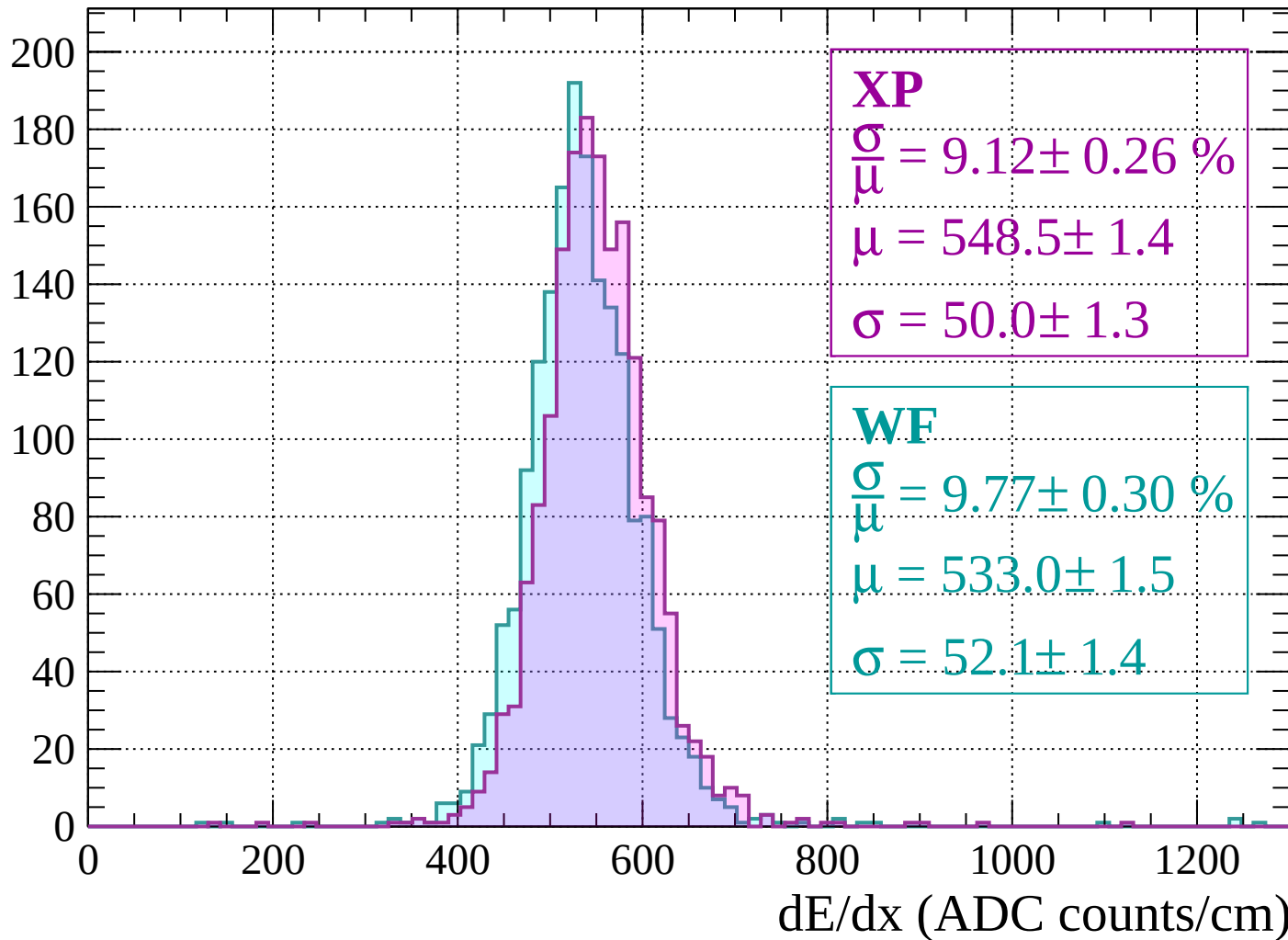
Energy loss | $-80 < p < -40$

Count



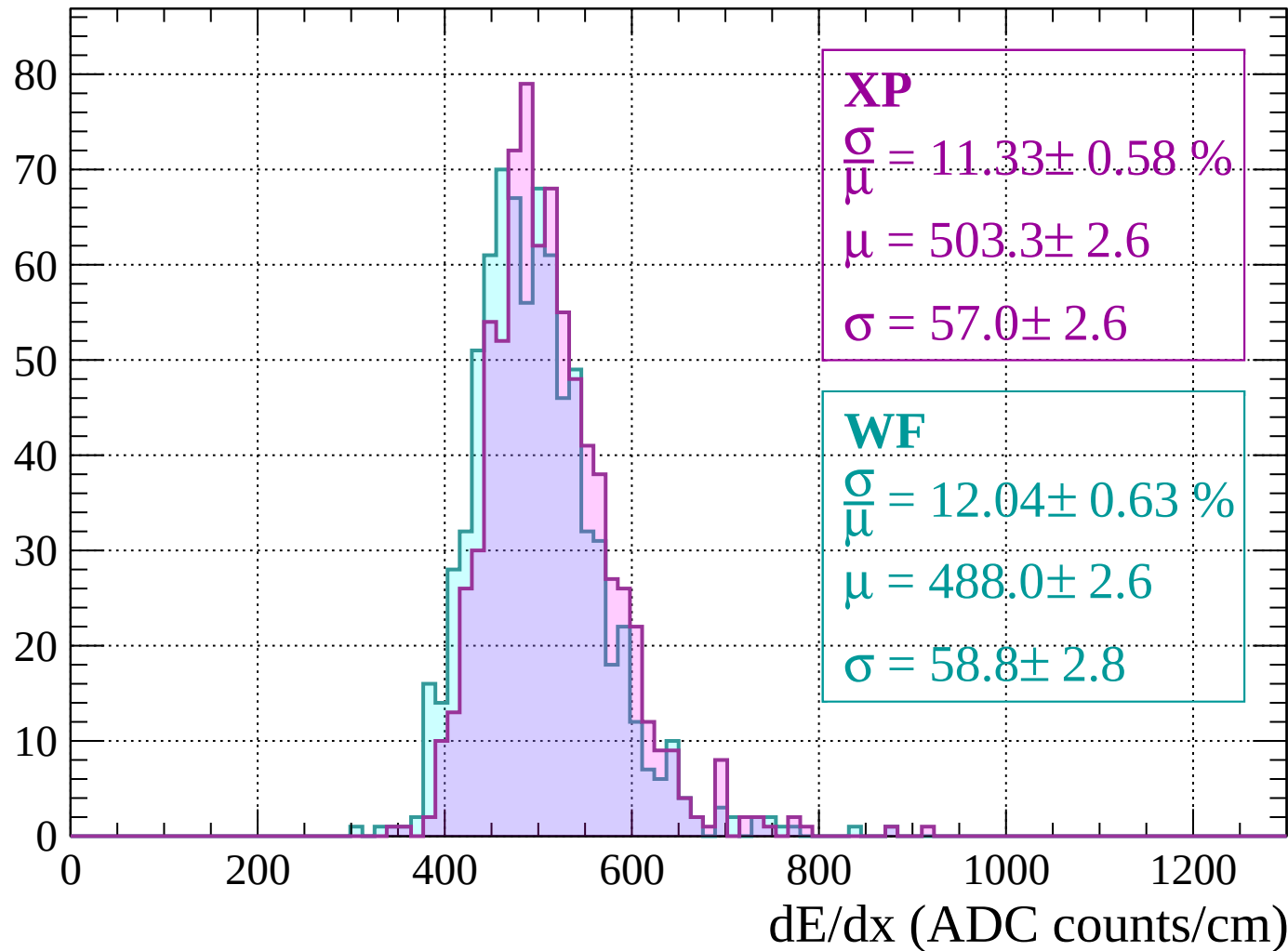
Energy loss | $-40 < p < 0$

Count



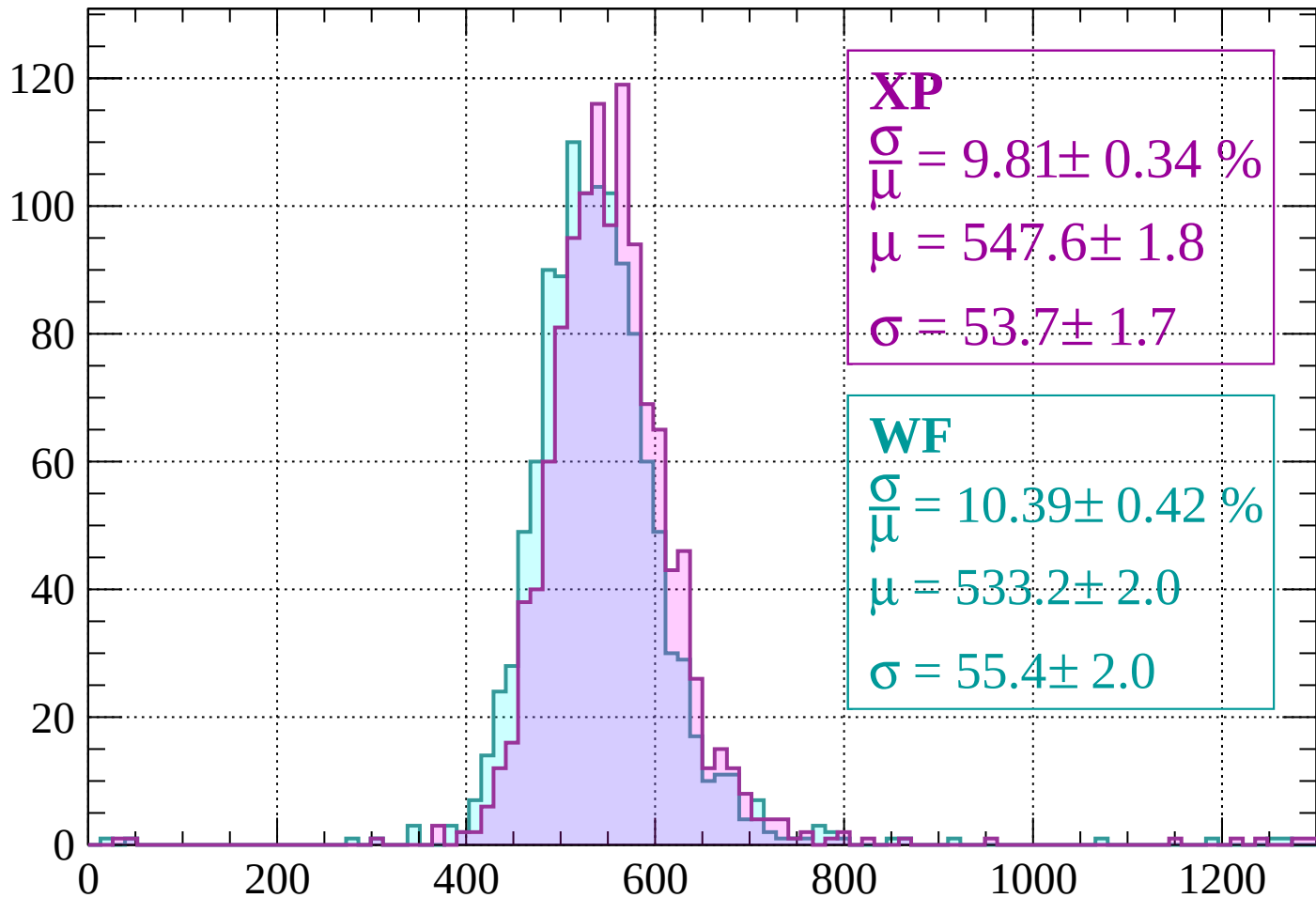
Energy loss | $0 < p < 40$

Count



Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 9.81 \pm 0.34 \%$$

$$\mu = 547.6 \pm 1.8$$

$$\sigma = 53.7 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 10.39 \pm 0.42 \%$$

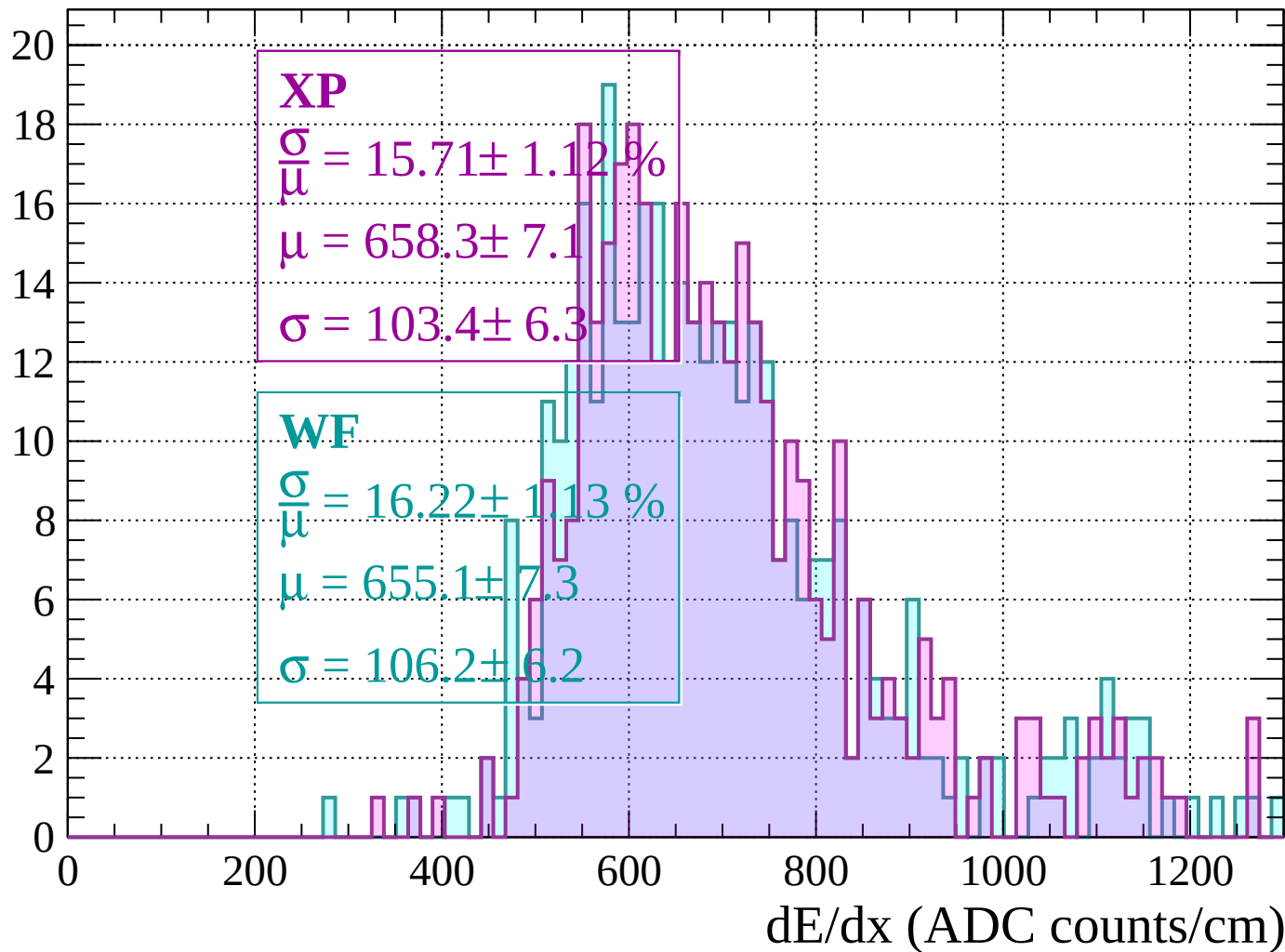
$$\mu = 533.2 \pm 2.0$$

$$\sigma = 55.4 \pm 2.0$$

dE/dx (ADC counts/cm)

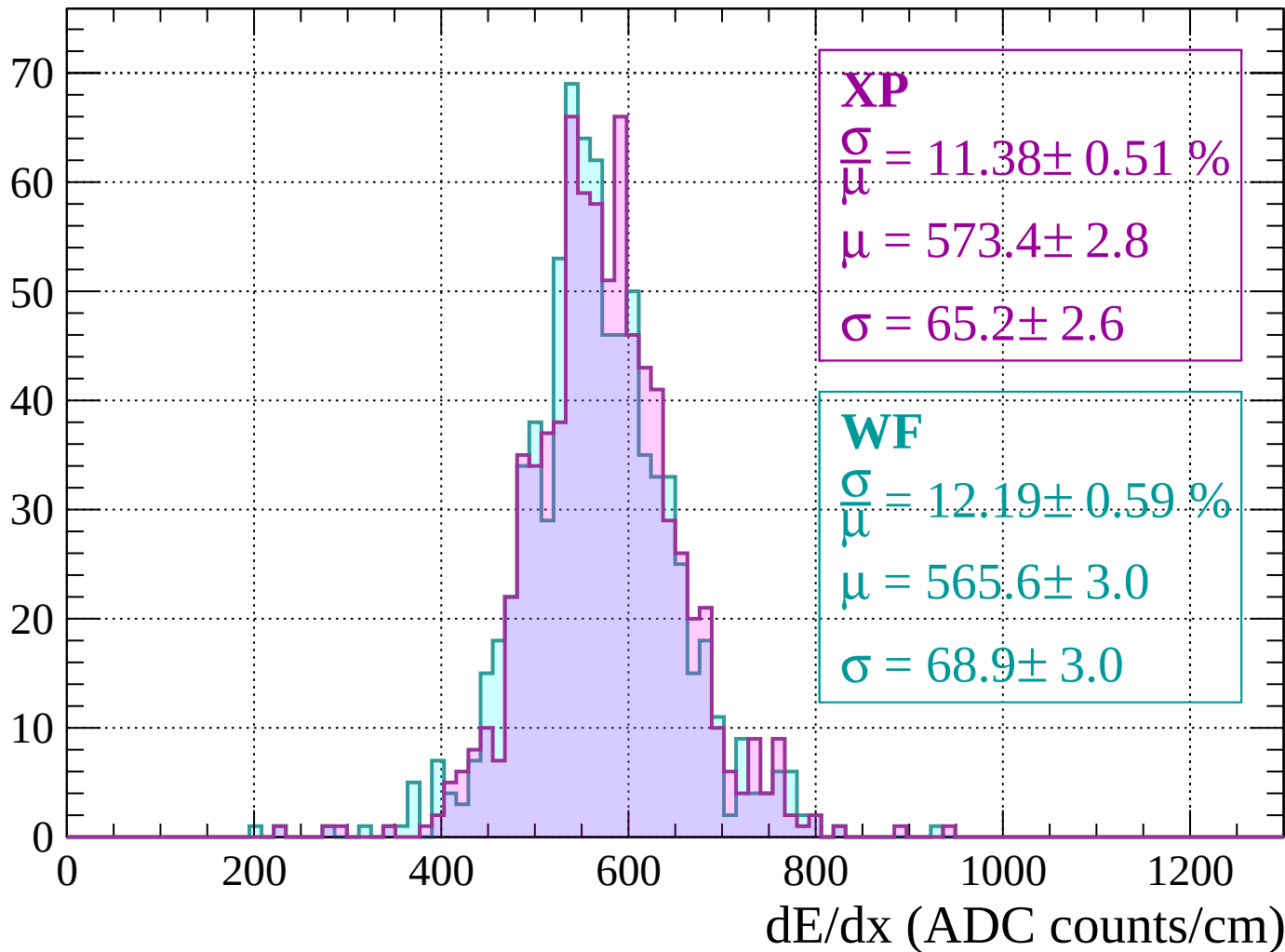
Energy loss | $80 < p < 120$

Count

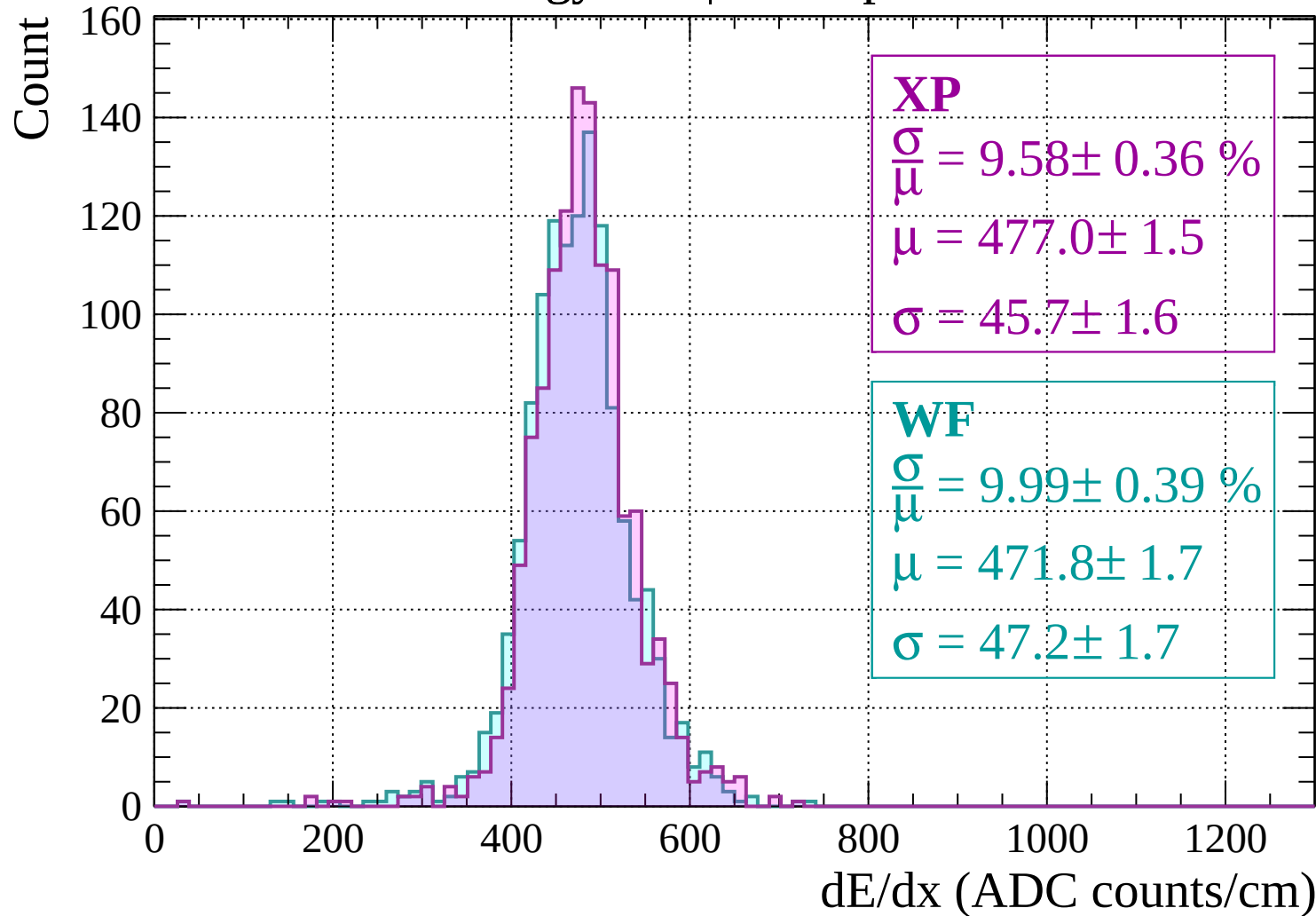


Energy loss | $120 < p < 160$

Count

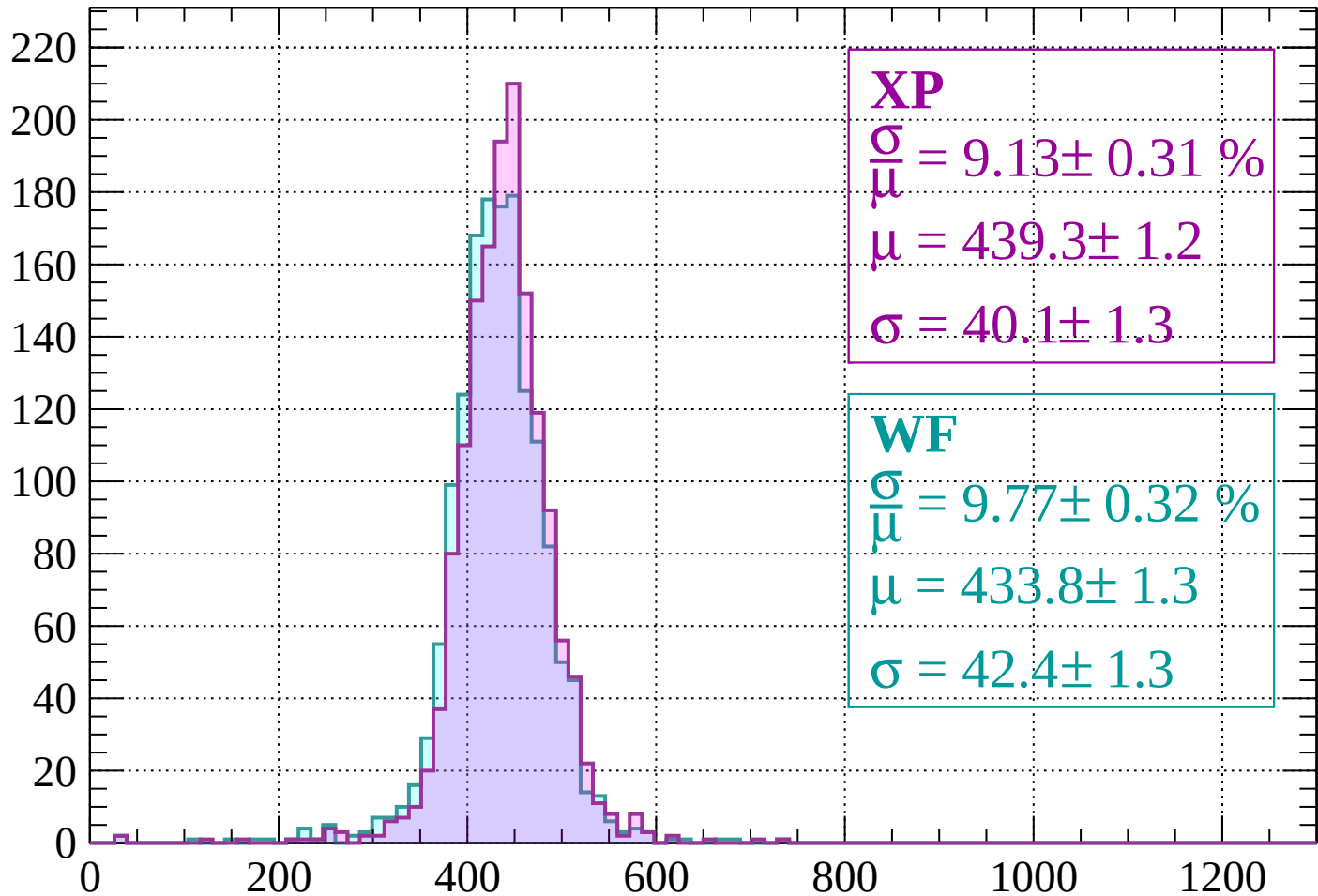


Energy loss | $160 < p < 200$



Energy loss | $200 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.31 \%$$

$$\mu = 439.3 \pm 1.2$$

$$\sigma = 40.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.77 \pm 0.32 \%$$

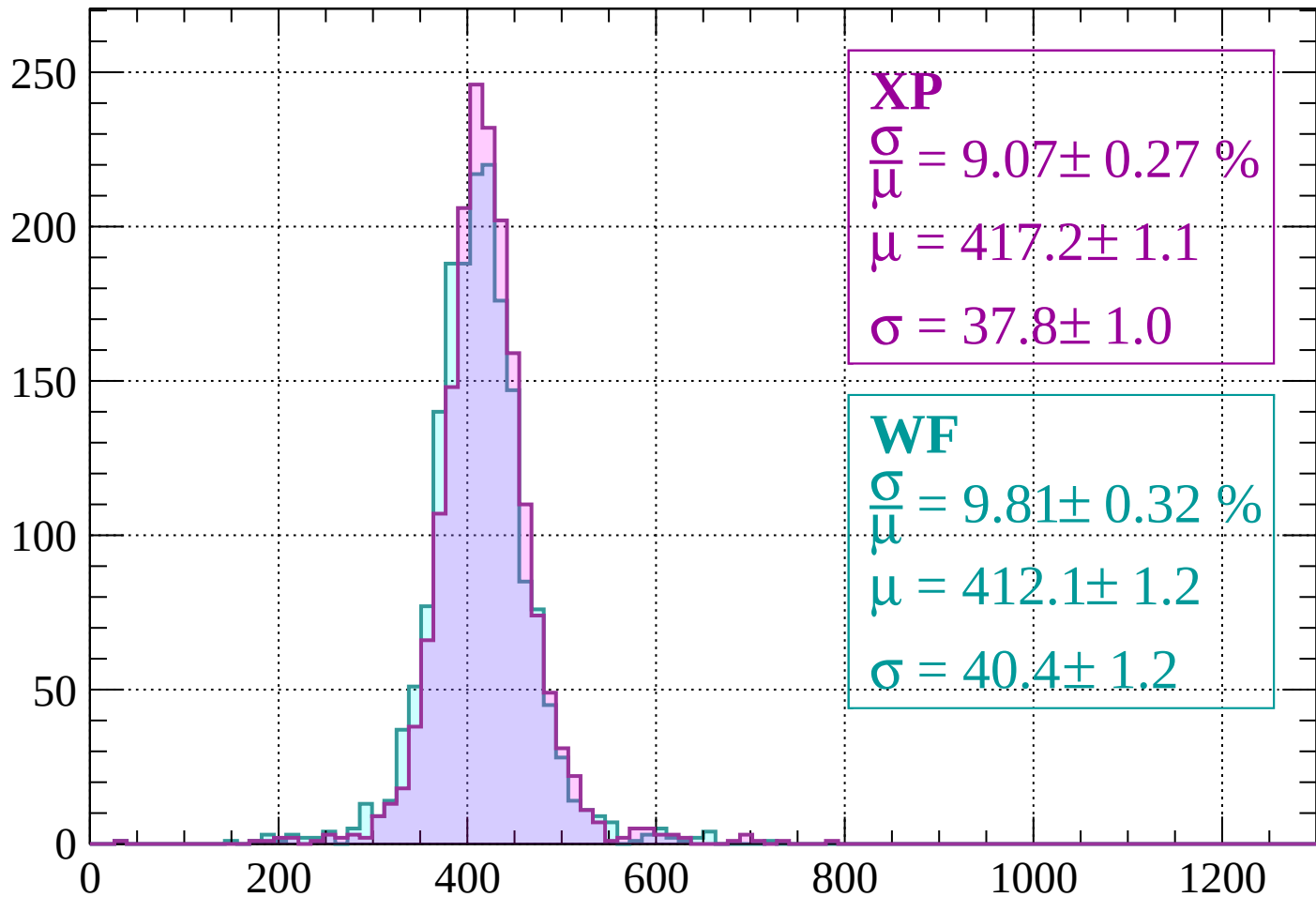
$$\mu = 433.8 \pm 1.3$$

$$\sigma = 42.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 9.07 \pm 0.27 \%$$

$$\mu = 417.2 \pm 1.1$$

$$\sigma = 37.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.81 \pm 0.32 \%$$

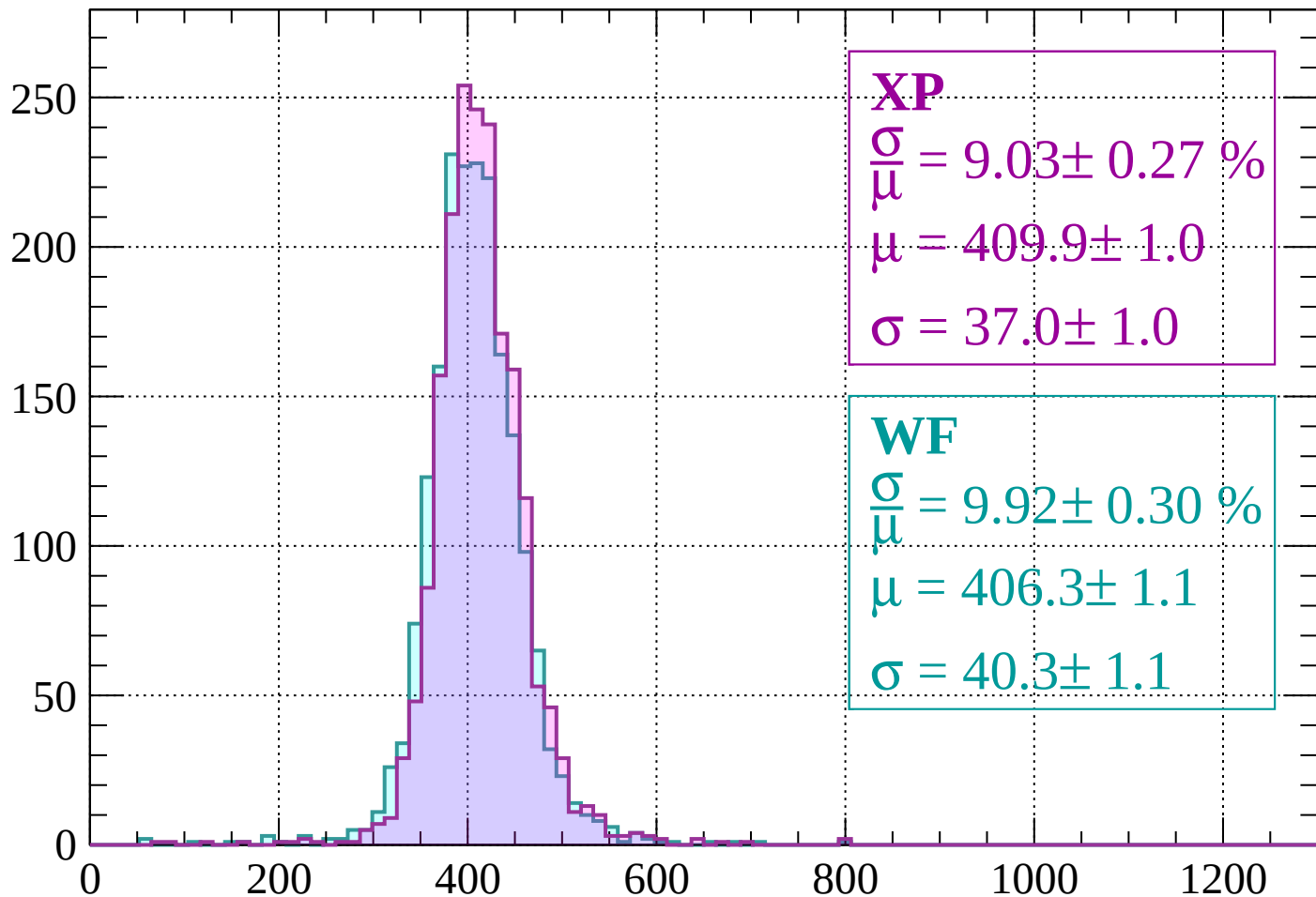
$$\mu = 412.1 \pm 1.2$$

$$\sigma = 40.4 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.03 \pm 0.27 \%$$

$$\mu = 409.9 \pm 1.0$$

$$\sigma = 37.0 \pm 1.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.92 \pm 0.30 \%$$

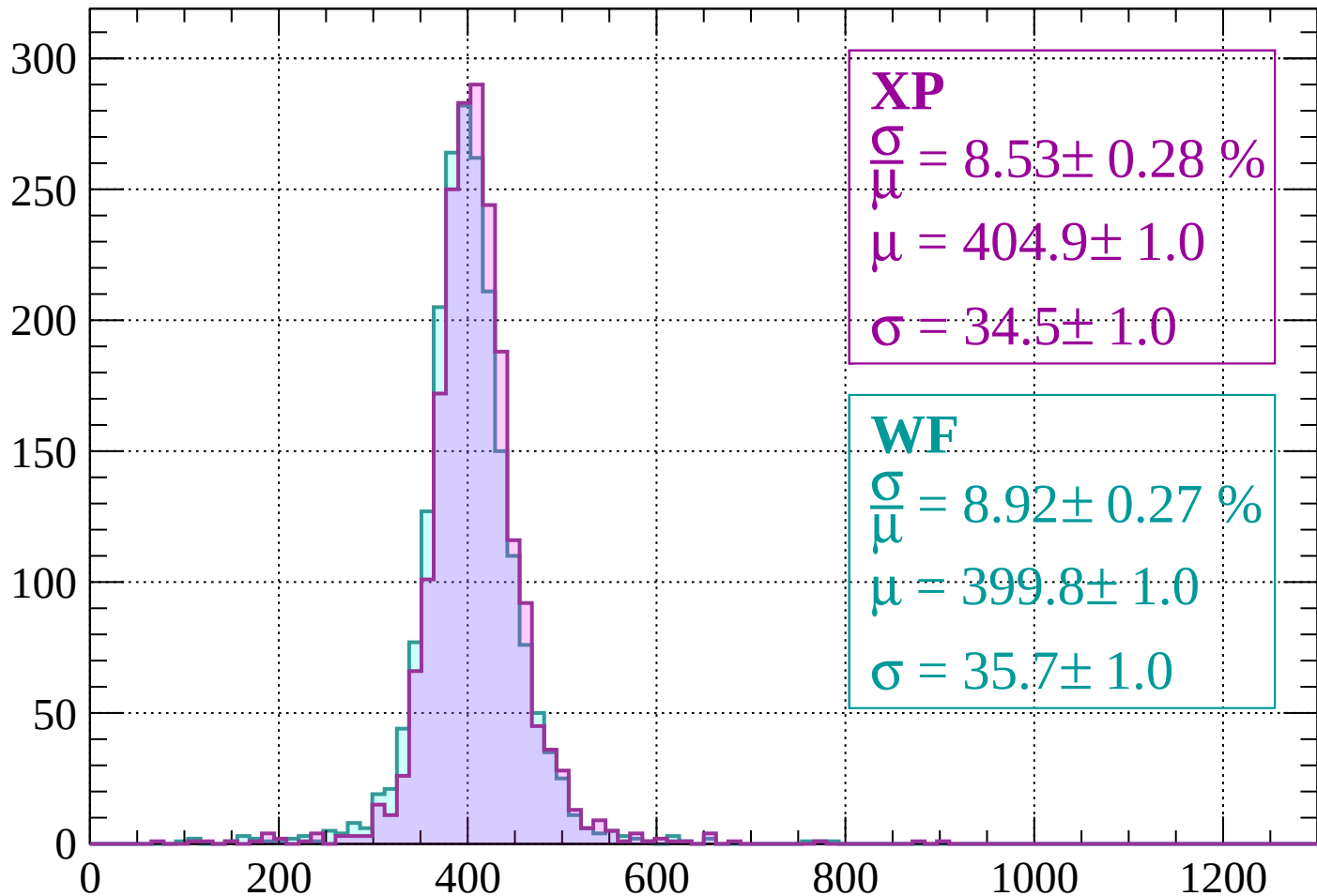
$$\mu = 406.3 \pm 1.1$$

$$\sigma = 40.3 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 8.53 \pm 0.28 \%$$

$$\mu = 404.9 \pm 1.0$$

$$\sigma = 34.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.92 \pm 0.27 \%$$

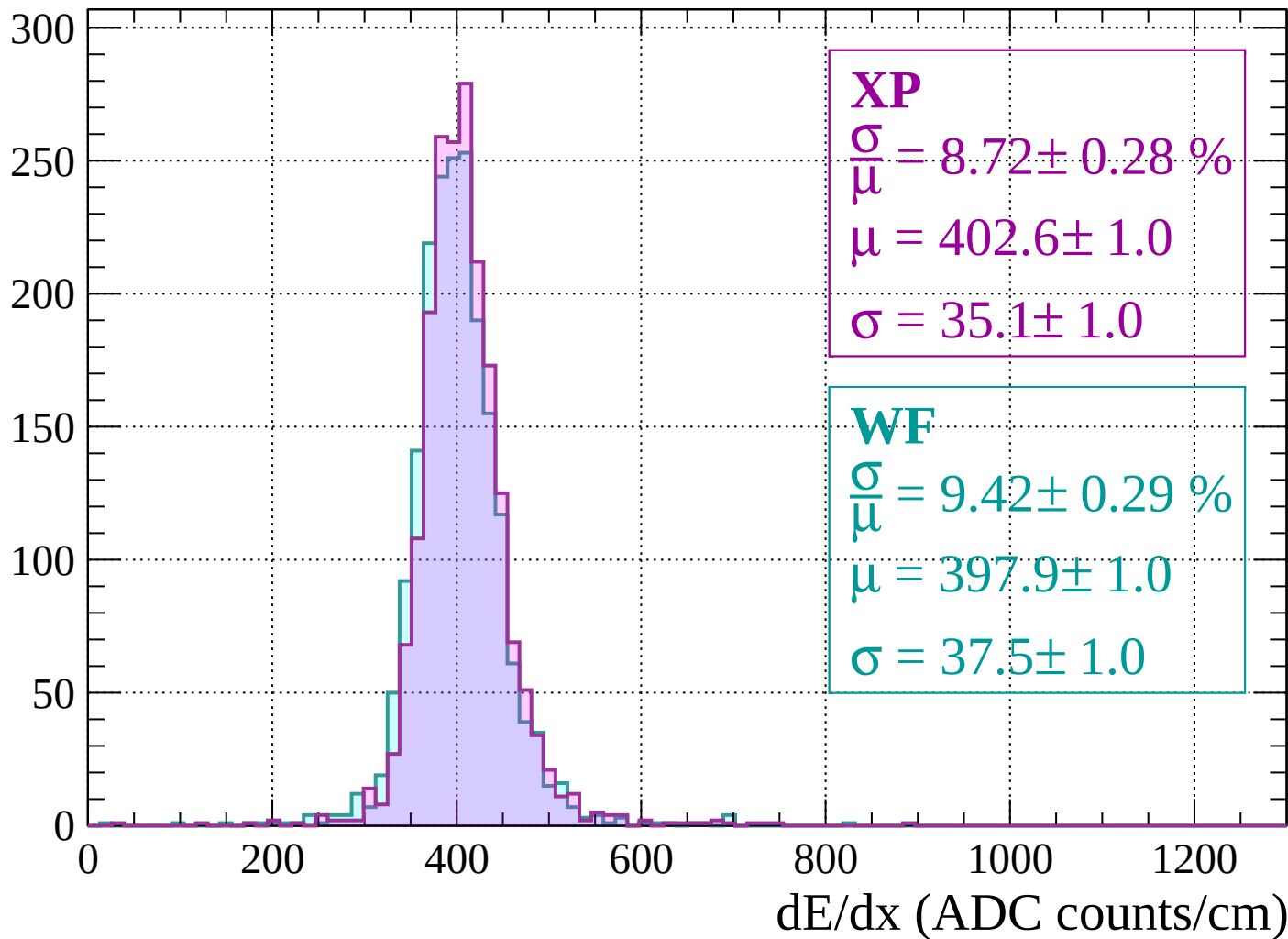
$$\mu = 399.8 \pm 1.0$$

$$\sigma = 35.7 \pm 1.0$$

dE/dx (ADC counts/cm)

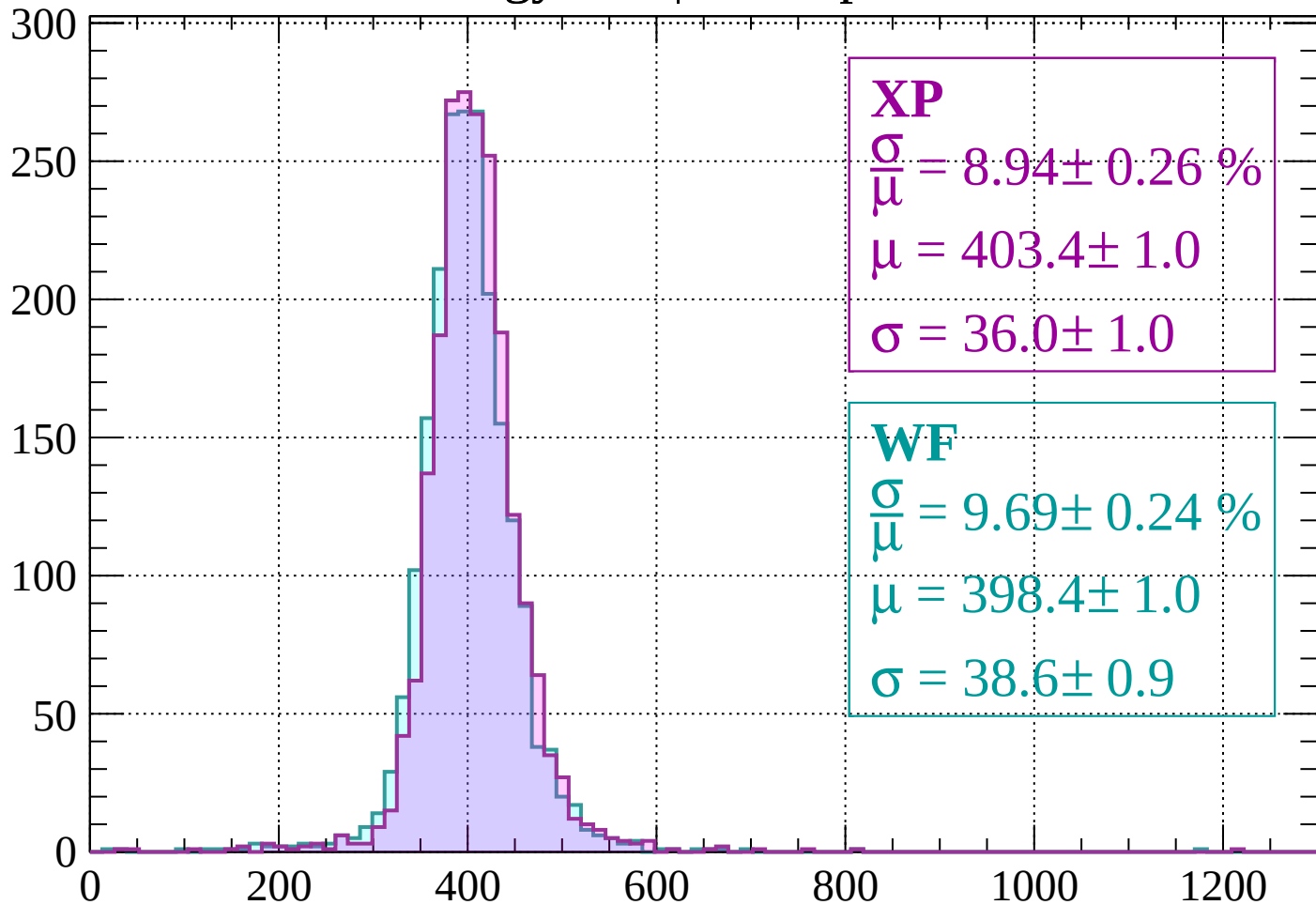
Energy loss | $360 < p < 400$

Count



Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 8.94 \pm 0.26 \%$$

$$\mu = 403.4 \pm 1.0$$

$$\sigma = 36.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.69 \pm 0.24 \%$$

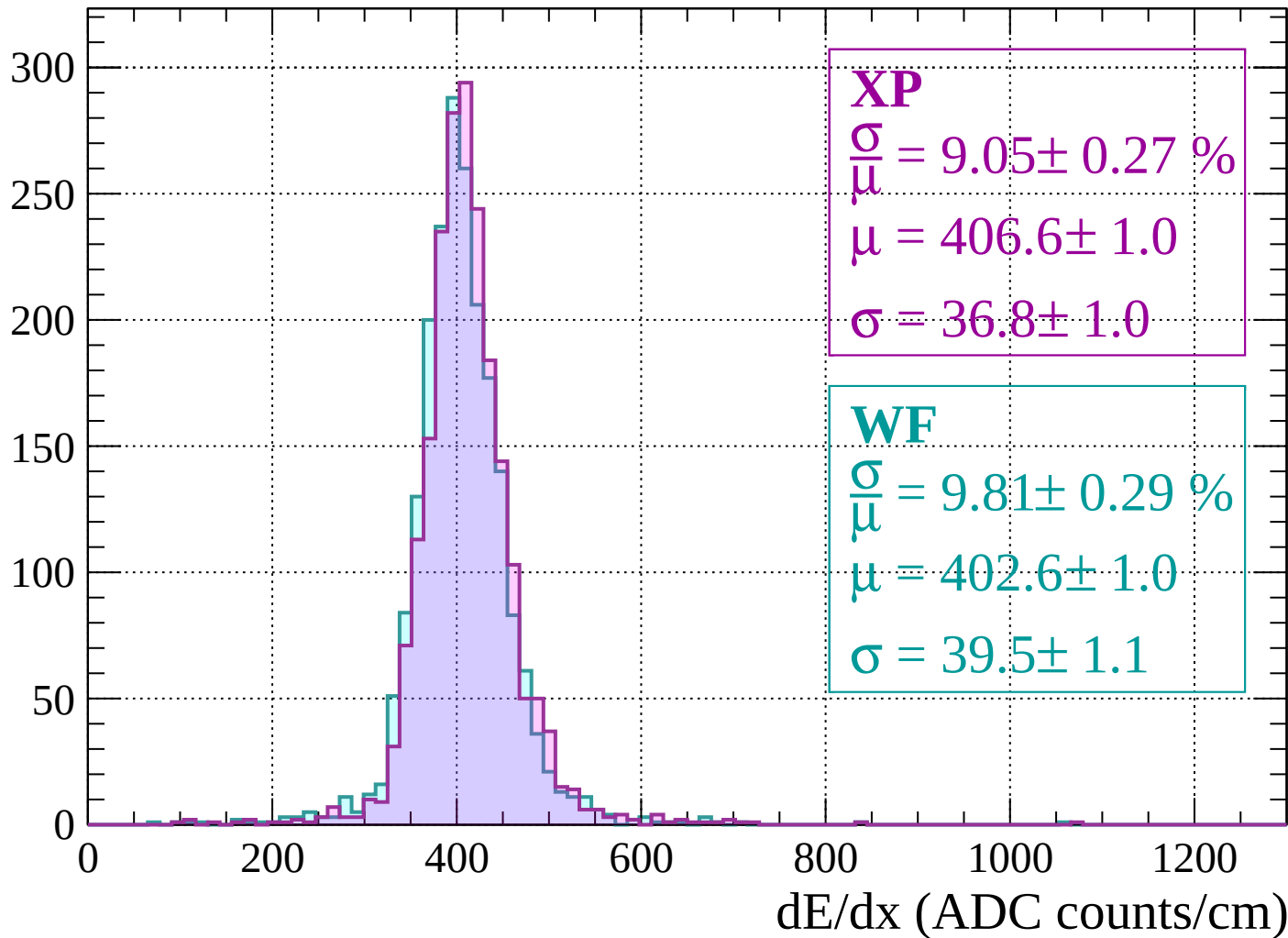
$$\mu = 398.4 \pm 1.0$$

$$\sigma = 38.6 \pm 0.9$$

dE/dx (ADC counts/cm)

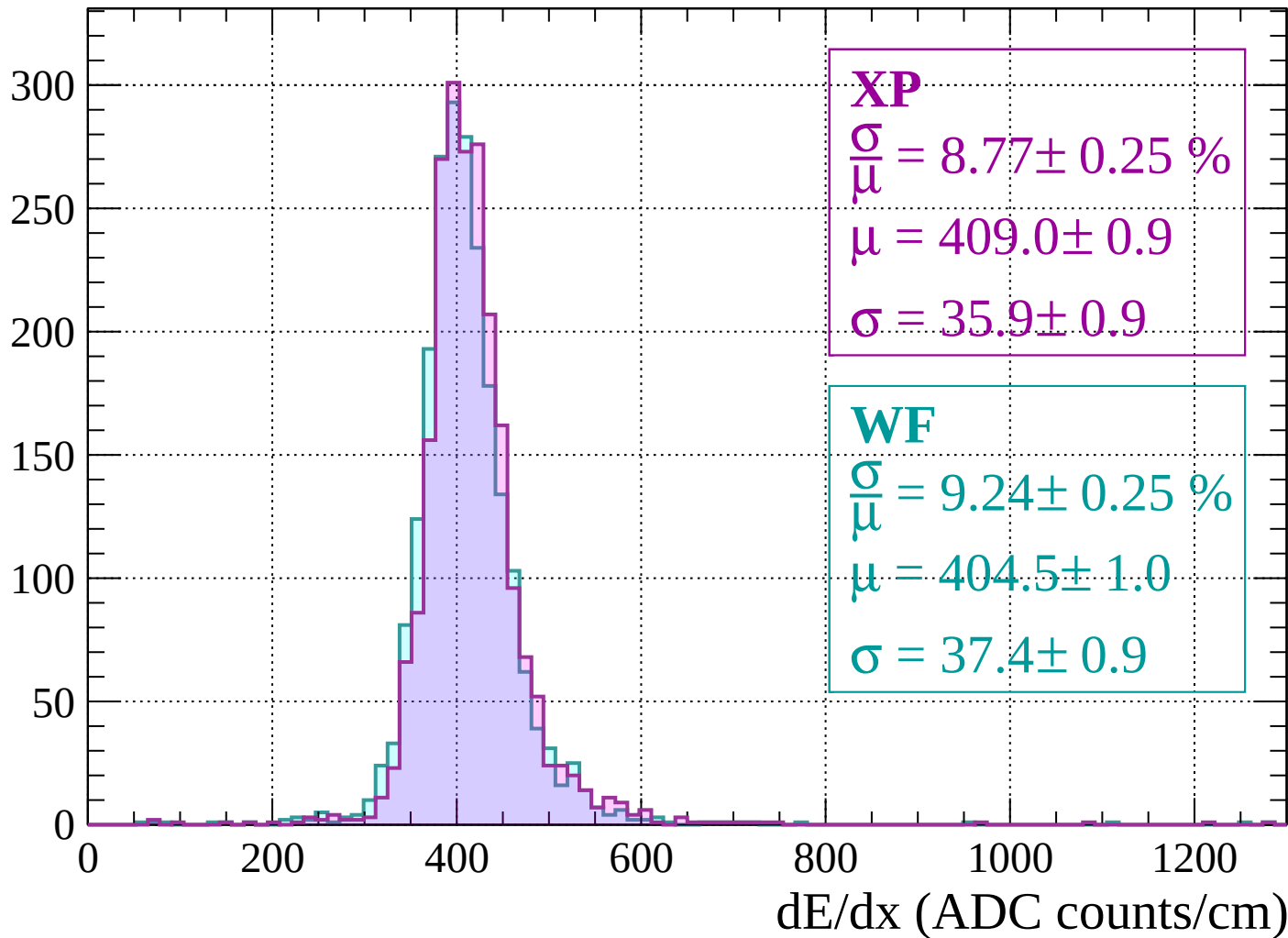
Energy loss | $440 < p < 480$

Count

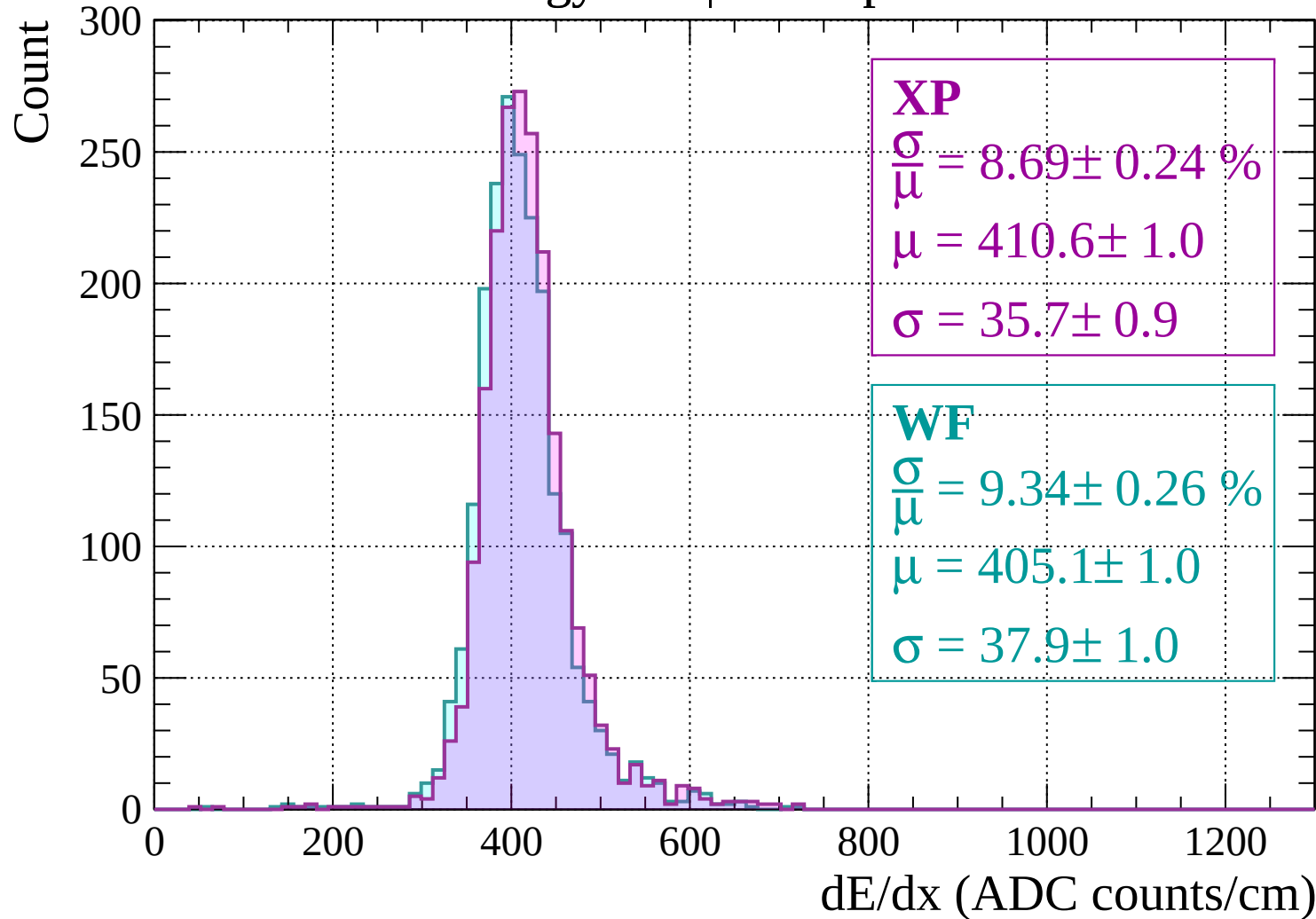


Energy loss | $480 < p < 520$

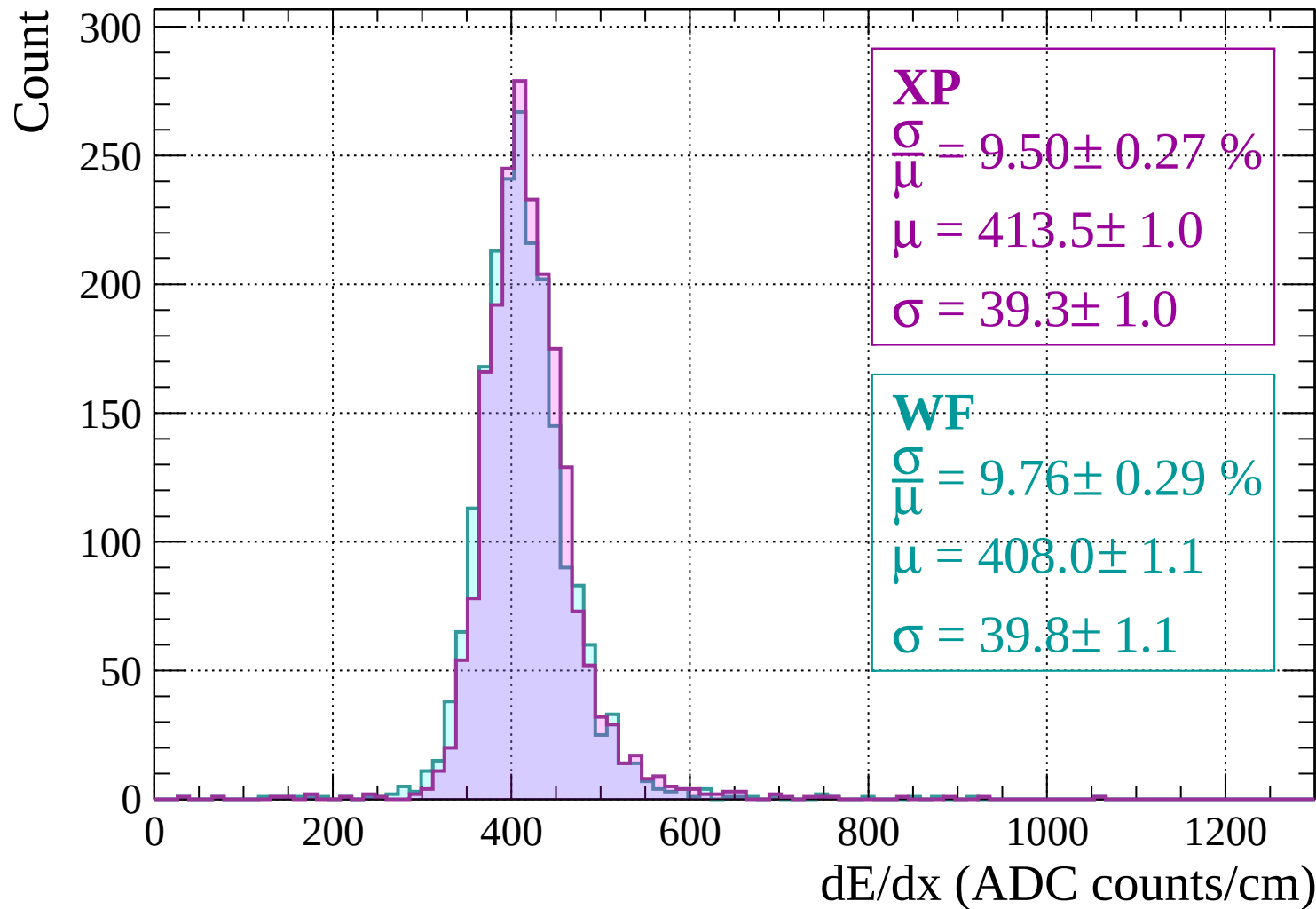
Count



Energy loss | $520 < p < 560$

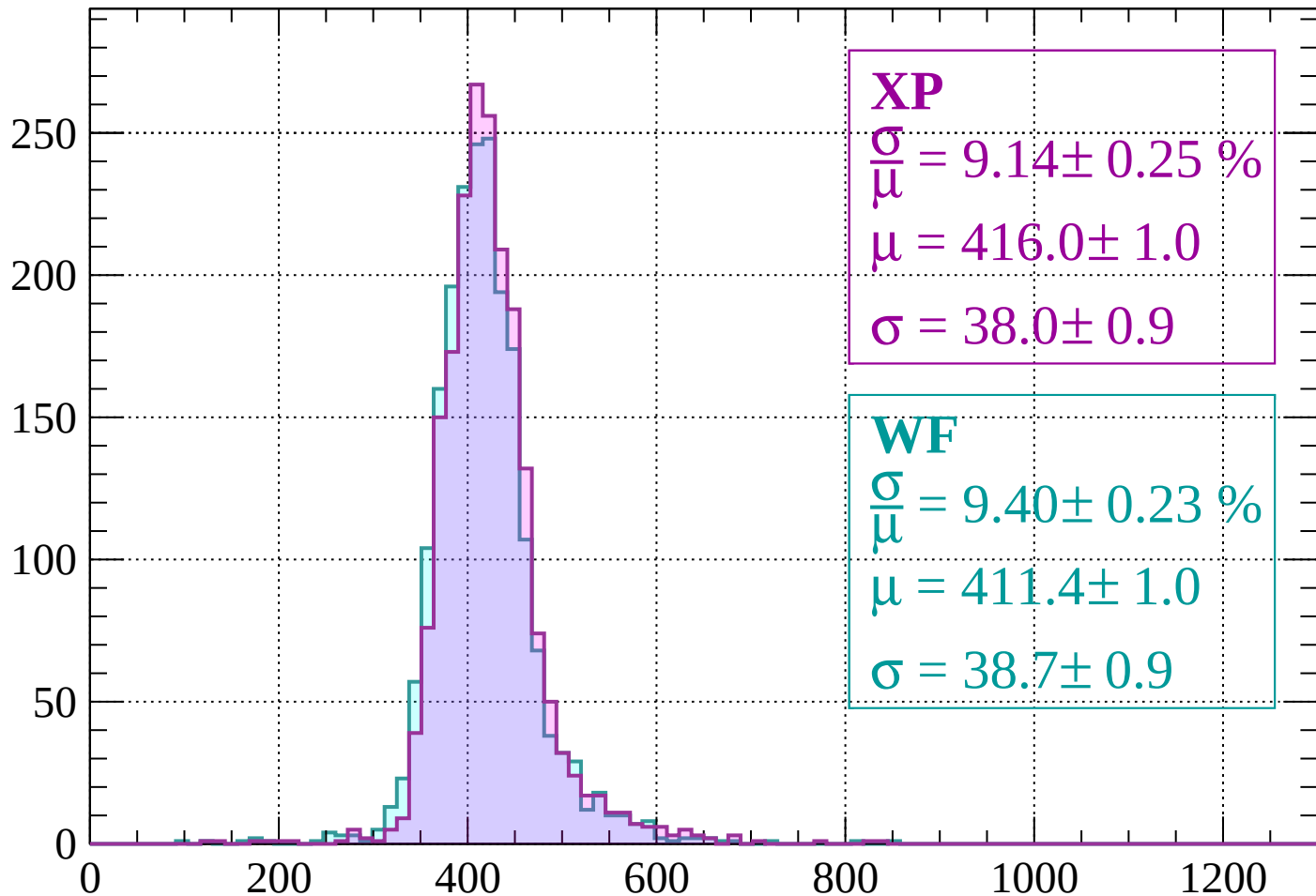


Energy loss | $560 < p < 600$



Energy loss | $600 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 9.14 \pm 0.25 \%$$

$$\mu = 416.0 \pm 1.0$$

$$\sigma = 38.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.23 \%$$

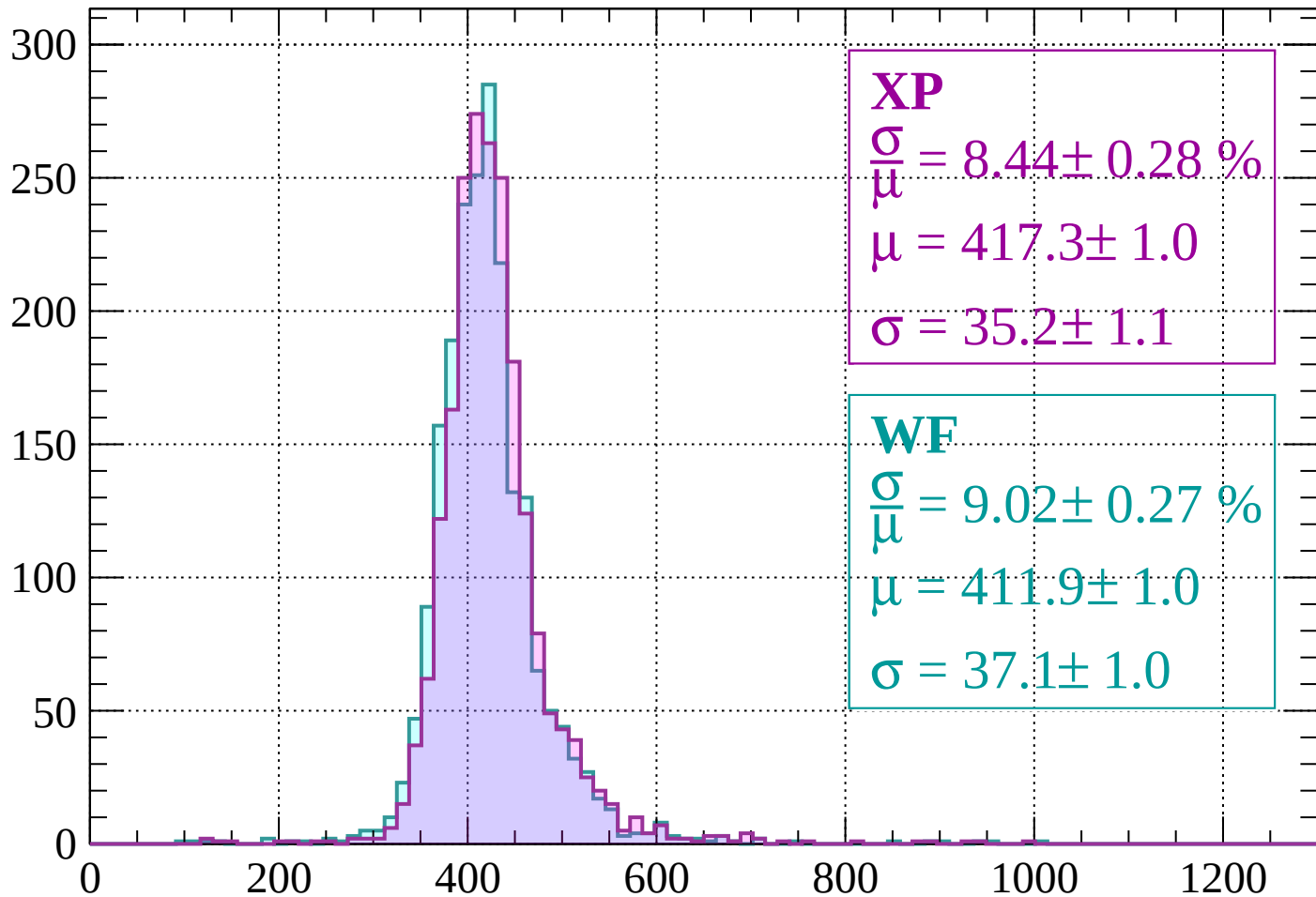
$$\mu = 411.4 \pm 1.0$$

$$\sigma = 38.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.44 \pm 0.28 \%$$

$$\mu = 417.3 \pm 1.0$$

$$\sigma = 35.2 \pm 1.1$$

WF

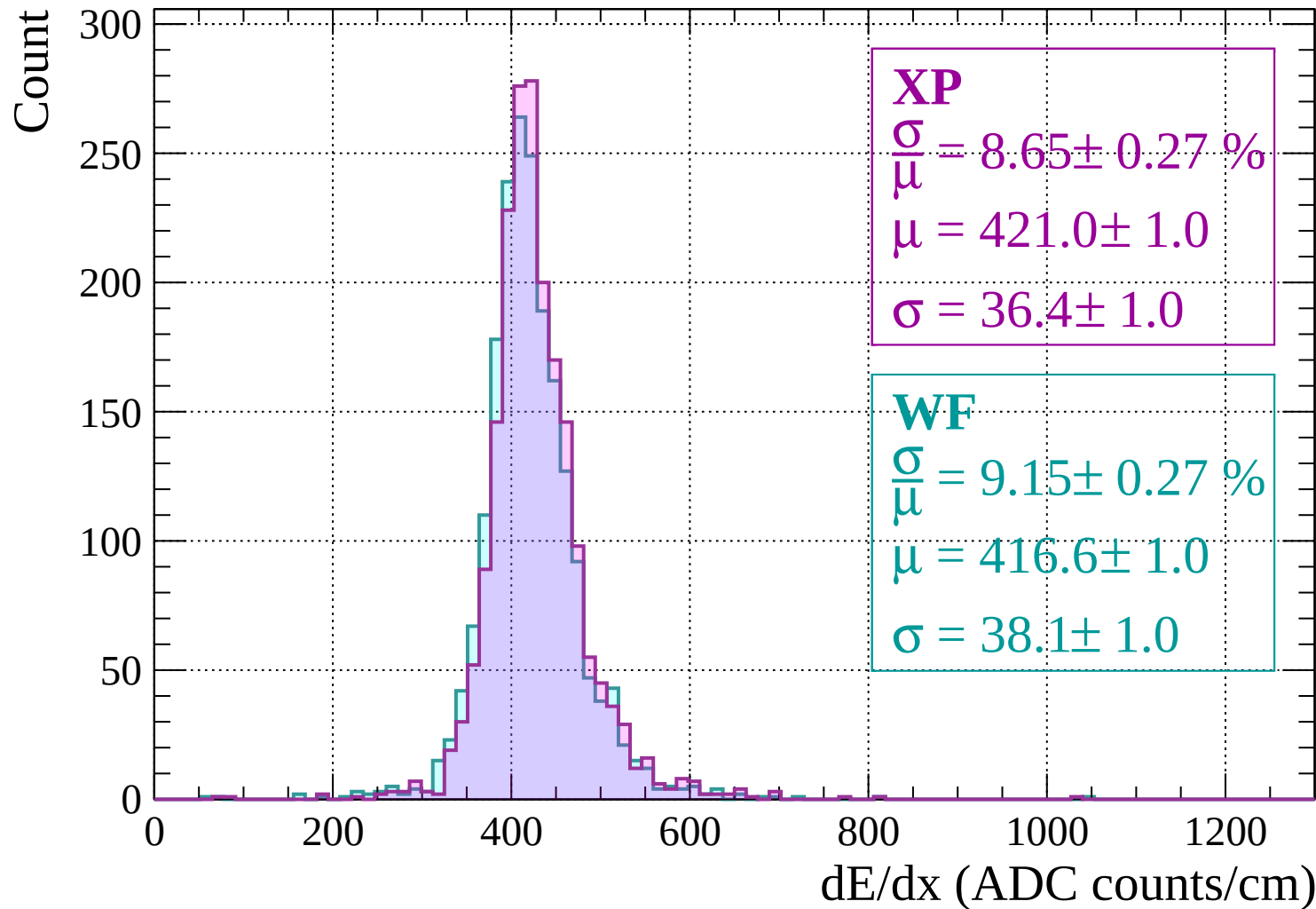
$$\frac{\sigma}{\bar{\mu}} = 9.02 \pm 0.27 \%$$

$$\mu = 411.9 \pm 1.0$$

$$\sigma = 37.1 \pm 1.0$$

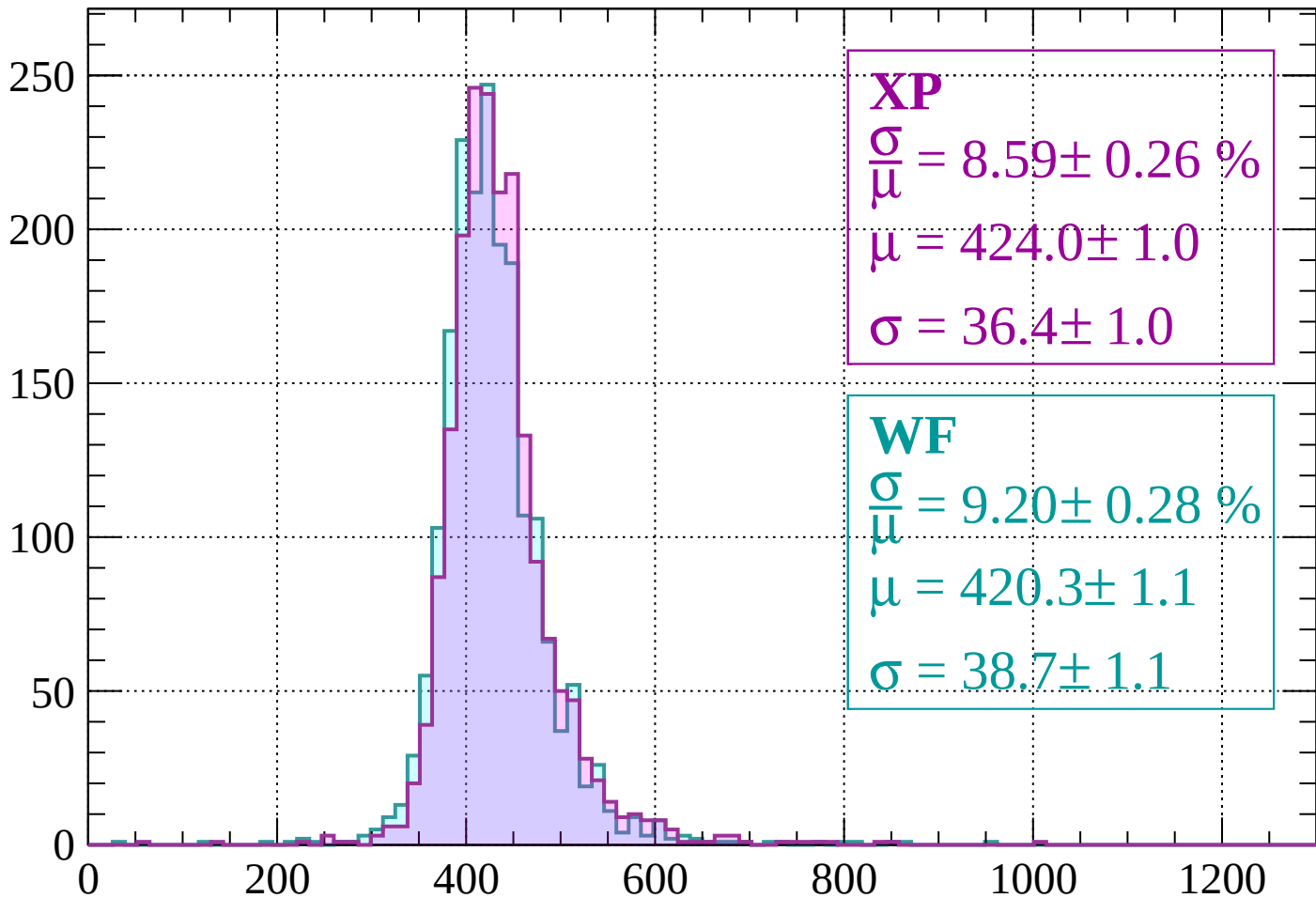
dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$



Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 8.59 \pm 0.26 \%$$

$$\mu = 424.0 \pm 1.0$$

$$\sigma = 36.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.20 \pm 0.28 \%$$

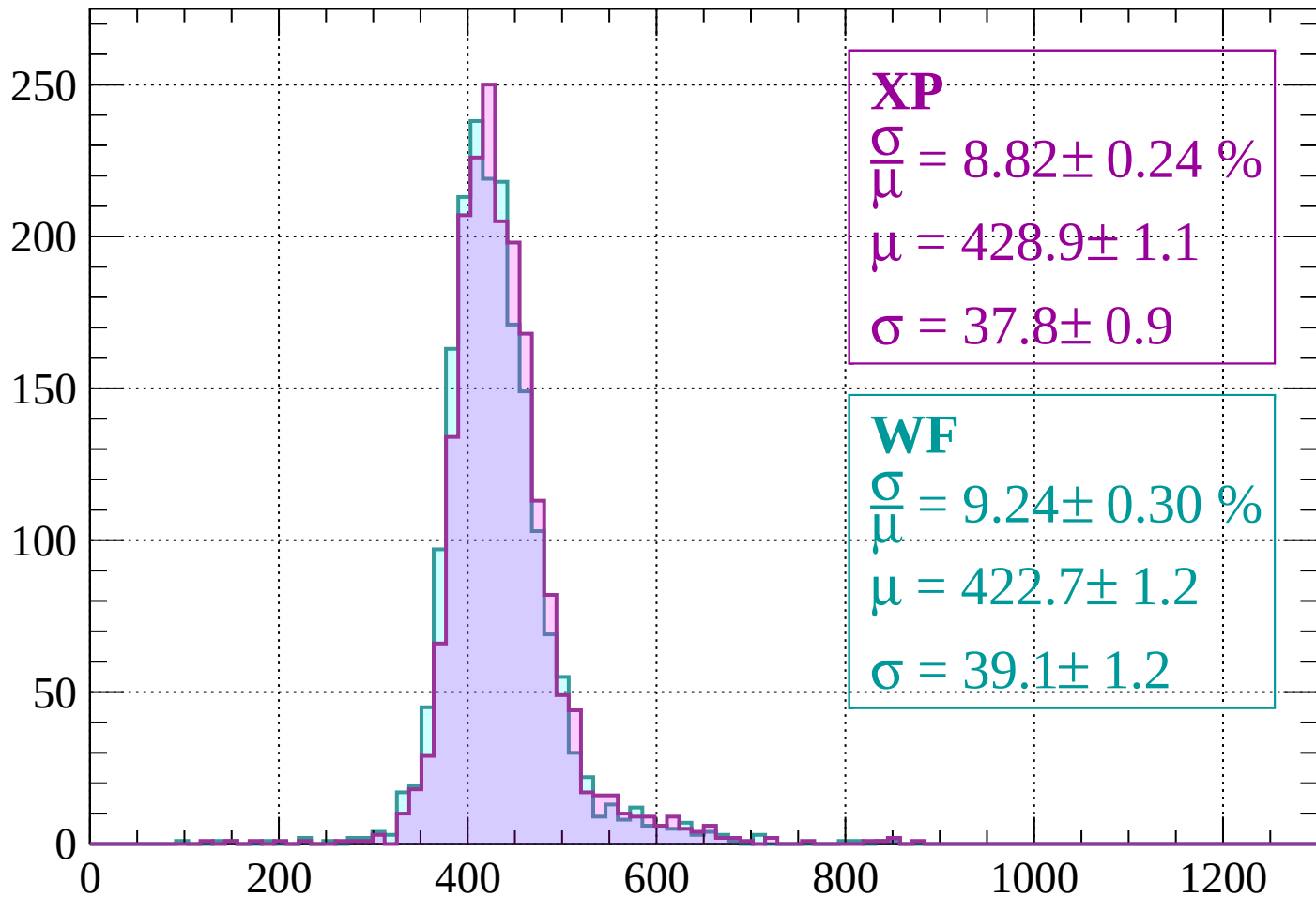
$$\mu = 420.3 \pm 1.1$$

$$\sigma = 38.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.82 \pm 0.24 \%$$

$$\mu = 428.9 \pm 1.1$$

$$\sigma = 37.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.24 \pm 0.30 \%$$

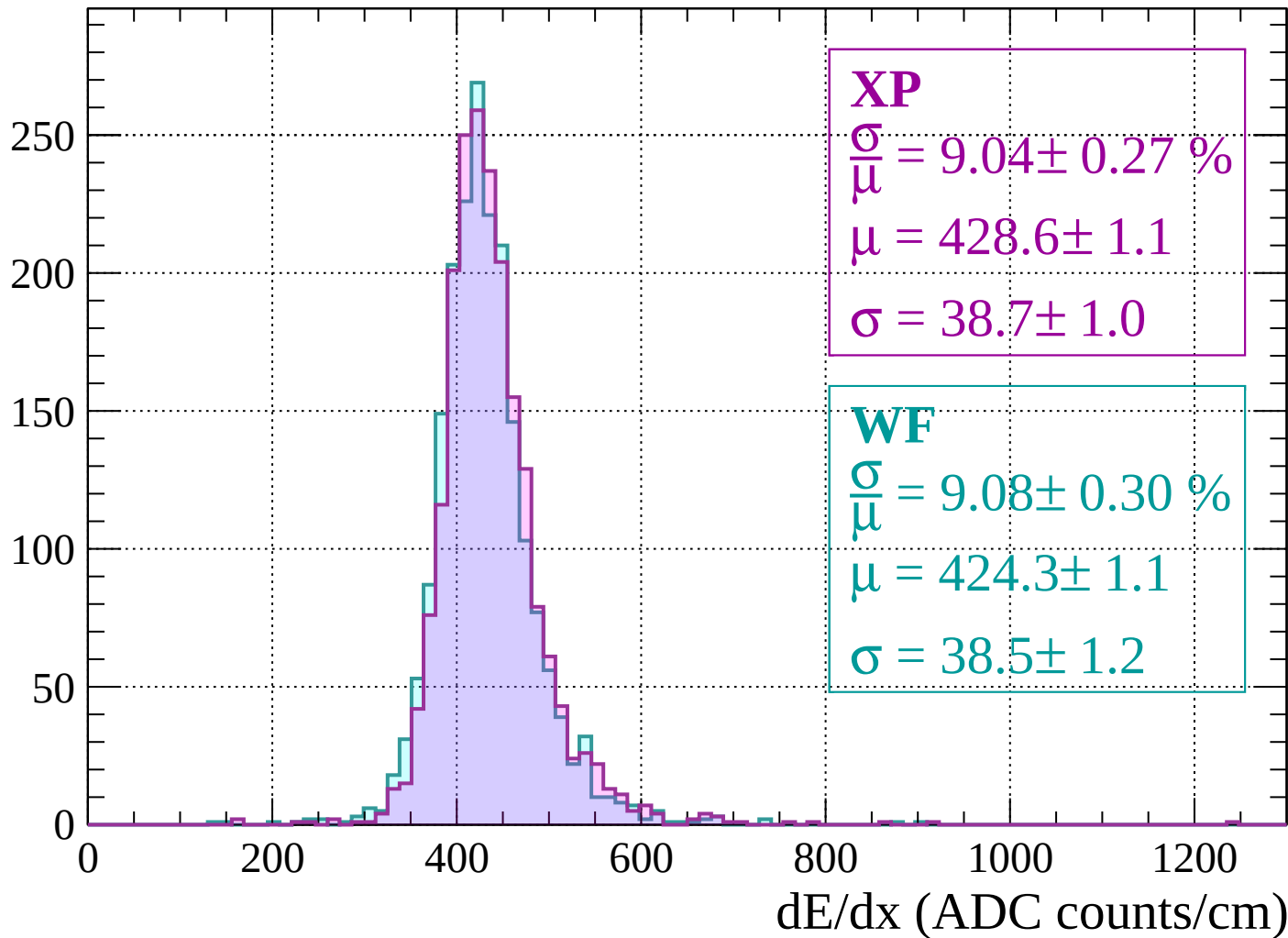
$$\mu = 422.7 \pm 1.2$$

$$\sigma = 39.1 \pm 1.2$$

dE/dx (ADC counts/cm)

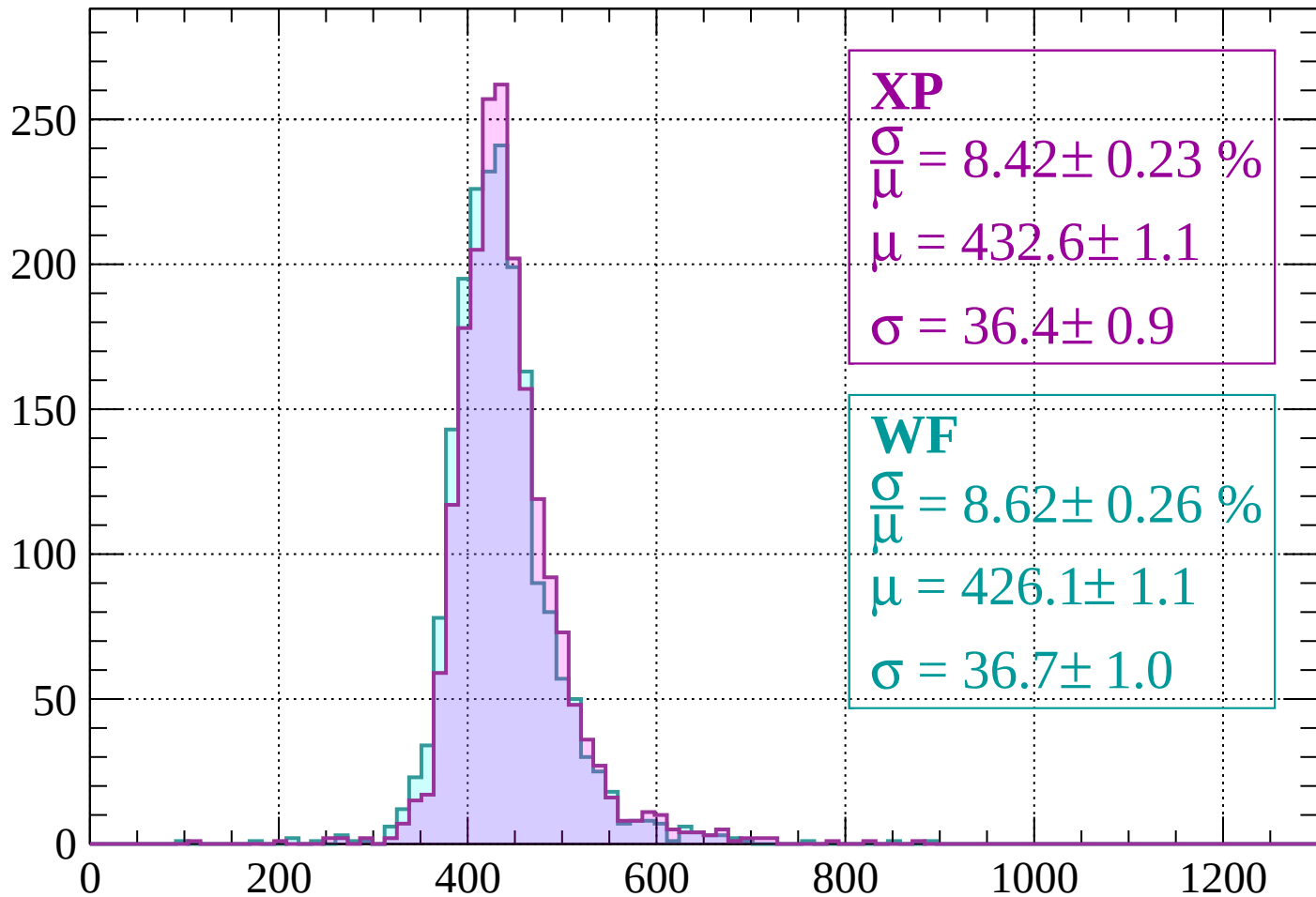
Energy loss | $800 < p < 840$

Count



Energy loss | $840 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.42 \pm 0.23 \%$$

$$\mu = 432.6 \pm 1.1$$

$$\sigma = 36.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.62 \pm 0.26 \%$$

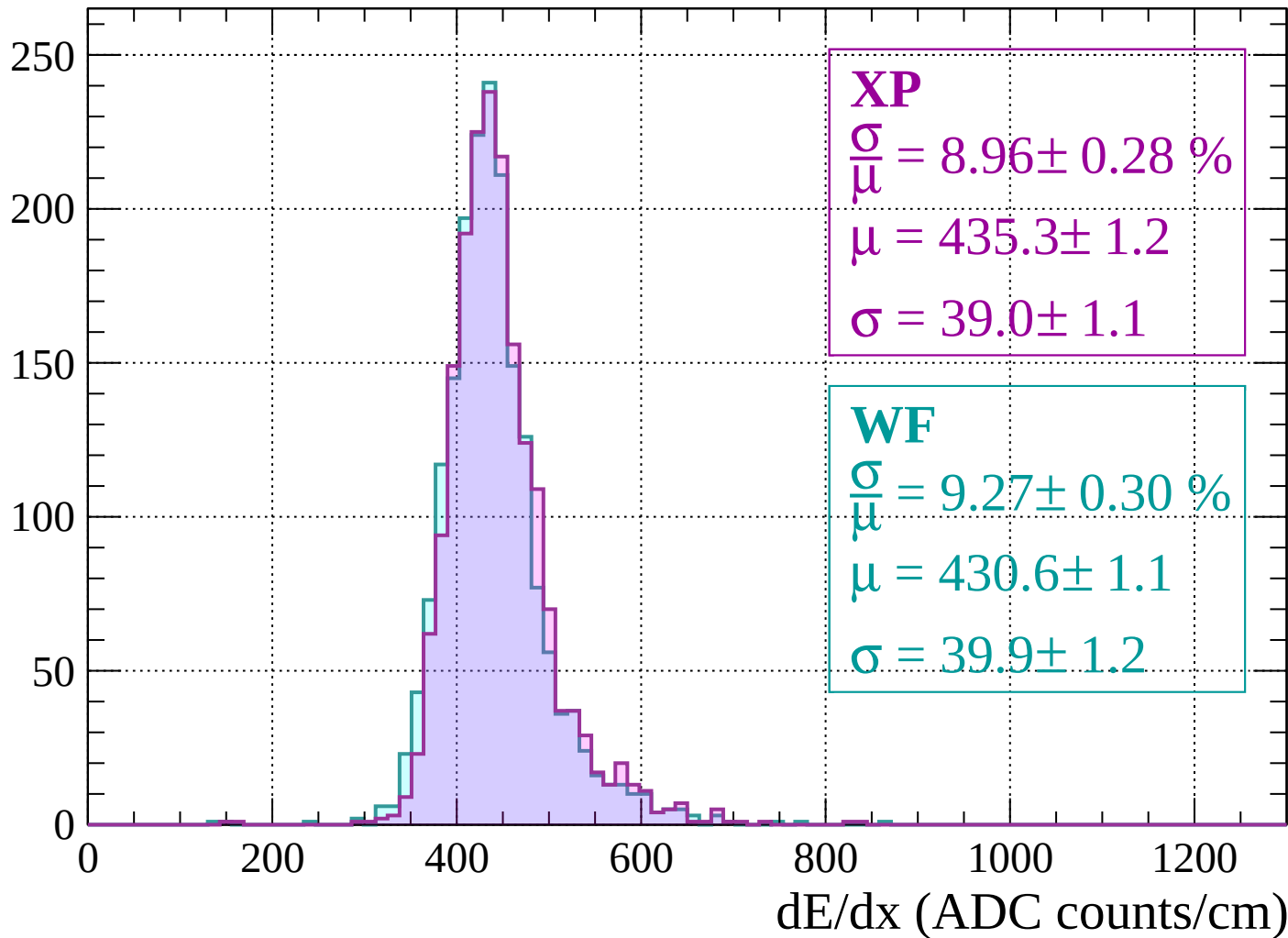
$$\mu = 426.1 \pm 1.1$$

$$\sigma = 36.7 \pm 1.0$$

dE/dx (ADC counts/cm)

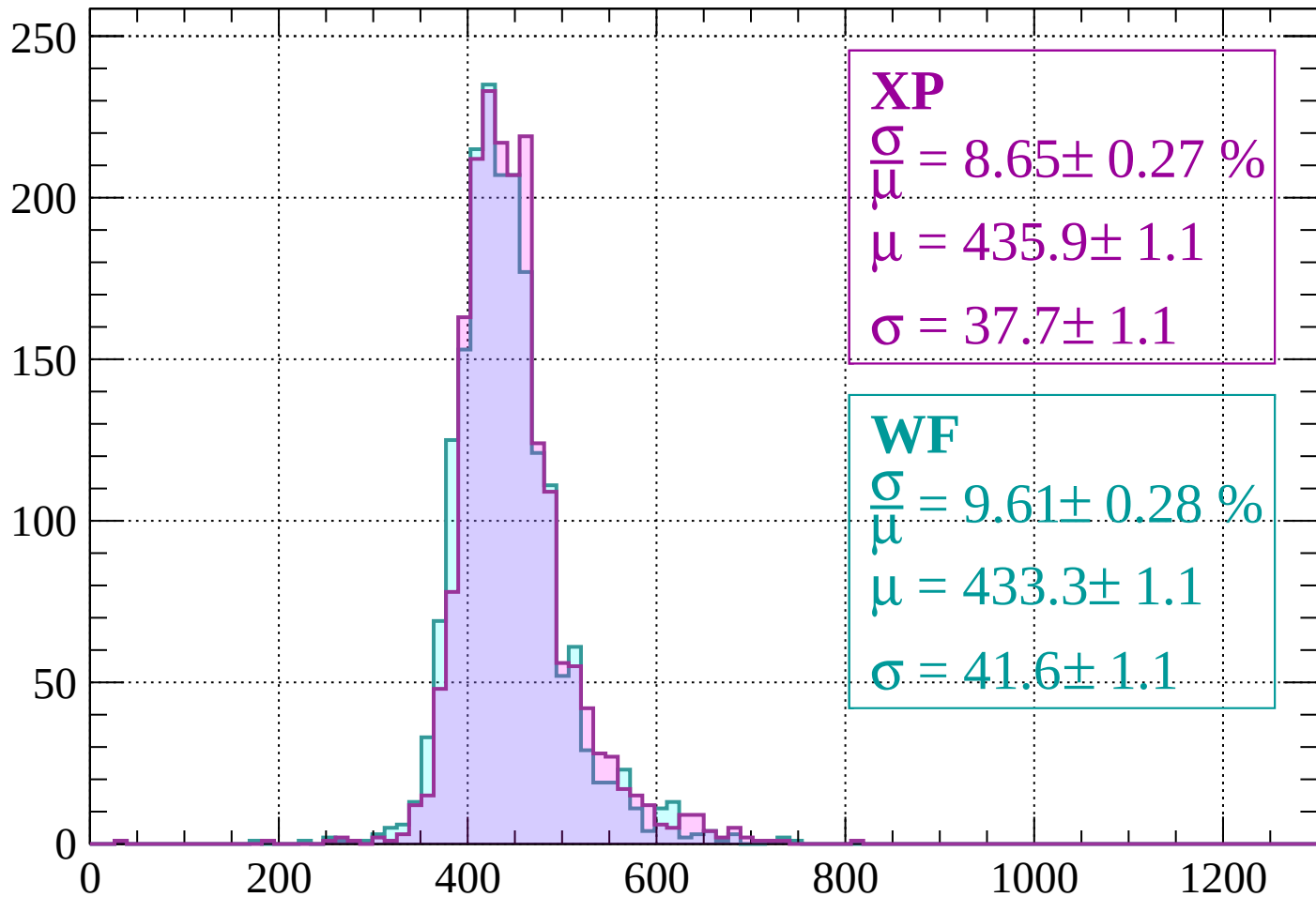
Energy loss | $880 < p < 920$

Count



Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.65 \pm 0.27 \%$$

$$\mu = 435.9 \pm 1.1$$

$$\sigma = 37.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.61 \pm 0.28 \%$$

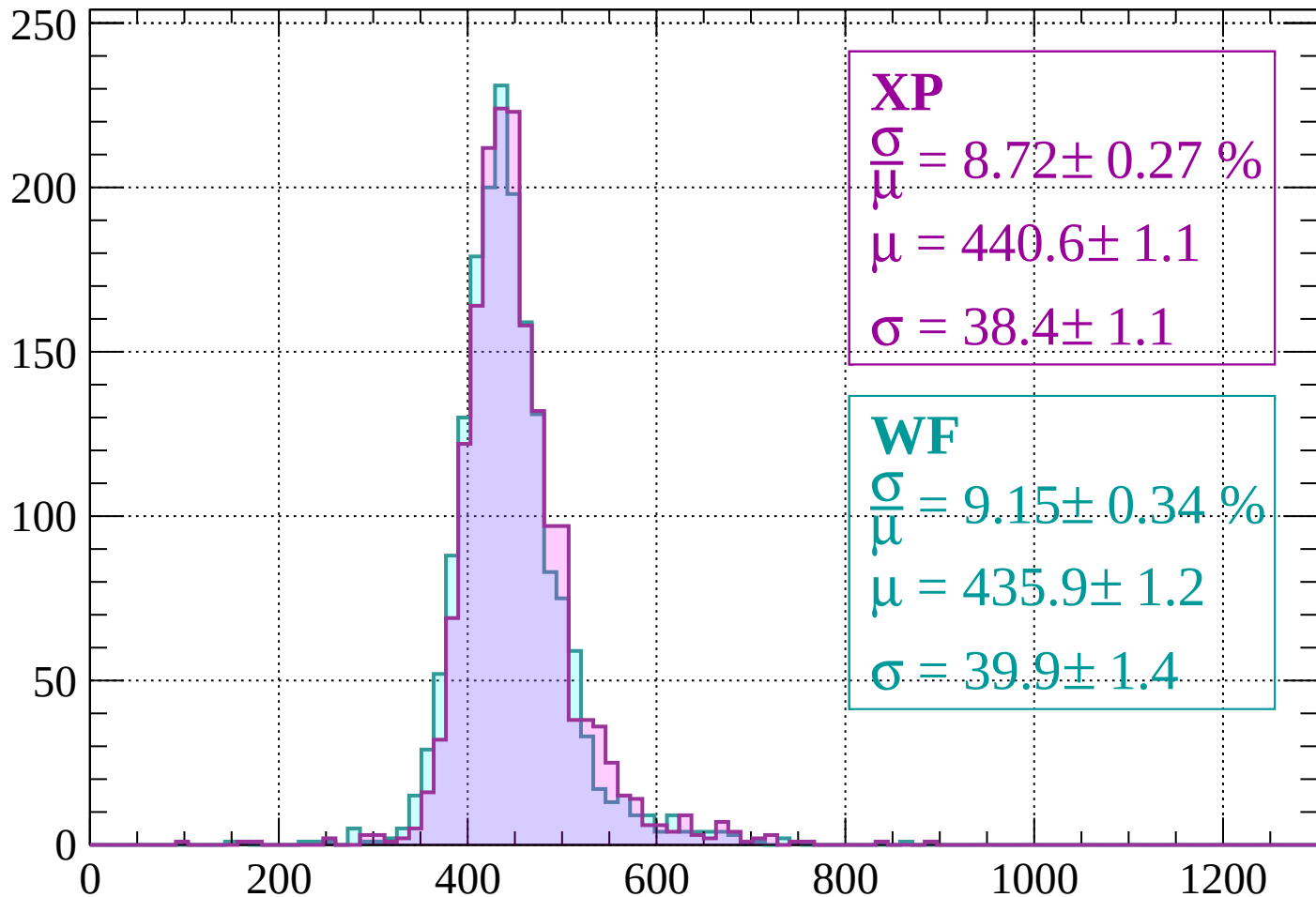
$$\mu = 433.3 \pm 1.1$$

$$\sigma = 41.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.72 \pm 0.27 \%$$

$$\mu = 440.6 \pm 1.1$$

$$\sigma = 38.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.34 \%$$

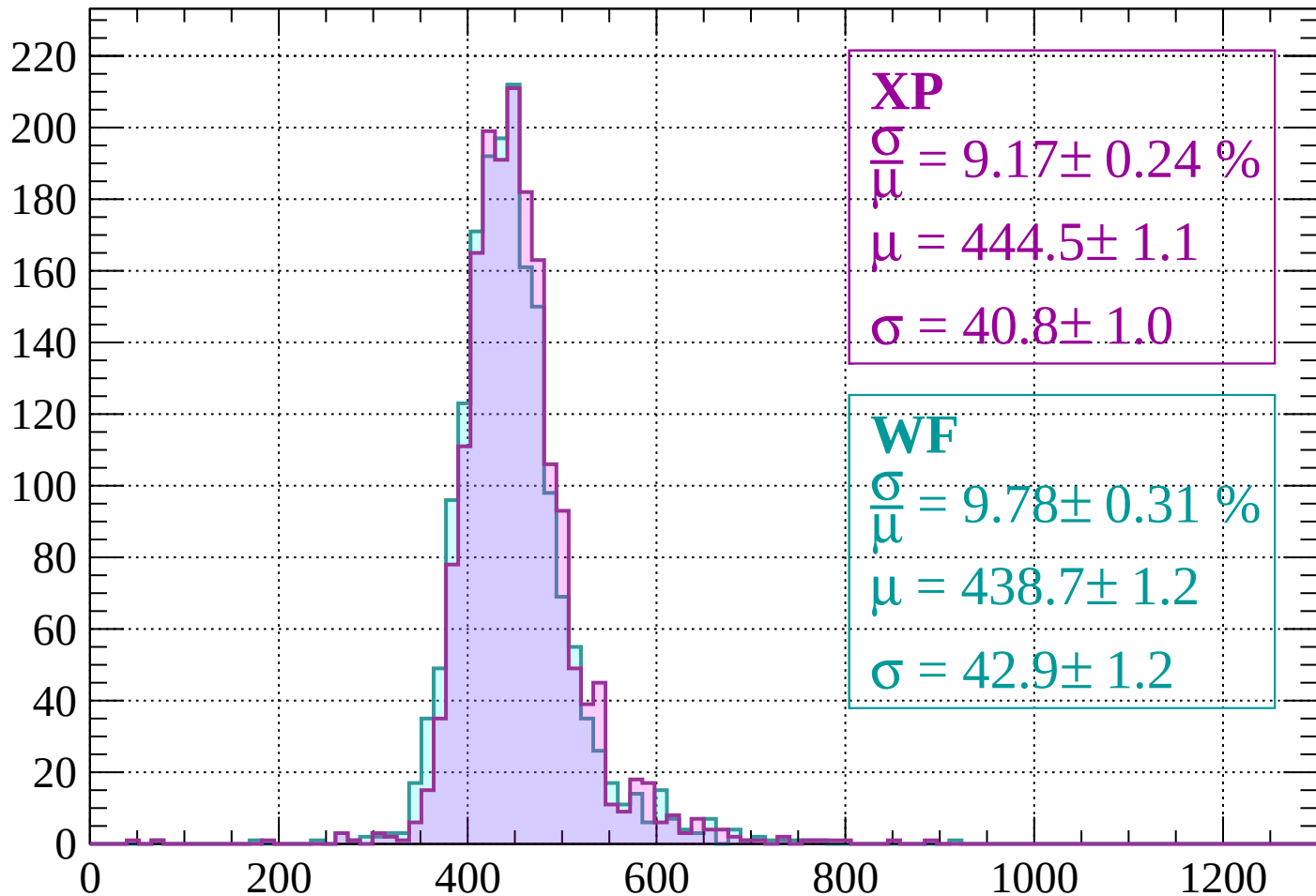
$$\mu = 435.9 \pm 1.2$$

$$\sigma = 39.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 9.17 \pm 0.24 \%$$

$$\mu = 444.5 \pm 1.1$$

$$\sigma = 40.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.78 \pm 0.31 \%$$

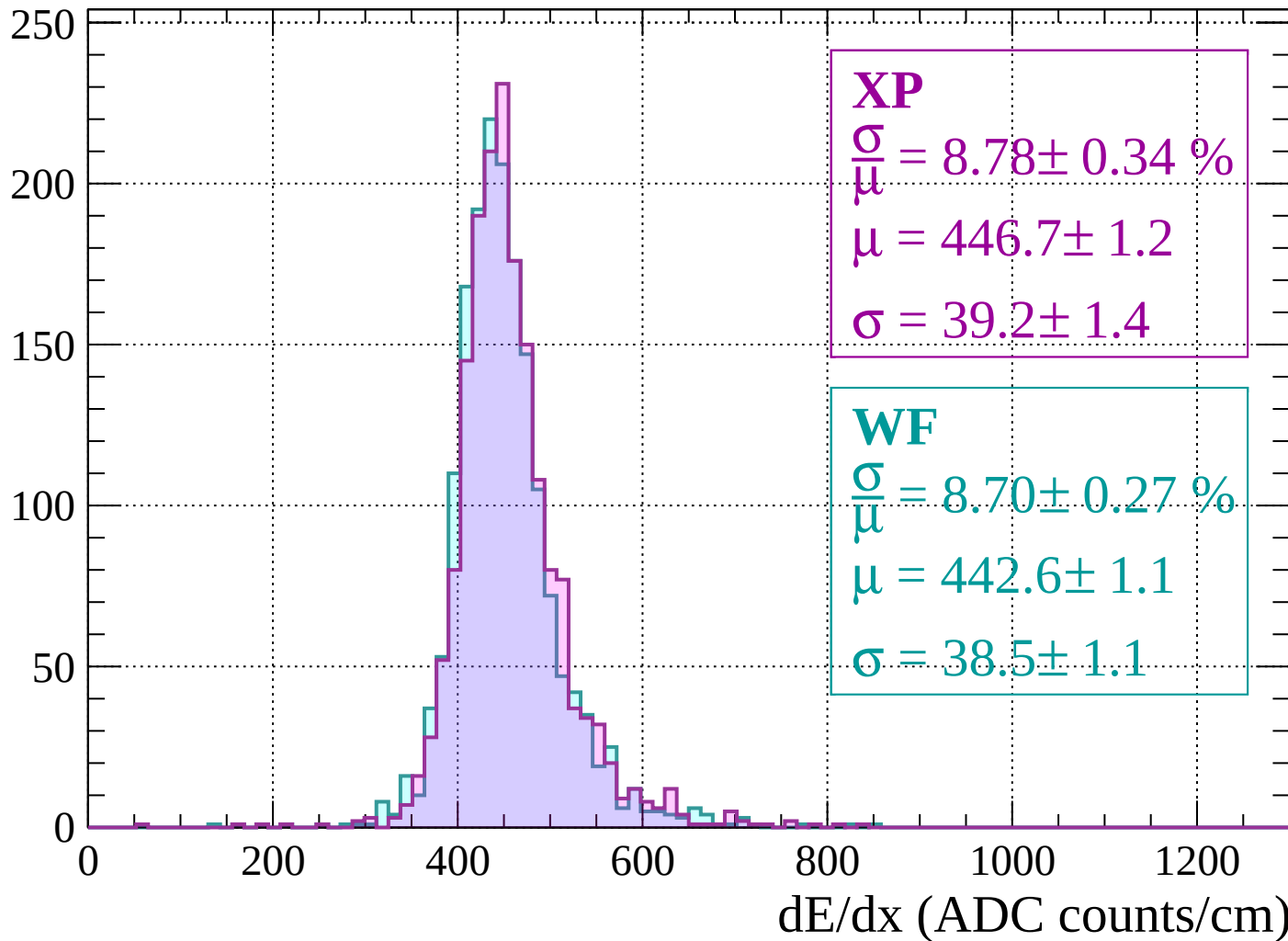
$$\mu = 438.7 \pm 1.2$$

$$\sigma = 42.9 \pm 1.2$$

dE/dx (ADC counts/cm)

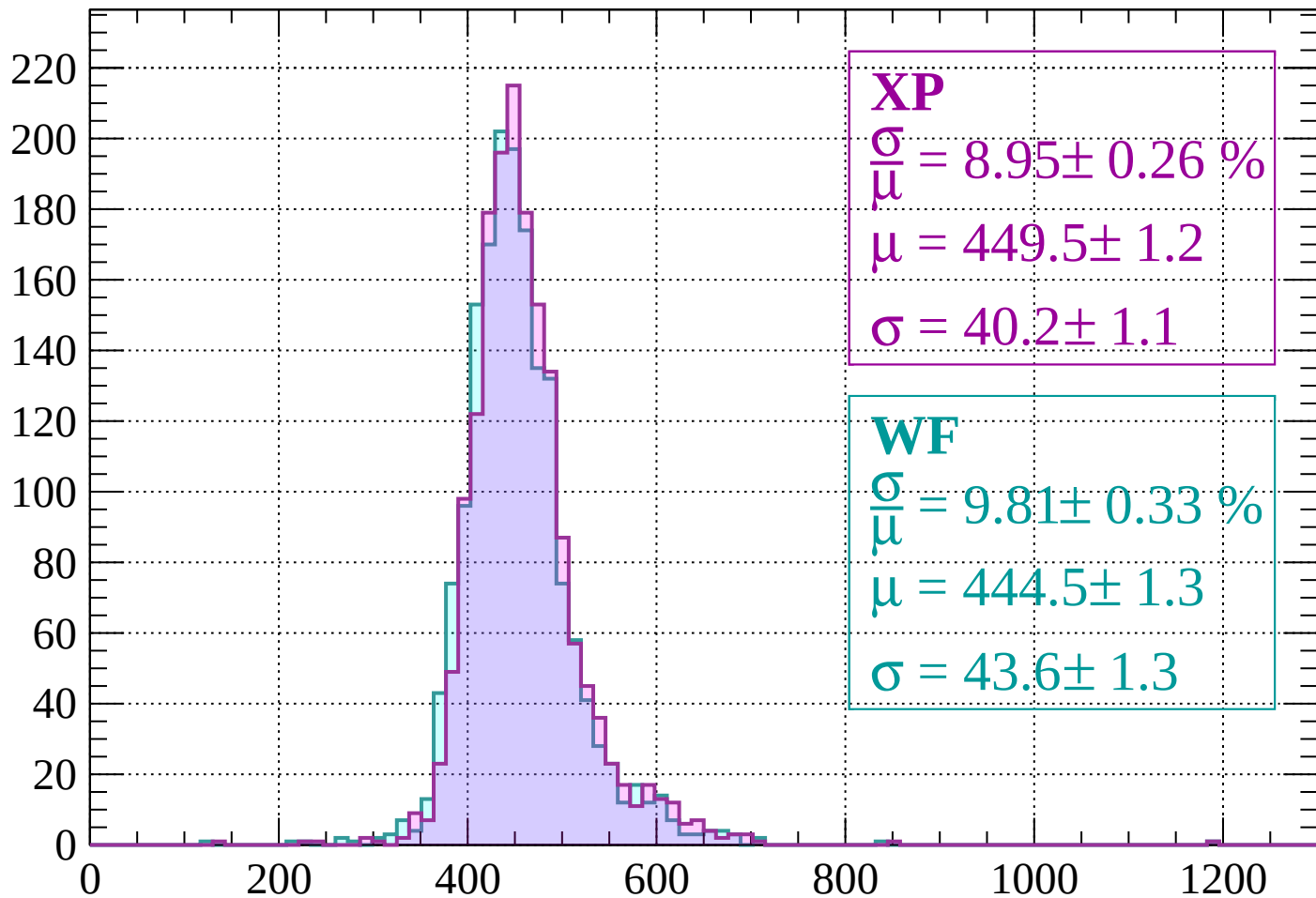
Energy loss | $1040 < p < 1080$

Count



Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.95 \pm 0.26 \%$$

$$\mu = 449.5 \pm 1.2$$

$$\sigma = 40.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.81 \pm 0.33 \%$$

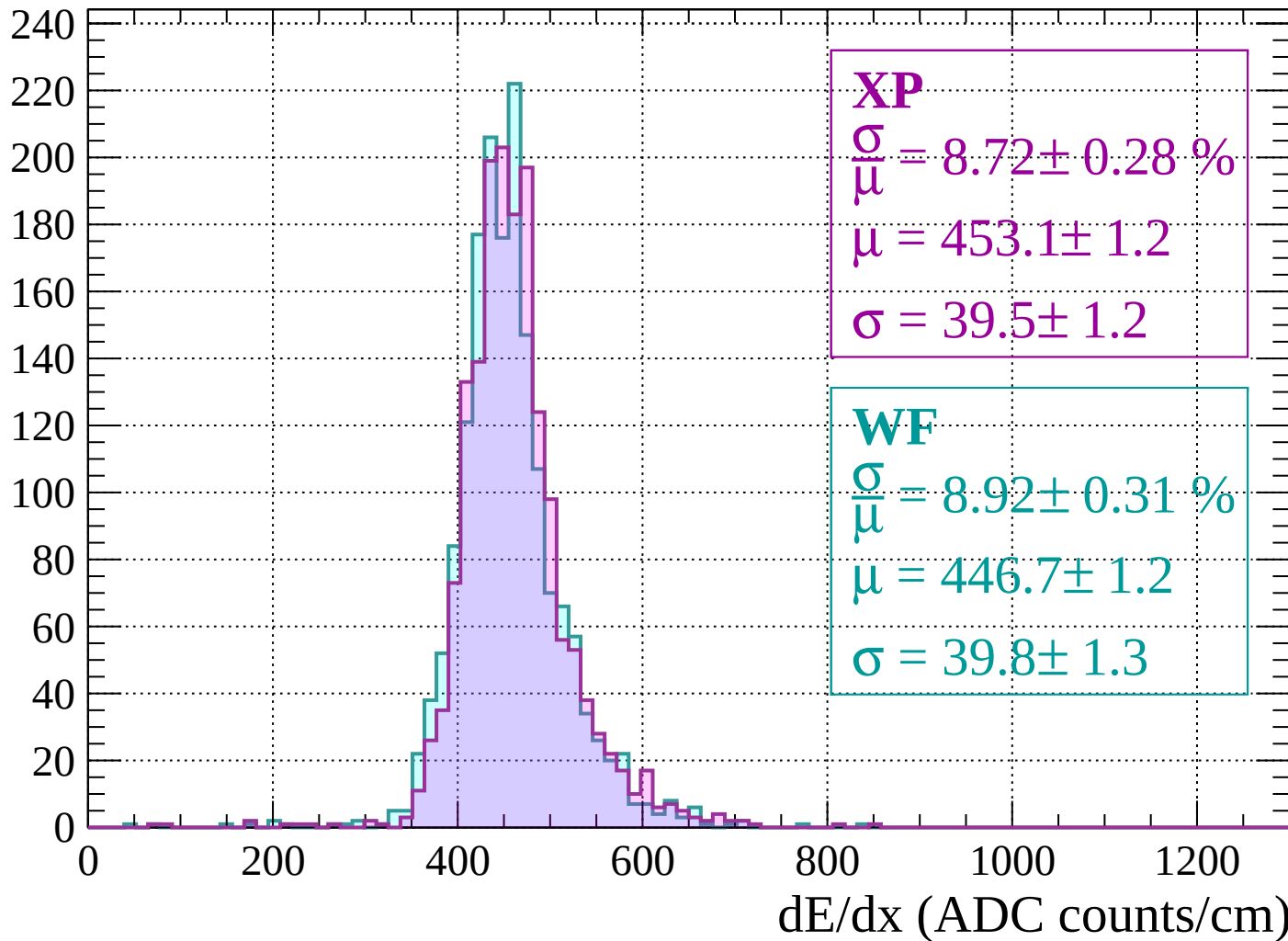
$$\mu = 444.5 \pm 1.3$$

$$\sigma = 43.6 \pm 1.3$$

dE/dx (ADC counts/cm)

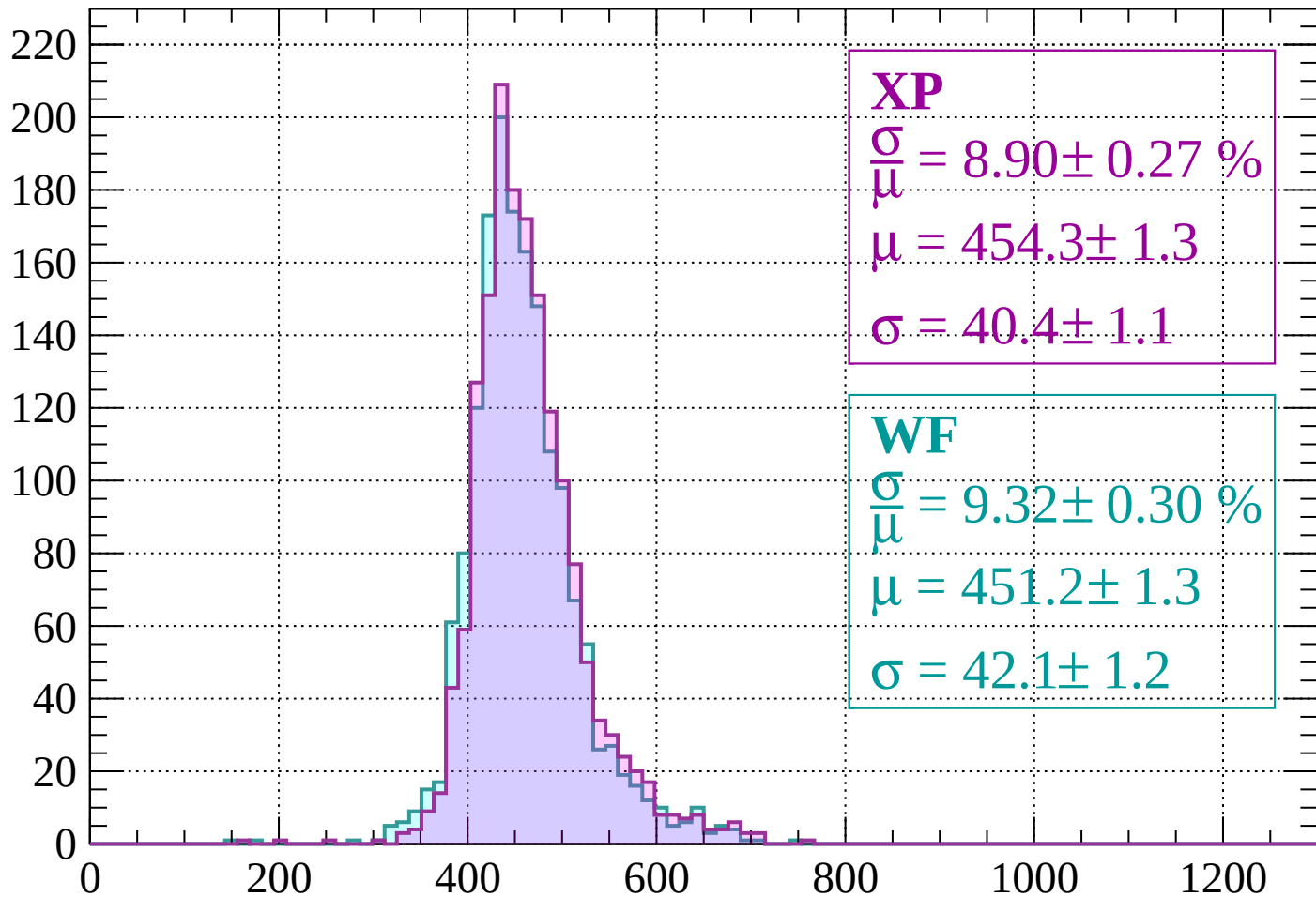
Energy loss | $1120 < p < 1160$

Count



Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 8.90 \pm 0.27 \%$$

$$\mu = 454.3 \pm 1.3$$

$$\sigma = 40.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.30 \%$$

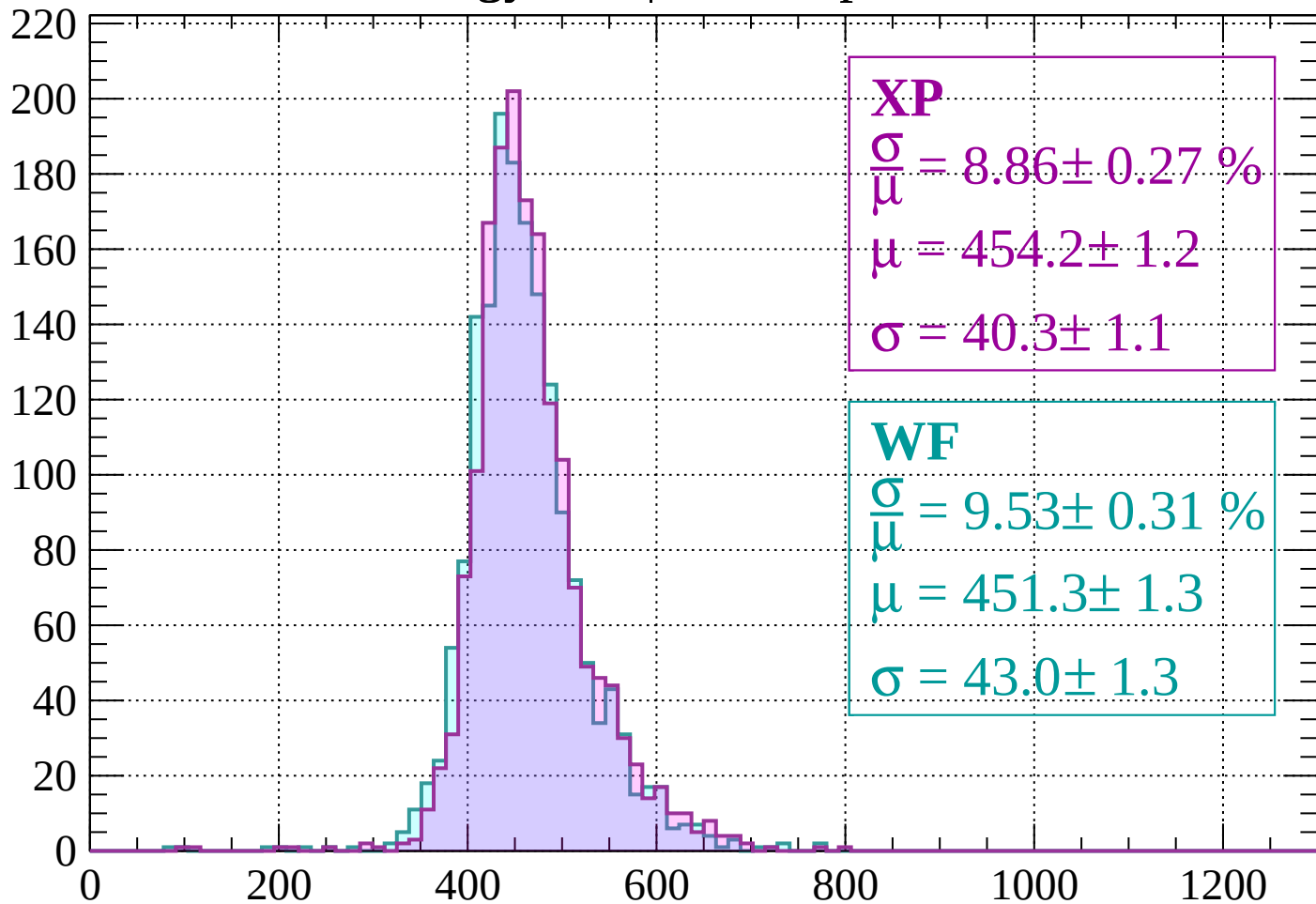
$$\mu = 451.2 \pm 1.3$$

$$\sigma = 42.1 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 8.86 \pm 0.27 \%$$

$$\mu = 454.2 \pm 1.2$$

$$\sigma = 40.3 \pm 1.1$$

WF

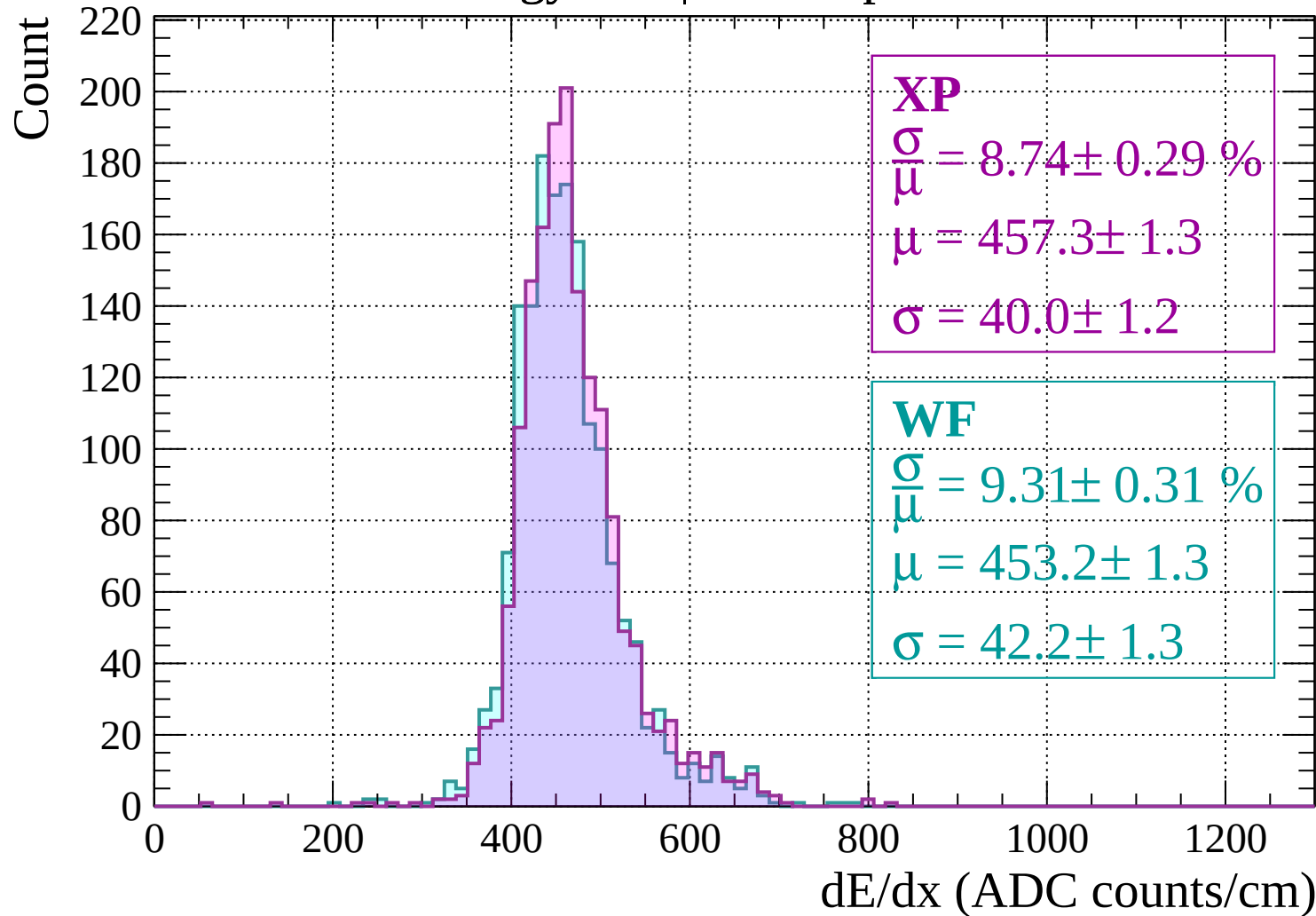
$$\frac{\sigma}{\mu} = 9.53 \pm 0.31 \%$$

$$\mu = 451.3 \pm 1.3$$

$$\sigma = 43.0 \pm 1.3$$

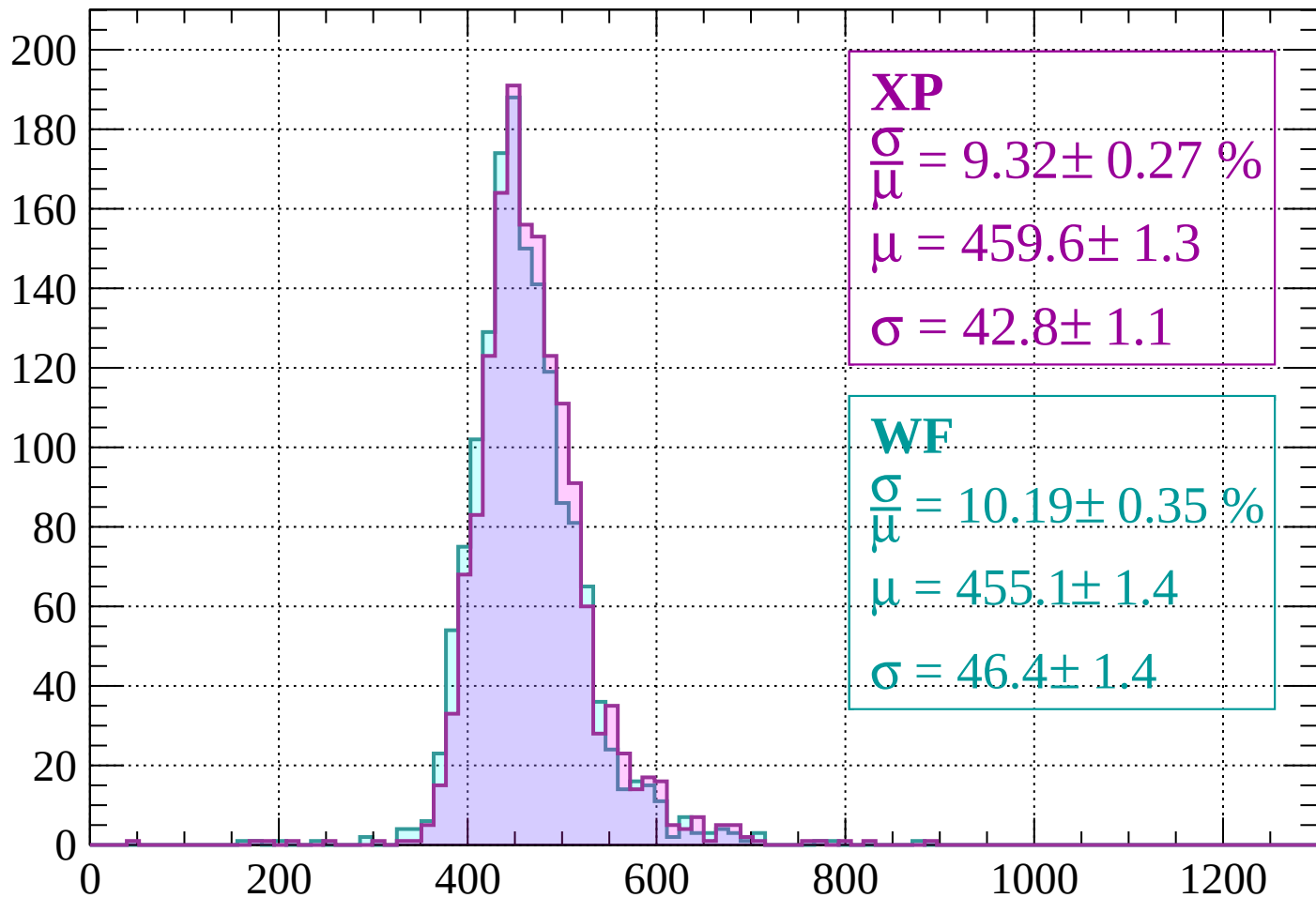
dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$



Energy loss | $1280 < p < 1320$

Count



XP

$$\frac{\sigma}{\mu} = 9.32 \pm 0.27 \%$$

$$\mu = 459.6 \pm 1.3$$

$$\sigma = 42.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.19 \pm 0.35 \%$$

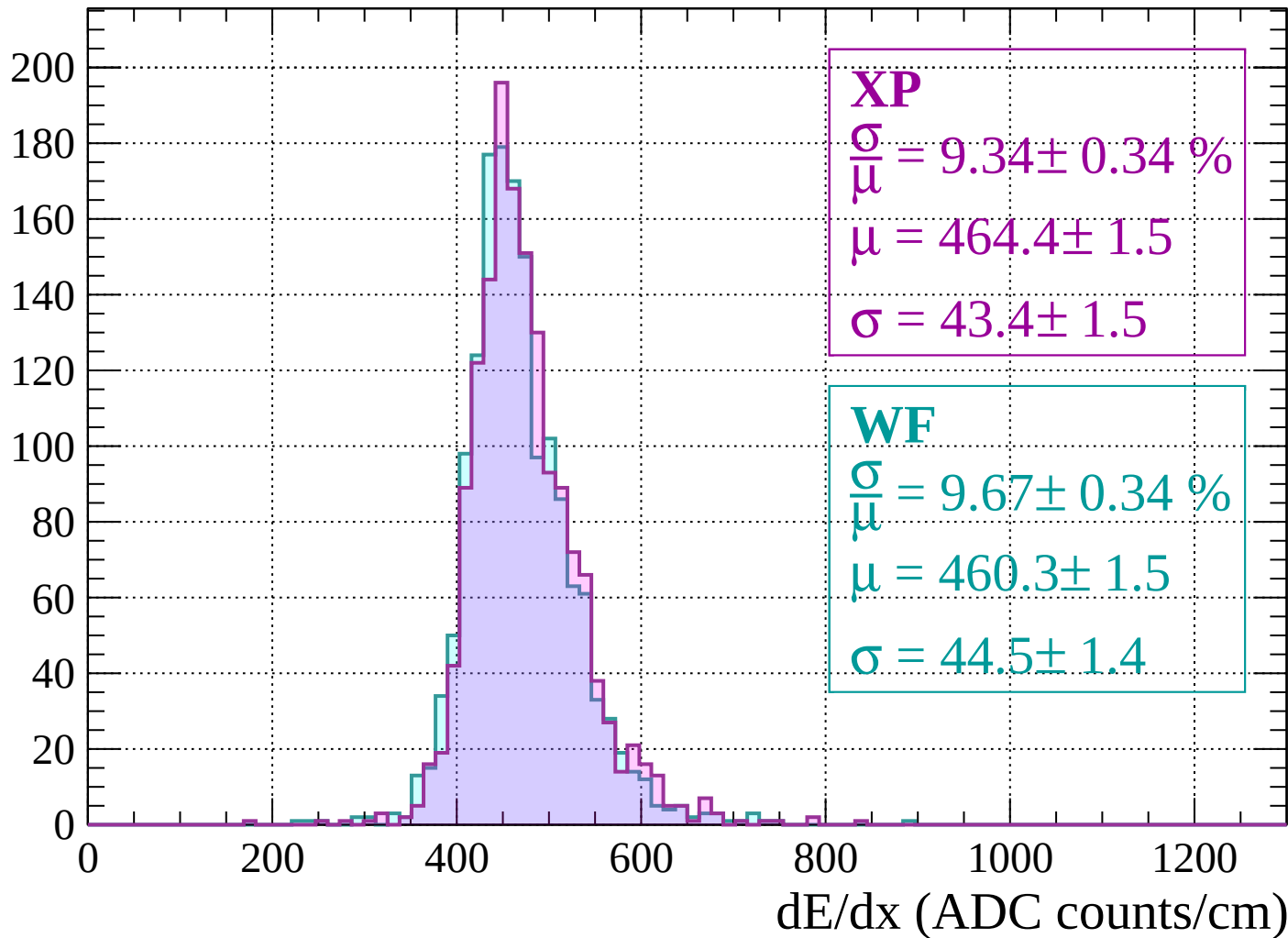
$$\mu = 455.1 \pm 1.4$$

$$\sigma = 46.4 \pm 1.4$$

dE/dx (ADC counts/cm)

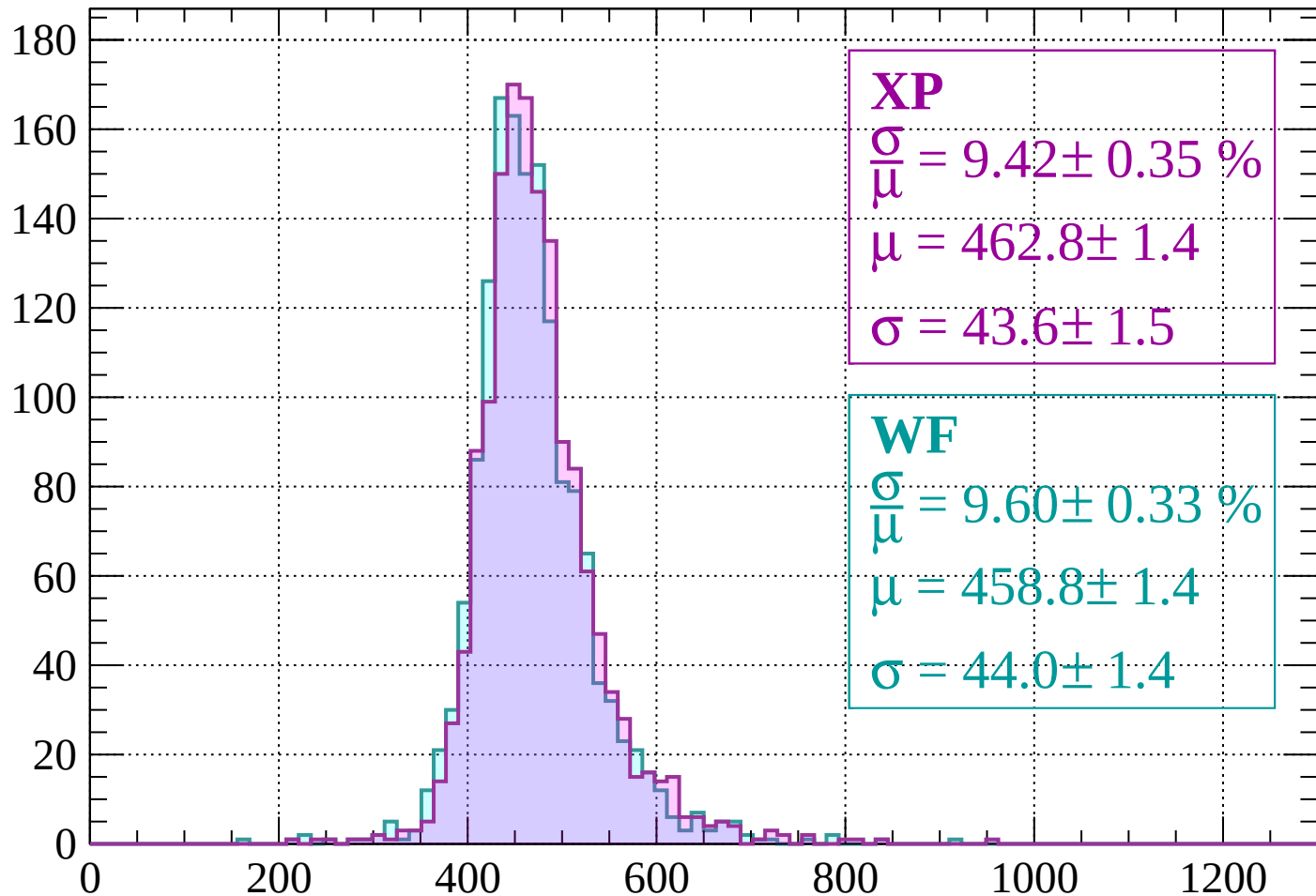
Energy loss | $1320 < p < 1360$

Count



Energy loss | $1360 < p < 1400$

Count



XP

$$\frac{\sigma}{\mu} = 9.42 \pm 0.35 \%$$

$$\mu = 462.8 \pm 1.4$$

$$\sigma = 43.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.60 \pm 0.33 \%$$

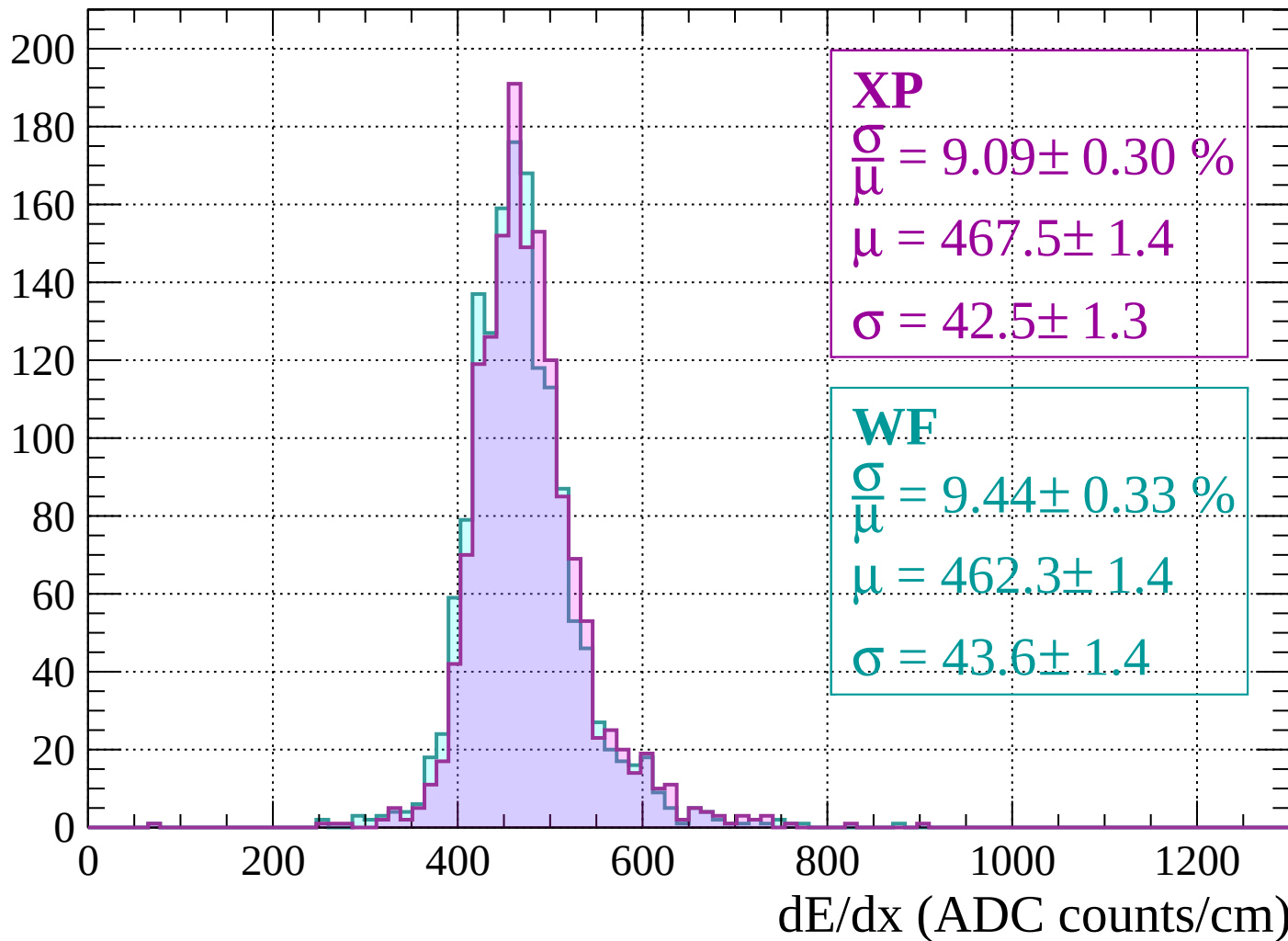
$$\mu = 458.8 \pm 1.4$$

$$\sigma = 44.0 \pm 1.4$$

dE/dx (ADC counts/cm)

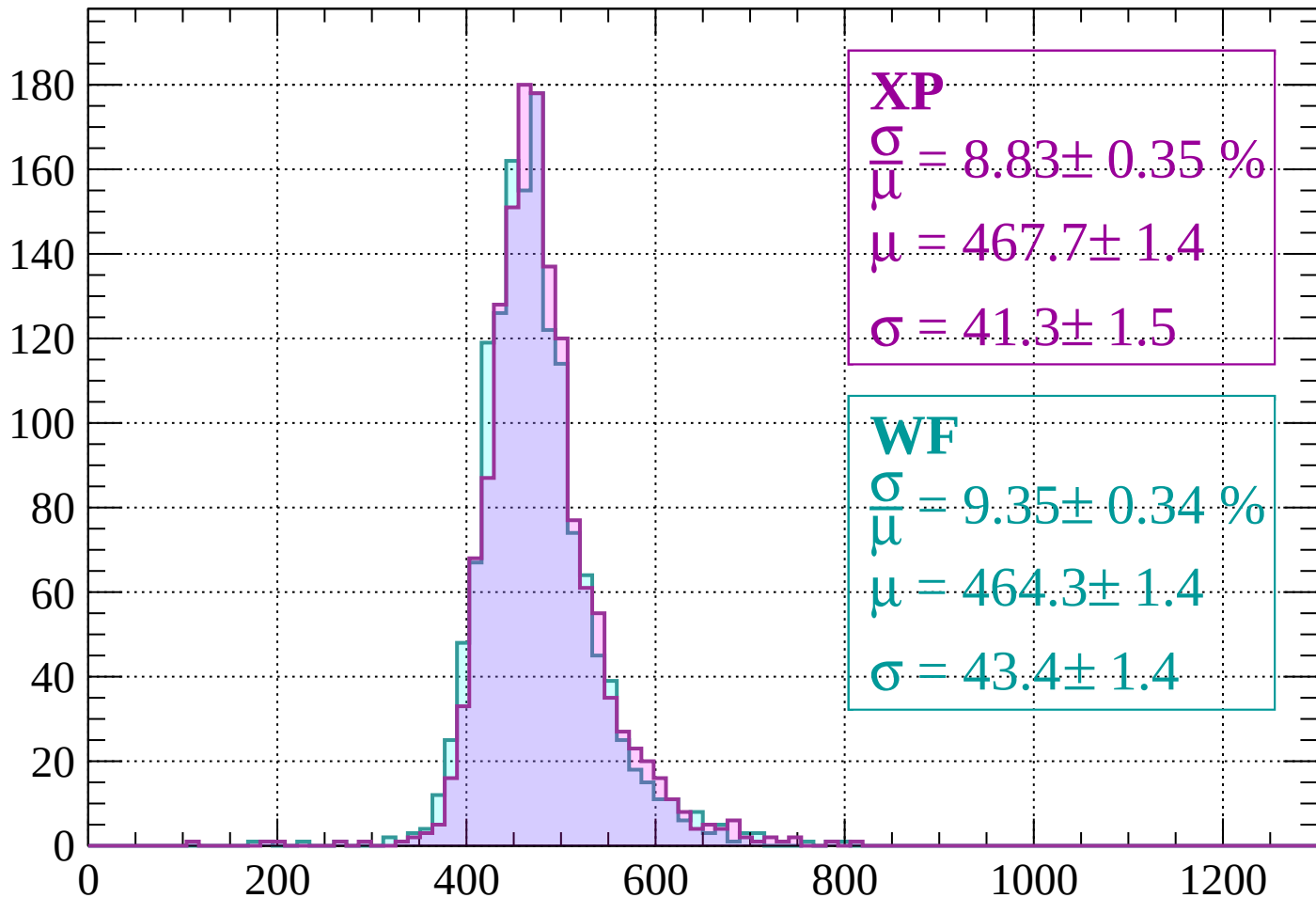
Energy loss | $1400 < p < 1440$

Count



Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.35 \%$$

$$\mu = 467.7 \pm 1.4$$

$$\sigma = 41.3 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.35 \pm 0.34 \%$$

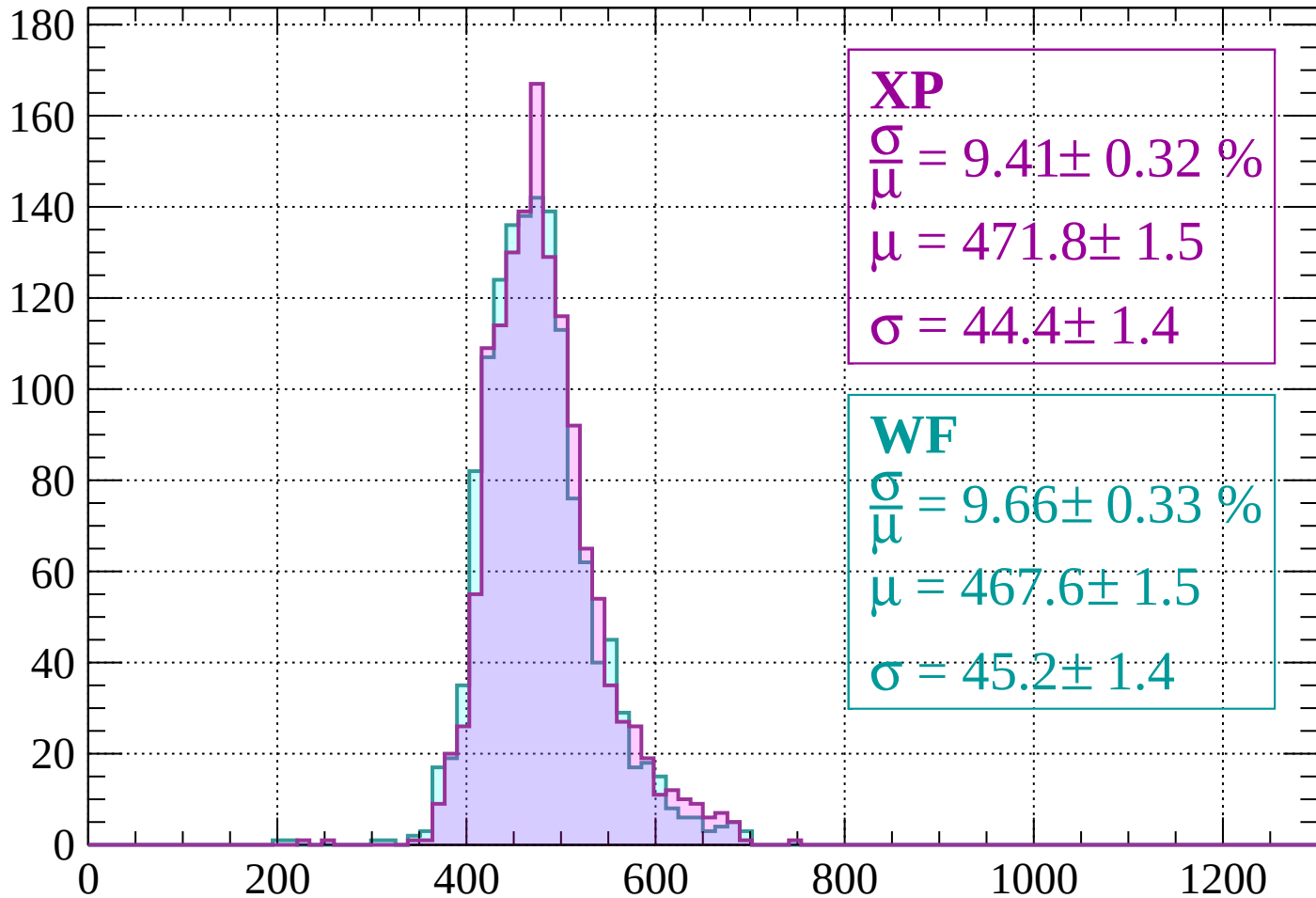
$$\mu = 464.3 \pm 1.4$$

$$\sigma = 43.4 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.32 \%$$

$$\mu = 471.8 \pm 1.5$$

$$\sigma = 44.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.33 \%$$

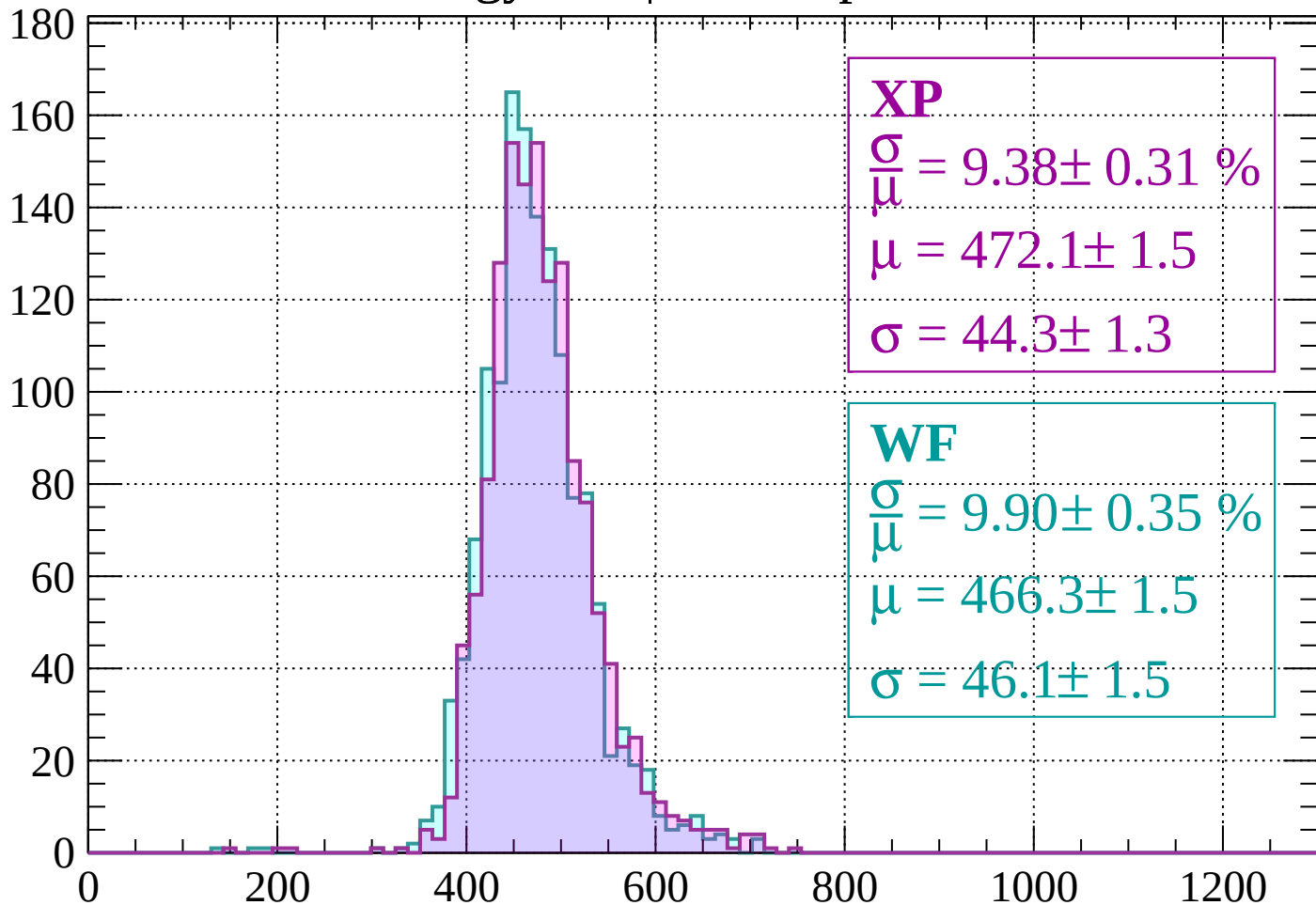
$$\mu = 467.6 \pm 1.5$$

$$\sigma = 45.2 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.38 \pm 0.31 \%$$

$$\mu = 472.1 \pm 1.5$$

$$\sigma = 44.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.90 \pm 0.35 \%$$

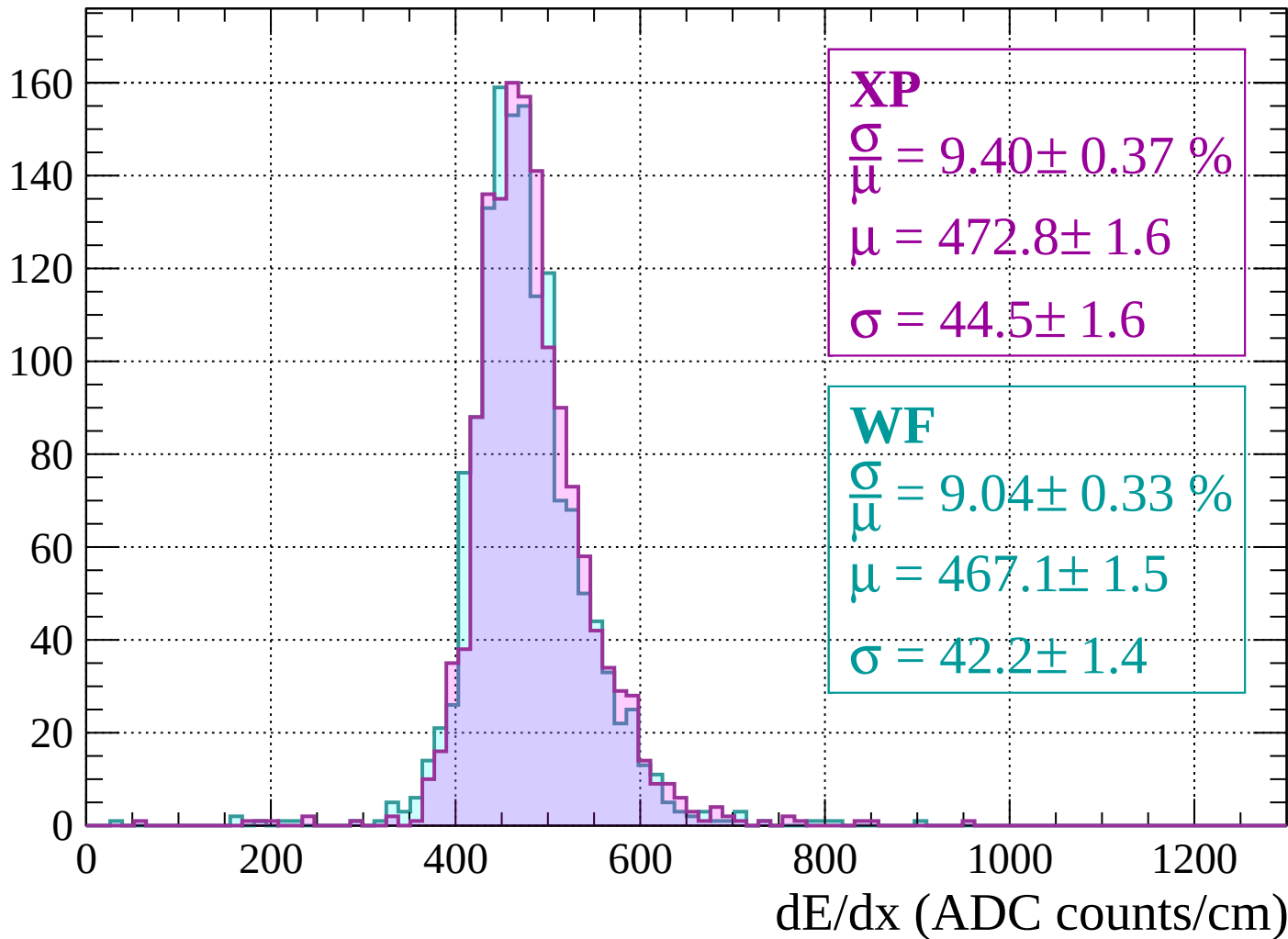
$$\mu = 466.3 \pm 1.5$$

$$\sigma = 46.1 \pm 1.5$$

dE/dx (ADC counts/cm)

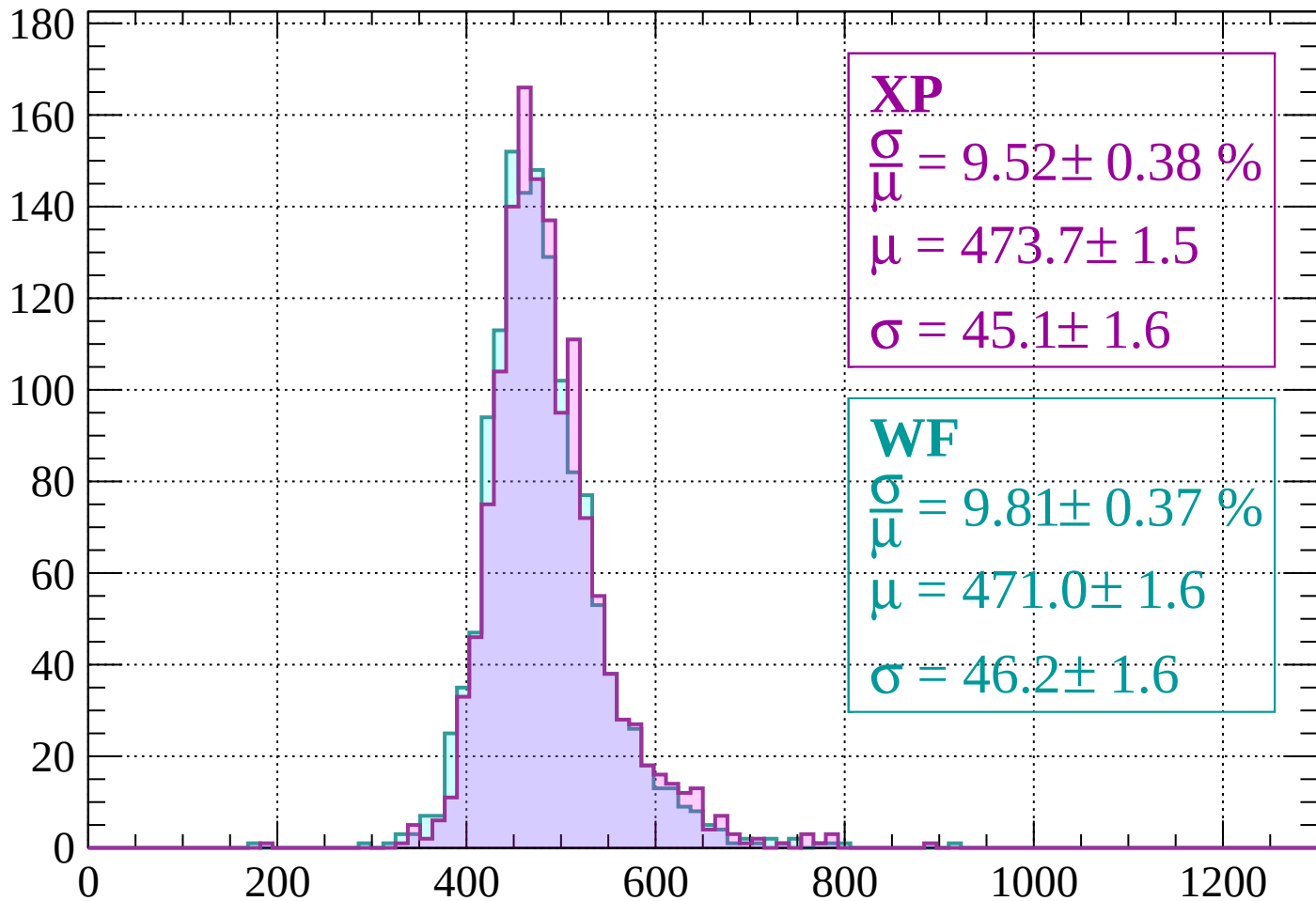
Energy loss | $1560 < p < 1600$

Count

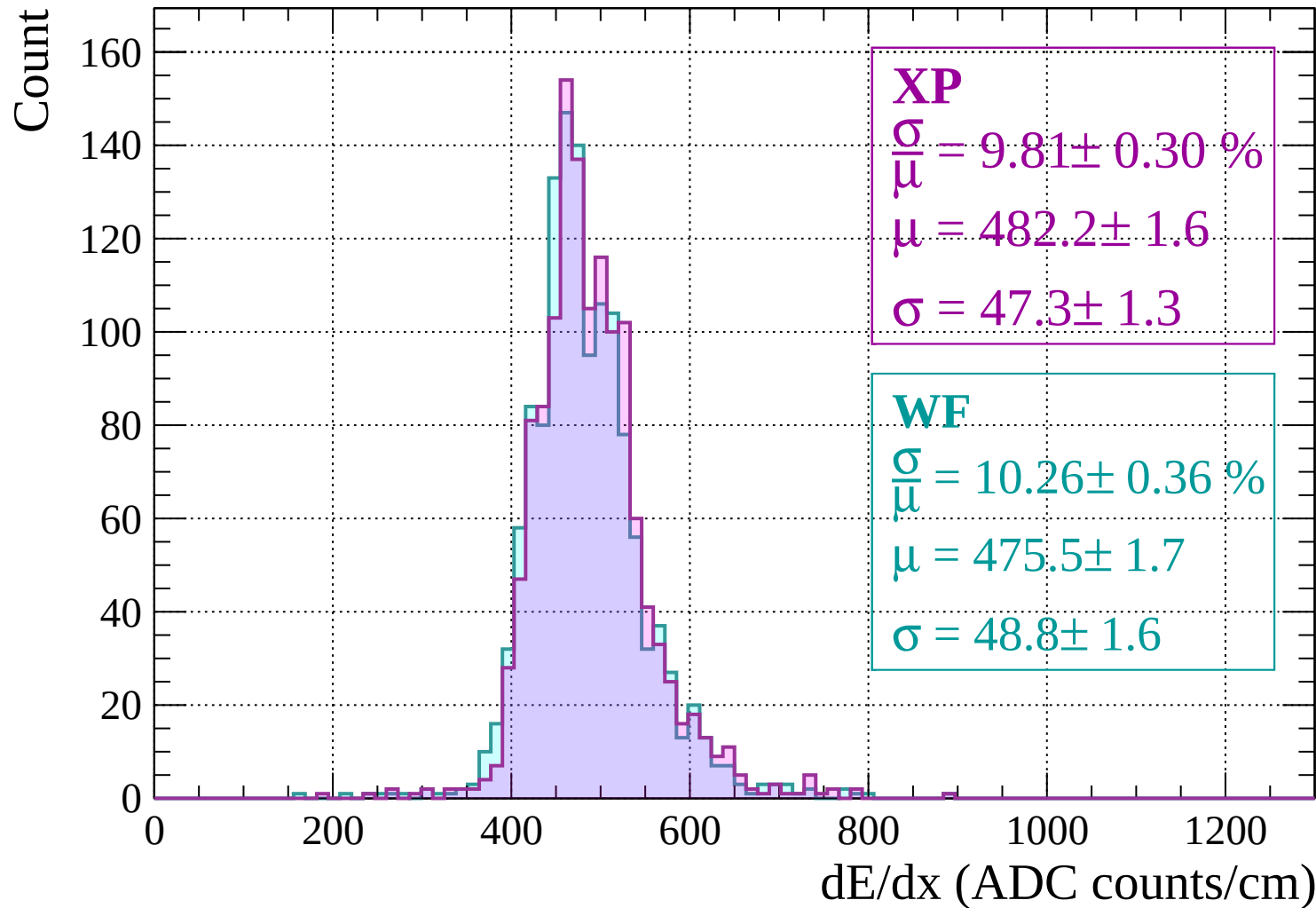


Energy loss | $1600 < p < 1640$

Count

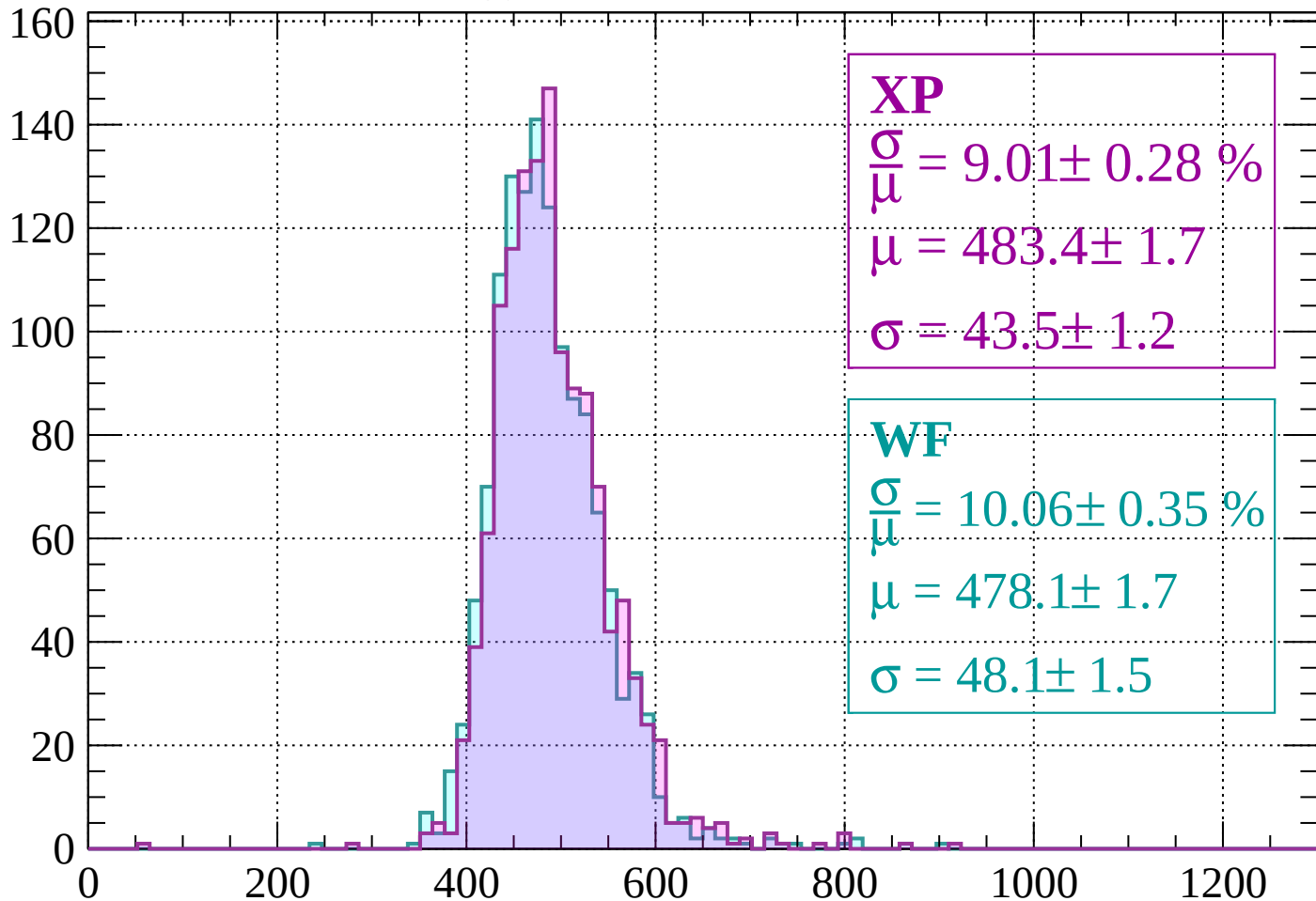


Energy loss | $1640 < p < 1680$



Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.28 \%$$

$$\mu = 483.4 \pm 1.7$$

$$\sigma = 43.5 \pm 1.2$$

WF

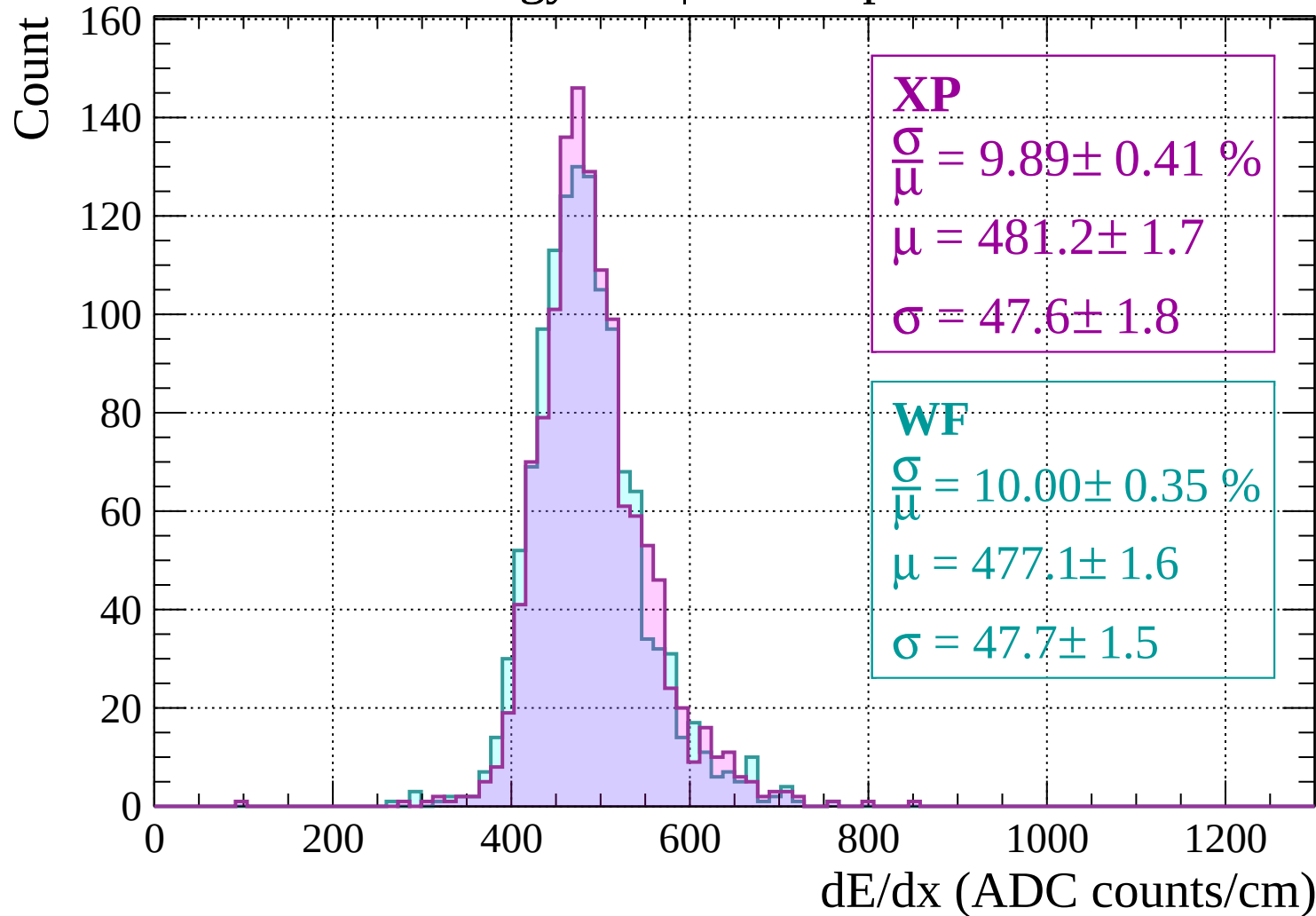
$$\frac{\sigma}{\mu} = 10.06 \pm 0.35 \%$$

$$\mu = 478.1 \pm 1.7$$

$$\sigma = 48.1 \pm 1.5$$

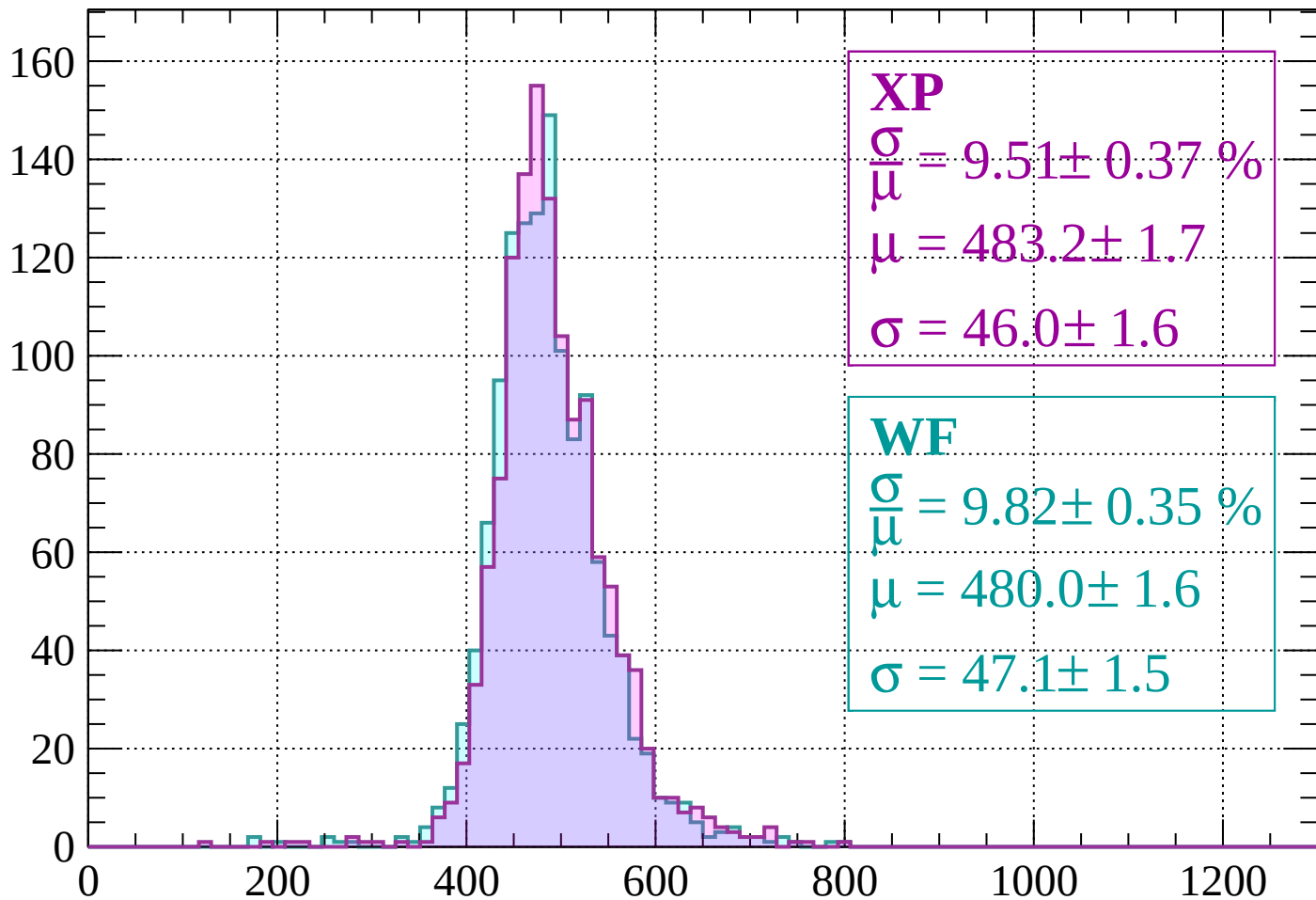
dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$



Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 9.51 \pm 0.37 \%$$

$$\mu = 483.2 \pm 1.7$$

$$\sigma = 46.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.82 \pm 0.35 \%$$

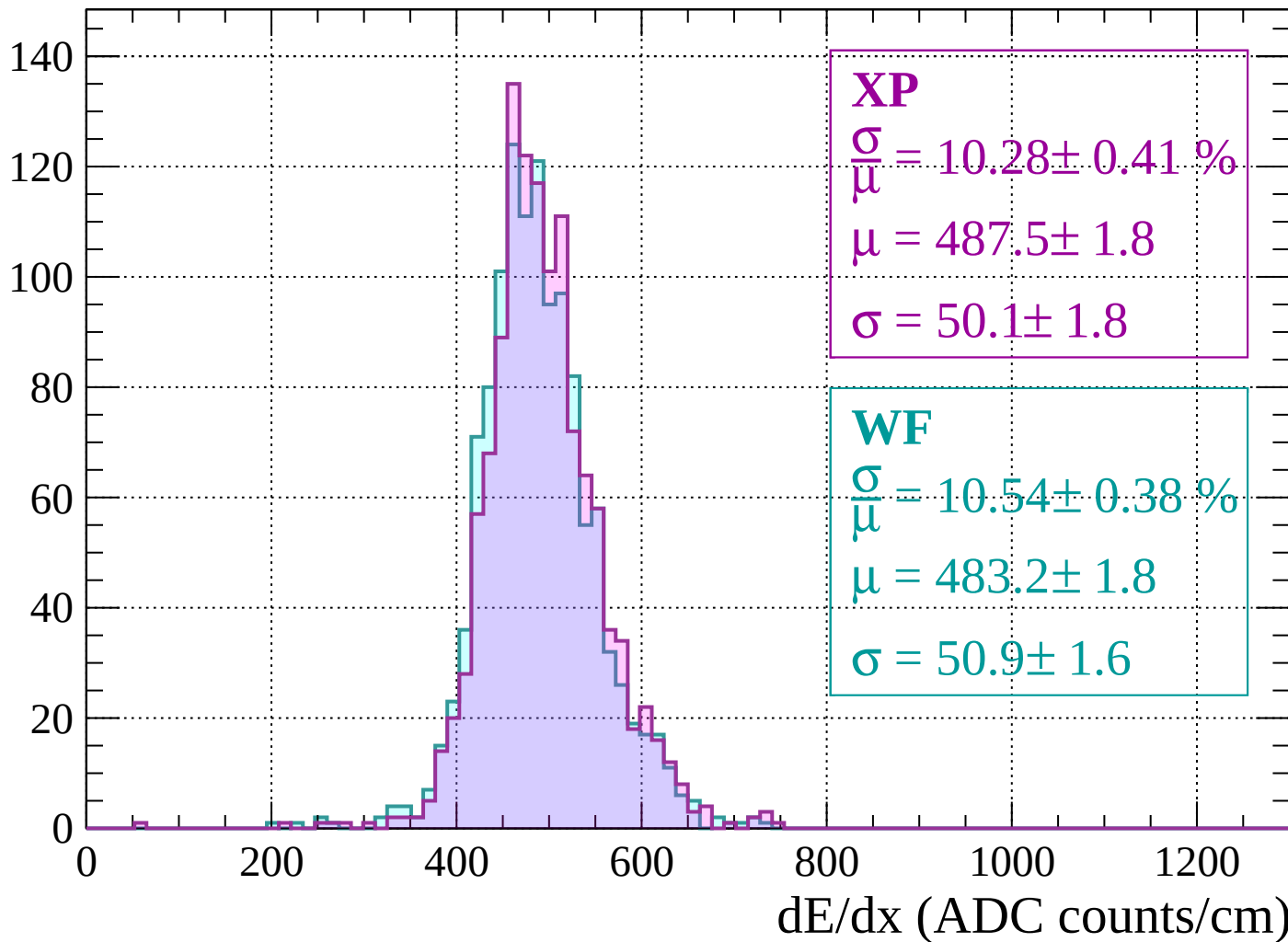
$$\mu = 480.0 \pm 1.6$$

$$\sigma = 47.1 \pm 1.5$$

dE/dx (ADC counts/cm)

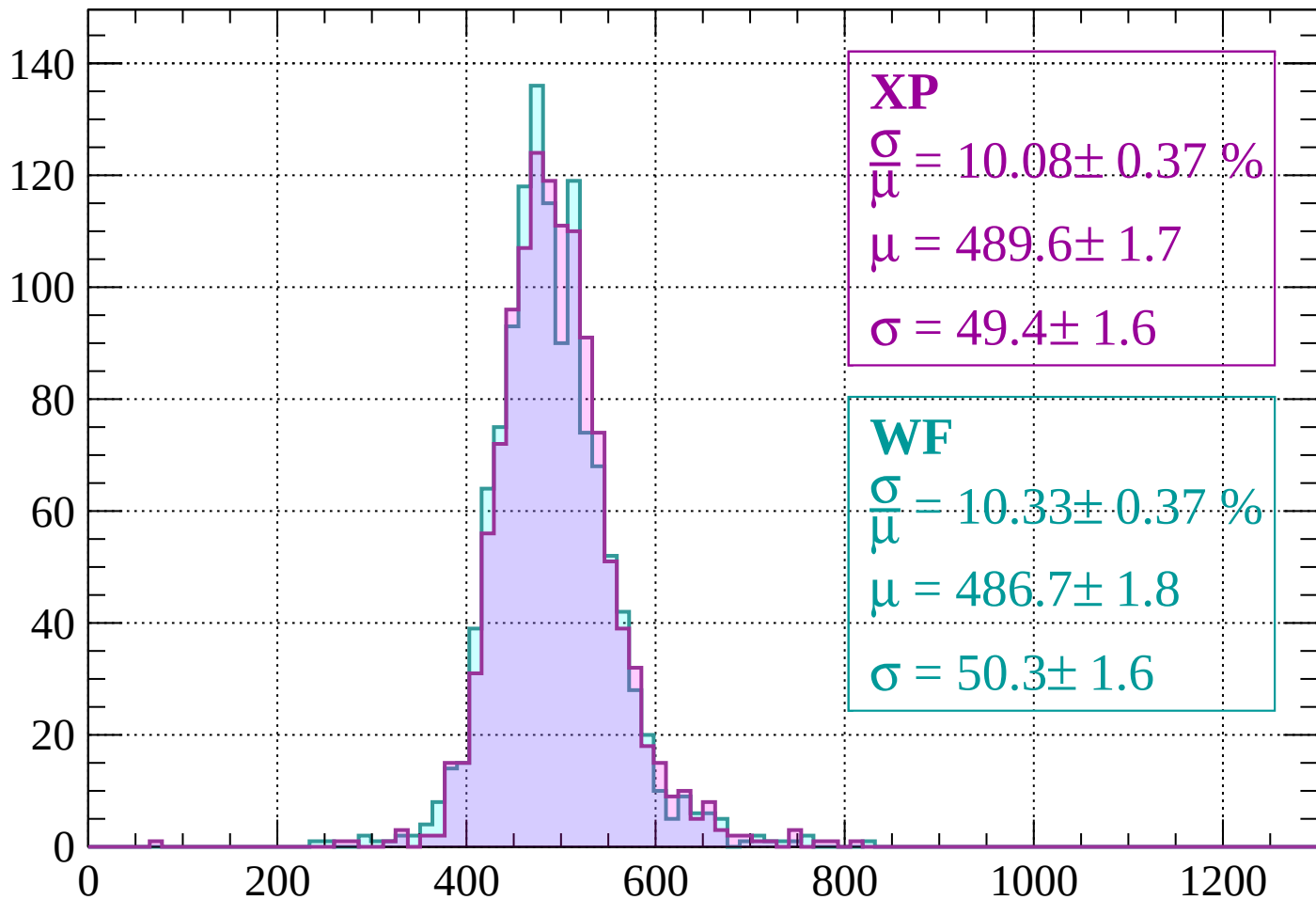
Energy loss | $1800 < p < 1840$

Count



Energy loss | $1840 < p < 1880$

Count



XP

$$\frac{\sigma}{\mu} = 10.08 \pm 0.37 \%$$

$$\mu = 489.6 \pm 1.7$$

$$\sigma = 49.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 10.33 \pm 0.37 \%$$

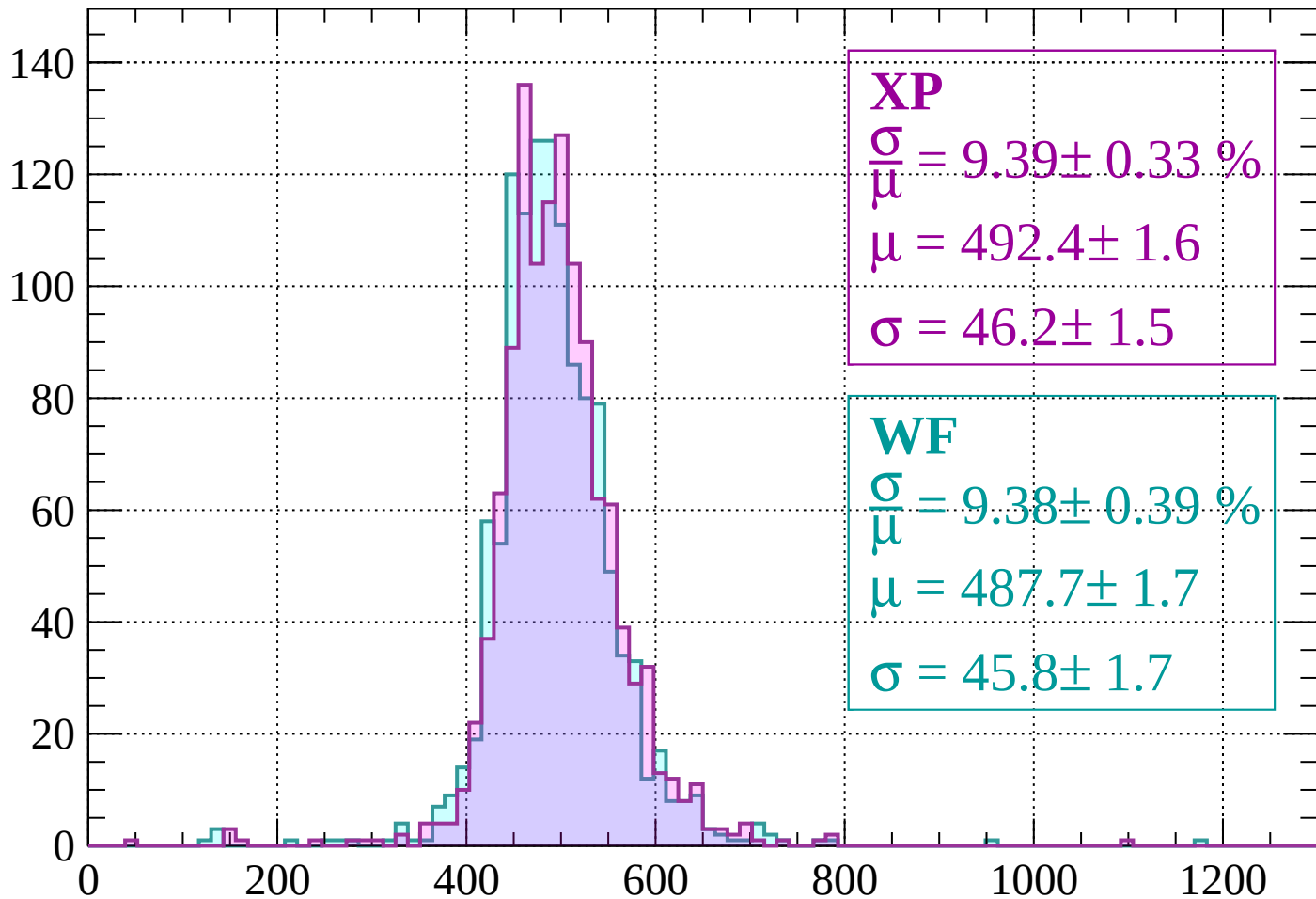
$$\mu = 486.7 \pm 1.8$$

$$\sigma = 50.3 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 9.39 \pm 0.33 \%$$

$$\mu = 492.4 \pm 1.6$$

$$\sigma = 46.2 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.38 \pm 0.39 \%$$

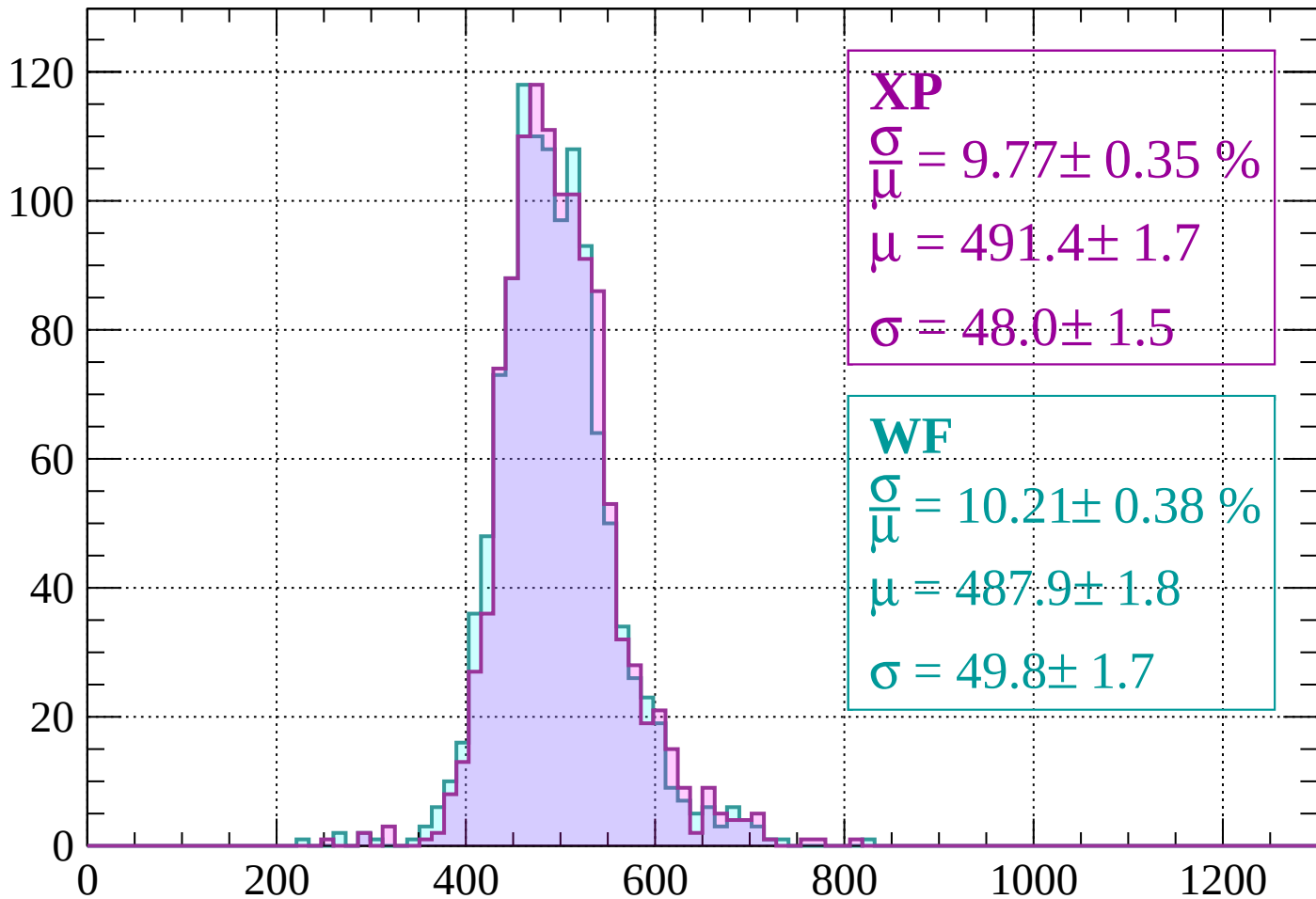
$$\mu = 487.7 \pm 1.7$$

$$\sigma = 45.8 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

Count



XP

$$\frac{\sigma}{\mu} = 9.77 \pm 0.35 \%$$

$$\mu = 491.4 \pm 1.7$$

$$\sigma = 48.0 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 10.21 \pm 0.38 \%$$

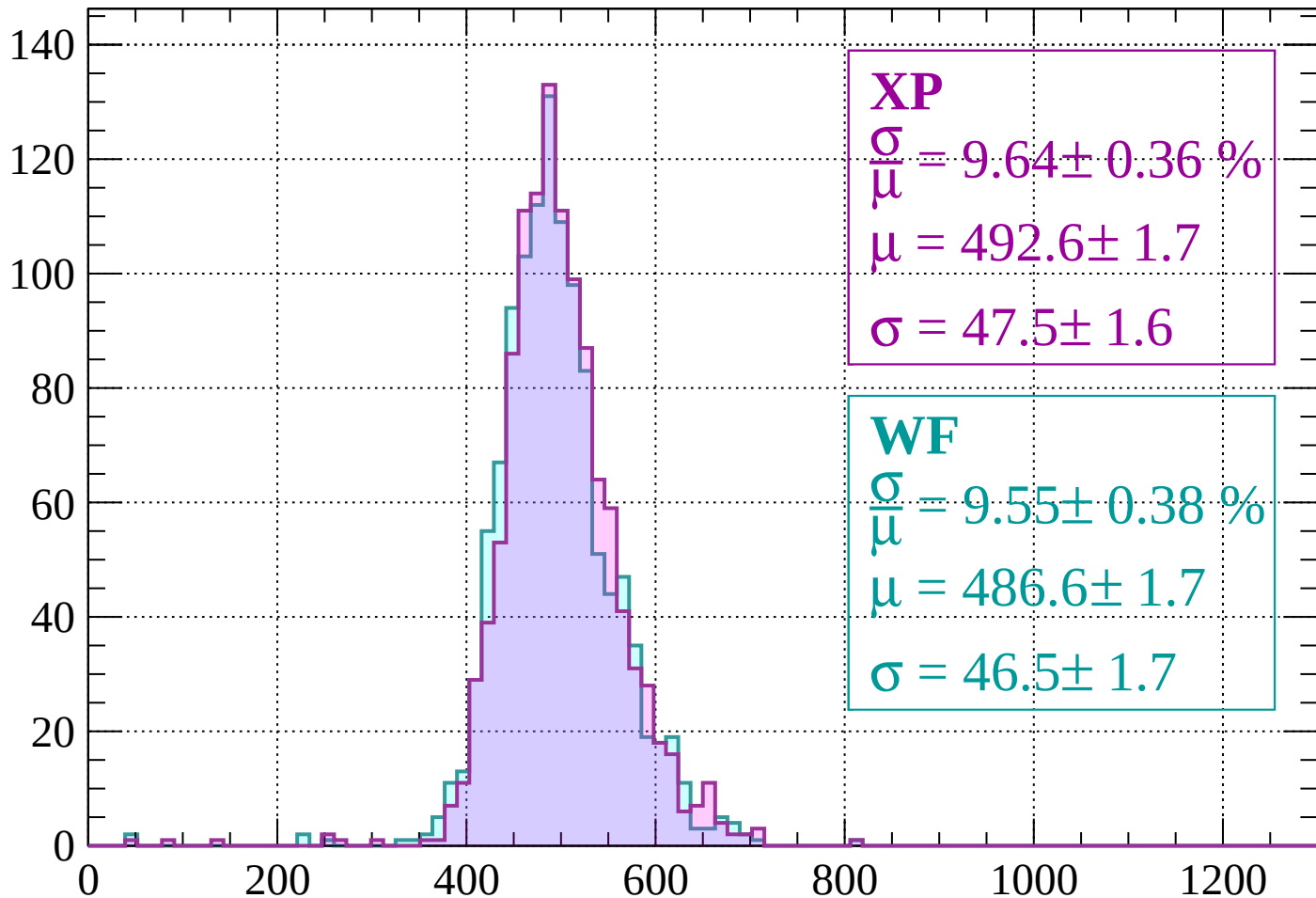
$$\mu = 487.9 \pm 1.8$$

$$\sigma = 49.8 \pm 1.7$$

dE/dx (ADC counts/cm)

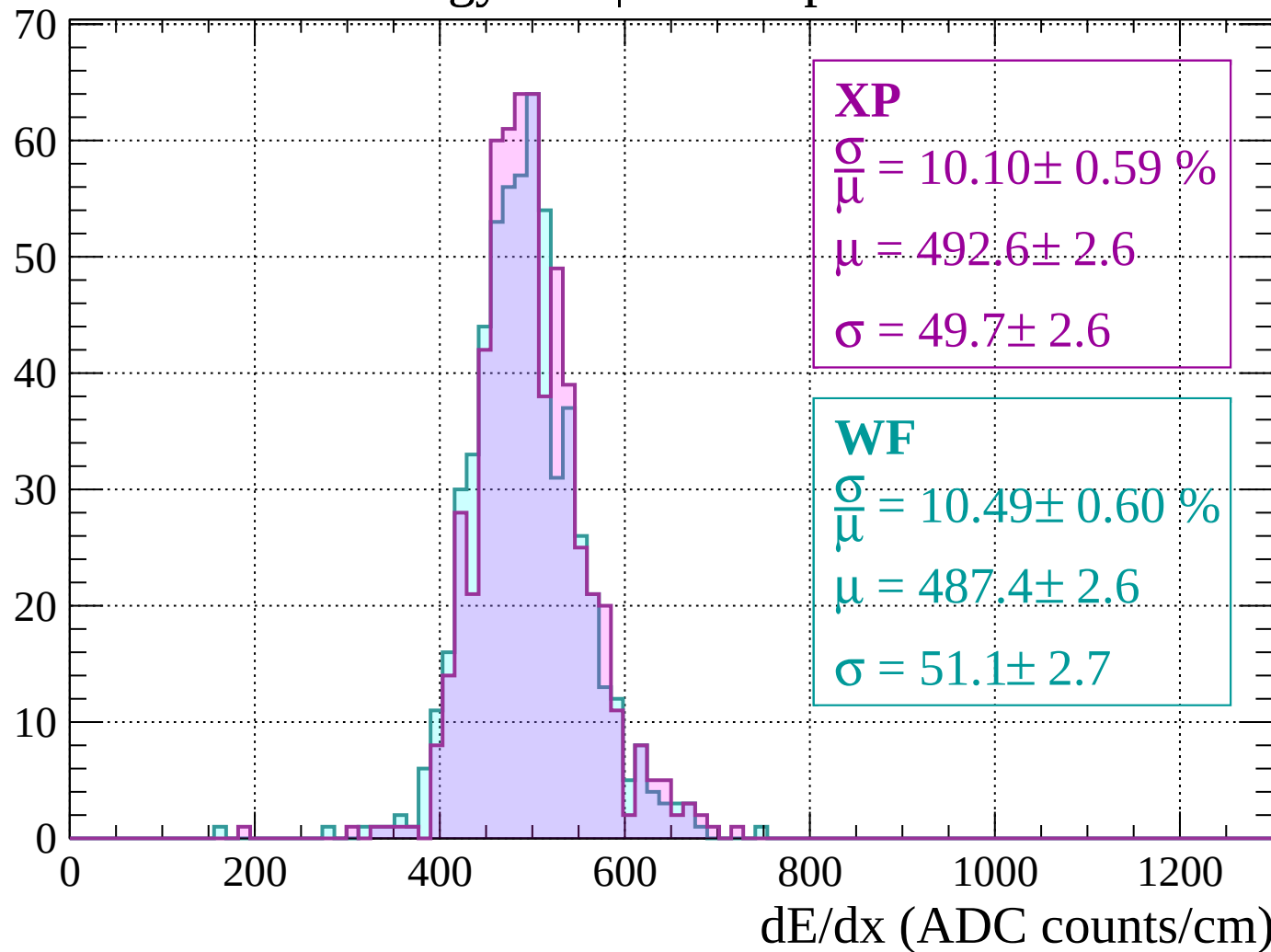
Energy loss | $1960 < p < 2000$

Count



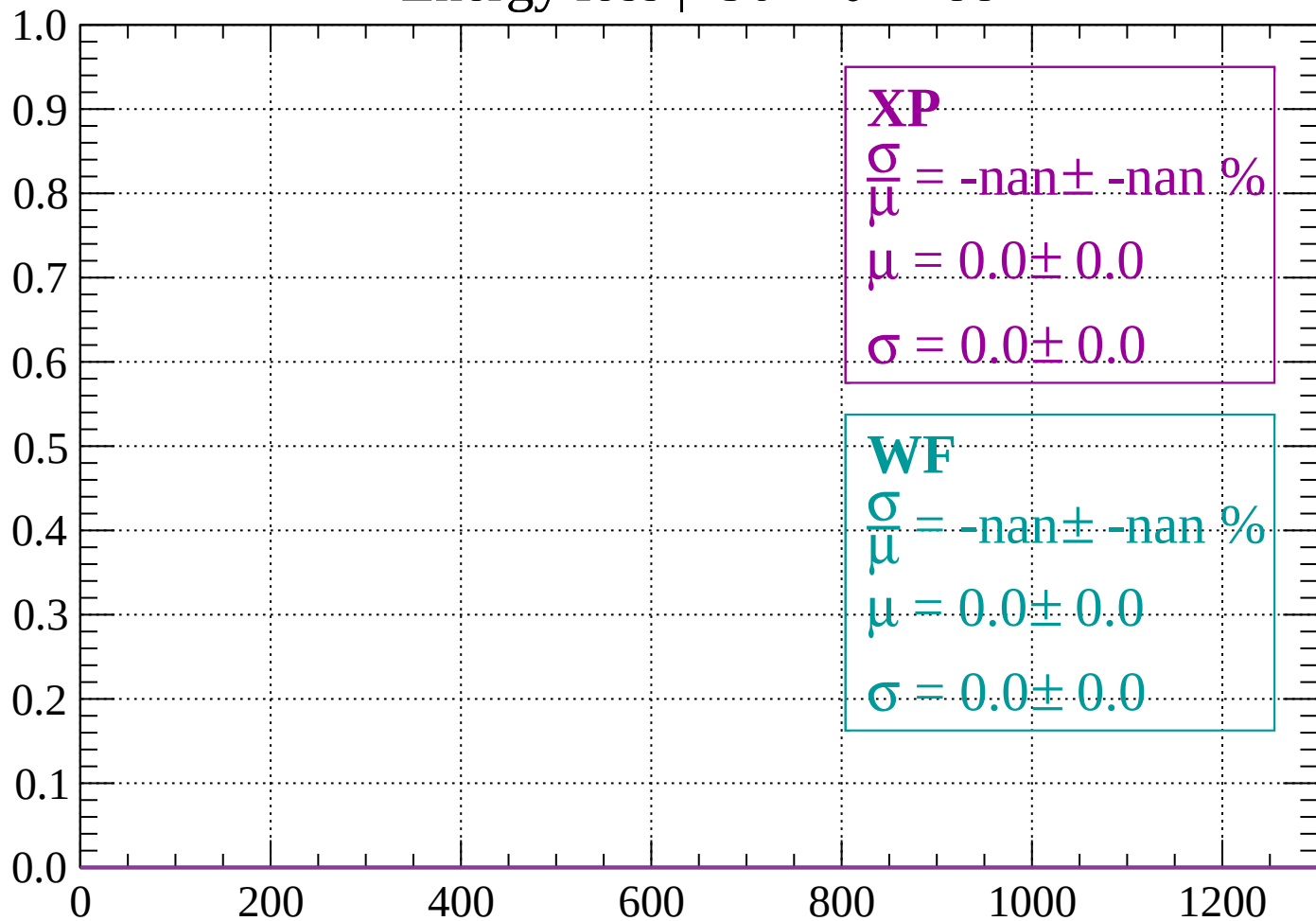
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

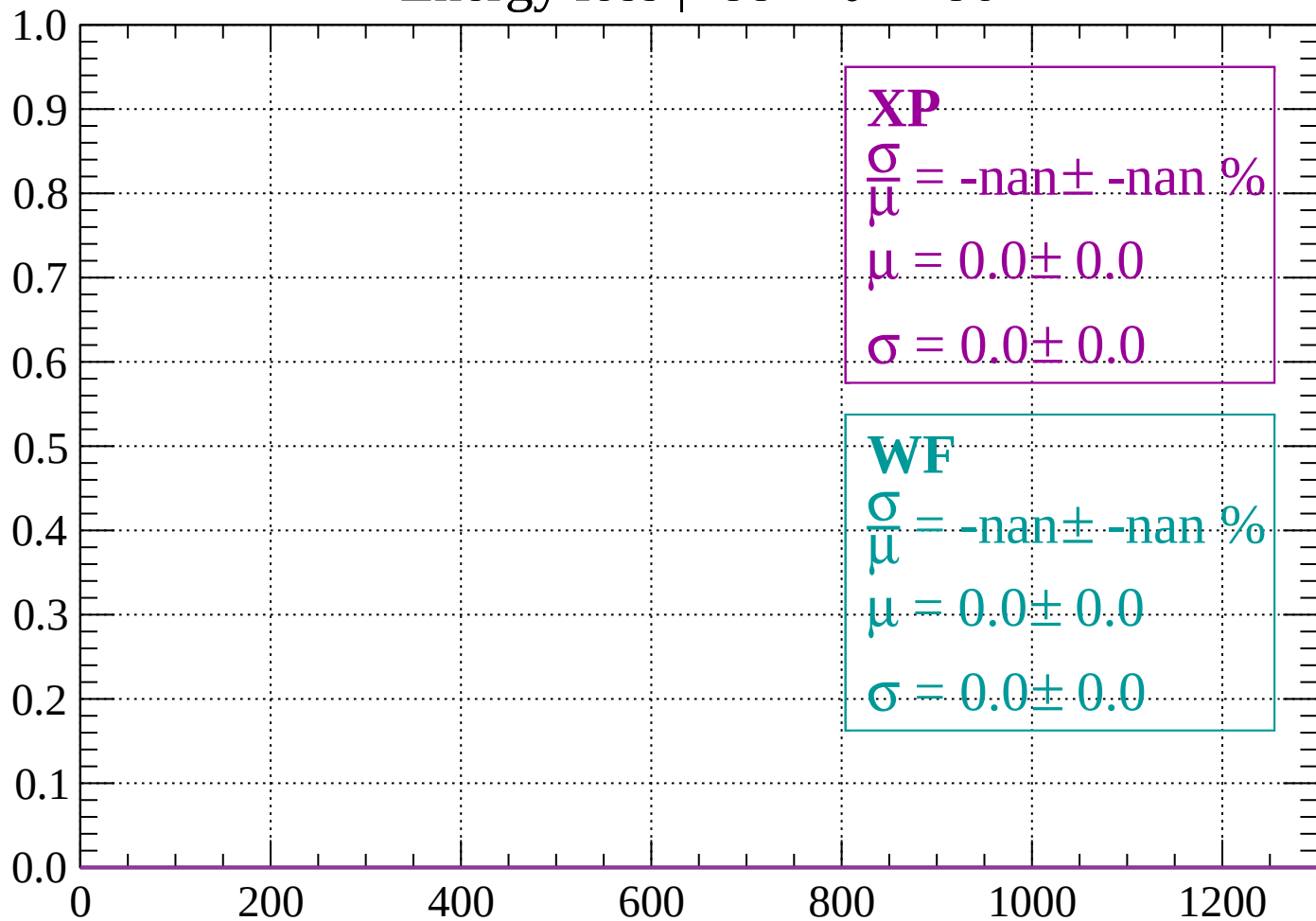
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

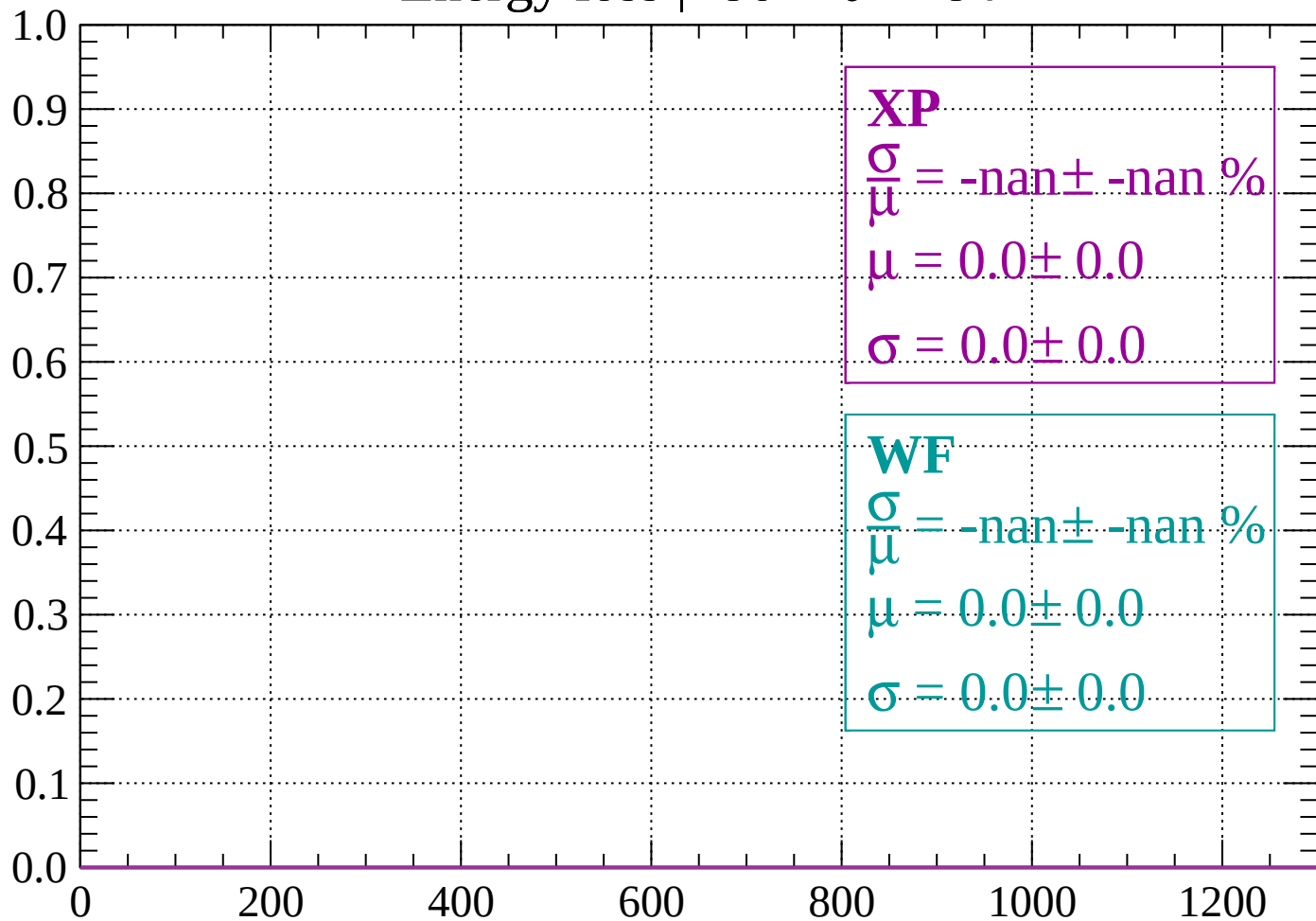
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

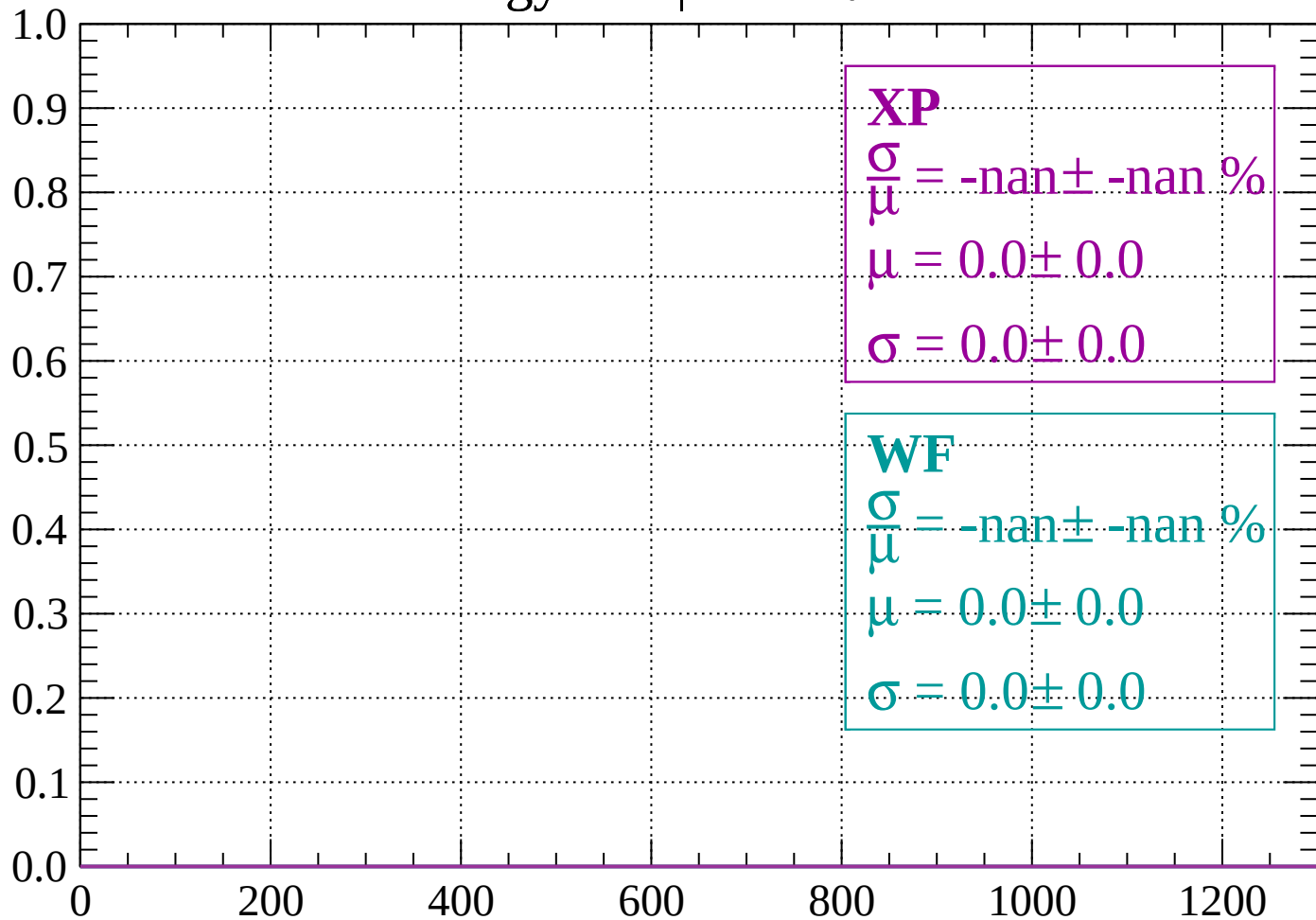
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

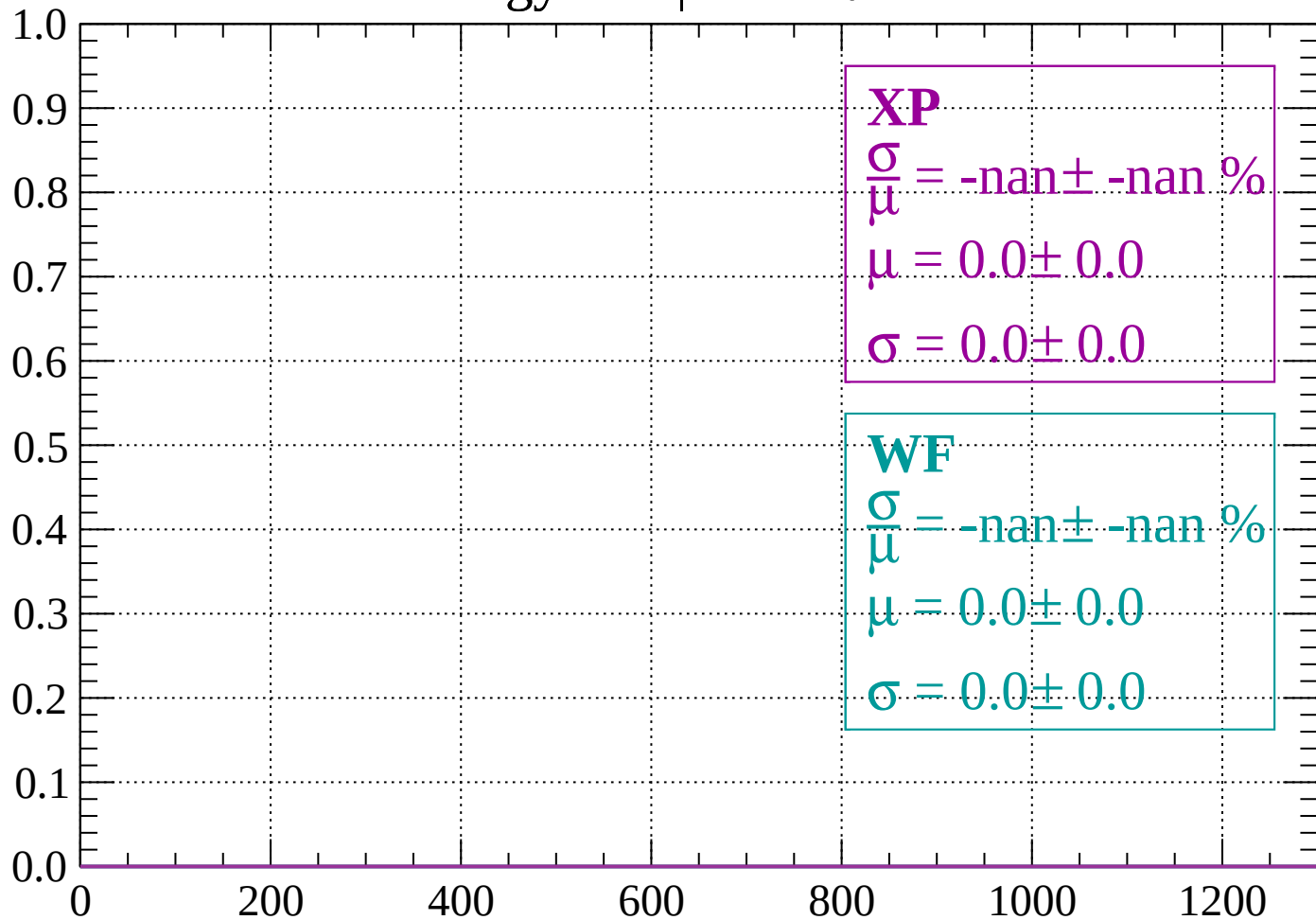
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

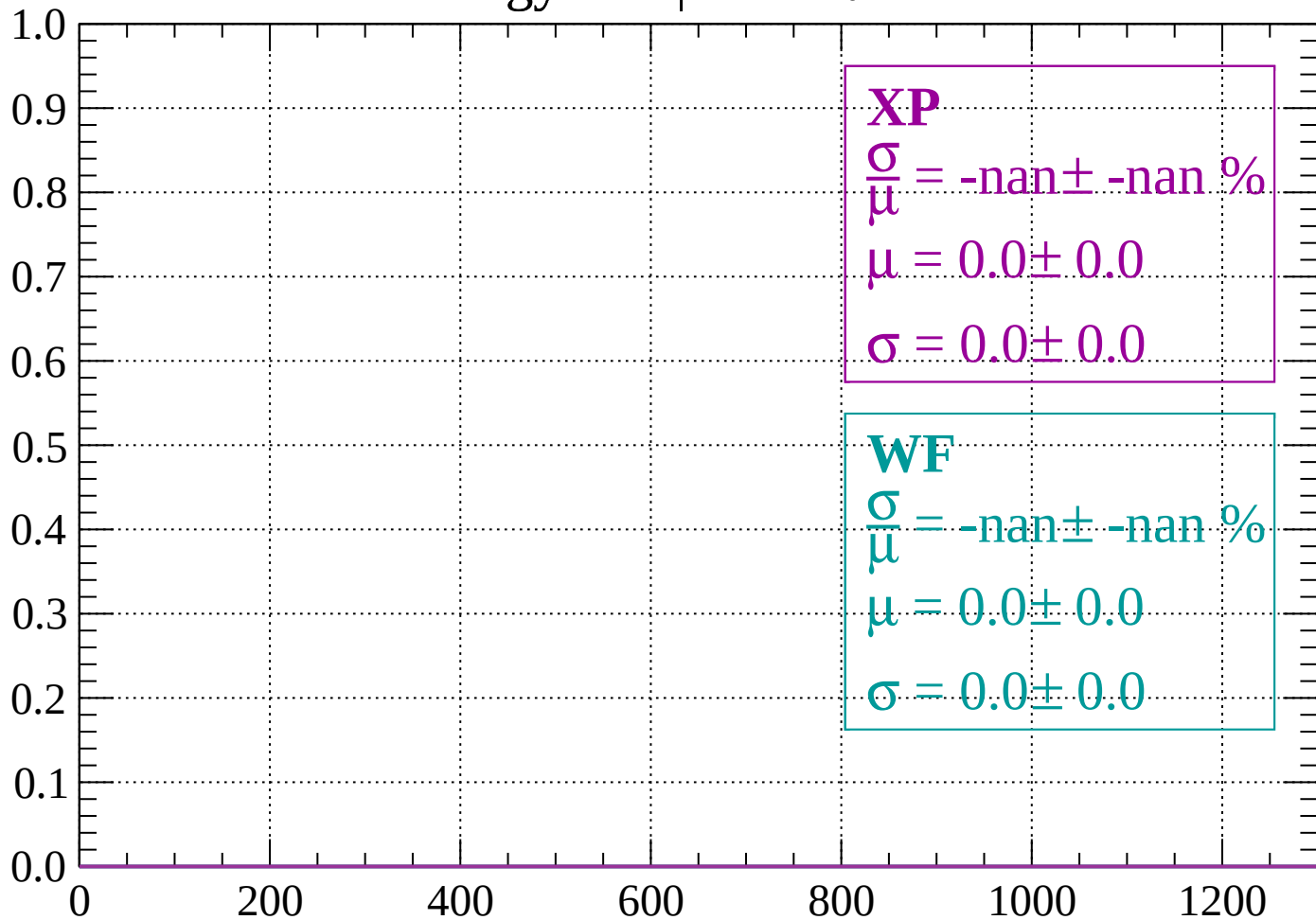
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

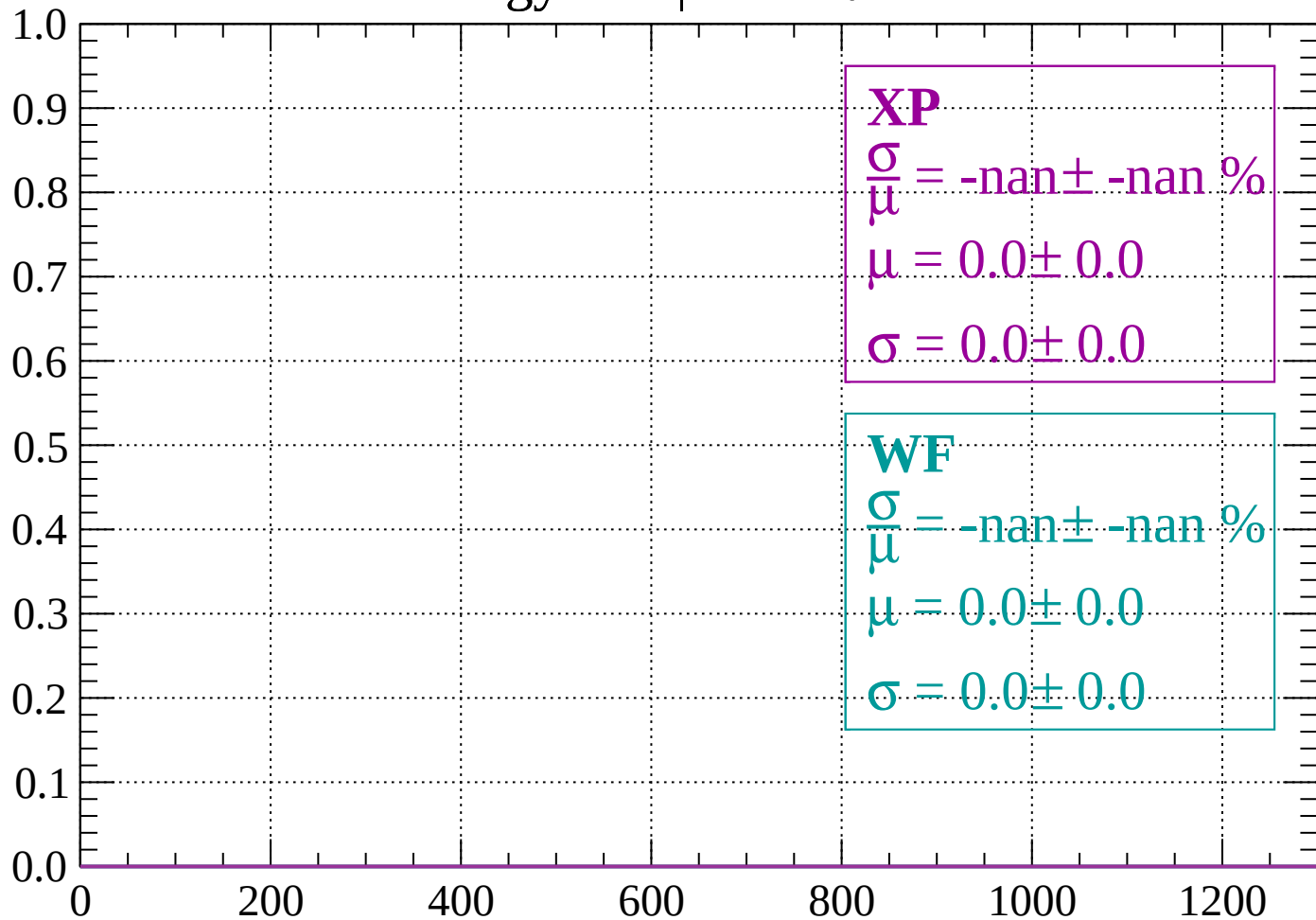
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

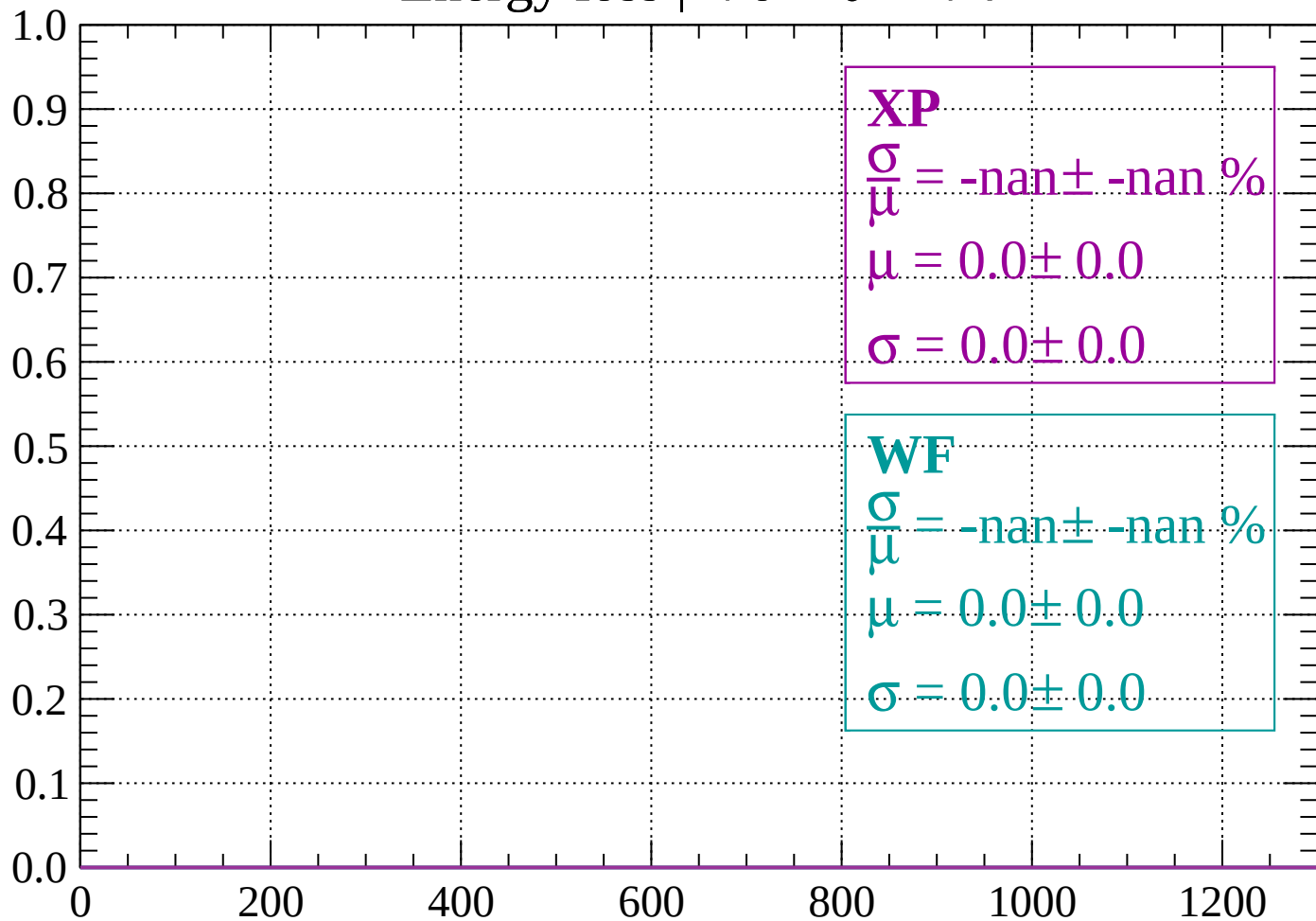
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

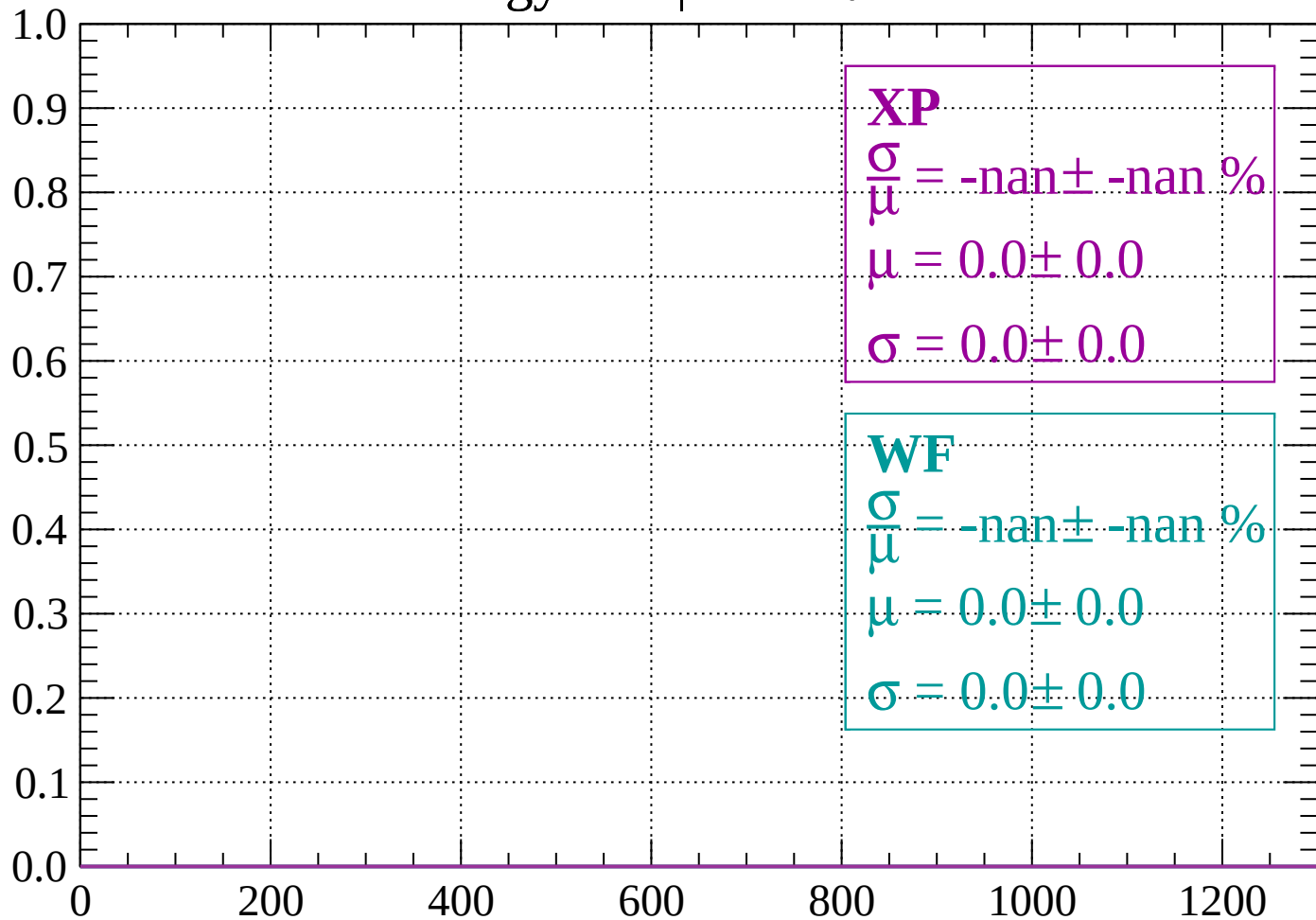
Count



dE/dx (ADC counts/cm)

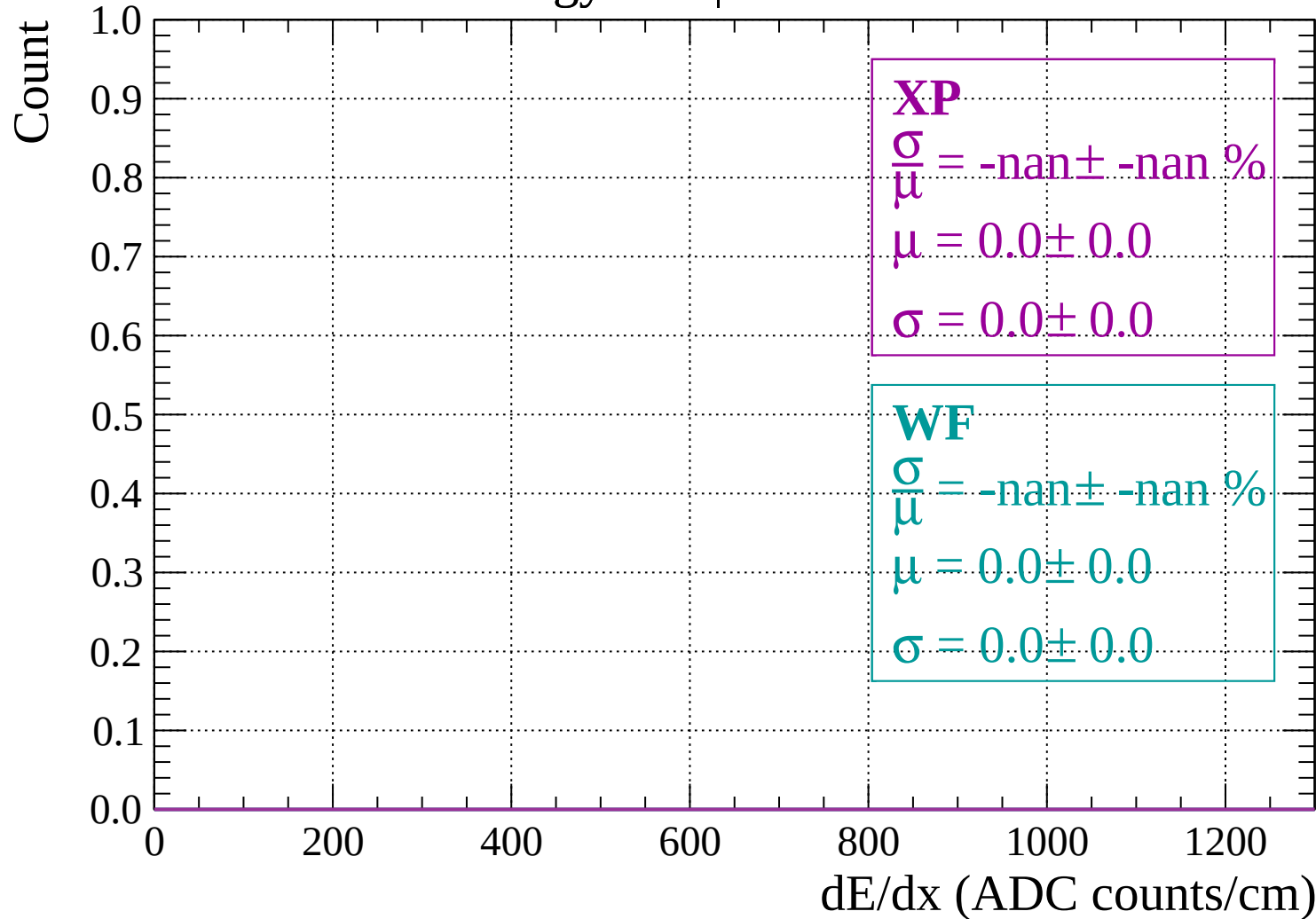
Energy loss | $-74 < \theta < -72$

Count



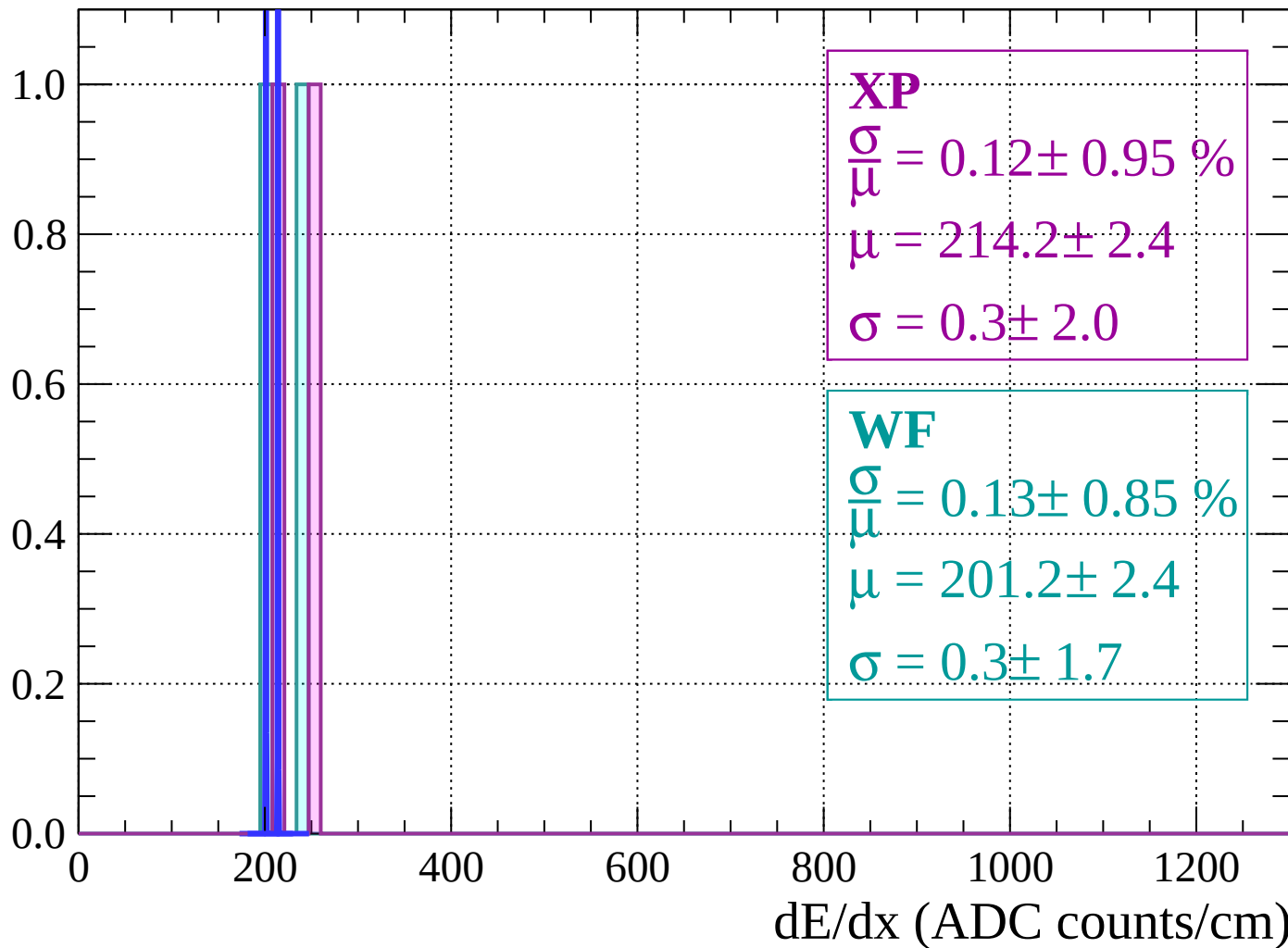
dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$



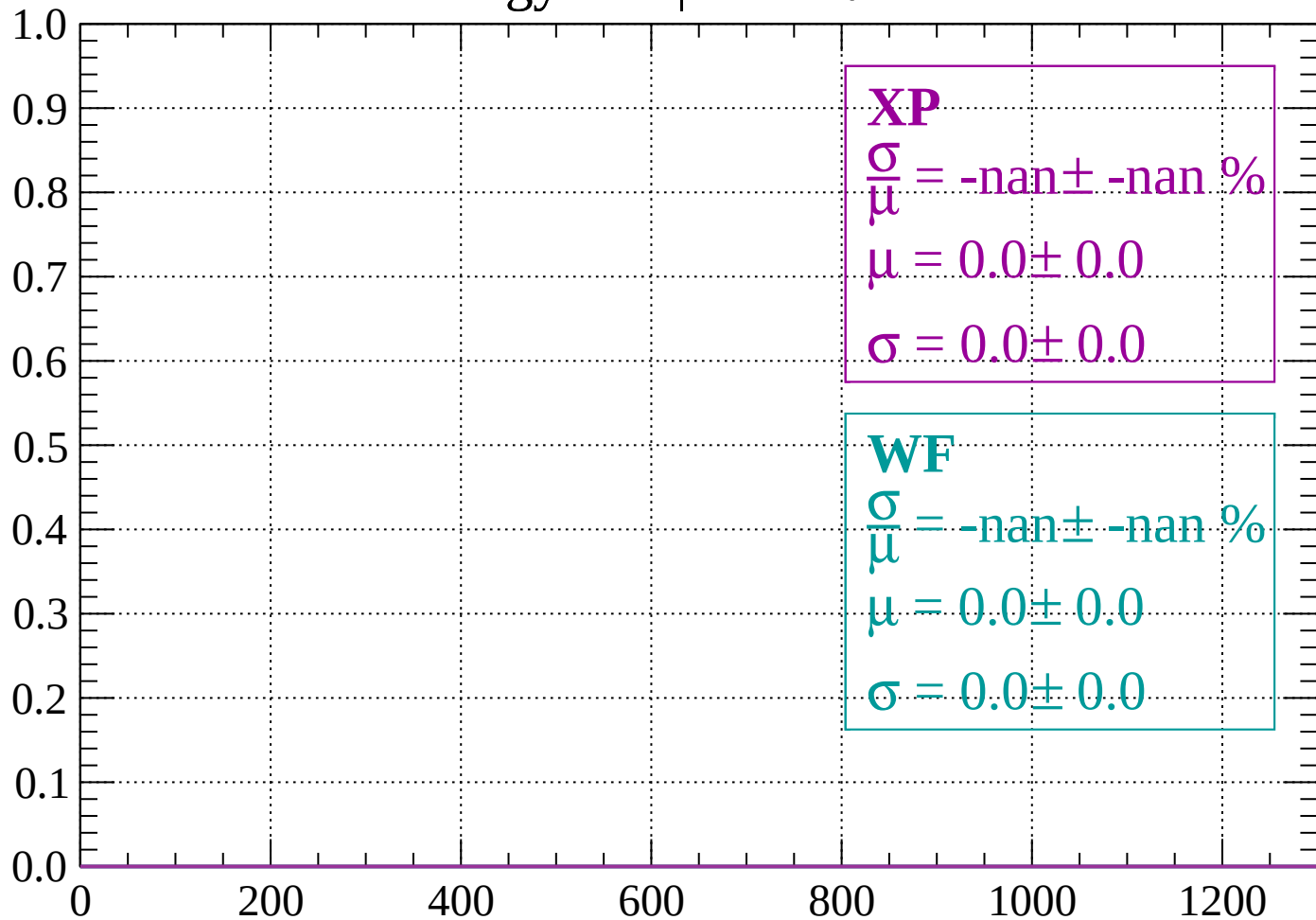
Energy loss | $-70 < \theta < -68$

Count



Energy loss | $-68 < \theta < -66$

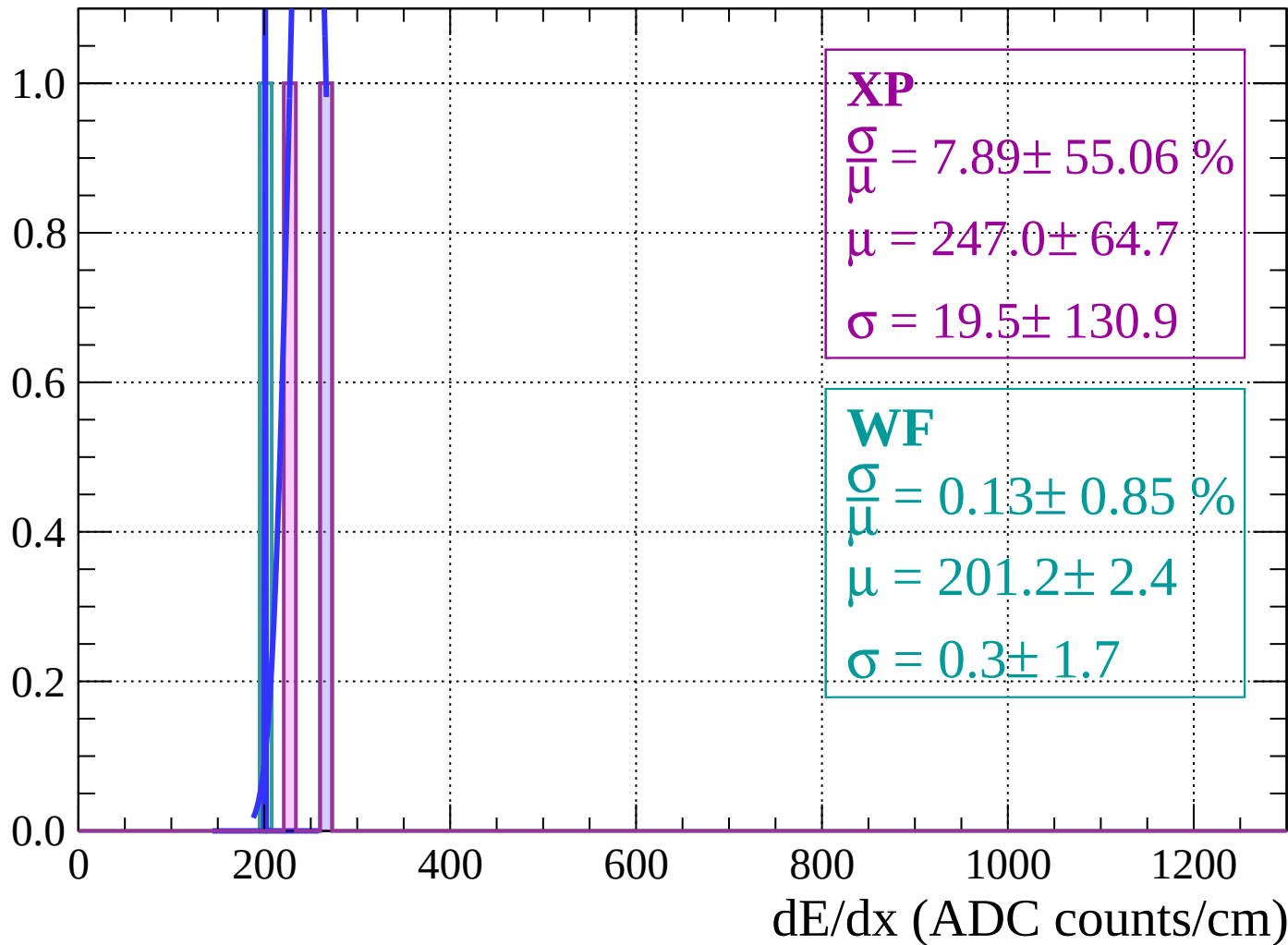
Count



dE/dx (ADC counts/cm)

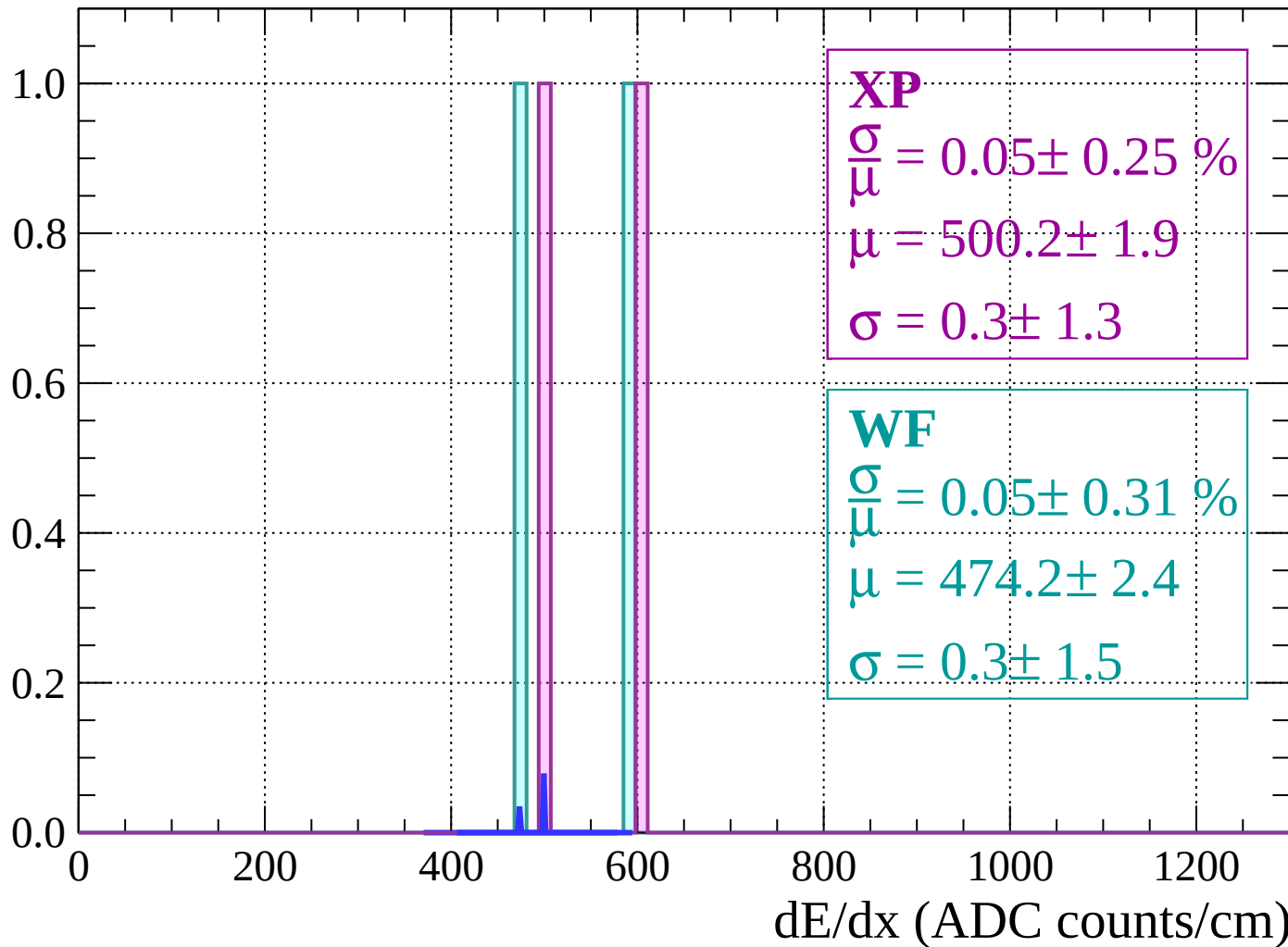
Energy loss | $-66 < \theta < -64$

Count



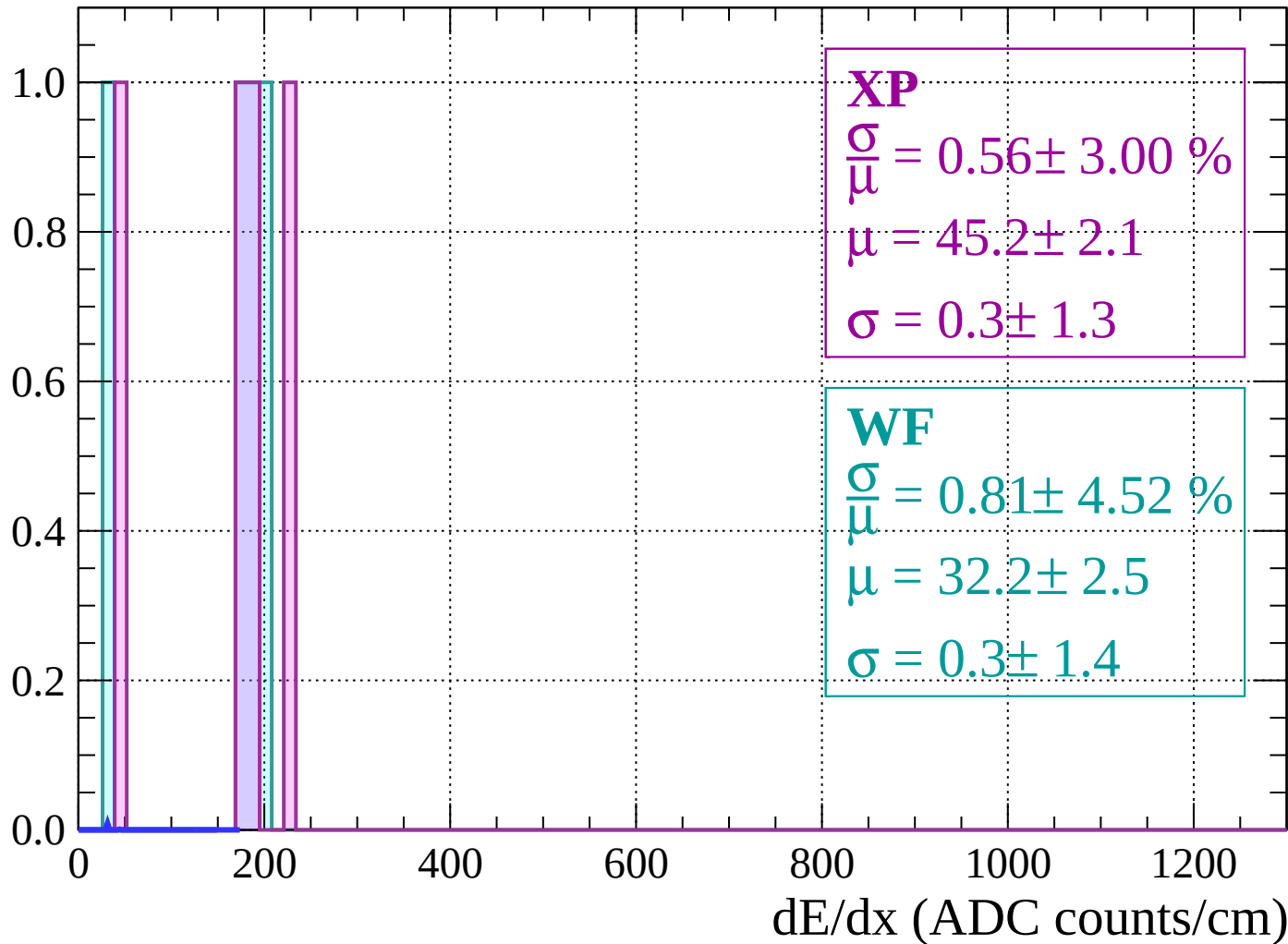
Energy loss | $-64 < \theta < -62$

Count



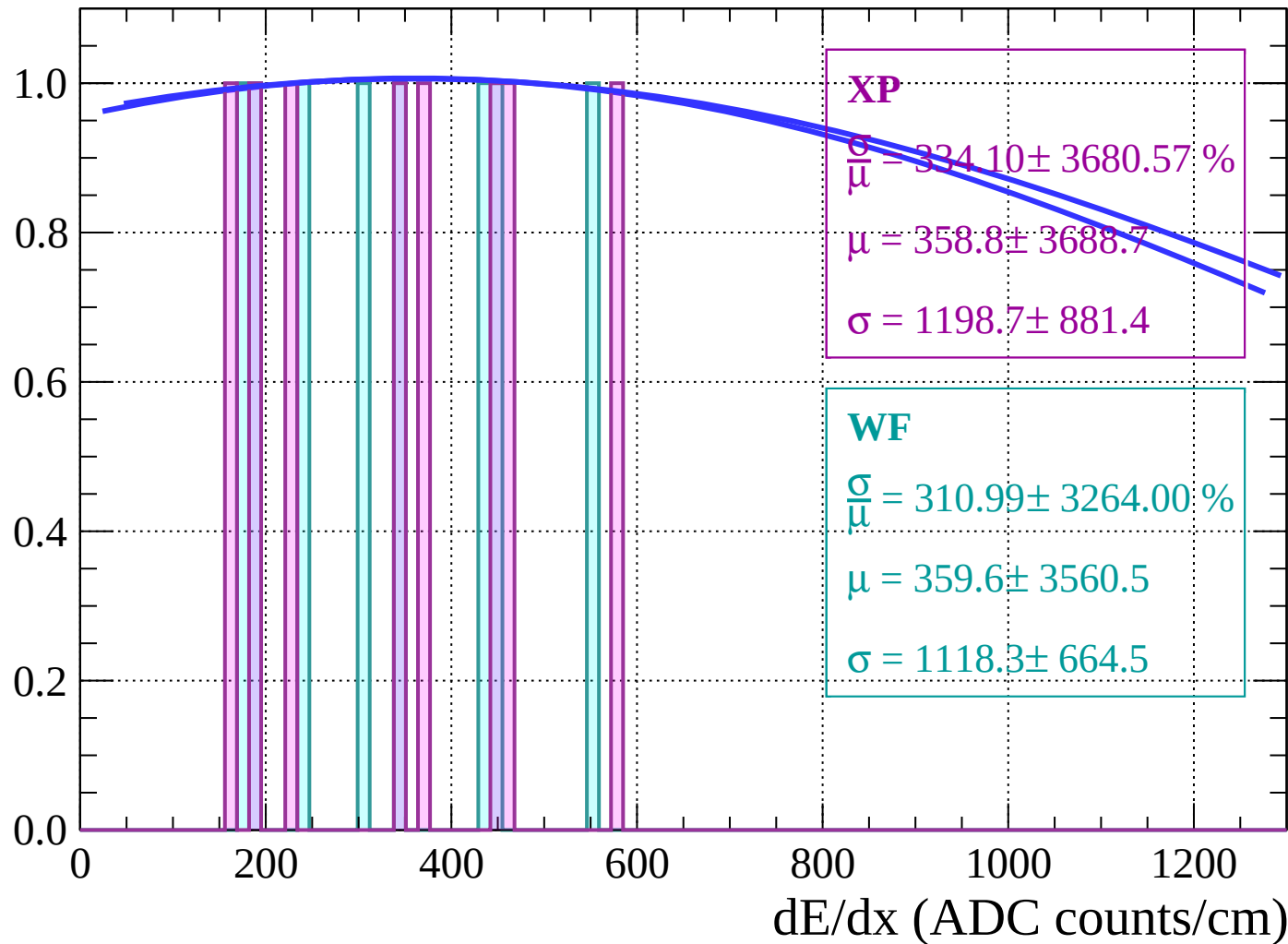
Energy loss | $-62 < \theta < -60$

Count



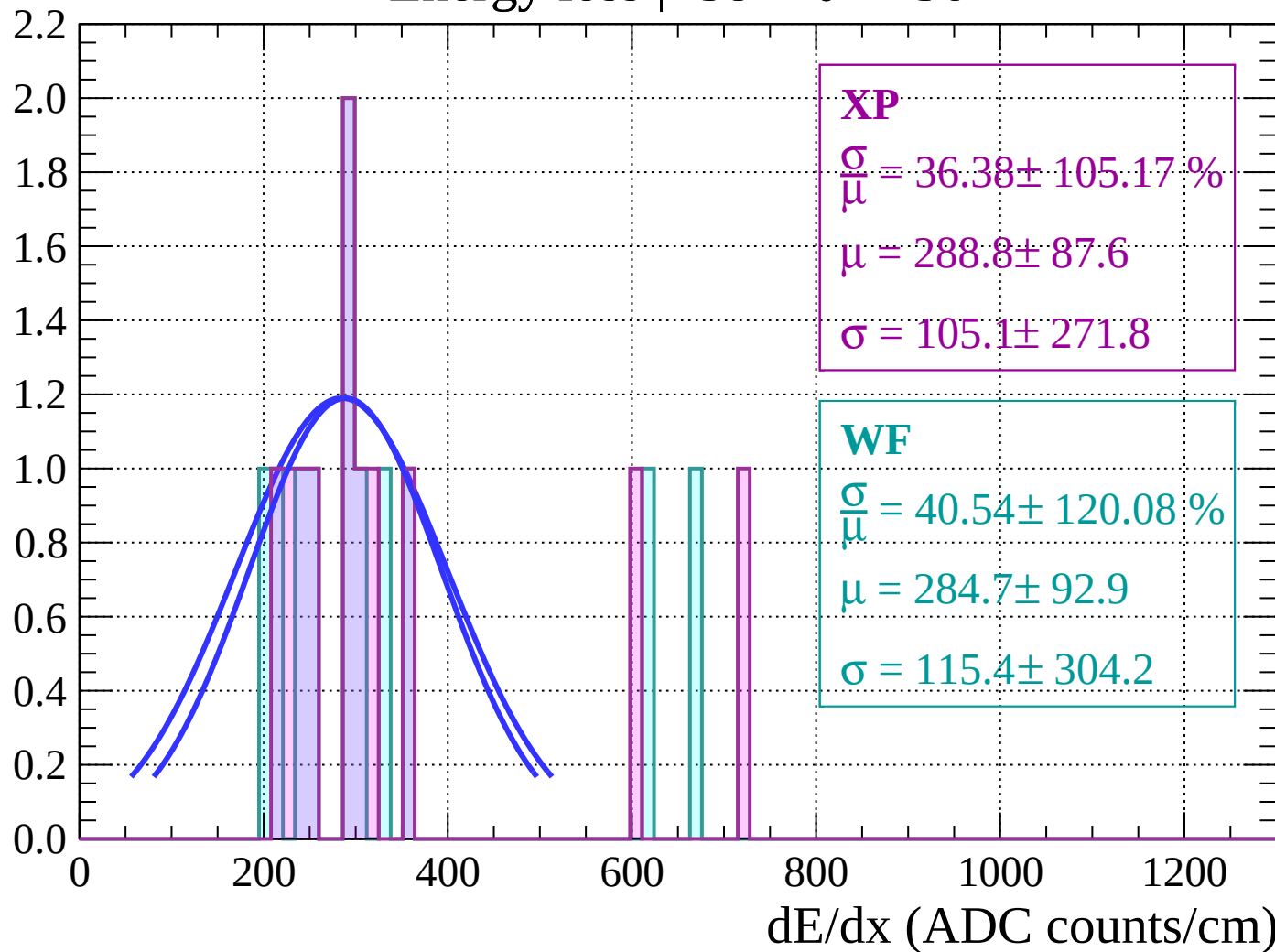
Energy loss | $-60 < \theta < -58$

Count

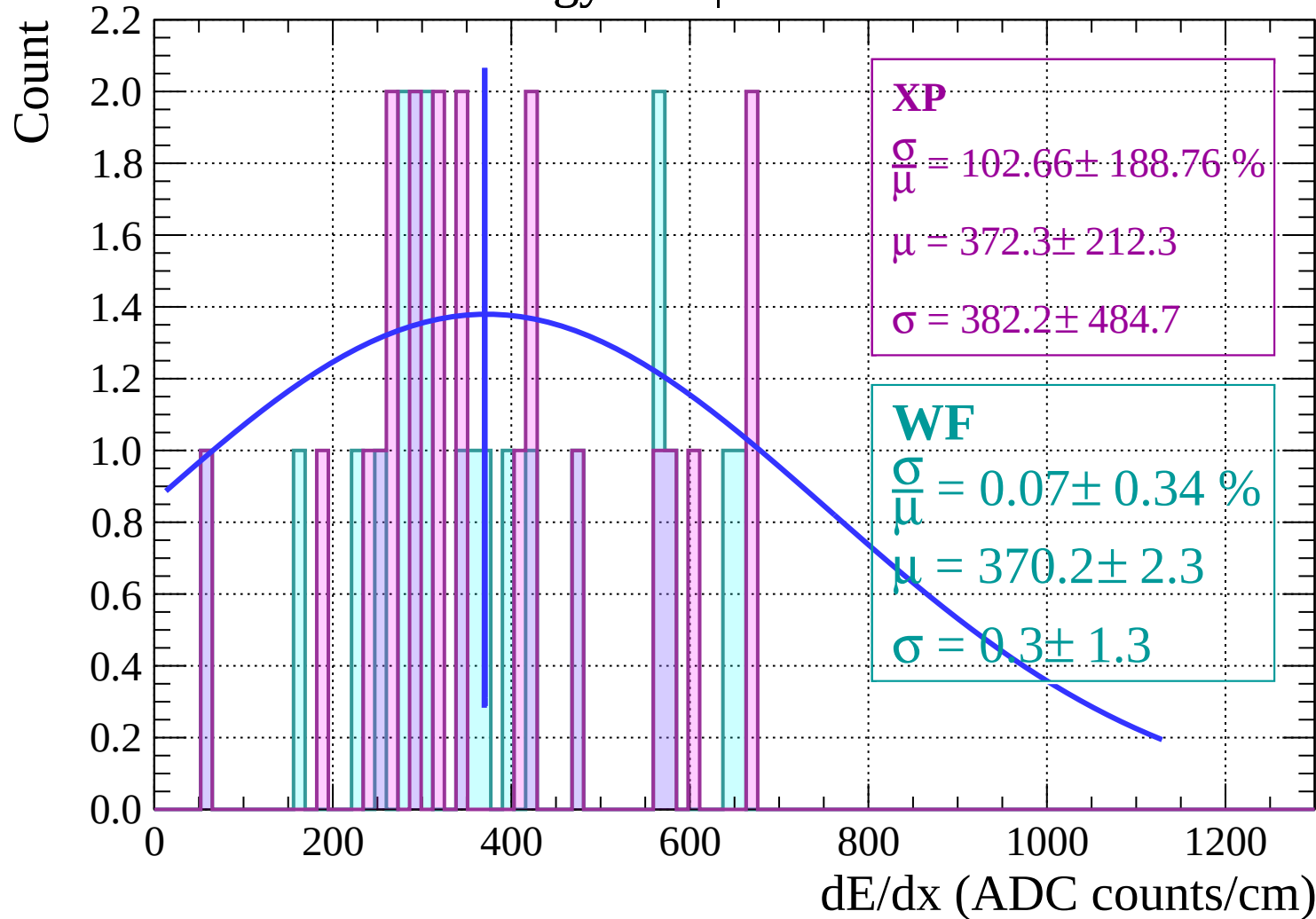


Energy loss | $-58 < \theta < -56$

Count

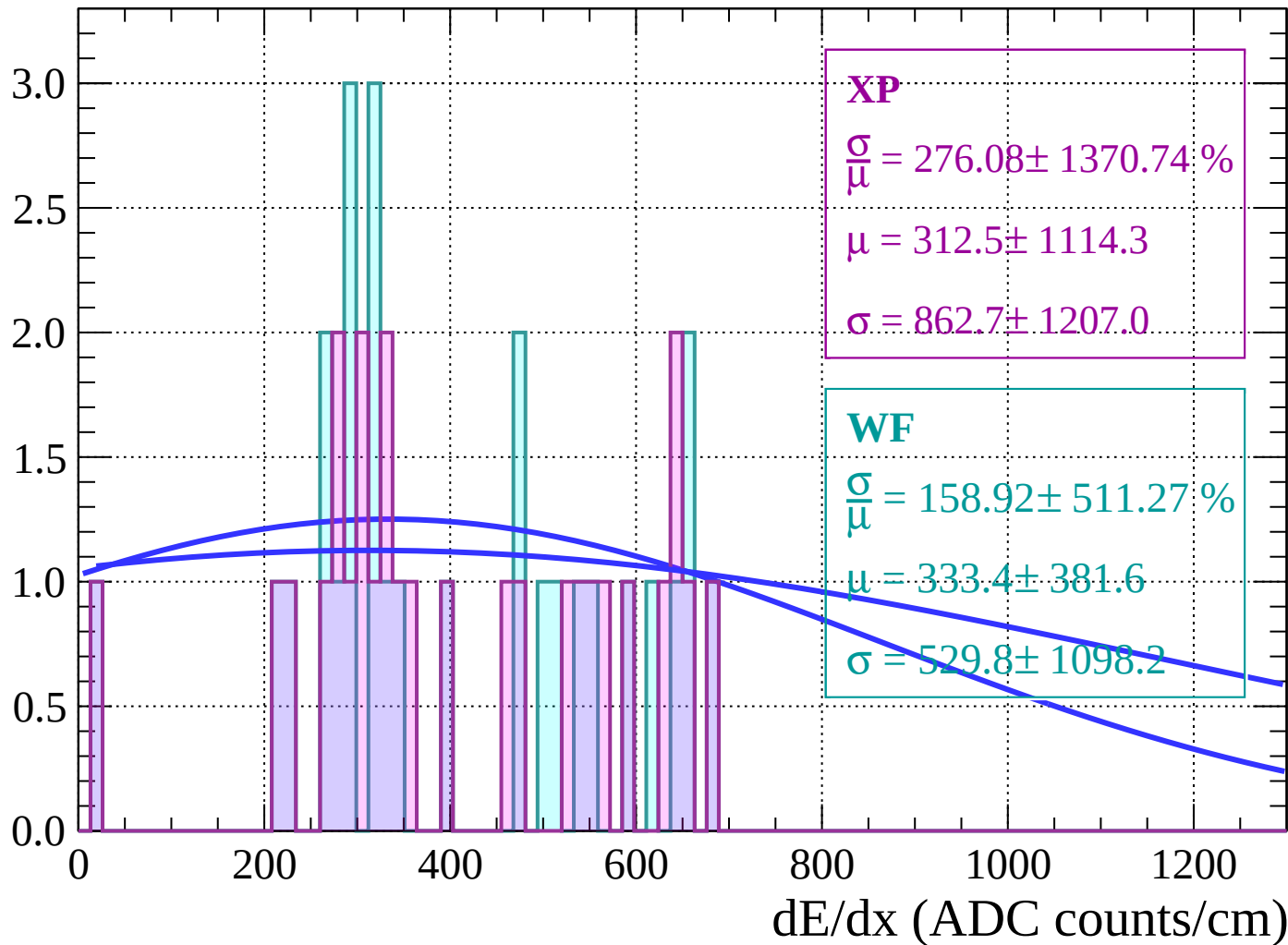


Energy loss | $-56 < \theta < -54$



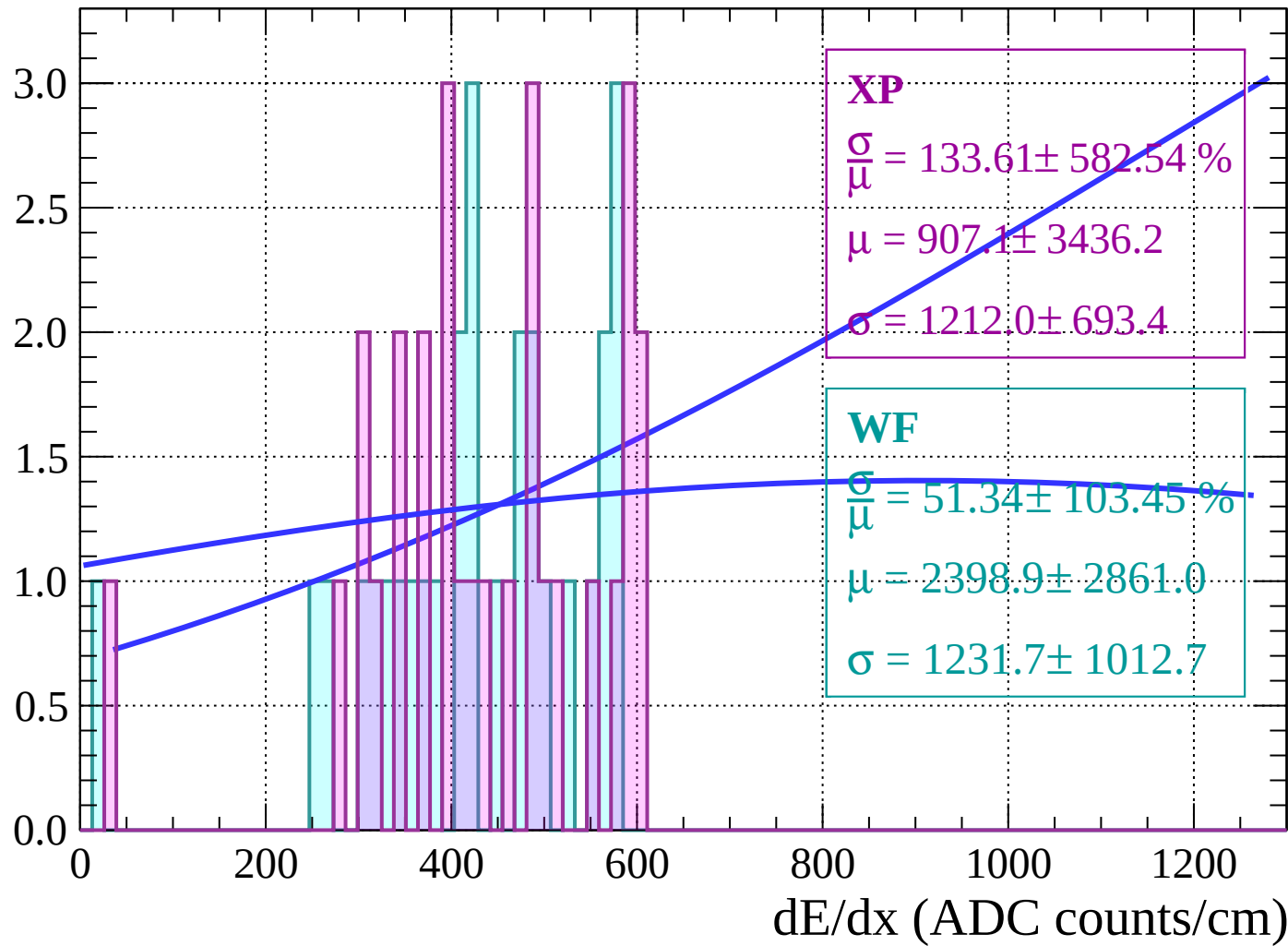
Energy loss | $-54 < \theta < -52$

Count



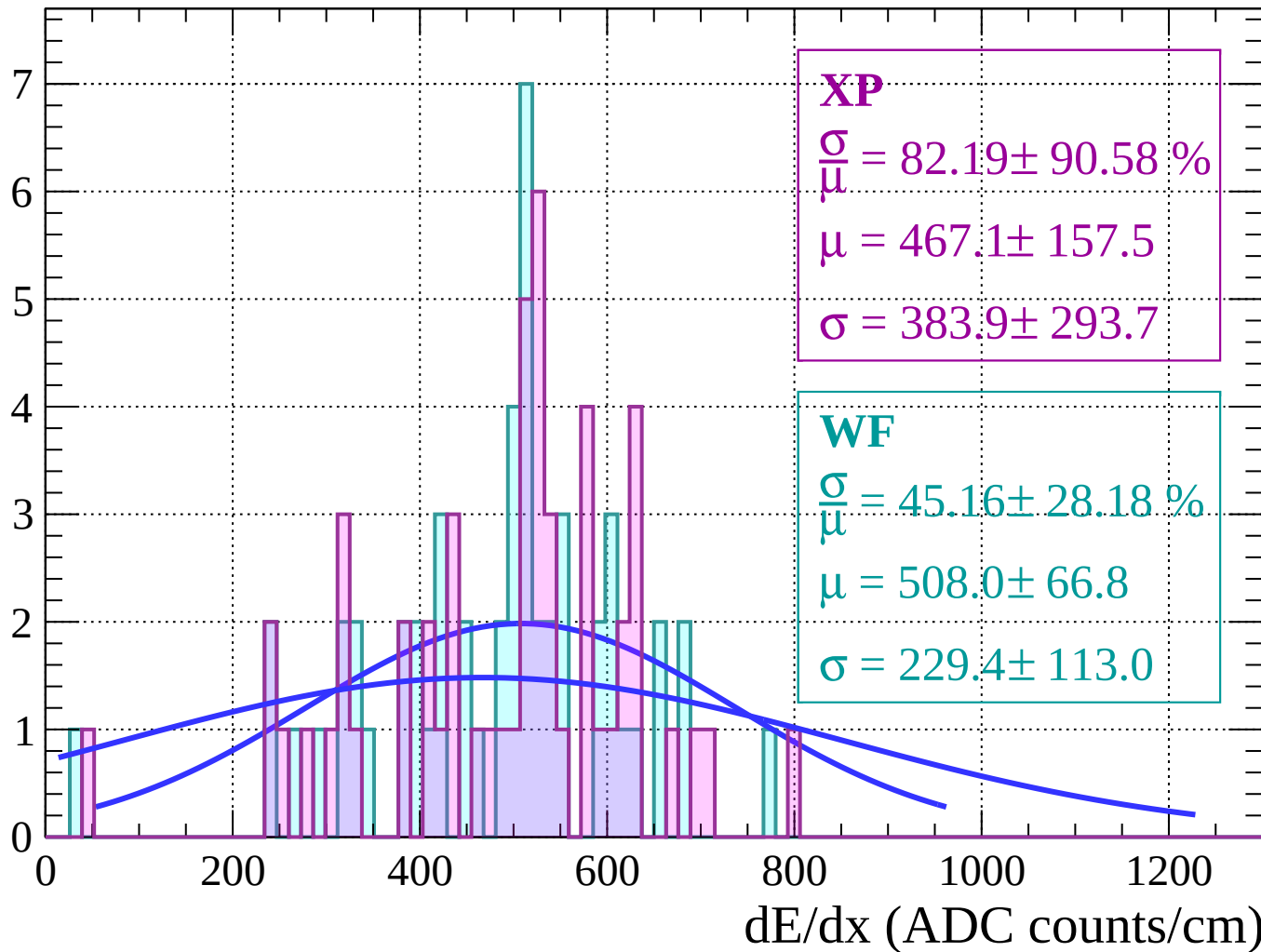
Energy loss | $-52 < \theta < -50$

Count



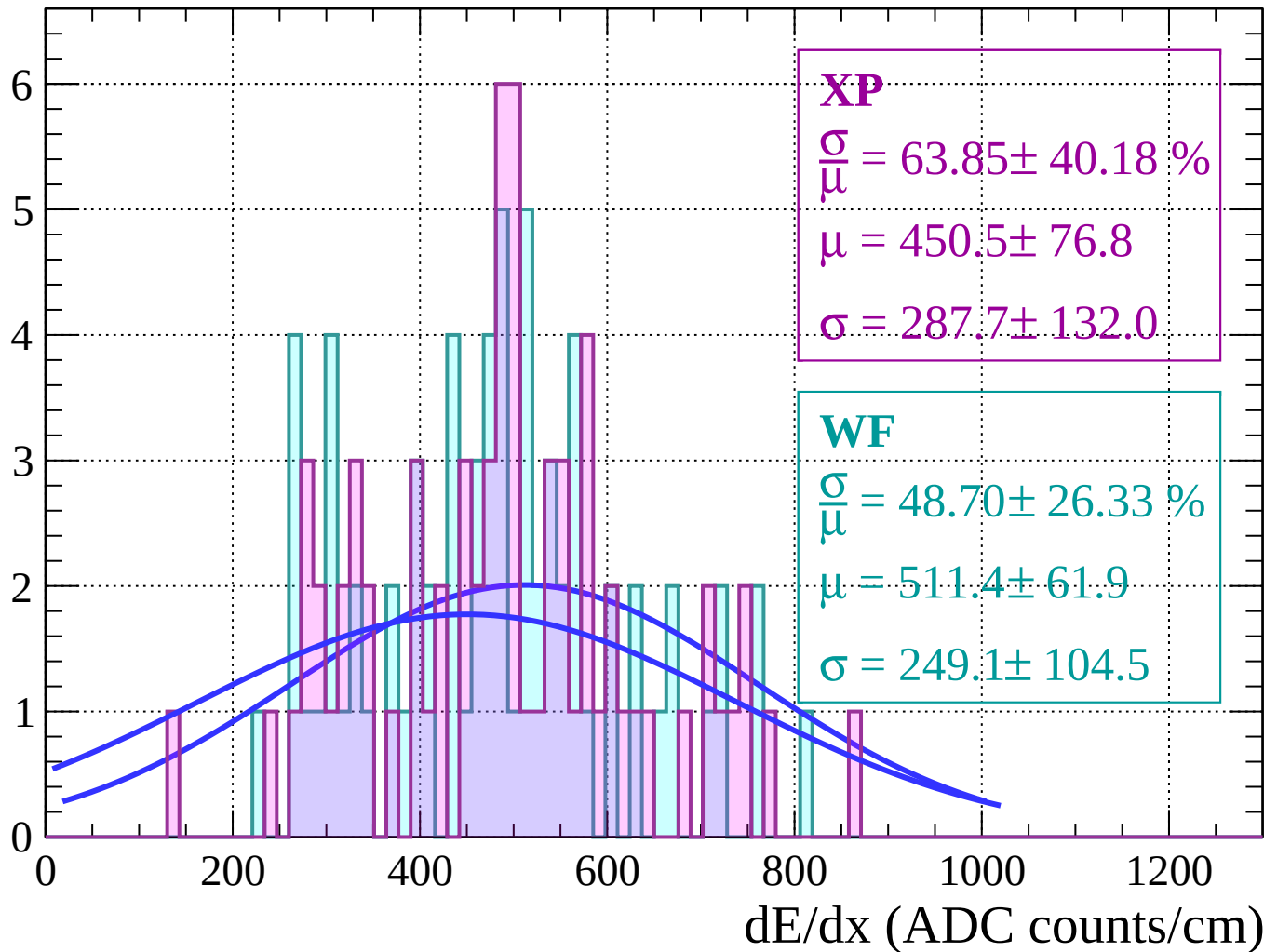
Energy loss | $-50 < \theta < -48$

Count



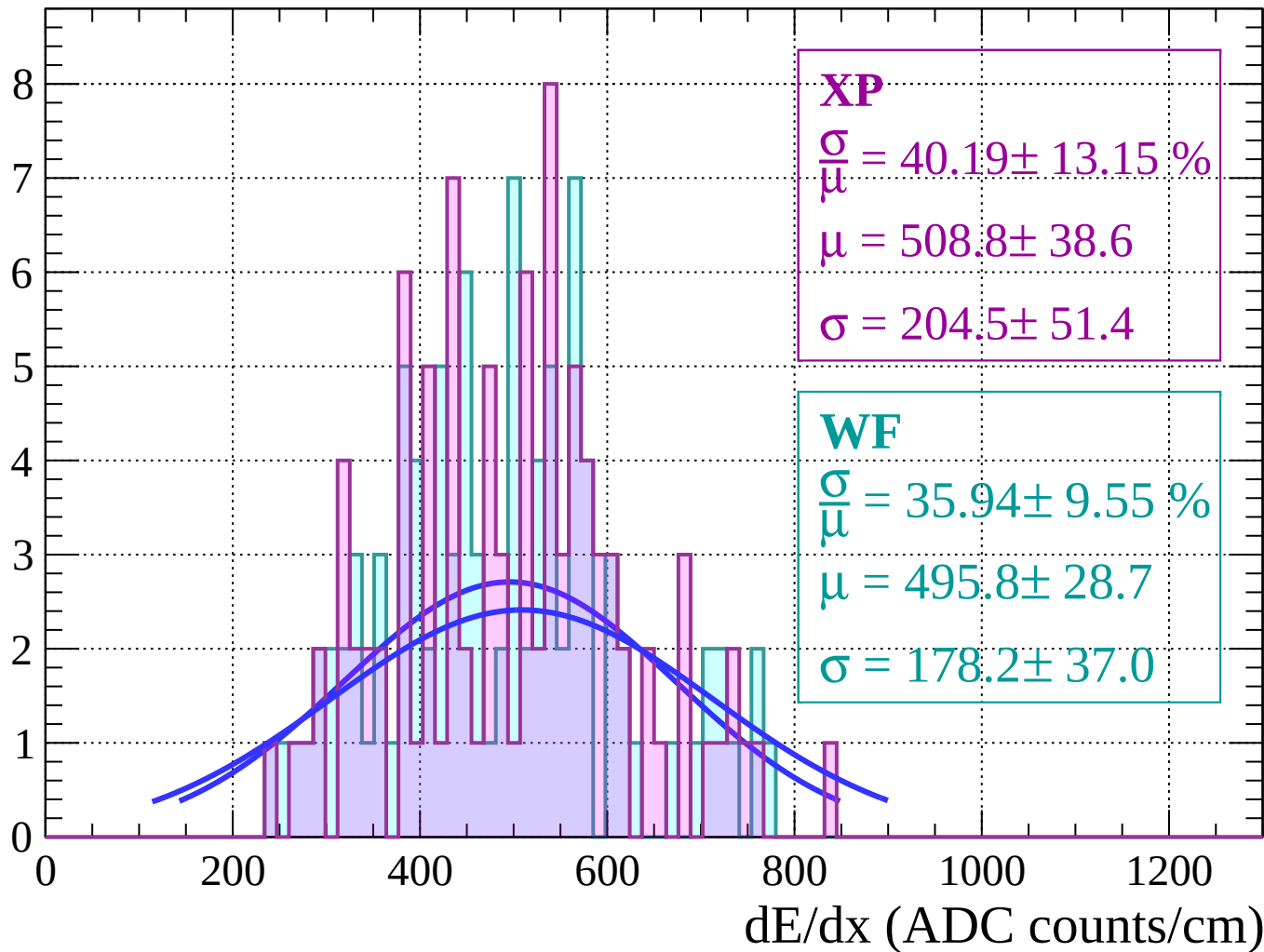
Energy loss | $-48 < \theta < -46$

Count



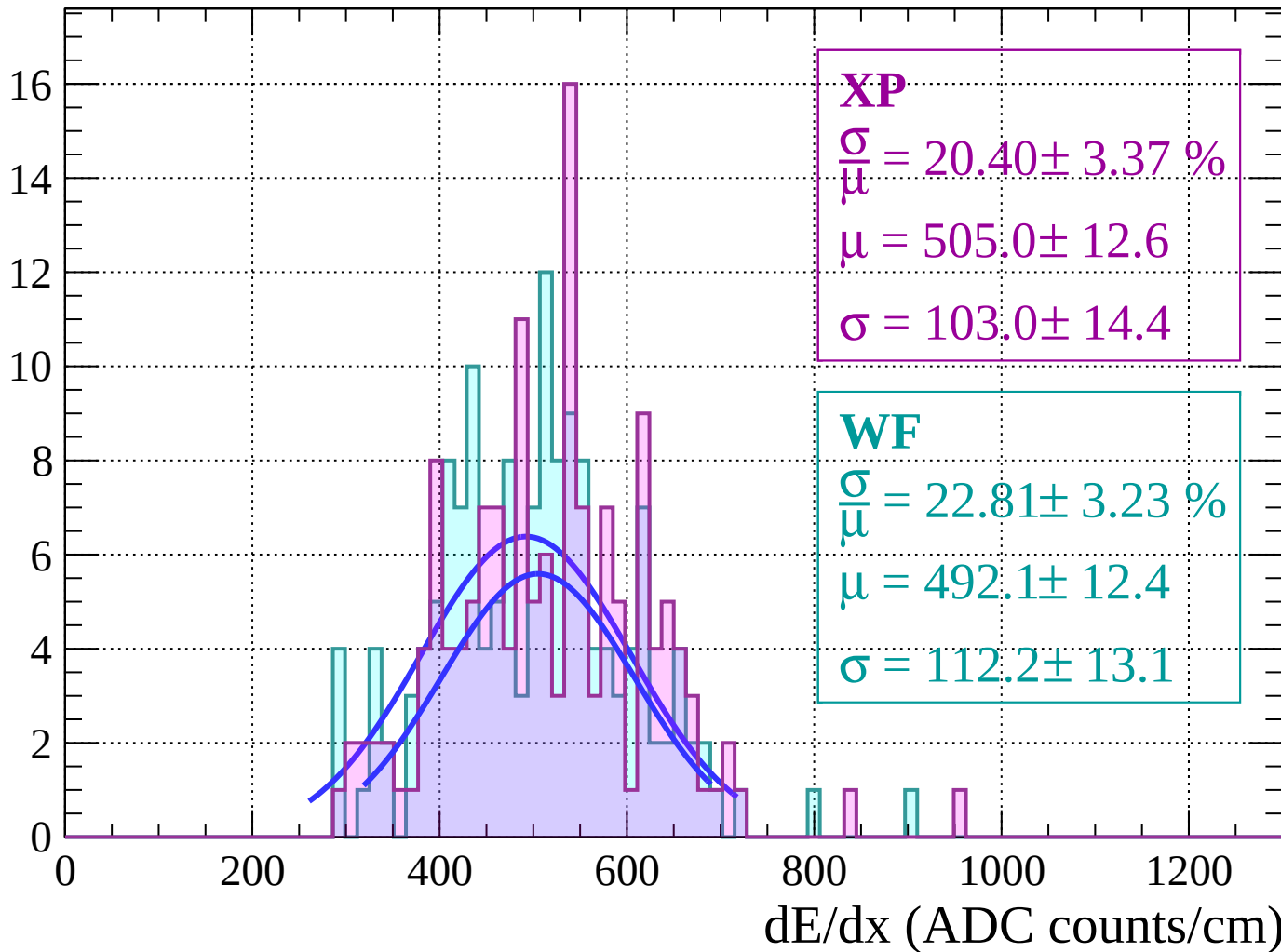
Energy loss | $-46 < \theta < -44$

Count



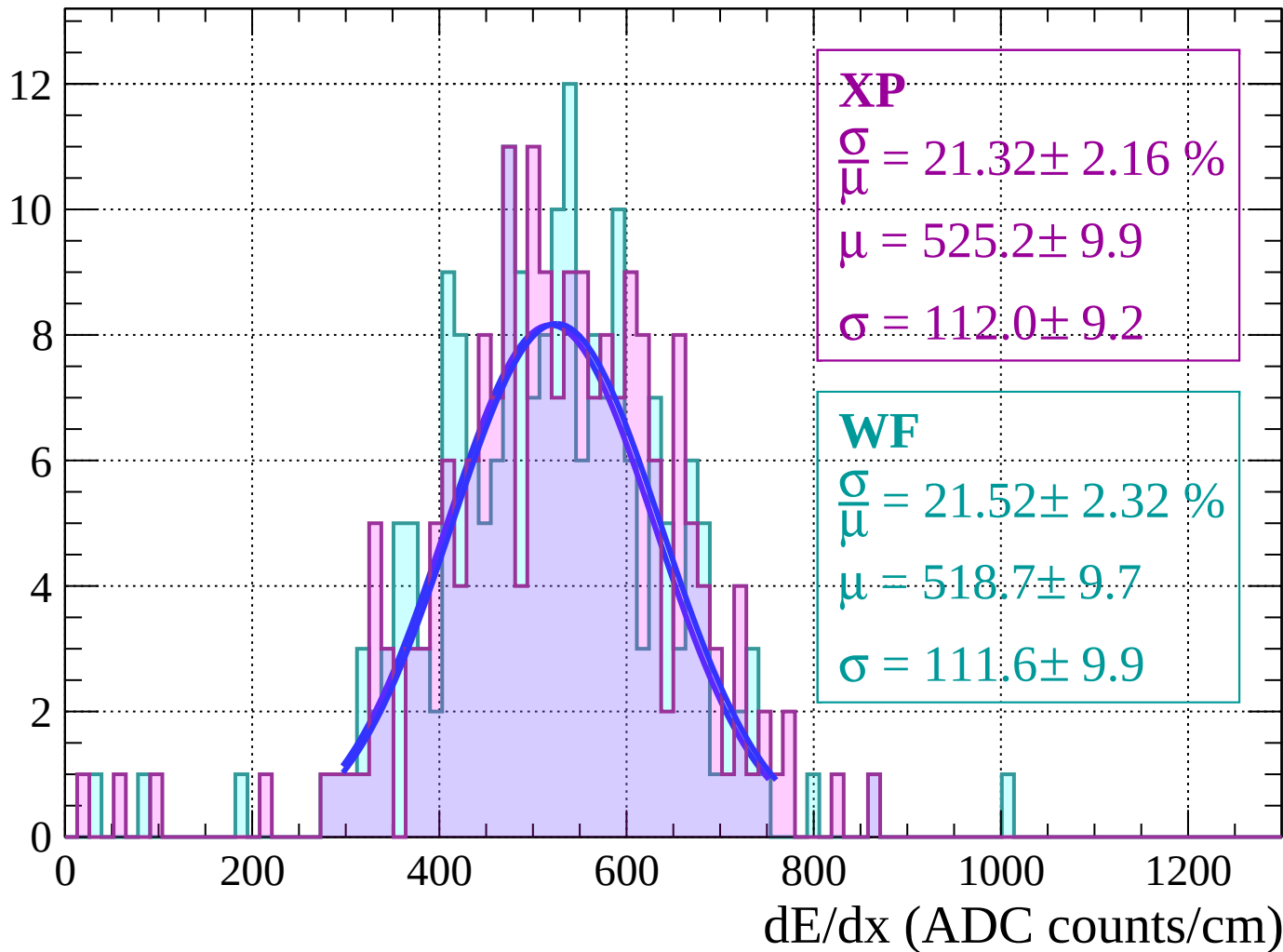
Energy loss | $-44 < \theta < -42$

Count



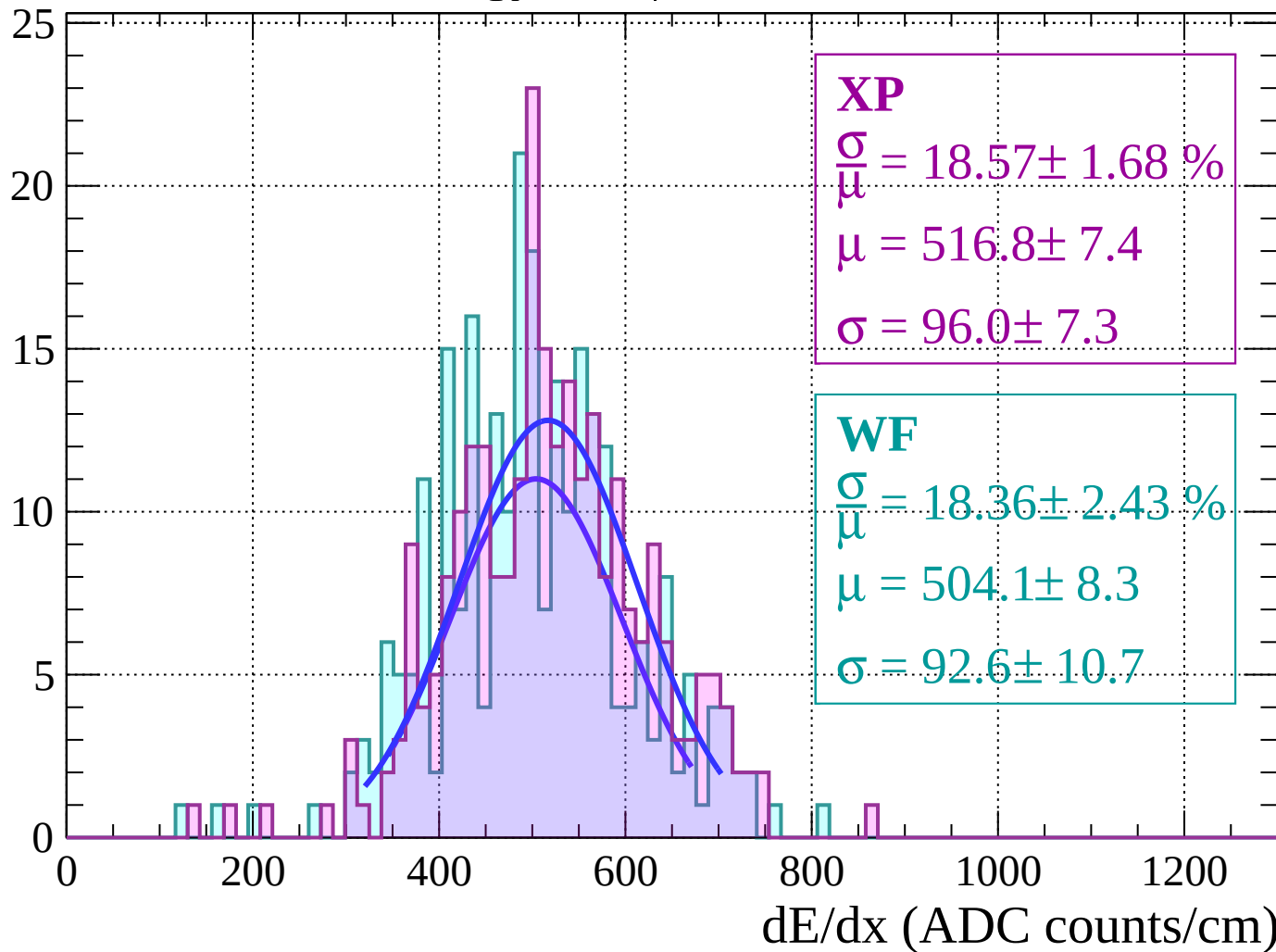
Energy loss | $-42 < \theta < -40$

Count



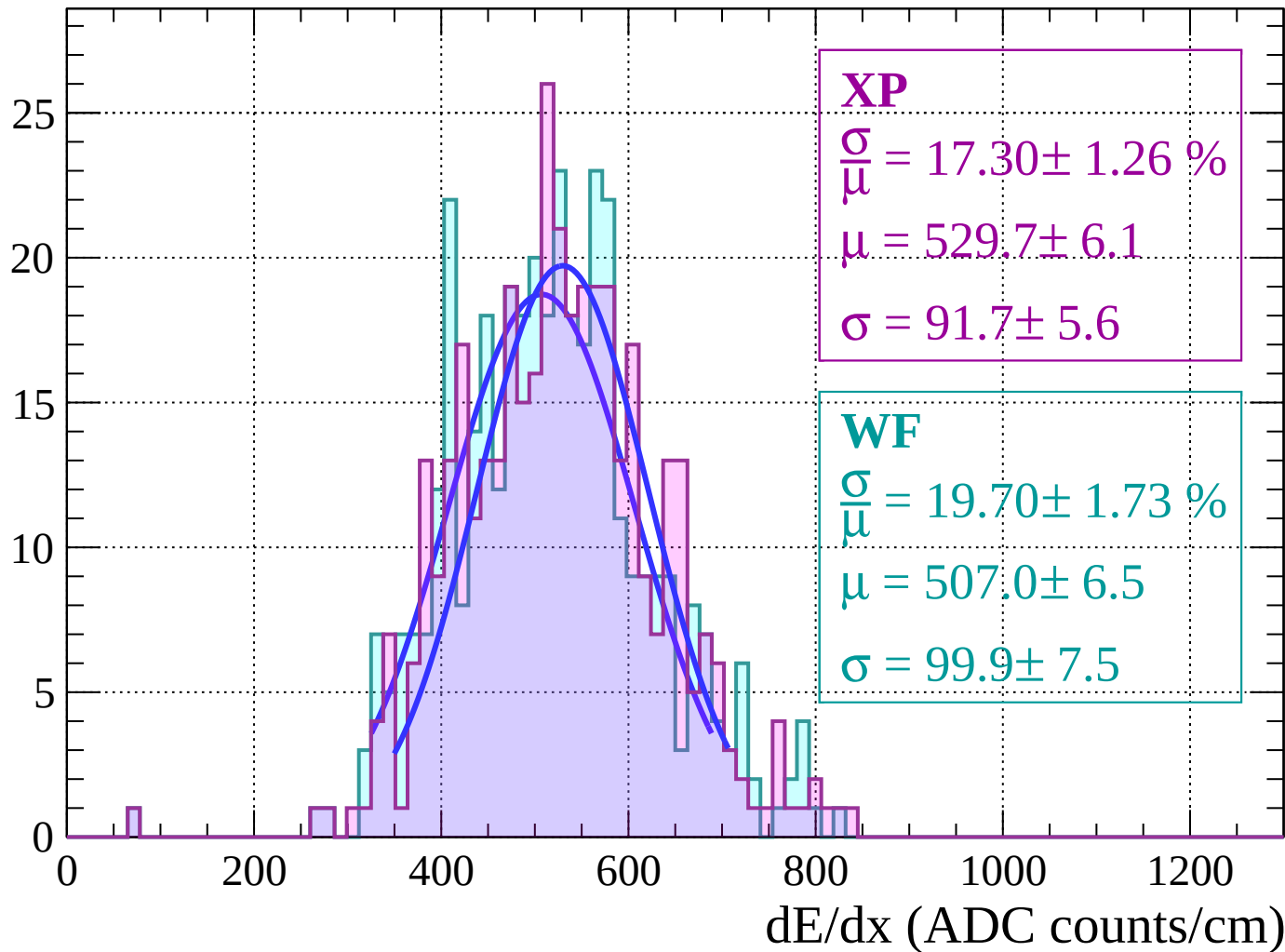
Energy loss | $-40 < \theta < -38$

Count



Energy loss $|-38 < \theta < -36$

Count



Energy loss $|-36 < \theta < -34$

Count

XP

$$\frac{\sigma}{\mu} = 17.69 \pm 0.97 \%$$

$$\mu = 528.2 \pm 5.0$$

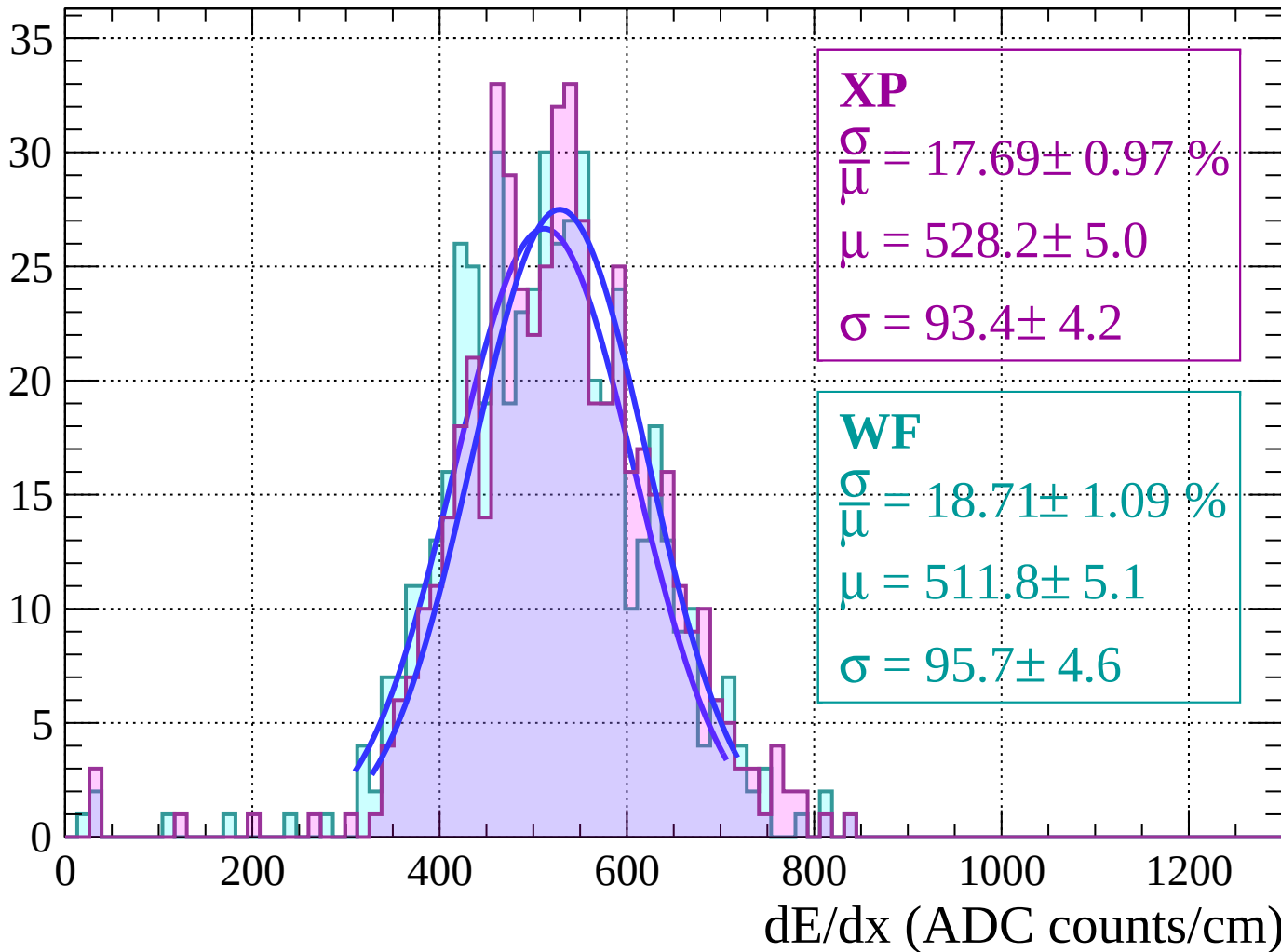
$$\sigma = 93.4 \pm 4.2$$

WF

$$\frac{\sigma}{\mu} = 18.71 \pm 1.09 \%$$

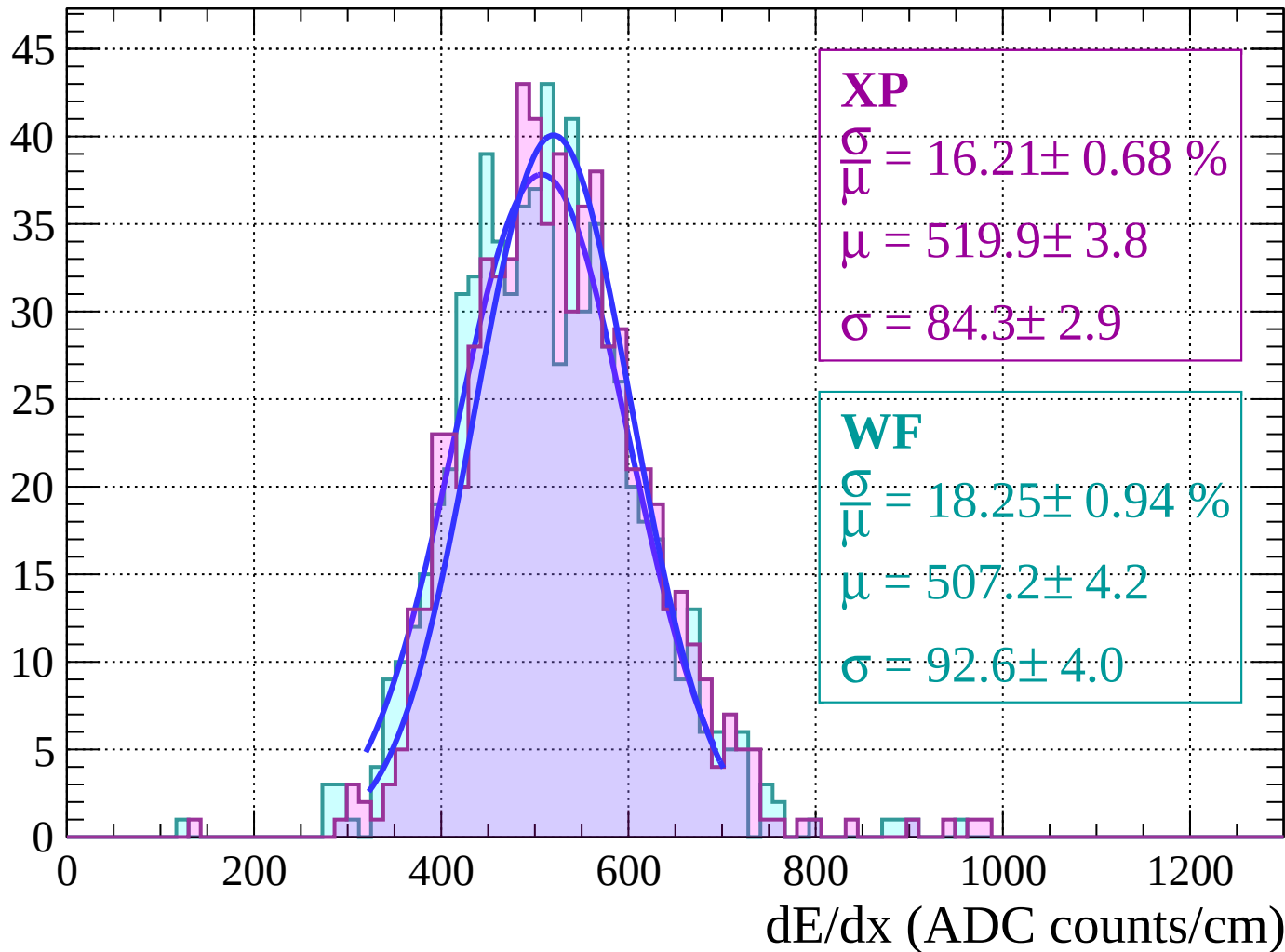
$$\mu = 511.8 \pm 5.1$$

$$\sigma = 95.7 \pm 4.6$$



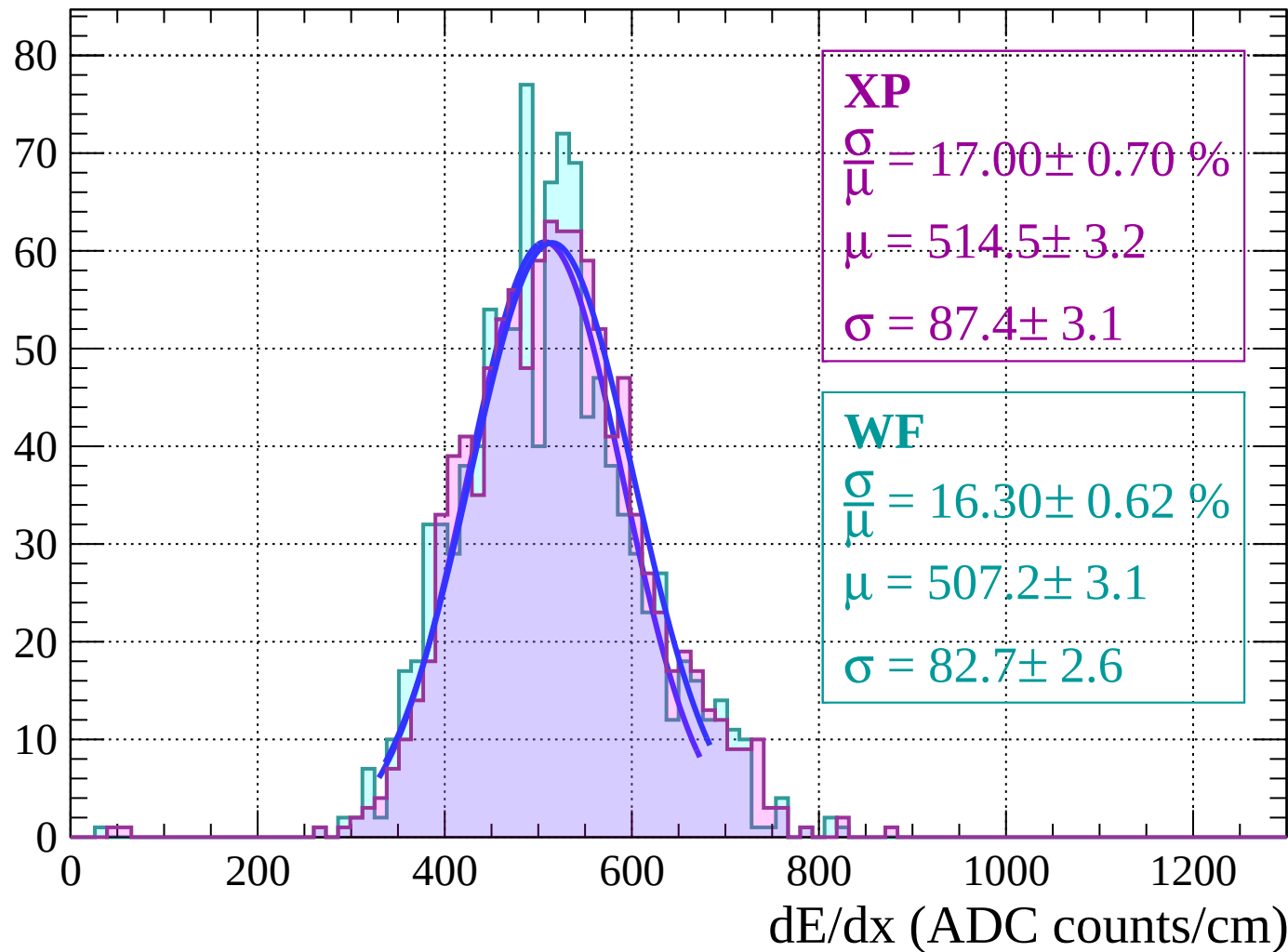
Energy loss $|-34 < \theta < -32$

Count



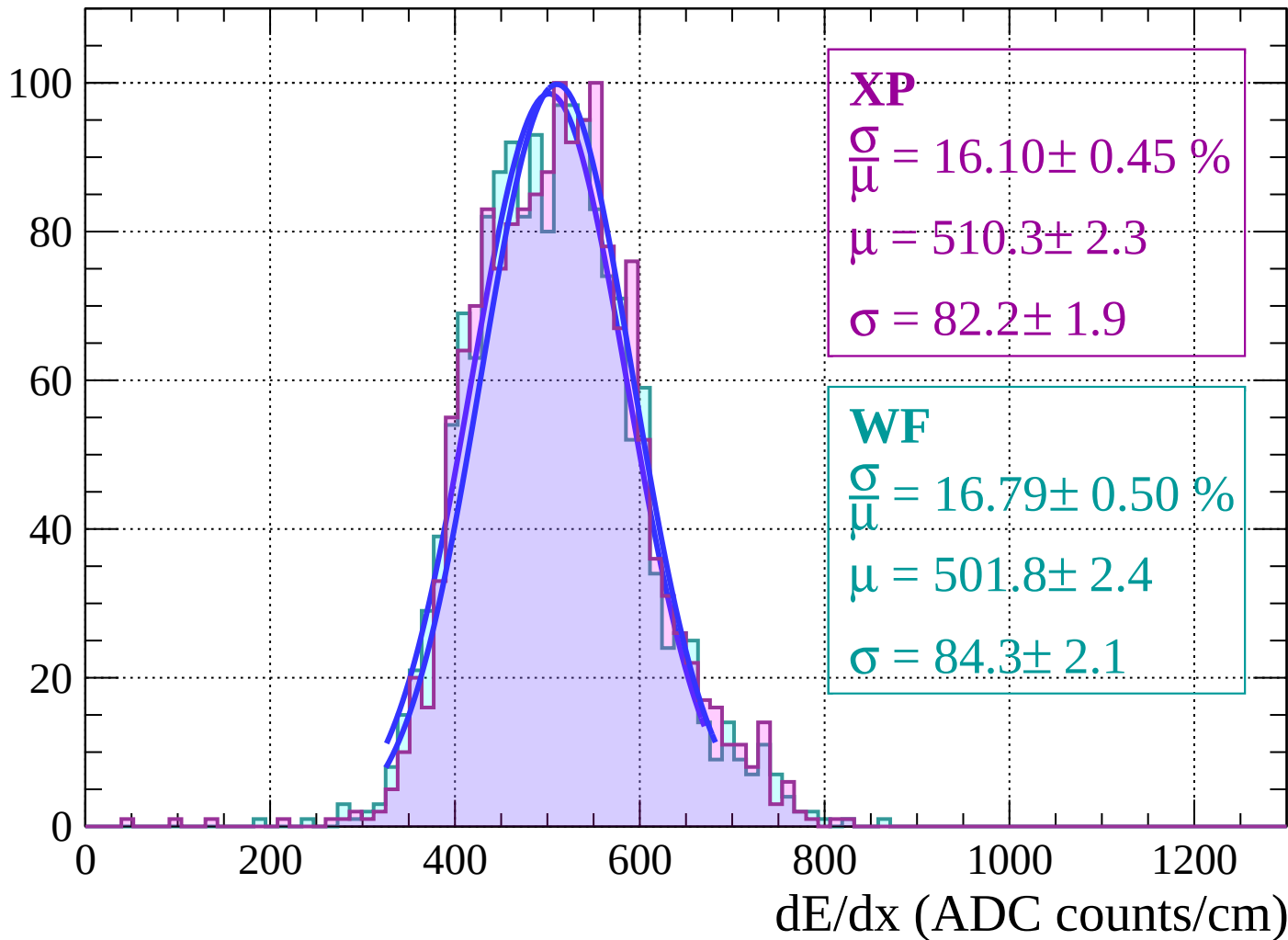
Energy loss | $-32 < \theta < -30$

Count



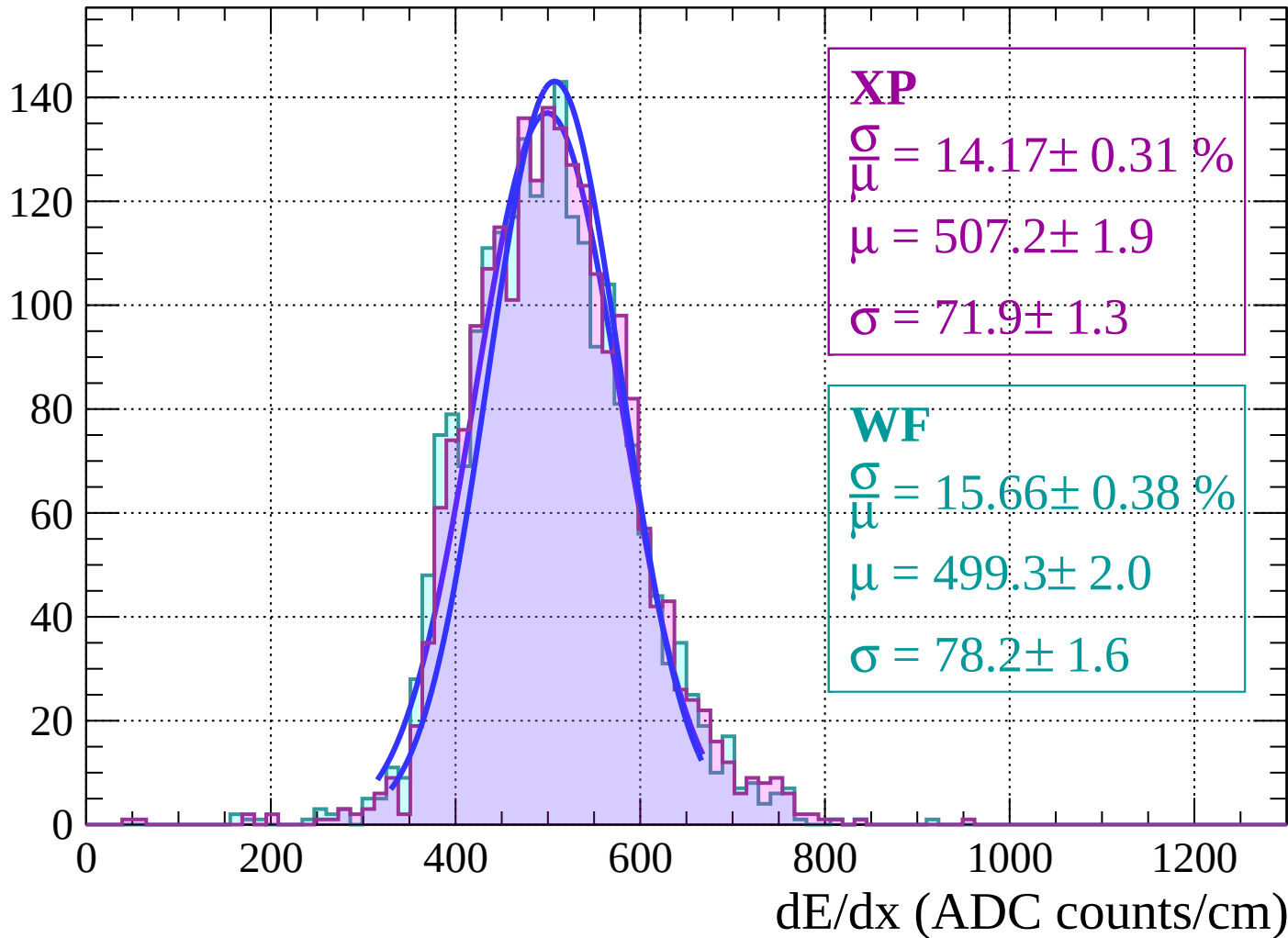
Energy loss | $-30 < \theta < -28$

Count

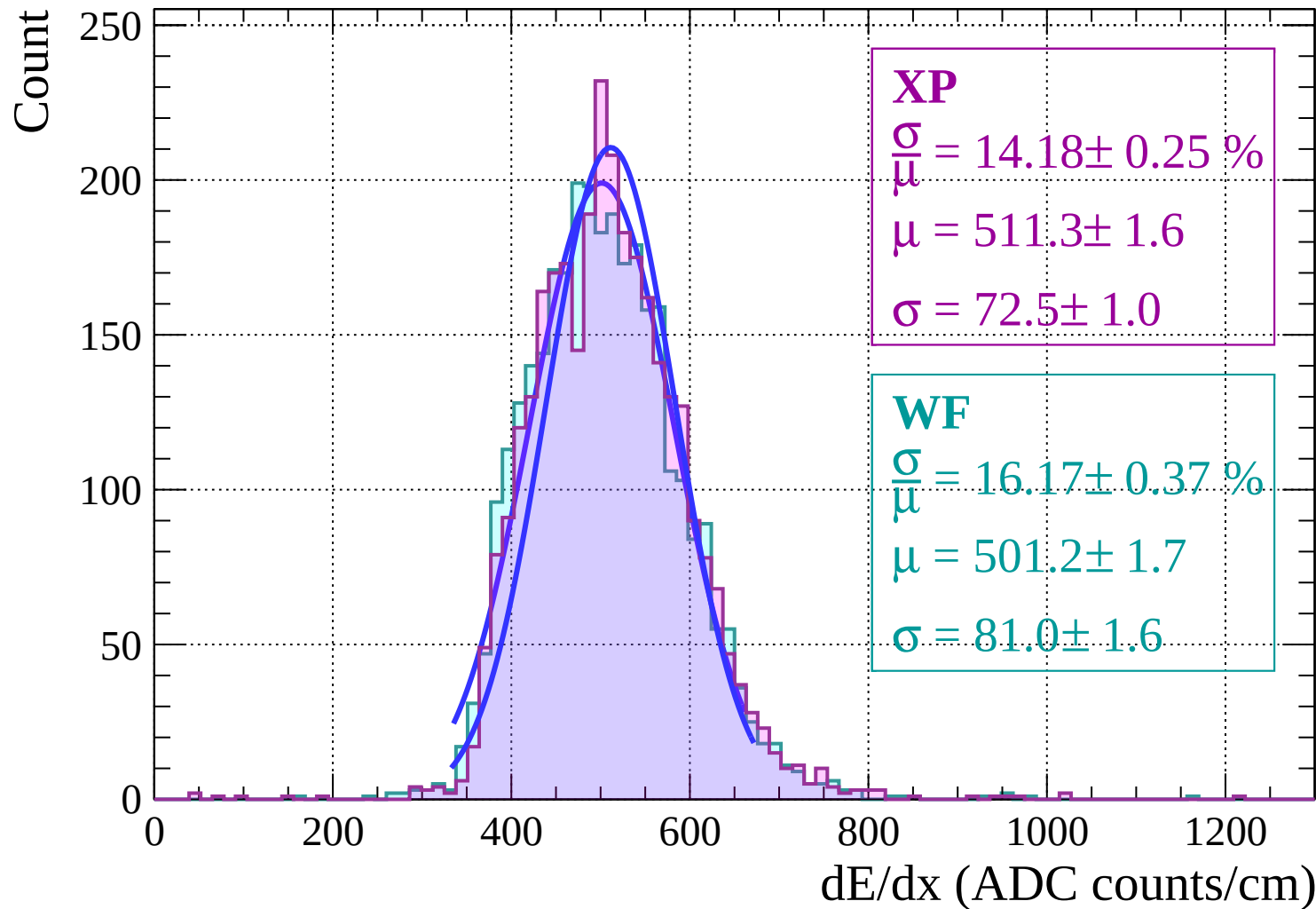


Energy loss $|-28 < \theta < -26$

Count

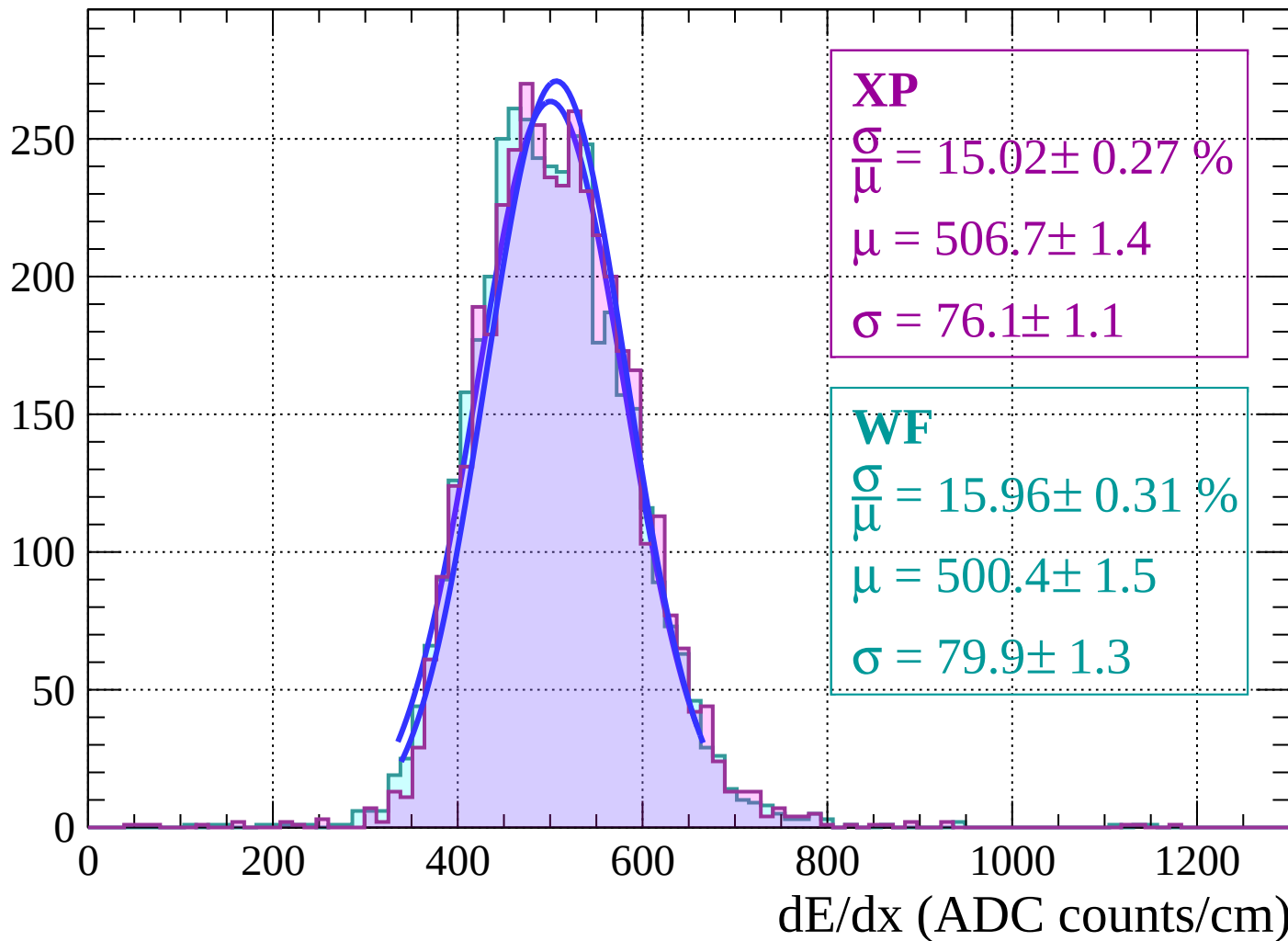


Energy loss | $-26 < \theta < -24$

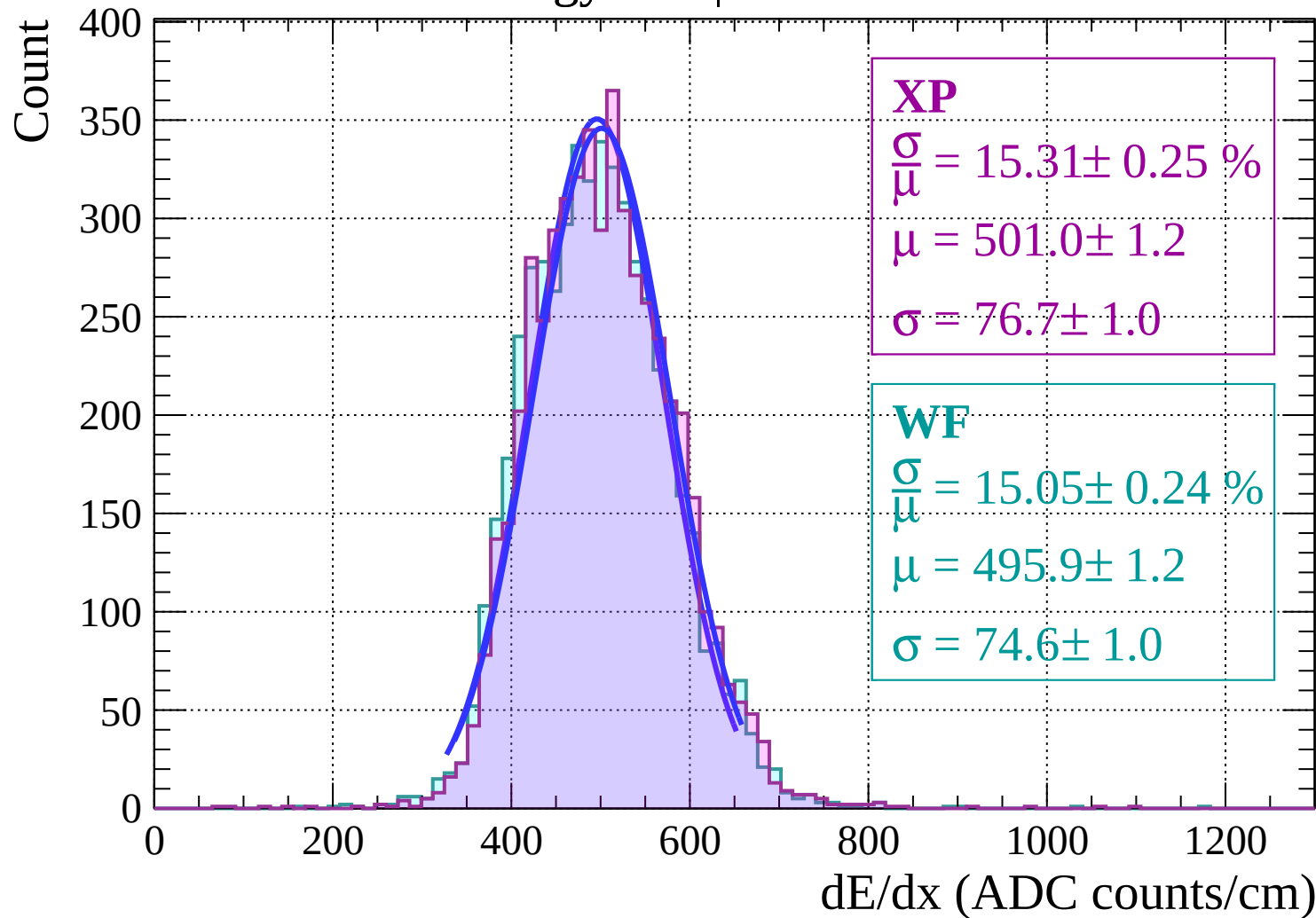


Energy loss $|-24 < \theta < -22$

Count

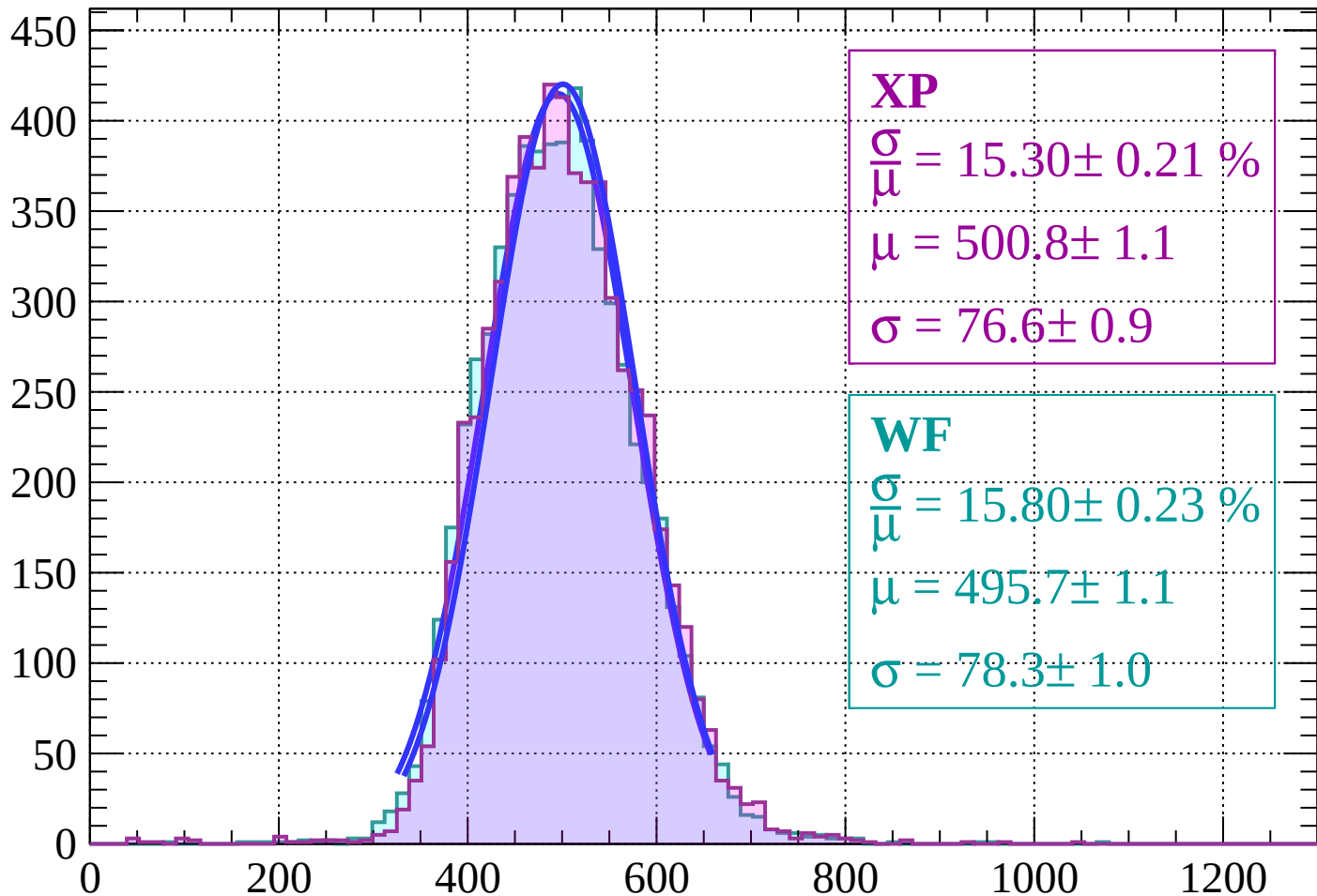


Energy loss $|-22 < \theta < -20$



Energy loss | $-20 < \theta < -18$

Count



XP

$$\frac{\sigma}{\mu} = 15.30 \pm 0.21 \%$$

$$\mu = 500.8 \pm 1.1$$

$$\sigma = 76.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 15.80 \pm 0.23 \%$$

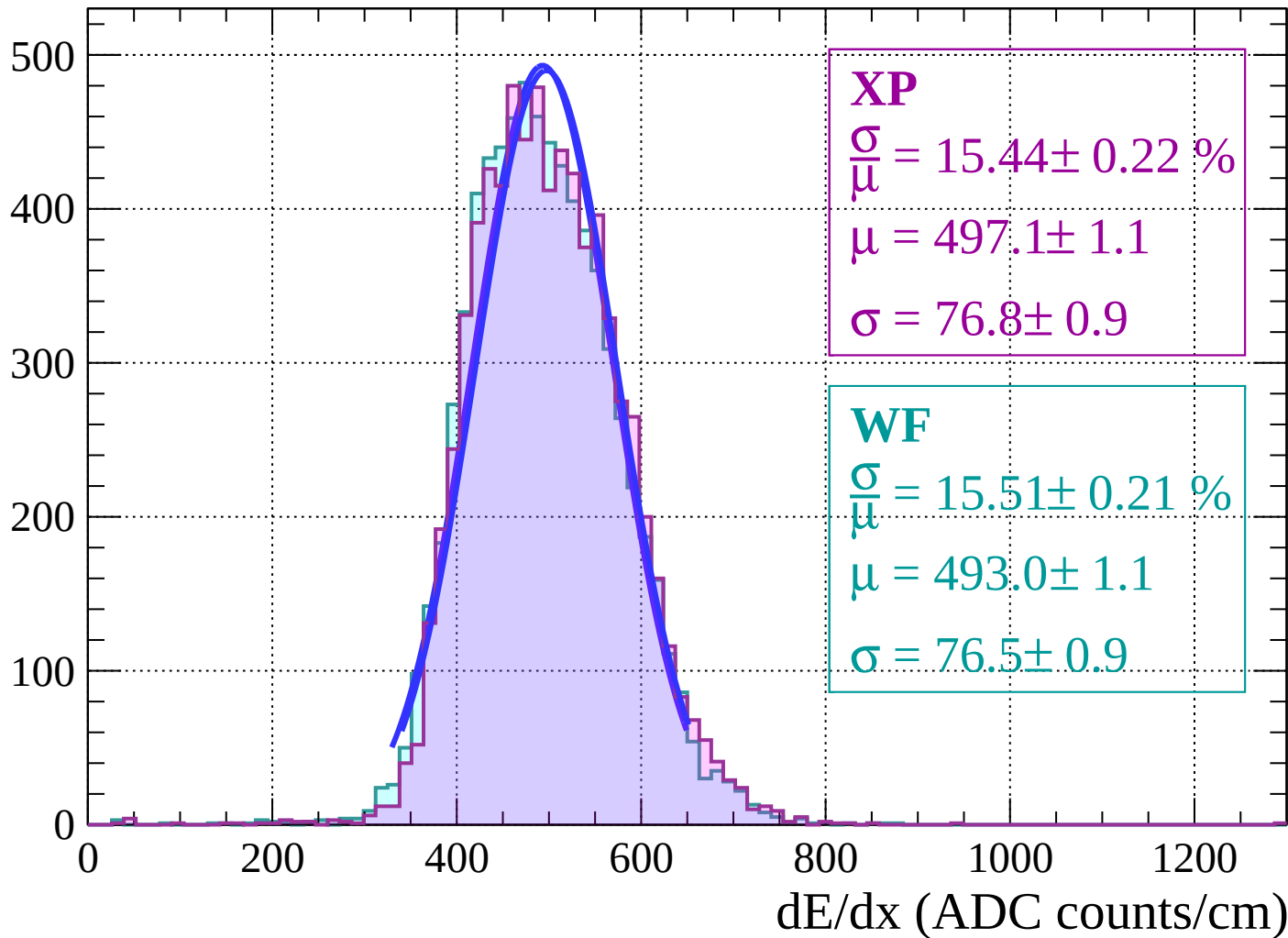
$$\mu = 495.7 \pm 1.1$$

$$\sigma = 78.3 \pm 1.0$$

dE/dx (ADC counts/cm)

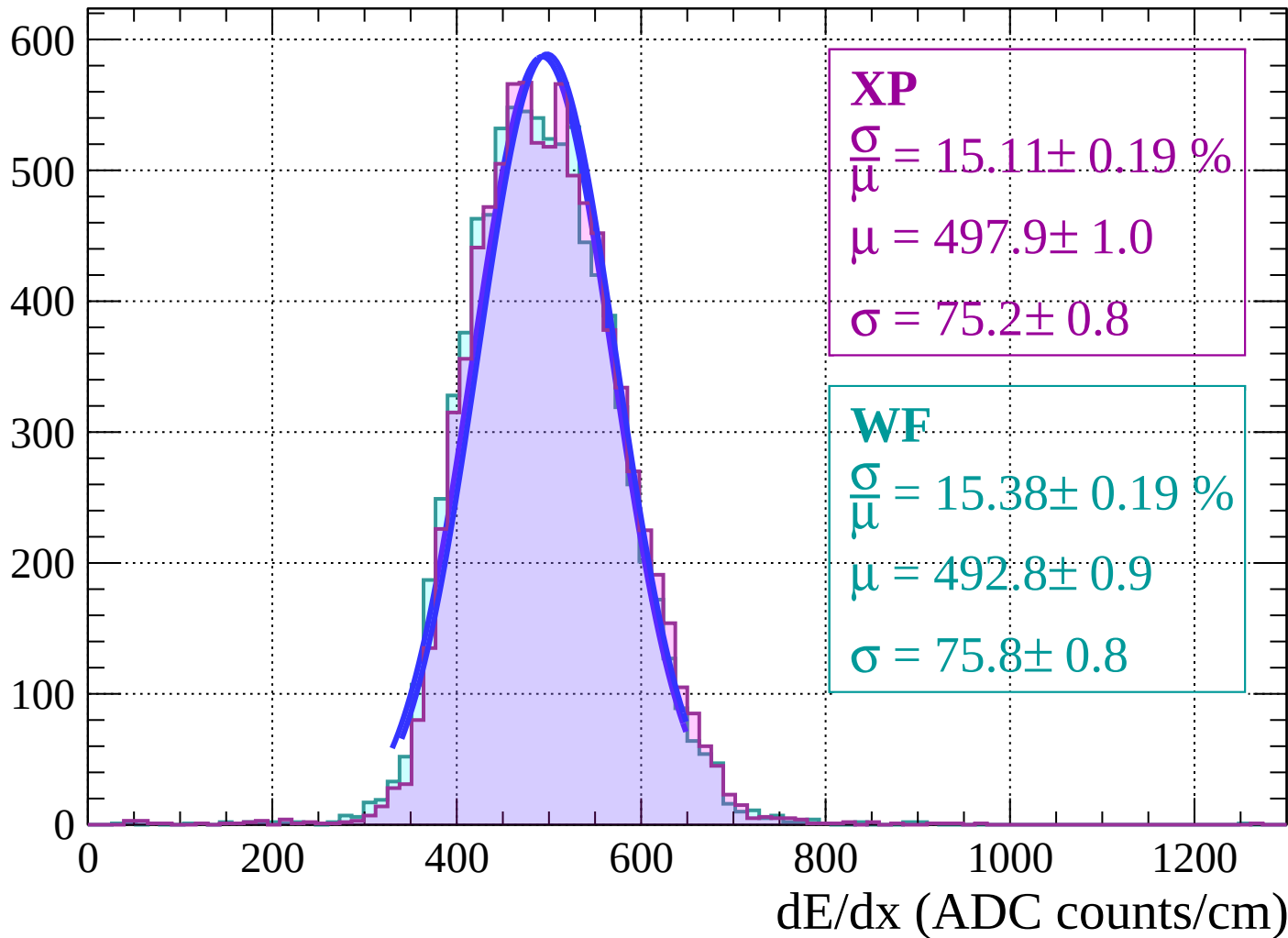
Energy loss | $-18 < \theta < -16$

Count

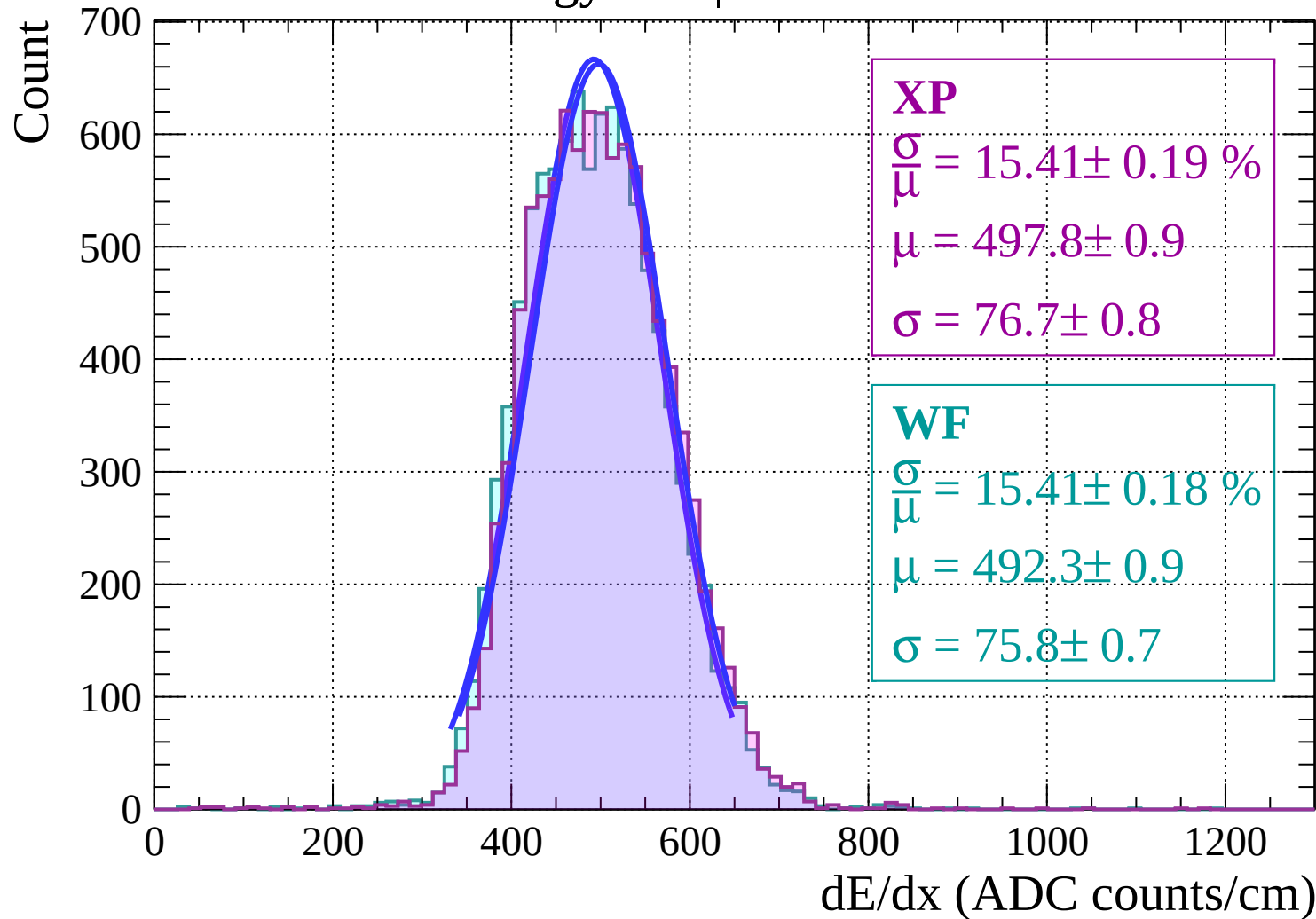


Energy loss | $-16 < \theta < -14$

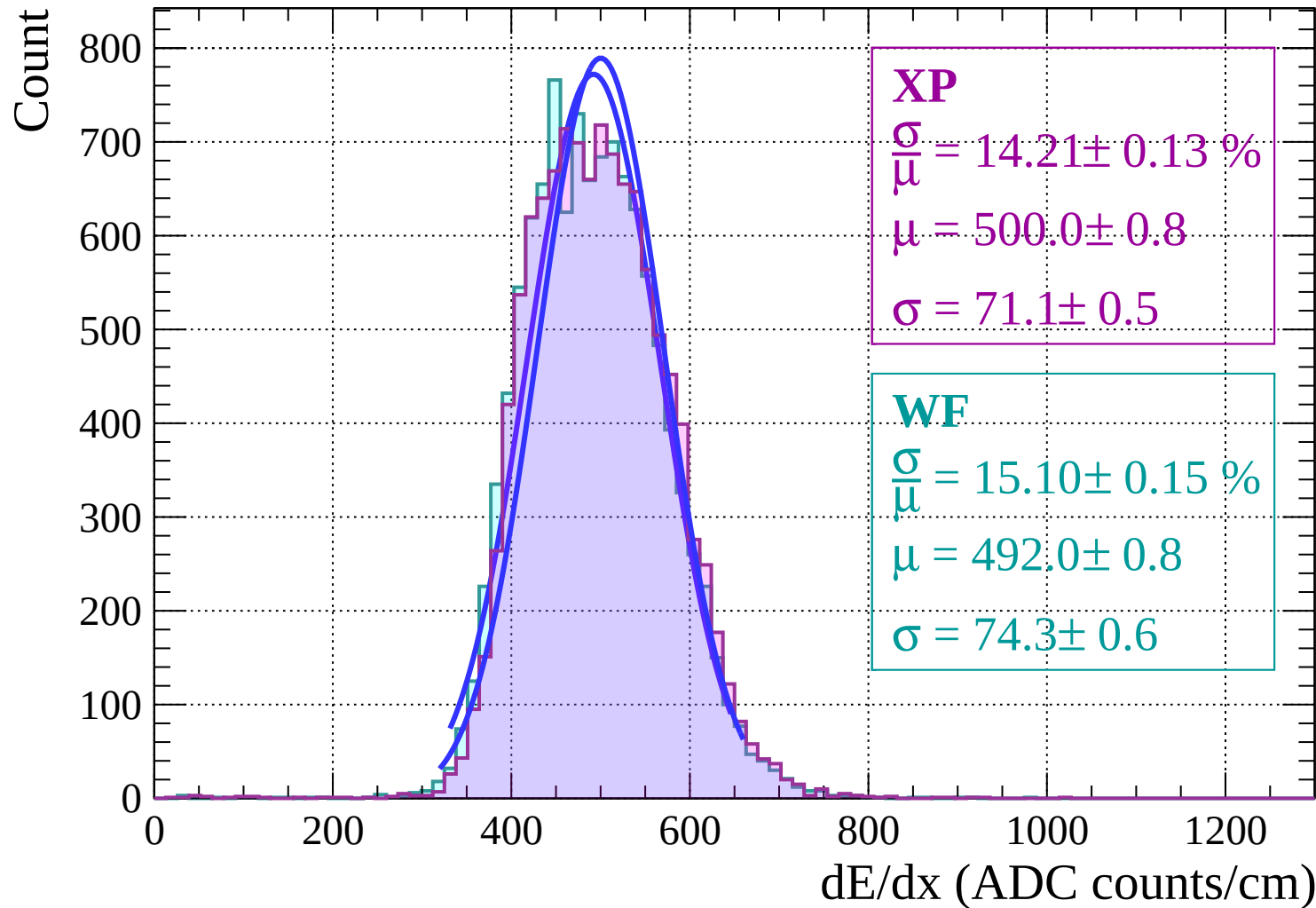
Count



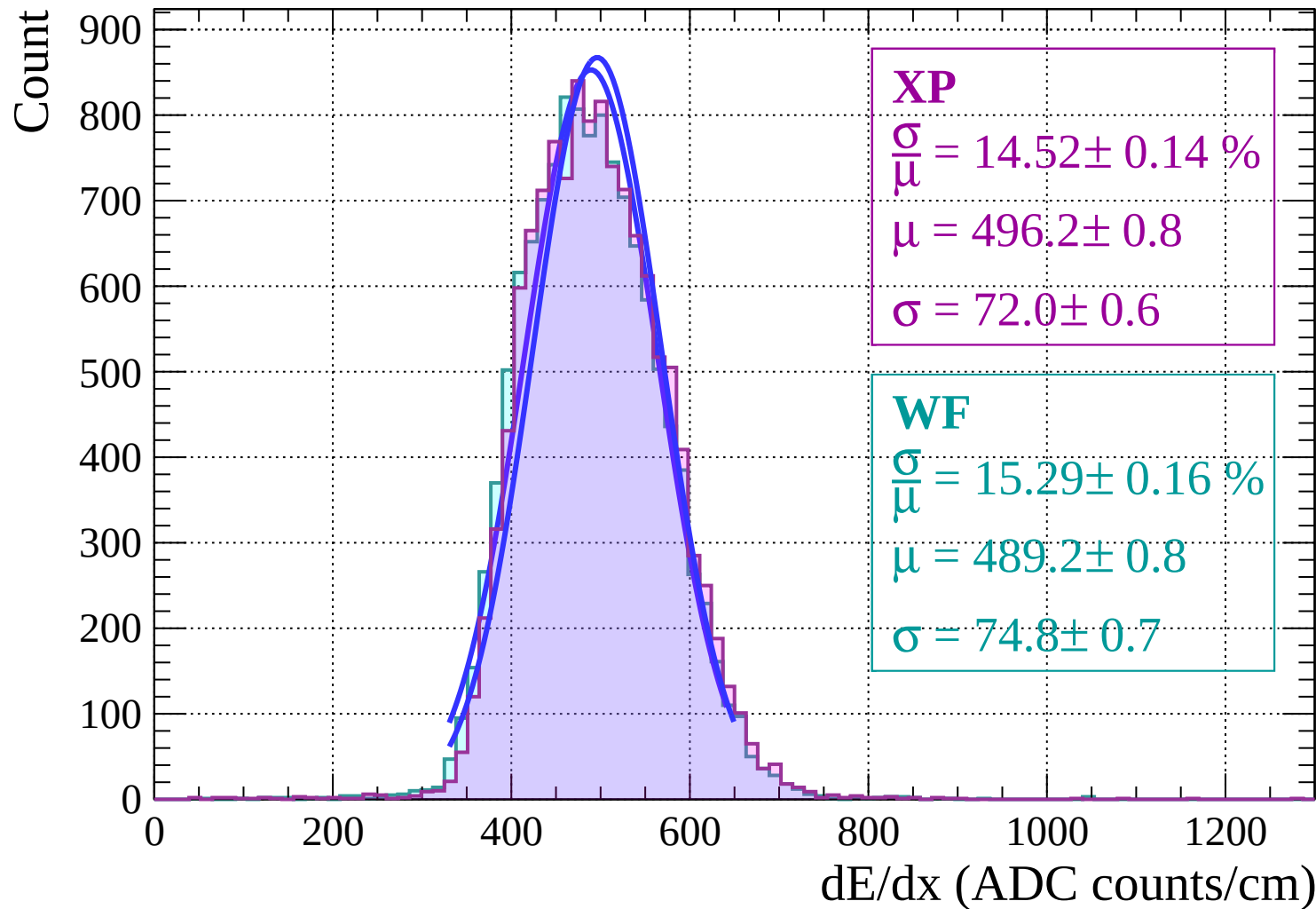
Energy loss $|-14 < \theta < -12$



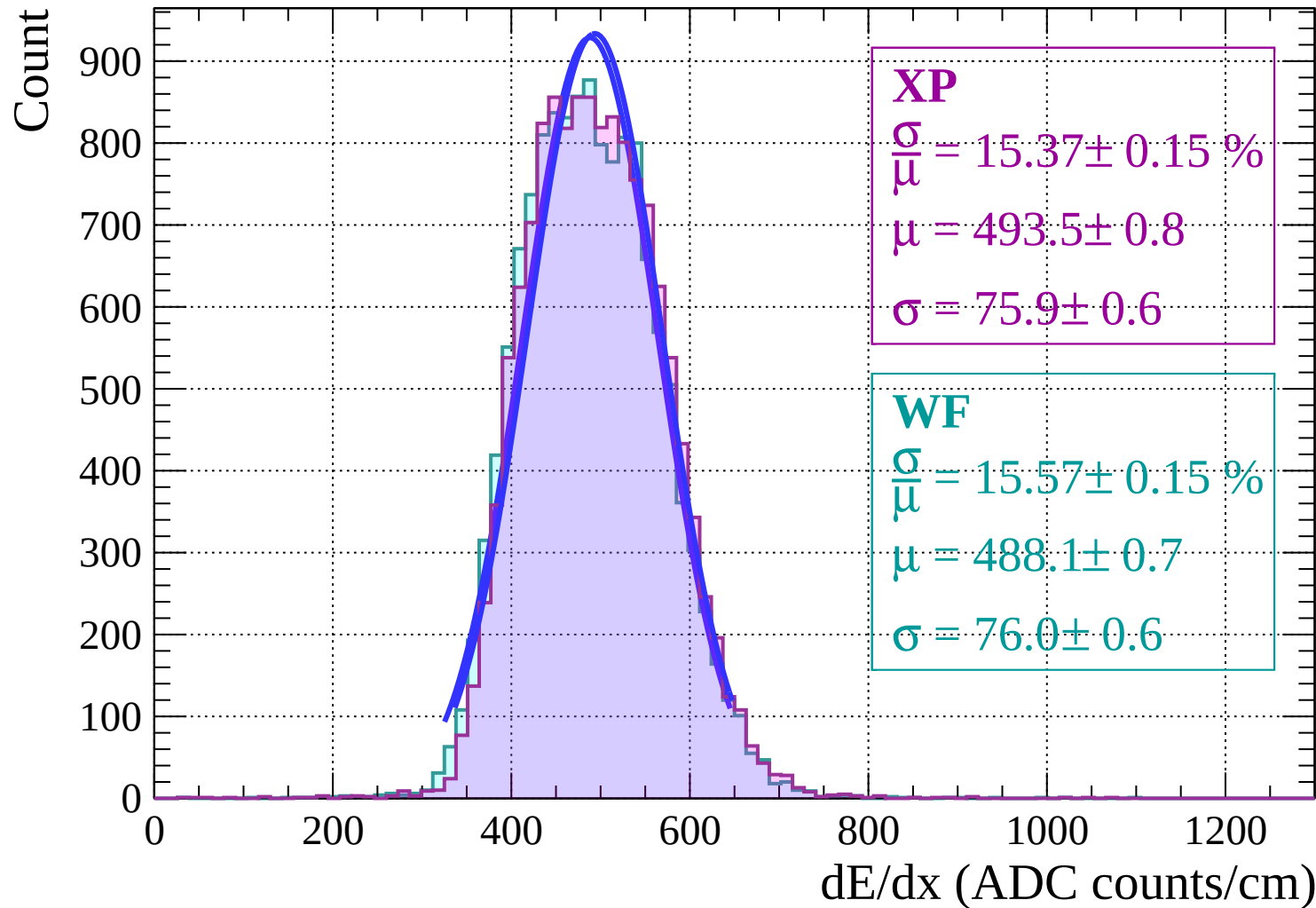
Energy loss | $-12 < \theta < -10$



Energy loss | $-10 < \theta < -8$

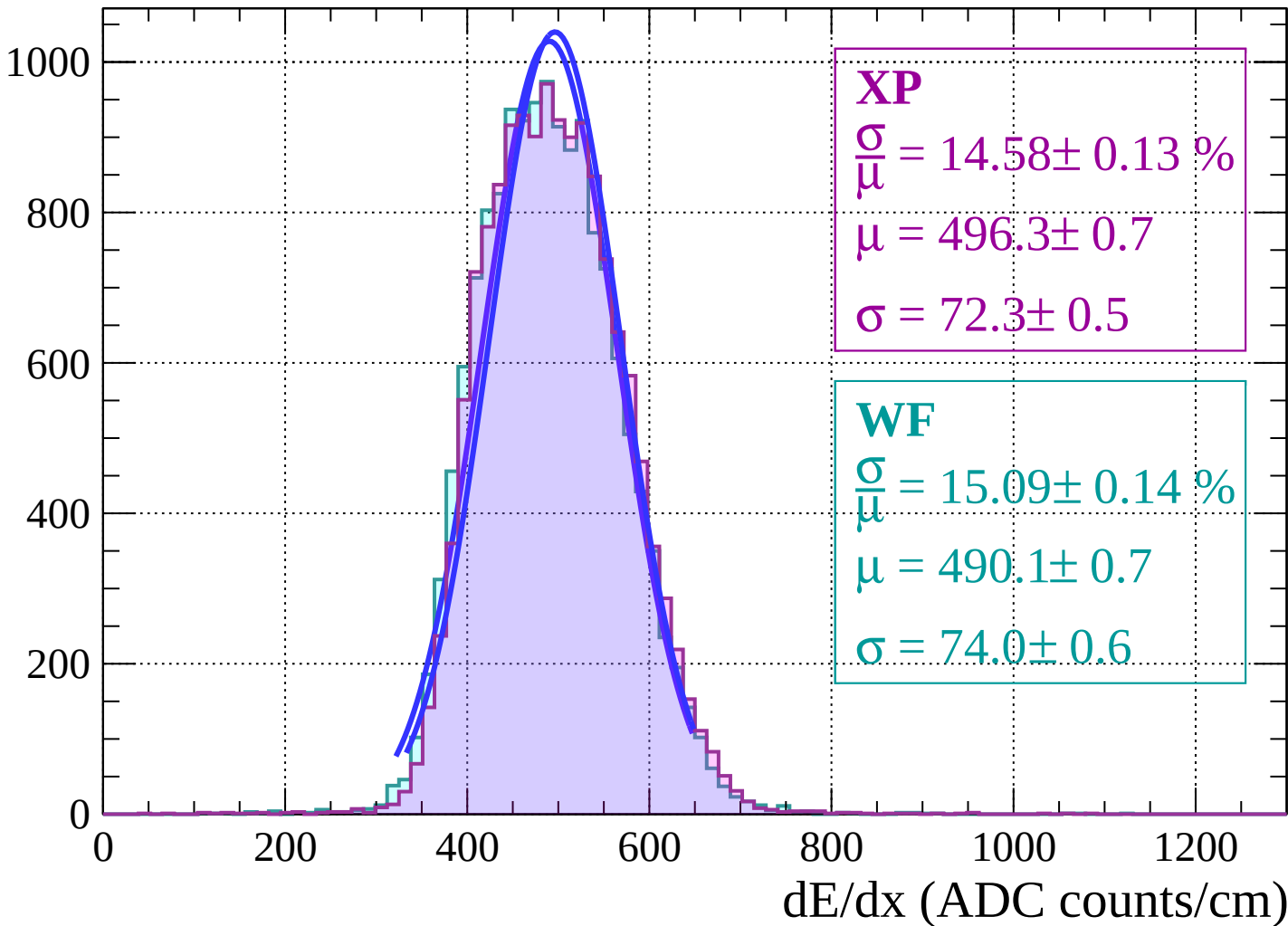


Energy loss | $-8 < \theta < -6$



Energy loss | $-6 < \theta < -4$

Count



Energy loss | $-4 < \theta < -2$

Count

1000

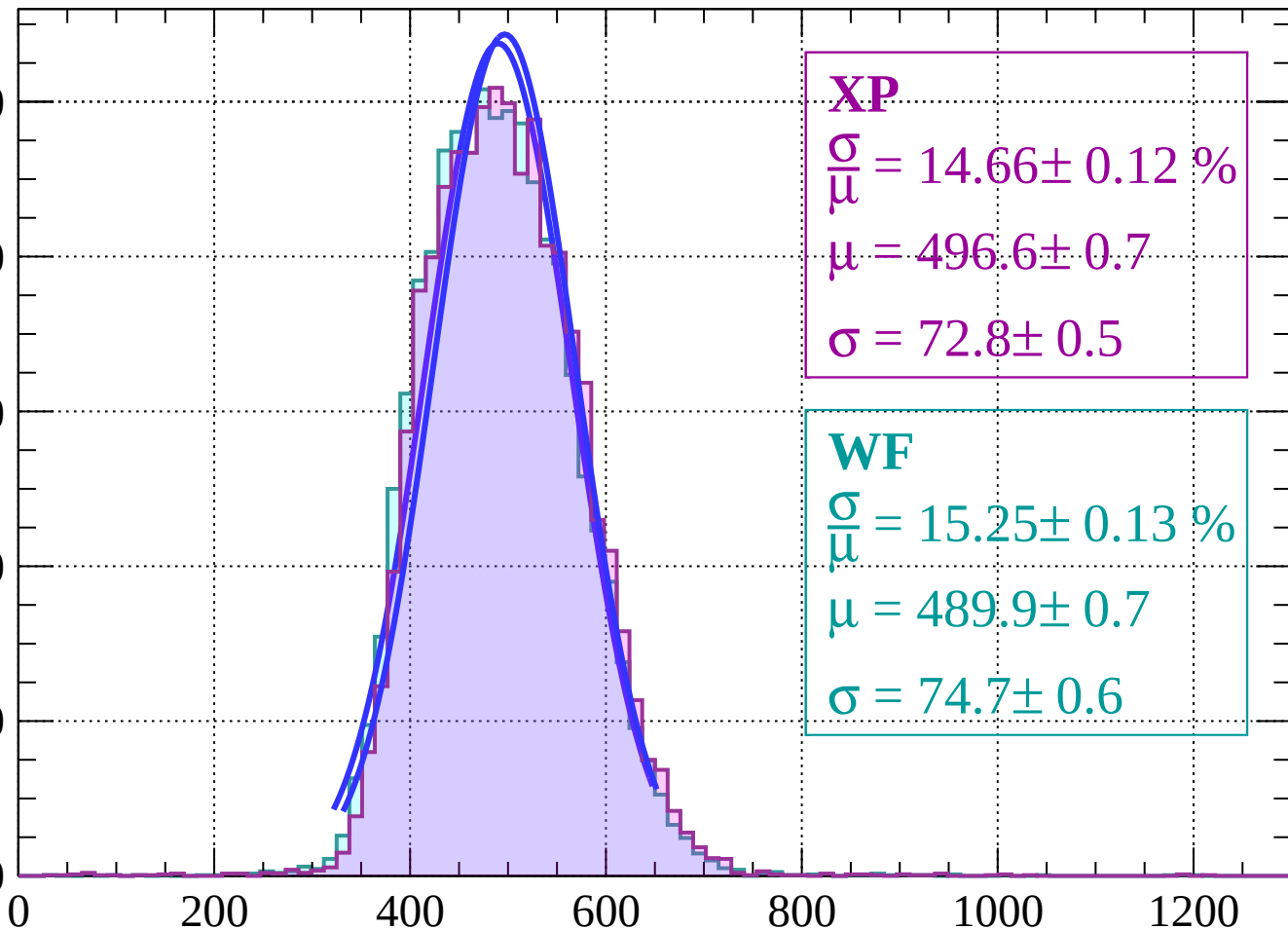
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.66 \pm 0.12 \%$$

$$\mu = 496.6 \pm 0.7$$

$$\sigma = 72.8 \pm 0.5$$

WF

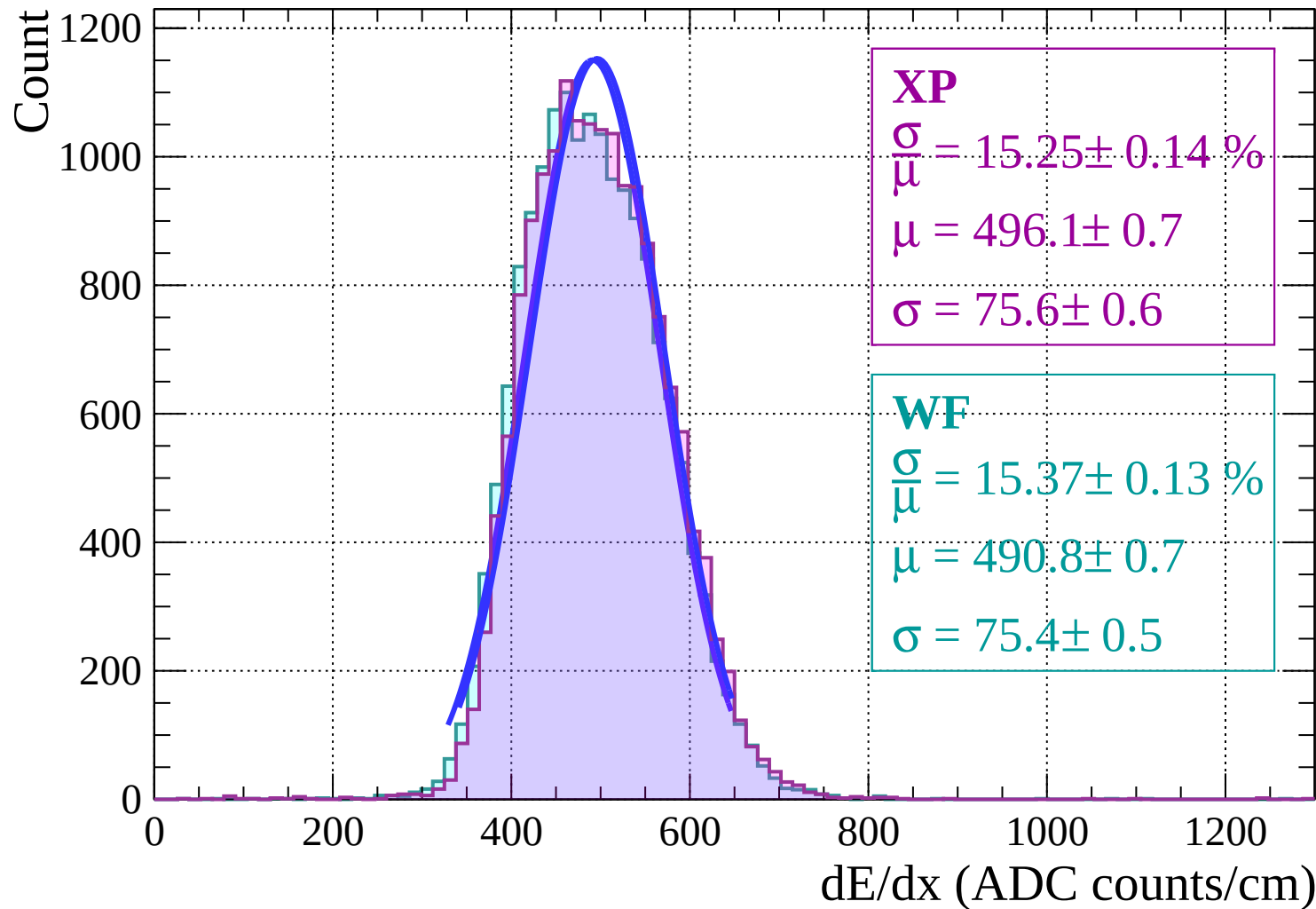
$$\frac{\sigma}{\mu} = 15.25 \pm 0.13 \%$$

$$\mu = 489.9 \pm 0.7$$

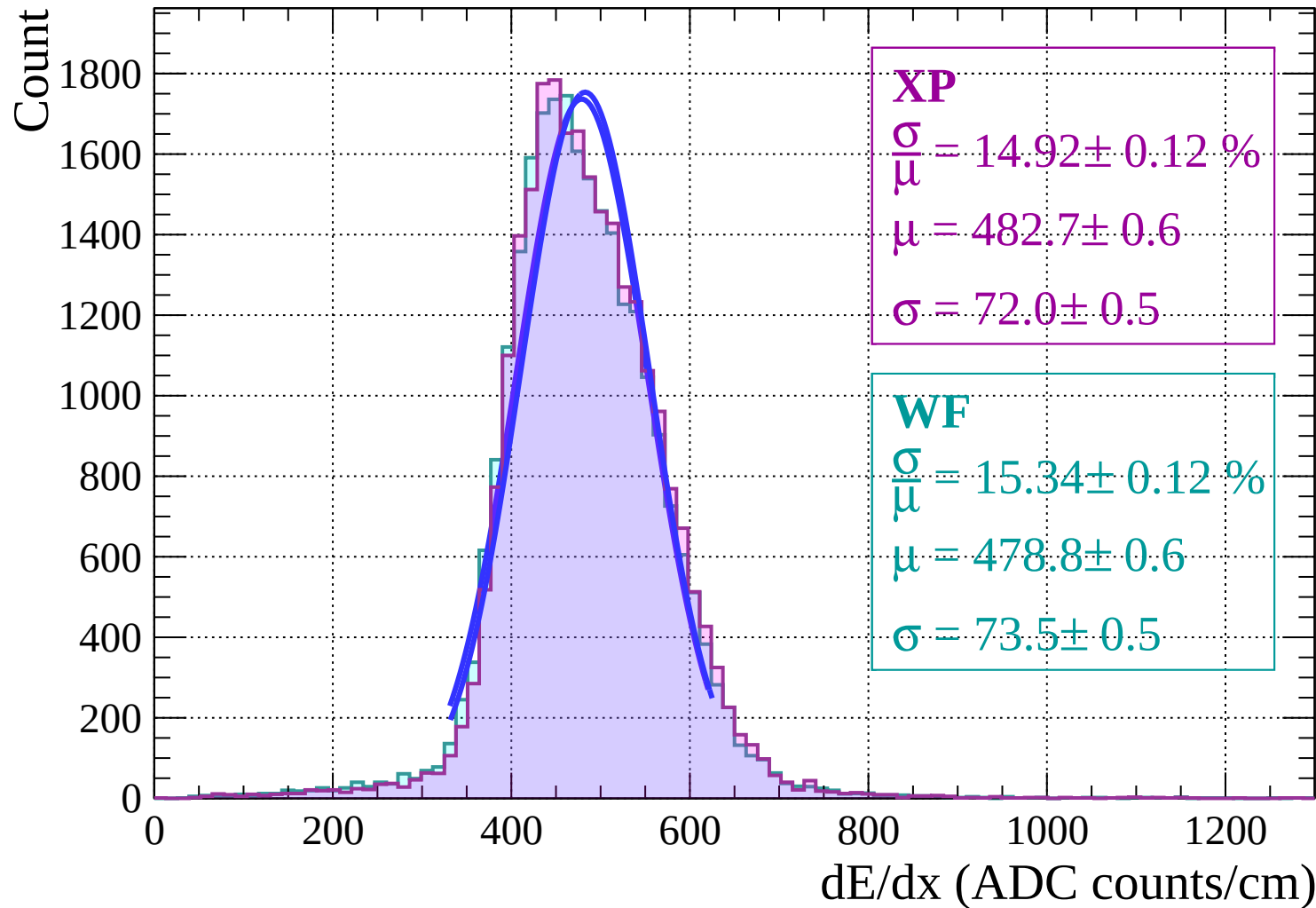
$$\sigma = 74.7 \pm 0.6$$

dE/dx (ADC counts/cm)

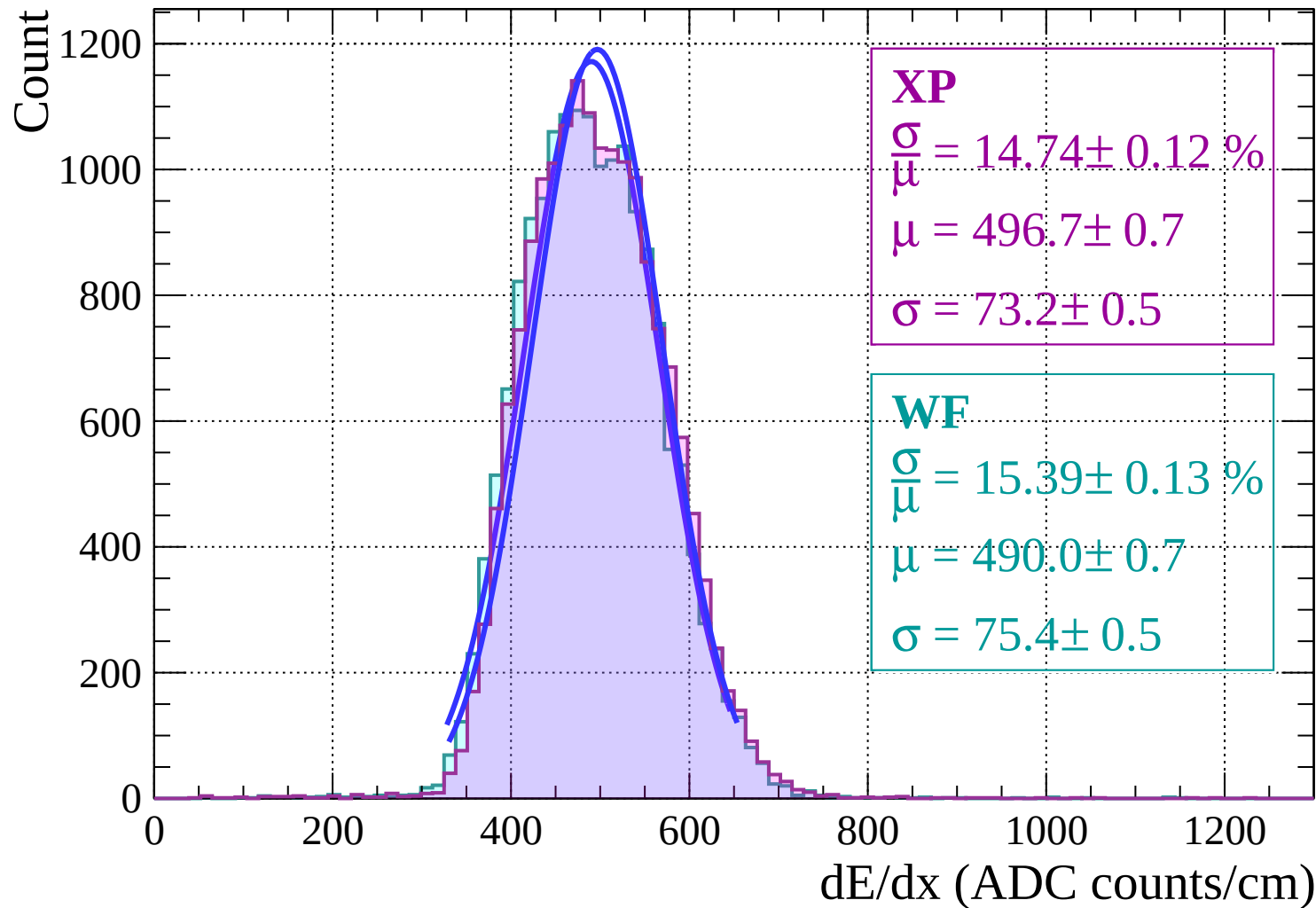
Energy loss | $-2 < \theta < 0$



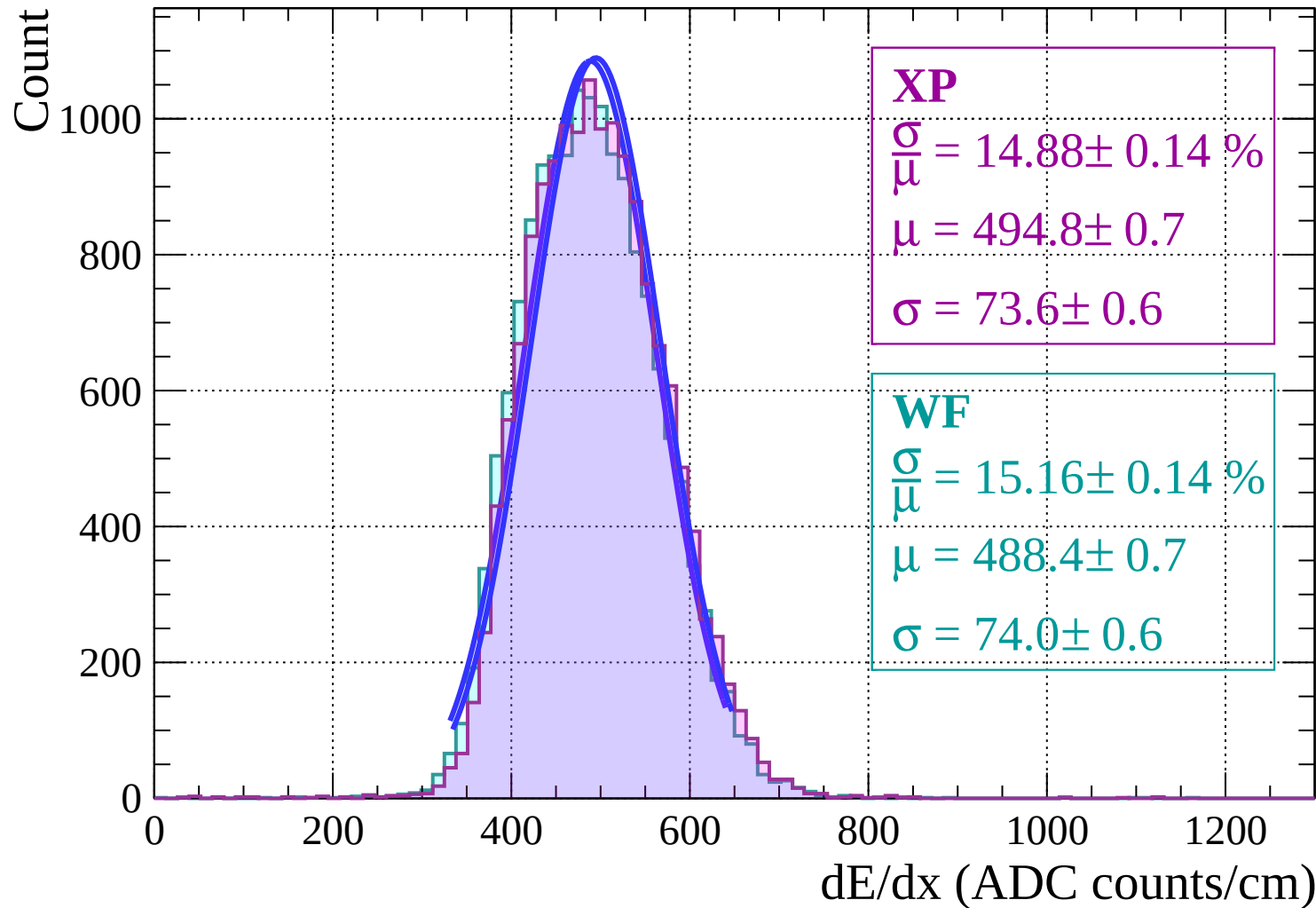
Energy loss | $0 < \theta < 2$



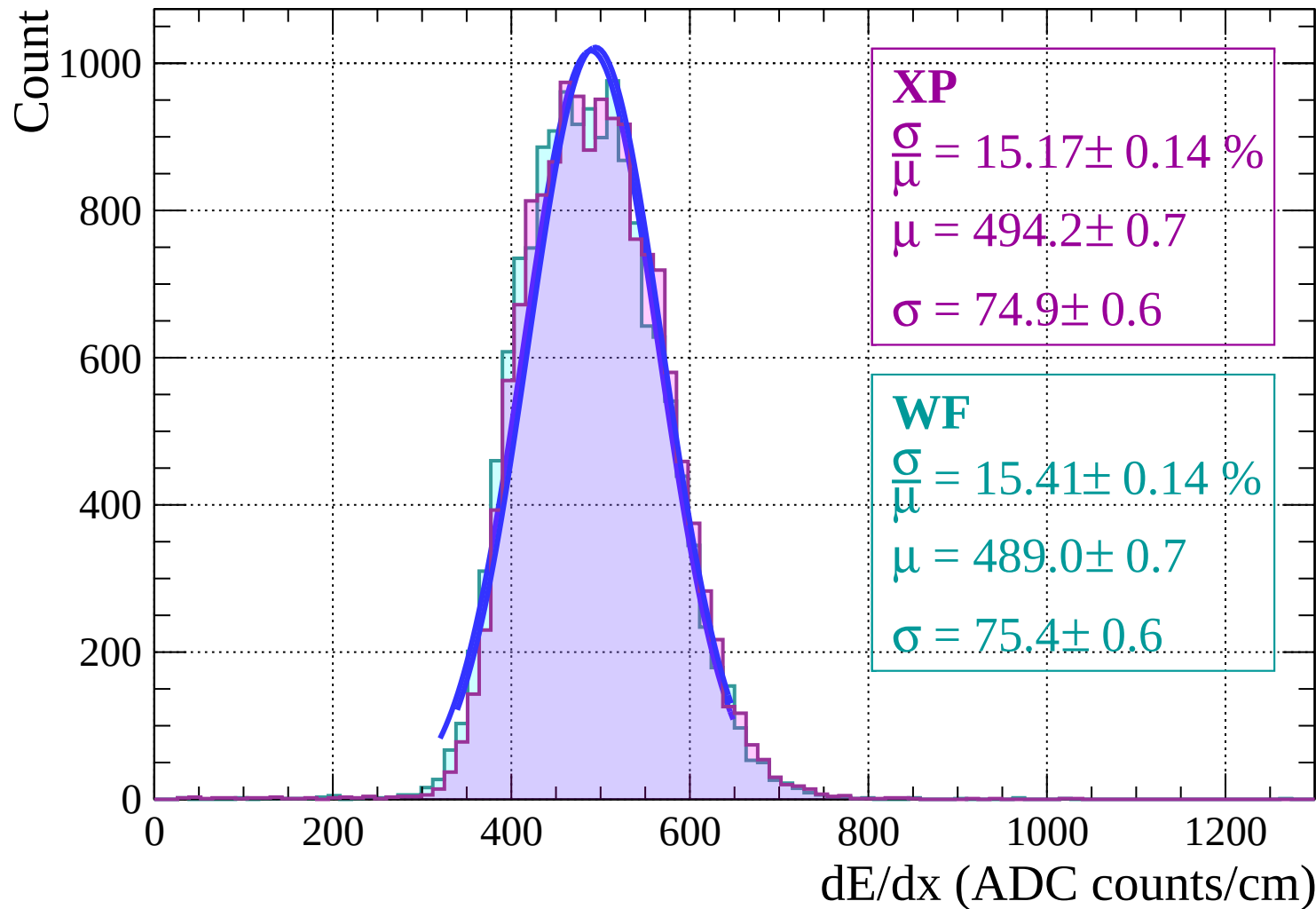
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$

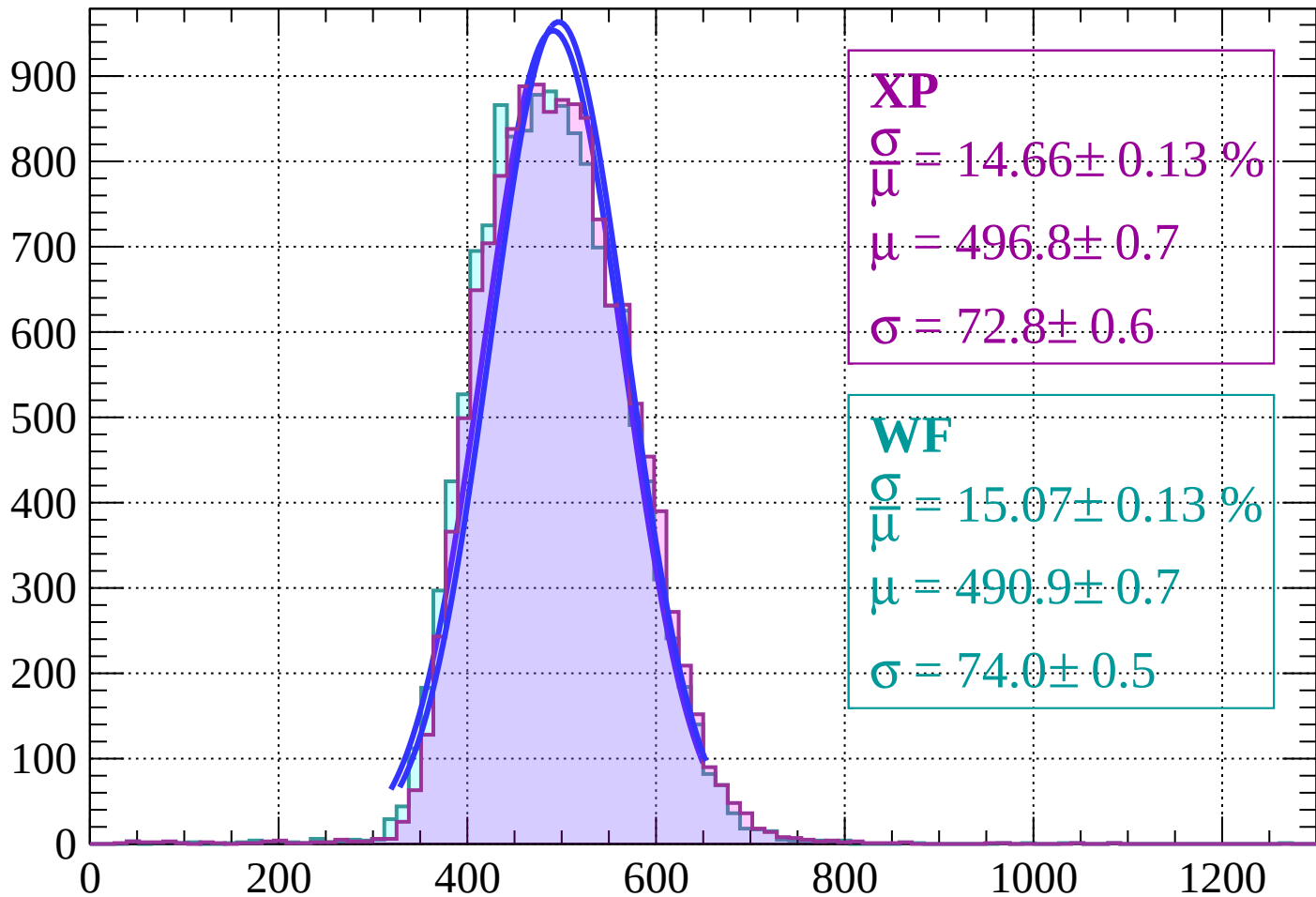


Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$

Count



XP

$$\frac{\sigma}{\mu} = 14.66 \pm 0.13 \%$$

$$\mu = 496.8 \pm 0.7$$

$$\sigma = 72.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 15.07 \pm 0.13 \%$$

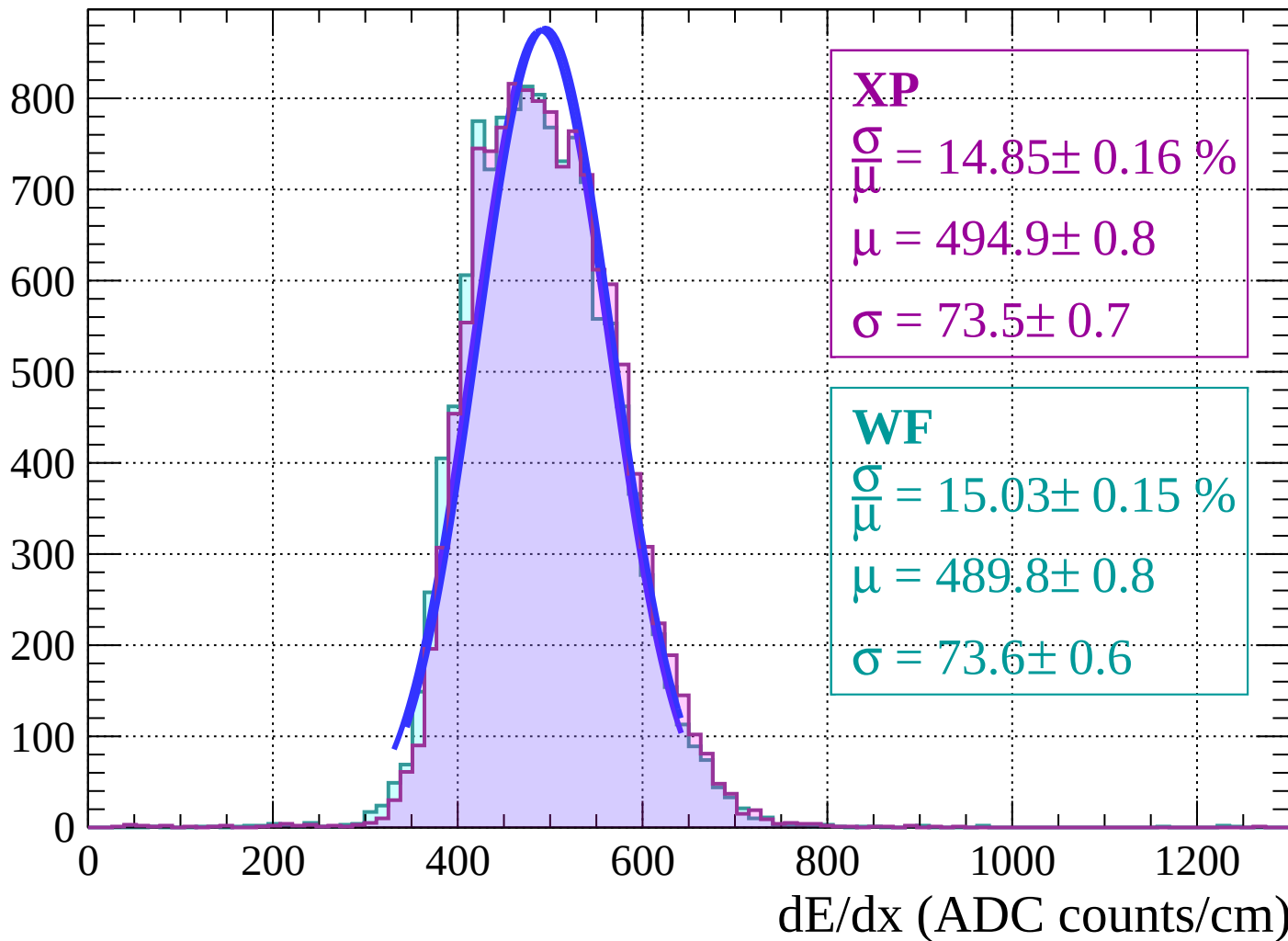
$$\mu = 490.9 \pm 0.7$$

$$\sigma = 74.0 \pm 0.5$$

dE/dx (ADC counts/cm)

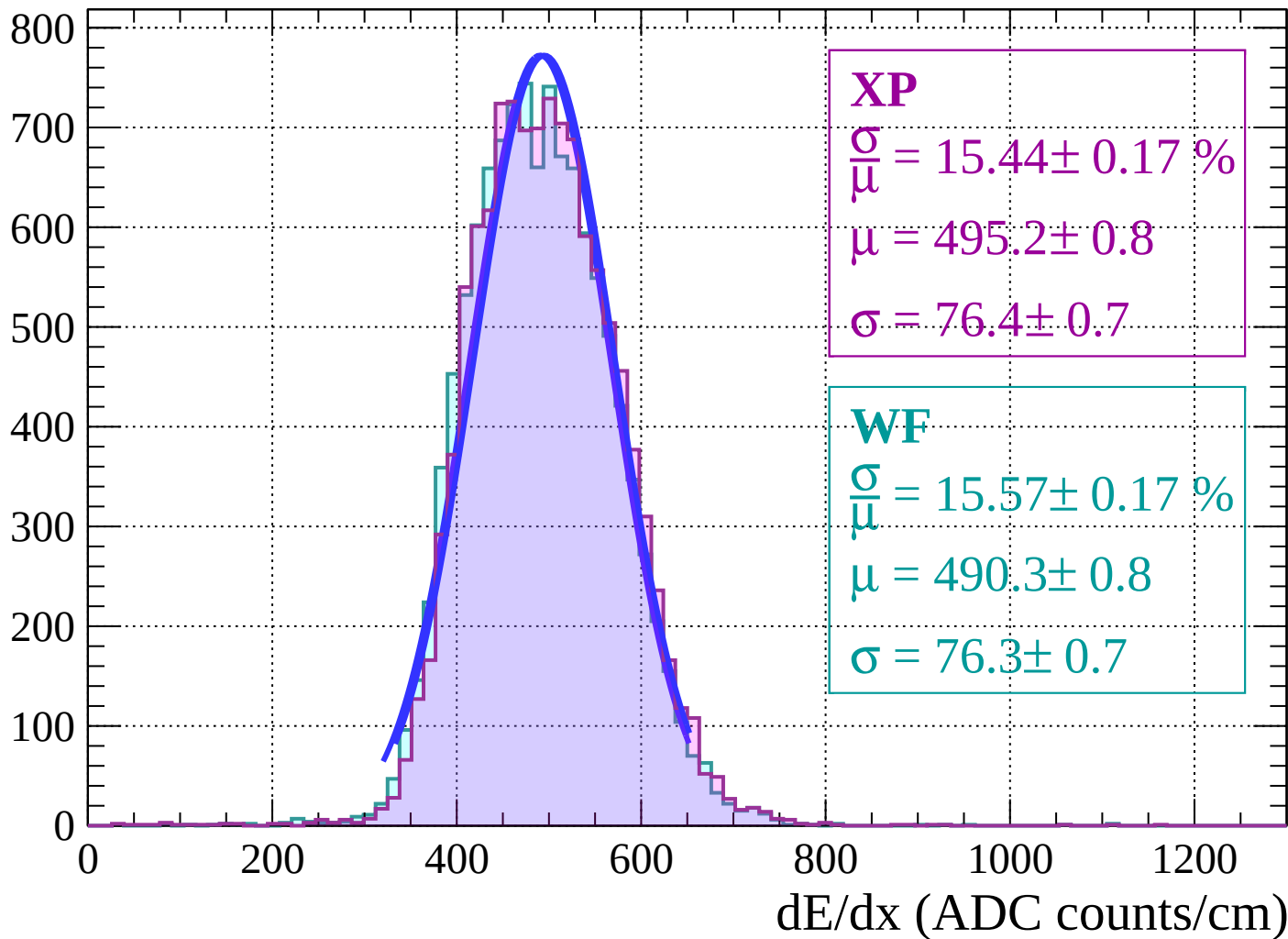
Energy loss | $10 < \theta < 12$

Count



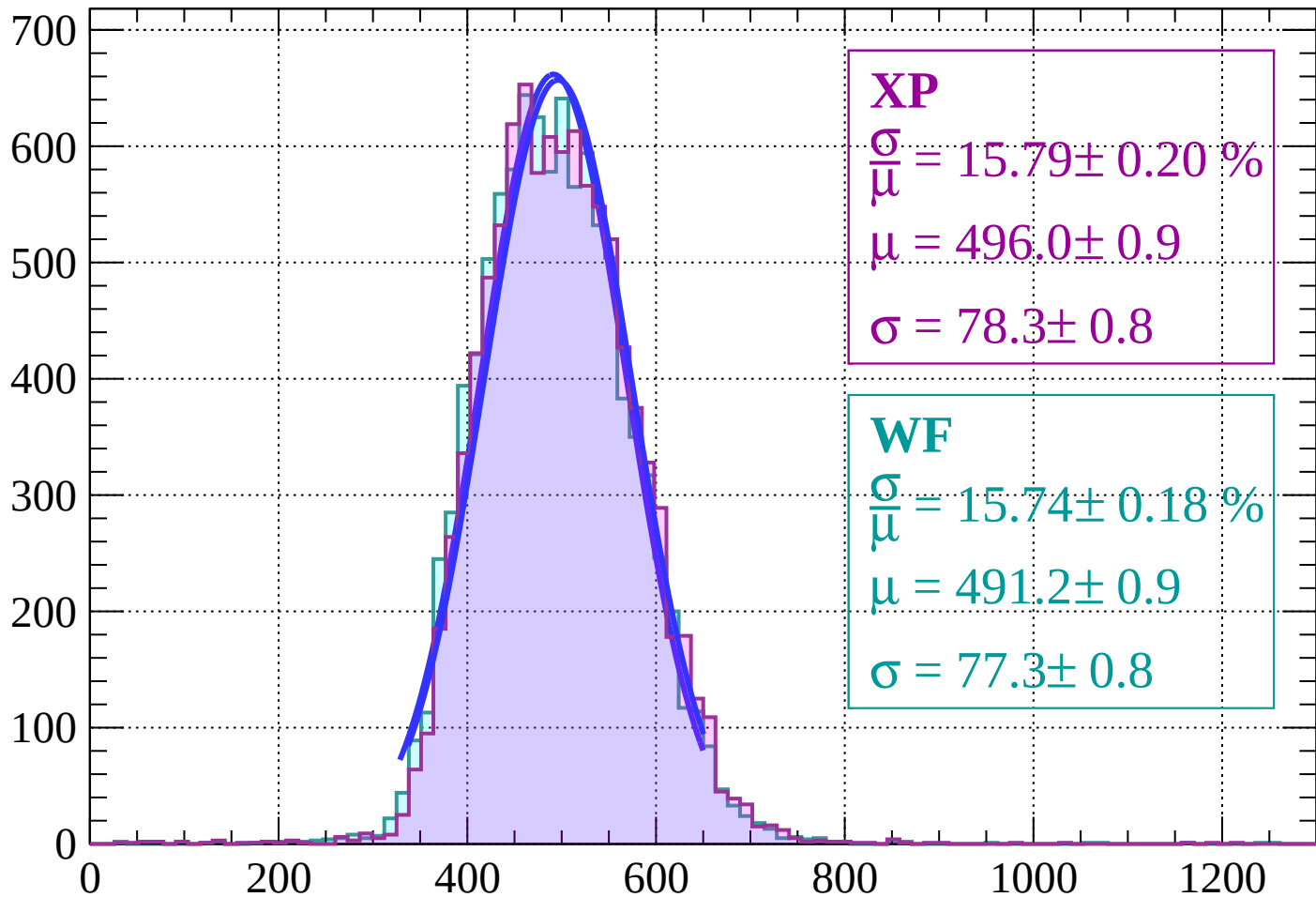
Energy loss | $12 < \theta < 14$

Count



Energy loss | $14 < \theta < 16$

Count



XP

$$\frac{\sigma}{\mu} = 15.79 \pm 0.20 \%$$

$$\mu = 496.0 \pm 0.9$$

$$\sigma = 78.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 15.74 \pm 0.18 \%$$

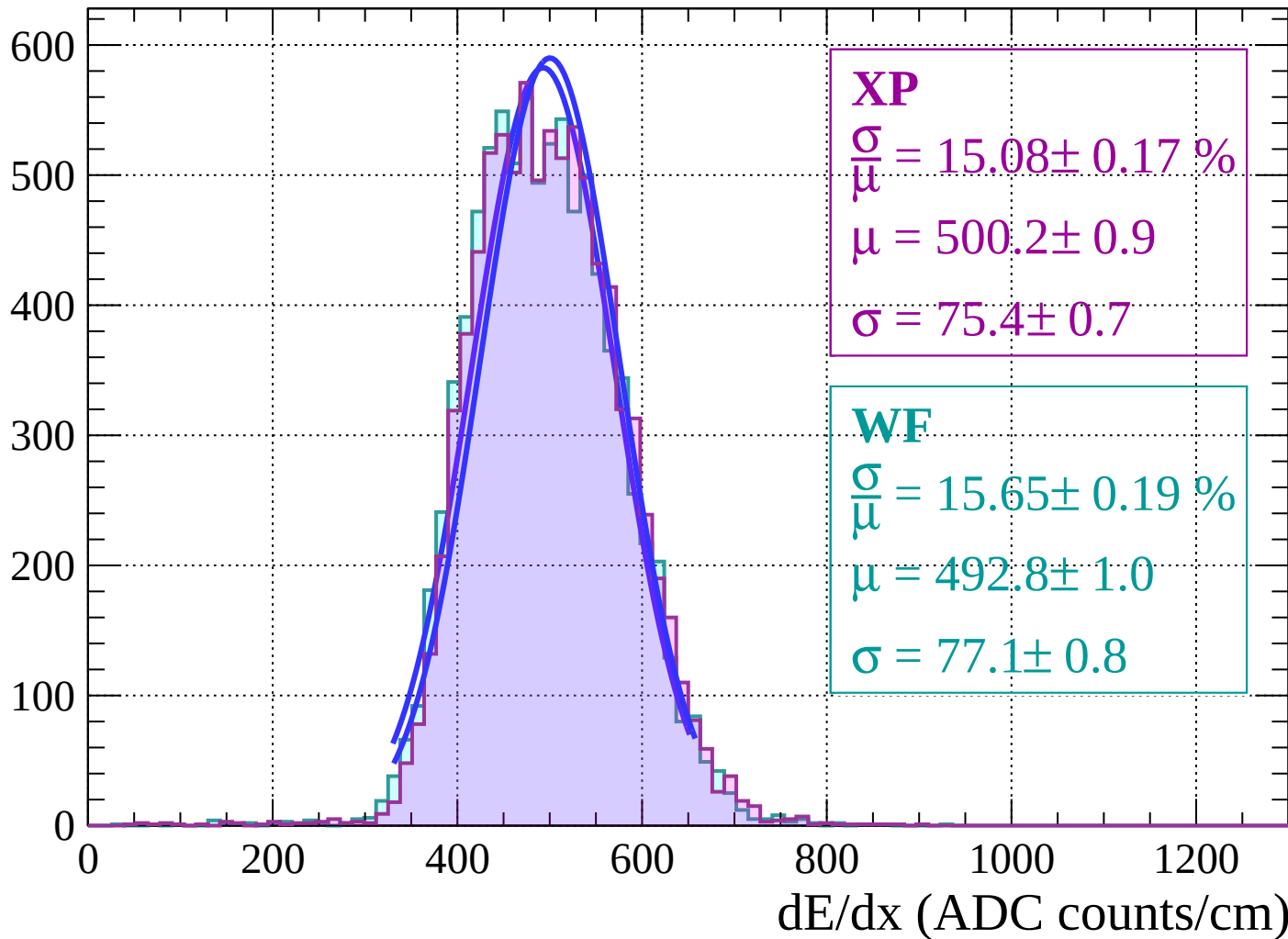
$$\mu = 491.2 \pm 0.9$$

$$\sigma = 77.3 \pm 0.8$$

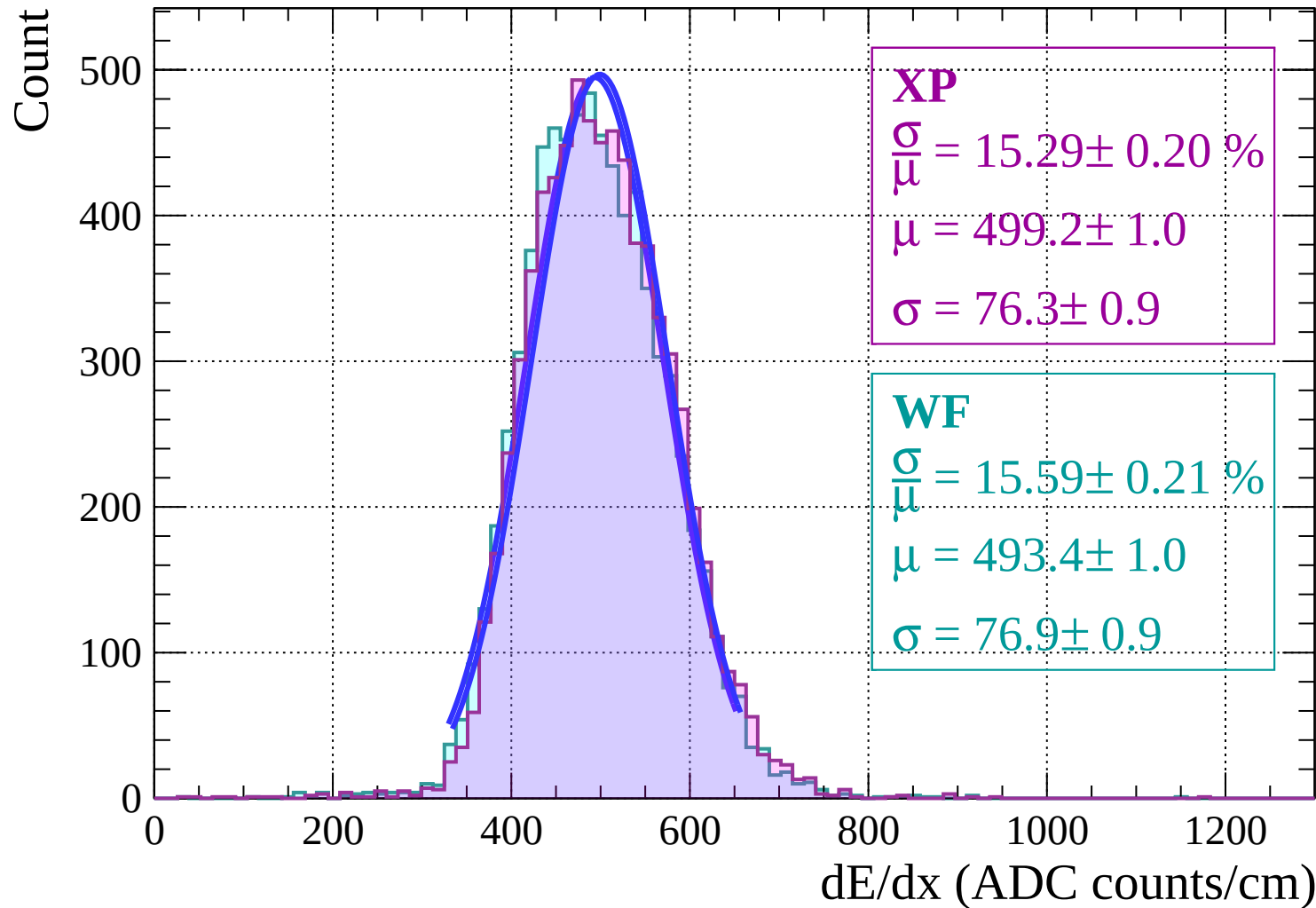
dE/dx (ADC counts/cm)

Energy loss | $16 < \theta < 18$

Count



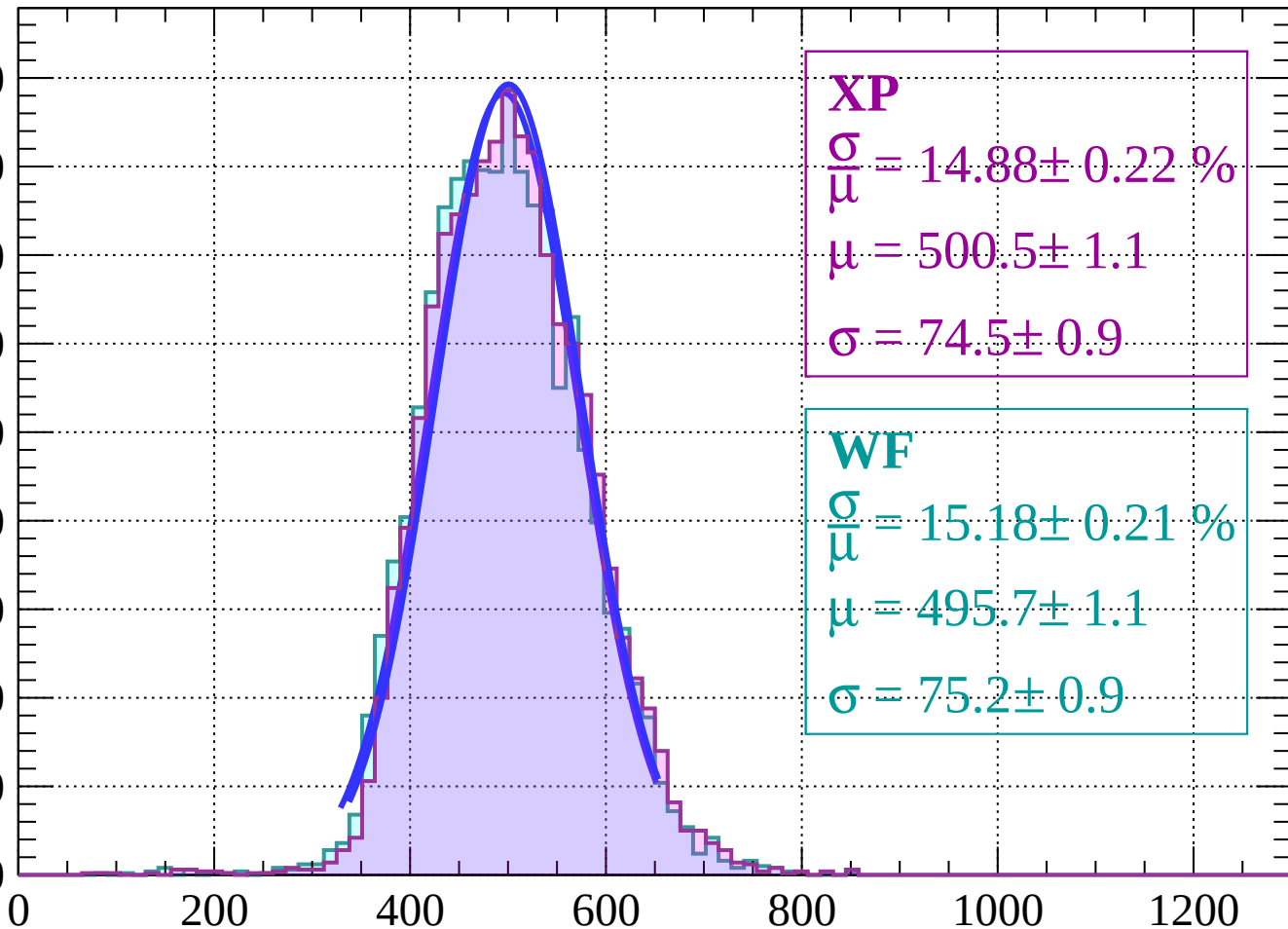
Energy loss | $18 < \theta < 20$



Energy loss | $20 < \theta < 22$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 14.88 \pm 0.22 \%$$

$$\mu = 500.5 \pm 1.1$$

$$\sigma = 74.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 15.18 \pm 0.21 \%$$

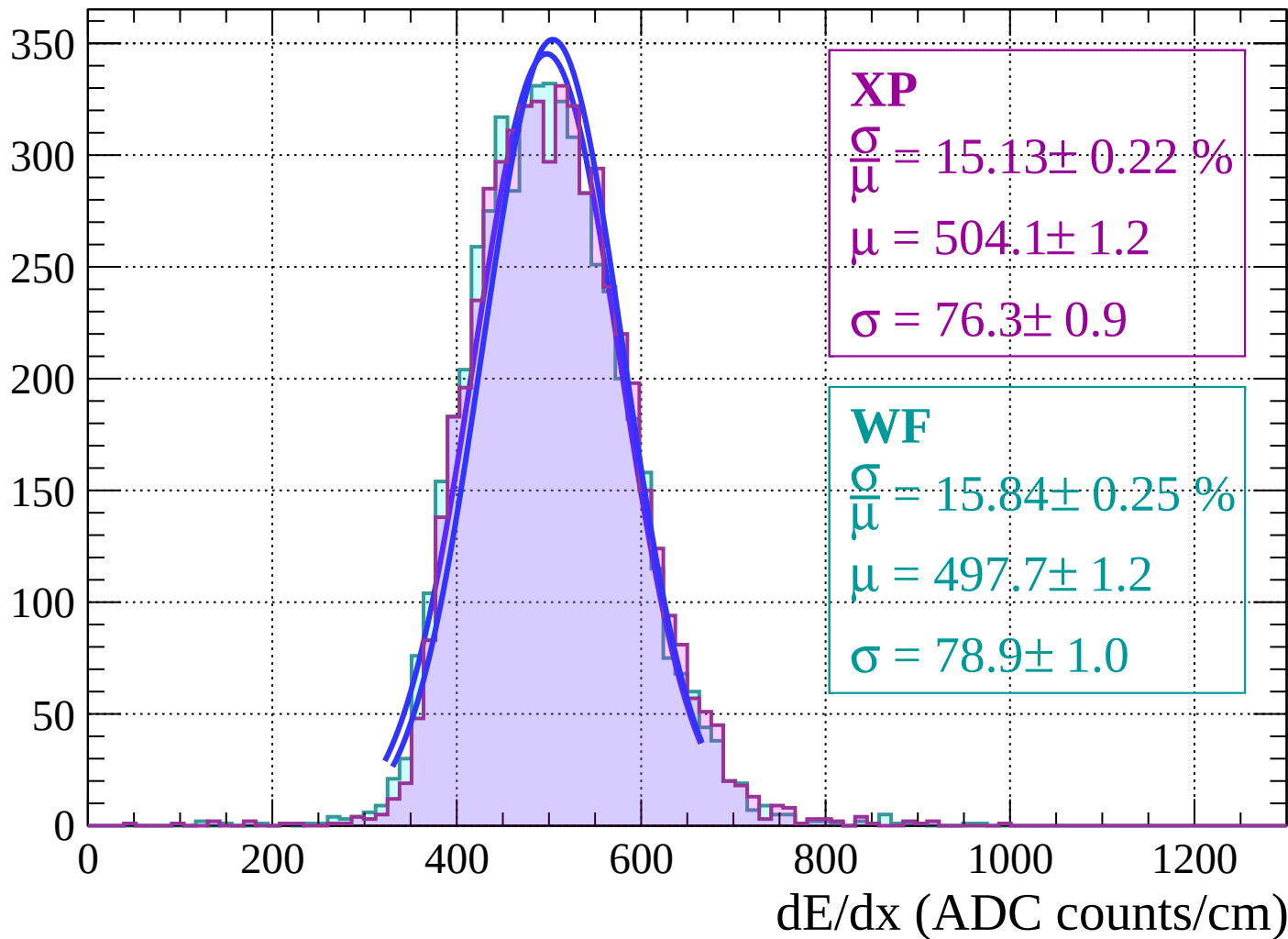
$$\mu = 495.7 \pm 1.1$$

$$\sigma = 75.2 \pm 0.9$$

dE/dx (ADC counts/cm)

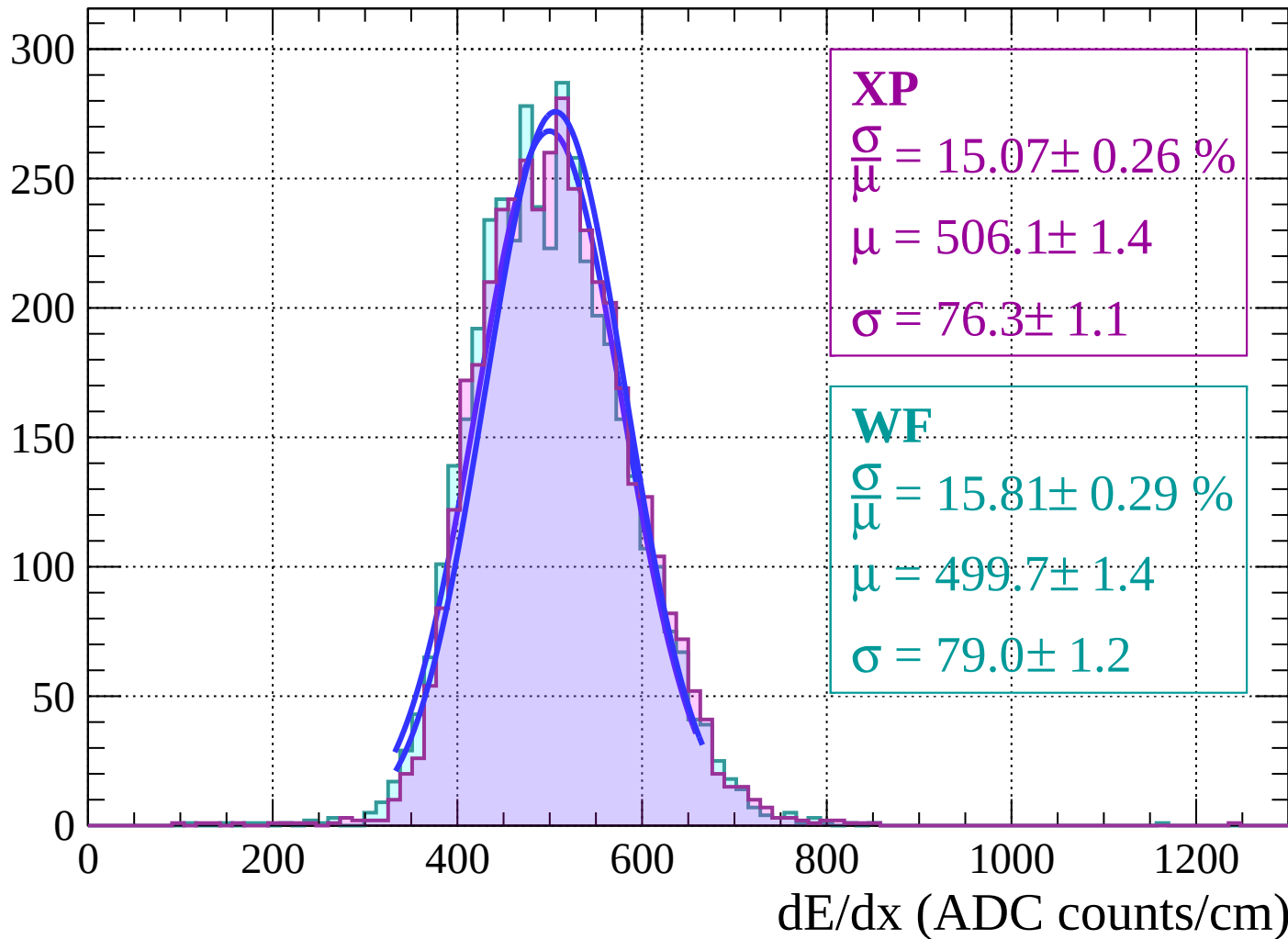
Energy loss | $22 < \theta < 24$

Count



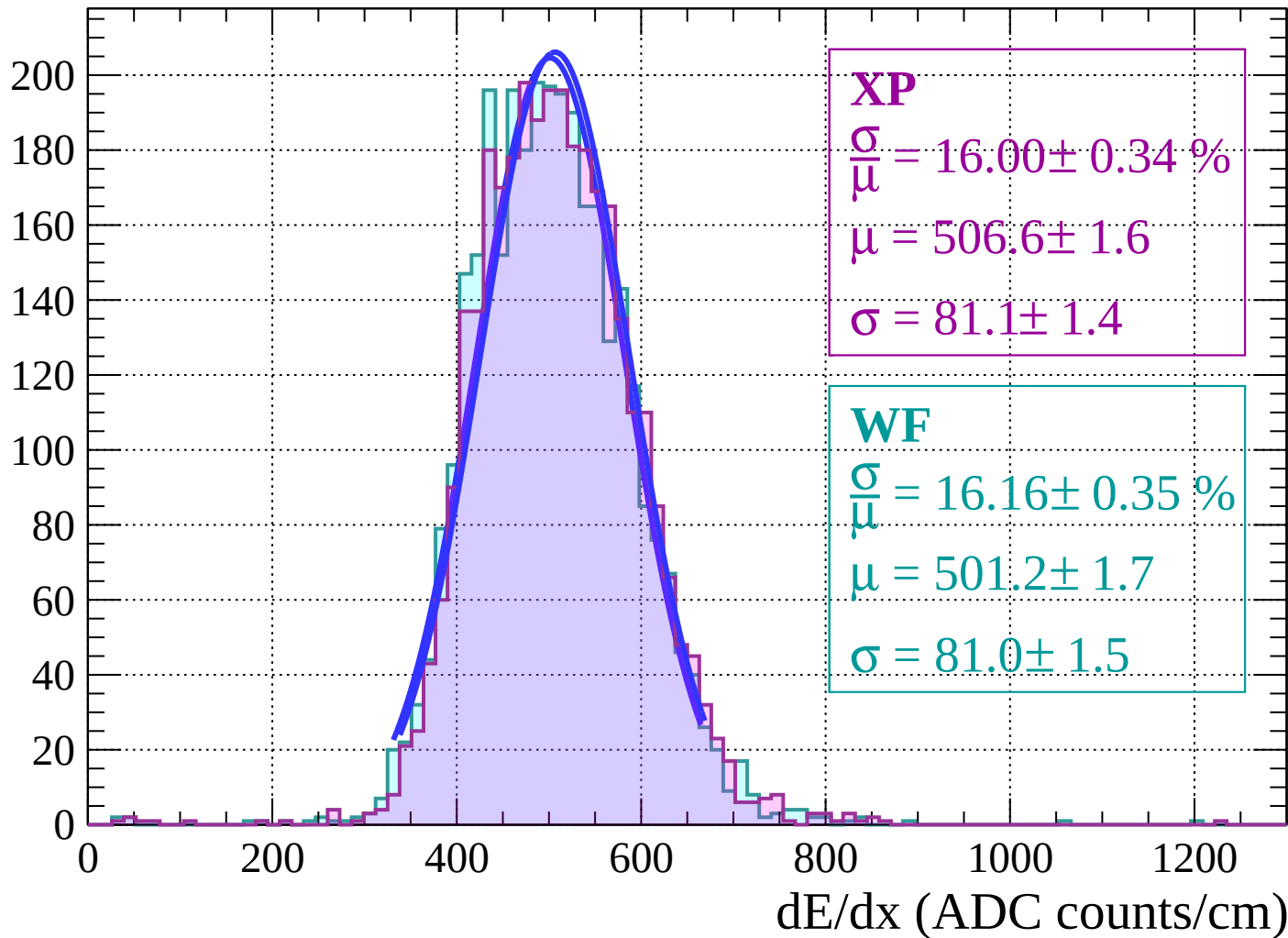
Energy loss | $24 < \theta < 26$

Count



Energy loss | $26 < \theta < 28$

Count



Energy loss | $28 < \theta < 30$

Count

140

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 16.28 \pm 0.37 \%$$

$$\mu = 510.7 \pm 2.0$$

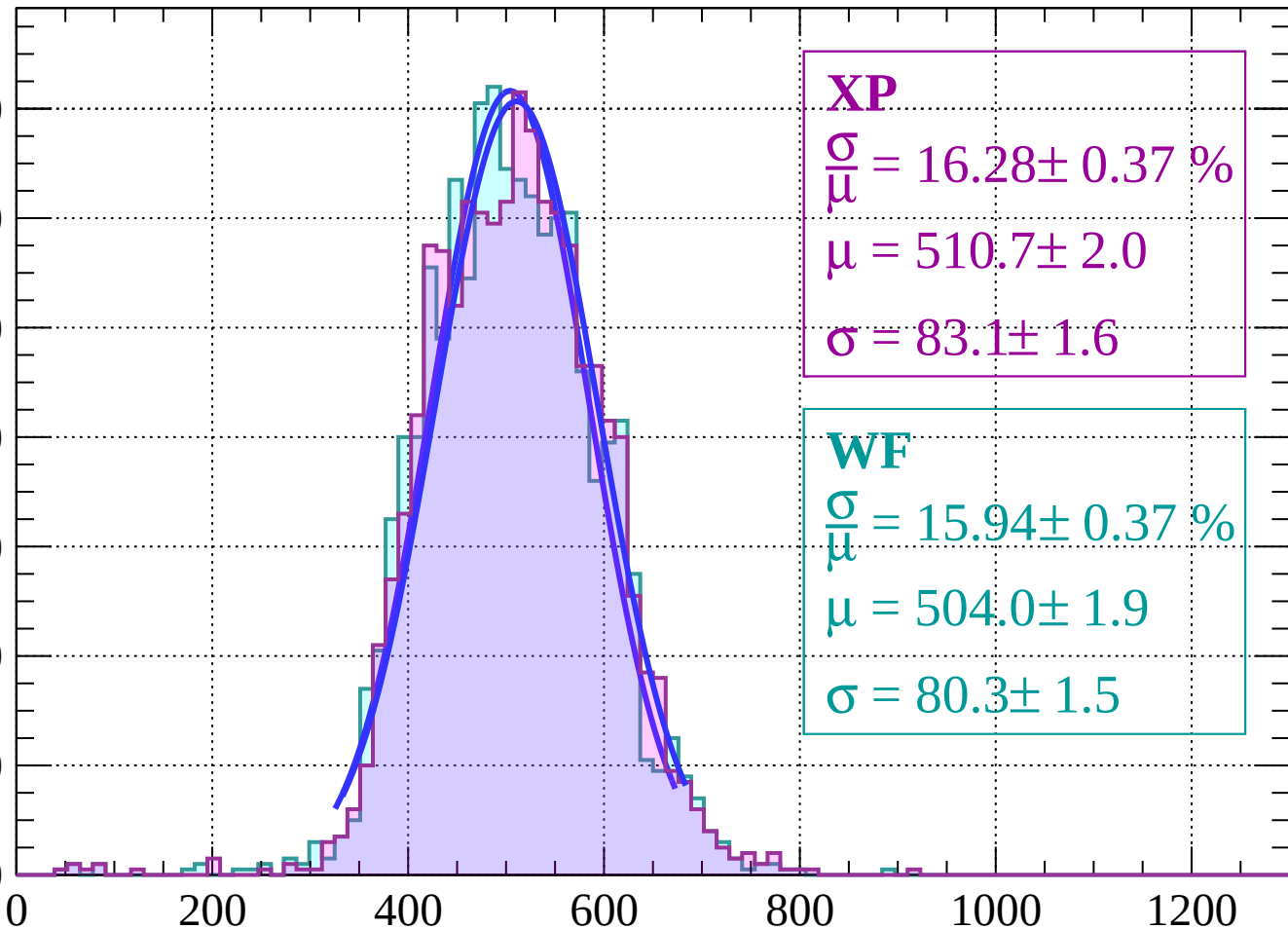
$$\sigma = 83.1 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 15.94 \pm 0.37 \%$$

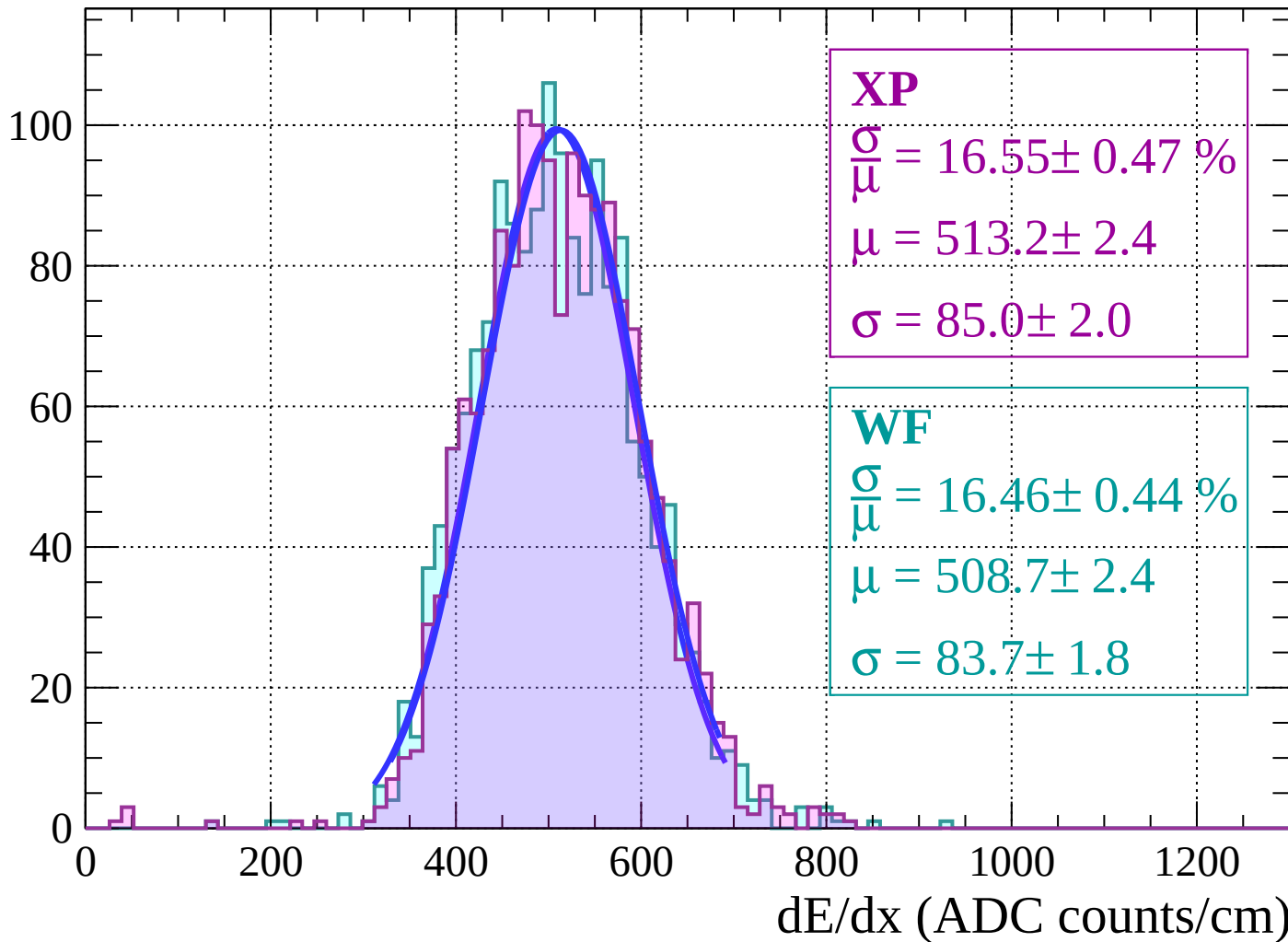
$$\mu = 504.0 \pm 1.9$$

$$\sigma = 80.3 \pm 1.5$$



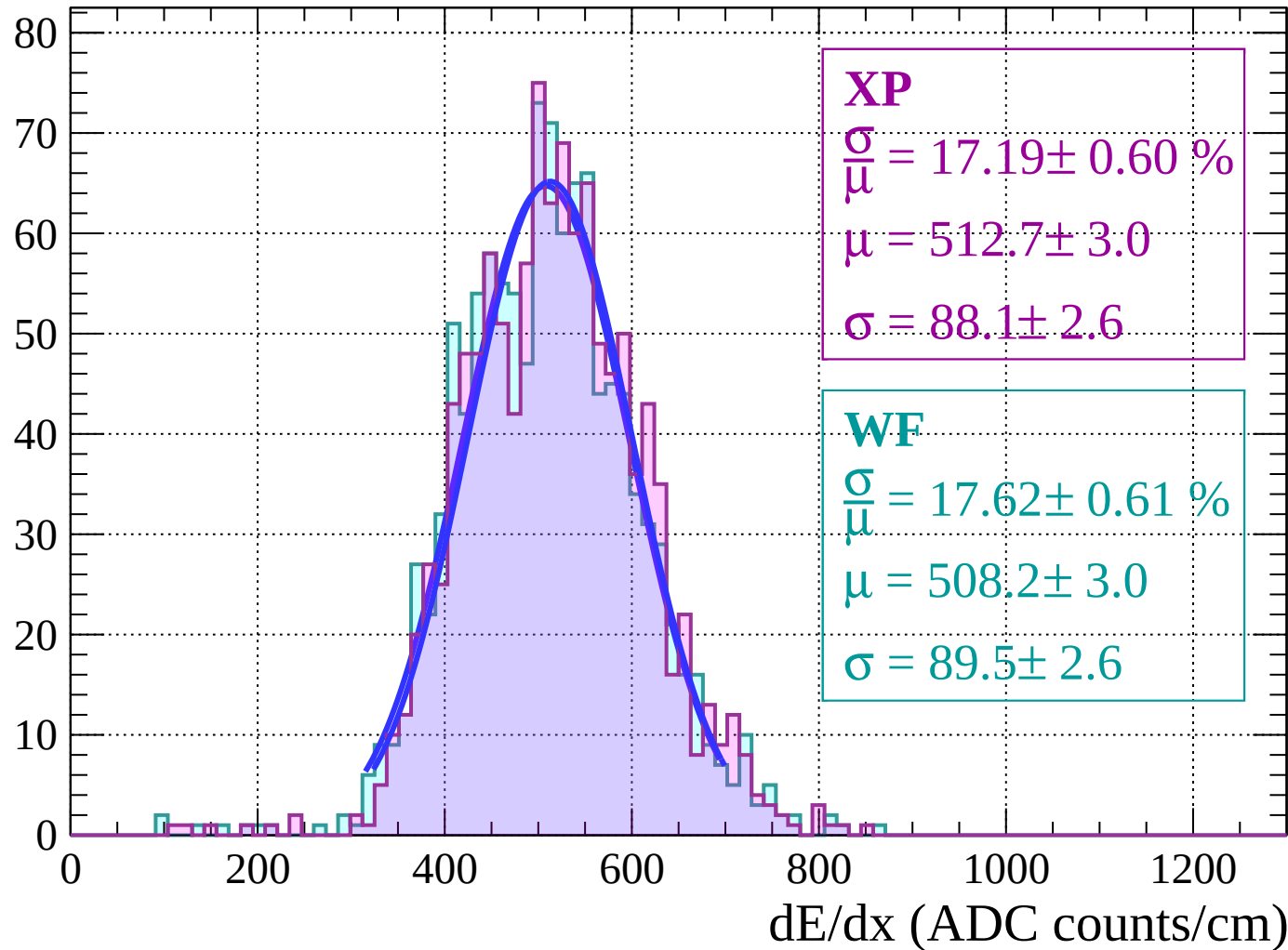
Energy loss | $30 < \theta < 32$

Count



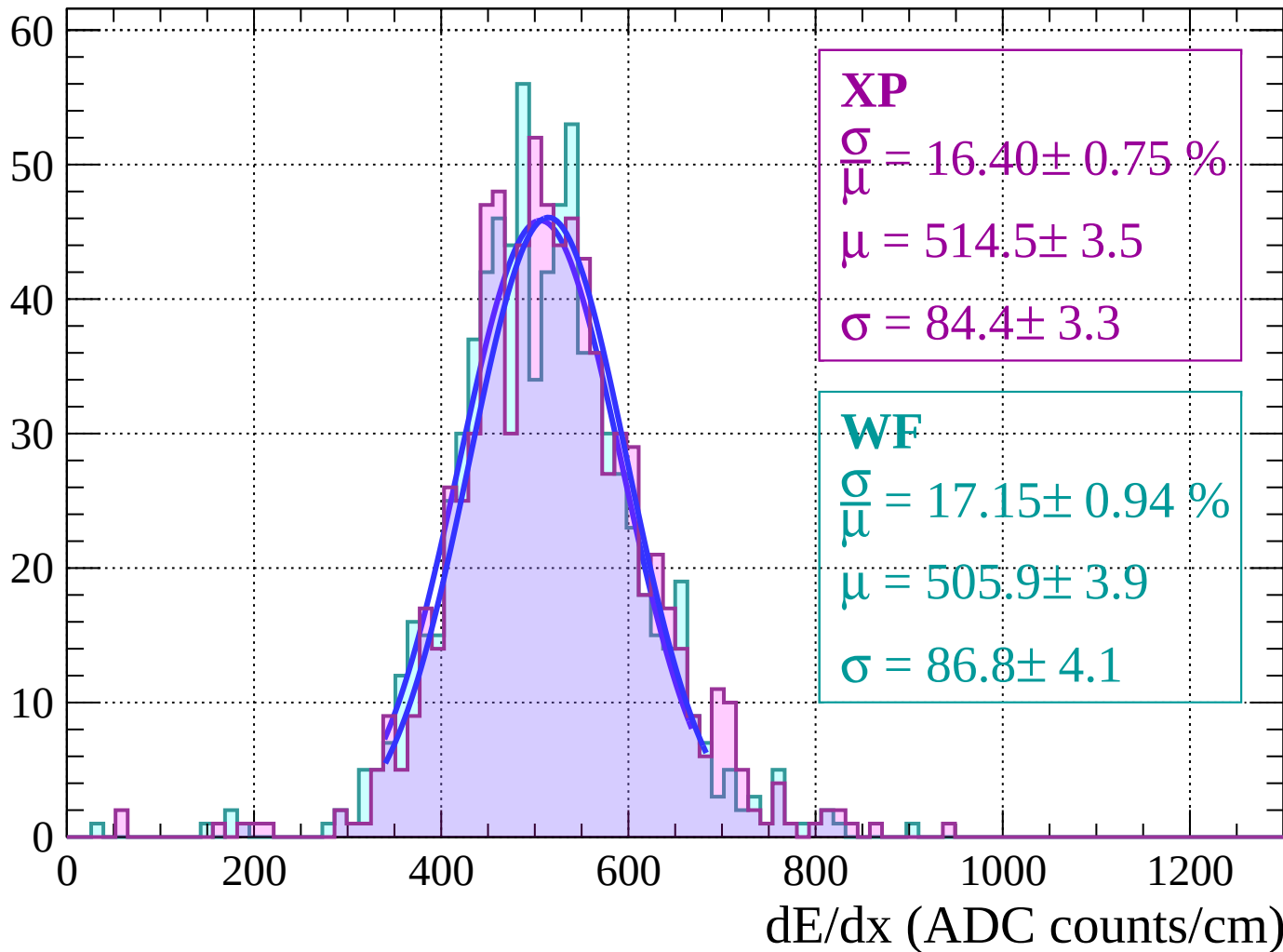
Energy loss | $32 < \theta < 34$

Count



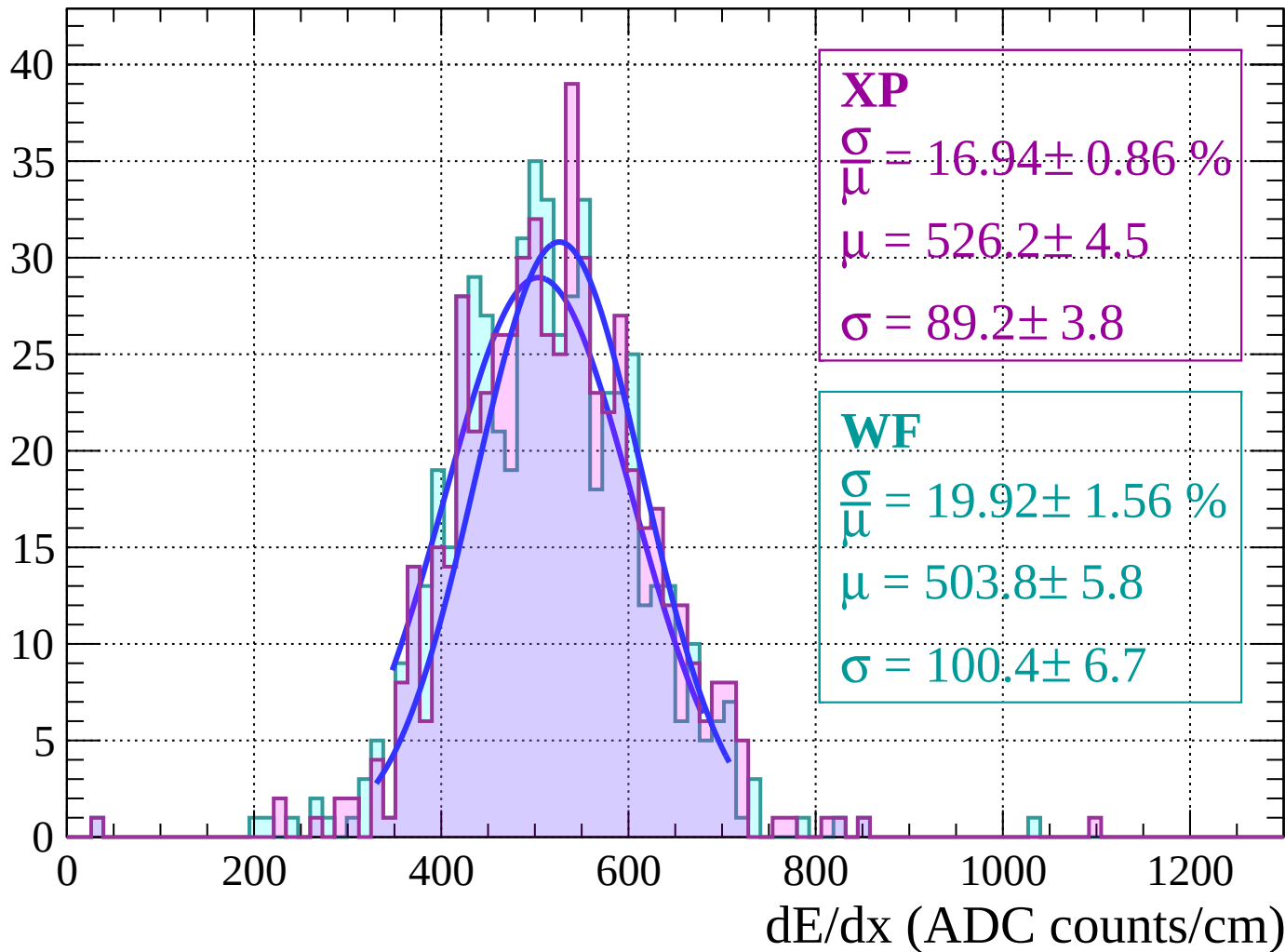
Energy loss | $34 < \theta < 36$

Count



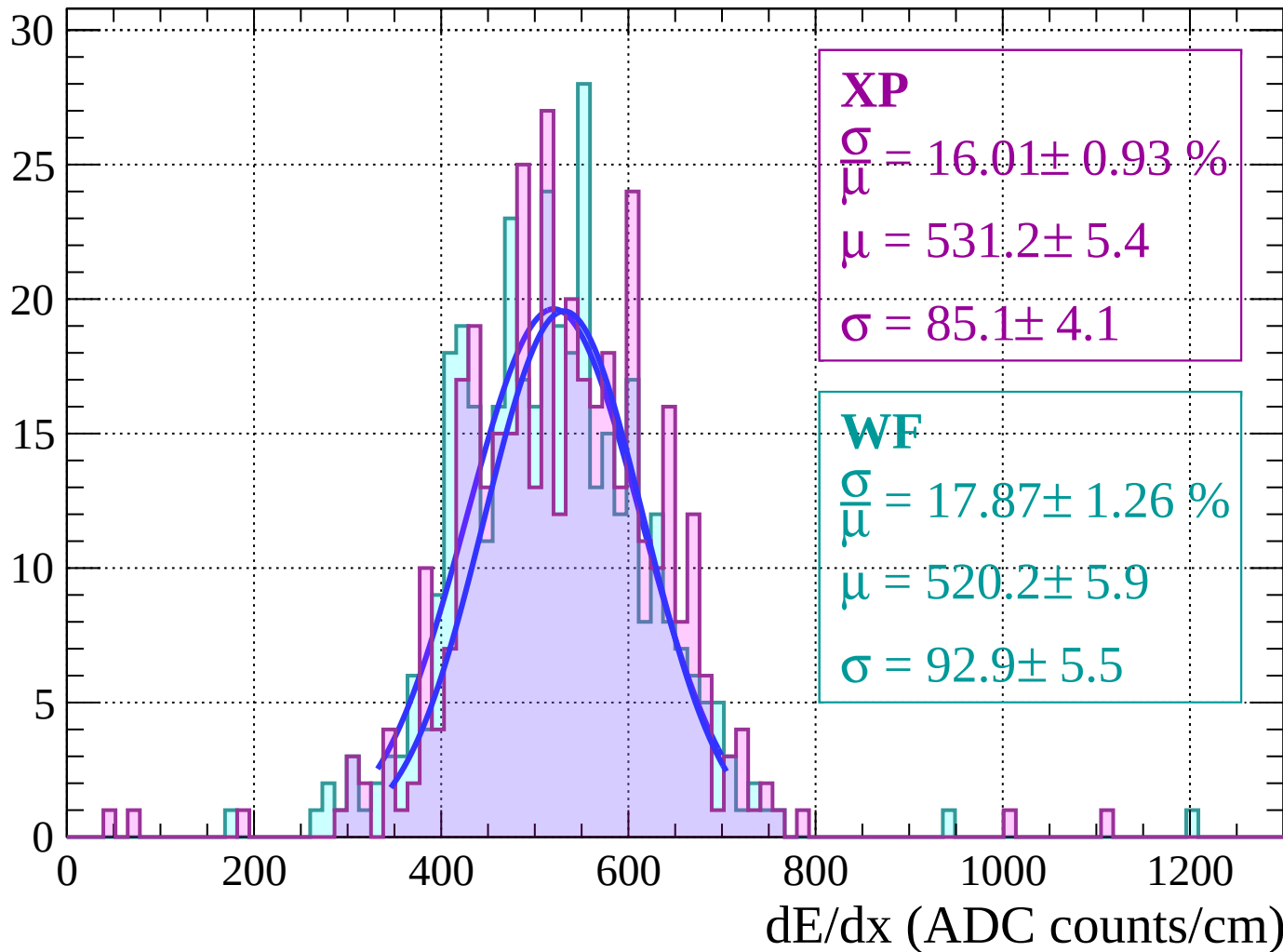
Energy loss | $36 < \theta < 38$

Count



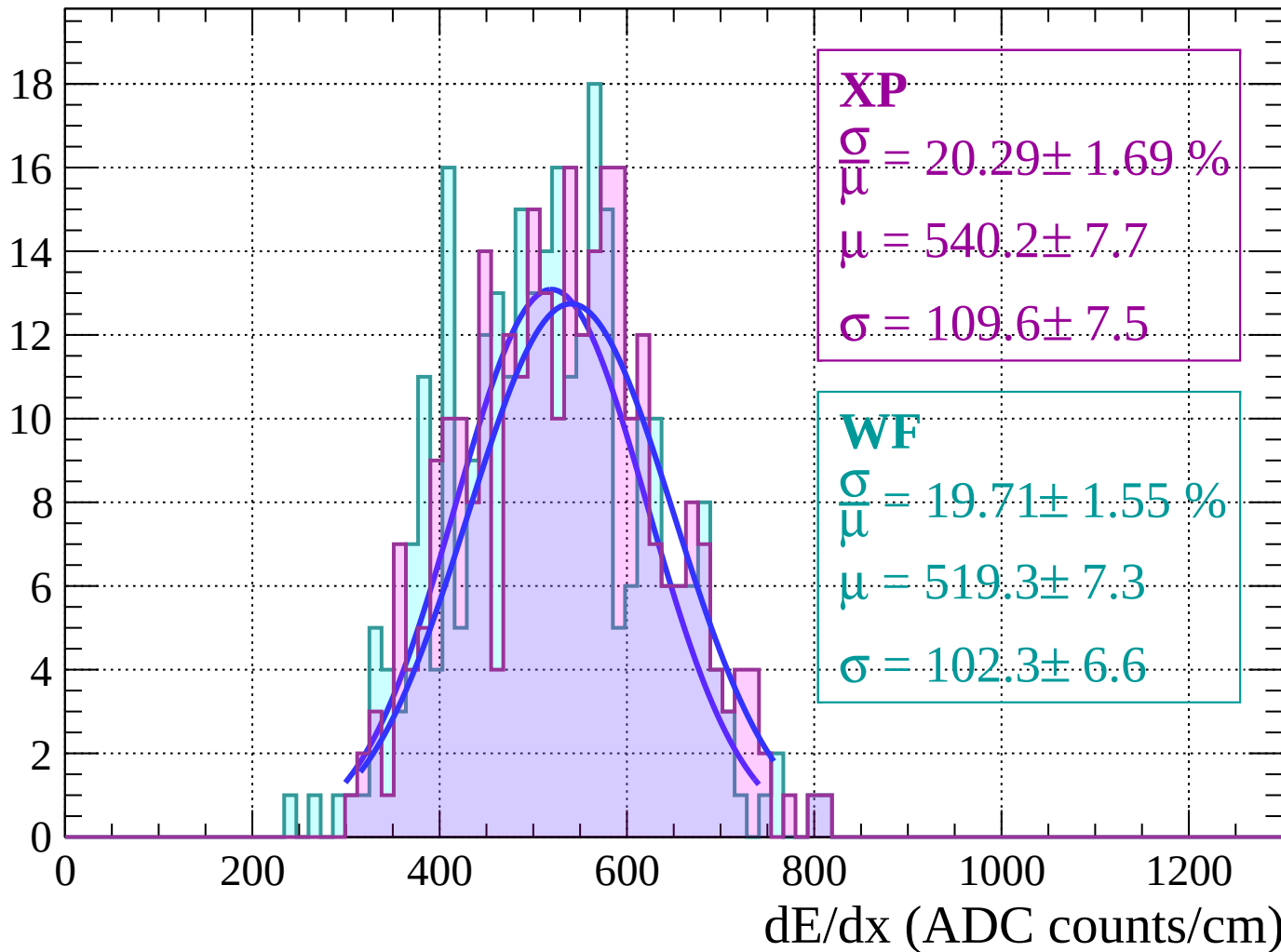
Energy loss | $38 < \theta < 40$

Count



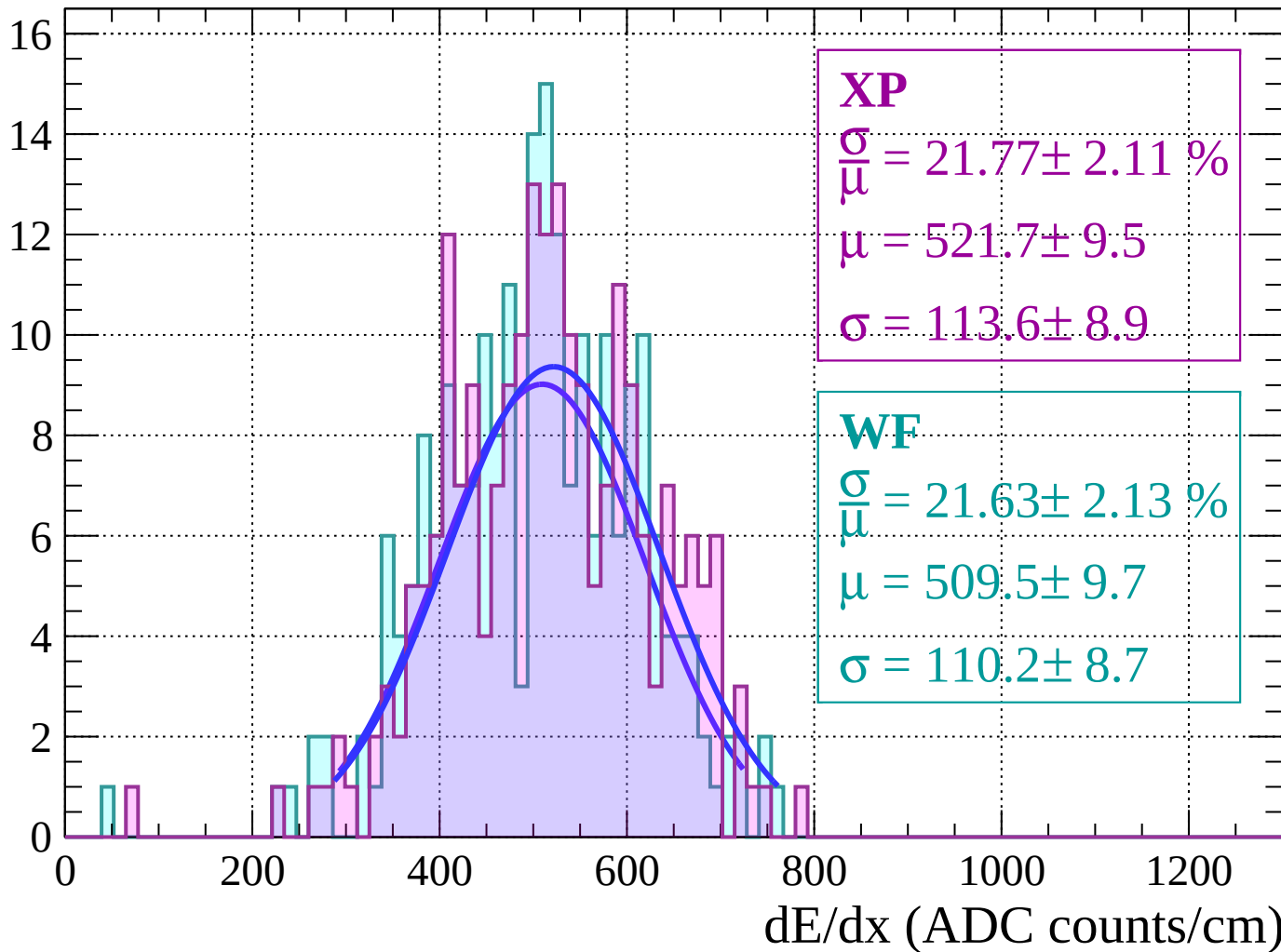
Energy loss | $40 < \theta < 42$

Count



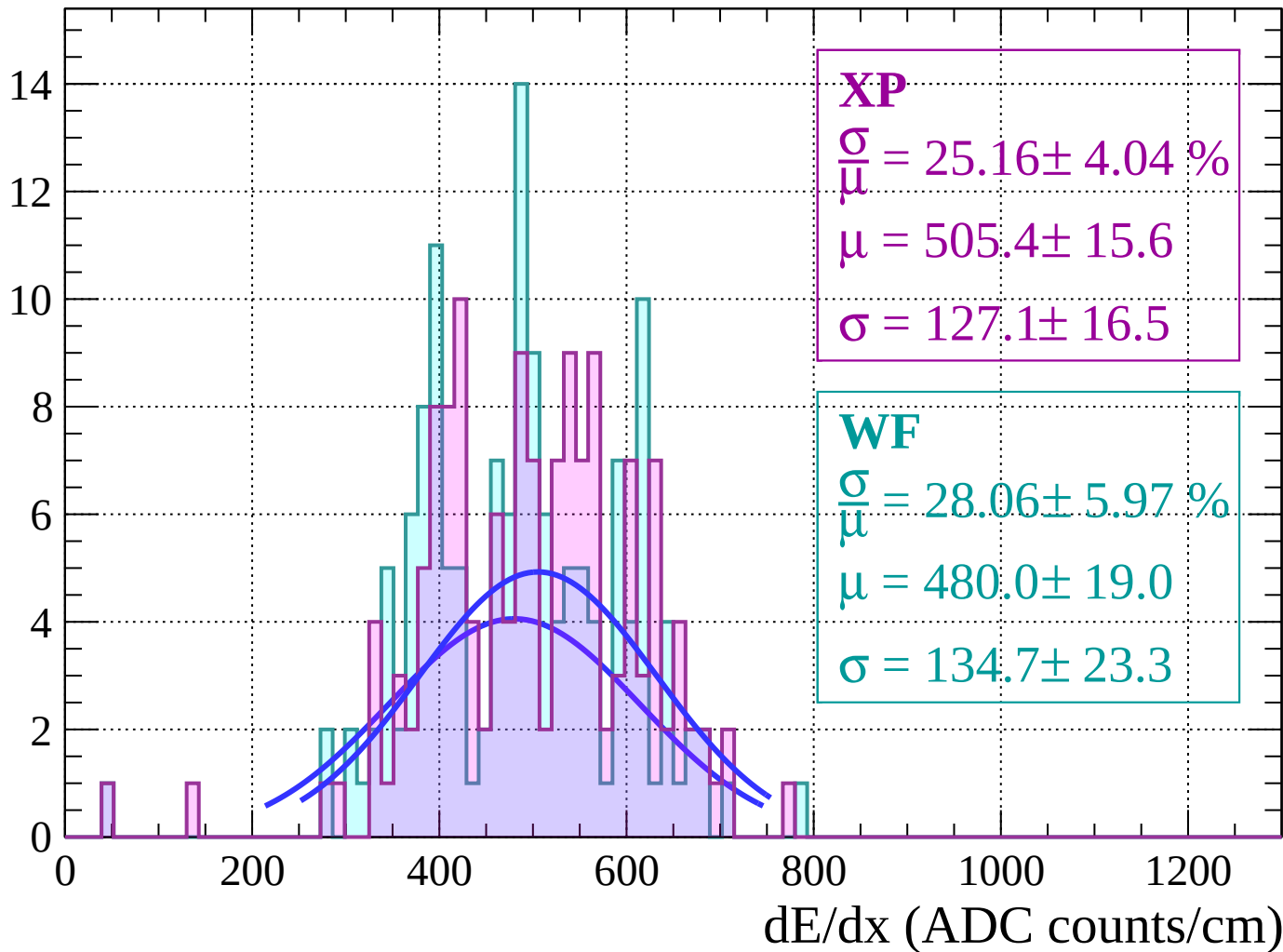
Energy loss | $42 < \theta < 44$

Count



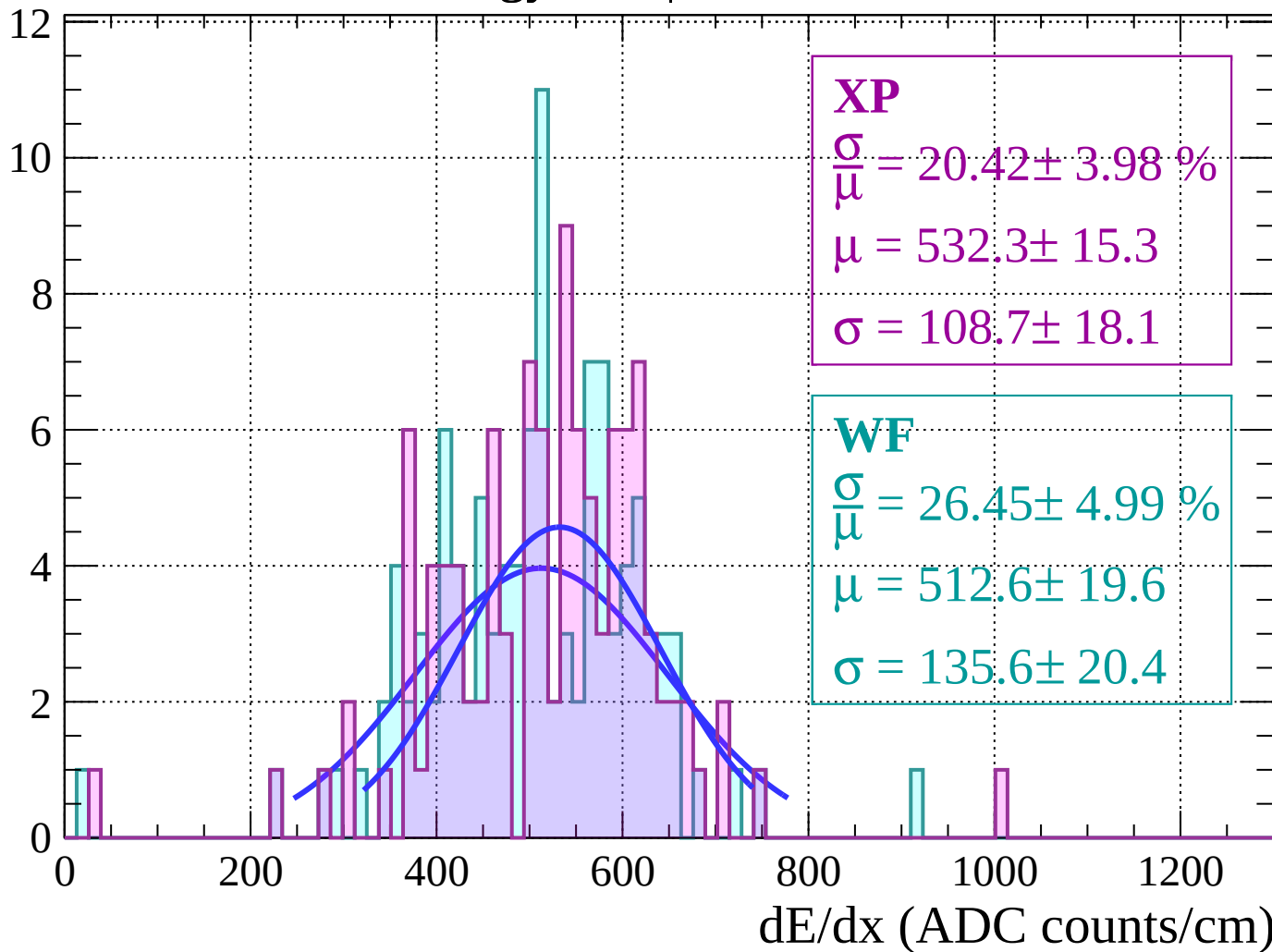
Energy loss | $44 < \theta < 46$

Count



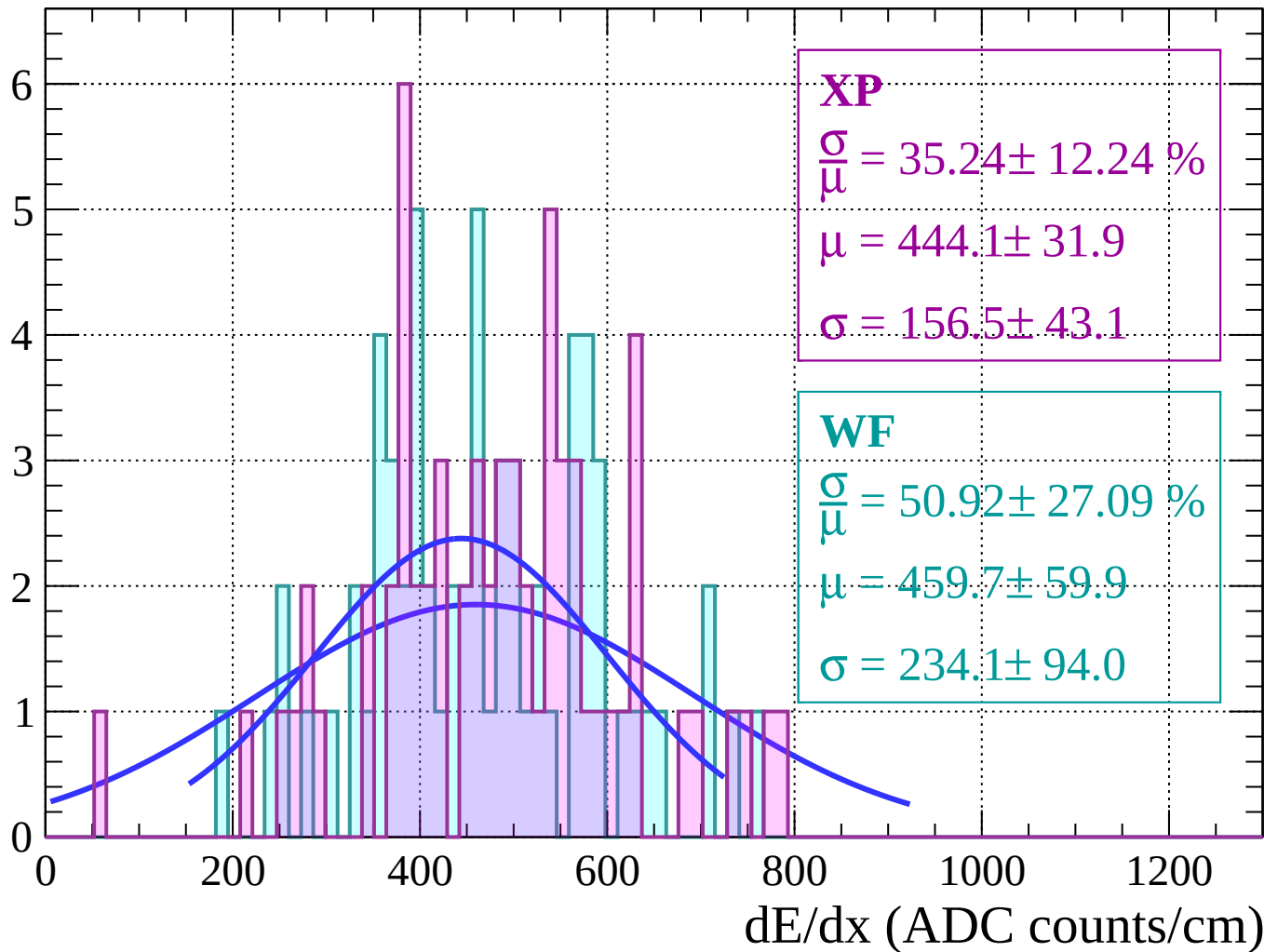
Energy loss | $46 < \theta < 48$

Count



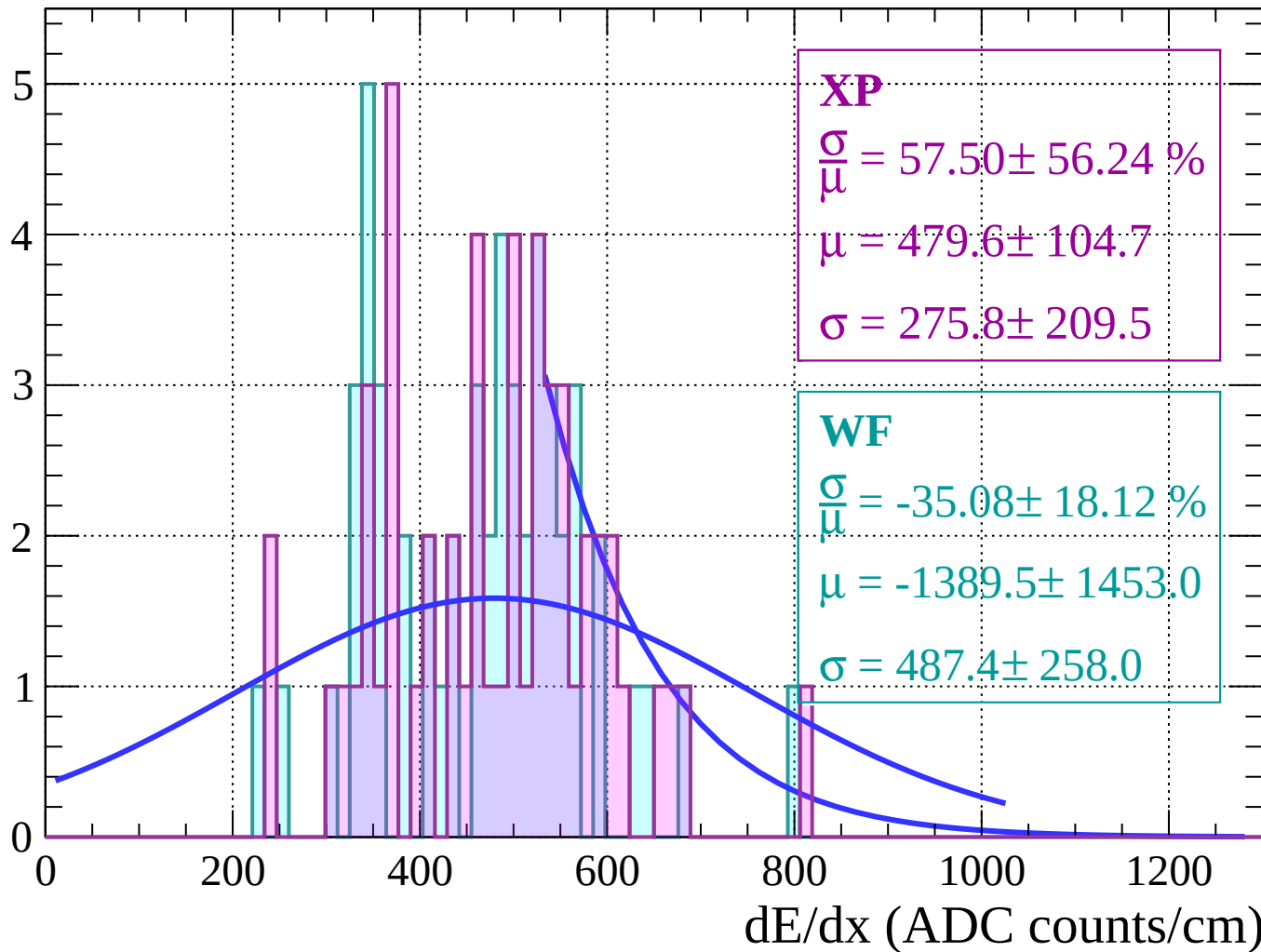
Energy loss | $48 < \theta < 50$

Count



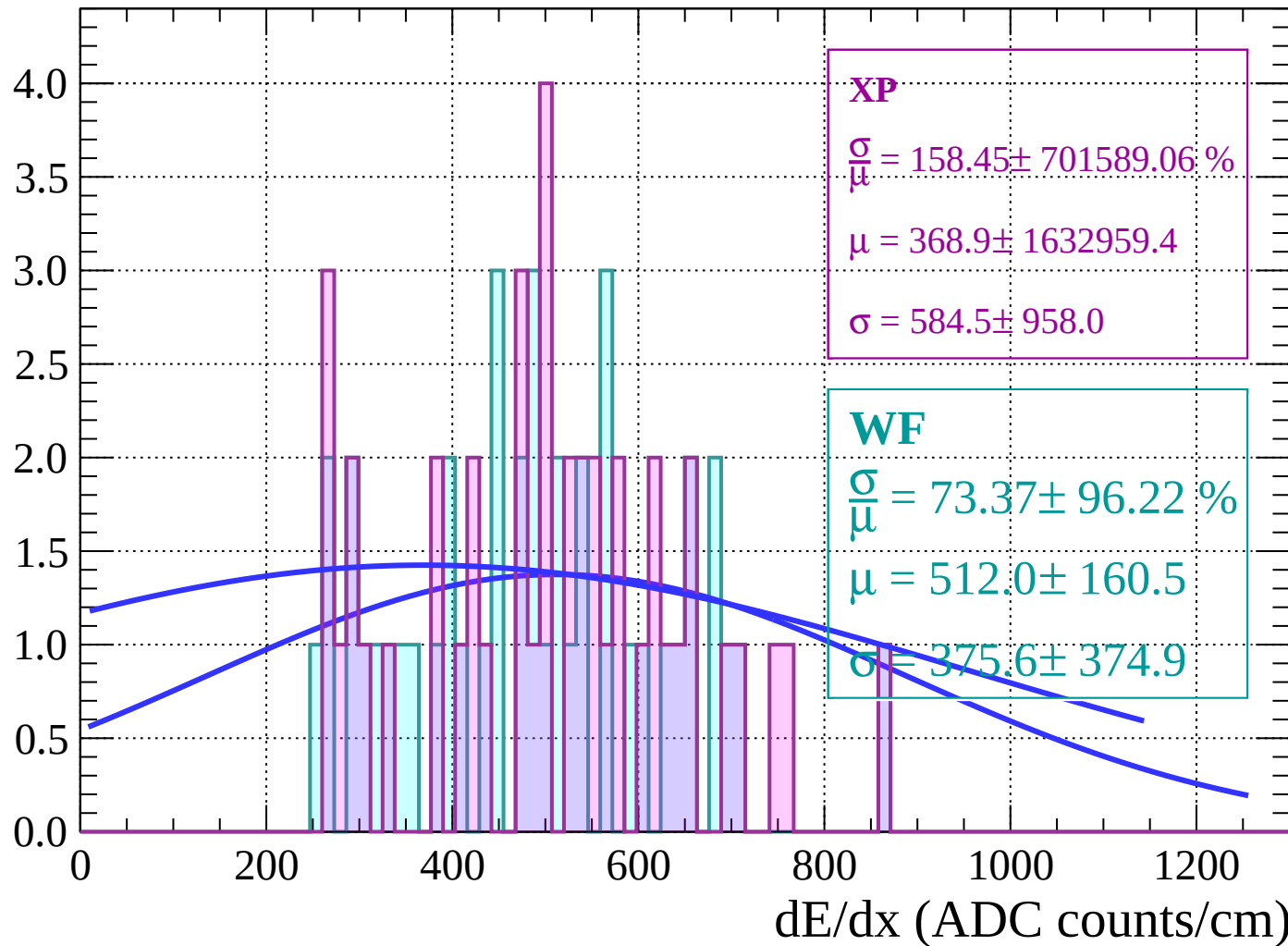
Energy loss | $50 < \theta < 52$

Count



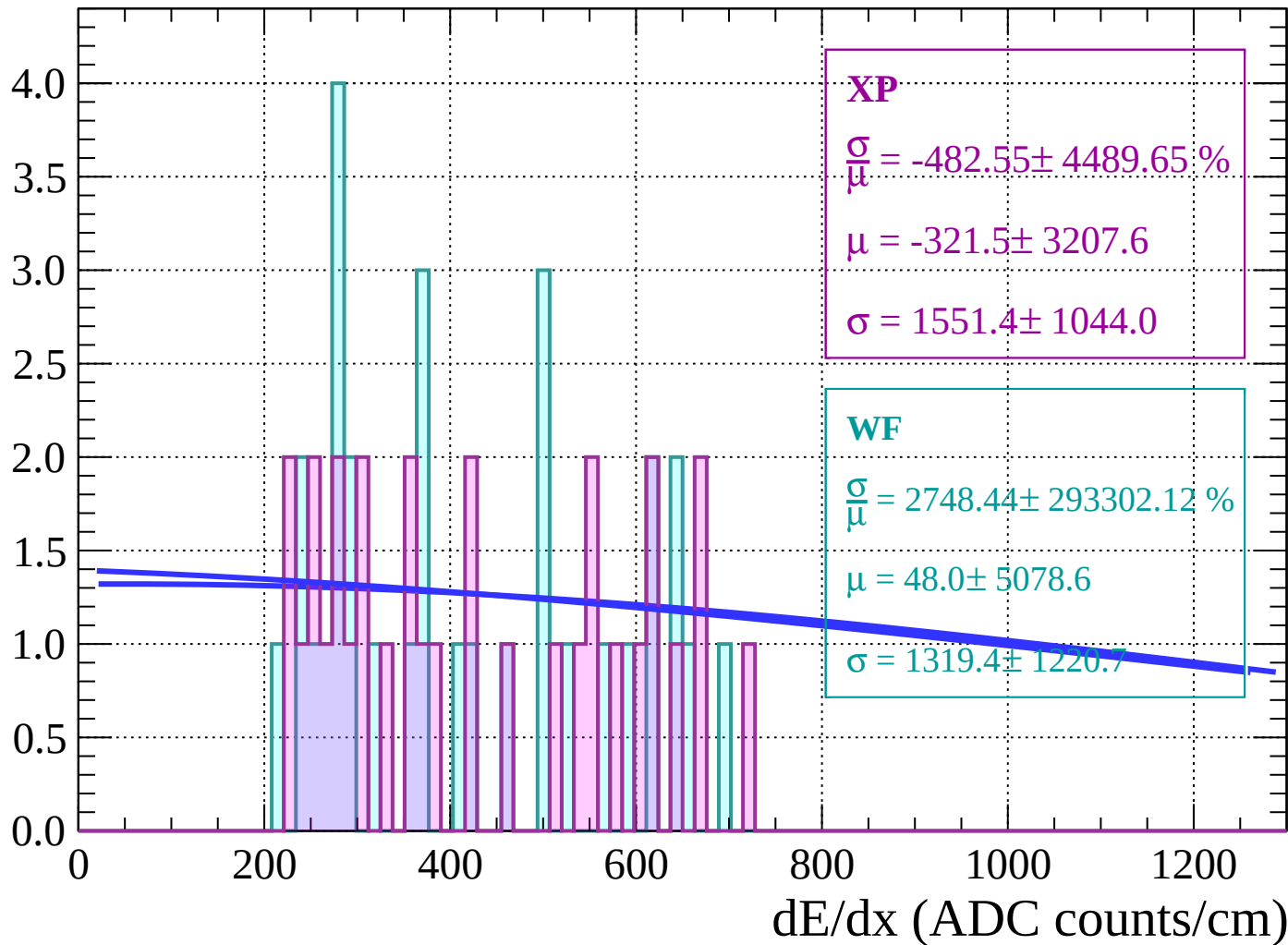
Energy loss | $52 < \theta < 54$

Count



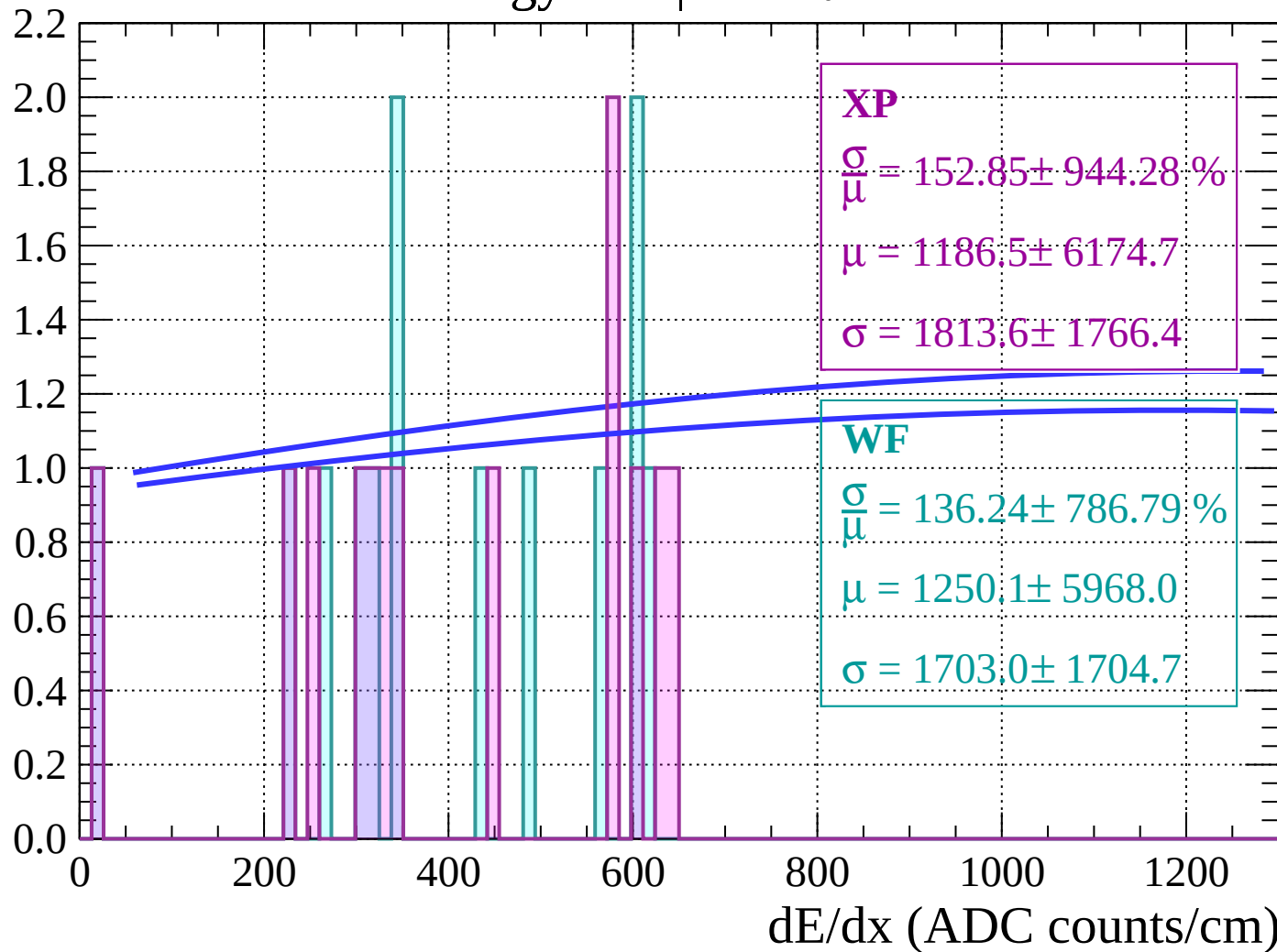
Energy loss | $54 < \theta < 56$

Count



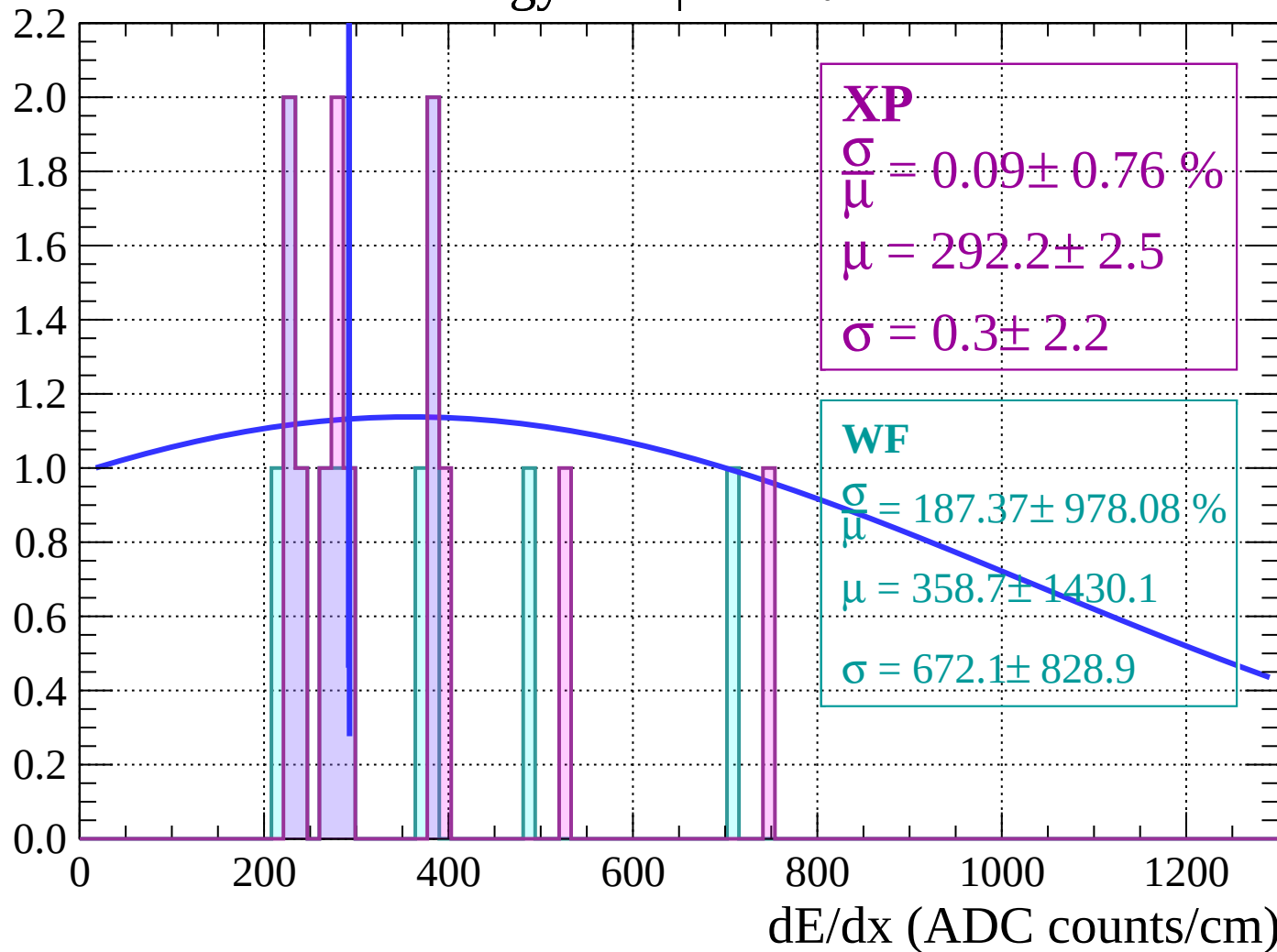
Energy loss | $56 < \theta < 58$

Count



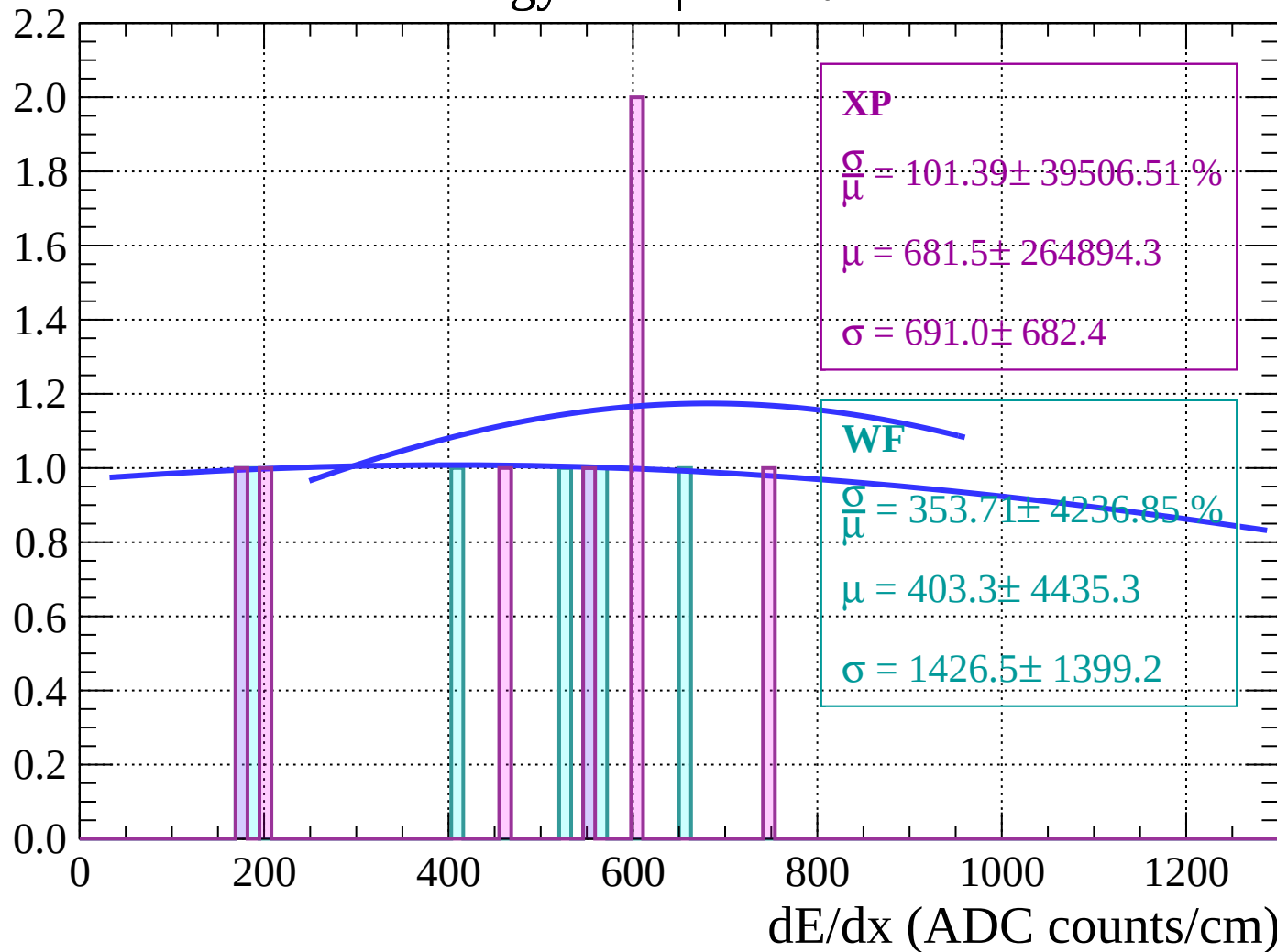
Energy loss | $58 < \theta < 60$

Count



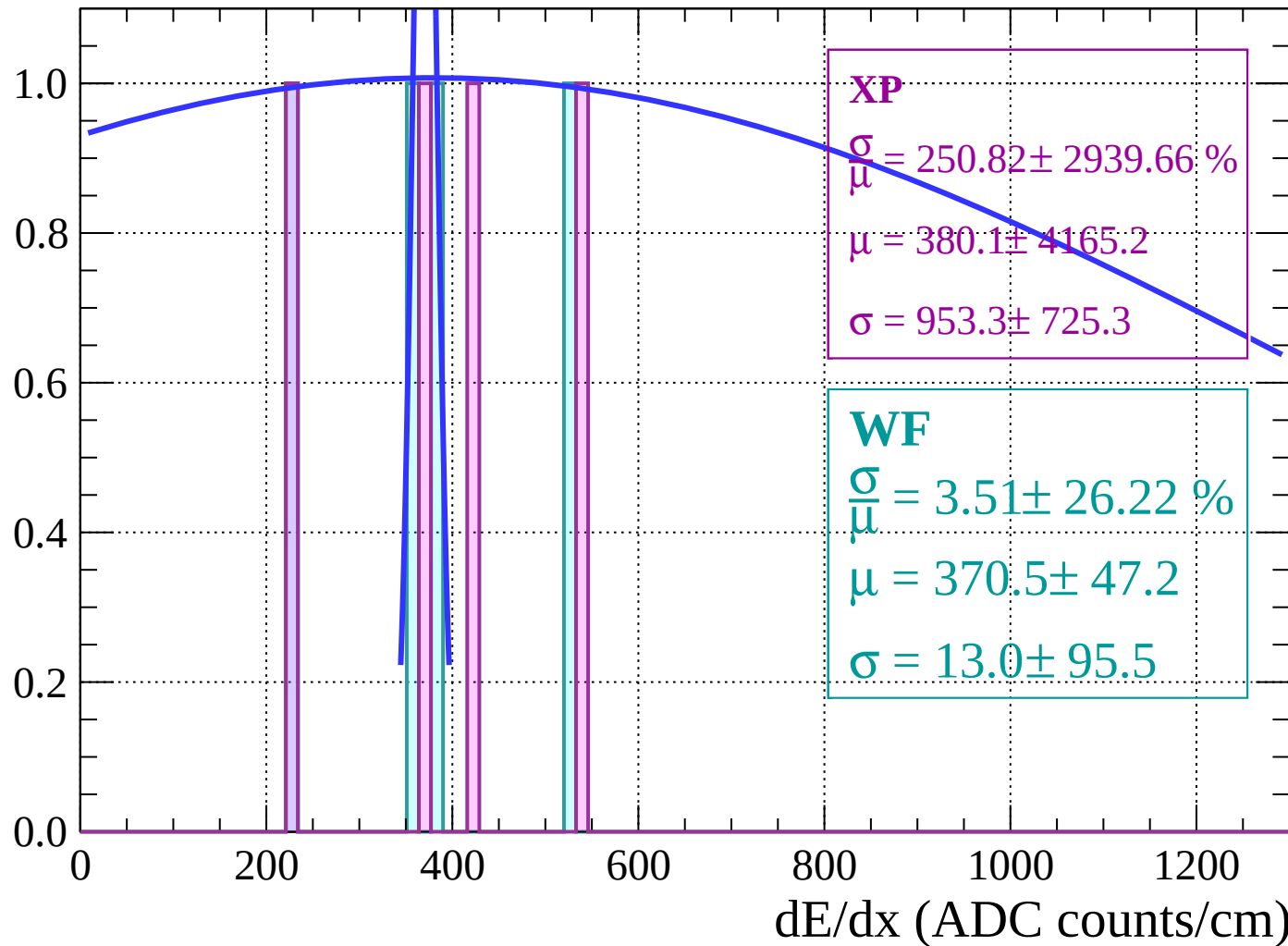
Energy loss | $60 < \theta < 62$

Count



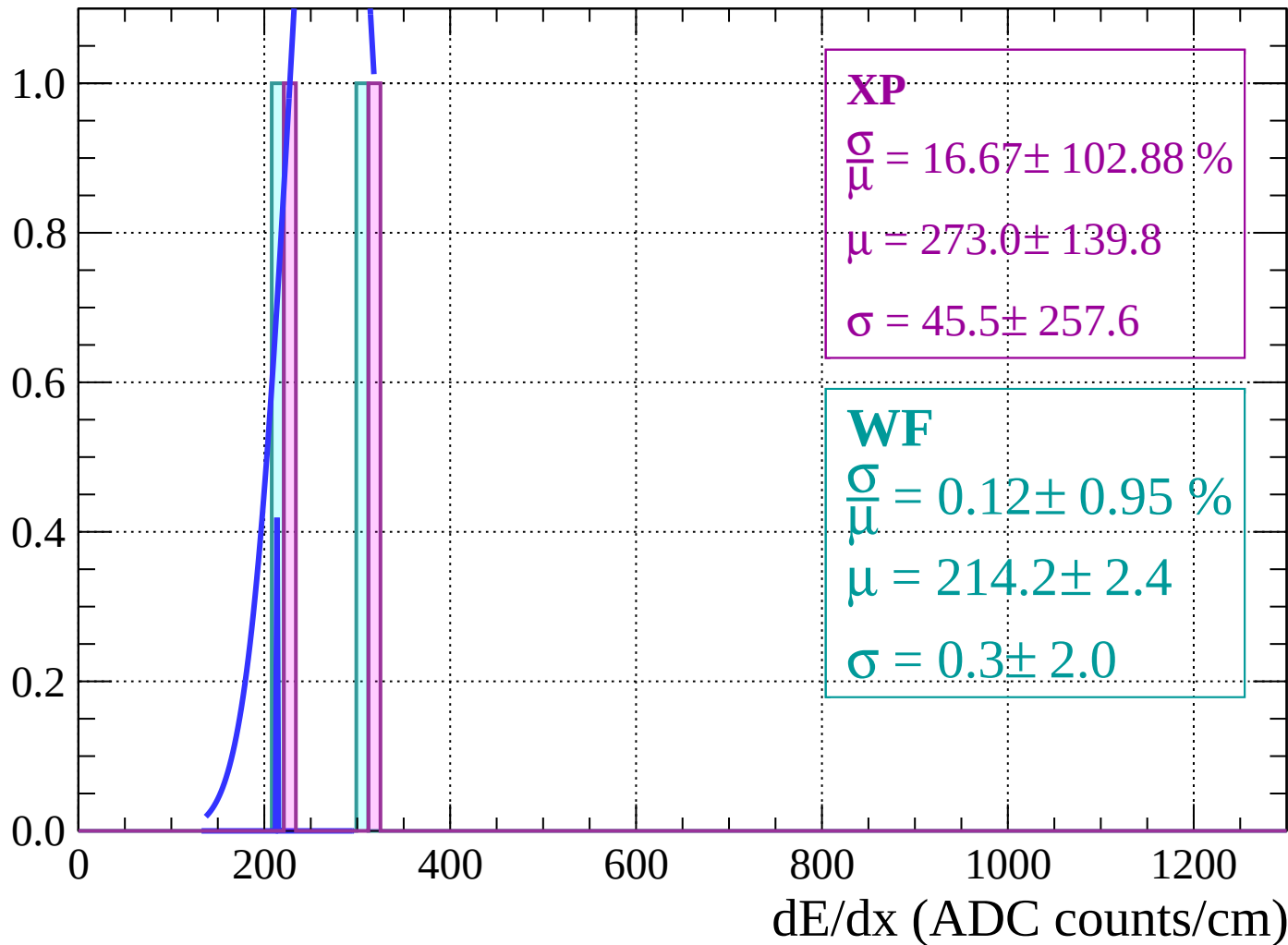
Energy loss | $62 < \theta < 64$

Count



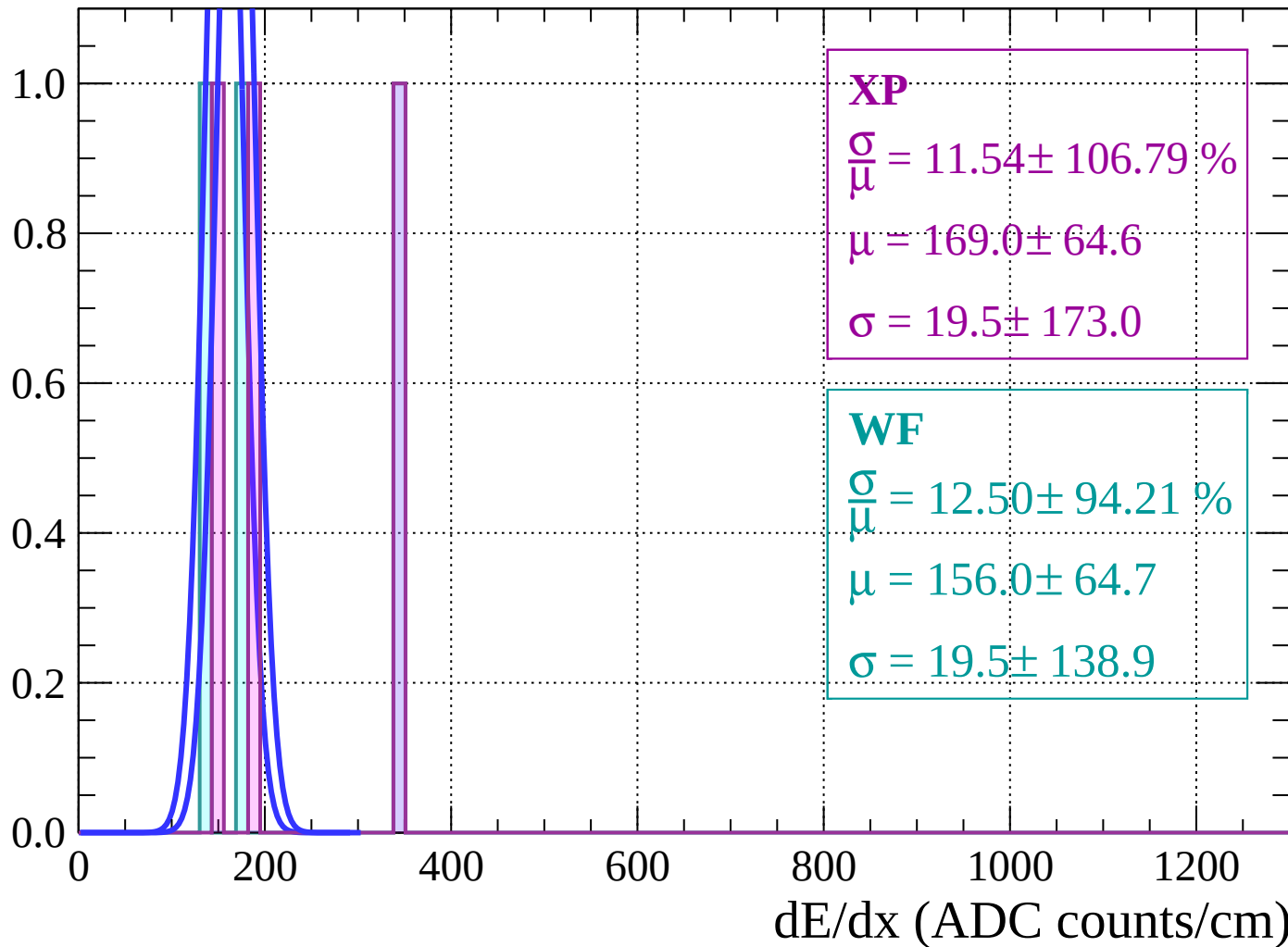
Energy loss | $64 < \theta < 66$

Count



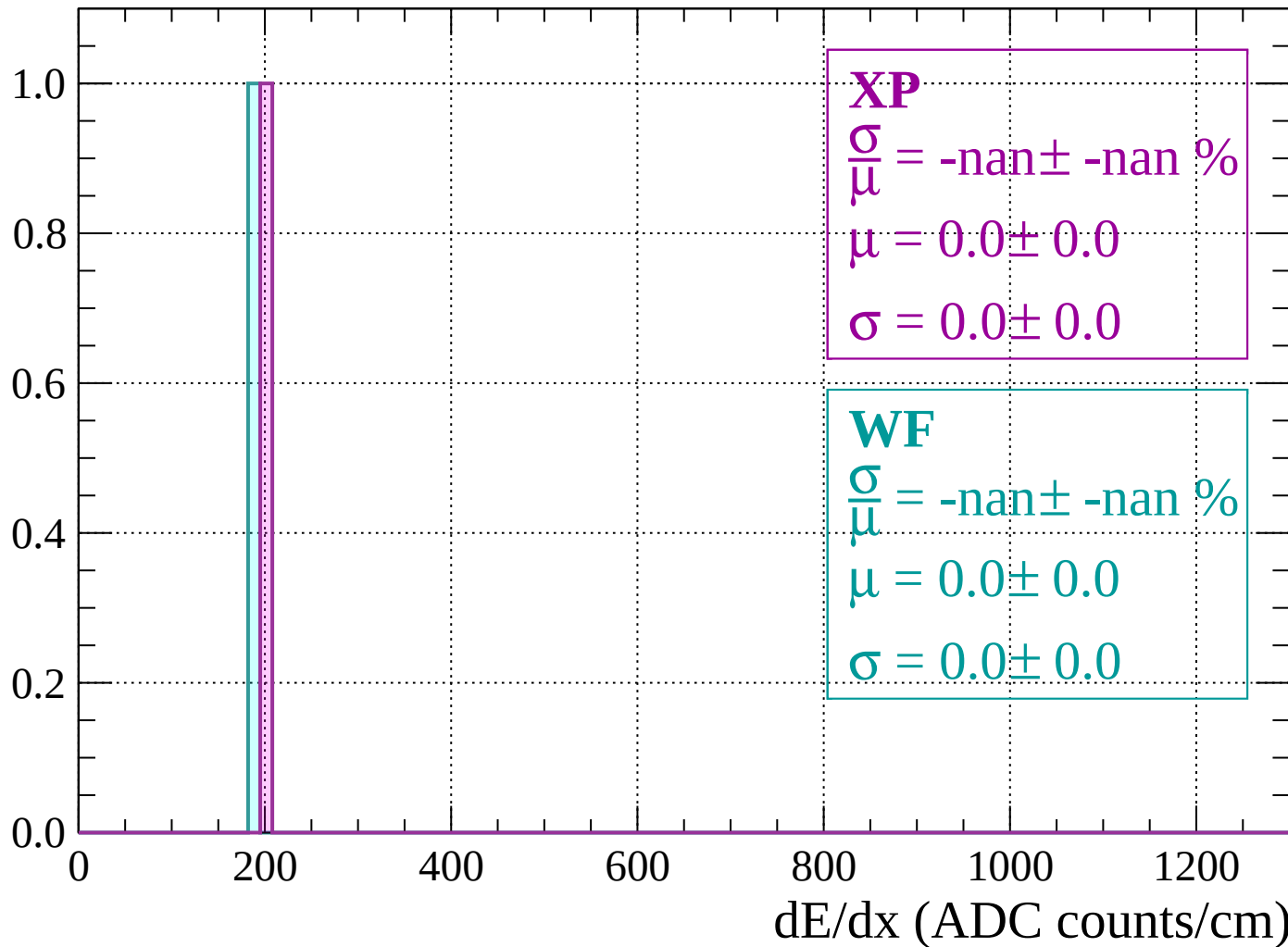
Energy loss | $66 < \theta < 68$

Count



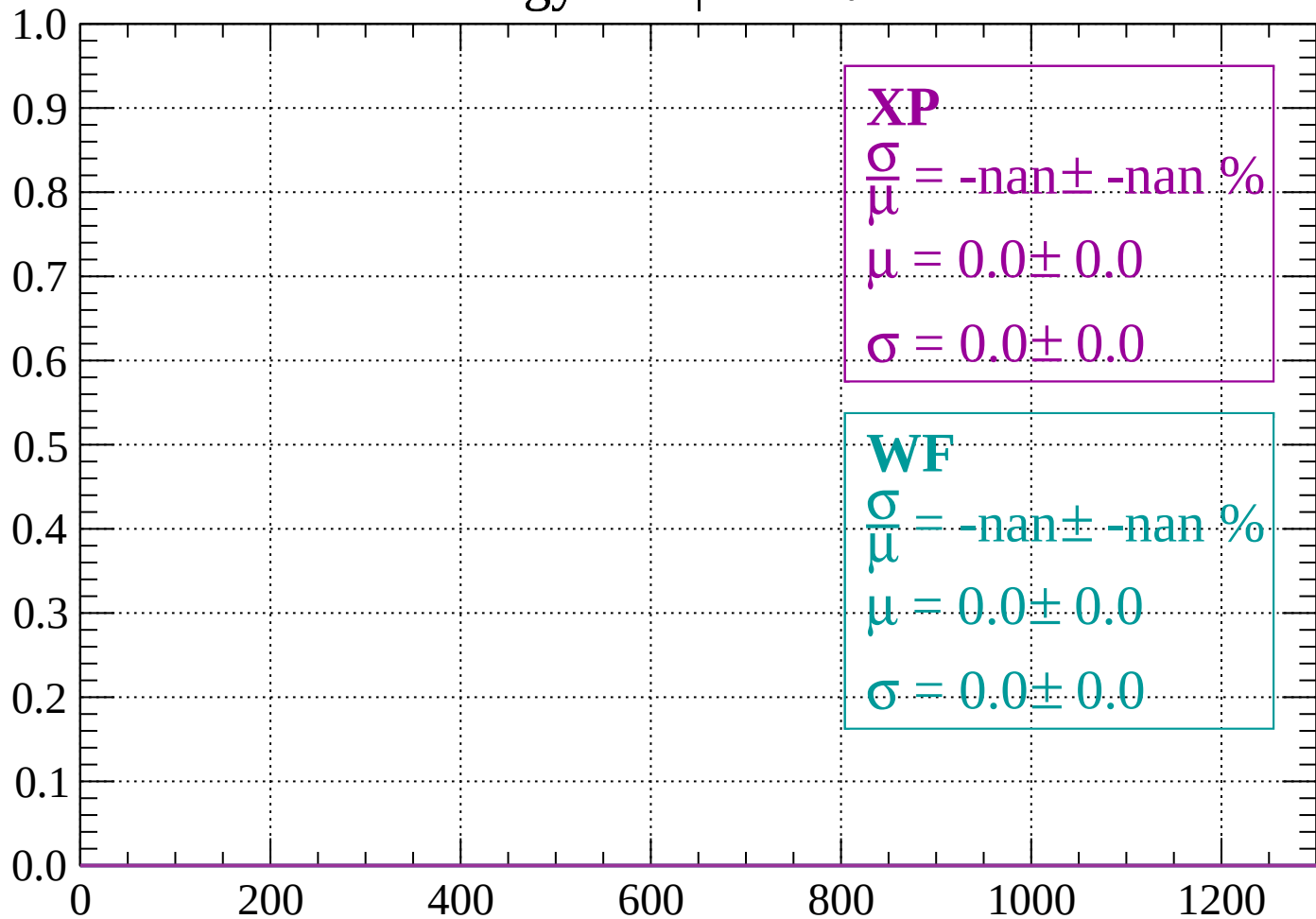
Energy loss | $68 < \theta < 70$

Count



Energy loss | $70 < \theta < 72$

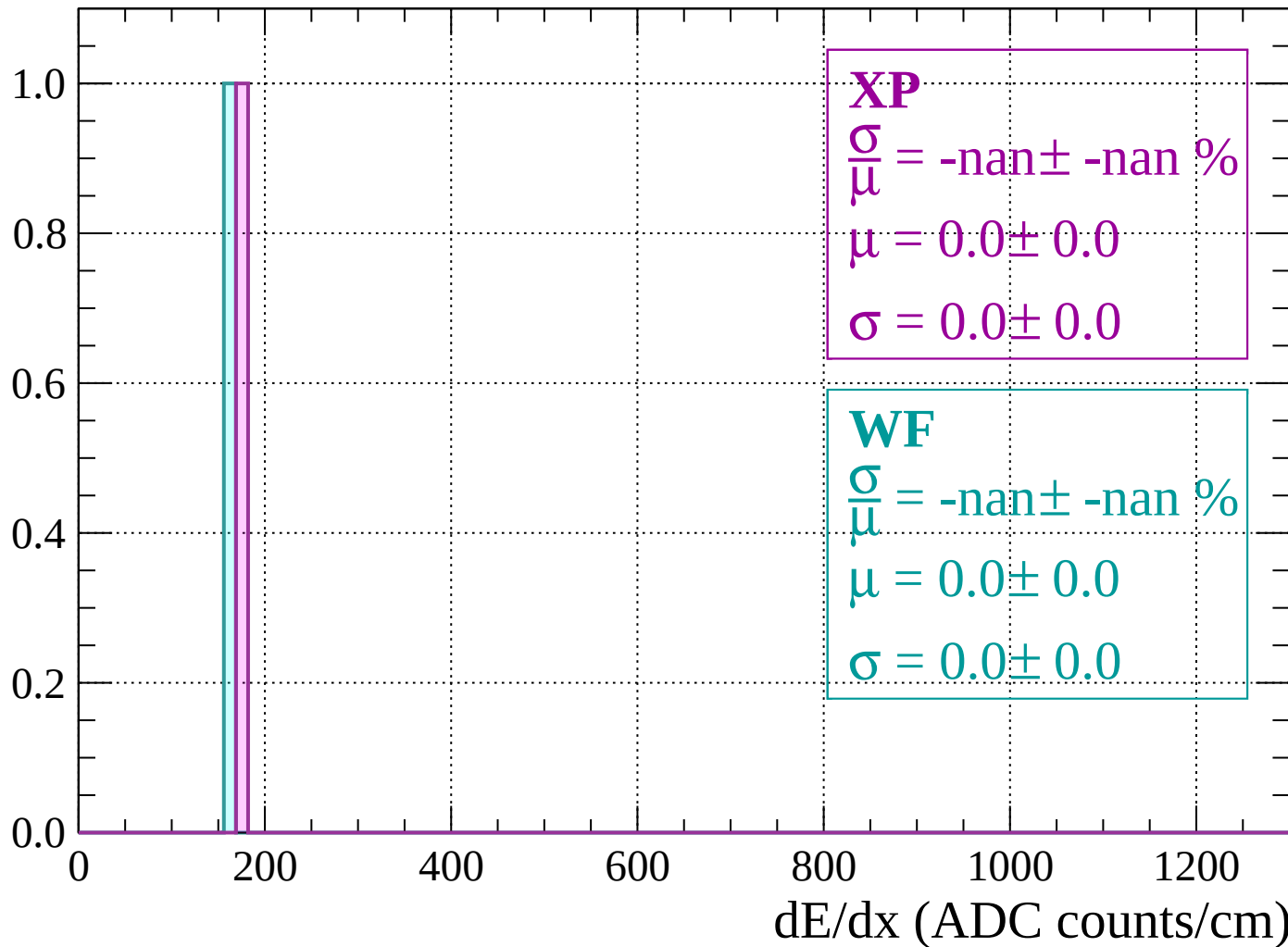
Count



dE/dx (ADC counts/cm)

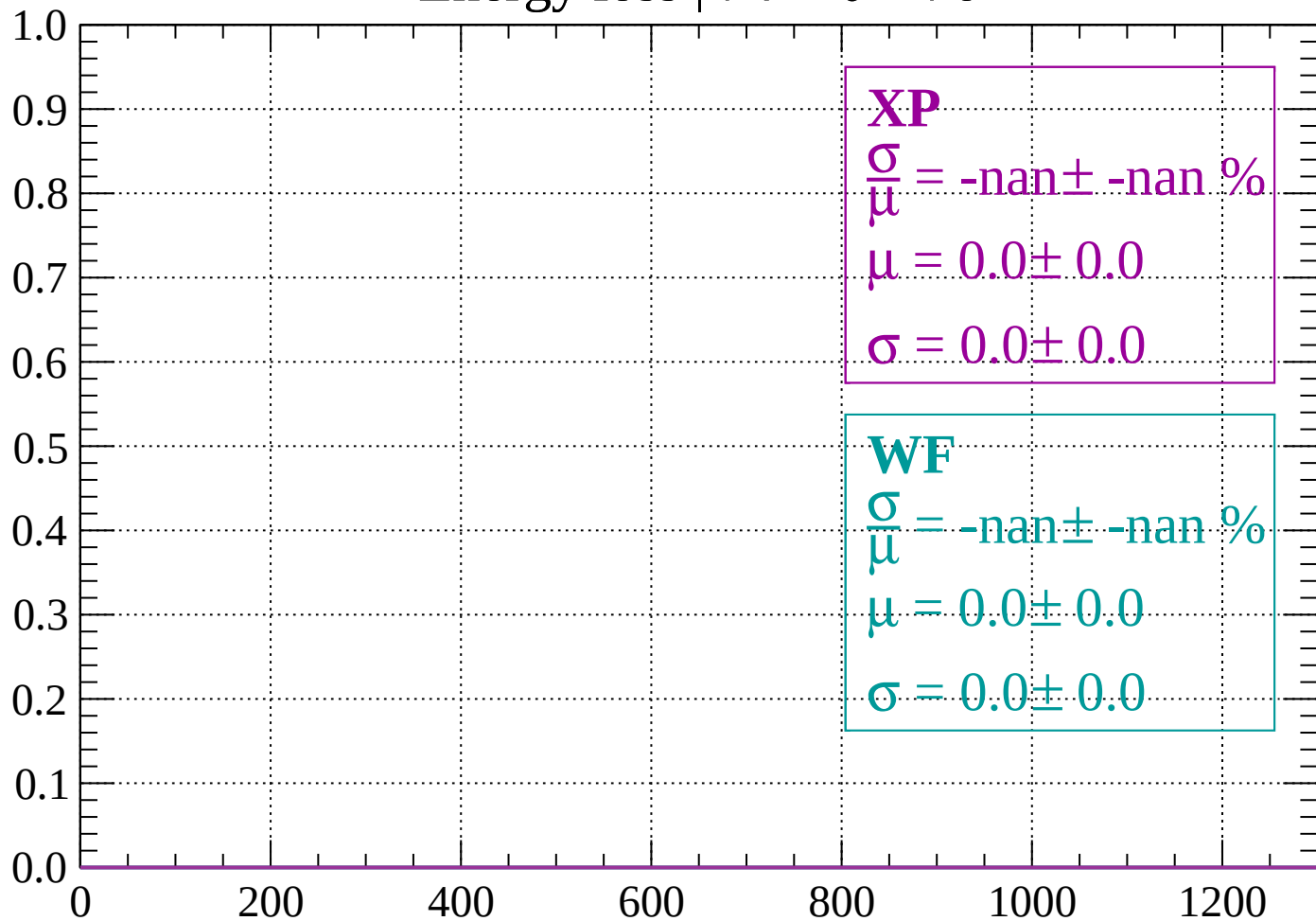
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

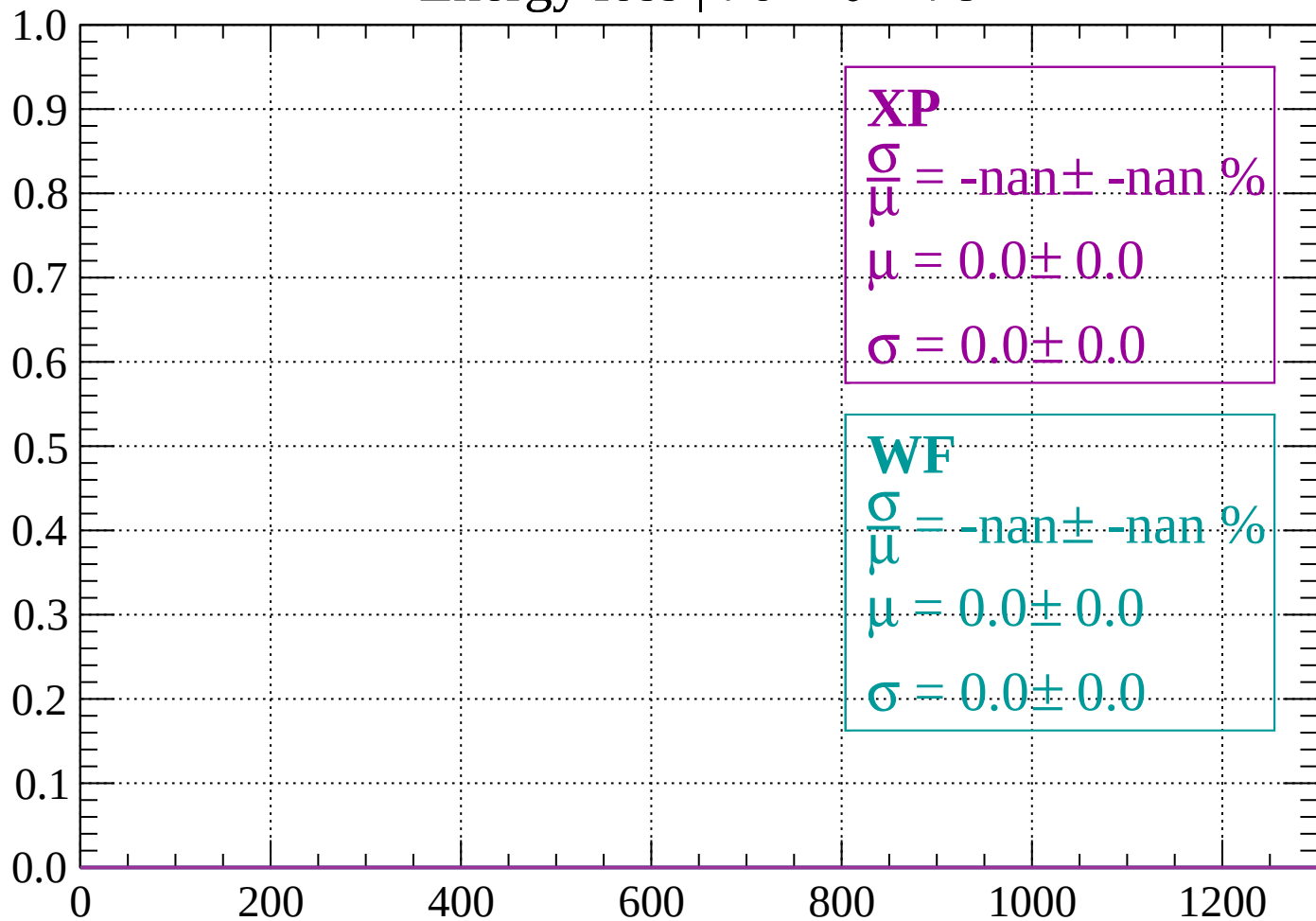
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

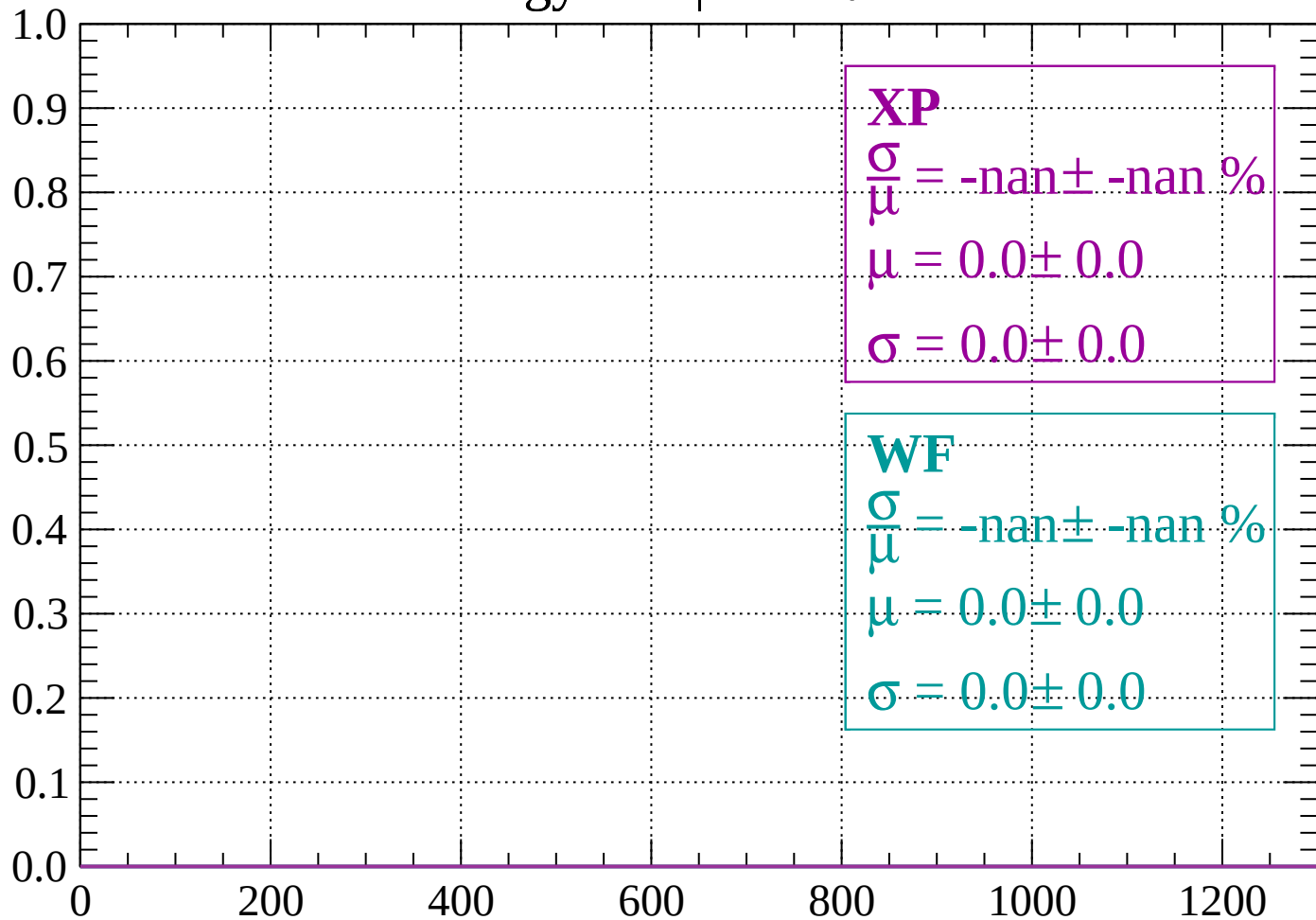
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

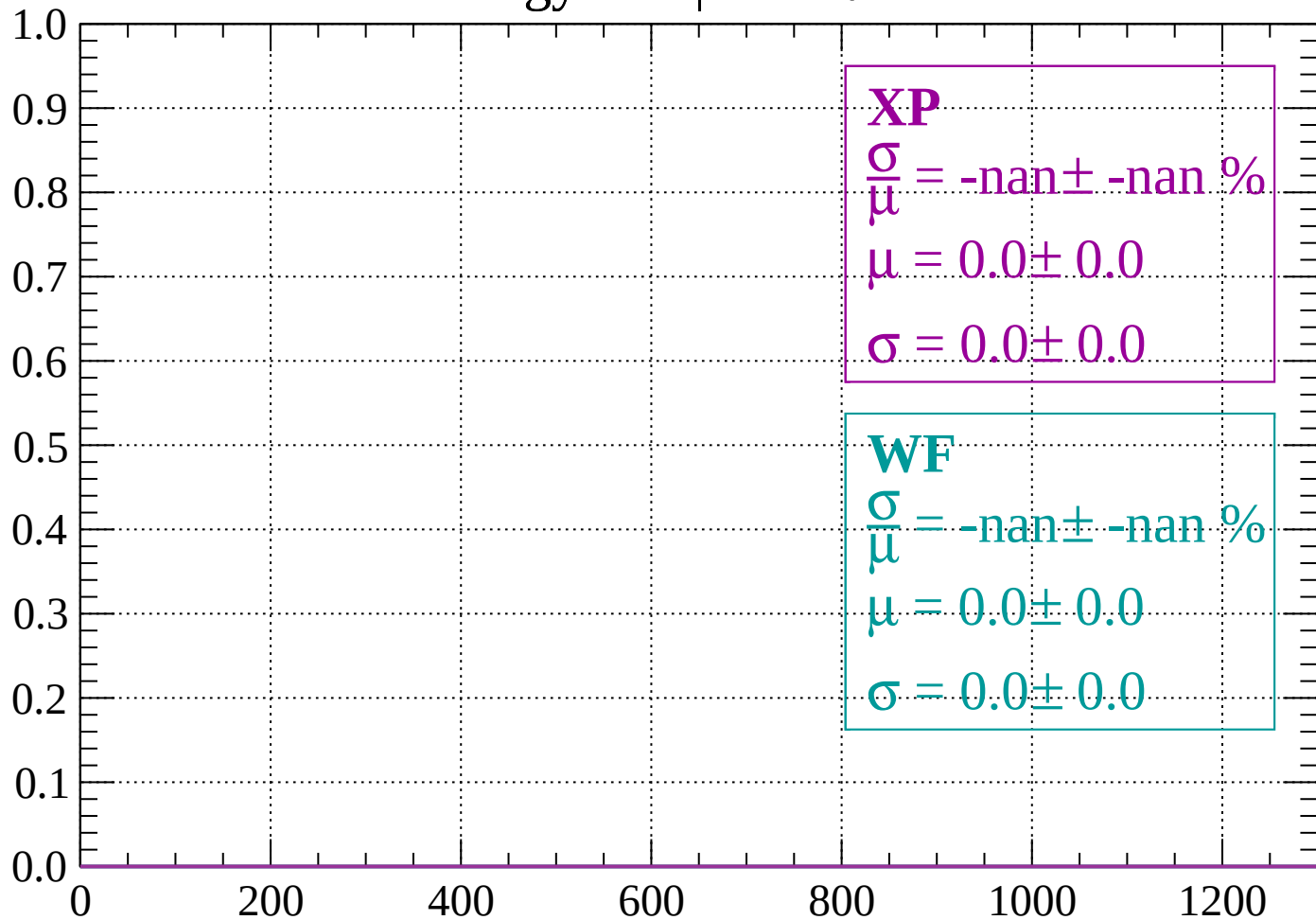
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

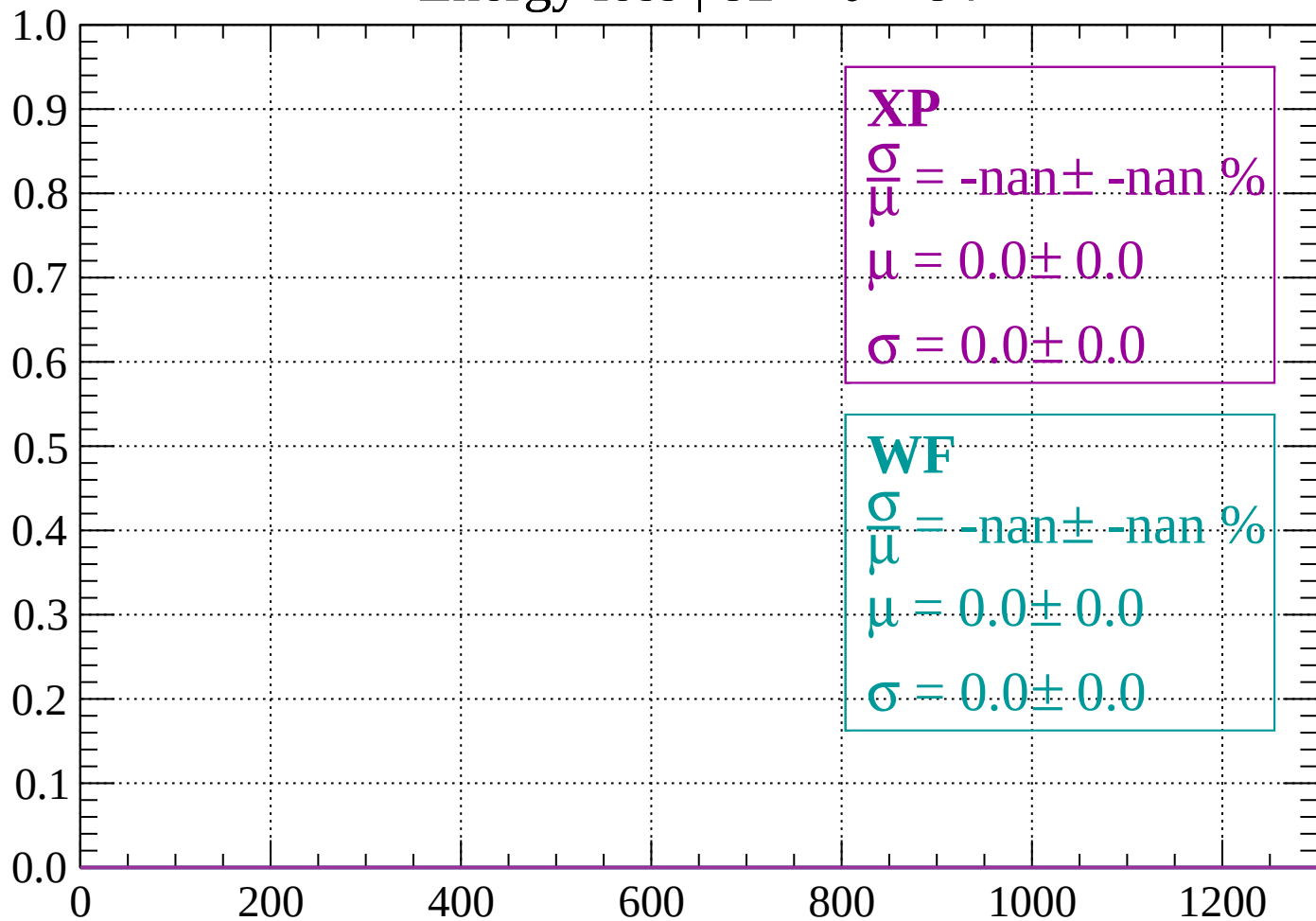
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

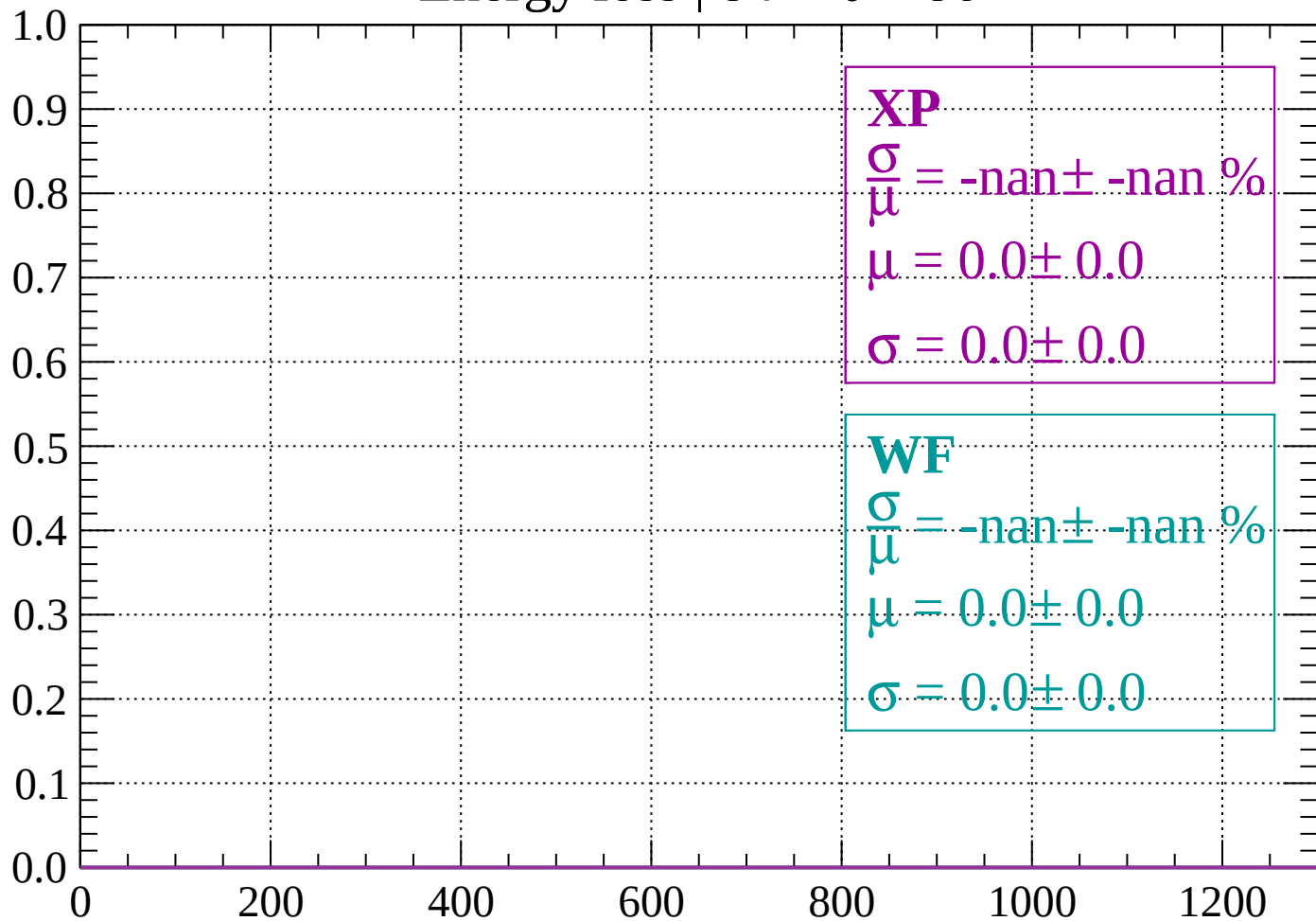
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

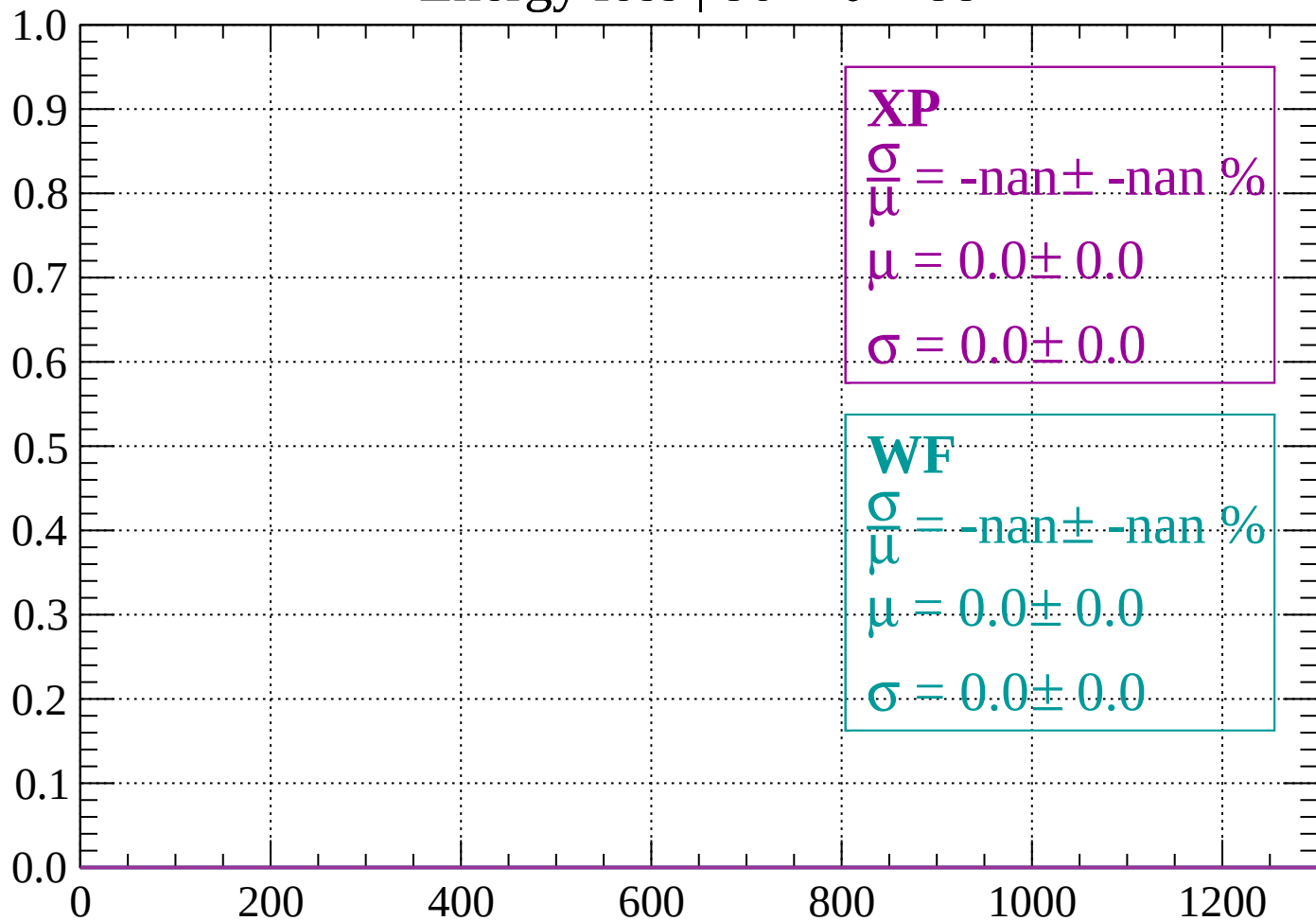
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

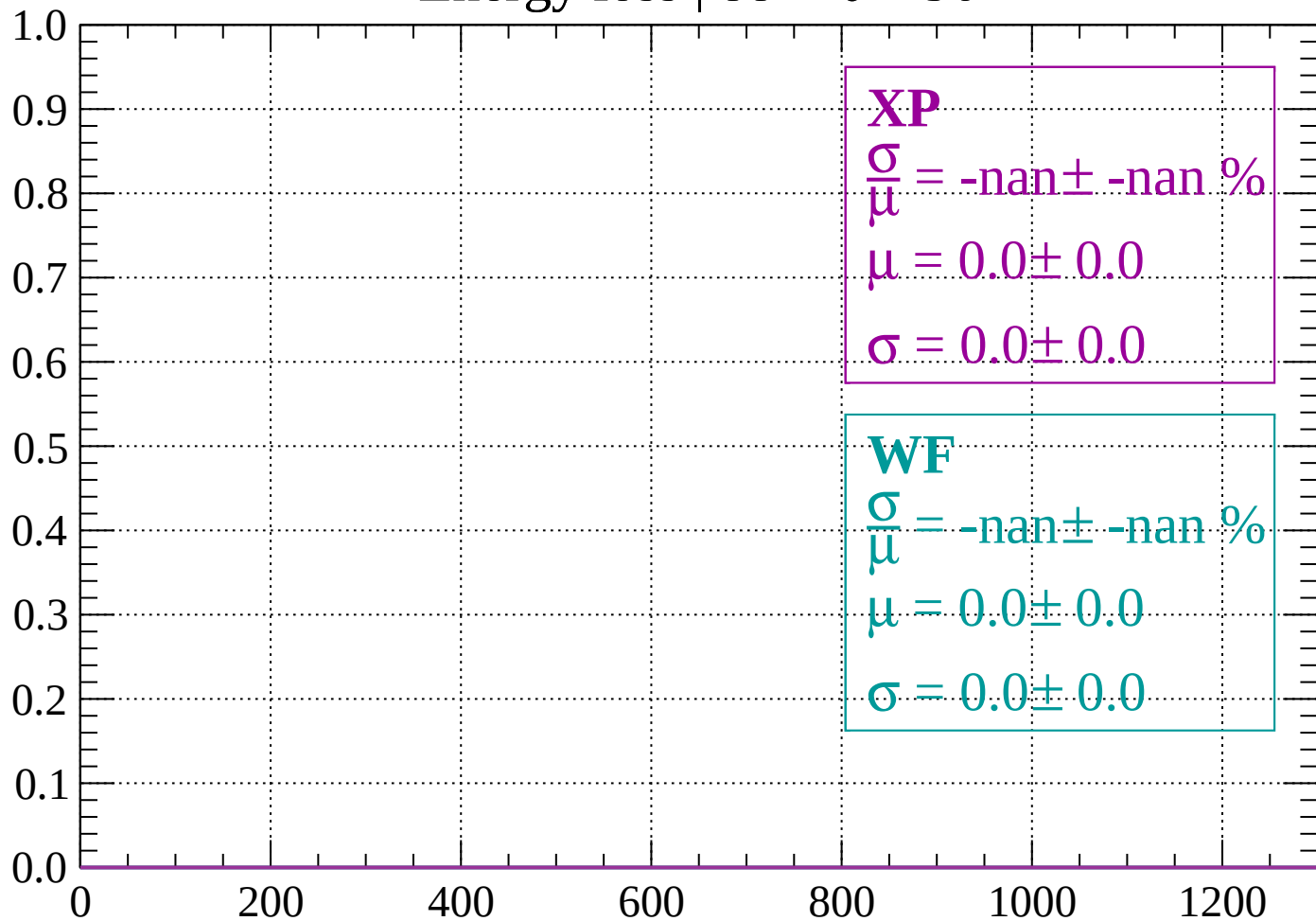
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

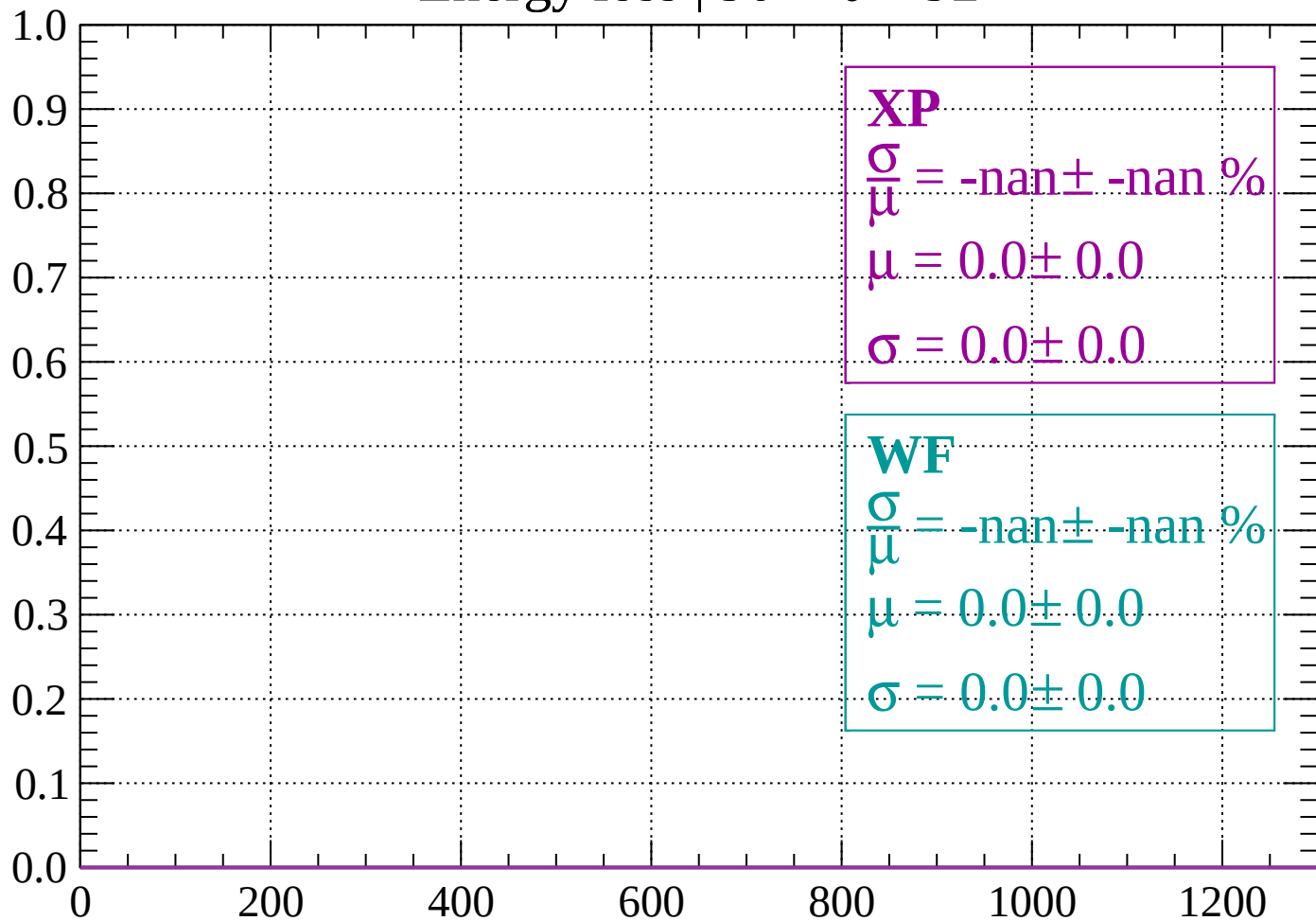
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

